



# Choupo

The Open Source Chemical Process Simulator

## Tutorials Guide

Choupo-dev

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## How to use this guide

---

Every Choupo tutorial is a complete, runnable case:

```
source etc/bashrc
runCase tutorials/<category>/<group>/<name>
```

This guide gives each tutorial the CONTEXT the case file cannot: the theory block at the head of each group states the physics and the equations you need (condensed – the Theory Guide carries the full derivations), and each entry tells you **what the case does** and **what to expect** before you run it. The per-tutorial text is assembled from the cases' own headers – the case file is always the source of truth, and reading it IS part of the exercise.

Three reading rules:

1. **Run first, read second, re-run third.** The log at verbosity 3 shows every Newton iteration, every K-value, every announcement – the glass box is the point.
2. **Expect the announcements.** Choupo never silently helps: initial guesses, bounds that bind, model fallbacks, unit overrides – all are printed. If a run is quiet about something, it did not happen.
3. **Golden numbers are commitments.** Cases carrying an **expected** file pin their KPIs; the validation AADs quoted in the entries are locked by the regression suite and re-verified on every change.

## Learning paths

---

**The process-simulation path.** Start with `flash01_benzene_toluene` (the Rachford–Rice loop, visible), then `flash` non-ideal cases (activity models bend the K-values), `distillation` (stages couple; Wang–Henke vs simultaneous), one `flowsheets` recycle case (tears and convergence), and close with a `gibbs` flame (equilibrium without mechanism). Then the batch side: an `ignition` case for stiffness – run the `_rk4` twin and watch the explicit method die – and a `ctrl` loop for feedback.

**The electrolytes path.** `pitzer01_nacl` ( $\gamma_{\pm}$ ,  $\varphi$ ,  $a_w$  of one salt), then `pitzer_gamma_hamer_wu` (the catalogue vs NBS data, zero refit), `pitzer_nacl_sp77_hot` (temperature, to 200°C), speciation (`scaling` cases: SI, pH, Davies-vs-Pitzer forking an engineering answer), the eNRTL series (`enrtl_mixed_nacl_ethanol_esteso` mixed-solvent; `enrtl_multisalt_nacl_kcl` two salts, two formulations, an experimental verdict), and the process payoff: evaporators and crystallisers where all of it sets duties and yields.

**The combustion path.** `gibbs06_h2_flame_radicals` (the radical pool at equilibrium; the  $c_p$  extrapolation trap), `ignition01_h2o2` (GRI kinetics, the stiffness lesson via the `_rk4` twin), `ignition02_h2air_slack` (quantitative:  $\tau_{\text{ign}}$  vs shock-tube data), `nox01_zeldovich_postflame` (thermal NO; kinetics meets Gibbs at 98.9%).

**The props-bench path.** `estimate_acetone` (CREATE: groups  $\rightarrow$  constants, error shown), a `scan` case (SEE), `vle01_consistency` (TEST the data), a `fit` case (Levenberg–Marquardt with identifiability), then the promote-and-use loop into a steady case. The Props Guide is this path’s companion.

## Category I: Steady-state flowsheets (choupoSolve)

---

Solved as one nonlinear system  $F(\mathbf{x}) = \mathbf{0}$ : units in sequence, recycles as tears, everything announced.

### absorption

**The theory you need.** Gas absorption transfers a solute from gas to liquid across stages; Henry's law ( $p_i = H_i x_i$ ) anchors the equilibrium at dilution. Expect the L/G ratio to control the operating line's slope, and a minimum solvent rate at the pinch with the equilibrium line. *Full derivation: Theory Guide, the absorption section.*

**absorber01\_NH3\_water**

tutorials/steady/absorption/absorber01\_NH3\_water



**What it does:** NH<sub>3</sub> absorption from air into water (Kremser, 6 stages)

**The story:** The textbook gas-absorption case: scrubbing ammonia out of an air stream with water, in a counter-current packed/plate column, solved by the Kremser group method (the built-in 'absorber').

Ammonia is very soluble in water (Henry's constant  $H \approx 0.99$  bar at 298 K  $\rightarrow K \approx 1$  at atmospheric pressure), so a modest water rate and a handful of stages recover most of it — unlike CO<sub>2</sub>/water ( $H \approx 1640$  bar), where water is a poor solvent.

Feed gas: 100 kmol/h, 10 % NH<sub>3</sub> / 90 % N<sub>2</sub> (N<sub>2</sub> = inert carrier) Solvent: 150 kmol/h pure water ( $L/V = 1.5$ ) Column: 6 theoretical stages, 298.15 K, 1 atm

The model is NON-ISOTHERMAL: ammonia dissolution is strongly exothermic ( $\Delta H \approx -34$  kJ/mol), so the heat of absorption heats the column. The energy balance is solved coupled with the mass balance.

Expected: liquid heats up  $\approx +17$  K (top 309 K, internal bulge  $\sim 318$  K, bottom 315 K — the classic absorber "temperature bulge") recovery\_NH<sub>3</sub>  $\approx 0.675$  (only  $\sim 68$  %, NOT the 97 % an isothermal Kremser would predict — because T rises, H rises, K rises, A falls: the exotherm self-limits the absorption. Real NH<sub>3</sub> / amine absorbers are cooled for this reason.) N<sub>2</sub> stays in the gas ( $K_{N_2} \gg 1 \rightarrow$  no absorption) NH<sub>3</sub> mass balance closes (in = clean-gas NH<sub>3</sub> + rich-liquid NH<sub>3</sub>)

Design question ("how many stages for 90 %?") is a DesignSpec on '\$stages', NOT a mode of this unit — the column has the stages it has; the recovery is what comes out.

**What to expect (golden-locked):**

| kind                                                             | unit/stream | key          | expected         |
|------------------------------------------------------------------|-------------|--------------|------------------|
| stream                                                           | cleanGas    | F            | 0.0259019084984  |
| stream                                                           | gasFeed     | F            | 0.02777777777778 |
| stream                                                           | richLiquid  | F            | 0.043542535946   |
| stream                                                           | solvent     | F            | 0.04166666666667 |
| kpi                                                              | scrubber    | A_N2         | 0.0027498297201  |
| kpi                                                              | scrubber    | A_NH3        | 1.53522727273    |
| kpi                                                              | scrubber    | F_cleanGas   | 0.0259019084984  |
| kpi                                                              | scrubber    | F_richLiquid | 0.043542535946   |
| kpi                                                              | scrubber    | K_N2         | 545.488322072    |
| kpi                                                              | scrubber    | K_NH3        | 0.977054034049   |
| kpi                                                              | scrubber    | L_in         | 0.04166666666667 |
| kpi                                                              | scrubber    | L_over_V     | 1.5              |
| kpi                                                              | scrubber    | T_bottom     | 315.417418114    |
| kpi                                                              | scrubber    | T_top        | 308.88795701     |
| ... and 10 more reference values in the case's <i>expected</i> . |             |              |                  |

**absorption01\_CO2\_water**

tutorials/steady/absorption/absorption01\_CO2\_water



**What it does:** Single-stage CO2 absorption in water (Henry's law)

**The story:** single-stage CO2 absorption in water, modelled via the v0.28 Henry's-law infrastructure.

Feed: a 5100 kmol/h mixed stream of CO2 / N2 / water already at 298.15 K and 5 bar, with mole fractions matching what you'd obtain by mixing 100 kmol/h of a 10/90 CO2/N2 sour-gas with 5000 kmol/h of liquid water (L/G ~ 50:1 mole). An IsothermalFlash then partitions the species between vapour and liquid.

Why a pre-mixed feed and not Mixer + Flash? The Mixer carries a liquid-enthalpy balance that needs Cp\_liq for every component, and CO2 is a permanent gas at 298 K with no liquid-Cp data (Tc = 304 K). Pre-mixing externally keeps the tutorial focused on the Henry's-law K-value selector — the point of the case.

Physics:  $H_{\text{CO2}}(298 \text{ K}) = 1.64\text{e}8 \text{ Pa} = 1640 \text{ bar}$  (Henry's law)  $K_{\text{CO2}} = H / P = 1.64\text{e}8 / 5\text{e}5 = 328$   $K_{\text{N2}} \gg 1$  (no Henry entry, no Antoine close to 298 K either — falls back to Raoult and Psat extrapolation)  $K_{\text{water}} \ll 1$  (most water stays in liquid;  $\text{Psat}(298) \sim 3 \text{ kPa}$ )

Observed result: cleanGas F = 66.0 kmol/h  $y_{\text{CO2}} = 0.123$   $y_{\text{N2}} = 0.871$   $y_{\text{water}} = 0.0063$  richLiquid F = 5034 kmol/h  $x_{\text{CO2}} = 3.75\text{e-}4$   $x_{\text{N2}} = 6.5\text{e-}3$   $x_{\text{water}} = 0.993$   $V/F = 0.013$   $K_{\text{CO2}} (y/x) = 328$  ? exactly  $H_{\text{CO2}}(298)/P = 1.64\text{e}8 / 5\text{e}5$   $K_{\text{water}} = 0.0063$  ?  $\text{Psat}_{\text{water}}(298)/P = 3.17 \text{ kPa} / 500 \text{ kPa}$  mass balance CO2: 10 in = 8.11 vapour + 1.89 liquid ~ 19 % of feed CO2 absorbed (a single-stage scrubber is far from perfect; an industrial column with 5-10 theoretical stages would recover > 95 %).

Toggling the GUI Units menu to Mass-basis shows the absorption as CO2 mass transfer between the two streams in kg/h, useful for scale-up arithmetic.

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key      | expected         |
|-----------------------------------------------------------------|-------------|----------|------------------|
| stream                                                          | cleanGas    | F        | 0.0164210205273  |
| stream                                                          | feed        | F        | 1.41666666667    |
| stream                                                          | richLiquid  | F        | 1.40024564614    |
| kpi                                                             | scrubber    | F_alpha  | 1.40024564614    |
| kpi                                                             | scrubber    | F_beta   | 0.0164210205273  |
| kpi                                                             | scrubber    | F_in     | 1.41666666667    |
| kpi                                                             | scrubber    | K_CO2    | 327.999999984    |
| kpi                                                             | scrubber    | K_N2     | 110.543208463    |
| kpi                                                             | scrubber    | K_water  | 0.00633349747981 |
| kpi                                                             | scrubber    | P        | 500000           |
| kpi                                                             | scrubber    | Q        | 0                |
| kpi                                                             | scrubber    | Q_kW     | 0                |
| kpi                                                             | scrubber    | T        | 298.15           |
| kpi                                                             | scrubber    | V_over_F | 0.0115913086075  |
| ... and 4 more reference values in the case's <i>expected</i> . |             |          |                  |

**acetone05\_luyben\_absorber**

tutorials/steady/absorption/acetone05\_luyben\_absorber



**What it does:** Luyben's acetone absorber: 9 stages, water washing acetone out of the hydrogen offgas – against his own acetone and bottoms

**The story:** The unit acetone04 was measured FOR. Luyben's absorber washes acetone out of the reactor's hydrogen-rich offgas with water: 9 stages at 1.5 atm, 20 kmol/h of water on top, 39.80 kmol/h of gas at the bottom.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a, Figure 1.

WHAT IS INDEPENDENT HERE, stated before the numbers because the last case in this family had to say the opposite. The gas inlet is Luyben's own ("Gas (to absorber)": 39.80 kmol/h, acetone 0.1147, IPA 0.0033) EXCEPT for its hydrogen and water, which were derived from the gas LEAVING using H2's inertness and a balance. So:

NOT independent – the outlet H2 and water (the inlet was built from them) INDEPENDENT – everything about the ACETONE, and the whole bottoms stream. Nothing in 0/ was built from those.

Luyben's answer, for the record, before this case is run:

bottoms 20.05 kmol/h acetone 0.1020 → 2.045 kmol/h absorbed offgas 39.76 kmol/h acetone 0.0634 → 2.521 kmol/h lost recovery 2.045 / 4.565 = 44.8 %

His own table closes on itself to 0.02 % in both acetone and water, which is two checks on a hand transcription and the reason those numbers are trusted as targets.

WHAT acetone04 PREDICTS FOR THIS UNIT. Measured there, on the same thermodynamics this case runs:  $\gamma_{\text{acetone}} = 10.27$  at 1 mol % in water and 318 K, giving  $K = 6.84$ . A model that holds acetone too weakly in the water will absorb LESS of it than Luyben does. That is the prediction; the run below is the test.

The thermodynamics is PREDICTIVE UNIFAC with H2 authorised outside the activity model – the same declared package as acetone03, for the same reasons, recorded in constant/thermoPhysPropDict.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected         |
|--------|-------------|---------------|------------------|
| stream | bottoms     | F             | 0.00574366573439 |
| stream | gasIn       | F             | 0.0110555555556  |
| stream | offgas      | F             | 0.0108674453767  |
| stream | washWater   | F             | 0.00555555555556 |
| kpi    | absorber    | A_H2          | 0.00044153375233 |
| kpi    | absorber    | A_acetone     | 0.0857422914441  |
| kpi    | absorber    | A_isopropanol | 4.51803004485    |
| kpi    | absorber    | A_water       | 8.0125706439     |
| kpi    | absorber    | F_cleanGas    | 0.0108674453767  |
| kpi    | absorber    | F_richLiquid  | 0.00574366573439 |
| kpi    | absorber    | K_H2          | 1138.10679288    |
| kpi    | absorber    | K_acetone     | 5.86073166871    |
| kpi    | absorber    | K_isopropanol | 0.11122382052    |
| kpi    | absorber    | K_water       | 0.0627155235376  |

... and 16 more reference values in the case's *expected*.

**extract01\_ethanol\_water\_benzene**

tutorials/steady/absorption/extract01\_ethanol\_water\_benzene



**What it does:** Counter-current LL extraction: recover ethanol from water with benzene

**The story:** Counter-current LL extraction column

Recover ETHANOL (the solute) from a dilute aqueous feed (WATER = the carrier) using BENZENE as the extracting SOLVENT. This is the classic liquid-liquid extraction problem: a multistage counter-current liquid-liquid cascade.

Topology (the ‘extractor’ unit op): the feed (water + ethanol) enters at stage 1; fresh solvent (benzene) enters at stage N; the two liquid phases pass counter-currently — the raffinate (water-rich, ethanol-depleted) flows  $1 \rightarrow N$ , the extract (benzene-rich, ethanol-enriched) flows  $N \rightarrow 1$ ; EXTRACT leaves stage 1, RAFFINATE leaves stage N.

Each theoretical stage is ONE liquid-liquid equilibrium, solved by the SHARED Gibbs-energy-minimisation LL flash (IsothermalFlash, phaseSet LL) — robust against the  $K = 1$  saddle that traps Newton-on-residual methods in LLE (CLAUDE.md sec.9). The cascade is torn by successive substitution (mirroring the distillation bubble-point loop).

Thermo: predictive UNIFAC for both liquid phases (water / ethanol / benzene groups inline), the same model used by the ternary LLE scan tutorial — no fitted pairs needed; self-DESCRIBING (inline manifest), UNIFAC records resolve from data/standards/ at runtime.

KPIs to watch: recovery\_ethanol (fraction of ethanol leaving in the extract), distribution\_ethanol ( $z_E / z_R$ ),  $F_{\text{extract}} / F_{\text{raffinate}}$ , massClosure (must be  $\sim 0$ ).

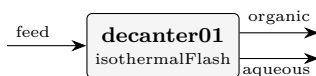
**What to expect (golden-locked):**

| kind   | unit/stream | key                  | expected         |
|--------|-------------|----------------------|------------------|
| stream | extract     | F                    | 0.0359293973028  |
| stream | feed        | F                    | 0.0277777777778  |
| stream | raffinate   | F                    | 0.0251772962492  |
| stream | solvent     | F                    | 0.0333333333333  |
| kpi    | extractor01 | F_extract            | 0.0359293973028  |
| kpi    | extractor01 | F_feed               | 0.0277777777778  |
| kpi    | extractor01 | F_raffinate          | 0.0251772962492  |
| kpi    | extractor01 | F_solvent            | 0.0333333333333  |
| kpi    | extractor01 | P                    | 101325           |
| kpi    | extractor01 | T                    | 298.15           |
| kpi    | extractor01 | converged            | 1                |
| kpi    | extractor01 | distribution_benzene | 295.27509389     |
| kpi    | extractor01 | distribution_ethanol | 0.555106921422   |
| kpi    | extractor01 | distribution_water   | 0.00706545462823 |

... and 13 more reference values in the case's *expected*.

**extraction01\_waterButanol**

tutorials/steady/absorption/extraction01\_waterButanol



**What it does:** Water/n-butanol LL flash at 298 K (experimental)

**The story:** Water + 1-butanol at 25 °C. This system has a well-known liquid-liquid miscibility gap below 125.5 °C (upper critical solution temperature). A 70/30 water/nButanol feed (inside the gap) → splits into a water-rich (aqueous) phase ~98 mol% water and a butanol-rich (organic) phase ~59 mol% nButanol / 41 mol% water.

Demonstrates: ‘phases (...)’ syntax with TWO liquid phases ‘phaseSet LL’ in the IsothermalFlash Michelsen TPD stability test to find a robust initial guess

*Water + 1-butanol at 25 °C and 1 atm. The pair has a well-known liquid-liquid miscibility gap below its upper critical solution temperature (125.5 °C), so a feed whose composition lies inside the gap spontaneously splits into two liquid phases. This case feeds a 70/30 water/nButanol mixture (inside the gap) to an isothermal decanter and lets the flash find the two-liquid split.*

*A single feed splits into two liquids, both at 298.15 K,  $v_f = 0$ :*

*The split is found by Michelsen TPD stability testing (to seed a robust initial guess) followed by direct Gibbs-energy minimisation — the LL path the engine uses precisely because a Newton/successive-substitution flash on a symmetric  $\gamma$ -model would collapse to the trivial  $K = 1$  saddle.*

*- phases (...) syntax with TWO liquid phases; - phaseSet LL in the IsothermalFlash; - the Michelsen TPD stability test as the initial-guess generator.*

*- The LL flash reuses the VL solver’s printout, so the two liquids are shown under Vapor flow V / Liquid flow L headings (and x/y/K labels). Read them as phase- $\alpha$  (organic) and phase- $\beta$  (aqueous), not as a vapour and a liquid. - The decanter is isothermal and adiabatic in reality (an enthalpy hand-check gives  $H_{in} \approx H_{out}$ ), so the physically correct duty is  $\approx 0$ . The KPI Q and any utility/cost the run reports for this LL split are a known artifact of the shared flash duty path — do not cost this unit from them.*

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected        |
|--------|-------------|--------------|-----------------|
| stream | aqueous     | F            | 0.0140665296362 |
| stream | feed        | F            | 0.0277777777778 |
| stream | organic     | F            | 0.0137112481416 |
| kpi    | decanter01  | F_alpha      | 0.0140665296362 |
| kpi    | decanter01  | F_beta       | 0.0137112481416 |
| kpi    | decanter01  | F_in         | 0.0277777777778 |
| kpi    | decanter01  | P            | 101325          |
| kpi    | decanter01  | T            | 298.15          |
| kpi    | decanter01  | betaFraction | 0.493604933098  |
| kpi    | decanter01  | phaseSet     | 1               |
| stream | aqueous     | T            | 298.15          |
| stream | feed        | T            | 298.15          |
| stream | organic     | T            | 298.15          |



**stripper01\_NH3\_water**

tutorials/steady/absorption/stripper01\_NH3\_water



**What it does:** NH3 stripping from water with air (Kremser, 6 stages, warm)

**The story:** The inverse of absorber01: instead of scrubbing NH3 OUT of a gas, here we strip NH3 OUT of water with a lean gas (air ~ N2). Counter-current Kremser; the rich liquid enters the top, the stripping gas the bottom.

Run warm (330 K): NH3 is less soluble hot (K rises), so the stripping factor  $S = K V/L$  exceeds 1 and the strip is efficient. But desorption is endothermic, so the column cools – which lowers K and self-limits the stripping (the mirror image of the absorber's exotherm).

Liquid: 100 kmol/h, 5 % NH3 in water, 330 K Gas: 100 kmol/h N2 ( $V/L = 1$ ), 330 K Column: 6 theoretical stages

Expected:  $S_{\text{NH3}} > 1 \rightarrow$  most of the NH3 stripped into the gas; stripped liquid nearly NH3-free; rich gas carries the NH3; mass closes 100 %.

**What to expect (golden-locked):**

| kind   | unit/stream    | key              | expected        |
|--------|----------------|------------------|-----------------|
| stream | liquidFeed     | F                | 0.0277777777778 |
| stream | richGas        | F                | 0.029140357434  |
| stream | strippedLiquid | F                | 0.0264151981216 |
| stream | strippingGas   | F                | 0.0277777777778 |
| kpi    | stripper       | F_richGas        | 0.029140357434  |
| kpi    | stripper       | F_strippedLiquid | 0.0264151981216 |
| kpi    | stripper       | K_NH3            | 3.62862883208   |
| kpi    | stripper       | K_water          | 0.170562530402  |
| kpi    | stripper       | L_in             | 0.0277777777778 |
| kpi    | stripper       | S_NH3            | 3.62862883208   |
| kpi    | stripper       | S_water          | 0.170562530402  |
| kpi    | stripper       | T_bottom         | 313.524974667   |
| kpi    | stripper       | T_top            | 315.973930965   |
| kpi    | stripper       | V_in             | 0.0277777777778 |

... and 10 more reference values in the case's *expected*.

**crystallisation**

**The theory you need.** Crystallisation removes solute when the solution crosses saturation:  $m > m_{\text{sat}}(T)$ , with  $K_{\text{sp}}(T)$  from  $\Delta G_{\text{diss}}$  and its  $T$ -dependence from  $\Delta H_{\text{diss}}$  (van 't Hoff). The heat of crystallisation is ION-DERIVED –  $\Delta H_{\text{cryst}} = -(\Delta H_{\text{soln}}^{\infty} + \bar{L}_2(m_{\text{sat}}))$  – never a stored constant (a solid's formation enthalpy is  $\sum \nu_i h_{f,i}^{\text{aq}} - \Delta H_{\text{soln}}$ , derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant,  $\gamma_{\pm}$  rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

**crystalliser01\_sugar**

tutorials/steady/crystallisation/crystalliser01\_sugar

**What it does:** Cooling crystalliser: sugar solution 80 → 20 C, yield by solubility**The story:** / flowsheetDict (LEAF node)

A hot sugar solution ( $z_{\text{sucrose}} 0.15$ ,  $\sim 3.35$  kg/kg water, below saturation at 80 C where  $c_{\text{sat}} \sim 3.8$ ) is cooled to 20 C ( $c_{\text{sat}} \sim 2.0$  kg/kg). The excess crystallises. Magma = crystals (s[]) + saturated mother liquor.

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected         |
|--------|-------------|------------------|------------------|
| stream | feed        | F                | 0.02777777777778 |
| stream | magma       | F                | 0.02777777777778 |
| kpi    | cryst       | Q_removed        | 271677.114808    |
| kpi    | cryst       | T_op             | 293.15           |
| kpi    | cryst       | c_sat            | 2.026            |
| kpi    | cryst       | crystal_flow     | 0.564469958333   |
| kpi    | cryst       | crystal_mol      | 0.00164906487154 |
| kpi    | cryst       | liquorFlow       | 0.0261287129062  |
| kpi    | cryst       | solute_dissolved | 0.861767541667   |
| kpi    | cryst       | solute_in        | 1.4262375        |
| kpi    | cryst       | yield            | 0.395775569169   |
| stream | feed        | T                | 353.15           |
| stream | magma       | T                | 293.15           |
| kpi    | cryst       | Q_kW             | -271.677114808   |

**crystalliser02\_msmpr**

tutorials/steady/crystallisation/crystalliser02\_msmpr

**What it does:** MSMPR sugar crystalliser: population balance by method of moments**The story:** / flowsheetDict (LEAF node)

Continuous MSMPR crystalliser. A hot near-saturated sugar solution is fed to a cooled, well-mixed vessel of working volume  $V$ ; the magma (crystals + mother liquor) is withdrawn at the same volumetric rate, so the residence time is  $\tau = V / Q$ . ‘model MSMPR’ selects the population-balance path (method of moments); the kinetics are referenced from constant/crystallisation.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.0277777777778   |
| stream | magma       | F            | 0.0277777777778   |
| kpi    | cryst       | CV           | 1                 |
| kpi    | cryst       | L_dominant   | 0.000223952804481 |
| kpi    | cryst       | L_meanMass   | 0.000298603739308 |
| kpi    | cryst       | L_meanNumber | 7.46509348269e-05 |
| kpi    | cryst       | Q_removed    | 269525.057151     |
| kpi    | cryst       | T_op         | 293.15            |
| kpi    | cryst       | c_sat        | 2.026             |
| kpi    | cryst       | crystal_flow | 0.396190602513    |
| kpi    | cryst       | crystal_mol  | 0.00115744690287  |
| kpi    | cryst       | growthRate   | 1.95272330048e-08 |
| kpi    | cryst       | liquorFlow   | 0.0266203308749   |
| kpi    | cryst       | magmaDensity | 302.920529908     |

... and 13 more reference values in the case's *expected*.

**crystalliser03\_pbe\_size\_independent**

tutorials/steady/crystallisation/crystalliser03\_pbe\_size\_independent

**What it does:** Crystalliser FVM-PBE: sanity vs MSMPR (size-independent growth)**The story:** / flowsheetDict (LEAF node)

Same physical setup as crystalliser02\_msmpr (hot sugar syrup, 5 m<sup>3</sup> vessel at 20 °C, k<sub>g</sub> 1e-7, k<sub>b</sub> 1e9) – only the solver method changes: ‘model FVM’ discretises the population balance  $n(L)$  on Nbins evenly-spaced size bins, marching the steady PBE forward in L with first-order upwind. With sizeFactor 0 (the default),  $G(L)$  is constant and the FVM result must match the analytical MSMPR exponential.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | feed        | F             | 0.02777777777778  |
| stream | magma       | F             | 0.02777777777778  |
| kpi    | cryst       | CV            | 0.999540291345    |
| kpi    | cryst       | L_dominant    | 0.000224609375    |
| kpi    | cryst       | L_meanMass    | 0.000298633437171 |
| kpi    | cryst       | L_meanNumber  | 7.48298114726e-05 |
| kpi    | cryst       | Q_removed     | 269576.1605       |
| kpi    | cryst       | T_op          | 293.15            |
| kpi    | cryst       | bins          | 256               |
| kpi    | cryst       | c_sat         | 2.026             |
| kpi    | cryst       | crystal_flow  | 0.400186610054    |
| kpi    | cryst       | crystal_mol   | 0.00116912099742  |
| kpi    | cryst       | growthRate_L0 | 1.90635340048e-08 |
| kpi    | cryst       | liquorFlow    | 0.0266086567804   |

... and 14 more reference values in the case's *expected*.

**crystalliser04\_size\_dependent\_growth**

tutorials/steady/crystallisation/crystalliser04\_size\_dependent\_growth

**What it does:** Crystalliser FVM-PBE with size-dependent growth (sizeFactor > 0)**The story:** / flowsheetDict (LEAF node)

Same hot-sugar feed and 5 m<sup>3</sup> vessel at 20 °C as crystalliser03, only ‘sizeFactor’ is non-zero — the size-dependent growth law that breaks the method-of-moments closure but is handled natively by the FVM.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | feed        | F             | 0.02777777777778  |
| stream | magma       | F             | 0.02777777777778  |
| kpi    | cryst       | CV            | 1.03562508094     |
| kpi    | cryst       | L_dominant    | 0.000232421875    |
| kpi    | cryst       | L_meanMass    | 0.000329374475484 |
| kpi    | cryst       | L_meanNumber  | 7.45169890057e-05 |
| kpi    | cryst       | Q_removed     | 269660.031771     |
| kpi    | cryst       | T_op          | 293.15            |
| kpi    | cryst       | bins          | 256               |
| kpi    | cryst       | c_sat         | 2.026             |
| kpi    | cryst       | crystal_flow  | 0.406744893212    |
| kpi    | cryst       | crystal_mol   | 0.00118828062534  |
| kpi    | cryst       | growthRate_L0 | 1.83025070562e-08 |
| kpi    | cryst       | liquorFlow    | 0.0265894971524   |

... and 14 more reference values in the case's *expected*.

**crystalliser05\_nacl\_pitzer**

tutorials/steady/crystallisation/crystalliser05\_nacl\_pitzer

**What it does:** NaCl crystalliser: saturation from the PITZER ion activity ( $K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$ ). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

**The story:** NaCl crystalliser whose SATURATION comes from the Pitzer ion activity ( $K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$ ), via the ‘electrolyte {}’ block – NaCl carries no solubility{} curve, so the electrolyte model is what makes this possible. The feed is a SUPERSATURATED brine (~7.6 mol/kg, as it would leave an evaporator); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess. Components live case-local in constant/components/; other curated records resolve from data/standards/ at runtime.

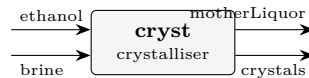
**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.0277777777778   |
| stream | magma       | F                | 0.0277777777778   |
| kpi    | cryst       | Ksp_activity     | 38.4846213082     |
| kpi    | cryst       | Q_removed        | 1251.26049958     |
| kpi    | cryst       | T_op             | 298.15            |
| kpi    | cryst       | c_sat            | 0.35905536        |
| kpi    | cryst       | crystal_flow     | 0.036683987968    |
| kpi    | cryst       | crystal_mol      | 0.000627720533333 |
| kpi    | cryst       | gamma_pm_sat     | 1.00970010562     |
| kpi    | cryst       | liquorFlow       | 0.0271500572444   |
| kpi    | cryst       | m_sat            | 6.144             |
| kpi    | cryst       | solute_dissolved | 0.158116012032    |
| kpi    | cryst       | solute_in        | 0.1948            |
| kpi    | cryst       | yield            | 0.18831616        |

...and 3 more reference values in the case's *expected*.

**crystalliser06\_nacl\_antisolvent**

tutorials/steady/crystallisation/crystalliser06\_nacl\_antisolvent



**What it does:** Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises  $\gamma_{\text{pm}}$ , and DROPS the saturation molality  $m_{\text{sat}}$  – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product  $K_{\text{sp}}$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\text{pm}}$ .

**The story:** ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity  $\gamma_{\text{pm}}$  (eNRTL mixed-solvent + Born), so the saturation molality  $m_{\text{sat}}$  COLLAPSES (6.14  $\rightarrow$  ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product  $K_{\text{sp}} = (\gamma_{\text{pm}} * m_{\text{sat}})^2$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\text{pm}}$  at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine ( $x_{\text{water}}$  0.900,  $x_{\text{NaCl}}$  0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol  $\rightarrow$  salt-free  $x_{\text{ethanol}}$  ~ 0.40. Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

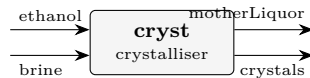
**What to expect (golden-locked):**

| kind   | unit/stream  | key               | expected          |
|--------|--------------|-------------------|-------------------|
| stream | brine        | F                 | 0.02777777777778  |
| stream | crystals     | F                 | 0.00202035108762  |
| stream | ethanol      | F                 | 0.01666666666667  |
| stream | motherLiquor | F                 | 0.0424240933568   |
| kpi    | cryst        | Ksp_activity      | 34.8439846653     |
| kpi    | cryst        | Q_removed         | 6559.87377671     |
| kpi    | cryst        | T_op              | 298.15            |
| kpi    | cryst        | antisolvent_xfrac | 0.399920015997    |
| kpi    | cryst        | c_sat             | 0.0517444790334   |
| kpi    | cryst        | cakeLiquidFlow    | 0.000329661969914 |
| kpi    | cryst        | cakeMoisture      | 0.1               |
| kpi    | cryst        | cakeWetness_pct   | 9.09090909091     |
| kpi    | cryst        | crystal_flow      | 0.0988038720389   |
| kpi    | cryst        | crystal_mol       | 0.00169068911771  |

... and 13 more reference values in the case's *expected*.

**crystalliser07\_nacl\_antisolvent\_cooled**

tutorials/steady/crystallisation/crystalliser07\_nacl\_antisolvent\_cooled



**What it does:** Antisolvent + COOLING crystallisation of NaCl: the same drowning-out brine + ethanol as crystalliser06, but the crystalliser is COOLED to 5 C. The eNRTL is now temperature-dependent ( $\tau(T)=\tau_{25}\cdot 298.15/T$ ,  $A_{DH}(T)$  via  $\epsilon_{sw}(T)/\rho_w(T)$ , Born at  $T$ ) and  $K_{sp}(T)$  shifts by van't Hoff with NaCl's enthalpy of solution – so cooling adds to the antisolvent's effect (yield 61%  $\rightarrow$  70%). Exercises the T-dependent electrolyte path.

**The story:** brine [cryst @ 5 C] crystals (wet NaCl cake) ethanol motherLiquor (saturated supernatant)

Identical drowning-out feed to crystalliser06 (saturated brine + ethanol, salt-free  $x_{ethanol} \sim 0.40$ ), but the crystalliser is COOLED to 5 C. Two levers now stack: ANTISOLVENT – ethanol lowers the dielectric  $\rightarrow$  mixed-solvent  $\gamma_{pm}(T)$ . COOLING – the electrolyte model is temperature-dependent:  $\tau(T) = \tau_{25} \cdot 298.15/T$  (the  $\Delta G/RT$  form, like the molecular NRTL), the Debye-Huckel  $A_{DH}(T)$  follows  $\epsilon_{sw}(T)/\rho_w(T)$ , the Born term is at  $T$ , and the saturation product  $K_{sp}(T)$  shifts by van't Hoff with NaCl's enthalpy of solution (dissolutionEnthalpy in NaCl.dat). Result:  $m_{sat} \sim 0.885 \rightarrow 0.681$  mol/kg, yield 61%  $\rightarrow$  70% vs the isothermal 06. (For NaCl alone cooling is weak – solubility is nearly T-flat; the antisolvent carries most of the effect. A salt with a large  $dH_{sol}$ , e.g. KNO<sub>3</sub>, would gain far more from cooling.)

Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

**What to expect (golden-locked):**

| kind   | unit/stream  | key               | expected          |
|--------|--------------|-------------------|-------------------|
| stream | brine        | F                 | 0.02777777777778  |
| stream | crystals     | F                 | 0.00231987381407  |
| stream | ethanol      | F                 | 0.01666666666667  |
| stream | motherLiquor | F                 | 0.0421245706304   |
| kpi    | cryst        | Ksp_activity      | 31.135192385      |
| kpi    | cryst        | Q_removed         | 82721.44242       |
| kpi    | cryst        | T_op              | 278.15            |
| kpi    | cryst        | antisolvent_xfrac | 0.399920015997    |
| kpi    | cryst        | c_sat             | 0.0398065230351   |
| kpi    | cryst        | cakeLiquidFlow    | 0.000380305320939 |
| kpi    | cryst        | cakeMoisture      | 0.1               |
| kpi    | cryst        | cakeWetness_pct   | 9.09090909091     |
| kpi    | cryst        | crystal_flow      | 0.113348382739    |
| kpi    | cryst        | crystal_mol       | 0.00193956849313  |

...and 13 more reference values in the case's *expected*.



**crystalliser08\_nacl\_antisolvent\_msmpr**

tutorials/steady/crystallisation/crystalliser08\_nacl\_antisolvent\_msmpr



**What it does:** Continuous MSMPR antisolvent crystallisation of NaCl: a saturated brine + ethanol fed to a well-mixed vessel (residence time  $\tau = V/Q$ ). The eNRTL antisolvent supersaturation  $S = c/c_{\text{sat}}(\text{eNRTL})$  drives nucleation + growth; the method of moments closes the population balance and returns the CRYSTAL-SIZE DISTRIBUTION  $n(L)$  (mean size, CV) – what the equilibrium model cannot give.

**The story:** CONTINUOUS MSMPR + ANTISOLVENT

brine [cryst MSMPR] magma (crystals WITH a size distribution  $n(L)$  ethanol + saturated mother liquor)

Same drowning-out feed as crystalliser06 (saturated brine + ethanol), but the vessel is a CONTINUOUS, well-mixed MSMPR crystalliser of working volume  $V$ ; the magma leaves at the same volumetric rate, so the residence time is  $\tau = V/Q$ .

Where the equilibrium model stops at the YIELD, the population balance goes on to the CRYSTAL-SIZE DISTRIBUTION: the antisolvent supersaturation  $S = c / c_{\text{sat}}(\text{eNRTL}, T, x_{\text{ethanol}})$  drives growth  $G = k_g (S-1)^g$  and nucleation  $B_0 = k_b (S-1)^b$ ; the method of moments gives  $n(L) = n_0 \exp(-L/G\tau)$ , the mean sizes ( $L_{43} = 4 G\tau$ ), the CV, and the magma PSD – plotted in the GUI.

The supersaturation here comes from the eNRTL ion-activity  $K_{\text{sp}}$  (the antisolvent lowers  $m_{\text{sat}}$ ), NOT a solubility curve – the MSMPR now reads the electrolyte saturation, the same helper the equilibrium model uses. Components + kinetics live under constant/; the eNRTL ion/ $\tau$  pair resolves from data/standards/ at runtime (self-DESCRIBING inline manifest).

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | brine       | F            | 0.02777777777778  |
| stream | ethanol     | F            | 0.01666666666667  |
| stream | magma       | F            | 0.04444444444444  |
| kpi    | cryst       | CV           | 1                 |
| kpi    | cryst       | L_dominant   | 2.02534396647e-05 |
| kpi    | cryst       | L_meanMass   | 2.70045862196e-05 |
| kpi    | cryst       | L_meanNumber | 6.75114655489e-06 |
| kpi    | cryst       | Q_kW         | -6.1566687571     |
| kpi    | cryst       | Q_removed    | 6156.6687571      |
| kpi    | cryst       | T_op         | 298.15            |
| kpi    | cryst       | c_sat        | 0.0517444790334   |
| kpi    | cryst       | crystal_flow | 0.092730856228    |
| kpi    | cryst       | crystal_mol  | 0.00158677029822  |
| kpi    | cryst       | growthRate   | 1.92664299147e-09 |

... and 15 more reference values in the case's *expected*.

**crystalliser09\_KHT\_KCl\_series**

tutorials/steady/crystallisation/crystalliser09\_KHT\_KCl\_series



**What it does:** Two crystallisers in series separating KCl from potassium bitartrate (KHT, cream of tartar) by manipulating T and adding ethanol. CRYST1 COOLS the hot feed to 5 C → KHT (sparingly soluble, steep solubility curve) crystallises while KCl stays dissolved. CRYST2 doses ETHANOL into the KCl-rich mother liquor → KCl drops out by drowning-out (eNRTL mixed-solvent). Two salts, two saturation routes (curve vs ion-activity), one package – a stress test of the property system.

**The story:** SELECTIVE crystallisation of two salts

feed (hot: KCl + KHT) [cryst1: COOL to 5 C] KHT\_cake (cream of tartar)

ethanol [cryst2: ANTISOLVENT] KCl\_cake

finalLiquor (EtOH+water+salt)

[recovery: DISTILLATION] recoveredEthanol (recycle-grade)

stillage (water + residual salt)

Two salts with very different solubilities are separated by two DIFFERENT levers, each crystalliser targeting one salt ('solute ...'):

CRYST1 cools the hot feed. Potassium bitartrate (KHT) is sparingly soluble with a STEEP solubility curve, so cooling 50 → 5 C precipitates most of it; KCl, far more soluble and only mildly T-sensitive, stays dissolved. KHT's saturation is its measured solubility CURVE  $c_{sat}(T)$  – no ion parameters.

CRYST2 doses ethanol into the KCl-rich mother liquor. KCl drops out by drowning-out: the eNRTL mixed-solvent term raises  $\gamma_{pm}$  as the dielectric falls, collapsing  $m_{sat}$ . KCl's saturation is the ion-activity  $K_{sp}$ .

The SAME package carries both routes; 'useElec' is decided per solute, so KHT uses its curve while KCl uses the eNRTL. Self-DESCRIBING (inline manifest); curated records resolve from data/standards/ at runtime.

Basis (per kg water): 55.5 water, 3.0 KCl (below the 5 C saturation ~3.9), and 0.078 KHT (saturated at 50 C).

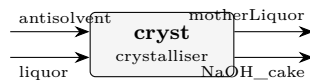
**What to expect (golden-locked):**

| kind   | unit/stream      | key            | expected          |
|--------|------------------|----------------|-------------------|
| stream | KCl_cake         | F              | 0.00138859586739  |
| stream | KHT_cake         | F              | 5.38404641059e-05 |
| stream | feed             | F              | 0.0277777777792   |
| stream | finalLiquor      | F              | 0.0541131192255   |
| stream | freshEthanol     | F              | 0.00416666666667  |
| stream | liquor1          | F              | 0.0277239373151   |
| stream | recoveredEthanol | F              | 0.0236111111111   |
| stream | stillage         | F              | 0.0305020081144   |
| kpi    | cryst1           | Q_kW           | -93.1162690166    |
| kpi    | cryst1           | Q_removed      | 93116.2690166     |
| kpi    | cryst1           | T_op           | 278.15            |
| kpi    | cryst1           | c_sat          | 0.0034013435      |
| kpi    | cryst1           | cakeLiquidFlow | 2.54671494088e-05 |
| kpi    | cryst1           | cakeMoisture   | 0.1               |

... and 57 more reference values in the case's *expected*.

**crystalliser10\_naoh\_antisolvent**

tutorials/steady/crystallisation/crystalliser10\_naoh\_antisolvent



**What it does:** NaOH antisolvent crystallisation (water + ethanol, eNRTL drowning-out) – the acceptance-case core

**The story:** H<sub>2</sub>O + NaOH + ethanol, every effect that IS calibrated active, every one that is NOT loudly announced: ACTIVE: drowning-out saturation (eNRTL mixed-solvent; Na-OH taus anchored to the Parker-slotted Pitzer surface), BPE-grade water activity, differential-K<sub>sp</sub> lineage, wet-cake split. ANNOUNCED OMISSIONS (honest): heat of dilution / crystallisation heat use the dH<sub>cryst</sub> constant (the eNRTL kernel is NOT calorimetrically fitted – per-kernel flag); water-ethanol heat of mixing gated until the H<sub>E</sub> refit; the solid is ANHYDROUS NaOH until the NaOH.H<sub>2</sub>O hydrate forms entry lands.

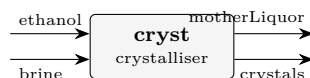
**What to expect (golden-locked):**

| kind   | unit/stream  | key                       | expected          |
|--------|--------------|---------------------------|-------------------|
| stream | NaOH_cake    | F                         | 0.00183771747926  |
| stream | antisolvent  | F                         | 0.00833333333333  |
| stream | liquor       | F                         | 0.01388888888889  |
| stream | motherLiquor | F                         | 0.020384504743    |
| kpi    | cryst        | K <sub>sp</sub> _activity | 7365.8380048      |
| kpi    | cryst        | Q_kW                      | 72.7276039114     |
| kpi    | cryst        | Q_removed                 | -72727.6039114    |
| kpi    | cryst        | T_op                      | 298.15            |
| kpi    | cryst        | antisolvent_xfrac         | 0.461538461538    |
| kpi    | cryst        | c_sat                     | 0.181200024556    |
| kpi    | cryst        | cakeLiquidFlow            | 0.000203756483719 |
| kpi    | cryst        | cakeMoisture              | 0.1               |
| kpi    | cryst        | cakeWetness_pct           | 9.09090909091     |
| kpi    | cryst        | crystal_flow              | 0.0653535379386   |

...and 13 more reference values in the case's *expected*.

**crystalliser11\_kcl\_antisolvent\_ethanol**

tutorials/steady/crystallisation/crystalliser11\_kcl\_antisolvent\_ethanol



**What it does:** Antisolvent (drowning-out) crystallisation of KCl: a SATURATED aqueous KCl brine (4.76 mol/kg) is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises  $\gamma_{\pm}$ , and DROPS the saturation molality – so KCl precipitates with NO cooling and NO evaporation. Saturation is the SAME ion-activity product  $K_{sp}$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\pm}$  at the feed's salt-free ethanol fraction. KCl-water-ethanol experimental reference: Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

**The story:** ANTISOLVENT (drowning-out) of KCl

[cryst] crystals (solid KCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual KCl)

The KCl counterpart of crystalliser06 (NaCl). KCl is the second salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)"; this case realises it.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity  $\gamma_{\pm}$  (eNRTL mixed-solvent + Born), so the KCl saturation molality  $m_{sat}$  COLLAPSES from its aqueous 4.76 mol/kg and KCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product  $K_{sp} = (\gamma_{\pm} \cdot m_{sat})^2$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\pm}$  at the feed's salt-free ethanol fraction.

Basis: 100 kmol/h saturated aqueous brine ( $x_{water}$  0.9210,  $x_{KCl}$  0.0790; 4.76 mol KCl / kg water) + 60.0 kmol/h ethanol  $\rightarrow$  salt-free  $x_{ethanol} \sim 0.39$ . Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

Experimental KCl-water-ethanol reference (the drowning-out anchor): Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

*The KCl counterpart of crystalliser06 (NaCl): a saturated aqueous KCl brine is mixed with ethanol, whose low dielectric collapses the KCl solubility, so KCl crystallises with no cooling and no evaporation. KCl is the salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)" — this case realises it.*

*Ethanol's relative permittivity (24.3) is far below water's (78.5). Dosing it lowers the mixed-solvent dielectric, which raises the mean ionic activity coefficient  $\gamma_{\pm}$  (eNRTL mixed-solvent + Born term), so the saturation molality collapses. The saturation is the same ion-activity product  $K_{sp} = (\gamma_{\pm} \cdot m_{sat})^2$  anchored in pure water, re-solved with the mixed-solvent  $\gamma_{\pm}$  at the feed's salt-free ethanol fraction — the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).*

*100 kmol/h saturated brine ( $x_{water}$  0.921,  $x_{KCl}$  0.079 = 4.76 mol/kg) + 60 kmol/h ethanol  $\rightarrow$  salt-free  $x_{ethanol} \approx 0.39$ :*

*The KCl-water-ethanol activity behaviour modelled here is measured in A. Lopes, F. Farelo & M. I. A. Ferra, "Activity Coefficients of Potassium Chloride in Water-Ethanol Mixtures", J. Solution Chem. 28 (1999) 117 — mean activity coefficients of KCl by emf in 5–20 % w/w ethanol, 25–45 °C. That  $\gamma_{\pm}$  rise with ethanol content is exactly the driver of the drowning-out captured above.*

*- The eNRTL  $\tau$  for K-Cl is the illustrative 1:1-alkali-chloride proxy (= Chen & Evans water-NaCl), so the numbers are qualitative. The Farelo data is the path to a quantitative refit of the KCl-water-ethanol interaction (the paper reports Pitzer  $\beta^0/\beta^1/C?$  per solvent composition). - The mass / yield is the reliable output. The crystalliser's heat term carries the shared electrolyte-enthalpy caveat (a dissolved salt's formation datum lives in its ions, so the stream enthalpy falls back to the sensible datum and is announced) — identical to crystalliser06.*

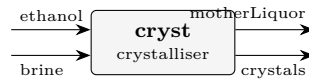
**What to expect (golden-locked):**

| kind   | unit/stream  | key               | expected          |
|--------|--------------|-------------------|-------------------|
| stream | brine        | F                 | 0.02777777777778  |
| stream | crystals     | F                 | 0.0021050886953   |
| stream | ethanol      | F                 | 0.01666666666667  |
| stream | motherLiquor | F                 | 0.0423393557491   |
| kpi    | cryst        | Ksp_activity      | 7.69319557877     |
| kpi    | cryst        | Q_kW              | -28.9281906957    |
| kpi    | cryst        | Q_removed         | 28928.1906957     |
| kpi    | cryst        | T_op              | 298.15            |
| kpi    | cryst        | antisolvent_xfrac | 0.394477317554    |
| kpi    | cryst        | c_sat             | 0.0311001943033   |
| kpi    | cryst        | cakeLiquidFlow    | 0.000423217143227 |
| kpi    | cryst        | cakeMoisture      | 0.1               |
| kpi    | cryst        | cakeWetness_pct   | 9.09090909091     |
| kpi    | cryst        | crystal_flow      | 0.125385206079    |

...and 13 more reference values in the case's *expected*.

**crystalliser12\_nacl\_pitzer\_antisolvent**

tutorials/steady/crystallisation/crystalliser12\_nacl\_pitzer\_antisolvent



**What it does:** Pitzer + antisolvent: the calorimetric  $L_{\phi}$  fit is AQUEOUS, so a mixed solvent drops it – and the duty basis says so

**The story:** crystalliser06\_nacl\_antisolvent – ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol’s low dielectric (24.3 vs water’s 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity  $\gamma_{\pm}$  (eNRTL mixed-solvent + Born), so the saturation molality  $m_{\text{sat}}$  COLLAPSES (6.14  $\rightarrow$  ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product  $K_{\text{sp}} = (\gamma_{\pm} \cdot m_{\text{sat}})^2$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\pm}$  at the feed’s salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine ( $x_{\text{water}}$  0.900,  $x_{\text{NaCl}}$  0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol  $\rightarrow$  salt-free  $x_{\text{ethanol}} \sim 0.40$ . Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output ‘magma’ form to a downstream ‘filter’.)

**What to expect (golden-locked):**

| kind   | unit/stream  | key               | expected        |
|--------|--------------|-------------------|-----------------|
| stream | brine        | F                 | 0.0277777777778 |
| stream | crystals     | F                 | 0               |
| stream | ethanol      | F                 | 0.0166666666667 |
| stream | motherLiquor | F                 | 0.0444444444444 |
| kpi    | cryst        | Ksp_activity      | 38.4846213082   |
| kpi    | cryst        | Q_kW              | -0              |
| kpi    | cryst        | Q_removed         | 0               |
| kpi    | cryst        | T_op              | 298.15          |
| kpi    | cryst        | antisolvent_xfrac | 0.399920015997  |
| kpi    | cryst        | c_sat             | 0.35905536      |
| kpi    | cryst        | cakeLiquidFlow    | 0               |
| kpi    | cryst        | cakeMoisture      | 0.1             |
| kpi    | cryst        | cakeWetness_pct   | 0               |
| kpi    | cryst        | crystal_flow      | 0               |

... and 13 more reference values in the case’s *expected*.

## overlay02\_nacl\_calorimetric

tutorials/steady/crystallisation/overlay02\_nacl\_calorimetric

**What it does:** NaCl crystalliser: saturation from the PITZER ion activity ( $K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$ ). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

**The story:** crystalliser05\_nacl\_pitzer (LEAF node)

NaCl crystalliser whose SATURATION comes from the Pitzer ion activity ( $K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$ ), via the ‘electrolyte { }’ block – NaCl carries no solubility{ } curve, so the electrolyte model is what makes this possible. The feed is a SUPERSATURATED brine (~7.6 mol/kg, as it would leave an evaporator); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess. Components live case-local in constant/components/; other curated records resolve from data/standards/ at runtime.

*A NaCl crystalliser (cloned from crystalliser05) whose case-local constant/components/NaCl.dat declares overlayOf NaCl; and recalibrates ONLY solidPhases.halite.calorimetric.dissolutionEnthalpy (3880 → 5880 J/mol).*

*The ThermoPackageBuilder deep-merges this delta over the standard record, so the crystalliser duty reflects it:  $Q_{removed} = 3690.997$  W here vs 2435.556 in crystalliser05 (ratio 1.515 = 5880/3880). Every other KPI ( $K_{sp\_activity}$ ,  $m_{sat}$ ,  $c_{sat}$ ,  $crystal\_mol$ ,  $\gamma_{pm}$ ) is byte-identical to crystalliser05 – logK25, dH, the dissolution reaction and the crystal block all still come from the standard.*

*This is the second half of the roadmap Phase A acceptance test: the SINGLE unified reader applies the overlay at ALL three component-loading sites (SpeciationSolver, Database, ThermoPackageBuilder), not just the SI path.*

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.0277777777778   |
| stream | magma       | F                | 0.0277777777778   |
| kpi    | cryst       | Ksp_activity     | 38.4846213082     |
| kpi    | cryst       | Q_kW             | -2.50670156624    |
| kpi    | cryst       | Q_removed        | 2506.70156624     |
| kpi    | cryst       | T_op             | 298.15            |
| kpi    | cryst       | c_sat            | 0.35905536        |
| kpi    | cryst       | crystal_flow     | 0.036683987968    |
| kpi    | cryst       | crystal_mol      | 0.000627720533333 |
| kpi    | cryst       | gamma_pm_sat     | 1.00970010562     |
| kpi    | cryst       | liquorFlow       | 0.0271500572444   |
| kpi    | cryst       | m_sat            | 6.144             |
| kpi    | cryst       | solute_dissolved | 0.158116012032    |
| kpi    | cryst       | solute_in        | 0.1948            |

... and 3 more reference values in the case's expected.

## distillation

**The theory you need.** A column is  $N$  flash stages coupled by counter-current flows. Each stage obeys MESH equations: Material balance, Equilibrium ( $y_i = K_i x_i$ ), Summation ( $\sum y = \sum x = 1$ ), and Heat balance. Choupo ships two solution strategies, selectable per column: Wang–Henke (bubble-point, solving the tridiagonal material balances stage by stage; robust for near-ideal systems, slow near azeotropes) and simultaneous (one Newton on ALL the MESH residuals; ~5–6 iterations even through azeotropes). Expect: reflux and boilup set the split; a minimum reflux below which no stage count reaches the spec; and azeotropes as fixed points no column can cross ( $x_D$  pinned at the azeotropic composition). *Full derivation: Theory Guide, the distillation chapter (McCabe–Thiele + MESH).*

**acetone06\_luyben\_column\_C1**

tutorials/steady/distillation/acetone06\_luyben\_column\_C1



**What it does:** Luyben's acetone column C1 on predictive UNIFAC: the test of a prediction written before the column was built

**The story:** THE COLUMN THAT TESTS A PREDICTION WRITTEN BEFORE IT EXISTED.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Column C1 of his Figure 1: 66 stages, feed on stage 54, reflux ratio 2.78, distillate 32.25 kmol/h, 1 atm. It takes the combined flash liquid plus absorber bottoms (his F1) and is asked to make the ACETONE PRODUCT at 99.9 mol %.

THE PREDICTION, quoted from the sibling props case rather than reinvented. 'props/compare/acetone04\_acetone\_water\_vle' measured the acetone/water VLE this column rests on BEFORE any column was built, and found that predictive UNIFAC puts a minimum-boiling azeotrope at  $x_{\text{acetone}} \sim 0.978$  where Luyben's UNIQUAC has only a pinch. It is a tiny feature – 0.044 K deep,  $|y - x|$  never above  $7.7 \times 10^{-4}$  – and acetone04 recorded, in advance and in writing:

"A column C1 built on this thermodynamics should be expected to asymptote near 97.9 % and fail his spec – and if it had been built before this was measured, the column would have been blamed for a defect that lives in the activity model."

So THIS case has a pass/fail criterion that is not its own golden: either the distillate stalls near the predicted ceiling, or the prediction was wrong. Both outcomes are results. Neither is a reason to touch the thermodynamics.

THE FEED IS LUYBEN'S OWN F1, not this project's upstream answer. acetone03 computes a flash liquid and acetone05 an absorber bottoms, and their sum is what physically feeds C1 – but wiring them in would test the column and two upstream units at once, and acetone05 already reports that it recovers a sixth of what the paper does. Feeding the PUBLISHED stream isolates the column: whatever it gets wrong here is the column and its thermodynamics. The composition in 0/F1 is his mole fractions times his 72.85 kmol/h.

WHAT IS NOT LUYBEN'S, and why each choice is the honest one:

UNIQUAC vs UNIFAC. He runs UNIQUAC; Choupo ships no fitted pair for any of these binaries and inventing one would convert unsourced into falsely sourced. Predictive UNIFAC is what is available, and acetone04 measured what it costs on the binary that decides this column.

The rigorous MESH ('model simultaneous'). Wang-Henke is the corpus default and is documented as unstable through an azeotrope, which is precisely the feature under test here. Choosing the method that CAN resolve the feature is not tuning; choosing the one that cannot would have hidden it.

Isopropanol's liquid heat capacity is Rowlinson-Bondi off the same Joback fragmentation as the rest of its record. Its measured accuracy on hydrogen-bonding molecules is 30-70 %, stated in the record itself. It is 5 mol % of the feed and it moves the DUTIES, not the split.

THIS COLUMN IS MODELLED DOWNSTREAM OF LUYBEN'S VENT, so it carries no hydrogen. His C1 has a partial condenser with a vent; Choupo's column has a total condenser and no vapour distillate. Both facts are stated before the reason, because the reason has to be argued rather than asserted:

- his vent is 0.0465 kmol/h at H2 0.2798, i.e. 0.0130 kmol/h of hydrogen, against the 0.0146 kmol/h that F1 carries. The vent removes essentially ALL of it. A column model with no vent and a total condenser therefore represents the column AFTER the vent, which is a statement about WHICH UNIT is being modelled – not an approximation to the physics of the one that is.

- what that costs is quantified rather than hidden: the same vent carries 0.0335 kmol/h of ACETONE (his own figure), 0.10 % of the product. Not modelling the vent therefore FLATTERS this column's acetone recovery by that much, and the comparison below must be read knowing it.

IT WAS RUN THE OTHER WAY FIRST, and the run is the reason this paragraph exists rather than a preference. With H2 in the feed the MESH converged (16 Newton iterations,  $\|F\|$  2.4e-11) to essentially



the same distillate –  $x_{\text{acetone}}$  0.9722 vs the shipped run’s goldened 0.9731 – and then the energy report REFUSED: hydrogen has no liquid heat capacity, because at 330 K it has no liquid. The refusal is right, and it is the honest form of the modelling error: a total condenser was silently asking a permanent gas to leave as a liquid. See this case’s CLAUDE.md.

#### What to expect (golden-locked):

| kind   | unit/stream    | key            | expected         |
|--------|----------------|----------------|------------------|
| stream | B1             | F              | 0.0112717083333  |
| stream | F1             | F              | 0.0202300416667  |
| stream | acetoneProduct | F              | 0.00895833333333 |
| kpi    | C1             | B              | 0.0112717083333  |
| kpi    | C1             | D              | 0.00895833333333 |
| kpi    | C1             | Q_condenser_kW | -997.746586798   |
| kpi    | C1             | Q_reboiler_kW  | 1030.49217736    |
| kpi    | C1             | R              | 2.78             |
| kpi    | C1             | T_bottom       | 353.606283382    |
| kpi    | C1             | T_top          | 329.375698025    |
| kpi    | C1             | feedStage      | 54               |
| kpi    | C1             | nFeeds         | 1                |
| kpi    | C1             | nSideDraws     | 0                |
| kpi    | C1             | nStages        | 66               |

...and 17 more reference values in the case’s *expected*.

**acetone07\_luyben\_column\_C2**

tutorials/steady/distillation/acetone07\_luyben\_column\_C2



**What it does:** Luyben's water column C2: the IPA recycle meets the azeotrope acetone01 measured

**The story:** THE SECOND COLUMN, AND THE SECOND PREDICTION WRITTEN BEFORE IT.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Column C2 of his Figure 1: 19 stages, feed on stage 16, reflux ratio 0.849, 1 atm. It takes C1's bottoms and makes the plant's WATER PRODUCT at the bottom (34.67 kmol/h, 99.9 mol % water) while sending the unconverted isopropanol back overhead as the RECYCLE (5.88 kmol/h at  $x_{\text{IPA}} 0.6500$ ).

THE PREDICTION, quoted from acetone01 rather than reinvented, and committed to the repository BEFORE this case was written:

"C2's overhead should cap at roughly  $x_{\text{IPA}} 0.59$  and cannot reach his 0.650. The gap is 0.06 in mole fraction, about 9 % relative, and it is a HARD ceiling rather than a shortfall in stages or reflux."

The reason is one measured number: acetone01 puts the UNIFAC IPA/water azeotrope at  $x_{\text{IPA}} 0.5901$  against Luyben's UNIQUAC value of 0.6732. A column cannot distil past a minimum-boiling azeotrope from below.

AND THE DIRECTION OF THIS ERROR IS THE OPPOSITE OF C1's, which is the part worth carrying away. C1's azeotrope is a UNIFAC INVENTION – Luyben's model has none in acetone/water – so it costs purity the paper does not lose. Here the azeotrope is REAL in both models and UNIFAC merely puts it in the wrong place. One failure is a spurious feature; the other is a misplaced real one. They are not the same defect and should not be reported as one.

THE FEED IS LUYBEN'S PUBLISHED B1, not acetone06's answer – the same choice and the same reason as in C1: it isolates the column. His B1 is 40.56 kmol/h at 370 K, A 0.0001 / IPA 0.0951 / W 0.9048. The CLOSED flowsheet (acetone08) is where the two columns finally see each other's real streams, and the difference between this case and that one is exactly the propagation the plant exists to show.

THE DISTILLATE RATE IS HIS, 5.88 kmol/h, and that matters here in a way it did not in C1. Fixing the rate means the column must send that much material overhead whatever its composition can be – so if the ceiling binds, it shows up as a WETTER recycle at the same flow, not as a smaller one. That is the honest way to pose it: the plant's IPA inventory has to go somewhere, and the question is what quality it comes back at.

**What to expect (golden-locked):**

| kind                                                             | unit/stream  | key            | expected         |
|------------------------------------------------------------------|--------------|----------------|------------------|
| stream                                                           | B1           | F              | 0.0112666944444  |
| stream                                                           | ipaRecycle   | F              | 0.00163333333333 |
| stream                                                           | waterProduct | F              | 0.00963336111111 |
| kpi                                                              | C2           | B              | 0.00963336111111 |
| kpi                                                              | C2           | D              | 0.00163333333333 |
| kpi                                                              | C2           | Q_condenser_kW | -120.936005804   |
| kpi                                                              | C2           | Q_reboiler_kW  | 114.900183788    |
| kpi                                                              | C2           | R              | 0.849            |
| kpi                                                              | C2           | T_bottom       | 366.174785539    |
| kpi                                                              | C2           | T_top          | 355.876498071    |
| kpi                                                              | C2           | feedStage      | 16               |
| kpi                                                              | C2           | nFeeds         | 1                |
| kpi                                                              | C2           | nSideDraws     | 0                |
| kpi                                                              | C2           | nStages        | 19               |
| ... and 17 more reference values in the case's <i>expected</i> . |              |                |                  |

column01\_benzene\_toluene

tutorials/steady/distillation/column01\_benzene\_toluene



**What it does:** Benzene/toluene distillation column (Wang-Henke)

**The story:** Benzene/toluene distillation — classical textbook problem with Raoult, no azeotrope, easy convergence. Used to validate the column code.

**What to expect (golden-locked):**

| kind                                                     | unit/stream | key                          | expected        |
|----------------------------------------------------------|-------------|------------------------------|-----------------|
| kpi                                                      | column01    | T_top                        | 354.21089412    |
| kpi                                                      | column01    | T_bottom                     | 382.893530951   |
| kpi                                                      | column01    | x_D_LK                       | 0.981245045999  |
| kpi                                                      | column01    | x_B_HK                       | 0.981245183716  |
| utility                                                  | column01    | heating.steamLP.duty_kW      | 1279.30480005   |
| utility                                                  | column01    | heating.steamLP.T            | 382.893530951   |
| utility                                                  | column01    | heating.steamLP.kg_s         | 0.585226349518  |
| utility                                                  | column01    | heating.steamLP.eur_h        | 55.265967362    |
| utility                                                  | column01    | cooling.coolingWater.duty_kW | -1281.04348002  |
| utility                                                  | column01    | cooling.coolingWater.T       | 354.21089412    |
| utility                                                  | column01    | cooling.coolingWater.kg_s    | 30.6469732063   |
| utility                                                  | column01    | cooling.coolingWater.eur_h   | 1.84470261124   |
| stream                                                   | bottoms     | F                            | 0.0138888888889 |
| stream                                                   | bottoms     | T                            | 382.893530951   |
| ... and 15 more reference values in the case's expected. |             |                              |                 |

**column02\_simultaneous**

tutorials/steady/distillation/column02\_simultaneous



**What it does:** Rigorous distillation by simultaneous MESH Newton (benzene/toluene; validates against column01)

**The story:** The SAME benzene/toluene column as column01, but solved by the rigorous SIMULTANEOUS-correction (MESH Newton) method instead of the sequential bubble-point (Wang-Henke). Selected by ‘model simultaneous;’.

Pedagogical point: on an ideal system both methods give the SAME answer — this case validates the MESH solver against column01. But the simultaneous Newton converges in ~6 iterations (quadratic) vs ~100 for the bubble-point, and — the reason it exists — it stays stable on azeotropic systems where the sequential method does not.

Expected (identical to column01):  $T_{\text{top}} \approx 354.2 \text{ K}$ ,  $T_{\text{bottom}} \approx 382.9 \text{ K}$ ,  $x_{\text{D}}(\text{benzene}) \approx 0.981$ ,  $x_{\text{B}}(\text{toluene}) \approx 0.981$ .

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected        |
|---------|-------------|-------------------------|-----------------|
| stream  | bottoms     | F                       | 0.0138888888889 |
| stream  | distillate  | F                       | 0.0138888888889 |
| stream  | feed        | F                       | 0.0277777777778 |
| kpi     | column01    | B                       | 0.0138888888889 |
| kpi     | column01    | D                       | 0.0138888888889 |
| kpi     | column01    | R                       | 2               |
| kpi     | column01    | T_bottom                | 382.893527916   |
| kpi     | column01    | T_top                   | 354.210890137   |
| kpi     | column01    | feedStage               | 8               |
| kpi     | column01    | nStages                 | 15              |
| kpi     | column01    | x_B_HK                  | 0.981245125322  |
| kpi     | column01    | x_D_LK                  | 0.981245125322  |
| utility | column01    | heating.steamLP.duty_kW | 1299.61040477   |
| utility | column01    | heating.steamLP.T       | 382.893527916   |

...and 13 more reference values in the case's *expected*.

**column03\_azeotrope\_mesh**

tutorials/steady/distillation/column03\_azeotrope\_mesh



**What it does:** Azeotropic distillation (ethanol/water, NRTL) by the rigorous simultaneous MESH method

**The story:** Ethanol/water (NRTL) distillation by the rigorous SIMULTANEOUS (MESH Newton) method. This is the payoff of the v0.48 solver: ethanol/water forms an azeotrope at  $x_{\text{eth}} \sim 0.894$  (1 atm), and the sequential bubble-point (Wang-Henke) is unstable here – on this very case it does NOT converge (200 outer iters). The MESH Newton converges in  $\sim 5$  iterations to a PHYSICAL distillate ( $x_{\text{eth}} \sim 0.80$ , below the azeotrope).

Feed: 100 kmol/h, 30/70 ethanol/water, 1 atm. The column feed enthalpy is set by `feedQuality = 1.0` (saturated liquid), NOT by the stream T: the stated  $T = 363$  K is  $\sim 7$  K above the 30/70 bubble point ( $\sim 355.5$  K at 1 atm), so the stream itself is slightly superheated — the MESH model uses `feedQuality`, so this does not affect the column solution. Column: 16 stages, feed at 8,  $R = 2.5$ ,  $D = 25$  kmol/h, method simultaneous.

Expected: converged;  $x_D(\text{ethanol}) \sim 0.80$  ( $< 0.894$  azeotrope – physical);  $T_{\text{top}} \sim 351.4$  K (the azeotropic temperature); mass closes 100%.

NOTE: asking for a distillate ABOVE the azeotrope ( $x_{\text{eth}} > 0.894$ ) is physically impossible – no method converges, and that is correct. The MESH method does not invent a non-physical "past the azeotrope" answer (which the bubble-point used to do).

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected         |
|---------|-------------|-------------------------|------------------|
| stream  | bottoms     | F                       | 0.0208333333333  |
| stream  | distillate  | F                       | 0.00694444444444 |
| stream  | feed        | F                       | 0.0277777777778  |
| kpi     | column      | B                       | 0.0208333333333  |
| kpi     | column      | D                       | 0.00694444444444 |
| kpi     | column      | R                       | 2.5              |
| kpi     | column      | T_bottom                | 357.713408957    |
| kpi     | column      | T_top                   | 351.388958309    |
| kpi     | column      | feedStage               | 8                |
| kpi     | column      | nStages                 | 16               |
| kpi     | column      | x_B_HK                  | 0.868205515093   |
| kpi     | column      | x_D_LK                  | 0.804616545279   |
| utility | column      | heating.steamLP.duty_kW | 935.650111764    |
| utility | column      | heating.steamLP.T       | 357.713408957    |

...and 13 more reference values in the case's *expected*.

**column04\_multifeed\_sidedraw**

tutorials/steady/distillation/column04\_multifeed\_sidedraw



**What it does:** Simultaneous MESH column: two feeds + a liquid side draw (rigorous MESH)

**The story:** Multiple feeds as FLOWSHEET STREAMS ('inputs ( feed feed2 )') + a liquid side draw. operation.feeds maps each input stream to a stage (information follows the streams → the plant-boundary mass + energy balance closes).

*The rigorous simultaneous MESH column extended to multiple feeds and side draws — the most-used features of a commercial simulator column, on the glass-box Newton solver.*

*A benzene / toluene / o-xylene split (light / intermediate / heavy):*

- primary feed — 100 kmol/h, benzene/toluene/o-xylene 0.40/0.40/0.20, enters high (stage 5); - second feed — 60 kmol/h, heavier (0.05/0.45/0.50), enters low (stage 11); - liquid side draw — 15 kmol/h pulled from stage 8, between the two feeds.

Mass closes ( $100 + 60 = 42 + 15 + 103 = 160$  kmol/h).  $Q_{\text{condenser}} = -1436$  kW,  $Q_{\text{reboiler}} = +1527$  kW. The side draw does exactly what it should — bleed the intermediate component where it is most enriched on the trays.

*The simultaneous MESH is a bubble-point-reduced Newton (CMO flows). The extension generalises the single fixed feed into a per-stage flow staircase:  $L[j]$ ,  $V[j]$  are rebuilt by a CMO march that adds each feed's liquid/vapour parts and subtracts each draw; every stage's material balance carries an  $F_j z_{\{F,j\}}$  source and  $(L_j + U_j)$ ,  $(V_j + W_j)$  outflows. The general balance reduces exactly to the old single-feed equations when there is one feed and no draw — so every existing distillation tutorial is unchanged (verified).*

*outputs ( distillate bottoms <draw1> <draw2> ... ) — side-draw streams follow distillate and bottoms, in stage order.*

**What to expect (golden-locked):**

| kind   | unit/stream     | key            | expected        |
|--------|-----------------|----------------|-----------------|
| stream | bottoms         | F              | 0.0286111111111 |
| stream | distillate      | F              | 0.0116666666667 |
| stream | feed            | F              | 0.0277777777778 |
| stream | intermediateCut | F              | 0.0041666666667 |
| kpi    | radfracLite     | B              | 0.0286111111111 |
| kpi    | radfracLite     | D              | 0.0116666666667 |
| kpi    | radfracLite     | Q_condenser_kW | -1435.80274013  |
| kpi    | radfracLite     | Q_reboiler_kW  | 1516.14164522   |
| kpi    | radfracLite     | R              | 3               |
| kpi    | radfracLite     | T_bottom       | 395.926996928   |
| kpi    | radfracLite     | T_top          | 354.978489481   |
| kpi    | radfracLite     | feedStage      | 5               |
| kpi    | radfracLite     | nFeeds         | 2               |
| kpi    | radfracLite     | nSideDraws     | 1               |

... and 17 more reference values in the case's *expected*.

**column05\_reactive\_methylacetate**

tutorials/steady/distillation/column05\_reactive\_methylacetate



**What it does:** Reactive distillation: methyl acetate synthesis (validation vs P?pken et al. 2001, run S-1)

**The story:** Both feeds are FLOWSHEET STREAMS ('inputs ( feed acidFeed )') – so the plant-boundary mass + energy balance sees them. 'operation.feeds' only maps each stream to a stage (no inline composition – the credo: information follows the streams).

*The simultaneous MESH column with an equilibrium reaction on the catalytic stages (rigorous column rung 4), compared to a published experiment.*

*Mole-conserving, so the CMO internal-flow profile is untouched. Each catalytic stage carries one extra unknown — the molar extent  $\xi_j$  — closed by the activity-based equilibrium residual  $\sum_i \nu_i \ln(\gamma_i x_i) = \ln K_a(T)$ , with  $\gamma_i$  from the liquid activity model and  $K_a(T)$  by van't Hoff:  $K_a(T) = K_a(298) \cdot \exp(-\Delta h^\circ_r/R \cdot (1/T - 1/298.15))$ . Converged by homotopy (solve non-reactive, then switch the reaction on).*

*> T. P?pken, S. Steinigeweg & J. Gmehling, "Synthesis and Hydrolysis of > Methyl Acetate by Reactive Distillation", Ind. Eng. Chem. Res. 40 (2001) 1566. Run S-1:  $N = 25$ , reactive stages 11–19, HOAc fed above / > MeOH below the zone, reflux 2.1,  $P \approx 1.017$  bar. Measured endpoints, cited, in > constant/experimental/methyl\_acetate\_RD\_profile.csv.*

*The equilibrium constant and thermodynamics are from the companion paper P?pken, G?tze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601:  $K_a(298) \approx 38.7$ ,  $\Delta h^\circ_r \approx -5.67$  kJ/mol  $\rightarrow K_a \approx 29$  in the reactive zone ( $\sim 65^\circ\text{C}$ ).*

*This is not a tight match, and the reason is physically important: with the correct equilibrium  $K_a$  ( $\approx 29$ ) the model gives the equilibrium ceiling (96 %), and it correctly brackets the measurement from above. The measured 87.6 % is \*below\* the equilibrium because the real column is kinetically limited — the paper's recommended model is an adsorption-based finite \*rate\* (per gram of catalyst), not equilibrium. An earlier run with an (incorrect, too low)  $K_a = 5.2$  gave 90.7 %, a closer \*number\* but for the wrong reason; using a wrong constant to fake agreement would be dishonest.*

*So the path to a bit-tight match is the kinetic rate model, not the thermo.*

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected          |
|--------|-------------|----------------|-------------------|
| stream | bottoms     | F              | 9.13888888889e-06 |
| stream | distillate  | F              | 8.36111111111e-06 |
| stream | feed        | F              | 8.94444444444e-06 |
| kpi    | rdColumn    | B              | 9.13888888889e-06 |
| kpi    | rdColumn    | D              | 8.36111111111e-06 |
| kpi    | rdColumn    | Q_condenser_kW | -0.790521316742   |
| kpi    | rdColumn    | Q_reboiler_kW  | 0.626813788646    |
| kpi    | rdColumn    | R              | 2.1               |
| kpi    | rdColumn    | T_bottom       | 363.229492539     |
| kpi    | rdColumn    | T_top          | 330.062558898     |
| kpi    | rdColumn    | conversion     | 0.960508806863    |
| kpi    | rdColumn    | feedStage      | 20                |
| kpi    | rdColumn    | nFeeds         | 2                 |
| kpi    | rdColumn    | nSideDraws     | 0                 |

*...and 17 more reference values in the case's expected.*

**column06\_fullMESH**

tutorials/steady/distillation/column06\_fullMESH



**What it does:** Full-MESH (Naphtali-Sandholm) distillation – energy-consistent  $V_j/L_j$ , seeded from the CMO solution (benzene/toluene; compare column02)

**The story:** The SAME benzene/toluene column as column02\_simultaneous, but solved by the FULL MESH (Naphtali-Sandholm), selected by ‘model fullMESH;’. The full MESH promotes the internal flows  $V_j$ ,  $L_j$  to UNKNOWNs and adds a per-stage total-mass + ENERGY balance, so the flows follow the REAL latent heats instead of the constant-molal-overflow (CMO) assumption behind the ‘simultaneous’ (bubble-point-reduced) method.

Robustness trick: the full MESH is SEEDED from the converged CMO profile — the cheap model bootstraps the rigorous one; ~4 Newton iterations from that seed.

Expected — close to, but NOT equal to, column02 (the gap IS the energy-balance correction the CMO drops):  $T_{\text{top}} \sim 354.3$  K,  $T_{\text{bottom}} \sim 382.8$  K,  $x_D(\text{benzene}) \sim 0.979$  (CMO: 0.981). Benzene/toluene latent heats are similar (30.8 vs 33.2 kJ/mol) so the gap is small here; on a wide-boiling column the full MESH earns its cost.

*The same benzene/toluene column as column02\_simultaneous, but solved by the full MESH (model fullMESH) instead of the bubble-point/CMO reduction.*

*The simultaneous method is a bubble-point-reduced MESH: it assumes constant molal overflow (CMO — equimolal latent heats), so the internal vapour/liquid flows  $V_j$ ,  $L_j$  are fixed by a mass march and only the compositions  $x_j$  and stage temperatures  $T_j$  are solved.*

*The full MESH (Naphtali–Sandholm) promotes  $V_j$  and  $L_j$  to unknowns and adds, per stage, the total-mass balance and the energy balance — so the flows follow the \*real\* latent heats instead of assuming them equal. The extra per-stage energy equation is the enthalpy residual the CMO never carried:*

*Total condenser:  $V_0 = 0$ ,  $L_0 = R \cdot D$ ; reboiler:  $L_{\{N-1\}} = B$ ; their energy balances give  $Q_{\text{cond}} / Q_{\text{reb}}$  as results.*

*The full-MESH is seeded from the converged CMO profile — the cheap model launches the rigorous one. Here it converges in ~4 Newton iterations from that seed (it would be far harder from a cold start).*

*Benzene ( $\Delta h_{\text{vap}} \approx 30.8$  kJ/mol) and toluene ( $\approx 33.2$  kJ/mol) have similar latent heats, so CMO is a good approximation and the two answers nearly coincide — the ~0.002 / ~0.1 K gap IS the energy-balance correction the full-MESH makes and the CMO drops. On a wide-boiling column (a depropaniser) the gap is much larger; that is where the full-MESH earns its cost.*

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0138888888889 |
| stream | distillate  | F              | 0.0138888888889 |
| stream | feed        | F              | 0.0277777777778 |
| kpi    | column06    | B              | 0.0138888888889 |
| kpi    | column06    | D              | 0.0138888888889 |
| kpi    | column06    | Q_condenser_kW | -1281.19539111  |
| kpi    | column06    | Q_reboiler_kW  | 1299.72437334   |
| kpi    | column06    | R              | 2               |
| kpi    | column06    | T_bottom       | 382.774927808   |
| kpi    | column06    | T_top          | 354.339103085   |
| kpi    | column06    | feedStage      | 8               |
| kpi    | column06    | nFeeds         | 1               |
| kpi    | column06    | nSideDraws     | 0               |
| kpi    | column06    | nStages        | 15              |

... and 13 more reference values in the case's *expected*.



column07\_naphtaliSandholm

tutorials/steady/distillation/column07\_naphtaliSandholm



**What it does:** Full-MESH (Naphtali-Sandholm) validation vs AIChE J. 17 (1971) 148, Problem 2: nC4/iC5/nC5

*Reproduces the ideal-hydrocarbon distillation of Naphtali & Sandholm, "Multicomponent Separation Calculations by Linearization", AIChE Journal 17(1), 148–153 (1971) — DOI [10.1002/aic.690170130](https://doi.org/10.1002/aic.690170130) — the paper that introduced the very full-MESH method Choupo’s model fullMESH implements.*

*n-butane / isopentane / n-pentane, 20 plates, liquid feed ( $T = 30\text{ }^{\circ}\text{F} = 272.04\text{ K}$ , 500 mol of each) on plate 10, reflux ratio 1000, 80 % of the feed recovered in the bottoms ( $B = 1200$ ). A liquid sidestream on plate 1 (the condenser) takes 30 % of the reflux.*

*Equivalent spec used here. The distillate ( $\approx 1$ ) and the condenser sidestream ( $\approx 299$ ) leave the \*total\* condenser with the same composition, so they are combined into ONE top product of 300 with reflux  $R = L_1/300 = 996.68/300 = 3.3223$ . This gives identical internal flows ( $L_1 = 996.68$ ,  $V_2 = 1296.68$ ), hence the identical column — without needing a condenser side draw.*

*Converges in 3 full-MESH Newton iterations (seeded from the CMO profile).*

*Products Choupo — top product (300) 99.99 % n-C<sub>4</sub> bottoms n-C<sub>4</sub> / i-C<sub>5</sub> / n-C<sub>5</sub> 200 / 500 / 500 B 1200*

*The profile shape — the flat n-butane-rich top, the sharp rise across the feed zone, the pentane-plateau, the reboiler climb — matches stage-by-stage. The ~1–2 K residual is thermo, not method: Choupo uses fitted Antoine vapour pressures, the paper used the K-value tables in Brown’s \*Unit Operations\* (1950).*

**What to expect (golden-locked):**

| kind                                                     | unit/stream | key            | expected        |
|----------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                   | feed        | F              | 0.416666666667  |
| kpi                                                      | column07    | B              | 0.333333333333  |
| kpi                                                      | column07    | D              | 0.0833333333333 |
| kpi                                                      | column07    | Q_condenser_kW | -8082.6120755   |
| kpi                                                      | column07    | Q_reboiler_kW  | 9374.37429576   |
| kpi                                                      | column07    | R              | 3.3223          |
| kpi                                                      | column07    | T_bottom       | 296.58946842    |
| kpi                                                      | column07    | T_top          | 272.663334648   |
| kpi                                                      | column07    | feedStage      | 10              |
| kpi                                                      | column07    | nFeeds         | 1               |
| kpi                                                      | column07    | nSideDraws     | 0               |
| kpi                                                      | column07    | nStages        | 20              |
| kpi                                                      | column07    | x_B_HK         | 0.416665691626  |
| kpi                                                      | column07    | x_D_LK         | 0.999874145869  |
| ... and 10 more reference values in the case’s expected. |             |                |                 |

**column08\_radfrac\_multidraw**

tutorials/steady/distillation/column08\_radfrac\_multidraw



**What it does:** rigorous column: multi-feed + multi-side-draw distillation (C3/C4/C5/C6), full-MESH

A C3/C4/C5/C6 depropanizer-style column at 16 bar, split four ways by the rigorous full-MESH, exercising the two most-used rigorous column features at once: TWO feeds (a light cut high, a heavy cut low) and TWO liquid side draws.

Two saturated-liquid feeds (100 kmol/h each, 290 K / 320 K): a light cut on stage 10, a heavy cut on stage 20. 30 stages,  $R = 4$ . Mass closes exactly ( $200 = 45 + 55 + 55 + 45$  kmol/h) and the energy balance closes: the feed-to-product enthalpy gap (1137.7 kW) equals the net duty  $Q_{\text{reb}} + Q_{\text{cond}} = 1942.3 - 804.6$  kW. Converges in a few full-MESH iterations from the CMO seed.

This case was first written at 1 atm, and the energy balance would not close — because at 1 atm and 290 K the light hydrocarbons (propane b.p. 231 K, n-butane 272 K) are above their boiling points: the feed streams flash to VAPOUR ( $vf = 1.0$ ), yet the column was told quality 1.0 (saturated liquid). The latent heat of the unrecognised vapour feed was exactly the missing  $\sim 1$  MW. At 16 bar the feeds are genuinely liquid ( $vf = 0$ , consistent with quality 1.0) and the balance closes — which is also why a real depropanizer runs at elevated pressure: so the propane distillate condenses against cooling water (here 321.6 K). A feed's thermal state must be specified consistently with its actual phase.

Both feeds are real flowsheet streams (inputs ( feed1 feed2 ), mapped to stages by operation.feeds), so the plant-boundary mass + energy balance sees them (\*information follows the streams\*). Side draws are outputs in stage order: outputs ( distillate bottoms drawC4 drawC5 ).

Self-consistently validated (mass + energy close, four cuts in volatility order) on the full-MESH solver that IS paper-validated against Naphtali & Sandholm (1971) in column07\_naphtaliSandholm. No external multi-feed+multi-draw single-column benchmark with published profiles is readily available in the open literature.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0125          |
| stream | distillate  | F              | 0.0125          |
| stream | drawC4      | F              | 0.0152777777778 |
| stream | drawC5      | F              | 0.0152777777778 |
| stream | feed1       | F              | 0.0277777777778 |
| stream | feed2       | F              | 0.0277777777778 |
| kpi    | radfrac     | B              | 0.0125          |
| kpi    | radfrac     | D              | 0.0125          |
| kpi    | radfrac     | Q_condenser_kW | -804.58622364   |
| kpi    | radfrac     | Q_reboiler_kW  | 1698.71150765   |
| kpi    | radfrac     | R              | 4               |
| kpi    | radfrac     | T_bottom       | 454.498489397   |
| kpi    | radfrac     | T_top          | 321.647346922   |
| kpi    | radfrac     | feedStage      | 10              |

... and 19 more reference values in the case's expected.

**column09\_tray\_hydraulics**

tutorials/steady/distillation/column09\_tray\_hydraulics



**What it does:** Sieve-tray hydraulics: the column that separates, sized so its trays do not flood

**The story:** column01, asked the question a converged MESH solution never answers: CAN THE TRAYS PASS THE TRAFFIC?

A converged column tells you what separates. It says nothing about whether the vapour rising through the holes will carry the liquid up with it (entrainment flooding), whether the liquid will back up the downcomer until it reaches the tray above (downcomer flooding), or whether it will simply rain through the holes instead of bubbling (weeping). A column that floods separates nothing, and the stage equations never notice.

Add an 'hydraulics {}' block and the column runs a RATING pass on its own converged profile. Two modes, the same equations in opposite directions:

give 'diameter' → RATING. How close to flood is each tray? omit 'diameter' → DESIGN. What diameter keeps every tray at or below 'floodFraction' of its flooding velocity? The widest tray sets the column. That is this case.

Nothing here is guessed. The orifice coefficient  $C_o$  is a chart in the literature (Liebson, Kelley & Bullington 1957, against hole-area ratio and plate-thickness/hole-diameter); it is a DECLARED number you own, printed back at you. The weep-point constant  $K_2$  is a chart too – leave it out and the column tells you the weep check was not performed rather than inventing it.

Read the printed table. The flow parameter  $F_{LV}$  climbs below the feed (more liquid, less vapour per unit mass), which lowers Fair's capacity parameter and therefore the flooding velocity – yet the stripping trays are not the tight ones here, because the vapour volumetric flow falls faster than the capacity. The column is sized by whichever tray loses that argument.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0138888888889 |
| stream | distillate  | F              | 0.0138888888889 |
| stream | feed        | F              | 0.0277777777778 |
| kpi    | column09    | B              | 0.0138888888889 |
| kpi    | column09    | D              | 0.0138888888889 |
| kpi    | column09    | L_rect         | 0.0277777777778 |
| kpi    | column09    | L_strip        | 0.0555555555556 |
| kpi    | column09    | Q_condenser_kW | -1281.04348002  |
| kpi    | column09    | Q_reboiler_kW  | 1279.30480005   |
| kpi    | column09    | R              | 2               |
| kpi    | column09    | T_bottom       | 382.893530951   |
| kpi    | column09    | T_top          | 354.21089412    |
| kpi    | column09    | V_rect         | 0.0416666666667 |
| kpi    | column09    | V_strip        | 0.0416666666667 |

... and 22 more reference values in the case's *expected*.

**column10\_flooding**

tutorials/steady/distillation/column10\_flooding



**What it does:** The same column, built too narrow: every tray floods – and it still converges

**The story:** the SAME column as column09, built 1.10 m wide instead of the 1.316 m that column09 designed for it.

Nothing about the separation changes. The distillate is still 98.12 % benzene, the bottoms still 98.12 % toluene, the MESH equations converge exactly as before, and every number in the stream table is identical. The column is a perfectly good column – on paper.

It floods. The worst tray runs at 114.5 % of its entrainment-flooding velocity: the vapour is fast enough to carry the liquid up to the tray above instead of letting it fall. In a real column the trays would fill, the pressure drop would run away, and the separation printed above would not happen at all.

This is the whole reason hydraulics exists as a separate pass. A converged solution is not a feasible column, and the stage equations have no way of knowing the difference. Widen it (column09), space the trays further apart, or cut the reflux – but do not believe the stream table until a tray can pass the traffic it carries.

Note the downcomer, too: the backup reaches 268.5 mm against a 275 mm limit. Narrow the column a little further and it would flood from below as well, for an entirely different reason.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0138888888889 |
| stream | distillate  | F              | 0.0138888888889 |
| stream | feed        | F              | 0.0277777777778 |
| kpi    | column10    | B              | 0.0138888888889 |
| kpi    | column10    | D              | 0.0138888888889 |
| kpi    | column10    | L_rect         | 0.0277777777778 |
| kpi    | column10    | L_strip        | 0.0555555555556 |
| kpi    | column10    | Q_condenser_kW | -1281.04348002  |
| kpi    | column10    | Q_reboiler_kW  | 1279.30480005   |
| kpi    | column10    | R              | 2               |
| kpi    | column10    | T_bottom       | 382.893530951   |
| kpi    | column10    | T_top          | 354.21089412    |
| kpi    | column10    | V_rect         | 0.0416666666667 |
| kpi    | column10    | V_strip        | 0.0416666666667 |

... and 22 more reference values in the case's *expected*.

**column11\_murphree**

tutorials/steady/distillation/column11\_murphree



**What it does:** Murphree tray efficiency: 14 REAL trays plus the reboiler (nStages 15, reboiler counted), not ideal stages – and the purity you actually get

**The story:** the same 15-stage benzene/toluene column as column09, with ONE line added: murphreeEfficiency 0.70.

Every column before this one is built from EQUILIBRIUM stages: the vapour leaving a tray is in equilibrium with the liquid leaving it. No real tray does that. The vapour spends a finite time in the froth and comes off somewhere between what entered it and what equilibrium would give. Murphree measured the shortfall on the vapour side:

$$y_j = E\_MV \cdot y_{*j} + (1 - E\_MV) \cdot y_{j+1}, y_{*j} = K_j x_j$$

With  $E\_MV = 1$  the tray is an ideal stage and nothing changes – every existing case still runs byte for byte. With  $E\_MV < 1$ , ‘nStages’ stops counting ideal stages and starts counting REAL TRAYS. Which matters, because the hydraulics pass below sizes those trays: until now we were dimensioning, to the millimetre, a tray we then assumed was perfect.

$E\_MV$  is a DECLARED number. It comes from a correlation (O’Connell), a chart, or plant data, and it is yours to own – exactly like the orifice coefficient  $C_o$  and the weep constant  $K2$  in the hydraulics block. Choupo will not invent it.

Compare against column09 (same column,  $E\_MV = 1$ ):

distillate benzene 98.12 % (ideal) → 95.05 % ( $E\_MV = 0.70$ ) bottoms toluene 98.12 % (ideal) → 95.05 %

Fourteen real trays do less than fourteen ideal ones. That is the whole content of a tray efficiency, and it is why a column designed on ideal stages and built on real trays comes up short. Sweep  $E\_MV$  and watch the limit close:

$E\_MV$  0.50 0.70 0.90 0.99 0.999 1.000  $x\_D$  0.9100 0.9505 0.9738 0.9806 0.9812 0.98125 <- the ideal stage

Nothing is switched on at  $E\_MV = 1$ ; the effective  $K$  collapses to  $K$  and the case reproduces column09 to the last bit.

NOTE: the efficiency is folded into an effective  $K$ , with the vapour from the tray below lagged one outer pass – the linear solve keeps its tridiagonal shape. At convergence the lag vanishes and the Murphree relation holds exactly. The MESH ‘simultaneous’ method REFUSES the keyword rather than quietly returning ideal stages: there the relation would have to enter the residual and Jacobian.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0138888888889 |
| stream | distillate  | F              | 0.0138888888889 |
| stream | feed        | F              | 0.0277777777778 |
| kpi    | column11    | B              | 0.0138888888889 |
| kpi    | column11    | D              | 0.0138888888889 |
| kpi    | column11    | L_rect         | 0.0277777777778 |
| kpi    | column11    | L_strip        | 0.0555555555556 |
| kpi    | column11    | Q_condenser_kW | -1284.29753602  |
| kpi    | column11    | Q_reboiler_kW  | 1281.01358292   |
| kpi    | column11    | R              | 2               |
| kpi    | column11    | T_bottom       | 381.492133108   |
| kpi    | column11    | T_top          | 355.136328304   |
| kpi    | column11    | V_rect         | 0.0416666666667 |
| kpi    | column11    | V_strip        | 0.0416666666667 |

...and 23 more reference values in the case's *expected*.

## column12\_stage\_is\_a\_flash

tutorials/steady/distillation/column12\_stage\_is\_a\_flash



**What it does:** An equilibrium stage IS a flash: a tray's own two products, recombined and re-flashed, must come back as the same tray

**The story:** AN EQUILIBRIUM STAGE IS A FLASH – the identity witness.

A column's MESH equations and a flash drum's Rachford-Rice are two DIFFERENT pieces of code that claim to compute the same thing: the equilibrium split of a mixture at (T, P). Nothing in the corpus made them say it to each other. This case does.

feed → [tower] → distillate → bottoms → stageLiquid (the liquid LEAVING stage 5) → stageVapour (the vapour LEAVING stage 5)

stageLiquid + stageVapour → [recombine] → stageMix stageMix → [restage] → checkLiquid, checkVapour

THE CLAIM. Stage 5 is an equilibrium stage: by definition the vapour leaving it is in equilibrium with the liquid leaving it. So if the two are put back together and the mixture is allowed to separate again, at the same pressure and with no heat added or removed, NOTHING must happen: the same temperature, the same two compositions, and a vapour fraction equal to the ratio of the two draws.

$$T(\text{checkLiquid}) == T(\text{stageLiquid}) == T(\text{stageVapour})$$

$$z(\text{checkLiquid}) == z(\text{stageLiquid})$$

$$z(\text{checkVapour}) == z(\text{stageVapour})$$

$$F(\text{checkVapour}) / (F(\text{checkLiquid}) + F(\text{checkVapour})) == W / (U + W)$$

Nothing above is a number anyone typed. Every one of them is produced by the run, on both sides, and the gate 'check\_stage\_identity' compares them. That is what makes this a witness rather than a demonstration.

WHY IT IS WORTH A CASE. The column asks the thermodynamic package for K-values per stage; the flash asks it for an equilibrium split. Those are different entry points and, for a package that resolves CHEMISTRY, they are different code. The effective K-value a reacting stage uses ('ThermoPackage::stageK', added 2026-08-04 for the sour-water column) is exactly the sort of definition that can be subtly wrong – wrong normalisation, wrong basis, the reaction counted twice – and still produce a column that converges to a plausible-looking profile. This identity is what catches it. The molecular case here is the CONTROL: it must hold on a system where nobody doubts the answer, before the same construction is trusted on one where they should.

HOW THE RECOMBINATION WORKS, and the one thing to understand about it. 'recombine' is a plain mixer, and a Choupo mixer MIXES – it does not do the vapour/liquid split ("mixers mix, flashes separate"). Handed a liquid and a vapour it conserves total enthalpy exactly and reports the result as a single dominant phase at whatever temperature carries that enthalpy. That temperature is FICTITIOUS – a nominal vapour read out at 182.9 K, far below its saturation – and the case does not use it for anything. What the mixer is doing here is carrying an ENTHALPY across one arrow, faithfully; 'restage' is an ADIABATIC flash, so enthalpy is precisely the quantity it reads back, and the fictitious temperature cancels out of the answer. The identity is therefore an ENERGY statement as well as an equilibrium one: it would fail if the column and the flash disagreed about the enthalpy of either phase.

WHY THE DRAWS ARE ON ONE STAGE AND NOT TWO. Both draws come off stage 5. A liquid draw from stage 5 and a vapour draw from stage 6 would also recombine into something – and it would not be an equilibrium mixture, so the flash would move it. The point of the case is that when the two phases genuinely leave the SAME stage, the flash cannot move them at all.

*What it demonstrates: that Choupo's two independent equilibrium routines — the column's MESH equations and the flash drum's Rachford-Rice — give the same answer, by making them say it to each other inside one run.*

*Stage 5 is an equilibrium stage, so the vapour leaving it is in equilibrium with the liquid leaving it. Put them back together, let them separate again at the same pressure with no heat added, and nothing may happen. The run must produce*

*Not one of those numbers is typed into the case. Both sides are computed, by different code, and `bin/cu-rate/check_stage_identity.py` compares them.*

*Three things at once, and each of them has been wrong in a simulator somewhere:*

*1. The K-value definition. The column asks the package for K-values per stage; the flash asks it for a split. A wrong normalisation or a wrong basis in one of them shifts the profile without ever making the column fail to converge. 2. The enthalpy of each phase. The recombination carries an `*enthalpy*` across one arrow and the adiabatic flash reads it back. If the column and the flash priced the vapour differently, the recovered temperature would be wrong even with perfect K-values. 3. That a two-phase inlet is priced as two phases. Building this case is what found `adiabaticFlash` pricing every inlet as a sub-cooled liquid regardless of its vapour fraction — silently, since no case in the corpus had ever fed it one.*

*The identity holds to about  $1e-7$  relative, not to machine precision, and the reason is visible in the log: the adiabatic flash stops its outer Newton at an energy residual of  $\sim 2.5e-4$  J/mol out of  $5.4e+4$  J/mol. That is the limit of the `*check*`, not of the physics. The gate is set at  $1e-6$  relative, which is a factor of ten of headroom over the observed agreement.*

#### What to expect (golden-locked):

| kind   | unit/stream | key   | expected          |
|--------|-------------|-------|-------------------|
| stream | bottoms     | F     | 0.008333333333333 |
| stream | checkLiquid | F     | 0.002777777761205 |
| stream | checkVapour | F     | 0.00277777794351  |
| stream | distillate  | F     | 0.0138888888889   |
| stream | feed        | F     | 0.0277777777778   |
| stream | stageLiquid | F     | 0.0027777777778   |
| stream | stageMix    | F     | 0.0055555555556   |
| stream | stageVapour | F     | 0.0027777777778   |
| kpi    | recombine   | F_out | 0.0055555555556   |
| kpi    | recombine   | P_out | 101325            |
| kpi    | recombine   | T_out | 182.884624593     |
| kpi    | recombine   | n_in  | 2                 |
| kpi    | recombine   | vf    | 1                 |
| kpi    | restage     | F_in  | 0.0055555555556   |

*... and 37 more reference values in the case's `expected`.*



## column13\_sour\_water\_stage\_identity

tutorials/steady/distillation/column13\_sour\_water\_stage\_identity



**What it does:** The stage identity under CHEMISTRY: a sour-water tray's own two products, recombined and re-flashed, must come back as the same tray

**The story:** THE STAGE IDENTITY, UNDER CHEMISTRY.

This is column12 ('column12\_stage\_is\_a\_flash') with one thing changed: the thermodynamic package resolves an aqueous EQUILIBRIUM. Read that case first – the construction, the mixer's fictitious temperature and the tolerance are all explained there and not repeated here.

feed --> [tower] --> distillate, bottoms --> stageLiquid, stageVapour (stage 2) stageLiquid + stageVapour --> [recombine] --> stageMix stageMix --> [restage] --> checkLiquid, checkVapour

WHY IT IS A DIFFERENT TEST FROM column12, and not a copy.

In a molecular world a stage's K-values and a flash's split come from the same handful of lines: 'ThermoPackage::stageK' forwards straight to 'Kvec', and the identity, while worth having, is checking arithmetic that was never in doubt.

Here it is not the same code. A package that resolves chemistry has no K-value to hand out: the volatility of NH<sub>3</sub> is set by how much of it is NH<sub>4</sub><sup>+</sup>, which depends on the pH, which depends on how much CO<sub>2</sub> has left. 'stageK' therefore solves the stage's own equilibrium at its own (T, P, z) and returns an effective APPARENT K – a definition, made by us, which nothing outside this case checks. A wrong normalisation, a basis confusion between apparent components and species, or the speciation counted twice would all produce a column that converges to a plausible profile. This case is what refuses it.

AND THE DEFINITION WAS WRONG ONCE, WHICH IS THE ARGUMENT FOR THE CASE. The first version read  $K = y/x$  off the flash. That is right whenever the stage is two-phase, and it hands back a column of ZEROS the moment the trial state is subsaturated – which a column solver's trial states usually are, on the way to the answer. Every bubble-point residual then reads -1, the Jacobian is singular in T, and the MESH stops without moving: "Newton iters: 0". A K-value is an INCIPIENT quantity, defined over a liquid whether or not a vapour happens to be there, so the subsaturated branch now uses the equilibrium partial pressures over the fully speciated liquid,  $K_i = (p_i/P)/x_i$ . The two branches agree where they meet, because at a two-phase state the partials sum to P.

WHAT MUST HOLD, all of it produced by the run and none of it authored:

$T(\text{checkLiquid}) == T(\text{stageLiquid}) == T(\text{stageVapour})$   
 $z(\text{checkLiquid}) == z(\text{stageLiquid})$  (apparent component basis)  
 $z(\text{checkVapour}) == z(\text{stageVapour})$   
 $F(\text{checkVapour})/(F(\text{checkLiquid})+F(\text{checkVapour})) == W/(U+W)$

and, in addition to column12, the thing only a reacting stage can be asked: the two liquids must carry the SAME SPECIATION. The column's stage liquid and the re-flashed liquid are the same material at the same state, so their ion inventories must agree species by species – not merely their apparent compositions. That is the check that would catch an effective K which happens to reproduce the apparent split while resting on a different chemistry underneath.

THE COST, said out loud. Every stage K-value evaluation is a full reactive flash, and the MESH Jacobian is numerical, so one Newton pass costs of the order of (stages x unknowns) reactive flashes. The column is FOUR stages for that reason and no other: this is an identity witness, not a design study. A student who wants a real sour-water stripper wants the S2 case, which does not exist yet.

WHAT THE COLUMN ACTUALLY DOES, since it is worth reading the profile. It converges in 7 Newton iterations to  $|F| \sim 8e-10$ , and the answer is the physics: NH<sub>3</sub> climbs from 0.0054 in the reboiler liquid to 0.30 in the overhead vapour while CO<sub>2</sub> strips out ahead of it, and T falls from 371.5 K at the bottom to 354.6 K at the top. No gammaPhi column can produce that profile, because in a molecular world ammonia's volatility does not depend on how much carbon dioxide has left the liquid.

*What it demonstrates: a distillation column solved under an aqueous equilibrium, and the proof that the effective  $K$ -value it uses per stage is the same equilibrium a flash computes — checked through two independent code paths, on the apparent components \*and\* on the ions.*

*Read column12\_stage\_is\_a\_flash first. This is the same construction; only the thermodynamic package differs.*

*Four stages, sour water ( $\text{NH}_3 + \text{CO}_2$  in water), 1 atm,  $R = 2$ ,  $D = 3$  kmol/h. It converges in 7 Newton iterations to  $\text{?F?} \approx 8\text{e-}10$ , and the profile is the physics:*

*No gammaPhi column can produce that profile, because in a molecular world ammonia's volatility does not depend on how much carbon dioxide has left the liquid. Here it does: the carbon dioxide a liquid still carries is what holds its ammonia as  $\text{NH}_4^+$ , so stripping the  $\text{CO}_2$  sets the ammonia free and the free ammonia then strips. (The usual shorthand for that is "stripping  $\text{CO}_2$  raises the pH" — which is true of a fixed solution and is \*not\* how it reads tray by tray here. See the per-tray table below: it is the carbonate loading that orders cleanly, not the pH.)*

*Both halves of stage 2 are drawn as real streams, recombined, and re-flashed at the same  $(T, P)$ . Nothing may happen — and nothing does, to about  $1\text{e-}9$  relative:*

*The ion inventories agreeing is the claim that a comparison of apparent compositions alone would miss: an effective  $K$  could reproduce the apparent split while resting on a different chemistry underneath.*

#### What to expect (golden-locked):

| kind   | unit/stream | key   | expected          |
|--------|-------------|-------|-------------------|
| stream | bottoms     | F     | 0.0259166666667   |
| stream | checkLiquid | F     | 0.00041666665957  |
| stream | checkVapour | F     | 0.000138888895986 |
| stream | distillate  | F     | 0.000833333333333 |
| stream | feed        | F     | 0.0273055555556   |
| stream | stageLiquid | F     | 0.000416666666667 |
| stream | stageMix    | F     | 0.000555555555556 |
| stream | stageVapour | F     | 0.000138888888889 |
| kpi    | recombine   | F_out | 0.000555555555556 |
| kpi    | recombine   | P_out | 101325            |
| kpi    | recombine   | T_out | 483.325341357     |
| kpi    | recombine   | n_in  | 2                 |
| kpi    | recombine   | vf    | 0                 |
| kpi    | restage     | F_in  | 0.000555555555556 |

*...and 37 more reference values in the case's expected.*

**column14\_klemola\_c4\_splitter**

tutorials/steady/distillation/column14\_klemola\_c4\_splitter



**What it does:** Klemola-Ilme industrial C4 splitter: 74 real trays at the measured  $E_{MV} = 1.191$ , products anchored on the plant's own key-component analysis

**The story:** an INDUSTRIAL i-butane/n-butane fractionator, 74 real trays, 2.9 m diameter, run on its own measured specification.

PRIMARY, and the only source used for any number below:

Klemola, K. T.; Ilme, J. K. Distillation Efficiencies of an Industrial-Scale i-Butane/n-Butane Fractionator. Ind. Eng. Chem. Res. 1996, 35 (12), 4579-4586. IE960390R.

The transcription record – Tables 1 and 2 number by number, the efficiency results, and the four things the paper does NOT publish – is docs/design/klemola-c4-splitter-reference-case.md. No number in this case is absent from that record; where the record says the paper is silent, this header DECLARES an assumption and says it is one.

**WHY THIS CASE EXISTS.** The corpus's other distillation tutorials are pinned on self-recorded goldens; the closest thing to an external anchor (acetone06/07) compares Choupo against another SIMULATION. This is the first column anchored on a measured plant. What it puts under test is the COLUMN MODEL – the MESH solve, the declared tray efficiency, the material balance – not an activity model: the system is nearly ideal, so a disagreement here cannot be blamed on thermodynamics and then forgotten.

### 1. WHAT IS TRANSCRIBED, PER NUMBER (record Sections 2 and 3, Tables 1 / 2)

number of trays 74 Table 1 → nStages 75 (see 2) feed tray 37 Table 2 → feedStage 37 (see 2) column top pressure 658.6 kPa Table 2 → P (see A2) feed pressure 892.67 kPa Table 2 → used in A1 only feed flow rate 26234 kg/h Table 2 → 0/feed (see 3) distillate flow rate 8115 kg/h Table 2 → distillateRate (see 4) reflux flow rate 92838 kg/h Table 2 → refluxRatio (see 5) bottoms flow rate 18119 kg/h Table 2 → NOT an input (see 6) Murphree tray efficiency 119.1 % text p.4580 → murphreeEfficiency reflux temperature 18.5 C Table 2 → NOT USED (see 7) boiler duty 10.240 MW Table 2 → NOT an input (see 8)

Table 1's tray geometry (2.900 m diameter, 0.600 m spacing, 4.9 m<sup>2</sup> active area, 772 valves, 18.82 % free hole area, ...) is transcribed in the record for the reader and is used by NOTHING here: no hydraulics block is declared, because the anchor is the separation, not the rating. Ballast V-1 valve trays are not the sieve tray Choupo's hydraulics pass models, and rating them with a sieve correlation would be a number with no source behind it.

### 2. STAGE COUNT AND FEED STAGE – and one AMBIGUITY, declared

Choupo's Wang-Henke cascade numbers stage 1 = TOP TRAY and stage N = REBOILER; the total condenser sits outside the stage list (the distillate is y<sub>1</sub>, the vapour leaving the top tray, condensed). So 74 real trays plus the reboiler gives

nStages 75 (= 74 trays + 1 reboiler) feedStage 37 (Table 2, taken as counted from the TOP)

**ASSUMPTION A0** – the paper prints "feed tray 37" and does not state the direction of its tray numbering; the record does not either, and the primary text was checked. Counted from the bottom instead, the feed would sit on tray 38 of 74 as Choupo numbers them: 37 vs 38 rectifying trays out of 74. Stage 1 = top is taken because it is Choupo's own convention and because the difference is one tray in the middle of a 74-tray column. Anyone who establishes the paper's direction should change this line and say so.

### 3. THE FEED – wt % to mole flows, arithmetic shown

Table 2 gives the feed as 26234 kg/h at these wt % (the paper's own units and its own rounding to 0.1 wt %). THE COLUMN SUMS TO 100.1, NOT 100 – that is the rounding, and it has to be resolved before anything can be converted. It is resolved by NORMALISING the wt % to unity and applying them to the

measured 26234 kg/h, so that the total mass flow – which the paper's own overall balance closes exactly ( $18119 + 8115 = 26234$ ) – is preserved. The alternative (applying the raw wt % literally) would invent 26 kg/h of feed.

MW from each component's own data/standards record.  $\text{kmol/h} = \text{kg/h} / \text{MW}$ .

component wt% wt% norm kg/h MW kmol/h x

```
propane 1.5 1.49850 393.1169 44.0960 8.915024 0.0196925 isoButane 29.4 29.37063 7705.0909 58.1222
132.567090 0.2928294 nButane 67.7 67.63237 17742.6753 58.1230 305.260832 0.6742952 1Butene 0.2 0.19980
52.4156 56.1063 0.934219 0.0020636 isoButene 0.2 0.19980 52.4156 56.1063 0.934219 0.0020636 trans2Butene
0.1 0.09990 26.2078 56.1063 0.467110 0.0010318 neopentane 0.1 0.09990 26.2078 72.1488 0.363246 0.0008024
isopentane 0.8 0.79920 209.6623 72.1488 2.905971 0.0064190 nPentane 0.1 0.09990 26.2078 72.1488 0.363246
0.0008024
```

TOTAL 100.1 100.00000 26234.0000 452.710957 1.0000000

mean feed MW =  $26234 / 452.710957 = 57.94867 \text{ kg/kmol}$

Those kmol/h are what 0/feed carries.

ASSUMPTION A1 – THE FEED THERMAL STATE. Table 2 gives the feed PRESSURE (892.67 kPa) and no feed temperature; record Section 5.2 names this gap and names "saturated liquid at the feed pressure" as the natural assumption. That is what is assumed here, and it has a consequence the case must carry rather than hide: a liquid saturated at 892.67 kPa is NOT saturated on a tray at 658.6 kPa – it flashes on the way in. Both steps were computed with THIS CASE'S OWN thermo (the dict in constant/, no other model), by two throwaway runs whose numbers are reproduced here so a reader can repeat them:

bubbleT of the feed composition at 892.67 kPa  $\rightarrow T = 334.4218 \text{ K}$  adiabaticFlash of that stream to 658.6 kPa  $\rightarrow V/F = 0.0929333445$   $T = 323.5875148 \text{ K}$

feedQuality  $q = 1 - V/F = 0.9070666555$

and 0/feed carries the COLUMN-INLET state (323.5875148 K, 658600 Pa, vaporFraction 0.0929333445), not the line state. The paper's 892.67 kPa therefore appears in this case only as the premise of A1, stated above. NOTE the arity cost, declared:  $q$  lives in two places (this key and the stream's vaporFraction) because the Wang-Henke path takes  $q$  as a number and does not read it off the inlet. They must be kept equal; the derivation above is how you check them.

Sensitivity, MEASURED rather than asserted – the competing assumption was run. Declaring instead "saturated liquid AT THE COLUMN PRESSURE" gives  $T = 323.2706 \text{ K}$  (bubble point at 658.6 kPa) and  $q = 1.0$  exactly, and the whole column moves by:

$x\_D(\text{i-butane}) 0.92571 \rightarrow 0.92576 (+5\text{e-}5)$   $x\_B(\text{i-butane}) 0.00352 \rightarrow 0.00350 (-2\text{e-}5)$

i.e. below the resolution of anything this case compares against. A1 is therefore chosen on FIDELITY TO THE RECORD, not on the answer, and the simpler assumption is available to anyone who prefers one derived number to three.

#### 4. DISTILLATE RATE – a mass spec the engine takes in moles

Table 2 gives 8115 kg/h; 'distillateRate' is a MOLAR flow. The conversion needs a mean MW, and the only one available is the one from the paper's own MEASURED top composition (top wt % of Table 2, which sums to 100.0 exactly):

component wt% kg/h MW kmol/h y

```
propane 5.3 430.0950 44.0960 9.753606 0.0686760 isoButane 93.5 7587.5250 58.1222 130.544353 0.9191740
nButane 0.2 16.2300 58.1230 0.279235 0.0019661 1Butene 0.4 32.4600 56.1063 0.578545 0.0040736 trans2Butene
0.6 48.6900 56.1063 0.867817 0.0061104 i-butene / neopentane / i-pentane / n-pentane: 0 wt % as printed
```

TOTAL 100.0 8115.0000 142.023556 1.0000000

mean distillate MW =  $8115 / 142.023556 = 57.13841 \text{ kg/kmol} \rightarrow \text{distillateRate } 142.023556 \text{ kmol/h}$

SAID PLAINLY: this input therefore borrows the measured distillate COMPOSITION, which is one of the quantities the case then predicts. A fully self-consistent conversion would iterate the mean MW against the

predicted composition. It is not iterated here, and the size of what is left on the table is MEASURED rather than waved at: the run's own converged/distillate comes out at mean MW 57.2186, i.e. +0.14 % on the 57.13841 used above, so the distillate leaves at 8126.4 kg/h against the Table-2 8115 kg/h. One iteration would close that; it is left open because the residue is smaller than the 0.1 wt % resolution of the composition it came from. If a future edit moves the distillate composition materially, this number must be recomputed with it.

#### 5. REFLUX RATIO – why the MASS ratio IS the MOLAR ratio here

Table 2 gives reflux 92838 kg/h and distillate 8115 kg/h, both MASS. The condenser is TOTAL, so the reflux and the distillate are drawn from one condensed stream and have IDENTICAL composition, hence identical mean MW.  $L/D$  in moles =  $(L\_mass/MW)/(D\_mass/MW) = L\_mass/D\_mass$  exactly – no composition-weighted conversion is needed, and none is done:

$$R = 92838 / 8115 = 11.440296 \text{ (11.4402957...)}$$

A NOTE ON A NUMBER IN CIRCULATION: 11.4441 has been quoted for this ratio. It is not what the division gives.  $8115 \times 11.4441 = 92859.9$  kg/h, 22 kg/h above the Table-2 reflux. This case uses 11.440296.

#### 6. BOTTOMS – an OUTPUT, and it does not land on 18119 kg/h

$B = F - D$  is the engine's mole balance:  $452.710957 - 142.023556 = 310.687401$  kmol/h. Priced at the PREDICTED bottoms composition that comes out at 18107.6 kg/h, 0.06 % under the Table-2 18119 kg/h. It cannot land on it exactly: F's molar total is fixed by the feed mass and the feed's measured composition, D's by the distillate mass and the distillate's measured composition, and the paper's three compositions do not close per component (record Section 5.4 – the wt % columns sum to 100.1, 100.0 and 100.3). The mass residue is the printing, not the column. 18119 kg/h is therefore transcribed above and used by nothing.

#### 7. REFLUX TEMPERATURE 18.5 C – transcribed, NOT USED, and why

18.5 C is far below the bubble point of the overhead at 658.6 kPa (about 41 C, and the run prints its own top-stage temperature). The plant's condenser therefore SUBCOOLS its reflux. Choupo's Wang-Henke path returns saturated reflux and has no subcooling keyword; declaring one would be inventing a model. The number is transcribed so nobody has to wonder whether it was overlooked, and the consequence is stated: the real column's internal liquid traffic in the top few trays is higher than the constant-molar-overflow value used here, by the amount the subcooled reflux condenses on contact. That is a KNOWN, UNMODELLED difference between this case and the plant, not a tuning knob.

#### 8. BOILER DUTY 10.240 MW – transcribed, and NOT claimed as an anchor

The run reports a reboiler duty of about 8.47 MW against the Table-2 10.240 MW. The case does NOT anchor on that, and the reason is in its own output rather than in an argument.

The Wang-Henke path solves M and E under CONSTANT MOLAR OVERFLOW: the internal traffic is fixed by R, D and q, and NO per-stage energy balance is solved. The duties are then reported by closing the column's overall enthalpy change, and the flowsheet's independent energy audit raises the first-law advisory on this run – 98.42 % closure, a residual of 2.96 kW, 0.017 % of the boundary enthalpy. That is small, and it is NOT evidence that the duty is right: the ledger and the duty are computed from the same three stream states, so agreement between them says the arithmetic is consistent, not that a column with no energy equation found the right vapour traffic. The companion arm column15 makes the point unmissable – the same three streams to within 0.8 kW of enthalpy, and its audit closes at -256.84 % with a 670.7 kW residual. Two runs of one column, one ledger claim each, three orders of magnitude apart. The corpus's other Wang-Henke columns sit further out still (column01/09/11's own energyBalance\_byUnit.csv), so this is a property of the solver path and of the reporting around it, and it predates this case by a long way.

Also unmodelled and worth naming beside the duty: the plant's subcooled reflux (7) and the condenser's own subcooling duty are not in this figure.

A duty comparison resting on any of that would be a number pretending to be evidence. 10.240 MW is transcribed; 8.47 MW is reported; neither is an anchor row.

#### 9. THERMO – mirroring the paper, not substituting for it

The paper states its own model: liquid activity by UNIFAC, vapour pressure by Antoine, vapour fugacity by the original Soave-Redlich-Kwong. All three are public in this tree, so constant/thermoPhysPropDict

declares exactly that and no comparison here hides a model swap. All nine components carry a UNIFAC group decomposition in data/standards/components/, and the only main-group pair the mixture needs ( $\text{CH}_2 \leftrightarrow \text{C}=\text{C}$ , for the three butenes) is in data/standards/parameters/UNIFAC/interactions.dat. There is no approximations{} block and none is needed: every component is priced by the model that is declared.

TWO CAVEATS THE RUN ANNOUNCES FOR ITSELF, repeated here so they are read before the numbers and not after: seven of the nine component records carry ‘reviewStatus interim’ (imported from CoolProp 7.2.0, not yet checked against primaries). The thermodynamics under this anchor is not itself anchored. propane’s Antoine fit is declared valid to 320.7 K and the stripping section runs to about 329 K, so its Psat is EXTRAPOLATED there. Propane is 1.5 wt % of the feed and leaves overhead; the extrapolation is announced by the engine on every run and is not corrected here.

10. METHOD – Wang-Henke, and it is not a preference

‘model simultaneous’ REFUSES a murphreeEfficiency other than 1: there the Murphree relation would have to enter the residual and the Jacobian, and it does not. The efficiency is implemented on the Wang-Henke path only, folded into an effective K with the vapour from the tray below lagged one outer pass, so this case is Wang-Henke because that is where the measured input is honoured. It costs iterations: the corpus’s other columns converge in tens of outer passes, this one in several hundred, because 74 trays of a close-boiling pair is a long cascade. maxOuterIter is raised accordingly.

11. E\_MV = 1.191 IS A MEASURED INPUT, AND IT IS ABOVE 1

The paper back-calculates a Murphree TRAY efficiency of 119.1 % (74 real trays behaving as 88 ideal ones; overall column efficiency 118.9 %). Above 100 % is not an error: E\_MV compares the vapour leaving a tray with equilibrium against that tray’s OUTLET liquid, and on a 2.9 m cross-flow tray the liquid enters rich and leaves lean, so the vapour has met liquid richer than the outlet. Nothing anywhere exceeds equilibrium. The engine ADMITS the value and announces it, and states what it does not check (the plug-flow ceiling, which needs the point efficiency and  $mV/L$  – neither is declared).

It is an INPUT. Predicting it needs the rate-based programme Choupo deliberately does not have; the paper spends its own length on that difficulty and its Table 4 shows ten published methods disagreeing across a factor of five. This case reads the measurement; it does not reproduce it.

The companion case column15\_klemola\_ideal\_trays is this column with murphreeEfficiency 1.0 and nothing else changed. The gap between the two is the pedagogical payload: how much of this separation the efficiency carries.

12. HAND ESTIMATE – what the specs imply, before the engine is trusted

Relative volatility of the keys from the same Antoine constants the case uses, at 325 K (mid-column):

$$\text{Psat}(\text{i-C}_4) = 7.2334 \text{ bar}, \text{Psat}(\text{n-C}_4) = 5.1636 \text{ bar} \rightarrow \alpha = 1.4008$$

FENSKE, on the measured key split (moles from the tables above):

$$S = (d_{\text{iC}_4}/b_{\text{iC}_4})(b_{\text{nC}_4}/d_{\text{nC}_4}) = (130.544/0.932)(304.898/0.279) = 1.5287\text{e}5 \quad N_{\text{min}} = \ln S / \ln \alpha = 11.937 / 0.33698 = 35.4 \text{ ideal stages}$$

UNDERWOOD, on the four real components ( $q = 1$  for this estimate):

$$\theta = 1.2483 \quad R_{\text{min}} = 7.626 \quad R / R_{\text{min}} = 11.440296 / 7.626 = 1.500$$

That last line is the strongest check available on the transcription: the reflux and distillate rates, taken from two separate lines of Table 2 and converted independently, land on exactly  $1.5 \times R_{\text{min}}$  – the textbook design point. Two mistranscribed numbers do not do that.

GILLILAND then gives  $X = 0.3066$ ,  $Y = 0.3761$ ,  $N = 57.4$  ideal stages – fewer than the 74 trays present, and fewer than the 88 ideal trays the paper’s own rigorous fit needs. The disagreement is expected and is not evidence against either: Gilliland is a correlation fitted across many systems and is known to be optimistic at high  $N/N_{\text{min}}$  on close-boiling pairs, and 88 is a fitted rigorous simulation, not a measurement. What the estimate establishes is the ORDER: this separation needs tens of stages, not five and not five hundred, and the declared reflux is comfortably above minimum. If the engine returned a near-perfect split on ten trays, or failed to separate at all on 74, the hand estimate would have caught it.

13. THE ANCHOR – and what a PASS on it does NOT mean

Compared: i-butane and n-butane, in BOTH products, against the Table-2 measurements. Only the keys. Record Section 5.4 forbids a trace-component anchor before this case was built – 0.1 wt % resolution on a 0.2 wt % number is  $\pm 25$  % before anything physical happens, and the traces do not close at all (propane +23 %, i-butene -100 %). The keys close to about 1 %, which is what a key-component anchor needs. The band on any anchor row belongs to the table's own rounding, not to the solver's tolerance.

This is an ENDPOINT anchor. The paper publishes no measured tray temperature profile and no measured tray composition profile – its figures are EFFICIENCY profiles from the back-calculation – so nothing here tests what the column model predicts INSIDE. The internals stay unanchored; the natural second case is a different primary (Vogelpohl's total-reflux profiles), never blended into this one.

What a PASS would mean: that the MESH solve plus a declared efficiency reproduces an industrial column's products from its own specification, on public thermodynamics. What it would NOT mean: that the internals are right (nothing here measures them), nor that Choupo could have PREDICTED the efficiency, nor that the trace components are modelled well.

#### 14. THE ANCHOR TABLE – observed, and it does NOT all agree

Predicted product compositions, converted back to wt % from this case's own converged/ streams, beside the Table-2 measurements. The "band" column is the measurement's OWN rounding: the paper prints to 0.1 wt %, so a printed value  $v$  carries  $\pm 0.05$  wt %.

DISTILLATE (predicted 8126.4 kg/h vs the 8115 kg/h spec – see 4)

component predicted measured band verdict

i-butane 94.033 93.5  $\pm 0.05$  +0.53 wt % HIGH n-butane 0.0015 0.2  $\pm 0.05$  LOW by  $\sim 130\times$

BOTTOMS (predicted 18107.6 kg/h vs the Table-2 18119 kg/h – see 6)

component predicted measured band verdict

i-butane 0.351 0.3  $\pm 0.05$  INSIDE the band n-butane 97.984 98.1  $\pm 0.05$  -0.12 wt %, just out

THREE OF THE FOUR KEY ROWS LAND WHERE A KEY-COMPONENT ANCHOR WANTS THEM, around the 1 % level at which record section 5.4 says the keys close at all. THE FOURTH DOES NOT, and it is not a rounding argument: the model rejects n-butane overhead about two orders of magnitude harder than the plant did. 0.2 wt % is at the printing resolution ( $\pm 25$  % before any physics), but  $130\times$  is not a resolution effect.

It is recorded, not explained and not tuned. What can be said without guessing: the disagreement is NOT the efficiency, because the control arm column15 at  $E\_MV = 1$  gives 0.0144 wt % – still an order of magnitude below the measurement. Under this model 74 IDEAL trays already over-separate this key, which is the opposite direction from the paper's own finding that 74 real trays behave as 88 ideal ones. Candidates a future slice could test, each of which this case declares as an assumption or a caveat above: the single column pressure (A2), the tray-numbering direction (A0), the unmodelled subcooled reflux (7), the absent per-stage energy balance (8), and seven interim component records (thermo dict). Whichever it is, an anchor that disagreed and said so is worth more than one that agreed after something was adjusted to make it.

THE OTHER FIVE COMPONENTS ARE NOT ANCHOR ROWS and are listed only so nobody reads their absence as concealment: propane 4.84 vs 5.3 wt % overhead and 0.00 vs 0.3 in the bottoms; trans-2-butene 0.00 vs 0.6 overhead; i-butene 0.58 vs 0 overhead. Record section 5.4 already showed these do not close in the SOURCE – propane +23 %, i-butene -100 %, trans-2-butene +155 % between the feed and the two products as printed – so a number computed here has nothing to be compared against. Anchoring on them would be measuring the printing.

#### 15. THE TWO ARMS, side by side

column14 ( $E\_MV = 1.191$ ) against column15 ( $E\_MV = 1.0$ ), everything else identical:

quantity  $E\_MV$  1.191  $E\_MV$  1.0 measured

distillate i-butane, wt % 94.033 93.975 93.5 distillate n-butane, wt % 0.0015 0.0144 0.2 bottoms i-butane, wt % 0.351 0.378 0.3 bottoms n-butane, wt % 97.984 97.977 98.1 top temperature, K 314.250 314.116 – reboiler temperature, K 329.437 329.435 – outer iterations 349 426 –

WHAT THE EFFICIENCY BUYS, read off that table: essentially nothing on the major components – i-butane overhead moves 0.06 wt %, n-butane in the bottoms 0.007 – and a real factor on the MINOR key in each product: the n-butane carried overhead drops about 10x, and the i-butane lost to the bottoms falls 7 %.

That shape is worth more than a single number. At 74 trays and  $R = 11.4$  this column is far past the point where separating power limits the major components; extra power goes entirely into the impurity that is left, which is exactly where a plant's money is. A student who expects a 19 % efficiency bonus to show up as a 19 % anything will not find it, and finding out why is the lesson.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0863020558333 |
| stream | distillate  | F              | 0.0394509877778 |
| stream | feed        | F              | 0.125753043611  |
| kpi    | splitter    | B              | 0.0863020558333 |
| kpi    | splitter    | D              | 0.0394509877778 |
| kpi    | splitter    | L_rect         | 0.45133097767   |
| kpi    | splitter    | L_strip        | 0.565397370357  |
| kpi    | splitter    | Q_condenser_kW | -8654.33073632  |
| kpi    | splitter    | Q_reboiler_kW  | 8467.13778877   |
| kpi    | splitter    | R              | 11.440296       |
| kpi    | splitter    | T_bottom       | 329.436820919   |
| kpi    | splitter    | T_top          | 314.249966433   |
| kpi    | splitter    | V_rect         | 0.490781965448  |
| kpi    | splitter    | V_strip        | 0.479095314524  |

*...and 16 more reference values in the case's **expected**.*



**column15\_klemola\_ideal\_trays**

tutorials/steady/distillation/column15\_klemola\_ideal\_trays



**What it does:** The Klemola-Ilme C4 splitter with IDEAL trays: the same column at  $E\_MV = 1$ , the control arm that says what the measured efficiency is worth

**The story:** the CONTROL ARM of the Klemola-Ilme anchor.

This is column14\_klemola\_c4\_splitter with ONE line changed:

murphreeEfficiency 1.191;  $\rightarrow$  murphreeEfficiency 1.0;

Everything else – the 74 trays, the feed on tray 37, the feed state, the pressure, the distillate rate, the reflux ratio, the thermo dict, the three 0/ files, the convergence controls – is byte-identical to that case, and if the two ever drift apart this case stops being a control. Read column14's flowsheetDict for the provenance of every number; it is the primary document and none of it is repeated here.

Klemola, K. T.; Ilme, J. K. Distillation Efficiencies of an Industrial-Scale i-Butane/n-Butane Fractionator. Ind. Eng. Chem. Res. 1996, 35 (12), 4579-4586.

WHY THIS IS A SEPARATE CASE AND NOT A SECOND ARM INSIDE column14

A Choupo case is ONE pass: one converged/ state, one KPI set, one golden. Two efficiencies could have been swept in a single case through system/outerDict, and that was rejected: a sweep publishes the DRIVER's KPI table, so the two arms' products would live in a sweep artefact instead of in two real converged/ directories, and the anchor – which is an ENDPOINT anchor, a comparison of product compositions – would have nowhere natural to sit. Two cases give two converged states, two goldens, and two independently sealable units of evidence, at the cost of the duplication named above.

WHAT THE GAP BETWEEN THE ARMS MEANS – and what it does NOT

$E\_MV = 1$  makes every one of the 74 trays an EQUILIBRIUM stage: the vapour leaving it is in equilibrium with the liquid leaving it. No real tray does that, and this plant's trays do BETTER than that – the paper back-calculates 119.1 %, because on a 2.9 m cross-flow tray the liquid is not well mixed, so the vapour has been in contact with liquid richer than the outlet.

The difference between this case's products and column14's is therefore the answer to a question a student can otherwise only be told: how much of an industrial separation is carried by the tray efficiency, as against the tray COUNT and the reflux? Run both and subtract. The paper's own version of the same arithmetic is that 74 real trays behave as 88 ideal ones.

WHAT IT IS NOT. This case is not an anchor and must never be compared with the Table-2 measurements as though it were one: it deliberately declares an efficiency the plant does not have. Its role is to be the OTHER END of a one-parameter difference. If it happened to sit closer to the measured products than column14 does, that would be a finding about the model, not a reason to prefer  $E\_MV = 1$  – 119.1 % is the measurement.

Note also that  $E\_MV = 1$  is the engine's DEFAULT, and the key is written out in full below anyway. An absent key would mean the same thing to the solver and something quite different to a reader: it would look like an oversight in a case whose entire subject is the efficiency.

WHAT THE TWO ARMS ACTUALLY GAVE

quantity  $E\_MV$  1.191  $E\_MV$  1.0 measured

distillate i-butane, wt % 94.033 93.975 93.5 distillate n-butane, wt % 0.0015 0.0144 0.2 bottoms i-butane, wt % 0.351 0.378 0.3 bottoms n-butane, wt % 97.984 97.977 98.1 top temperature, K 314.250 314.116 – reboiler temperature, K 329.437 329.435 – outer iterations 349 426 –

The efficiency moves the MAJOR components by almost nothing (i-butane overhead by 0.06 wt %) and the MINOR key in each product by a real factor (n-butane carried overhead drops about 10x; i-butane lost to the bottoms falls 7 %). At 74 trays and  $R = 11.4$  the column is far past the point where separating power limits the majors, so extra power goes entirely into the impurity that is left. column14's flowsheetDict section 15 argues that at length; this is the same table so that either case can be read alone.

A FINDING THIS ARM IS WHAT ESTABLISHED, and it belongs here because this is the case that establishes it: at  $E_{MV} = 1$  the model ALREADY rejects n-butane overhead an order of magnitude harder than the plant measured (0.0144 wt % against 0.2). So the anchor's one clear disagreement is not caused by the declared efficiency – turning the efficiency off does not recover the measurement, it only halves the distance in the log. Under this model 74 ideal trays over-separate this key, which runs opposite to the paper's own conclusion that 74 real trays behave as 88 ideal ones. Recorded; not explained here and not tuned away.

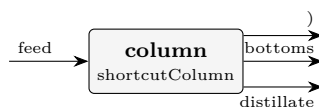
ONE MORE OBSERVATION, and it is about the ENGINE rather than the column. These two runs produce product streams whose tabulated enthalpies agree to 0.8 kW, and their flowsheet energy audits disagree wildly: column14 closes at 98.42 % (residual 2.96 kW) and this arm at -256.84 % (residual 670.7 kW). A difference that large between two nearly identical states does not come from the physics of the two arms; it is in how the energy report prices the products, and this pair is a compact reproducer for anyone who goes looking. Neither case claims a duty anchor – see column14's section 8.

#### What to expect (golden-locked):

| kind                                                            | unit/stream | key            | expected        |
|-----------------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                          | bottoms     | F              | 0.0863020558333 |
| stream                                                          | distillate  | F              | 0.0394509877778 |
| stream                                                          | feed        | F              | 0.125753043611  |
| kpi                                                             | splitter    | B              | 0.0863020558333 |
| kpi                                                             | splitter    | D              | 0.0394509877778 |
| kpi                                                             | splitter    | L_rect         | 0.45133097767   |
| kpi                                                             | splitter    | L_strip        | 0.565397370357  |
| kpi                                                             | splitter    | Q_condenser_kW | -8659.56425253  |
| kpi                                                             | splitter    | Q_reboiler_kW  | 8471.59434507   |
| kpi                                                             | splitter    | R              | 11.440296       |
| kpi                                                             | splitter    | T_bottom       | 329.435279168   |
| kpi                                                             | splitter    | T_top          | 314.116313789   |
| kpi                                                             | splitter    | V_rect         | 0.490781965448  |
| kpi                                                             | splitter    | V_strip        | 0.479095314524  |
| ...and 15 more reference values in the case's <i>expected</i> . |             |                |                 |

**shortcut01\_benzene\_toluene**

tutorials/steady/distillation/shortcut01\_benzene\_toluene



**What it does:** Shortcut (FUG) design of a benzene/toluene column

**The story:** Preliminary column design by the Fenske-Underwood-Gilliland (FUG) shortcut method (the built-in ‘shortcutColumn’). The same benzene/ toluene separation as the rigorous ‘column01’, but designed in closed form: you specify the key recoveries and the reflux, and FUG returns the minimum stages, the minimum reflux, the actual stage count and the feed location.

Feed: 100 kmol/h, 50/50 benzene/toluene, saturated liquid, 1 atm. Keys: LK = benzene, HK = toluene ( $\alpha \approx 2.5$ ). Spec: 99 % of the benzene to the distillate, 1 % of the toluene. Reflux:  $R = 1.3 \cdot R_{\min}$ .

Expected ( $\alpha \approx 2.5$ ):  $N_{\min} \approx 10.1$  (Fenske, total reflux)  $R_{\min} \approx 1.3$  (Underwood)  $N \approx 22$  (Gilliland, at  $R = 1.3 R_{\min}$ ) distillate ~99 % benzene, bottoms ~99 % toluene; mass closes 100 %.

Compare with the rigorous column01 (Wang-Henke): FUG is a fast design estimate; the rigorous column is the verification. FUG does NOT handle azeotropes — it assumes constant relative volatility.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected        |
|--------|-------------|---------------|-----------------|
| stream | bottoms     | F             | 0.0138888888889 |
| stream | distillate  | F             | 0.0138888888889 |
| stream | feed        | F             | 0.0277777777778 |
| kpi    | column      | B             | 0.0138888888889 |
| kpi    | column      | D             | 0.0138888888889 |
| kpi    | column      | N_min         | 10.067785029    |
| kpi    | column      | N_theoretical | 21.5880731317   |
| kpi    | column      | R             | 1.68248609289   |
| kpi    | column      | R_min         | 1.29422007145   |
| kpi    | column      | alpha_LK_HK   | 2.49137883567   |
| kpi    | column      | feed_stage    | 10.7940365659   |
| kpi    | column      | theta         | 1.42716041594   |
| kpi    | column      | x_B_HK        | 0.99            |
| kpi    | column      | x_D_LK        | 0.99            |

...and 3 more reference values in the case's *expected*.

**stripper01\_sour\_water**

tutorials/steady/distillation/stripper01\_sour\_water



**What it does:** S2: a sour-water stripper over the chemistry – eight reactive trays, the CO<sub>2</sub>-strips-first mechanism measured across the whole column.

**The story:** S2 of the sour-water programme: a REAL multi-stage column over the chemistry, not a two-stage identity rig.

column13 proved the machinery: one reactive flash per stage per residual, the apparent components the state, the ions internal to each tray. This case is what the machinery exists FOR – a sour-water stripper: feed near the top, most of the trays below it stripping, the reboiler making the vapour that carries NH<sub>3</sub> and CO<sub>2</sub> overhead.

THE MECHANISM IS THE LESSON (docs/design/sour-water-stripper-scope.md ?4.3): stripping CO<sub>2</sub> RAISES the liquid pH down the column, and the rising pH converts NH<sub>4</sub><sup>+</sup> back to free NH<sub>3</sub>, which then strips. The separation and the speciation are the same problem. Eight trays instead of column13's four give the mechanism room to show, and it shows STRICTLY, measured on this converged profile: the carbonate loading falls tray by tray from 0.524 mol/kg (top) to 0.024 (reboiler) – over an order of magnitude – while the free-ammonia fraction rises 0.802 → 0.947 with no exception, and the pH climbs 8.46 → 8.80 below the top tray. (The top tray itself sits ~7 K colder than tray 2 and its pH reads higher for the temperature of its own K<sub>w</sub> – comparing pH across that jump compares two water dissociations, which is why the pH claim starts at tray 2.) check\_tray\_chemistry's ordering claim is tested across this whole span rather than across four points.

WHAT BUILDING IT FOUND, because the first honest run is an instrument (both fixes are the engine's, recorded in the scope doc ?6d): a MESH initialisation ramps T across the column, and an 8-stage ramp visits temperatures a 4-stage ramp never did – including ones above the feed's two-phase band, where the reactive Newton has no interior V/F and stalled. stageK now prices such a trial INCIPIENT over the hypothetical speciated liquid (announced once), the same construction its subsaturated branch always used – continuous K, and the aid shapes the path, never the answer (a converged stage sits at saturation, where the constructions agree); the per-tray chemistry report used to FLASH each converged tray liquid – a liquid pinned by the MESH to its own bubble point, i.e. the flash's degenerate corner – and at reflux 0.5 five trays of eight printed NaN for chemistry the package resolves easily. The report now asks the right question (speciate the LIQUID), and NaN is reserved for a tray whose speciation itself fails.

WHAT THIS CASE DECLARES, said plainly (the column12/13 rule): 'model simultaneous;' – the rigorous MESH, whose per-stage equilibrium is 'ThermoPackage::stageK': for this reactive package, one reactive flash per stage per residual evaluation. Nested, not a species-basis MESH – the Jacobian sees apparent components only. the per-tray chemistry (pH, ionic strength, every molality) rides in reports/unitOperations/tower/profile.csv – engine-owned, the same surface column13 writes.

DATA SCOPE: sealed (bin/choupo-import); records mirrored under constant/ and sha256-verified by constant/propertyManifest.

*Eight reactive trays over a sour water (2.5 mol% NH<sub>3</sub>, 1 mol% CO<sub>2</sub>), feed on tray 2, reflux 0.5, distillate 5 of 100 kmol/h. Converges in 9 MESH Newton iterations; the overhead carries 33.4 mol% NH<sub>3</sub> at 355 K, the bottoms leave at 99.1 % water. This is S2 of the sour-water programme (docs/design/sour-water-stripper-scope.md ?6d) – the case the effective stage K (ThermoPackage::stageK) was built \*for\*, after column13 proved the machinery on an identity rig.*

*A molecular column cannot express this separation. The volatility of NH<sub>3</sub> is set by a chemistry the column itself moves:*

- 1. CO<sub>2</sub> strips first – it is the more volatile solute, and the carbonate loading of the liquid falls tray by tray, across an order of magnitude (0.52 → 0.024 mol CO<sub>2</sub>-species per mol NH<sub>3</sub>-species).*
- 2. The falling carbonate frees the ammonia. The CO<sub>2</sub> a liquid still carries is what holds its ammonia protonated as NH<sub>4</sub><sup>+</sup>; as it leaves, the pH climbs (8.46 → 8.80 below the top tray) and the free-ammonia fraction rises 0.802 →*

0.947 — strictly, with no exception, on every tray. 3. The freed ammonia then strips. The whole cascade is visible in reports/unitOperations/tower/profile.csv: per-tray pH, ionic strength and every species molality, engine-owned.

The surge — the mechanism's own signature. Free ammonia is not merely stripped; it is *produced* by the deprotonation the falling carbonate allows. Just below the feed the production outruns the stripping:  $m_{\text{NH3aq}}$  rises from 1.316 (tray 2, the feed tray) to 1.424 (tray 3) and only then falls strictly to the reboiler (0.464). A first instinct would be to exempt this from the "everything is stripped downward" check; instead it is pinned as its own claim (check\_tray\_chemistry T5), because hiding a mechanism's signature to make a monotonicity check pass would be the check defeating its subject. Sabotage-verified: move the feed to tray 5 and T5 alone fires.

The top tray sits  $\sim 7$  K colder than tray 2, so its pH reads higher for the temperature of its own  $K_w$  — comparing pH across that jump compares two water dissociations, which is why the pH claim starts at tray 2.

\* The MESH initial ramp vs the two-phase band. An 8-stage linear  $T$  ramp visits temperatures a 4-stage ramp never did — including  $T_f + 15$  K, *above* the feed's two-phase band, where the reactive Newton has no interior  $V/F$  and stalled before the first iteration printed. stageK now catches the typed ReactiveVLE::NonConvergence and prices such a trial *incipient* over the hypothetical speciated liquid (announced once) — the same  $K_i = (p_i^{\text{eq}}/P)/x_i$  its subsaturated branch always used, so the  $K$  surface is continuous across the band. A converged stage sits at saturation, where the constructions agree: the aid shapes the path, never the answer. \* The per-tray chemistry report flashed a liquid pinned to its own bubble point. The MESH's residual ( $\Sigma y = 1$ ) places every converged tray liquid exactly at its bubble point — the two-phase flash's degenerate corner — and at reflux 0.5 five trays of eight printed NaN for chemistry the package resolves easily. The report now asks the right question (speciateReactiveAsLiquid); NaN is reserved for a tray whose speciation itself fails.

#### What to expect (golden-locked):

| kind   | unit/stream | key            | expected         |
|--------|-------------|----------------|------------------|
| stream | bottoms     | F              | 0.0263888888889  |
| stream | distillate  | F              | 0.00138888888889 |
| stream | feed        | F              | 0.0277777777778  |
| kpi    | tower       | B              | 0.0263888888889  |
| kpi    | tower       | D              | 0.00138888888889 |
| kpi    | tower       | Q_condenser_kW | -51.4325432244   |
| kpi    | tower       | Q_reboiler_kW  | 71.9165526448    |
| kpi    | tower       | R              | 0.5              |
| kpi    | tower       | T_bottom       | 370.531146763    |
| kpi    | tower       | T_top          | 355.393273713    |
| kpi    | tower       | feedStage      | 2                |
| kpi    | tower       | nFeeds         | 1                |
| kpi    | tower       | nSideDraws     | 0                |
| kpi    | tower       | nStages        | 8                |

...and 13 more reference values in the case's *expected*.

**stripper02\_sour\_water\_h2s**

tutorials/steady/distillation/stripper02\_sour\_water\_h2s



**What it does:** The three-family sour water:  $\text{CO}_2$  strips overhead while  $\text{H}_2\text{S}$  concentrates downward and the pH FALLS – the selectivity lesson the two-family stripper cannot teach.

**The story:** the THREE-family sour water:  $\text{NH}_3 + \text{CO}_2 + \text{H}_2\text{S}$ , and the SELECTIVITY lesson the two-family stripper cannot teach.

stripper01 shows the mechanism:  $\text{CO}_2$  strips, pH rises, ammonia frees. Add the third volatile weak electrolyte and the story INVERTS in the most instructive way, measured on this converged profile:

$\text{CO}_2$  still strips overhead, hard: its loading falls from 0.195 to  $1\text{e-}5$  (per mol of ammonia species) between tray 2 and the reboiler;  $\text{H}_2\text{S}$  does NOT follow – its loading RISES down the column, 0.18 to 0.51: the  $\text{H}_2\text{S}/\text{CO}_2$  selectivity of a sour-water stripper at these conditions favours  $\text{CO}_2$  removal, and the sulfide concentrates in the liquid heading for the bottoms; and the pH FALLS down the column ( $8.49 \rightarrow 7.61$ ), the OPPOSITE of stripper01's climb: the departing ammonia and the accumulating sulfide acidify faster than the departing  $\text{CO}_2$  alkalises.

A student who has read stripper01 should predict a rising pH here and be wrong – that is the point. The chemistry is the same machinery (three networks at once: ammonia, carbonate, sulfide – the  $\text{H}_2\text{S}$  bridge is 'aqueousSpeciation sulfide',  $\text{HS}^- + \text{H}^+$ , declared on the component), and the per-tray profile carries all three families' molalities.

Same column as stripper01 otherwise: 8 reactive trays, feed on tray 2, reflux 0.5,  $D = 5$  of 100 kmol/h, 'model simultaneous' (one reactive flash per stage per residual evaluation; the Jacobian sees apparent components only).

DATA SCOPE: sealed (bin/choupo-import); records mirrored under constant/ and sha256-verified by constant/propertyManifest.

*The three-family sour water:  $\text{NH}_3$  (2.5) +  $\text{CO}_2$  (1.0) +  $\text{H}_2\text{S}$  (0.5 kmol/h) in 96 kmol/h water, over the same 8 reactive trays as stripper01. Three speciation networks run at once on every tray — ammonia, carbonate, sulfide — and the per-tray profile carries all of them.*

*stripper01 teaches:  $\text{CO}_2$  strips  $\rightarrow$  pH rises  $\rightarrow \text{NH}_4^+$  deprotonates  $\rightarrow$  ammonia strips. Add  $\text{H}_2\text{S}$  and a student should predict the same rising pH. The converged profile says otherwise:*

*$\text{CO}_2$  is the more volatile acid gas at these conditions, so the stripper is selective: the carbonate leaves overhead while the sulfide stays with the liquid — and with the ammonia also leaving, the residual  $\text{H}_2\text{S}$  acidifies the bottoms faster than the departing  $\text{CO}_2$  can alkalise it. The pH direction is not a law of strippers; it is an outcome of \*which\* acid gas wins, and this pair of cases shows both outcomes with the same machinery.*

*\* reports/unitOperations/tower/profile.csv — per-tray pH, ionic strength and every species molality of all three families ( $m_{\text{H}_2\text{Saq}}$ ,  $m_{\text{HS}}$  beside the carbonate and ammonia columns). \* The distillate: an acid-gas overhead ( $\text{NH}_3 + \text{CO}_2 + \text{a little H}_2\text{S}$ ) — the real design question of a sour-water unit (where each contaminant ends up) is answered per tray, not just at the ends. \* A tray whose  $\text{H}_2\text{S}$  has stripped to numerical zero reports the sulfide molalities as 0 exactly — a solved tray with an empty family is a zero, never a NaN (NaN stays reserved for a tray whose speciation itself failed).*

*model simultaneous; — one reactive flash per stage per residual evaluation; the Jacobian sees apparent components only. Sealed (bin/choupo-import).*

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected         |
|--------|-------------|----------------|------------------|
| stream | bottoms     | F              | 0.0263888888889  |
| stream | distillate  | F              | 0.00138888888889 |
| stream | feed        | F              | 0.0277777777778  |
| kpi    | tower       | B              | 0.0263888888889  |
| kpi    | tower       | D              | 0.00138888888889 |
| kpi    | tower       | Q_condenser_kW | -52.339216667    |
| kpi    | tower       | Q_reboiler_kW  | 76.4475297294    |
| kpi    | tower       | R              | 0.5              |
| kpi    | tower       | T_bottom       | 372.031545214    |
| kpi    | tower       | T_top          | 356.430868411    |
| kpi    | tower       | feedStage      | 2                |
| kpi    | tower       | nFeeds         | 1                |
| kpi    | tower       | nSideDraws     | 0                |
| kpi    | tower       | nStages        | 8                |

...and 23 more reference values in the case's *expected*.

**drying**

**evapDryer01\_nacl**

tutorials/steady/drying/evapDryer01\_nacl



**What it does:** Evaporative dryer: free surface water on NaCl crystals evaporated into dry nitrogen, adiabatic, the binding limit announced

**The story:** FREE WATER, NOT BOUND WATER.

‘evaporativeDryer’ is the honest counterpart of ‘solidDryer’: where that one models a **HYGROSCOPIC** material whose moisture is GAB-sorbed and dries toward an equilibrium isotherm, this one models a **NON**-sorbing crystal whose moisture is free surface water. There is no isotherm, so there is no equilibrium moisture to stop at – the drying stops at whichever of three real limits binds **FIRST**, and the unit **ANNOUNCES** which:

1. the free water runs out; 2. the exhaust reaches saturation (the constant-rate air-capacity limit); 3. the gas cannot supply the latent + sensible heat.

Limits 2 and 3 are coupled – evaporation cools the gas, which lowers its carrying capacity – and the unit bisects on the removed water until the two agree, rather than applying them in sequence and hoping.

Wet salt: 20 kmol/h NaCl crystals (1169 kg/h) carrying 15 kmol/h of free water. Dry nitrogen at 400 K. The gas flow is deliberately generous, so the binding limit here is the **FIRST** one: the water runs out. Halve the nitrogen and the announcement changes – that is the case to run next.

**What to expect (golden-locked):**

| kind   | unit/stream  | key              | expected          |
|--------|--------------|------------------|-------------------|
| stream | drySolid     | F                | 0.005555555555556 |
| stream | hotGas       | F                | 0.1668055555556   |
| stream | humidExhaust | F                | 0.1709722222222   |
| stream | wetSolid     | F                | 0.009722222222222 |
| kpi    | dryer        | T_out            | 360.675490975     |
| kpi    | dryer        | X_final          | 0                 |
| kpi    | dryer        | X_initial        | 0.231198665298    |
| kpi    | dryer        | air_in_kmol_h    | 600.5             |
| kpi    | dryer        | drySolid_flow    | 0.3246666666667   |
| kpi    | dryer        | exhaust_humidity | 0.0392946211155   |
| kpi    | dryer        | moisture_pct_wb  | 0                 |
| kpi    | dryer        | water_removed    | 0.0750625         |
| stream | drySolid     | T                | 360.675490975     |
| stream | hotGas       | T                | 400               |

...and 2 more reference values in the case's *expected*.



**evapDryer02\_energy\_limited**

tutorials/steady/drying/evapDryer02\_energy\_limited



**What it does:** The same dryer starved of gas: 40 kmol/h instead of 600, so the ENERGY floor binds and 16.6 wt% of the water stays on the crystal

**The story:** FREE WATER, NOT BOUND WATER.

‘evaporativeDryer’ is the honest counterpart of ‘solidDryer’: where that one models a HYGROSCOPIC material whose moisture is GAB-sorbed and dries toward an equilibrium isotherm, this one models a NON-sorbing crystal whose moisture is free surface water. There is no isotherm, so there is no equilibrium moisture to stop at – the drying stops at whichever of three real limits binds FIRST, and the unit ANNOUNCES which:

1. the free water runs out; 2. the exhaust reaches saturation (the constant-rate air-capacity limit); 3. the gas cannot supply the latent + sensible heat.

Limits 2 and 3 are coupled – evaporation cools the gas, which lowers its carrying capacity – and the unit bisects on the removed water until the two agree, rather than applying them in sequence and hoping.

THIS CASE IS THE STARVED ONE. Identical to evapDryer01\_nacl except for the gas: 40 kmol/h instead of 600. The announcement changes to [energy-limited], a WARNING says the gas cannot supply the drying heat, T\_out is floored at the wet solid’s own temperature, and 16.6 wt% of the water stays on the crystal – a dryer that does not dry, reported as such instead of returning a dry solid it cannot produce.

Read the exhaust humidity beside it: 0.95, exactly the cap. Both limits are within rounding of each other here, which is what the coupling means – the gas cools as it evaporates, and running out of heat and running out of room are the same wall approached from two sides.

**What to expect (golden-locked):**

| kind   | unit/stream  | key              | expected         |
|--------|--------------|------------------|------------------|
| stream | drySolid     | F                | 0.00914216942312 |
| stream | hotGas       | F                | 0.01125          |
| stream | humidExhaust | F                | 0.0118300527991  |
| stream | wetSolid     | F                | 0.00972222222222 |
| kpi    | dryer        | T_out            | 310.725305009    |
| kpi    | dryer        | X_final          | 0.199012881389   |
| kpi    | dryer        | X_initial        | 0.231198665298   |
| kpi    | dryer        | air_in_kmol_h    | 40.5             |
| kpi    | dryer        | drySolid_flow    | 0.389279515491   |
| kpi    | dryer        | exhaust_humidity | 0.949998302456   |
| kpi    | dryer        | moisture_pct_wb  | 16.5980603276    |
| kpi    | dryer        | water_removed    | 0.0104496511758  |
| stream | drySolid     | T                | 310.725305009    |
| stream | hotGas       | T                | 400              |

... and 2 more reference values in the case’s *expected*.

**solidDryer01\_sugar**

tutorials/steady/drying/solidDryer01\_sugar

**What it does:** Solid dryer: wet sugar powder dried to sorption equilibrium**The story:** / flowsheetDict (LEAF node)

Wet sugar dried by a REAL hot-air STREAM (no airTemperature/RH parameters, no phantom duty): the hot air BRINGS the heat (its sensible cooling) and CARRIES AWAY the moisture (a humid exhaust). Adiabatic → the outlet T is a RESULT; the plant energy balance closes with real streams.

Wet solid: 10 kmol/h sucrose (3423 kg/h) + 38 kmol/h bound water ( $X_{in}=0.20$ ). Hot air: 800 kmol/h at 420 K, DRY (2 mol% water → RH ~0.4%) – hot dry air, like a real sugar dryer. Its low RH sets a low GAB equilibrium moisture, so the sugar dries to  $X_{eq} \sim 0.001$  (~0.1 wt%); the exhaust leaves carrying the evaporated water (838 > 800 kmol/h, the pickup visible in the molar balance).

**What to expect (golden-locked):**

| kind   | unit/stream  | key             | expected         |
|--------|--------------|-----------------|------------------|
| stream | drySolid     | F               | 0.00283583138708 |
| stream | hotAir       | F               | 0.222222222229   |
| stream | humidExhaust | F               | 0.23271974852    |
| stream | wetSolid     | F               | 0.0133333576778  |
| kpi    | solidDryer   | T_out           | 347.5126791      |
| kpi    | solidDryer   | X_equilibrium   | 0.00109945367934 |
| kpi    | solidDryer   | X_final         | 0.00109945367934 |
| kpi    | solidDryer   | X_initial       | 0.199991235822   |
| kpi    | solidDryer   | air_in_kmol_h   | 800.000000025    |
| kpi    | solidDryer   | drySolid_flow   | 0.951878730237   |
| kpi    | solidDryer   | moisture_pct_wb | 0.109824620851   |
| kpi    | solidDryer   | water_activity  | 0.00431158404412 |
| kpi    | solidDryer   | water_removed   | 0.189112936127   |
| stream | drySolid     | T               | 347.5126791      |

...and 3 more reference values in the case's *expected*.

**sprayDryer01\_sugar**

tutorials/steady/drying/sprayDryer01\_sugar



**What it does:** Spray dryer: 40 wt% sugar solution → powder (rotary atomiser, Ranz-Marshall)

**The story:** flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z\_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | dryingAir   | F                | 0.0138888888889   |
| stream | exhaustAir  | F                | 0.0148058483113   |
| stream | feed        | F                | 0.000957611111111 |
| stream | powder      | F                | 4.06516887181e-05 |
| kpi    | dryer       | Nu               | 2.24583682227     |
| kpi    | dryer       | Pr               | 0.746977616919    |
| kpi    | dryer       | Re               | 0.203916784363    |
| kpi    | dryer       | Sc               | 0.6026290984      |
| kpi    | dryer       | Sh               | 2.22885536787     |
| kpi    | dryer       | T_out            | 372.025243986     |
| kpi    | dryer       | T_wetbulb        | 318.053221267     |
| kpi    | dryer       | d_droplet        | 6.04023981348e-05 |
| kpi    | dryer       | d_droplet_micron | 60.4023981348     |
| kpi    | dryer       | d_particle       | 3.80926671659e-05 |

...and 36 more reference values in the case's *expected*.

**sprayDryer02\_residence\_sweep**

tutorials/steady/drying/sprayDryer02\_residence\_sweep



**What it does:** Spray dryer: chamber-height sweep → the drying curve (residence =  $L/v$ ; constant + falling rate → equilibrium)

**The story:** sprayDryer01's sugar dryer with the drying-chamber HEIGHT swept by system/outerDict (0.01 → 0.3 m, 40 pts): residence time =  $L / v_{\text{particle}}$  traces the DRYING CURVE – constant-rate then falling-rate, down to the equilibrium moisture.

The base flowsheet is sprayDryer01\_sugar's, unchanged: one unit, two feeds, two products.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ;  $z_{\text{sucrose}} = 0.03390$

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | dryingAir   | F             | 0.01388888888889  |
| stream | exhaustAir  | F             | 0.0148058483112   |
| stream | feed        | F             | 0.000957611111111 |
| stream | powder      | F             | 4.06516887566e-05 |
| kpi    | dryer       | Nu            | 2.23180639411     |
| kpi    | dryer       | Pr            | 0.746977616919    |
| kpi    | dryer       | Re            | 0.181305061943    |
| kpi    | dryer       | Sc            | 0.6026290984      |
| kpi    | dryer       | Sh            | 2.21579410729     |
| kpi    | dryer       | T_out         | 372.025243986     |
| kpi    | dryer       | T_wetbulb     | 318.053221267     |
| kpi    | dryer       | X_critical    | 0.15              |
| kpi    | dryer       | X_equilibrium | 0.0132756494842   |
| kpi    | dryer       | X_final       | 0.0132756495467   |

...and 36 more reference values in the case's *expected*.

sprayDryer03\_pressure\_nozzle

tutorials/steady/drying/sprayDryer03\_pressure\_nozzle



**What it does:** Spray dryer: 40 wt% sugar solution → powder (PRESSURE-SWIRL nozzle at 40 bar,  $d_{32}=9575 \cdot dP^{-1/3}$ ; Ranz-Marshall)

**The story:** spray dryer: pressure-swirl-nozzle atomisation ( $d_{32}$  from nozzle  $dP$ ).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ;  $z_{\text{sucrose}} = 0.03390$

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a PRESSURE-SWIRL nozzle at 40 bar ( $d_{32} = 9575 \cdot dP^{-1/3}$  um, the class-average design curve) instead of a wheel.

What to expect (golden-locked):

| kind                                                    | unit/stream | key           | expected          |
|---------------------------------------------------------|-------------|---------------|-------------------|
| stream                                                  | dryingAir   | F             | 0.0138888888889   |
| stream                                                  | exhaustAir  | F             | 0.0148058483113   |
| stream                                                  | feed        | F             | 0.000957611111111 |
| stream                                                  | powder      | F             | 4.06516887181e-05 |
| kpi                                                     | dryer       | Nu            | 2.24532614778     |
| kpi                                                     | dryer       | Pr            | 0.746977616919    |
| kpi                                                     | dryer       | Re            | 0.203070475508    |
| kpi                                                     | dryer       | Sc            | 0.6026290984      |
| kpi                                                     | dryer       | Sh            | 2.2283799688      |
| kpi                                                     | dryer       | T_out         | 372.025243986     |
| kpi                                                     | dryer       | T_wetbulb     | 318.053221267     |
| kpi                                                     | dryer       | X_critical    | 0.15              |
| kpi                                                     | dryer       | X_equilibrium | 0.0132756494842   |
| kpi                                                     | dryer       | X_final       | 0.0132756494842   |
| ...and 36 more reference values in the case's expected. |             |               |                   |

**sprayDryer04\_profiles**

tutorials/steady/drying/sprayDryer04\_profiles



**What it does:** Spray dryer DISTRIBUTED: axial profiles of moisture, size and temperature along the chamber

**The story:** spray dryer: axial profiles of moisture, droplet size and T along the chamber.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z\_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | dryingAir   | F             | 0.0138888888889   |
| stream | exhaustAir  | F             | 0.0148058483113   |
| stream | feed        | F             | 0.000957611111111 |
| stream | powder      | F             | 4.06516887181e-05 |
| kpi    | dryer       | Nu            | 2.24583682227     |
| kpi    | dryer       | Pr            | 0.746977616919    |
| kpi    | dryer       | Re            | 0.203916784363    |
| kpi    | dryer       | Sc            | 0.6026290984      |
| kpi    | dryer       | Sh            | 2.22885536787     |
| kpi    | dryer       | T_out         | 372.025243986     |
| kpi    | dryer       | T_wetbulb     | 318.053221267     |
| kpi    | dryer       | X_critical    | 0.15              |
| kpi    | dryer       | X_equilibrium | 0.0132756494842   |
| kpi    | dryer       | X_final       | 0.0132756494842   |

...and 36 more reference values in the case's *expected*.

**sprayDryer05\_whey**

tutorials/steady/drying/sprayDryer05\_whey



**What it does:** Spray dryer: whey (prob 5.2) – 500 kg/h,  $X_0=5$ , rotary co-current, sorption-isotherm residual moisture

**The story:** spray dryer: whey feed (500 kg/h,  $X_0=5$ ), rotary co-current, sorption-isotherm residual.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ;  $z_{\text{sucrose}} = 0.03390$

dryingAir: ~500 kmol/h of hot AIR (N<sub>2</sub>/O<sub>2</sub>/Ar) at 180 C – sized to carry the whey water. Whey solids modelled as sucrose (lactose proxy, a disaccharide MW~342).

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | dryingAir   | F             | 0.138888888889    |
| stream | exhaustAir  | F             | 0.14529654835     |
| stream | feed        | F             | 0.00649261111111  |
| stream | powder      | F             | 8.49516499934e-05 |
| kpi    | dryer       | Nu            | 2.33731434528     |
| kpi    | dryer       | Pr            | 0.745230561117    |
| kpi    | dryer       | Re            | 0.384509154435    |
| kpi    | dryer       | Sc            | 0.602916461126    |
| kpi    | dryer       | Sh            | 2.31430911286     |
| kpi    | dryer       | T_out         | 382.7774359       |
| kpi    | dryer       | T_wetbulb     | 315.824008236     |
| kpi    | dryer       | X_critical    | 0.5               |
| kpi    | dryer       | X_equilibrium | 0.0134981947613   |
| kpi    | dryer       | X_final       | 0.0134981947613   |

... and 36 more reference values in the case's *expected*.

**sprayDryer06\_rea**

tutorials/steady/drying/sprayDryer06\_rea



**What it does:** Spray dryer REA (model chen): Reaction Engineering Approach kinetics – smooth rate, no critical-moisture break

**The story:** spray dryer: Reaction Engineering Approach (REA) drying kinetics (Chen model).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z\_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | dryingAir   | F             | 0.0138888888889   |
| stream | exhaustAir  | F             | 0.0148058483113   |
| stream | feed        | F             | 0.000957611111111 |
| stream | powder      | F             | 4.06516887181e-05 |
| kpi    | dryer       | Nu            | 2.24583682227     |
| kpi    | dryer       | Pr            | 0.746977616919    |
| kpi    | dryer       | Re            | 0.203916784363    |
| kpi    | dryer       | Sc            | 0.6026290984      |
| kpi    | dryer       | Sh            | 2.22885536787     |
| kpi    | dryer       | T_out         | 372.025243986     |
| kpi    | dryer       | T_wetbulb     | 318.053221267     |
| kpi    | dryer       | X_critical    | 0.15              |
| kpi    | dryer       | X_equilibrium | 0.0132756494842   |
| kpi    | dryer       | X_final       | 0.0132756494842   |

... and 36 more reference values in the case's *expected*.



**sprayDryer07\_design**

tutorials/steady/drying/sprayDryer07\_design



**What it does:** Spray dryer DESIGN: give targets (powder size, residence), the sizing pass inverts to the geometry

**The story:** spray dryer: DESIGN mode – targets in, the sizing pass inverts to geometry.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z\_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N<sub>2</sub>/O<sub>2</sub>/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | dryingAir   | F             | 0.0138888888889   |
| stream | exhaustAir  | F             | 0.0148058483113   |
| stream | feed        | F             | 0.000957611111111 |
| stream | powder      | F             | 4.06516887181e-05 |
| kpi    | dryer       | Nu            | 2.24583682227     |
| kpi    | dryer       | Pr            | 0.746977616919    |
| kpi    | dryer       | Re            | 0.203916784363    |
| kpi    | dryer       | Sc            | 0.6026290984      |
| kpi    | dryer       | Sh            | 2.22885536787     |
| kpi    | dryer       | T_out         | 372.025243986     |
| kpi    | dryer       | T_wetbulb     | 318.053221267     |
| kpi    | dryer       | X_critical    | 0.15              |
| kpi    | dryer       | X_equilibrium | 0.0132756494842   |
| kpi    | dryer       | X_final       | 0.0132756494842   |

...and 36 more reference values in the case's *expected*.

**economics**

**The theory you need.** Sizing (stirred tanks, shell-and-tube areas) feeds costing (Guthrie modules); prices are CASE data (constant/economics, dated and cited) – the engine refuses to invent a price. Expect honest refusal when a price is missing, never a silent default.

**economics01\_esterification\_dcf**

tutorials/steady/economics/economics01\_esterification\_dcf



**What it does:** Esterification plant: CAPEX (Guthrie) → OPEX (COM\_d) → discounted-cash-flow appraisal (NPV/IRR/payback)

**The story:** A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed → reactor (CSTR) → heater → separator (flash) |→ liqProd (ethyl acetate rich) |→ vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges ( $V$  in  $[0.3, 520]$  m<sup>3</sup>,  $A$  in  $[10, 1000]$  m<sup>2</sup>) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing → costing → economics; the economics pass prints the COM\_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

*A fixed (designed) ethyl-acetate plant taken all the way from equipment sizing to a discounted-cash-flow economic appraisal:*

1. *sizing* — StirredTank for the CSTR, ShellTubeHX for the heater, sized from each unit's KPIs.
2. *costing* — Guthrie/Turton purchased + bare-module + total-module cost, CEPCI-updated to the target year, in EUR.
3. *economics* — the new EconomicsPass: it builds the CAPEX ladder (FCI, working capital, TCI), the Turton cost-of-manufacture COM\_d, revenue, a straight-line-depreciation cash-flow timeline, then NPV, IRR (bisection bracket + Newton polish), and discounted + simple payback.

*Every cost factor and DCF assumption is dict-owned and printed — read the appraisal block and you can re-derive every number by hand.*

*Depreciation enters the DCF exactly once (it is \*not\* in COM\_d) — the classic textbook double-count is avoided. Working capital is an outflow at startup and is recovered untaxed at end-of-life.*

*Market/time-specific data — product and feed prices, labour rate, FX — lives only in constant/economics, each value dated and primary-cited (an index/bulletin/agency named in a provenance{} block). The EconomicsPass refuses to run if a required price is absent, with a remedy message; set economics.refuseOnMissingPrice 0; in the postDict to switch to defaulted-but-loud. No price ever defaults to a literal in code.*

*The economics pass also publishes its headline scalars into result.kpis["economics"] (IRR, NPV, paybackYears, FCI, TCI, COM\_d), which is what lets a sweep or optimisation read economics as an ordinary response — see economics02\_irr\_sweep.*

**What to expect (golden-locked):**

| kind   | unit/stream | key     | expected       |
|--------|-------------|---------|----------------|
| stream | feed        | F       | 0.138888888889 |
| stream | hotStream   | F       | 0.138888888889 |
| stream | liqProd     | F       | 0.126899267211 |
| stream | reactorOut  | F       | 0.138888888889 |
| stream | vapProd     | F       | 0.011989621678 |
| kpi    | economics   | COM_d   | 221241889.353  |
| kpi    | economics   | FCI     | 407855.018165  |
| kpi    | economics   | NPV     | -964539867.664 |
| kpi    | economics   | TCI     | 469033.27089   |
| kpi    | economics   | WC      | 61178.2527248  |
| kpi    | economics   | revenue | 22620564.309   |
| kpi    | heater      | F       | 0.138888888889 |
| kpi    | heater      | P       | 101325         |
| kpi    | heater      | Q       | 300000         |

... and 43 more reference values in the case's *expected*.

## economics02\_irr\_sweep

tutorials/steady/economics/economics02\_irr\_sweep



**What it does:** Sweep reactor volume; read economics.IRR/NPV/payback per point (SweepDriver runs the post chain)

**The story:** A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed  $\rightarrow$  reactor (CSTR)  $\rightarrow$  heater  $\rightarrow$  separator (flash)  $\rightarrow$  liqProd (ethyl acetate rich)  $\rightarrow$  vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges ( $V$  in  $[0.3, 520]$   $\text{m}^3$ ,  $A$  in  $[10, 1000]$   $\text{m}^2$ ) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing  $\rightarrow$  costing  $\rightarrow$  economics; the economics pass prints the COM\_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

*The same esterification plant as economics01, but driven by a sensitivity sweep over the reactor volume that reads economic responses at every point:*

*The SweepDriver now runs the full post-processing chain (sizing  $\rightarrow$  costing  $\rightarrow$  economics) on each converged point, so the headline economic scalars the EconomicsPass publishes into result.kpis["economics"] resolve as ordinary unit.kpi responses. A sweep can finally \*see cost\*:*

*A bigger reactor raises conversion (more product, more revenue) but raises CAPEX and feed throughput — the IRR falls monotonically over this range, a trade-off invisible from a single CAPEX number.*

*> The range floors at 4  $\text{m}^3$  on purpose: below  $\sim 3 \text{ m}^3$  the lower conversion > drops the mixture bubble point and the fixed-duty heater would cross the > saturation dome (a heater is sensible-only) — the deliberate validity edge.*

**What to expect (golden-locked):**

| kind   | unit/stream | key     | expected       |
|--------|-------------|---------|----------------|
| stream | feed        | F       | 0.138888888889 |
| stream | hotStream   | F       | 0.138888888889 |
| stream | liqProd     | F       | 0.125841050463 |
| stream | reactorOut  | F       | 0.138888888889 |
| stream | vapProd     | F       | 0.013047838426 |
| kpi    | economics   | COM_d   | 221264557.696  |
| kpi    | economics   | FCI     | 460663.338135  |
| kpi    | economics   | NPV     | -952396923.312 |
| kpi    | economics   | TCI     | 529762.838855  |
| kpi    | economics   | WC      | 69099.5007202  |
| kpi    | economics   | revenue | 25155181.8032  |
| kpi    | heater      | F       | 0.138888888889 |
| kpi    | heater      | P       | 101325         |
| kpi    | heater      | Q       | 300000         |

... and 43 more reference values in the case's *expected*.

## evaporation

**The theory you need.** An evaporator boils solvent off a solution. Two pieces of physics beyond a flash: boiling-point elevation,  $T_{\text{boil}} = T_{\text{sat}}(P) + \text{BPE}$  with the BPE from the electrolyte activity of water ( $\ln a_w = -\varphi \nu m M_w / 1000$ ); and the heat of dilution/concentration (the relative apparent molar enthalpy  $L_\varphi(m, T)$ ) entering the duty. Expect: the BPE to GROW as the brine concentrates, steam economy below 1 for a single effect, and – for NaCl – the full Silvester–Pitzer  $T$ -dependence of  $\gamma_{\pm}$  shaping the answer between 25 and 200°C. *Full derivation: Theory Guide, the electrolyte chapters.*

## evaporator01\_brine

tutorials/steady/evaporation/evaporator01\_brine



**What it does:** Single-effect evaporator: 35 g/L brine at 1 bar; the fixed steam duty/UA evaporates ~82 % of the feed water ( $V/F=0.82$ ), enriching NaCl ~5.5x ( $x \ 0.011 \rightarrow 0.059$ ) – duty-driven, not a target (BPE included)

**The story:** single-effect brine concentrator (NaCl-water), PHYSICAL-EQUIPMENT model (v0.29+).

We say WHAT THE HARDWARE IS (heat-exchange area, overall HTC, operating pressure on the process side, heating-steam temperature) and the simulator tells us WHAT IT DOES ( $V/F$ , duty  $Q$ ,  $T_{\text{boil}}$  with BPE, steam demand, economy). There is no "vaporFraction" or "duty" spec to choose — the equipment shape itself IS the spec.

Feed: seawater-equivalent 35 g/L NaCl in water, 100 kmol/h (~ 1.8 kg/s of seawater, the volume of a small medium-capacity desalination unit), 25 °C. At  $MW_{\text{NaCl}} = 58.44$  g/mol and  $\rho_{\text{water}} \sim 1$  g/cm<sup>3</sup>:

$35 \text{ g/L} / 58.44 \text{ g/mol} = 0.599 \text{ mol NaCl per litre water} = 0.599 \text{ mol/kg water (dilute approx)}$   $x_{\text{NaCl\_feed}} \approx 0.599 / (0.599 + 55.5) = 0.01067$

Heating side: saturated steam at 130 °C (= 403.15 K,  $P \approx 2.7$  bar).

Equipment: heat-exchange area = 20 m<sup>2</sup> overall  $U = 2000 \text{ W/(m}^2\cdot\text{K)}$  operating pressure = 1 bar (process side)

Expected order-of-magnitude (computed by the solver, not specified):  $T_{\text{sat}}(\text{water}, 1 \text{ bar}) \approx 373 \text{ K}$  (~ 100 °C)  $\Delta T_{\text{drive}}$  (steam - process)  $\approx 30 \text{ K}$   $Q = U \cdot A \cdot \Delta T \approx 2000 \cdot 20 \cdot 30 \approx 1200 \text{ kW}$   $V_{\text{evaporated}} \approx Q / \Delta H_{\text{vap}} \approx 110 \text{ kmol/h} \approx 1980 \text{ kg/h water}$   $V/F$  (mol-basis)  $\approx 0.82$  (the CONVERGED run; the rough hand-estimate above is order-of-magnitude only) Steam consumption  $\approx Q / \Delta H_{\text{vap\_at\_steam}} \approx 2000 \text{ kg/h}$  Economy  $\approx V_{\text{evap}} / F_{\text{steam}} \approx 0.99$   $x_{\text{NaCl\_outlet}} \approx 0.059$  (~5.5x the feed, matching the run header)

Industrial relevance: this is the building block of multi-effect evaporators in seawater desalination (MED, MSF), brine crystallisers, and the sugar industry. Real plants run under vacuum ( $T_{\text{boil}} \sim 60$  °C at 0.2 bar) to minimise scaling and use BPE-loaded multi-effect cascades to recover the latent heat — see evaporator02\_triple\_effect\_sugar.

### What to expect (golden-locked):

| kind   | unit/stream        | key           | expected         |
|--------|--------------------|---------------|------------------|
| stream | concentrated       | F             | 0.00498887089392 |
| stream | condensate         | F             | 0.0277777777778  |
| stream | feed               | F             | 0.0277777777778  |
| stream | saturatedSteam130C | F             | 0.0277777777778  |
| stream | vapour             | F             | 0.0227889068839  |
| kpi    | evap               | A             | 20               |
| kpi    | evap               | A_m2          | 20               |
| kpi    | evap               | BPE           | 1.79842779271    |
| kpi    | evap               | F_conc        | 0.00498887089392 |
| kpi    | evap               | F_conc_kmol_h | 17.9599352181    |
| kpi    | evap               | F_conc_mass   | 0.101856029987   |
| kpi    | evap               | F_cond        | 0.500416666667   |
| kpi    | evap               | F_cond_kg_h   | 1801.5           |
| kpi    | evap               | F_in          | 0.0277777777778  |

... and 24 more reference values in the case's *expected*.



Closing the inter-effect mass balance turns the cascade into a coupled system (one constraint per link) that wants an outer DesignSpec wrap: manipulate (T\_boil\_i) or (A\_i) to enforce "V\_i produced == V\_i consumed at effect i+1's chest". That is the v0.30 multi-effect tutorial.

**What to expect (golden-locked):**

| kind                                                            | unit/stream         | key  | expected        |
|-----------------------------------------------------------------|---------------------|------|-----------------|
| stream                                                          | L1                  | F    | 0.243400997575  |
| stream                                                          | L2                  | F    | 0.150822370746  |
| stream                                                          | L3                  | F    | 0.0533879418767 |
| stream                                                          | V1                  | F    | 0.0870984081237 |
| stream                                                          | V2                  | F    | 0.0925786268285 |
| stream                                                          | V3                  | F    | 0.0974344288697 |
| stream                                                          | cond1               | F    | 0.1375          |
| stream                                                          | cond2               | F    | 0.0870984081237 |
| stream                                                          | cond3               | F    | 0.0925786268285 |
| stream                                                          | feed                | F    | 0.330499405699  |
| stream                                                          | utilitySteam_200kPa | F    | 0.1375          |
| kpi                                                             | effect1             | A    | 100             |
| kpi                                                             | effect1             | A_m2 | 100             |
| kpi                                                             | effect1             | BPE  | 0.107198891506  |
| ...and 92 more reference values in the case's <i>expected</i> . |                     |      |                 |

**evaporator03\_co\_current**

tutorials/steady/evaporation/evaporator03\_co\_current



**What it does:** Co-current forward-feed triple-effect evaporator, user-defined components (textbook problem 10.38, case a)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. ('sucrose.dat' does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

triple-effect evaporator, forward-feed, user-defined components, textbook Example 10.38 (a).

Problem (textbook gives): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) P<sub>3</sub> such that effect 3 boils at 325 K => P<sub>3</sub> = 13.5 kPa U<sub>1</sub> 3100 W/(m<sup>2</sup>·K) U<sub>2</sub> 2500 W/(m<sup>2</sup>·K) U<sub>3</sub> 1700 W/(m<sup>2</sup>·K) Spec productConc = 40 wt% solids Equal areas A<sub>1</sub> = A<sub>2</sub> = A<sub>3</sub> = A

Question (in system/outerDict): find A, P<sub>1</sub>, P<sub>2</sub> and the live- steam consumption.

Wiring (liquid + vapour BOTH flow 1 → 2 → 3):

feed (294 K, 10 %) productConc (40 %) | ^ v | +-----+ L1to2 +-----+ L2to3 +-----+ | effect1  
|----->| effect2 |----->| effect3 | | P\_1 | | P\_2 | | P\_3 | +---+---+ +---+---+ +---+---+ ^ ^  
liveSteam V1 (from 1) V2 (from 2) (170 kPa) chest of 2 chest of 3

V<sub>3</sub> → condenser (outside the dict) cond1, cond2, cond3 → condensate terminals

Notes for the student: Thermo properties (MW, Cp, Antoine, ...) of 'agua' and 'solidos' live in constant/-components/<name>.dat — these are USER-DEFINED components, NOT from the standard catalogue. Open them to see the textbook's Cp = 4.18 kJ/(K·kg) assumption baked in, and to confirm BPE is set to zero (no K<sub>b</sub> on 'agua'). Forward-feed with COLD feed (294 K) penalises effect 1: most of its duty goes into sensible heating, not evaporation. Case (b) of the textbook pre-heats the feed to 355 K — swap T<sub>feed</sub> below to compare economies.

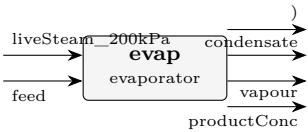
**What to expect (golden-locked):**

| kind   | unit/stream | key  | expected        |
|--------|-------------|------|-----------------|
| stream | L1to2       | F    | 0.0469536109199 |
| stream | L2to3       | F    | 0.0295647943065 |
| stream | V1          | F    | 0.0161887935578 |
| stream | V2          | F    | 0.0173888166134 |
| stream | V3          | F    | 0.018462595509  |
| stream | cond1       | F    | 0.0262363636123 |
| stream | cond2       | F    | 0.0161887935578 |
| stream | cond3       | F    | 0.0173888166134 |
| stream | feed        | F    | 0.0631424044778 |
| stream | liveSteam   | F    | 0.0262363636123 |
| stream | productConc | F    | 0.0111021987975 |
| kpi    | effect1     | A    | 16.5383245281   |
| kpi    | effect1     | A_m2 | 16.5383245281   |
| kpi    | effect1     | BPE  | 0               |

... and 92 more reference values in the case's *expected*.

evaporator04\_orange\_juice\_simple

tutorials/steady/evaporation/evaporator04\_orange\_juice\_simple



**What it does:** Single-effect evaporator concentrating orange juice 10% to 50% solids (textbook Example 10.2 a-c, without BPE)

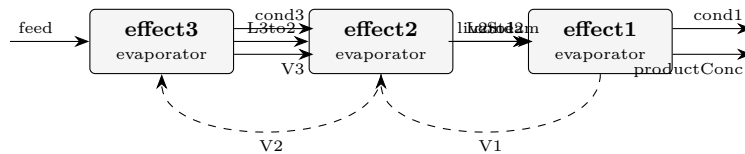
**What to expect (golden-locked):**

| kind                                                    | unit/stream      | key           | expected        |
|---------------------------------------------------------|------------------|---------------|-----------------|
| stream                                                  | condensate       | F             | 0.360686102898  |
| stream                                                  | feed             | F             | 0.350791135988  |
| stream                                                  | liveSteam_200kPa | F             | 0.360686102898  |
| stream                                                  | productConc      | F             | 0.0424059629635 |
| stream                                                  | vapour           | F             | 0.308385173024  |
| kpi                                                     | evap             | A             | 75.5133330137   |
| kpi                                                     | evap             | A_m2          | 75.5133330137   |
| kpi                                                     | evap             | BPE           | 0               |
| kpi                                                     | evap             | F_conc        | 0.0424059629635 |
| kpi                                                     | evap             | F_conc_kmol_h | 152.661466669   |
| kpi                                                     | evap             | F_conc_mass   | 1.38888555242   |
| kpi                                                     | evap             | F_cond        | 6.49776014371   |
| kpi                                                     | evap             | F_cond_kg_h   | 23391.9365174   |
| kpi                                                     | evap             | F_in          | 0.350791135988  |
| ...and 24 more reference values in the case's expected. |                  |               |                 |



**evaporator05\_counter\_current**

tutorials/steady/evaporation/evaporator05\_counter\_current



**What it does:** Counter-current backward-feed triple-effect evaporator (textbook problem 10.38 variant)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. ('sucrose.dat' does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

COUNTER-CURRENT (BACKWARD-FEED) triple-effect evaporator, Mode-2, credo-pure.

Problem (textbook gives, textbook problem 10.38 numerics, counter-current variant): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) entering EFFECT 1 (hot end) P\_3 (cold end) such that  $T_3 = 325 \text{ K} \rightarrow 13.5 \text{ kPa}$   $U_1 3100 \text{ W/(m}^2\cdot\text{K)}$   $U_2 2500 \text{ W/(m}^2\cdot\text{K)}$   $U_3 1700 \text{ W/(m}^2\cdot\text{K)}$  Spec productConc = 40 wt% solids (exits from effect 1) Equal areas  $A_1 = A_2 = A_3 = A$  (single \$A by reference)

Wiring (LIQUID flows  $3 \rightarrow 2 \rightarrow 1$  ; VAPOUR flows  $1 \rightarrow 2 \rightarrow 3$  ; OPPOSITE directions => counter-current):

```
feed (294 K, 10 %) productConc (40 %, exits effect 1) | ^ v | +-----+ L3to2 +-----+ L2to1 +-----+ |
effect3 |<-----| effect2 |<-----| effect1 | | COLD | | MID | | HOT | +---+---+ +---+---+ +---+---+
^ ^ ^ | | V2 (from 2) V1 (from 1) liveSteam chest of 3 chest of 2 (170 kPa)
```

v V3 → condenser cond1, cond2, cond3 → condensate terminals

Both V\_1 and V\_2 flow OPPOSITE to the liquid: effect 1 produces V\_1 which feeds effect 2's chest; effect 2 produces V\_2 which feeds effect 3's chest. In the directed graph this creates a CYCLE:

```
effect3 -L3to2-→ effect2 -L2to1-→ effect1 ^ | | V2 V1 | | v +- effect2 <- effect1 (V1 chest) +
```

So we DECLARE V\_1 and V\_2 as TEAR STREAMS. The flowsheet executive runs effect3 → effect2 → effect1 once per pass, with the initial V\_1 / V\_2 guesses from the streams block; at end of pass the newly-produced V\_1, V\_2 are compared with the old. Wegstein (or plain direct substitution) iterates until they match.

In Mode 2 each effect computes T\_boil and P\_op locally from the chest T and the hardware (A, U) — so the only thing that the Wegstein loop actually iterates is V\_i.F (the produced flows). T and P\_op stabilise IMMEDIATELY (single pass). Convergence in ~3-5 Wegstein passes.

Design unknowns (in outerDict): \$A common heat-exchange area \$F\_steam live-steam flow

Constraints: productConc.F\_mass = 0.3125 kg/s (40 wt%) effect3.P = 13.5 kPa (textbook terminal vacuum)

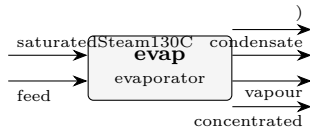
**What to expect (golden-locked):**

| kind   | unit/stream | key  | expected        |
|--------|-------------|------|-----------------|
| stream | L2to1       | F    | 0.0319496046345 |
| stream | L3to2       | F    | 0.0496904316429 |
| stream | V1          | F    | 0.020847035226  |
| stream | V2          | F    | 0.0177408270084 |
| stream | V3          | F    | 0.0134519728349 |
| stream | cond1       | F    | 0.0227907734916 |
| stream | cond2       | F    | 0.0208470355984 |
| stream | cond3       | F    | 0.0177408244446 |
| stream | feed        | F    | 0.0631424044778 |
| stream | liveSteam   | F    | 0.0227907734916 |
| stream | productConc | F    | 0.0111025694085 |
| kpi    | effect1     | A    | 17.1719452001   |
| kpi    | effect1     | A_m2 | 17.1719452001   |
| kpi    | effect1     | BPE  | 0               |

...and 92 more reference values in the case's *expected*.

evaporator06\_nacl\_pitzer

tutorials/steady/evaporation/evaporator06\_nacl\_pitzer



**What it does:** Single-effect NaCl evaporator with the boiling-point elevation from the PITZER electrolyte model (a\_w), not the ideal K\_b\*m. Concentrated brine → BPE several degrees, the real value.

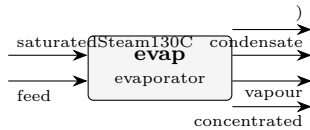
**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the PITZER water activity (a\_w), not the ideal K\_b\*m: concentrated brine is a REAL electrolyte, and the BPE follows from the osmotic coefficient. The thermo arrives via the aqueousNaCl\_pitzer property package (see constant/thermoPhysPropDict) – swap to evaporator07’s eNRTL to compare.

What to expect (golden-locked):

| kind                                                     | unit/stream        | key           | expected         |
|----------------------------------------------------------|--------------------|---------------|------------------|
| stream                                                   | concentrated       | F             | 0.0203403273098  |
| stream                                                   | condensate         | F             | 0.0125           |
| stream                                                   | feed               | F             | 0.0277777777778  |
| stream                                                   | saturatedSteam130C | F             | 0.0125           |
| stream                                                   | vapour             | F             | 0.00743745046799 |
| kpi                                                      | evap               | A             | 20               |
| kpi                                                      | evap               | A_m2          | 20               |
| kpi                                                      | evap               | BPE           | 5.06216808829    |
| kpi                                                      | evap               | F_conc        | 0.0203403273098  |
| kpi                                                      | evap               | F_conc_kmol_h | 73.2251783152    |
| kpi                                                      | evap               | F_conc_mass   | 0.422576829819   |
| kpi                                                      | evap               | F_cond        | 0.2251875        |
| kpi                                                      | evap               | F_cond_kg_h   | 810.675          |
| kpi                                                      | evap               | F_in          | 0.0277777777778  |
| ... and 26 more reference values in the case’s expected. |                    |               |                  |

evaporator07\_nacl\_enrtl

tutorials/steady/evaporation/evaporator07\_nacl\_enrtl



**What it does:** Single-effect NaCl evaporator with the boiling-point elevation from the eNRTL water activity (electrolyte-NRTL) – the same slot as model pitzer, one word changed. eNRTL is the model for salt in a non-ideal molecular solvent.

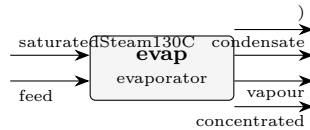
**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the eNRTL water activity (a\_w), not the ideal K\_b\*m – the SAME flowsheet as evaporator06, with only the electrolyte model swapped (pitzer → eNRTL): run both and compare the BPE the two G<sup>E</sup> surfaces predict.

What to expect (golden-locked):

| kind                                                            | unit/stream        | key           | expected         |
|-----------------------------------------------------------------|--------------------|---------------|------------------|
| stream                                                          | concentrated       | F             | 0.0200845615144  |
| stream                                                          | condensate         | F             | 0.0125           |
| stream                                                          | feed               | F             | 0.0277777777778  |
| stream                                                          | saturatedSteam130C | F             | 0.0125           |
| stream                                                          | vapour             | F             | 0.00769321626341 |
| kpi                                                             | evap               | A             | 20               |
| kpi                                                             | evap               | A_m2          | 20               |
| kpi                                                             | evap               | BPE           | 3.86220786107    |
| kpi                                                             | evap               | F_conc        | 0.0200845615144  |
| kpi                                                             | evap               | F_conc_kmol_h | 72.3044214517    |
| kpi                                                             | evap               | F_conc_mass   | 0.417969209015   |
| kpi                                                             | evap               | F_cond        | 0.2251875        |
| kpi                                                             | evap               | F_cond_kg_h   | 810.675          |
| kpi                                                             | evap               | F_in          | 0.0277777777778  |
| ...and 25 more reference values in the case's <i>expected</i> . |                    |               |                  |

**evaporator08\_naoh\_dilution\_heat**

tutorials/steady/evaporation/evaporator08\_naoh\_dilution\_heat



**What it does:** Single-effect NaOH evaporator: the heat of dilution IN the duty (fitted  $L_{\phi}$ , Parker-validated)

**The story:** the heat of dilution, IN the duty

Single-effect caustic (NaOH) evaporator – the McCabe enthalpy-concentration classic. NaOH is the FIRST calorimetrically fitted salt (Silvester-Pitzer  $\delta\beta/\delta T$  vs Parker NSRDS-NBS 2, rel-AAD 4.97%), so this evaporator's duty CLIMBS THE MEASURED  $L_{\phi}$  CURVE: concentrating caustic absorbs extra heat beyond sensible+latent, and the run prints it as its own named line:

heat of dilution = X kW ( $L_{\phi}$ : m a  $\rightarrow$  b mol/kg, calorimetric fit)

Contrast evaporator06 (whose CASE-LOCAL pair overlay omits the T-slots, so ITS NaCl runs calorimetricFit false – the standards Na-Cl record is fitted since the Parker refit): same machinery, but there the dilution heat is OMITTED and the run says so loudly. The feed (m  $\sim$  2.0) concentrates to m  $\sim$  4-5 mol/kg – inside the Parker fit window ( $\leq 6$ ), so no extrapolation flag; push the steam rate up and watch the [electrolyte] EXTRAPOLATED advisory fire.

**What to expect (golden-locked):**

| kind   | unit/stream        | key           | expected        |
|--------|--------------------|---------------|-----------------|
| stream | concentrated       | F             | 0.0118365226457 |
| stream | condensate         | F             | 0.0208333333333 |
| stream | feed               | F             | 0.0277777777778 |
| stream | saturatedSteam130C | F             | 0.0208333333333 |
| stream | vapour             | F             | 0.0159412551321 |
| kpi    | evap               | A             | 20              |
| kpi    | evap               | A_m2          | 20              |
| kpi    | evap               | BPE           | 6.45099255003   |
| kpi    | evap               | F_conc        | 0.0118365226457 |
| kpi    | evap               | F_conc_kmol_h | 42.6114815246   |
| kpi    | evap               | F_conc_mass   | 0.234606344352  |
| kpi    | evap               | F_cond        | 0.3753125       |
| kpi    | evap               | F_cond_kg_h   | 1351.125        |
| kpi    | evap               | F_in          | 0.0277777777778 |

... and 26 more reference values in the case's *expected*.

**evaporator09\_nacl\_sucrose\_brine**

tutorials/steady/evaporation/evaporator09\_nacl\_sucrose\_brine



**What it does:** Evaporating a brine that also carries sugar: the electrolyte BPE is computed on the salt molality, and the rest is announced, not absorbed

**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the PITZER water activity ( $a_w$ ), not the ideal  $K_{b*m}$ : concentrated brine is a REAL electrolyte, and the BPE follows from the osmotic coefficient. The thermo arrives via the aqueousNaCl\_pitzer property package (see constant/thermoPhysPropDict) – swap to evaporator07's eNRTL to compare.

**What to expect (golden-locked):**

| kind   | unit/stream        | key           | expected         |
|--------|--------------------|---------------|------------------|
| stream | concentrated       | F             | 0.0211538965752  |
| stream | condensate         | F             | 0.0125           |
| stream | feed               | F             | 0.0281111111111  |
| stream | saturatedSteam130C | F             | 0.0125           |
| stream | vapour             | F             | 0.00695721453587 |
| kpi    | evap               | A             | 20               |
| kpi    | evap               | A_m2          | 20               |
| kpi    | evap               | BPE           | 4.91270305537    |
| kpi    | evap               | F_conc        | 0.0211538965752  |
| kpi    | evap               | F_conc_kmol_h | 76.1540276709    |
| kpi    | evap               | F_conc_mass   | 0.545327280136   |
| kpi    | evap               | F_cond        | 0.2251875        |
| kpi    | evap               | F_cond_kg_h   | 810.675          |
| kpi    | evap               | F_in          | 0.0281111111111  |

... and 26 more reference values in the case's *expected*.

**flash**

**The theory you need.** An isothermal flash splits a feed of composition  $z_i$  into vapour  $y_i$  and liquid  $x_i$  at fixed  $(T, P)$ . Phase equilibrium is  $y_i = K_i x_i$  with  $K_i = \gamma_i P_i^{\text{sat}} / P$  ( $\gamma$ - $\varphi$  formulation), and the vapour fraction  $\beta$  solves the Rachford–Rice equation

$$\sum_i \frac{z_i (K_i - 1)}{1 + \beta (K_i - 1)} = 0,$$

monotone in  $\beta$  – Newton converges in a handful of iterations, all of them visible at verbosity 3. Expect: a bubble point when the equation's root hits  $\beta = 0$ , a dew point at  $\beta = 1$ , and non-ideal systems ( $\gamma_i \neq 1$ : NRTL/Wilson/UNIQUAC/UNIFAC) bending  $K_i$  with composition. Liquid–liquid and three-phase cases replace successive-substitution with direct Gibbs-energy minimisation (the symmetric  $\gamma$ -model trap: Newton collapses onto the trivial  $K = 1$  root). *Full derivation: Theory Guide, the VLE and flash chapters.*

**adiabaticFlash01\_benzene\_toluene**

tutorials/steady/flash/adiabaticFlash01\_benzene\_toluene



**What it does:** Adiabatic flash, benzene/toluene 40/60, 380K/3bar  $\rightarrow$  1atm

**The story:** Adiabatic flash: sub-cooled 40/60 benzene/toluene liquid at 380 K, 3 bar expanded to 1 atm.  $\Delta T$  and  $V/F$  come from the energy balance.

The outlet pressure is a plain editable value – tweak it in the GUI's Properties box (transient tinkering) and re-run, or edit it here.

**What to expect (golden-locked):**

| kind   | unit/stream | key        | expected          |
|--------|-------------|------------|-------------------|
| kpi    | flashAdiab  | T_out      | 368.661803634     |
| kpi    | flashAdiab  | V_over_F   | 0.0518811133842   |
| stream | feed        | F          | 0.0277777777778   |
| stream | liquid      | F          | 0.0263366357393   |
| stream | vapor       | F          | 0.00144114203845  |
| stream | feed        | T          | 380               |
| stream | liquid      | T          | 368.661803634     |
| stream | vapor       | T          | 368.661803634     |
| kpi    | flashAdiab  | F_in       | 0.0277777777778   |
| kpi    | flashAdiab  | F_liquid   | 0.0263366357393   |
| kpi    | flashAdiab  | F_vapor    | 0.00144114203845  |
| kpi    | flashAdiab  | H_residual | -0.00215364377073 |
| kpi    | flashAdiab  | P          | 101325            |
| kpi    | flashAdiab  | T_drop     | 11.3381963657     |

... and 1 more reference values in the case's *expected*.

**analysis01\_water\_analysis\_inlet**

tutorials/steady/flash/analysis01\_water\_analysis\_inlet



**What it does:** A laboratory water analysis as a first-class inlet: mg/L with alkalinity as CaCO<sub>3</sub>, a measured density, and a +0.78 % ion imbalance reconciled onto chloride within a declared 10 % limit – measurement, reconciled inventory and equilibrium recorded as three separate layers

**The story:** A LABORATORY ANALYSIS AS AN INLET.

The inlet of this flash is not a composition somebody worked out: it is a water analysis as a laboratory sends it – mg/L, alkalinity reported as CaCO<sub>3</sub>, a measured density, per-line uncertainties, and an ion balance that is off by 0.78 %.

THREE LAYERS, SEPARATELY RECORDED, and the separation is the whole point:

1 0/feed the MEASUREMENT. Never rewritten by a run (ratified 2026-08-09, O1) – a reconciliation that edited it would destroy the thing it was reconciling, and the next run would reconcile the reconciliation. 2 converged/feed calculated { analysisReconciliation { ? } conservedInventory { ? } } the RECONCILED conserved inventory, with the imbalance before, the imbalance after, and every correction that was applied. This is the layer the engine did not have: an answer never says what it was corrected from. 3 the flash the EQUILIBRIUM solved from layer 2.

WHAT IS MEASURED HERE (the numbers the gate re-derives independently):

ion balance BEFORE +0.7813 % (cations minus anions, over the sum) ion balance AFTER 0.0000 % chloride adjusted +5.1208 % of the measured 45.0 mg/L limit declared 10 % – exceeded, it REFUSES with the numbers

FIVE REFUSALS GUARD IT, each by name, and check\_aqueous\_analysis.py fires every one through the real reader: no DENSITY ROUTE – mg/L is per volume, a stream is a flow; an ITERATIVE density is a declared A1 gap, refused rather than done silently; a correction OVER THE LIMIT – past the declared bound it stops being a reconciliation and becomes the analysis rewritten to fit; pH NAMED AS THE BALANCING VARIABLE – pH may participate only under an explicit declared rule, and that grammar is A2's; 'method weightedLeastSquares;' – THE NAMED NEXT SLICE. Refused by name rather than silently approximated by the one method A1 has; TWO MATERIAL FORMS in one file – the seventh canonical form is exclusive with the other six, same as they are with each other.

And with no 'reconciliation {}' block at all: an analysis already closing within the tolerance passes through with an announcement stating what was measured; one that does not REFUSES, naming the two legal remedies. What never happens is an adjustment nobody asked for.

Contract: docs/design/aqueous-analysis-inlet-scope.md (slice A1). Compare: flash18\_water\_analysis\_basis – the same water as an INVENTORY, where charge closure is a contract and a violation is refused.



**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected        |
|--------|-------------|--------------|-----------------|
| stream | feed        | F            | 0.0307824830015 |
| stream | liquid      | F            | 0.0307824830015 |
| stream | vapor       | F            | 0               |
| kpi    | flash01     | F_alpha      | 0.0307824830015 |
| kpi    | flash01     | F_beta       | 0               |
| kpi    | flash01     | F_in         | 0.0307824830015 |
| kpi    | flash01     | P            | 101325          |
| kpi    | flash01     | T            | 313.15          |
| kpi    | flash01     | V_over_F     | 0               |
| kpi    | flash01     | pH           | 7.90774504802   |
| kpi    | flash01     | p_eq_sum_atm | 0.0747733964818 |
| kpi    | flash01     | phaseSet     | 0               |
| stream | feed        | T            | 313.15          |
| stream | liquid      | T            | 313.15          |

...and 1 more reference values in the case's *expected*.

**analysis02\_weighted\_least\_squares**

tutorials/steady/flash/analysis02\_weighted\_least\_squares



**What it does:** The same laboratory water as analysis01, reconciled by constrained weighted least squares over four measured quantities under electroneutrality plus an elemental-conservation redundancy: the correction spreads across all four at under 0.29 sigma each instead of shoving 5.12 % into chloride

**The story:** THE SAME WATER, RECONCILED PROPERLY.

analysis01 closes a +0.78 % ion balance by moving ONE ion: chloride absorbs the whole residue and shifts 5.12 %. That is a legitimate convention and it is what a water chemist does by hand – but it is a CHOICE dressed as an answer. Nothing about the analysis says chloride was the wrong number; what the sheet actually says is that FOUR quantities were measured, each with its own uncertainty, and that they cannot all be right at once.

A2 asks the question the uncertainties make askable:

what is the composition CLOSEST to what was measured – measuring closeness in each quantity’s OWN declared uncertainty – that satisfies the conservation laws exactly?

That is constrained weighted least squares, and it is a one-way PRE-STEP:

laboratory analysis → reconciliation → ONE conserved admissible composition → aqueous equilibrium / speciation

The arrow never reverses. The reconciler lives in its own translation unit (src/streams/AnalysisReconciler.{H,cpp}) which includes NOTHING from thermo/, so it cannot name a speciation solver, an activity model or a thermodynamic package – the equilibrium is structurally unable to reach back into the measurements and make the chemistry easier to satisfy.

THE SHEET. A calcium bicarbonate groundwater, the same water analysis01 carries, plus one line a real laboratory almost always also sends:

Ca 84.0 mg/L ICP-OES 2 % Cl 45.0 mg/L IC 5 % alkalinity 143.0 mg/L as CaCO<sub>3</sub> titration 4 % totalHardness 208.0 mg/L as CaCO<sub>3</sub> EDTA titration 3 % <– NEW

The hardness line is the interesting one. It measures the SAME CALCIUM the ICP measured, by a completely different method, and the two disagree by 0.84 % – which is unremarkable laboratory life and exactly why nobody can say which is right. It is declared ‘element Ca;’: it carries NO material of its own (that would count the calcium twice), it is a REDUNDANT determination, and its only effect is the law that ties it to the Ca row.

Note what is NOT declared on it. ‘alkalinity ... as CaCO<sub>3</sub>’ needs ‘perFormulaUnit 2’ because nothing in the formula CaCO<sub>3</sub> says a titration counts two bicarbonate per unit – that is a CONVENTION. ‘totalHardness { element Ca; as CaCO<sub>3</sub>; }’ needs no such key and is refused if it carries one: how many Ca there are in a CaCO<sub>3</sub> is ARITHMETIC, and the formula already says it.

TWO LAWS ARE ENFORCED, and both are declared rather than assumed:

electroneutrality  $\sum z_i n_i = 0$  elementalConservation the measured Ca total = the Ca the species rows carry

...plus non-negativity, which is not in the list because it is not optional.

WHAT THE FIT DOES. Two residues have to disappear – +0.0650 meq/L of charge and +0.0177 mmol/L of calcium disagreement – and the fit spends them where the measurements are weakest:

Ca -0.2735 sigma (-0.547 %) Cl +0.1562 sigma (+0.781 %) alkalinity +0.2813 sigma (+1.125 %) totalHardness +0.1000 sigma (+0.300 %)  $\sum ((x-m)/\text{sigma})^2 = 0.1883$

FOUR quantities move, none by a third of its own uncertainty, against analysis01's single 5.12 % shove. The declared limit is 3 sigma and the worst correction is 0.28 – and the run REFUSES with the numbers if any quantity would go past it.

READ 'converged/feed' FOR THE POINT OF THE SLICE. Every corrected quantity states what was reported, what it became, its sigma and weight, its correction IN SIGMA, which law caused it (exactly – the per-law parts sum to the correction, they are not shares invented afterwards), and how much of that law's own residue it removed. And the block opens with a legend saying that NOTHING in it is a chemistry result: the ions and the solved pH are in 'speciation {}', keyed by species rather than by the author's sheet labels, and carrying no 'reportedValue' because nothing in them was reported.

Provenance: a REPRESENTATIVE analysis composed for this witness, not a transcription of a published one. The disagreements are deliberate and are what the case exists to exercise.

Contract: docs/design/aqueous-analysis-inlet-scope.md (slice A2). Compare: analysis01\_water\_analysis\_inlet – the same water, one ion moved; flash18\_water\_analysis\_basis – the same water as an INVENTORY, where charge closure is a contract and a violation is refused.

#### What to expect (golden-locked):

| kind   | unit/stream | key          | expected        |
|--------|-------------|--------------|-----------------|
| stream | feed        | F            | 0.0307825083878 |
| stream | liquid      | F            | 0.0307825083878 |
| stream | vapor       | F            | 0               |
| kpi    | flash01     | F_alpha      | 0.0307825083878 |
| kpi    | flash01     | F_beta       | 0               |
| kpi    | flash01     | F_in         | 0.0307825083878 |
| kpi    | flash01     | P            | 101325          |
| kpi    | flash01     | T            | 313.15          |
| kpi    | flash01     | V_over_F     | 0               |
| kpi    | flash01     | pH           | 7.90867882055   |
| kpi    | flash01     | p_eq_sum_atm | 0.074797651023  |
| kpi    | flash01     | phaseSet     | 0               |
| stream | feed        | T            | 313.15          |
| stream | liquid      | T            | 313.15          |

... and 1 more reference values in the case's *expected*.

**bubbleT01\_ethanol\_water**

tutorials/steady/flash/bubbleT01\_ethanol\_water



**What it does:** Bubble-T of 30/70 ethanol/water at 1 atm (NRTL)

**The story:** Bubble-T of a 30/70 ethanol/water liquid at 1 atm (NRTL gamma, non-ideal). A saturation calc, not a flash: the unit solves  $\sum(K_i x_i) = 1$  for T at the given P (no product streams). Expected:  $T_{\text{bubble}} \sim 354.4$  K with  $y_{\text{ethanol}} \sim 0.58$  – ethanol enriched in the incipient vapour well beyond its 0.30 in the liquid.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| kpi    | bubble01    | P         | 101325           |
| kpi    | bubble01    | T_bubble  | 354.424819099    |
| kpi    | bubble01    | converged | 1                |
| kpi    | bubble01    | y_ethanol | 0.581915403358   |
| kpi    | bubble01    | y_water   | 0.418084596646   |
| stream | feed        | T         | 350              |

**bubbleT02\_uniquac\_ethanol\_water**

tutorials/steady/flash/bubbleT02\_uniquac\_ethanol\_water



**What it does:** Bubble-T of 30/70 ethanol/water at 1 atm (UNIQUAC) – validates the UNIQUAC model against the NRTL/Wilson twins

**The story:** Bubble-T of a 30/70 ethanol/water liquid at 1 atm (UNIQUAC).

Same conditions as bubbleT01 (NRTL): a strongly non-ideal binary whose bubble temperature and vapour composition are set by the activity coefficients. The UNIQUAC result should land close to the NRTL/Wilson twins ( $T_{\text{bubble}} \sim 353.5$  K, the vapour enriched in ethanol towards the azeotrope,  $y_{\text{ethanol}} \sim 0.58$ ).

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| kpi    | bubble02    | P         | 101325           |
| kpi    | bubble02    | T_bubble  | 353.469206954    |
| kpi    | bubble02    | converged | 1                |
| kpi    | bubble02    | y_ethanol | 0.582711434226   |
| kpi    | bubble02    | y_water   | 0.417288565785   |
| stream | feed        | T         | 350              |

**dewT01\_ethanol\_water**

tutorials/steady/flash/dewT01\_ethanol\_water



**What it does:** Dew-T of 30/70 ethanol/water vapour at 1 atm (NRTL)

**The story:** Dew-T of a 30/70 ethanol/water vapour at 1 atm (NRTL gamma, non-ideal). The mirror of bubbleT01: the unit solves  $\sum(y_i / K_i) = 1$  for T at the given P (a saturation calc, no product streams); the incipient liquid is water-enriched. Dew-T sits ABOVE the bubble-T of the same overall composition – the two bound the two-phase window.

**What to expect (golden-locked):**

| kind   | unit/stream | key | expected         |
|--------|-------------|-----|------------------|
| stream | feed        | F   | 0.02777777777778 |
| stream | feed        | T   | 370              |

**flash01\_benzene\_toluene**

tutorials/steady/flash/flash01\_benzene\_toluene



**What it does:** Isothermal flash benzene/toluene at 370 K, 1 bar (Raoult)

**The story:** THE first tutorial: isothermal flash of a 40/60 benzene/toluene feed (100 kmol/h) at 370 K, 1 bar, ideal Raoult K-values ( $K_i = P_{sat,i}/P$ ). 370 K sits inside the two-phase window (bubble ~368 K, dew ~374 K), so the feed splits into a benzene-enriched vapour and a toluene-enriched liquid – the Rachford-Rice solve is printed iteration by iteration at verbosity 3.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| kpi    | flash01     | V_over_F  | 0.30398357314    |
| kpi    | flash01     | T         | 370              |
| stream | vapor       | F         | 0.00844398814279 |
| stream | liquid      | F         | 0.019333789635   |
| stream | feed        | F         | 0.0277777777778  |
| stream | feed        | T         | 370              |
| stream | liquid      | T         | 370              |
| stream | vapor       | T         | 370              |
| kpi    | flash01     | F_alpha   | 0.019333789635   |
| kpi    | flash01     | F_beta    | 0.00844398814279 |
| kpi    | flash01     | F_in      | 0.0277777777778  |
| kpi    | flash01     | K_benzene | 1.65136809583    |
| kpi    | flash01     | K_toluene | 0.673501829388   |
| kpi    | flash01     | P         | 100000           |

...and 3 more reference values in the case's *expected*.

**flash02\_ethanol\_water**

tutorials/steady/flash/flash02\_ethanol\_water



**What it does:** Isothermal flash 50/50 ethanol/water at 355 K (NRTL)

**The story:** isothermal flash, ethanol/water, NRTL

NOTE (golden re-recorded): the operation pressure formerly read a BARE 'P 1.01325;' (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to O(1e5), so the flash silently reported a fully superheated vapour ( $V/F = 1$ ,  $\text{phaseSet} = 0$ ) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the 'bar' unit restores the physically-correct partial vaporisation; the 'expected' golden was re-recorded accordingly (was the wrong all-vapour result).

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.00562323865754 |
| stream | vapor       | F         | 0.0221545391202  |
| kpi    | flash02     | F_alpha   | 0.00562323865754 |
| kpi    | flash02     | F_beta    | 0.0221545391202  |
| kpi    | flash02     | F_in      | 0.02777777777778 |
| kpi    | flash02     | P         | 101325           |
| kpi    | flash02     | T         | 355              |
| kpi    | flash02     | V_over_F  | 0.797563408329   |
| kpi    | flash02     | phaseSet  | 0                |
| stream | feed        | T         | 355              |
| stream | liquid      | T         | 355              |
| stream | vapor       | T         | 355              |
| kpi    | flash02     | K_ethanol | 2.22741301451    |

... and 3 more reference values in the case's *expected*.



**flash03\_wilson\_ethanol\_water**

tutorials/steady/flash/flash03\_wilson\_ethanol\_water



**What it does:** Isothermal flash 50/50 ethanol/water at 355 K (Wilson activity model)

**The story:** same conditions as flash02, but Wilson activity model

NOTE (golden re-recorded): the operation pressure formerly read a BARE 'P 1.01325;' (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to O(1e5), so the flash silently reported a fully superheated vapour (V/F = 1, phaseSet = 0) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the 'bar' unit restores the physically-correct partial vaporisation; the 'expected' golden was re-recorded accordingly (was the wrong all-vapour result).

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.0063767199082  |
| stream | vapor       | F         | 0.0214010578696  |
| kpi    | flash03     | F_alpha   | 0.0063767199082  |
| kpi    | flash03     | F_beta    | 0.0214010578696  |
| kpi    | flash03     | F_in      | 0.02777777777778 |
| kpi    | flash03     | P         | 101325           |
| kpi    | flash03     | T         | 355              |
| kpi    | flash03     | V_over_F  | 0.770438083305   |
| kpi    | flash03     | phaseSet  | 0                |
| stream | feed        | T         | 355              |
| stream | liquid      | T         | 355              |
| stream | vapor       | T         | 355              |
| kpi    | flash03     | K_ethanol | 2.07217514574    |

...and 3 more reference values in the case's *expected*.

**flash04\_molarFlows\_input**

tutorials/steady/flash/flash04\_molarFlows\_input



**What it does:** Isothermal flash, feed declared via molarFlows (benzene 40 + toluene 60 kmol/h)

**The story:** same benzene/toluene flash as flash01, but the feed is declared via per-species molar flows.

Instead of writing  $F = 100 \text{ kmol/h} + \text{composition } \{ \text{benzene } 0.40; \text{toluene } 0.60; \}$ , this case writes the absolute molar flow of each species:

molarFlows { benzene 40 kmol/h; toluene 60 kmol/h; }

The parser sums to  $F_{\text{total}} = 100 \text{ kmol/h}$  and derives  $x_{\text{benzene}} = 0.4$ ,  $x_{\text{toluene}} = 0.6$ . Physically equivalent to flash01 – and the result table is identical to printer precision ( $V/F = 0.304$ ,  $L = 69.6 \text{ kmol/h}$ ,  $V = 30.4 \text{ kmol/h}$ ).

Demonstrates the (d) form of stream declaration; see userGuide "Composition basis" for the five accepted forms.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.019333789635   |
| stream | vapor       | F         | 0.00844398814279 |
| kpi    | flash04     | F_alpha   | 0.019333789635   |
| kpi    | flash04     | F_beta    | 0.00844398814279 |
| kpi    | flash04     | F_in      | 0.02777777777778 |
| kpi    | flash04     | P         | 100000           |
| kpi    | flash04     | T         | 370              |
| kpi    | flash04     | V_over_F  | 0.30398357314    |
| kpi    | flash04     | phaseSet  | 0                |
| stream | feed        | T         | 370              |
| stream | liquid      | T         | 370              |
| stream | vapor       | T         | 370              |
| kpi    | flash04     | K_benzene | 1.65136809583    |

...and 3 more reference values in the case's *expected*.

**flash05\_ternary\_massFlows**

tutorials/steady/flash/flash05\_ternary\_massFlows



**What it does:** Ternary flash benzene/toluene/n-hexane, feed in mass flows (5000 kg/h)

**The story:** ternary benzene/toluene/n-hexane flash, feed declared via per-species MASS flows.

This is the (e) form of stream input from the v0.27 grammar (see userGuide "Composition basis"): instead of writing F total plus a composition, the student writes the absolute mass flow of each species and the parser sums + converts to molar:

$$m\_i \text{ [kg/s]} \rightarrow F\_mol,i \text{ [kmol/s]} = m\_i / MW\_i \quad F\_total \text{ [kmol/s]} = \text{Sigma } F\_mol,i \quad x\_i = F\_mol,i / F\_total$$

Feed composition by mass (5000 kg/h total): benzene 1500 kg/h (30 wt%) MW = 78.11 kg/kmol toluene 2500 kg/h (50 wt%) MW = 92.14 kg/kmol n-hexane 1000 kg/h (20 wt%) MW = 86.18 kg/kmol

Conversion to molar:  $F\_benzene = 1500 / 78.11 = 19.203$  kmol/h  $F\_toluene = 2500 / 92.14 = 27.131$  kmol/h  $F\_nHexane = 1000 / 86.18 = 11.604$  kmol/h  $F\_total = 57.938$  kmol/h  $x\_benzene = 0.3315$  ;  $x\_toluene = 0.4682$  ;  $x\_nHexane = 0.2003$

At T = 365 K, P = 1 bar the n-hexane (Tb = 342 K) is well above its boiling point and goes preferentially to the vapour; toluene (Tb = 384 K) is sub-cooled and stays preferentially in the liquid; benzene (Tb = 353 K) partitions. Result:

molar basis F = 57.94 kmol/h V/F = 0.494 L = 29.34 kmol/h V = 28.60 kmol/h

mass basis F = 5000.0 kg/h V/F<sub>mass</sub> = 0.487 L = 2567.3 kg/h V = 2432.7 kg/h

Mass fractions tell the partition story directly:

feed: w<sub>benzene</sub> 0.300 w<sub>toluene</sub> 0.500 w<sub>nHexane</sub> 0.200 liquid: w<sub>benzene</sub> 0.244 w<sub>toluene</sub> 0.624 w<sub>nHexane</sub> 0.133 vapor: w<sub>benzene</sub> 0.360 w<sub>toluene</sub> 0.369 w<sub>nHexane</sub> 0.271

Mass balance closes per species — e.g. n-hexane: feed: 5000 \* 0.200 = 1000 kg/h liquid: 2567 \* 0.133 = 341 kg/h vapor: 2433 \* 0.271 = 659 kg/h → 341 + 659 = 1000.

Toggle the units menu in the GUI to Mass-basis to see the F and composition flip back to kg/h + w<sub>i</sub>, matching the input statement symmetrically.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.0160941453232  |
| stream | liquid      | F         | 0.00814913402453 |
| stream | vapor       | F         | 0.00794501129864 |
| kpi    | flash05     | F_alpha   | 0.00814913402453 |
| kpi    | flash05     | F_beta    | 0.00794501129864 |
| kpi    | flash05     | F_in      | 0.0160941453232  |
| kpi    | flash05     | P         | 100000           |
| kpi    | flash05     | T         | 365              |
| kpi    | flash05     | V_over_F  | 0.493658478851   |
| kpi    | flash05     | phaseSet  | 0                |
| stream | feed        | T         | 365              |
| stream | liquid      | T         | 365              |
| stream | vapor       | T         | 365              |
| kpi    | flash05     | K_benzene | 1.43515088002    |

... and 4 more reference values in the case's *expected*.

**flash06\_sweep\_T**

tutorials/steady/flash/flash06\_sweep\_T



**What it does:** Flash benzene/toluene at 1 bar: T swept 360→385 K across the bubble→dew window (V/F S-curve + duty)

**The story:** the SAME 40/60 benzene/toluene isothermal flash as flash01 (100 kmol/h, 1 bar, Raoult); the T value below is only the BASE point – system/outerDict sweeps it 360 → 385 K across the bubble→dew window (~368..374 K) and reads the V/F S-curve + duty as responses.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.0110077140444  |
| stream | vapor       | F         | 0.0167700637334  |
| kpi    | flash01     | F_alpha   | 0.0110077140444  |
| kpi    | flash01     | F_beta    | 0.0167700637334  |
| kpi    | flash01     | F_in      | 0.02777777777778 |
| kpi    | flash01     | K_benzene | 1.74456473693    |
| kpi    | flash01     | K_toluene | 0.716223951412   |
| kpi    | flash01     | P         | 100000           |
| kpi    | flash01     | Q         | 284075.447071    |
| kpi    | flash01     | Q_kW      | 284.075447071    |
| kpi    | flash01     | T         | 372              |
| kpi    | flash01     | V_over_F  | 0.603722294402   |
| kpi    | flash01     | phaseSet  | 0                |

*...and 7 more reference values in the case's expected.*

flash07\_gridsweep\_TP

tutorials/steady/flash/flash07\_gridsweep\_TP



**What it does:** 2-D grid sweep: flash V/F over T x P (the flash envelope)

**The story:** one isothermal flash, benzene/toluene 40/60.

The base point is 370 K, 1 bar. The gridSweep in system/outerDict overrides BOTH operation.T and operation.P across a grid, so V/F traces the flash envelope (V/F=0 bubble line, V/F=1 dew line) over the T x P plane.

**What to expect (golden-locked):**

| kind | unit/stream      | key                   | expected |
|------|------------------|-----------------------|----------|
| csv  | gridsweep_TP.csv | 8:flashDrum.V_over_F  | 0.235178 |
| csv  | gridsweep_TP.csv | 23:flashDrum.V_over_F | 0.303984 |
| csv  | gridsweep_TP.csv | 38:flashDrum.V_over_F | 0.641142 |
| csv  | gridsweep_TP.csv | 46:flashDrum.V_over_F | 0.385974 |
| csv  | gridsweep_TP.csv | 0:flashDrum.V_over_F  | 1        |
| csv  | gridsweep_TP.csv | 1:flashDrum.V_over_F  | 0        |

**flash08\_co2\_water\_package**

tutorials/steady/flash/flash08\_co2\_water\_package

**What it does:** Carbonated water flash whose thermo is an INLINE property-package MANIFEST: constant/thermoPhysPropDict IS the full record (components, methods per phase, solvent+solutes, the CO2-water Henry pair declared+verified). CO2 dissolves by the unsymmetric Henry convention; water stays Raoult. Read the run log: the assembly announces itself.

**The story:** the property-package convention, minimal.

Soda water at 298 K: 1 bar flash of a CO2-rich feed. The THERMO is not a pile of flat keys – the case’s constant/thermoPhysPropDict IS the full INLINE manifest (name flash08\_co2Water) and the ThermoPackageBuilder assembles everything from it: liquid = solution.henryDilute (water on the Raoult rung, CO2 on the infinite-dilution Henry rung – ONE Gibbs surface, two conventions), vapour = ideal gas, the CO2-water Henry pair declared and VERIFIED at assembly (delete the pair path from the package and the run REFUSES, naming the file). Expect  $K_{CO2} = H(298)/P \sim 1.6e3$  (Sander): a huge K sets the vapour’s PURITY, not the split – the vapour is nearly pure CO2 ( $y_{CO2} \sim 0.97$ ) but its AMOUNT is tiny ( $V/F \sim 4e-4$ ), so only  $\sim 40\%$  of the fed CO2 leaves in it; the liquid keeps  $x_{CO2} \sim y_{CO2}/K \sim 6e-4$ .

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected          |
|--------|-------------|----------|-------------------|
| stream | flatWater   | F        | 0.000277662593734 |
| stream | gasOff      | F        | 1.15184043824e-07 |
| stream | sodaWater   | F        | 0.000277777777778 |
| kpi    | node        | F_alpha  | 0.000277662593734 |
| kpi    | node        | F_beta   | 1.15184043824e-07 |
| kpi    | node        | F_in     | 0.000277777777778 |
| kpi    | node        | P        | 101325            |
| kpi    | node        | Q        | 0                 |
| kpi    | node        | Q_kW     | 0                 |
| kpi    | node        | T        | 298.15            |
| kpi    | node        | V_over_F | 0.000414662557767 |
| kpi    | node        | phaseSet | 0                 |
| stream | flatWater   | T        | 298.15            |
| stream | gasOff      | T        | 298.15            |

...and 3 more reference values in the case’s *expected*.

**flash09\_n2ch4\_stryjek**

tutorials/steady/flash/flash09\_n2ch4\_stryjek

**What it does:** phi-phi VLE VALIDATION vs Stryjek, Chappelaar & Kobayashi (1974), J. Chem. Eng. Data 19:334 (DOI 10.1021/je60063a023), Table III, -190 F = 149.82 K. The case carries its thermo as an INLINE manifest: liquid+vapour both eos.SRK (one cubic, two roots:  $K = \phi_L/\phi_V$ ) with the N2-CH4 interaction constant  $k_{ij} = 0.0289$  (a small measured correction to the mixing rule, from the Knapp/DECHEMA data compilation). Measured at 299 psia:  $x_{N2}=0.1503$ ,  $y_{N2}=0.4579$  – compare the Flash Result table in the log with these measured values (agreement ~2%).

**The story:** the phi-phi world pinned to a MEASURED isotherm

Nitrogen-methane at 149.82 K (-190.00 F), 20.616 bar (299 psia) – a point of Stryjek, Chappelaar & Kobayashi (1974), J. Chem. Eng. Data 19:334, Table III (DOI 10.1021/je60063a023). MEASURED there:  $x_{N2} = 0.1503$   $y_{N2} = 0.4579$   $K_{N2} = 3.047$   $K_{CH4} = 0.638$  The feed  $z_{N2} = 0.30$  sits inside the envelope, so the flash must split into exactly those phases if the thermodynamics is right. The K-values come from  $K = \phi_L/\phi_V$  – the SAME SRK cubic evaluated at its liquid and vapour roots, with  $k_{ij}(N2,CH4) = 0.0289$  – a small MEASURED correction to the cubic mixing rule, from the Knapp/DECHEMA VLE data compilation (announced in this log). Neighbouring measured points bracket the model:  $K_{N2} = 3.047$  at 299 psia and 2.241 at 396.5 psia – the engine interpolates between them consistently (try P 22 bar in this dict:  $K_{N2} \sim 2.9$ ). This is the quantitative anchor of the phi-phi world, the same standard as Wiebe-1934 for the Henry world.

THE AGREEMENT IS EXECUTABLE (2026-08-10). Everything above was prose: the golden pinned F/P/T/Q – all SELF-RECORDED – so a model that drifted off Stryjek's isotherm would still have passed while this header went on claiming a measured anchor. 'expected' now carries two 'anchor' rows holding the PAPER's  $K_{N2} = 3.047$  and  $K_{CH4} = 0.638$ ; they are compared on every run and are never refreshed by 'record' (a published number is not the model's to rewrite). Measured agreement:  $K_{N2} +2.25\%$ ,  $K_{CH4} -1.56\%$ ,  $x_{N2} -0.52\%$ ,  $y_{N2} +1.73\%$  – inside the declared 3 % band, with margin.

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected          |
|--------|-------------|----------|-------------------|
| stream | feed        | F        | 0.000277777777778 |
| stream | liq         | F        | 0.000145628387466 |
| stream | vap         | F        | 0.000132149390312 |
| kpi    | node        | F_alpha  | 0.000145628387466 |
| kpi    | node        | F_beta   | 0.000132149390312 |
| kpi    | node        | F_in     | 0.000277777777778 |
| kpi    | node        | P        | 2061600           |
| kpi    | node        | Q        | 0                 |
| kpi    | node        | Q_kW     | 0                 |
| kpi    | node        | T        | 149.82            |
| kpi    | node        | V_over_F | 0.475737805121    |
| kpi    | node        | phaseSet | 0                 |
| kpi    | node        | K_N2     | 3.047             |
| kpi    | node        | K_CH4    | 0.638             |

... and 3 more reference values in the case's *expected*.

**flash09\_nh3\_water\_reactive**

tutorials/steady/flash/flash09\_nh3\_water\_reactive



**What it does:** Reactive NH<sub>3</sub>/water flash: SIMULTANEOUS aqueous speciation (NH<sub>4</sub><sup>+</sup>/H<sup>+</sup>/OH<sup>-</sup>, pH solved) + VLE at 368 K, 1 atm – ions never reach the vapour or the streams

**The story:** the volatile weak electrolyte, done honestly.

Ammonia in water IONISES (NH<sub>3</sub> + H<sub>2</sub>O  $\leftrightarrow$  NH<sub>4</sub><sup>+</sup> + OH<sup>-</sup>) and VOLATILISES (Henry) at the same time: the ionised fraction is unavailable to the vapour, so speciation and phase split must be solved TOGETHER – never flash-once/speciate-once. The flash unit implements NO chemistry: it delegates to the package's reactive equilibrium (unit-local speciation, property authority section 6b). Streams – including 0/ and converged/ – carry ONLY the apparent components NH<sub>3</sub> and water; the aqueous species (NH<sub>4</sub><sup>+</sup>, H<sup>+</sup>, OH<sup>-</sup>) exist in the internal state and the report.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected         |
|--------|-------------|--------------|------------------|
| stream | feed        | F            | 0.02777777777778 |
| stream | liquid      | F            | 0.0224533469429  |
| stream | vapor       | F            | 0.00532443083488 |
| kpi    | flash01     | F_alpha      | 0.0224533469429  |
| kpi    | flash01     | F_beta       | 0.00532443083488 |
| kpi    | flash01     | F_in         | 0.02777777777778 |
| kpi    | flash01     | P            | 101325           |
| kpi    | flash01     | Q            | 0                |
| kpi    | flash01     | Q_kW         | 0                |
| kpi    | flash01     | T            | 368.15           |
| kpi    | flash01     | V_over_F     | 0.191679510056   |
| kpi    | flash01     | pH           | 9.899823444      |
| kpi    | flash01     | p_eq_sum_atm | 1                |
| kpi    | flash01     | phaseSet     | 0                |

...and 5 more reference values in the case's *expected*.



**flash10\_acetic\_water\_reactive**

tutorials/steady/flash/flash10\_acetic\_water\_reactive



**What it does:** Reactive acetic acid/water flash: SIMULTANEOUS aqueous speciation (Acetate-/H+/OH-, pH solved) + VLE – the weak ACID mirror of flash09; ions never reach the vapour or the streams

**The story:** the volatile weak ACID, done honestly.

Acetic acid in water IONISES ( $\text{CH}_3\text{COOH} \leftrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$ ,  $\text{pK}_a$  4.756) and VOLATILISES at the same time – and its VAPOUR DIMERISES ( $2 \text{ HAc} = (\text{HAc})_2$ , carboxylic hydrogen bonding), so even the saturation check must price the dimer's own partial pressure. Speciation and phase behaviour are solved TOGETHER: the flash unit implements no chemistry, it delegates to the package's reactive equilibrium (unit-local speciation, section 6b). Streams – 0/ and converged/ included – carry ONLY the apparent components aceticAcid and water; the aqueous species ( $\text{CH}_3\text{COO}^-$ ,  $\text{H}^+$ ,  $\text{OH}^-$ ) exist in the internal state and the report.

THE OPERATING POINT IS DELIBERATELY SUBSATURATED (T fixed at 368.15 K, 1 atm;  $\text{sum } p_{\text{eq}} = 0.818 \text{ atm} < P$ ): the honest regime for a dilute, barely-volatile acid on the Davies rung. The unit is an ISOTHERMAL (T,P) flash – T is the specification, not a result. Temperature still ACTS, through BOTH van't Hoff legs (dissolution  $\text{dH} = -51.5 \text{ kJ/mol}$ , dimerisation  $\text{dH} = -63.2 \text{ kJ/mol}$ ), and the flash publishes what it moves as KPIs: pH,  $p_{\text{eq\_sum\_atm}}$ ,  $p_{\text{mono\_atm}}$ ,  $p_{\text{dimer\_atm}}$ , dimer\_share. flash11\_acetic\_T\_sweep sweeps T across the subsaturated window so the student SEES the total volatility climb while the dimer share falls; flash13\_acetic\_ethanol\_vacuum\_flash crosses into the TWO-PHASE regime, where the vapour mole balance is re-weighted by the dimer explicitly (apparent moles =  $n_{\text{mono}} + 2 n_{\text{dim}}$ ) and its association heat is priced into the duty – never a silent approximation.

**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected         |
|--------|-------------|-------------|------------------|
| stream | feed        | F           | 0.02777777777778 |
| stream | liquid      | F           | 0.02777777777778 |
| stream | vapor       | F           | 0                |
| kpi    | flash01     | F_alpha     | 0.02777777777778 |
| kpi    | flash01     | F_beta      | 0                |
| kpi    | flash01     | F_in        | 0.02777777777778 |
| kpi    | flash01     | P           | 101325           |
| kpi    | flash01     | Q           | 0                |
| kpi    | flash01     | Q_kW        | 0                |
| kpi    | flash01     | T           | 368.15           |
| kpi    | flash01     | V_over_F    | 0                |
| kpi    | flash01     | dimer_share | 0.0948608286627  |
| kpi    | flash01     | pH          | 2.15260602547    |
| kpi    | flash01     | p_dimer_atm | 0.00389347151965 |

...and 6 more reference values in the case's *expected*.

**flash10\_ch4propane\_pcsaft**

tutorials/steady/flash/flash10\_ch4propane\_pcsaft

**What it does:** PC-SAFT phi-phi binary flash, CH<sub>4</sub>/propane at 273.15 K / 20 bar,  $k_{ij} = 0$  (announced): the PREDICTIVE baseline of the SAFT world –  $K = \phi_L/\phi_V$  from ONE  $a_{res}$  surface at its two density roots. Sanity anchor carried by the case itself: propane’s curated Antoine gives  $P_{sat}(273.15\text{ K}) = 4.7\text{ bar}$ , so the Raoult-magnitude  $K_{propane} \sim 0.24$  – the PC-SAFT  $K_{propane} = 0.32$  sits near it (the deviation IS the fugacity correction of a liquid holding ~11% dissolved CH<sub>4</sub>, visible);  $K_{CH_4} \gg 1$  (supercritical light key). The published- $k_{ij}$  refinement is a NAMED follow-up: it needs the cited pair record (data/standards/parameters/PCSAFT/), never an invented number.

**The story:** the PC-SAFT phi-phi world on a BINARY,  $k_{ij} = 0$

Methane/propane, 273.15 K, 20 bar,  $z = 0.50/0.50$ . CH<sub>4</sub> is supercritical ( $T_c = 190.6\text{ K}$ ) – no Raoult route exists for it; the phi-phi  $K$  comes from the SAME PC-SAFT  $a_{res}$  at the liquid and vapour density roots. With NO binary correction ( $k_{ij} = 0$ , announced by the EoS), everything is PREDICTED from the two components’ ( $m$ ,  $\sigma$ ,  $\epsilon/k$ ) alone (Gross & Sadowski 2001, Table 1; CH<sub>4</sub> and propane parameters independently cross-checked against the NIST teqp reference implementation’s test values).

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected          |
|--------|-------------|----------|-------------------|
| stream | feed        | F        | 0.000277777777778 |
| stream | liq         | F        | 9.97838549688e-05 |
| stream | vap         | F        | 0.000177993922809 |
| kpi    | node        | F_alpha  | 9.97838549688e-05 |
| kpi    | node        | F_beta   | 0.000177993922809 |
| kpi    | node        | F_in     | 0.000277777777778 |
| kpi    | node        | P        | 2000000           |
| kpi    | node        | Q        | 0                 |
| kpi    | node        | Q_kW     | 0                 |
| kpi    | node        | T        | 273.15            |
| kpi    | node        | V_over_F | 0.640778122112    |
| kpi    | node        | phaseSet | 0                 |
| stream | feed        | T        | 273.15            |
| stream | liq         | T        | 273.15            |

... and 3 more reference values in the case’s *expected*.

**flash11\_acetic\_T\_sweep**

tutorials/steady/flash/flash11\_acetic\_T\_sweep



**What it does:** Reactive acetic/water flash: T swept 328→368 K in the subsaturated window (p<sub>eq</sub> climbs, dimer share falls, pH drifts)

**The story:** the same reactive flash as flash10, but the temperature specification is OWNED by system/outerDict: the sweep driver rewrites units[0].operation.T pass by pass (setScalarAtPath – the dict IS the schema) and records the flash’s reactive KPIs at each point. Topology, chemistry, and feed are identical to flash10\_acetic\_water\_reactive; see that case’s header for the physics and for why the two-phase corner is refused.

**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected         |
|--------|-------------|-------------|------------------|
| stream | feed        | F           | 0.02777777777778 |
| stream | liquid      | F           | 0.02777777777778 |
| stream | vapor       | F           | 0                |
| kpi    | flash01     | F_alpha     | 0.02777777777778 |
| kpi    | flash01     | F_beta      | 0                |
| kpi    | flash01     | F_in        | 0.02777777777778 |
| kpi    | flash01     | P           | 101325           |
| kpi    | flash01     | Q           | -43266.66666667  |
| kpi    | flash01     | Q_kW        | -43.26666666667  |
| kpi    | flash01     | T           | 348.15           |
| kpi    | flash01     | V_over_F    | 0                |
| kpi    | flash01     | dimer_share | 0.115426638388   |
| kpi    | flash01     | pH          | 2.15093435013    |
| kpi    | flash01     | p_dimer_atm | 0.00184409797792 |

...and 10 more reference values in the case’s *expected*.

**flash12\_nh3\_acetic\_ethanol\_reactive**

tutorials/steady/flash/flash12\_nh3\_acetic\_ethanol\_reactive



**What it does:** Reactive 4-component flash:  $\text{NH}_3 + \text{HAc}$  neutralise into an ammonium-acetate buffer ( $\text{pH} \sim \text{pKa}_{\text{HAc}}$ ); ethanol classified nonionising from its record fact, ideal VLE authorised; subsaturated at 358.15 K

**The story:** One isothermal flash, four components, two aqueous families + one authorised molecular co-volatile:

feed (2 HAc + 1  $\text{NH}_3$  + 1 EtOH + 96  $\text{H}_2\text{O}$ , kmol/h)  $\rightarrow$  [flash01, 358.15 K / 1 atm]  $\rightarrow$  liquid (everything) + vapor (0)

The chemistry does the interesting work in the LIQUID:  $\text{NH}_3 + \text{HAc} \rightarrow \text{NH}_4^+ + \text{Ac}^-$  consumes the base completely ( $K = K_{\text{a\_HAc}}/K_{\text{a\_NH}_4} \sim 10^{4.5}$ ), leaving an equimolar  $\text{Ac}^-/\text{HAc}$  buffer at  $\text{pH} \sim 4.6$ . The saturation check prices each volatile over THAT speciated liquid – ionised acid and ammonium are unavailable to the vapour, the acetic dimer is priced, and ethanol contributes  $x \cdot \text{psat}$  (authorised ideal). KPIs: pH,  $p_{\text{eq\_sum\_atm}}$ ,  $p_{\text{mono\_atm}}$  /  $p_{\text{dimer\_atm}}$  /  $\text{dimer\_share}$ ,  $p_{\text{ethanol\_atm}}$ .

**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected          |
|--------|-------------|-------------|-------------------|
| stream | feed        | F           | 0.02777777777778  |
| stream | liquid      | F           | 0.02777777777778  |
| stream | vapor       | F           | 0                 |
| kpi    | flash01     | F_alpha     | 0.02777777777778  |
| kpi    | flash01     | F_beta      | 0                 |
| kpi    | flash01     | F_in        | 0.02777777777778  |
| kpi    | flash01     | P           | 101325            |
| kpi    | flash01     | Q           | 0                 |
| kpi    | flash01     | Q_kW        | 0                 |
| kpi    | flash01     | T           | 358.15            |
| kpi    | flash01     | V_over_F    | 0                 |
| kpi    | flash01     | dimer_share | 0.0226239722903   |
| kpi    | flash01     | pH          | 4.6182465052      |
| kpi    | flash01     | p_dimer_atm | 0.000106738278327 |

...and 7 more reference values in the case's *expected*.

**flash13\_acetic\_ethanol\_vacuum\_flash**

tutorials/steady/flash/flash13\_acetic\_ethanol\_vacuum\_flash



**What it does:** Reactive VACUUM flash on mixed-solvent v1 (NRTL backbone + Davies ions):  $V/F = 0.58$  at 358.15 K / 0.65 atm,  $\gamma_{\text{EtOH}} = 2.97$ , the buffer holds the ammonia, dimer heat priced exactly

**The story:** One isothermal VACUUM flash, four components, REAL vapour flow:

feed (2 HAc + 1 NH<sub>3</sub> + 12 EtOH + 85 H<sub>2</sub>O, kmol/h)  $\rightarrow$  [flash01, 358.15 K / 0.65 atm]  $\rightarrow$  liquid (42.3 kmol/h, pH 4.98) + vapor (57.7 kmol/h)

On the NRTL backbone the feed's bubble pressure is  $\sim 1.1$  atm ( $\gamma_{\text{EtOH}} = 2.97$  triples the ideal ethanol volatility): 0.65 atm strips well over half the feed without wringing the liquid dry. The vapour balance carries the acetic dimer explicitly ( $v_{\text{HAc}} = t_{\text{mono}} + 2 t_{\text{dim}}$ ,  $K_{\text{dim}}(T)$  van't Hoff) and the buffer chemistry keeps the ammonia liquid-bound. KPIs: pH,  $p_{\text{mono\_atm}}$ ,  $p_{\text{dimer\_atm}}$ ,  $\text{dimer\_share}$ ,  $p_{\text{ethanol\_atm}}$ ,  $p_{\text{eq\_sum\_atm}}$  (=  $P$ , two-phase).

**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected          |
|--------|-------------|-------------|-------------------|
| stream | feed        | F           | 0.02777777777778  |
| stream | liquid      | F           | 0.011761964504    |
| stream | vapor       | F           | 0.0160158132738   |
| kpi    | flash01     | F_alpha     | 0.011761964504    |
| kpi    | flash01     | F_beta      | 0.0160158132738   |
| kpi    | flash01     | F_in        | 0.02777777777778  |
| kpi    | flash01     | P           | 65861.25          |
| kpi    | flash01     | Q           | 665815.899842     |
| kpi    | flash01     | Q_kW        | 665.815899842     |
| kpi    | flash01     | T           | 358.15            |
| kpi    | flash01     | V_over_F    | 0.576569277856    |
| kpi    | flash01     | dimer_share | 0.0276466273131   |
| kpi    | flash01     | pH          | 4.98104745363     |
| kpi    | flash01     | p_dimer_atm | 0.000161043019564 |

...and 15 more reference values in the case's *expected*.

**flash14\_calcite\_carbonate\_basis**

tutorials/steady/flash/flash14\_calcite\_carbonate\_basis



**What it does:** Reactive calcite/carbonate flash: the FIRST component whose aqueous bridge spans TWO master families ( $\text{CaCO}_3 \rightarrow \text{Ca}+2$  AND  $\text{HCO}_3^-$ ), so the component-to-species map is a matrix, not a renaming

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.0275033888889   |
| stream | liquid      | F            | 0.0269197611327   |
| stream | vapor       | F            | 0.000583627756178 |
| kpi    | flash01     | F_alpha      | 0.0269197611327   |
| kpi    | flash01     | F_beta       | 0.000583627756178 |
| kpi    | flash01     | F_in         | 0.0275033888889   |
| kpi    | flash01     | P            | 101325            |
| kpi    | flash01     | T            | 313.15            |
| kpi    | flash01     | V_over_F     | 0.0212202124813   |
| kpi    | flash01     | pH           | 6.01561221041     |
| kpi    | flash01     | p_eq_sum_atm | 1                 |
| kpi    | flash01     | phaseSet     | 0                 |
| stream | feed        | T            | 313.15            |
| stream | liquid      | T            | 313.15            |

... and 4 more reference values in the case's *expected*.

**flash15\_refused\_salt\_basis\_rank**

tutorials/steady/flash/flash15\_refused\_salt\_basis\_rank



**What it does:** DELIBERATE REFUSAL: K/Na x Cl/SO<sub>4</sub> – four salt components onto four ion families with rank 3, so the component basis has exactly  $(c-1)(a-1) = 1$  degree of freedom and the converged state cannot be read back as these components uniquely

**flash16\_calcite\_precipitation**

tutorials/steady/flash/flash16\_calcite\_precipitation



**What it does:** Reactive flash that PRECIPITATES: the calcite/carbonate basis of flash14 fed past saturation, with calcite admitted – the speciation drives SI to 0 and the freed proton drops the pH from 6.63 to 6.02 – ONE proton per mole of calcite, and exactly once: the crystal is neutral, so it moves no charge, and the acidification arrives through the master balances that electroneutrality closes

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.0275083333333   |
| stream | liquid      | F            | 0.0269247487752   |
| stream | vapor       | F            | 0.000583584558179 |
| kpi    | flash01     | F_alpha      | 0.0269247487752   |
| kpi    | flash01     | F_beta       | 0.000583584558179 |
| kpi    | flash01     | F_in         | 0.0275083333333   |
| kpi    | flash01     | P            | 101325            |
| kpi    | flash01     | T            | 313.15            |
| kpi    | flash01     | V_over_F     | 0.0212148279253   |
| kpi    | flash01     | pH           | 6.02023176857     |
| kpi    | flash01     | p_eq_sum_atm | 1                 |
| kpi    | flash01     | phaseSet     | 0                 |
| stream | feed        | T            | 313.15            |
| stream | liquid      | T            | 313.15            |

...and 6 more reference values in the case's *expected*.



**flash17\_two\_liquids\_reactive**

tutorials/steady/flash/flash17\_two\_liquids\_reactive



**What it does:** Reactive flash with TWO liquids: vapour + speciated aqueous + benzene-rich organic at 313.15 K / 0.35 atm – V/F = 0.324, organic 1.7 % of the liquid, pH 2.89

**The story:** One isothermal flash, four components, THREE phases:

feed (60 H<sub>2</sub>O + 20 benzene + 8 EtOH + 0.1 HAc, kmol/h) → [flash01, 313.15 K / 0.35 atm] → liquid (59.6 kmol/h: aqueous + organic) + vapor (28.5 kmol/h)

The two liquids leave as ONE apparent liquid stream. That is deliberate and it is the same contract the speciation already lives under: the split is INTERNAL state – reported in full, never persisted – exactly as the ions are. A stream file carries apparent components; it does not carry phases, and it does not carry species.

KPIs: V/F, pH, p\_benzene\_atm, p\_ethanol\_atm, p\_eq\_sum\_atm (= P, two-phase), p\_mono\_atm / p\_dimer\_atm / dimer\_share (the acetic vapour dimer).

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected         |
|--------|-------------|---------------|------------------|
| stream | feed        | F             | 0.0244722222222  |
| stream | liquid      | F             | 0.0165455900723  |
| stream | vapor       | F             | 0.00792663214995 |
| kpi    | flash01     | F_alpha       | 0.0165455900723  |
| kpi    | flash01     | F_beta        | 0.00792663214995 |
| kpi    | flash01     | F_in          | 0.0244722222222  |
| kpi    | flash01     | P             | 35463.75         |
| kpi    | flash01     | Q             | 86.2488491825    |
| kpi    | flash01     | Q_kW          | 0.0862488491825  |
| kpi    | flash01     | T             | 313.15           |
| kpi    | flash01     | V_over_F      | 0.323903243358   |
| kpi    | flash01     | dimer_share   | 0.00665763422177 |
| kpi    | flash01     | pH            | 2.89010173914    |
| kpi    | flash01     | p_benzene_atm | 0.232353913068   |

... and 16 more reference values in the case's *expected*.

**flash18\_water\_analysis\_basis**

tutorials/steady/flash/flash18\_water\_analysis\_basis



**What it does:** A stream written in the AQUEOUS-SPECIES basis: a calcium-bicarbonate water analysis (Ca 0.0122, HCO<sub>3</sub> 0.0244 kmol/h) inverted onto the case's components by  $m = A n$  – the engine formulates the salts, the author does not

**The story:** Everything else in this corpus writes a stream in apparent components. This one writes it in the AQUEOUS SPECIES a laboratory actually measures, and lets the engine do the formulation:

```
speciesMolarFlows { network carbonate; basis analytical; Ca 0.0122; HCO3 0.0244; water 97.0; }
```

–  $m = A n$ , against the components' DECLARED bridges –

```
componentMolarFlows { CO2 0.0122; CaCO3 0.0122; water 97.0; }
```

**WHY IT MATTERS.** Converting an analysis into components by hand is a choice of LABELS, not a calculation: the same ions can be formulated as different component sets, and picking one silently is the arbitrariness the basis-rank test exists to refuse. Done here, the inversion is performed by the engine and REFUSED when it is not unique – the same refusal flash15\_refused\_salt\_basis\_rank raises, reached from the input side.

THREE REFUSALS GUARD IT, and check\_stream\_basis.py exercises all three: an analysis that does not close in CHARGE (a measurement to fix, never a residue for the solved pH to absorb); a 'network' naming a chemistry set this system does not declare; a missing 'basis' – because analytical and stoichiometric sets differ exactly in whether charge must close (H/OH are excluded mediators), so there is no safe default.

The answer is IDENTICAL to the same water written in components: the species basis is a change of coordinates, not a change of model. pH 7.757 – a natural calcium-bicarbonate water.

Contract: docs/design/aqueous-stream-basis-proposal.md.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected        |
|--------|-------------|--------------|-----------------|
| stream | feed        | F            | 0.026951222222  |
| stream | liquid      | F            | 0.026951222222  |
| stream | vapor       | F            | 0               |
| kpi    | flash01     | F_alpha      | 0.026951222222  |
| kpi    | flash01     | F_beta       | 0               |
| kpi    | flash01     | F_in         | 0.026951222222  |
| kpi    | flash01     | P            | 101325          |
| kpi    | flash01     | T            | 313.15          |
| kpi    | flash01     | V_over_F     | 0               |
| kpi    | flash01     | pH           | 7.75731867771   |
| kpi    | flash01     | p_eq_sum_atm | 0.0881612782912 |
| kpi    | flash01     | phaseSet     | 0               |
| stream | feed        | T            | 313.15          |
| stream | liquid      | T            | 313.15          |

*... and 1 more reference values in the case's expected.*

**flash19\_organic\_and\_precipitate**

tutorials/steady/flash/flash19\_organic\_and\_precipitate



**What it does:** FOUR phases at once: vapour + speciated aqueous + benzene-rich organic + solid calcite. flash16's feed plus benzene and ethanol, nothing else changed – the co-solvent moves the precipitation from 729 to 1096 mg/kg

*One isothermal flash at 313.15 K / 1 atm, and four phases coexist:*

*The three condensed phases leave through one bottoms nozzle and are named inside that stream's phases {} block; the vapour is its own stream.*

*Until it was written, the reactive path had been exercised with an organic liquid (flash17) and with a precipitate (flash16) — but never both at once. The file format supported it and the solver was believed to; no run proved it. This is that run, and the joint residual closes to = 6.6e-14.*

*The feed is flash16's, unchanged, plus benzene 2.0 and ethanol 1.0 kmol/h. Same water, same CO<sub>2</sub>, same CaCO<sub>3</sub>, same T, same P. So the two cases differ by exactly those two components:*

*Adding a co-solvent pushed half again as much calcite out of solution.*

*That number is not a mixed-solvent antisolvent calculation, and a student who reads it as one will be wrong about why.*

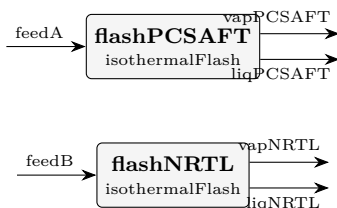
**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | feed        | F             | 0.0283416666667   |
| stream | liquid      | F             | 0.0275385746121   |
| stream | vapor       | F             | 0.000803092054611 |
| kpi    | flash01     | F_alpha       | 0.0275385746121   |
| kpi    | flash01     | F_beta        | 0.000803092054611 |
| kpi    | flash01     | F_in          | 0.0283416666667   |
| kpi    | flash01     | P             | 101325            |
| kpi    | flash01     | T             | 313.15            |
| kpi    | flash01     | V_over_F      | 0.0283360913124   |
| kpi    | flash01     | pH            | 6.10723545836     |
| kpi    | flash01     | p_benzene_atm | 0.238716176589    |
| kpi    | flash01     | p_eq_sum_atm  | 1                 |
| kpi    | flash01     | p_ethanol_atm | 0.011504063078    |
| kpi    | flash01     | phaseSet      | 0                 |

*...and 10 more reference values in the case's expected.*

**flash20\_ethanol\_water\_pcsaft**

tutorials/steady/flash/flash20\_ethanol\_water\_pcsaft



**What it does:** Ethanol/water flash: predictive PC-SAFT (assoc,  $k_{ij}=0$ ) vs fitted NRTL

**The story:** flash20 – ethanol/water VLE, PREDICTIVE vs FITTED on the same state

The same 30/70 mol% ethanol/water feed flashed at 358.15 K / 1 atm through TWO thermodynamic worlds, side by side:

flashPCSAFT the GLOBAL world: phi-phi PC-SAFT with the ASSOCIATION term active on both components (2B water + 2B ethanol, Gross & Sadowski 2002 Table 1 – the paper lists water with TWO sites; the mixture witness of the association programme, validation item 4). The cross-association follows the Wolbach-Sandler combining rules and  $k_{ij} = 0$  (announced) – so the MIXTURE behaviour is fully PREDICTED: nothing in this route was fitted to any ethanol/water datum. Only the PURE parameters are fits.

flashNRTL a per-unit thermo {} override: the classic gamma-phi route on the curated NRTL pair (fitted TO ethanol/water VLE data) + Antoine. This is the case's experimental anchor: the fitted route reproduces the measured azeotropic system, so the DIFFERENCE between the two units' K-values and splits is an honest measurement of what pure-component association physics alone can and cannot predict for a strongly hydrogen-bonded pair.

What the student compares, measured at the feed composition  $z$  (illustrative – what is golden are the CONVERGED K's: PC-SAFT 11.47 / 0.581 vs NRTL 3.896 / 0.602 for ethanol / water):

predictive PC-SAFT fitted NRTL (gamma-phi) K\_ethanol (at  $z$ ) 2.185 2.238 K\_water (at  $z$ ) 0.769 0.695  
V/F 0.649 0.512  $x_{\text{ethanol}}$  /  $y_{\text{ethanol}}$  0.038 / 0.441 0.121 / 0.471

The predictive route lands the K's within ~2 % (ethanol) and ~11 % (water) of the data-fitted route – from PURE-component parameters alone, a genuinely strong showing for a hydrogen-bonded pair – and the residual K\_water error still moves V/F from 0.51 to 0.65: the pedagogical point, stated up front rather than hidden. Predictive means "no mixture data used", not "as accurate as a fit"; the fitted gamma-phi route stays the design tool, the predictive EoS is the screen for systems with NO data to fit.

This case is ALSO the witness that catches a curation error the pure witnesses cannot: with water mis-curved as 4C (the same eps/kappa, the wrong scheme) the pure density read +0.6 % – fine – while THIS flash collapsed to  $K_{\text{water}} = 0.0044$ . The scheme is part of the fit.

A model boundary note: the two units run DIFFERENT thermo worlds on the same global component set (the general heterogeneous-solver contract); each stream leaves in its own unit's world.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feedA       | F         | 0.02777777777778 |
| stream | feedB       | F         | 0.02777777777778 |
| stream | liqNRTL     | F         | 0.0135537277127  |
| stream | liqPCSAFT   | F         | 0.00975103940319 |
| stream | vapNRTL     | F         | 0.014224050065   |
| stream | vapPCSAFT   | F         | 0.0180267383746  |
| kpi    | flashNRTL   | F_alpha   | 0.0135537277127  |
| kpi    | flashNRTL   | F_beta    | 0.014224050065   |
| kpi    | flashNRTL   | F_in      | 0.02777777777778 |
| kpi    | flashNRTL   | K_ethanol | 3.89559152547    |
| kpi    | flashNRTL   | K_water   | 0.602023174315   |
| kpi    | flashNRTL   | P         | 101325           |
| kpi    | flashNRTL   | Q         | 0                |
| kpi    | flashNRTL   | Q_kW      | 0                |

...and 23 more reference values in the case's *expected*.

**flash21\_brine\_co2\_pitzer**

tutorials/steady/flash/flash21\_brine\_co2\_pitzer



**What it does:** NaCl brine + CO<sub>2</sub> reactive flash under the pitzerHMW ionic rung – seam S5(b) witness: the declared model IS the served model.

**The story:** the S5(b) witness: a 2 mol/kg NaCl brine with dissolved CO<sub>2</sub>, flashed at 313.15 K / 1 atm, with the aqueous speciation priced by pitzerHMW (curated Na-Cl / Na-HCO<sub>3</sub> / Na-CO<sub>3</sub> / Na-OH / H-Cl pairs) instead of the charge-only Davies correlation that has no business at  $I = 2$ . The salt stays in the liquid by construction (nonvolatile, complete dissociation through its declared bridge); CO<sub>2</sub> partitions against its ionised fraction exactly as in flash09's ammonia lesson – speciation and phase split solved TOGETHER, never in sequence.

WHAT THE MODEL CHOICE MOVES, measured at build time (2026-08-24) against an otherwise identical davies twin: the liquid pH (3.84 here vs 3.96 under davies – the charge-only bracket is out of its band at  $I = 2$  and the Davies advisory says so on that run), while V/F barely moves (0.0211 vs 0.0210): the vapour is set by the NEUTRAL CO<sub>2</sub>aq, which Davies prices at exactly  $\gamma = 1$  and HMW at its real specific-interaction value.

A CAVEAT THIS CASE ANNOUNCES AND DOES NOT HIDE: the energy report tries to re-resolve every stream's own equilibrium, and the VAPOUR stream (93 % CO<sub>2</sub>, 7 % water, zero salt) read as an aqueous liquid has an implied molality near 770 mol/kg – outside any aqueous model's domain. pitzerHMW fails there loudly (singular Jacobian) and the report falls back to the carried state, announced in the caveat block. Davies returns a number at the same absurd state, which is not a virtue: a charge-only correlation cannot know it is out of its depth. The answer's own streams are unaffected.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.02777777777778  |
| stream | liquid      | F            | 0.0271904240052   |
| stream | vapor       | F            | 0.000587353772572 |
| kpi    | flash01     | F_alpha      | 0.0271904240052   |
| kpi    | flash01     | F_beta       | 0.000587353772572 |
| kpi    | flash01     | F_in         | 0.02777777777778  |
| kpi    | flash01     | K_CO2        | 3299.5218183      |
| kpi    | flash01     | K_NaCl       | 0                 |
| kpi    | flash01     | K_water      | 0.0696646141595   |
| kpi    | flash01     | P            | 101325            |
| kpi    | flash01     | T            | 313.15            |
| kpi    | flash01     | V_over_F     | 0.0211447358126   |
| kpi    | flash01     | pH           | 3.83985178539     |
| kpi    | flash01     | p_eq_sum_atm | 1                 |

...and 4 more reference values in the case's *expected*.

**flash21\_freeze\_concentration**

tutorials/steady/flash/flash21\_freeze\_concentration



**What it does:** Freeze-concentration SLE flash: ice crystallises from a water/ethanol liquor at -15 °C; the liquor concentrates until  $\gamma_w \cdot x_w = \exp(-dG_{fus}/RT)$

*What this witnesses (migration S4b, 2026-08-09): the flash's LIQUID+SOLID branch. A C3 uniform phases ( ? ) declaration — one liquid, one crystallizing solid (component water;), no vapour — dispatches to the SLE solve, where the ice phase becomes a SolidCandidate on THE ONE solid-equilibrium mechanism (solidEq::equilibrate, ruling R1's active-set + simultaneous damped Newton). The candidate's lnSI is  $\ln(x_w \cdot f_{eff\_liq} / f_{eff\_solid})$  and its remove pulls water from the liquid inventory — the one ledger.*

*The acceptance is a closed form, not a golden alone. At equilibrium the  $K = 1$  condition collapses to the freezing-point-depression identity*

$$\gamma_w \cdot x_w = \exp(-dG_{fus} / (R \cdot T)), \quad dG_{fus} = dH_{fus} \cdot (1 - T/T_{fus})$$

*with every quantity from the record's own data (water.dat Hfus 6008 J/mol, triple point 273.16 K). At 258.15 K the right side is 0.857434120855; the run publishes the liquor's activity as KPI a<sub>water</sub> = 0.857434120931 — agreement to 9e-11. check\_solid\_service arm A10 recomputes the closed form independently in Python on every suite run.*

*The numbers: feed 90/10 water/ethanol (mol) at 25 °C, 1 atm; the freezer operates at -15 °C. Ice forms at solidFraction 0.476 mol per mol of feed — a function of the operating (T, P) alone, not of the feed temperature; the liquor concentrates to  $x_{ethanol} \approx 0.19$ , exactly where the NRTL water activity meets the saturation value. The liquor and crystal leave through ONE outlet (liquor): its overall material equals the feed and the phases {} block decomposes it; vent is the structurally-required second outlet and carries  $F = 0$  (no vapour phase is declared).*

*The duty (gap closed 2026-08-09 by docs/design/tp-stream-energy-coherence.md slice 1, R-E3/R-E4): h<sub>formation</sub> gained its gas-natural → solid view ( $H_f - H_{vap}(298) - H_{fus} + ?cp\_solid$ , from the record's own declared transition data), and the flash prices every crystallised mole on that solid formation rung — the SAME leg the balance report reads. The warm feed makes the duty visible as physics:  $Q \approx -167.4$  kW = sensible cooling of 100 kmol/h by 40 K ( $\approx -88$  kW) + fusion of 47.6 kmol/h of ice ( $\approx -79$  kW). Cooling is negative by the sign convention; gate arm A10 asserts the published  $Q$  is finite and negative.*

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquor      | F         | 0.02777777777778 |
| stream | vent        | F         | 0                |
| kpi    | freezer01   | F_alpha   | 0.02777777777778 |
| kpi    | freezer01   | F_beta    | 0                |
| kpi    | freezer01   | F_in      | 0.02777777777778 |
| kpi    | freezer01   | P         | 101325           |
| kpi    | freezer01   | Q         | -167415.991569   |
| kpi    | freezer01   | Q_kW      | -167.415991569   |
| kpi    | freezer01   | T         | 258.15           |
| kpi    | freezer01   | V_over_F  | 0                |
| kpi    | freezer01   | aEq_water | 0.857434120855   |
| kpi    | freezer01   | a_water   | 0.857434120931   |
| kpi    | freezer01   | phaseSet  | 0                |

*... and 5 more reference values in the case's expected.*

flash\_polymer\_devolatilisation

tutorials/steady/flash/flash\_polymer\_devolatilisation



**What it does:** Devolatilise a polystyrene melt: flash off residual toluene; the polymer (fixed-MW pseudo-component, role nonvolatile) stays liquid

**The story:** polymer melt + residual solvent → flash (devolatiliser).

The flash boils off the residual toluene (vapour) and leaves the polymer in the liquid. polystyrene is ‘role nonvolatile;’, so its  $K = 0$ : it stays 100 % in the liquid by construction (no  $P_{\text{sat}}$  needed for a nonvolatile lump).

What to expect (golden-locked):

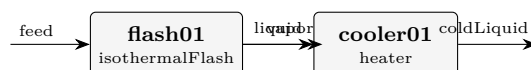
| kind   | unit/stream | key           | expected          |
|--------|-------------|---------------|-------------------|
| stream | liquid      | F             | 0.00246249927126  |
| stream | meltFeed    | F             | 0.00277777777778  |
| stream | vapor       | F             | 0.000315278506519 |
| kpi    | devol01     | F_alpha       | 0.00246249927126  |
| kpi    | devol01     | F_beta        | 0.000315278506519 |
| kpi    | devol01     | F_in          | 0.00277777777778  |
| kpi    | devol01     | K_polystyrene | 0                 |
| kpi    | devol01     | K_toluene     | 24.2878386162     |
| kpi    | devol01     | P             | 20000             |
| kpi    | devol01     | Q             | 0                 |
| kpi    | devol01     | Q_kW          | 0                 |
| kpi    | devol01     | T             | 450               |
| kpi    | devol01     | V_over_F      | 0.113500262347    |
| kpi    | devol01     | phaseSet      | 0                 |

...and 3 more reference values in the case's *expected*.



**flash\_pseudoComponent\_petCut**

tutorials/steady/flash/flash\_pseudoComponent\_petCut



**What it does:** Flash of a petroleum pseudo-component (Riazi-Daubert lump) + toluene, with a heater closing the energy balance

**The story:** feed → isothermal flash → heater on the liquid.

The flash splits the feed into vapour + liquid (the pseudo-component participates in VLE through its critical{} + Ambrose-Walton Psat). The heater unit then REMOVES sensible heat from the liquid (Q -30 kW – a heater with negative duty is a cooler); energyBalance reports the implied duty, and it CLOSES because the lump carries gasIdeal + liquidPure Cp. No reaction → no formation datum needed (matching the lump's omitted standardThermochemistry).

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | coldLiquid  | F         | 0.00758956122005 |
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.00758956122005 |
| stream | vapor       | F         | 0.0201882165577  |
| kpi    | cooler01    | F         | 0.00758956122005 |
| kpi    | cooler01    | P         | 100000           |
| kpi    | cooler01    | Q         | -30000           |
| kpi    | cooler01    | Q_kW      | -30              |
| kpi    | cooler01    | Q_per_mol | -3952.79768226   |
| kpi    | cooler01    | T_in      | 420              |
| kpi    | cooler01    | T_out     | 405.894884539    |
| kpi    | cooler01    | dT        | -14.1051154607   |
| kpi    | cooler01    | vf_out    | 0                |
| kpi    | flash01     | F_alpha   | 0.00758956122005 |

...and 16 more reference values in the case's *expected*.

**leaf01\_flash**

tutorials/steady/flash/leaf01\_flash

**What it does:** Leaf-node form: isothermal flash benzene/toluene (fractal base case)**The story:** / flowsheetDict (LEAF NODE – fractal base case)

A leaf carries ‘type’ + ‘operation’ (the unit) + ‘boundary’ (its interface) + ‘streams’ (the default feeds, so it runs isolated). No ‘units (...)’ list. Compare flash01\_benzene\_toluene, which writes the SAME flash as a one-element units list – identical physics, two roupas.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.02777777777778 |
| stream | liquid      | F         | 0.019333789635   |
| stream | vapor       | F         | 0.00844398814279 |
| kpi    | flash       | F_alpha   | 0.019333789635   |
| kpi    | flash       | F_beta    | 0.00844398814279 |
| kpi    | flash       | F_in      | 0.02777777777778 |
| kpi    | flash       | P         | 100000           |
| kpi    | flash       | T         | 370              |
| kpi    | flash       | V_over_F  | 0.30398357314    |
| kpi    | flash       | phaseSet  | 0                |
| stream | feed        | T         | 370              |
| stream | liquid      | T         | 370              |
| stream | vapor       | T         | 370              |
| kpi    | flash       | K_benzene | 1.65136809583    |

... and 3 more reference values in the case's *expected*.

**marcilla01\_lls\_tie\_triangle**

tutorials/steady/flash/marcilla01\_lls\_tie\_triangle



**What it does:** LLS anchor contact vs Marcilla 1995 (Table 10A row 4 charge): brine + solid halite reproduced; the organic vertex is NOT – the corpus UNIFAC (VLE table) keeps water/1-butanol miscible, a named model finding, never tuned

**The story:** THE FLAGSHIP LLS WITNESS (solid-equilibrium migration S5).

One isothermal flash at the paper’s own conditions: 25.0 C, 1 atm. The feed is Marcilla, Ruiz & Olaya’s Table 10A initial mixture, row 4 – a published experimental charge, not an invented one – and the anchor is the tie-triangle that charge produced (Table 10B row 4): an aqueous brine, a butanol-rich organic liquid, and solid NaCl, all simultaneously at equilibrium. PRIMARY: A. Marcilla, F. Ruiz, M. M. Olaya, Fluid Phase Equilibria 105 (1995) 71-91, DOI 10.1016/0378-3812(94)02595-R.

What the engine must do at once here, through the ONE mechanism each: dissolve NaCl onto the Na/Cl master families (the typed dissociatesTo bridge), split the molecular backbone into two liquids by equality of activity, and hold the halite complementarity through the solid- equilibrium service while that split resolves. No unit below declares any of that chemistry – the case does (thermoPhysPropDict + chemistryDict), and the flash reads it.

WHAT THIS CASE CURRENTLY WITNESSES, honestly (2026-08-10): the brine + solid halite half of the anchor, and a MODEL FINDING on the other half. The corpus’ UNIFAC is the VLE table (Hansen 1991), and under it the water/1-butanol pair is fully miscible at 25 C – g\_mix stays convex (independently confirmed by a propertyScanBinary sweep), so the Gibbs decision correctly reports ONE liquid where the paper measures two. The organic vertex of the tie-triangle is therefore NOT reproduced, and no number here was tuned to change that. The named remedy is curation: a UNIFAC-LLE parameter set (Magnussen, Rasmussen & Fredenslund, Ind. Eng. Chem. Process Des. Dev. 20 (1981) 331 – the parameterisation created precisely because the VLE table misses binodals), or fitted NRTL pairs for water/1-butanol and ethanol/1-butanol. See README.md.

PRIMARY: A. Marcilla, F. Ru?z, M. M. Olaya, \*Fluid Phase Equilibria\* 105 (1995) 71–91, DOI 10.1016/0378-3812(94)02595-R. Water–ethanol–1-butanol–NaCl at 25.0 ± 0.1 °C. Anchor values staged in [docs/design/solid-migration-witness-data.md](../docs/design/solid-migration-witness-data.md) ?2 (reviewStatus interim; this case cites the primary).

The feed is the paper’s own Table 10A row 4 initial mixture — a published experimental charge, not an invented one (wt%):

That charge produced the paper’s Table 10B row 4 tie-triangle — three phases simultaneously at equilibrium (wt%):

(The lever-rule check on the paper’s own numbers closes to  $\Sigma\lambda = 1.010$ .)

Davies ions on water-referenced molality + UNIFAC molecular backbone (the corpus table — the VLE parameterisation of Hansen et al. 1991) + halite complementarity through the solid-equilibrium service + a declared WET organic (water is a member; its cross-liquid equality carries the ionic  $a_w$  factor).

Reproduced: the saturated brine + solid halite coexistence. The run converges with solid NaCl present and a single fluid liquid; Davies is announced out of its validity band ( $I \approx 3.6$  mol/kg against a  $\sim 0.5$  band), so the predicted saturation is \*indicative\* — the model holds  $\sim 3.6$  mol/kg against Pinho & Macedo’s measured 6.14, and that gap is Davies’, reported.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected          |
|--------|-------------|----------------|-------------------|
| stream | feed        | F              | 0.000849885277778 |
| stream | liquid      | F              | 0.000849885277778 |
| stream | vapor       | F              | 0                 |
| kpi    | flash01     | F_alpha        | 0.000849885277778 |
| kpi    | flash01     | F_beta         | 0                 |
| kpi    | flash01     | F_in           | 0.000849885277778 |
| kpi    | flash01     | P              | 101325            |
| kpi    | flash01     | T              | 298.15            |
| kpi    | flash01     | V_over_F       | 0                 |
| kpi    | flash01     | pH             | 7.35672575928     |
| kpi    | flash01     | p_eq_sum_atm   | 0.0415604200493   |
| kpi    | flash01     | p_ethanol_atm  | 0.0111811211227   |
| kpi    | flash01     | p_nButanol_atm | 0.00389708966162  |
| kpi    | flash01     | phaseSet       | 0                 |

...and 3 more reference values in the case's *expected*.

**marcilla02\_lls\_ternary\_invariant**

tutorials/steady/flash/marcilla02\_lls\_ternary\_invariant



**What it does:** The ternary LLS invariant vs Marcilla 1995 Table 4B: aqueous brine + WET butanol-rich organic + solid halite, all three at once – the first PRESENT wet-organic witness (UNIQUE backbone, Winkelman pair)

**The story:** THE TERNARY LLS INVARIANT (solid-equilibrium migration S5b).

Water + 1-butanol + NaCl at 25.0 C, 1 atm: with solid NaCl present the two-liquid tie-line is INVARIANT ( $F = 0$  at fixed  $T, P$ ), so the paper reports ONE tie-triangle for the whole three-phase region – Table 4B: aqueous 73.14 / 0.72 / 26.14 (W / B / NaCl, wt%) organic 7.53 / 92.28 / 0.19 solid NaCl PRIMARY: A. Marcilla, F. Ruiz, M. M. Olaya, Fluid Phase Equilibria 105 (1995) 71-91, DOI 10.1016/0378-3812(94)02595-R.

This is the state marcilla01 could not reach: there the UNIFAC (VLE) backbone kept water/1-butanol miscible. Here the backbone is UNIQUE on the CITED Winkelman 2009 water/1-butanol set – the same values vll01 consumes inline, where they demonstrably open the gap – so the wet organic (water a declared member) is PRESENT, and all three phases coexist through their one mechanism each: the halite complementarity, the activity-equality split with the ionic  $a_w$  factor on water, and the Davies speciation of the brine.

PRIMARY: A. Marcilla, F. Ruiz, M. M. Olaya, \*Fluid Phase Equilibria\* 105 (1995) 71-91, DOI 10.1016/0378-3812(94)02595-R, Table 4B — water + 1-butanol + NaCl at  $25.0 \pm 0.1$  °C. With solid NaCl present the two-liquid tie-line is invariant, so one tie-triangle describes the whole three-phase region. Anchors staged in docs/design/solid-migration-witness-data.md ?2a (interim).

This is the state marcilla01 could not reach (its UNIFAC-VLE backbone keeps water/1-butanol miscible). The backbone here is UNIQUE on the cited Winkelman 2009 water/1-butanol set (case-local record — the same values vll01\_waterButanol consumes inline, where they demonstrably open the gap). The engine's first PRESENT wet organic: all three phases coexist, each through its one mechanism — halite complementarity, activity-equality split with the ionic  $a_w$  factor on water, Davies speciation of the brine.

Every deviation is a NAMED model approximation, announced by the run:

\* Aqueous NaCl low (16.2 vs 26.1 wt%): Davies saturates halite at 3.65 mol/kg against the measured 6.14 — the same number marcilla01 measured, seven times outside Davies' band, announced. \* Aqueous butanol high (7.7 vs 0.72 wt%): Davies prices neutral solutes at  $\gamma = 1$ , so the brine's salting-out of butanol — an order of magnitude in the anchor — is invisible to the ionic leg. \* Organic too wet (13.6 vs 7.53 wt% water): Davies'  $\varphi = 1$  water activity ( $a_w \approx 0.88$  vs the real  $\approx 0.75$  over saturated brine) under-prices the salt's pull on the water, so too much crosses.

The internal identities the gate pins: the cross-liquid equality residual including water's  $a_w$  factor is at machine precision ( $\sim 1e-15$  per member), and the dissolved salt over the AQUEOUS water alone reproduces the Davies saturation molality — the conversion-basis bug this case caught (a total-water basis overstated the crystal by exactly the organic water's share, while every mass balance still closed: a wrong basis conserves mass perfectly).

Energy balance: UNAVAILABLE (the apparent salt has no elements-datum route in this world — named diagnostic; mass and element balances close).

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected          |
|--------|-------------|----------------|-------------------|
| stream | feed        | F              | 0.000820413055556 |
| stream | liquid      | F              | 0.000820413055556 |
| stream | vapor       | F              | 0                 |
| kpi    | flash01     | F_alpha        | 0.000820413055556 |
| kpi    | flash01     | F_beta         | 0                 |
| kpi    | flash01     | F_in           | 0.000820413055556 |
| kpi    | flash01     | P              | 101325            |
| kpi    | flash01     | T              | 298.15            |
| kpi    | flash01     | V_over_F       | 0                 |
| kpi    | flash01     | pH             | 7.35672525831     |
| kpi    | flash01     | p_eq_sum_atm   | 0.032687682758    |
| kpi    | flash01     | p_nButanol_atm | 0.00579877206855  |
| kpi    | flash01     | phaseSet       | 0                 |
| stream | feed        | T              | 298.15            |

...and 2 more reference values in the case's *expected*.

**vllc01\_waterButanol**

tutorials/steady/flash/vllc01\_waterButanol



**What it does:** Liquid-liquid flash of water + 1-butanol at 320 K, 1.013 bar – the real miscibility gap from the cited Winkelman et al. (2009) UNIQUAC LLE fit: water-rich ~98.3 mol% water, butanol-rich ~46 mol% butanol (their Figs 1-2). See README.

**The story:** liquid-liquid flash of water + 1-butanol with the cited Winkelman et al. (2009) UNIQUAC LLE parameters.

Water + 1-butanol is THE textbook partially-miscible binary. At 320 K and 1.013 bar an 80/20 water/butanol mixture is well below the bubble point (water boils at 373 K, 1-butanol near 390 K), so the equilibrium is GENUINELY two LIQUIDS – no vapour. A simple ‘phaseSet LL’ flash (Michelsen TPD pre-check, then direct Gibbs-energy minimisation, multi-start Nelder-Mead) finds the split.

THE PARAMETERS (cited, NOT fabricated – see constant/thermoPhysPropDict and README.md): the UNIQUAC interaction parameters are the LLE-specific fit of

J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, Fluid Phase Equilibria 284 (2009) 71-79, Table 2 (p.73); structural r,q from their Table 1 (p.72).

with the full temperature dependence  $A_{ij}(T) = a_{ij} + b_{ij}T + c_{ij}T^2$  (their Eq. 10),  $\tau_{ij} = \exp(-A_{ij}/T)$  (their Eq. 7), valid 273-363 K. The numbers are inline here; the record is STAGED CASE-LOCALLY (reviewStatus interim) under marcilla02’s constant/parameters/UNIQUAC/, NOT curated into data/standards/ – promotion is the maintainer’s batch review.

a REAL liquid-liquid miscibility gap, reproduced from a cited primary source – the water-rich phase is ~98.3 mol% water (~1.7 mol% butanol) and the butanol-rich phase is ~54 mol% water (~46 mol% butanol), matching Winkelman’s Figs 1-2; the IsothermalFlash LL branch: a Michelsen TPD stability test that flags the feed as unstable, then a direct Gibbs-minimisation that resolves the two coexisting compositions; honest phase labelling – the two output streams are named by their actual composition (waterRich / butanolRich), with alphaRich/betaRich hints steering which liquid lands in which slot.

HISTORICAL NOTE: an earlier version of this case carried only a DECHEMA VLE-fit NRTL set, which (being a VLE fit) has NO miscibility gap, so the flash found a single liquid – it was kept as an honest "VLE-fit-misses-LLE" lesson pending a cited LLE set. Winkelman et al. (2009) IS that cited set, so the case now shows the genuine split. (The VLE-blind NRTL lesson lives on in the NRTL catalogue file’s provenance.)

*Water + 1-butanol (n-butanol) is \*the\* textbook partially-miscible binary. Below the bubble point it splits into two coexisting liquids: a water-rich phase carrying only a couple of mol % butanol, and a butanol-rich phase holding roughly half water. So a flash of a feed \*inside the miscibility gap\* should give two liquids — and here it does.*

*At 320 K, 1.013 bar (well below the bubble point, so no vapour) a feed of 80 mol % water / 20 mol % butanol splits into:*

*with split fraction  $\beta \approx 0.41$  into the butanol-rich phase. The solver reports the Michelsen TPD instability ( $tm_{min} \approx -0.45$ , \*unstable = YES\*) and then a direct Gibbs-energy minimisation that lands the two compositions.*

*An independent re-implementation of the UNIQUAC  $\gamma$  (fresh code, isoactivity Newton) reproduces the engine’s split to 4 decimals across 298–350 K, confirming the parameters are entered correctly. The agreement with the published figures is essentially exact — the UNIQUAC fit is doing its job.*

*constant/thermoPhysPropDict carries the LLE-specific UNIQUAC for water/1-butanol, cited per value:*

- Primary source: J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, *"Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"*, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013. - Binary interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation  $A_{ij}(T) = a_{ij} +$

$b_{ij} \cdot T + c_{ij} \cdot T^2$  (their Eq. 10) and  $\tau_{ij} = \exp(-A_{ij}/T)$  (their Eq. 7): -  $A_{1,3}$  (butanol $\rightarrow$ water):  $a = 155.31$ ,  $b = 1.0822$ ,  $10^2 \cdot c = -43.711$  -  $A_{3,1}$  (water $\rightarrow$ butanol):  $a = -579.36$ ,  $b = 2.7517$ ,  $10^2 \cdot c = -6.7700$  - Structural  $r$ ,  $q$  — their Table 1 (p.72) (value Magnussen et al.): 1-butanol  $r = 3.9243$ ,  $q = 3.668$ ; water  $r = 0.9200$ ,  $q = 1.400$ .

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected        |
|--------|-------------|--------------|-----------------|
| stream | butanolRich | F            | 0.0114431411907 |
| stream | feed        | F            | 0.0277777777778 |
| stream | waterRich   | F            | 0.016334636587  |
| kpi    | decanter01  | F_alpha      | 0.016334636587  |
| kpi    | decanter01  | F_beta       | 0.0114431411907 |
| kpi    | decanter01  | F_in         | 0.0277777777778 |
| kpi    | decanter01  | P            | 101300          |
| kpi    | decanter01  | T            | 320             |
| kpi    | decanter01  | betaFraction | 0.411953082867  |
| kpi    | decanter01  | phaseSet     | 1               |
| stream | butanolRich | T            | 320             |
| stream | feed        | T            | 320             |
| stream | waterRich   | T            | 320             |



**vlle03\_audit\_artificial**

tutorials/steady/flash/vlle03\_audit\_artificial



**What it does:** Algorithm-audit synthetic VLE: 3 components with identical  $P_{\text{sat}}$  and exaggerated NRTL. Should converge to 3-phase by construction.

**The story:** algorithm-correctness audit for the VLE Gibbs-min branch.

Three synthetic components A, B, C with identical Antoine ( $P_{\text{sat}} \approx 0.5$  bar at 350 K). NRTL parameters chosen to GUARANTEE three-phase coexistence at  $T = 350$  K,  $P = 1$  bar:

A-B:  $\tau_{AB} = \tau_{BA} = 5$  (very strong, near-immiscible) A-C:  $\tau_{AC} = \tau_{CA} = 5$  (very strong, near-immiscible) B-C:  $\tau_{BC} = \tau_{CB} = 0$  (ideal, fully miscible)

Expected behaviour by symmetry and construction: one liquid phase rich in A (the "isolated" component); another liquid phase rich in B+C (they pair together); a vapour phase carrying some of each ( $P_{\text{sat}}$  is equal so all are equally volatile,  $\ln(P/P_{\text{sat}}) \approx +0.301$  at 1 bar).

Equimolar feed (1/3, 1/3, 1/3). This case is the litmus test for the VLE solver: if it cannot find 3-phase here, the bug is in the algorithm, not in any real-system NRTL parameter set.

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected         |
|---------|-------------|-------------------------|------------------|
| stream  | liquidA     | F                       | 0.011307571384   |
| stream  | feed        | F                       | 0.0277777777778  |
| stream  | liquidB     | F                       | 0.0113091924076  |
| stream  | vapor       | F                       | 0.00516101398624 |
| kpi     | decanter01  | F_alpha                 | 0.0164702063938  |
| kpi     | decanter01  | F_beta                  | 0.011307571384   |
| kpi     | decanter01  | F_in                    | 0.0277777777778  |
| kpi     | decanter01  | F_vapor                 | 0.00516101398624 |
| kpi     | decanter01  | P                       | 100000           |
| kpi     | decanter01  | T                       | 350              |
| kpi     | decanter01  | V_over_F                | 0.407072569823   |
| kpi     | decanter01  | beta_vapor              | 0.185796503505   |
| kpi     | decanter01  | phaseSet                | 2                |
| utility | decanter01  | heating.steamLP.duty_kW | 4.38784508636    |

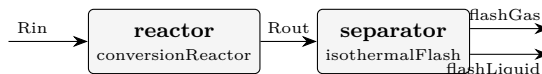
...and 12 more reference values in the case's *expected*.

**flowsheets**

**The theory you need.** Recycle turns a flowsheet into a fixed-point problem: guess the tear stream, march the units, compare. Successive substitution converges linearly (and diverges past unit gain); Wegstein accelerates it; Newton-on-tears solves the tear residuals with a Jacobian. Choupo announces the tear choice, every iteration's residual, and any bound that binds – no silent crutch. Expect: purge streams controlling inert build-up (close the loop with no purge and watch the accumulation), and convergence behaviour that CHANGES with the tear location. *Full derivation: Theory Guide, the flowsheet-convergence chapter.*

**acetone03\_luyben\_reaction\_section**

tutorials/steady/flowsheets/acetone03\_luyben\_reaction\_section



**What it does:** Luyben's acetone REACTION SECTION: reactor → cooler → flash, against his Rout and the flash split

**The story:** Three units of Luyben's Figure 1 wired together – reactor, cooler, flash – and the first piece of the plant whose answer DOES depend on the mixture thermodynamics. acetone02 isolated the reaction; this adds the phase split that sends hydrogen and acetone one way and the aqueous liquid the other.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Feed Rin as derived in acetone02 (57.82 kmol/h, IPA 0.668 / water 0.332, 389 K, 2.6 atm).

WHAT IS INDEPENDENT HERE, and it is the point. acetone02's outlet composition agreed with the paper because the feed had been DERIVED from that same outlet – a closed loop, and its golden says so. The flash split is not: nothing in 0/Rin was built from it. Luyben's Figure 1 gives

flash vapour 39.76 kmol/h H2 0.8742 A 0.0634 IPA 0.0002 W 0.0622 flash liquid 52.84 kmol/h (92.6 - 39.76; his F1 of 72.85 kmol/h is this stream AFTER the absorber bottoms of 20.05 kmol/h rejoin it)

and those are a real test of the thermodynamics this case declares.

THE THERMODYNAMICS IS DECLARED AND WEAK, said before the numbers rather than after. Luyben runs UNIQUAC; Choupo ships no UNIQUAC pair for any of these binaries, so this case runs PREDICTIVE UNIFAC on a lake-estimated isopropanol. The sibling props case measures what that costs on the same binary: the IPA boiling point is 4.5 K high and the IPA/water azeotrope lands at x 0.590 against a published 0.673. A separation built on that will not reproduce a separation built on fitted pairs, and the interesting question is HOW MUCH, not WHETHER.

Plus the one declared approximation, in constant/thermoPhysPropDict: H2 runs outside the activity model at gamma = 1, authorised by name, recorded in the answer.

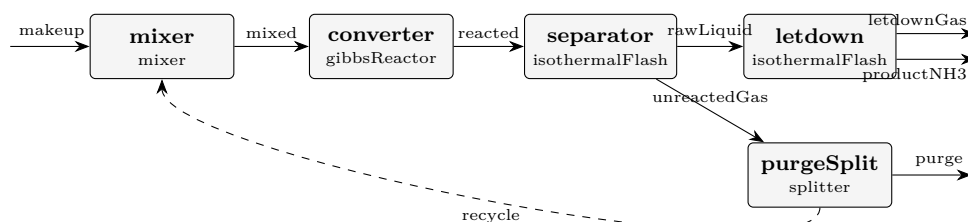
**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected         |
|--------|-------------|------------------|------------------|
| stream | Rin         | F                | 0.01606111111111 |
| stream | Rout        | F                | 0.02571861111111 |
| stream | flashGas    | F                | 0.0123053585397  |
| stream | flashLiquid | F                | 0.0134132525714  |
| kpi    | reactor     | F_in_kmol_h      | 57.82            |
| kpi    | reactor     | F_out_kmol_h     | 92.587           |
| kpi    | reactor     | Q_kW             | 840.263234652    |
| kpi    | reactor     | T                | 623              |
| kpi    | reactor     | conversion       | 0.9              |
| kpi    | reactor     | dHrxn_kJ_per_mol | 48.7173210118    |
| kpi    | reactor     | extent_kmol_h    | 34.767           |
| kpi    | separator   | F_alpha          | 0.0134132525714  |
| kpi    | separator   | F_beta           | 0.0123053585397  |
| kpi    | separator   | F_in             | 0.02571861111111 |

... and 20 more reference values in the case's *expected*.

**ammonia01\_synthesis\_loop**

tutorials/steady/flowsheets/ammonia01\_synthesis\_loop



**What it does:** Complete ammonia synthesis loop (Haber-Bosch) with recycle and purge: makeup 1:3 N<sub>2</sub>:H<sub>2</sub> (+ Ar inert) → mix with recycle → SRK Gibbs converter at 700 K / 200 bar (real-gas fugacity) → chill-flash at 250 K to condense the NH<sub>3</sub> product → split the unreacted gas into a small Ar-bleeding PURGE and the recycle. The recycle tear + purge are the whole point: high overall N<sub>2</sub>/H<sub>2</sub> utilisation despite a modest per-pass conversion, with the inert Ar held at steady state by the purge.

**The story:** the classic Haber-Bosch loop, recycle + purge

makeup (N<sub>2</sub>:H<sub>2</sub> = 1:3 + trace Ar) [mixer] recycle [converter] gibbsReactor (SRK) N<sub>2</sub> + 3H<sub>2</sub> ↔ 2NH<sub>3</sub>, 700 K, 200 bar (fugacity-corrected equilibrium ~30% NH<sub>3</sub>) [separator] isothermalFlash 250 K, 200 bar (chill: NH<sub>3</sub> condenses out) rawLiquid unreactedGas [purgeSplit] splitter [letdown] flash 20 bar purge recycle (bleeds Ar) (~95% back to the mixer) productNH<sub>3</sub> letdownGas (97.8% NH<sub>3</sub>) (H<sub>2</sub>-rich)

**THE LOOP.** Per-pass conversion is equilibrium-limited (~30% NH<sub>3</sub> at the reactor outlet), but the RECYCLE lifts overall N<sub>2</sub>/H<sub>2</sub> utilisation high – the reason the loop exists. The reactor equilibrium AND the separator both use SRK FUGACITY (real gas at 200 bar): switch the package to idealGas and the conversion falls. Mass closes to 100% on N, H and Ar.

**THE PURGE.** Ar enters with the makeup and would accumulate without end; the purge bleeds it, holding Ar at steady state (Ar<sub>in</sub> = Ar<sub>out</sub>). Delete the purge and the recycle fills with Ar until the loop chokes – try it.

**THE LET-DOWN.** At 200 bar the SRK flash leaves a lot of N<sub>2</sub>/H<sub>2</sub> DISSOLVED in the liquid (a real high-pressure effect, over-stated here because there is no N<sub>2</sub>/H<sub>2</sub>-NH<sub>3</sub> binary interaction parameter – SRK over-predicts light-gas solubility). So the raw separator liquid is only ~80% NH<sub>3</sub>. A let-down drum (flash to 20 bar) boils the dissolved gas off → 97.8% NH<sub>3</sub> product.

**HONEST SIMPLIFICATIONS.** (1) The loop is isobaric at 200 bar; the makeup compression and the feed/effluent heat integration are folded into the unit duties (reported), not drawn as separate machines. (2) The let-down gas (letdownGas, H<sub>2</sub>-rich) is RECOMPRESSED AND RECYCLED in a real plant; here it exits as a secondary stream to keep ONE clean recycle tear – so the H<sub>2</sub> loss reads higher than industrial. Recovering it (a recompressor + a second tear) is the efficiency refinement. The topology keeps the loop's ESSENCE: reaction, separation, recycle, purge, product finishing.

**What to expect (golden-locked):**

| kind   | unit/stream  | key          | expected          |
|--------|--------------|--------------|-------------------|
| stream | letdownGas   | F            | 1.06533052494e-06 |
| stream | makeup       | F            | 0.000277777777778 |
| stream | mixed        | F            | 0.000562684679632 |
| stream | productNH3   | F            | 0.000131750445273 |
| stream | purge        | F            | 1.49951001055e-05 |
| stream | rawLiquid    | F            | 0.000132815775798 |
| stream | reacted      | F            | 0.000432717777908 |
| stream | recycle      | F            | 0.000284906902004 |
| stream | unreactedGas | F            | 0.000299902002109 |
| kpi    | converter    | F_in_kmol_h  | 2.02566484668     |
| kpi    | converter    | F_out_kmol_h | 1.55778400047     |
| kpi    | converter    | N_in_mol_s   | 0.562684679632    |
| kpi    | converter    | N_out_mol_s  | 0.432717777908    |
| kpi    | converter    | P            | 20000000          |

...and 55 more reference values in the case's *expected*.

**ammonia02\_full\_plant**

tutorials/steady/flowsheets/ammonia02\_full\_plant

**What it does:** Full-scale ammonia plant (~59 t/h NH<sub>3</sub>, 1400 t/day): explicit syngas + recycle COMPRESSORS, feed-effluent HEAT EXCHANGER, ADIABATIC converter (the reaction heats itself – exothermic), water cooler + refrigerated separator, product let-down, and the recycle + purge. SRK fugacity throughout; dissolved H<sub>2</sub>/N<sub>2</sub> by measured Henry constants (Wiebe 1934, Larson-Black 1925 via IUPAC SDS 5/6).

**The story:** the complete Haber-Bosch plant, ~50 t/h NH<sub>3</sub>

makeup 25 bar [makeupComp 25→200] recycle [recycleComp] [mixer] mixed (cool) cold side [FEHE] hot effluent (heat recovery) preheatedFeed (~600 K) [converter] gibbsReactor mode ADIABATIC N<sub>2</sub>+3H<sub>2</sub>↔2NH<sub>3</sub> exothermic → self-heats to ~720 K hotEffluent hot side of FEHE → cooledEffluent [waterCooler] (cooling water) ~310 K [separator] flash 250 K/200 bar (refrig.) rawLiquid unreactedGas [letdown] [purgeSplit] productNH<sub>3</sub> gas purge recycle (tear)

THE REACTOR HEAT. N<sub>2</sub>+3H<sub>2</sub>→2NH<sub>3</sub> is EXOTHERMIC – there is NO external reactor heating. The converter runs ADIABATIC: the exotherm raises the gas T (feed ~705 K in, ~848 K out), and the feed-effluent exchanger (FEHE) recovers that heat to preheat the incoming feed to catalyst light-off. Self-sustaining once lit. The compressors set the 200-bar synthesis pressure; the water cooler (cooling-water utility) + refrigerated flash condense the NH<sub>3</sub>. Every machine is explicit and its duty/work is reported.

CAPACITY ~59 t/h NH<sub>3</sub> (~1400 t/day) at 98.9% product purity – a world-scale single train.

DISSOLVED GASES BY HENRY'S LAW. H<sub>2</sub> and N<sub>2</sub> are supercritical at the separator (no physical Psat), so their dissolved amounts follow MEASURED Henry constants (pairs H<sub>2</sub>-NH<sub>3</sub> from Wiebe 1934, N<sub>2</sub>-NH<sub>3</sub> from Larson & Black 1925 as evaluated in IUPAC SDS Vol 5/6), with the SRK vapour fugacity at 200 bar:  $K = H(T)/(\phi_V P)$ . The raw liquid carries ~0.5% H<sub>2</sub> + 0.2% N<sub>2</sub> – physical, vs the ~13% a Psat extrapolated above T<sub>c</sub> used to fake. KNOWN RESIDUAL: Ar has no measured Ar-NH<sub>3</sub> Henry pair yet, so its dissolved ~1% is a Raoult-extrapolation artefact (~7x over by analogy with N<sub>2</sub>) – the product's main impurity is overstated.

THE SIMULATION STRATEGY (how you build a plant like this). 1. PICK THE THERMO BY THE PHYSICS, per phase ("the two worlds"): the 200-bar GAS side needs real-gas fugacity → a cubic EoS (SRK) declared once in the package; the reactor folds phi into its equilibrium automatically. The DISSOLVED gases are supercritical solutes in a condensable solvent → the unsymmetric convention (Henry / Krichevsky-Kasarnovsky), NOT Raoult: pair files H<sub>2</sub>-NH<sub>3</sub> + N<sub>2</sub>-NH<sub>3</sub> fitted to measured solubility, 'solvent NH<sub>3</sub>'; in the package, 'role solute;' overlays on H<sub>2</sub>/N<sub>2</sub> in this case's constant/. 2. TEAR THE LOOP WHERE INFORMATION FLOWS BACKWARD: one MASS tear (recycle, guessed then Newton-converged) and one ENERGY tear (hotEffluent – the FEHE preheats the feed with the reactor's own product heat, so the converter's outlet is needed upstream of the converter). Both declared in solverDict tearStreams; the solver announces every iteration. 3. LET THE EXOTHERM DO THE HEATING: the converter is ADIABATIC (mode adiabatic; T is a guess, not a spec) – the reaction self-heats 705 → 848 K and the FEHE recycles that heat to light off the feed. No phantom reactor heater exists anywhere. 4. MAKE EVERY MACHINE EXPLICIT so it can be SIZED AND COSTED: compressors by shaft power, exchangers by area, the converter + drums by vessel design bases (space velocity / residence time) in system/postDict → Guthrie costs → the DCF economics in constant/economics. Every unit operation carries its own sizing – the doctrine. 5. VALIDATE AGAINST MEASUREMENT AT EVERY JOINT: the reactor equilibrium against the ideal-vs-SRK shift (gibbs10), the dissolved gas against Wiebe 1934 / Larson-Black 1925 (x\_H<sub>2</sub> at 400 atm: 0.0287 computed vs 0.0282 measured), the balances against 100% element closure.

HONEST SIMPLIFICATIONS. (1) The converter is a SINGLE adiabatic bed, so it runs hotter (~575 C) than a real multi-bed converter with interbed quench cooling (which stays near the 400-500 C equilibrium sweet spot for more NH<sub>3</sub> per pass). (2) The makeup compressor is one stage (real: multi-stage with intercooling), so its discharge is hot. (3) The let-down gas and the loop pressure drops are as in ammonia01. (4) 'water' is in the package ONLY as the cooling-water utility fluid (it is not a reaction species).

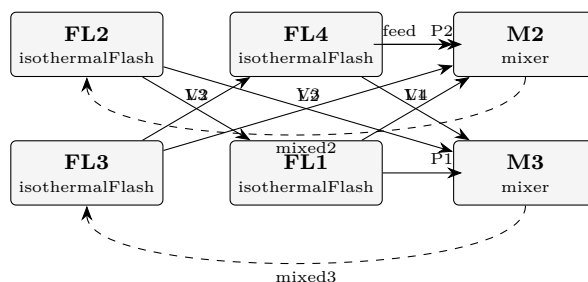
**What to expect (golden-locked):**

| kind   | unit/stream     | key | expected         |
|--------|-----------------|-----|------------------|
| stream | chilledEffluent | F   | 10.5545178888    |
| stream | compMakeup      | F   | 2.2222222222     |
| stream | compRecycle     | F   | 9.29181627432    |
| stream | cooledEffluent  | F   | 10.5545178888    |
| stream | cw              | F   | 138.888888889    |
| stream | cwOut           | F   | 138.888888889    |
| stream | hotEffluent     | F   | 10.554505292     |
| stream | letdownGas      | F   | 0.00745588546829 |
| stream | makeup          | F   | 2.2222222222     |
| stream | mixed           | F   | 11.5140384965    |
| stream | preheatedFeed   | F   | 11.5140384965    |
| stream | productNH3      | F   | 0.967859425136   |
| stream | purge           | F   | 0.287376077346   |
| stream | rawLiquid       | F   | 0.975315310604   |

*... and 147 more reference values in the case's **expected**.*

**cavett01\_recycle\_train**

tutorials/steady/flowsheets/cavett01\_recycle\_train



**What it does:** Cavett (1963) four-flash recycle benchmark on the primary Rosen-Pauls spec: 2 tears, Newton 7 it (Wegstein defeated). NO published product comparison is carried – see the flowsheetDict header

**The story:** / flowsheetDict – the Cavett problem on its PRIMARY specification (Cavett 1963; Rosen & Pauls 1977; Rosen, CACHE News Fall 2005; Seader & Henley 1998, Exercise 10.35).

The flowsheet is a four-theoretical-stage NEAR-ISOTHERMAL distillation (pressure, not temperature, changes stage to stage – the Gunther arrangement, US 3,575,077): FL1 is the partial condenser, FL4 the partial reboiler, and the FEED enters at the second stage. Each stage's liquid cascades DOWN; each stage's vapour climbs back UP – those three up-going vapours are the recycles R1, R2, R3 of the literature.

P1 (vapour product) FL1 (100 F, 800 psia) V2 L1 (R1) feed M2 FL2 (120 F, 270 psia) V3 L2 (R2) M3 FL3 (96 F, 49 psia) V4 L3 (R3) FL4 (85 F, 13 psia) P2 (liquid product)

HISTORY OF THIS DICT: the first commit of this case carried a reconstructed wiring (feed into FL1's mixer; V4 into FL2's) built from paraphrases, with the header saying so. The primary source then became available and showed BOTH edges wrong – the feed enters FL2 and V4 recycles to FL3. The reconstruction is gone; what you read is the documented flowsheet. A topology built from paraphrase and one built from the source differed in 2 edges out of 11 – worth remembering when citing anything from memory.

THE MISSING ANCHOR, named rather than implied (2026-08-10). This case's controlDict used to advertise "SRK vs published APR/FLOWTRAN products". It carries NO such comparison: no published product table is staged anywhere in the case, and nothing checks a Choupo number against one. What IS anchored here is the SPECIFICATION – the feed, the four stage conditions and the topology, all from the primary – plus the solver behaviour (Wegstein defeated on the real conditions; Newton converges in 7 iterations). That is a reproduction of the PROBLEM, not agreement with a published ANSWER, and the description now says so.

To become a real external anchor this case needs Rosen & Pauls' (or FLOWTRAN's) product compositions transcribed from the primary and pinned as 'anchor' rows in 'expected' – published values, never refreshed by '–record'. Until someone reads that paper, the gap stays visible: a claim of agreement nobody can check is worse than an admitted absence.

TWO tears still close all three loops: cut the two mixer outlets and every stream is computable in the declared order (validated by the sequential-plan contract, not asserted here).

**What to expect (golden-locked):**

| kind   | unit/stream | key     | expected      |
|--------|-------------|---------|---------------|
| stream | L1          | F       | 23.6955550394 |
| stream | L2          | F       | 11.4542917866 |
| stream | L3          | F       | 5.71821815965 |
| stream | P1          | F       | 3.68430363786 |
| stream | P2          | F       | 3.91030744687 |
| stream | V2          | F       | 27.3798586772 |
| stream | V3          | F       | 7.54398434784 |
| stream | V4          | F       | 1.80791071278 |
| stream | feed        | F       | 7.59461111111 |
| stream | mixed2      | F       | 38.8341504983 |
| stream | mixed3      | F       | 13.2622024993 |
| kpi    | FL1         | F_alpha | 23.6955550394 |
| kpi    | FL1         | F_beta  | 3.68430363786 |
| kpi    | FL1         | F_in    | 27.3798586772 |

*... and 134 more reference values in the case's **expected**.*



**composite01\_two\_flashes**

tutorials/steady/flowsheets/composite01\_two\_flashes

**What it does:** Composite node: two flashes in series (fractal step 2)**The story:** / flowsheetDict (COMPOSITE node)

children : the child unit ops (folders flash1/, flash2/ – leaf nodes). connections : the wiring. ‘from’/‘to’ endpoints are either a bare name (this node’s boundary inlet/outlet) or ‘<child>/<port>’. The engine flattens to namespaced streams (flash1.liquid → flash2.feed).

Two flashes in series, each a GENUINE vapour/liquid split. flash1 (370 K) splits the feed; its liquid (benzene-lean) feeds flash2 (373 K), which is still below that liquid’s dew point, so flash2 produces a real vapour AND a real liquid ‘bottoms’ – not a total vaporiser.

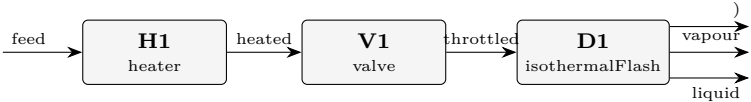
**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected         |
|--------|-------------|----------|------------------|
| stream | bottoms     | F        | 0.00370154472079 |
| stream | feed        | F        | 0.0277777777778  |
| stream | vapor1      | F        | 0.0212163063101  |
| stream | vapor2      | F        | 0.00285992674691 |
| kpi    | flash1      | F_alpha  | 0.0065614714677  |
| kpi    | flash1      | F_beta   | 0.0212163063101  |
| kpi    | flash1      | F_in     | 0.0277777777778  |
| kpi    | flash1      | P        | 100000           |
| kpi    | flash1      | T        | 370              |
| kpi    | flash1      | V_over_F | 0.763787027163   |
| kpi    | flash1      | phaseSet | 0                |
| kpi    | flash2      | F_alpha  | 0.00370154472079 |
| kpi    | flash2      | F_beta   | 0.00285992674691 |
| kpi    | flash2      | F_in     | 0.0065614714677  |

... and 26 more reference values in the case’s *expected*.

credo01\_valve\_heater\_drum

tutorials/steady/flowsheets/credo01\_valve\_heater\_drum



**What it does:** Info-follows-streams credo: heater (Q) + valve (P) set the state; the flash drum (no knobs) just separates

**The story:** "Information follows the streams" — every change of state is a **VISIBLE** box on the stream, and the separator has no operating knobs. Read the chain top-to-bottom, the way the plant is built:

feed (10 bar liquid) → [H1 heater : Q] the heat box. Its only knob is the duty; T<sub>out</sub> | is a result. Stays liquid at 10 bar (well below | the bubble point). → [V1 valve : P] the pressure box. Its only knob is the | downstream pressure; the let-down to 1 bar is | isenthalpic, so the hot liquid flashes and T and | vf come out as RESULTS. → [D1 flash : {} ] the separator. NO knobs — ‘operation {}’ is | empty. It inherits T and P from the stream V1 | hands it and just splits the phases. → liquid + vapour

Compare with flash01\_benzene\_toluene, where the same separation is asked as ‘isothermalFlash; operation { T 370; P 1 bar; }’. Those two numbers on the flash were, physically, a hidden heater (T) and a hidden valve (P). Here they are visible boxes, and the drum is honestly knob-less.

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected        |
|---------|-------------|-------------------------|-----------------|
| kpi     | H1          | T <sub>out</sub>        | 413.08978161    |
| kpi     | V1          | T <sub>out</sub>        | 369.30464285    |
| kpi     | V1          | vf                      | 0.20555173      |
| kpi     | V1          | dP                      | 900000          |
| stream  | liquid      | F                       | 0.0220680075    |
| stream  | vapour      | F                       | 0.0057097703    |
| utility | H1          | heating.steamMP.duty_kW | 260             |
| utility | H1          | heating.steamMP.T       | 413.089781612   |
| utility | H1          | heating.steamMP.kg_s    | 0.1305876444    |
| utility | H1          | heating.steamMP.eur_h   | 14.04           |
| stream  | feed        | F                       | 0.0277777777778 |
| stream  | feed        | T                       | 350             |
| stream  | heated      | F                       | 0.0277777777778 |
| stream  | heated      | T                       | 413.089781612   |

...and 28 more reference values in the case's *expected*.

**plant01\_two\_sectors**

tutorials/steady/flowsheets/plant01\_two\_sectors

**What it does:** Recursive plant: factory of two composite sectors (fractal step 3)**The story:** the RECURSIVE node (fractal step 3).

A FACTORY whose children are SECTORS that are themselves composite (each a flowsheet of one flash).  
 Recursion: plant  $\rightarrow$  sector  $\rightarrow$  unit. Tests nested namespacing (secA.fa, secB.fb) and multi-level cabling  
 (feed  $\rightarrow$  secA  $\rightarrow$  secA/liquid  $\rightarrow$  secB). The whole plant runs from one command.

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected         |
|---------|-------------|-------------------------|------------------|
| stream  | bottoms     | F                       | 0                |
| stream  | feed        | F                       | 0.02777777777778 |
| stream  | vaporA      | F                       | 0.0212163063101  |
| stream  | vaporB      | F                       | 0.0065614714677  |
| utility | secA.fa     | heating.steamLP.duty_kW | 686.964038241    |
| utility | secA.fa     | heating.steamLP.T       | 370              |
| utility | secA.fa     | heating.steamLP.kg_s    | 0.314256193157   |
| utility | secA.fa     | heating.steamLP.eur_h   | 29.676846452     |
| utility | secB.fb     | heating.steamLP.duty_kW | 226.733908578    |
| utility | secB.fb     | heating.steamLP.T       | 378              |
| utility | secB.fb     | heating.steamLP.kg_s    | 0.103720909688   |
| utility | secB.fb     | heating.steamLP.eur_h   | 9.79490485056    |
| stream  | bottoms     | T                       | 378              |
| stream  | feed        | T                       | 365              |

... and 2 more reference values in the case's *expected*.

**plant02\_basename\_twins**

tutorials/steady/flowsheets/plant02\_basename\_twins

**What it does:** basename twins: a top-level ‘feed’ and a DISTINCT sector-internal ‘A.feed’ coexist

**The story:** the ADVERSARIAL alias case of design forum #87-P0.

A top-level stream ‘feed’ (the domain inlet) and a sector-INTERNAL edge also named ‘feed’ (A.feed – a completely different stream between A’s two heaters) coexist. Identity is the PATH, never the basename: the manifest must carry BOTH (0/feed and 0/A/feed). The old basename filter silently dropped the top-level one from the manifest, so a legitimate case reported ORPHAN on its own inlet.

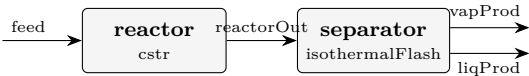
**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected          |
|---------|-------------|-------------------------|-------------------|
| stream  | feed        | F                       | 0.001000000000002 |
| stream  | hot         | F                       | 0.001000000000002 |
| stream  | out         | F                       | 0.001000000000002 |
| kpi     | h           | F                       | 0.001000000000002 |
| kpi     | h           | P                       | 100000            |
| kpi     | h           | Q                       | 2000              |
| kpi     | h           | Q_kW                    | 2                 |
| kpi     | h           | Q_per_mol               | 1999.99999997     |
| kpi     | h           | T_in                    | 298.15            |
| kpi     | h           | T_out                   | 366.647115308     |
| kpi     | h           | dT                      | 68.497115308      |
| utility | A.h1        | heating.steamMP.duty_kW | 1                 |
| utility | A.h1        | heating.steamMP.T       | 400.742542906     |
| utility | A.h1        | heating.steamMP.kg_s    | 0.000502260170768 |

...and 13 more reference values in the case’s *expected*.

process01\_reactor\_flash

tutorials/steady/flowsheets/process01\_reactor\_flash



**What it does:** Two-unit flowsheet: CSTR → isothermal flash

**The story:** Two-unit flowsheet: feed → CSTR → flash → (liquid, vapour)

feed reactorOut reactor separator (CSTR) (T 355, P 1 bar) liqProd vapProd

Reaction kinetics are in constant/reactions Solver defaults are in system/solverDict

What to expect (golden-locked):

| kind                                                    | unit/stream | key             | expected           |
|---------------------------------------------------------|-------------|-----------------|--------------------|
| stream                                                  | feed        | F               | 0.0002777777777778 |
| stream                                                  | liqProd     | F               | 0.000152263837157  |
| stream                                                  | reactorOut  | F               | 0.0002777777777778 |
| stream                                                  | vapProd     | F               | 0.00012551394062   |
| kpi                                                     | reactor     | Da_kTau         | 1.10929132154      |
| kpi                                                     | reactor     | F_in_kmol_h     | 1                  |
| kpi                                                     | reactor     | F_out_kmol_h    | 1                  |
| kpi                                                     | reactor     | Q_kW            | -1.87948262728     |
| kpi                                                     | reactor     | T               | 350                |
| kpi                                                     | reactor     | V_R             | 0.005              |
| kpi                                                     | reactor     | X_limiting      | 0.52590711876      |
| kpi                                                     | reactor     | k               | 0.00357499942014   |
| kpi                                                     | reactor     | minus_r_ethanol | 14.6085310767      |
| kpi                                                     | reactor     | tau_s           | 310.291329081      |
| ...and 22 more reference values in the case's expected. |             |                 |                    |

process02\_reactor\_hx\_flash

tutorials/steady/flowsheets/process02\_reactor\_hx\_flash



**What it does:** Three-unit flowsheet: CSTR → heater → flash

**The story:** Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~359 K, 0.9 % vap) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator] vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

What to expect (golden-locked):

| kind                                                            | unit/stream | key       | expected          |
|-----------------------------------------------------------------|-------------|-----------|-------------------|
| stream                                                          | feed        | F         | 0.000277777777778 |
| stream                                                          | hotStream   | F         | 0.000277777777778 |
| stream                                                          | liqProd     | F         | 0.000262168078511 |
| stream                                                          | reactorOut  | F         | 0.000277777777778 |
| stream                                                          | vapProd     | F         | 1.56096992667e-05 |
| kpi                                                             | heater      | F         | 0.000277777777778 |
| kpi                                                             | heater      | P         | 101325            |
| kpi                                                             | heater      | Q         | 400               |
| kpi                                                             | heater      | Q_kW      | 0.4               |
| kpi                                                             | heater      | Q_per_mol | 1440              |
| kpi                                                             | heater      | T_in      | 350               |
| kpi                                                             | heater      | T_out     | 361.879907822     |
| kpi                                                             | heater      | dT        | 11.8799078219     |
| kpi                                                             | heater      | vf_out    | 0.00022200059932  |
| ...and 37 more reference values in the case's <i>expected</i> . |             |           |                   |

process02\_with\_design

tutorials/steady/flowsheets/process02\_with\_design



**What it does:** process02 with full reports: balances + sizing + Guthrie costing

**The story:** Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~359 K, 0.9 % vap) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator] vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

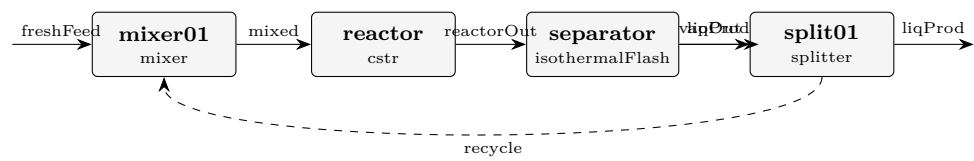
What to expect (golden-locked):

| kind   | unit/stream | key       | expected          |
|--------|-------------|-----------|-------------------|
| stream | feed        | F         | 0.000277777777778 |
| stream | hotStream   | F         | 0.000277777777778 |
| stream | liqProd     | F         | 0.000262168078511 |
| stream | reactorOut  | F         | 0.000277777777778 |
| stream | vapProd     | F         | 1.56096992667e-05 |
| kpi    | heater      | F         | 0.000277777777778 |
| kpi    | heater      | P         | 101325            |
| kpi    | heater      | Q         | 400               |
| kpi    | heater      | Q_kW      | 0.4               |
| kpi    | heater      | Q_per_mol | 1440              |
| kpi    | heater      | T_in      | 350               |
| kpi    | heater      | T_out     | 361.879907822     |
| kpi    | heater      | dT        | 11.8799078219     |
| kpi    | heater      | vf_out    | 0.00022200059932  |

...and 37 more reference values in the case's *expected*.

process03\_recycle

tutorials/steady/flowsheets/process03\_recycle



**What it does:** Recycle loop: the tear stream is solved by Newton (5 iters,  $|r|2\sim5e-6$ )

**The story:** the classic Reactor-Separator-Recycle architecture

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge

(the splitter is fed by liqProd)

Tear stream: ‘recycle’ — declared explicitly; the Flowsheet runs NEWTON on the tear variables (F, z..., T) until the post-pass value matches the input guess to a tolerance. (‘recycleSolver Wegstein;’ selects the fixed-point accelerator instead — try both and compare the iteration counts.)

The CSTR converts ~25% per pass (small volume); without recycle most ethanol would simply leave in liqProd. With ~80% recycle, the overall ethanol conversion approaches 100% at steady state.

What to expect (golden-locked):

| kind                                                    | unit/stream | key         | expected          |
|---------------------------------------------------------|-------------|-------------|-------------------|
| stream                                                  | freshFeed   | F           | 0.000277777777778 |
| stream                                                  | liqOut      | F           | 0.000229958363658 |
| stream                                                  | liqProd     | F           | 9.19833454634e-05 |
| stream                                                  | mixed       | F           | 0.000415752593523 |
| stream                                                  | reactorOut  | F           | 0.000415752593523 |
| stream                                                  | recycle     | F           | 0.000137975018195 |
| stream                                                  | vapProd     | F           | 0.000185794229864 |
| kpi                                                     | mixer01     | F_out       | 0.000415752593523 |
| kpi                                                     | mixer01     | P_out       | 100000            |
| kpi                                                     | mixer01     | T_out       | 354.51289855      |
| kpi                                                     | mixer01     | n_in        | 2                 |
| kpi                                                     | mixer01     | vf          | 0                 |
| kpi                                                     | reactor     | Da_kTau     | 4.22484004006     |
| kpi                                                     | reactor     | F_in_kmol_h | 1.49670933668     |
| ...and 33 more reference values in the case's expected. |             |             |                   |



**process04\_research\_workflow**

tutorials/steady/flowsheets/process04\_research\_workflow



**What it does:** Case-local binary pair file (research workflow pattern)

**The story:** isothermal flash of ethanol/water at 355 K, 1 atm

Identical to flash02\_ethanol\_water, but this case demonstrates the RESEARCH WORKFLOW: parameters loaded from case-local parameters file rather than declared inline. See README.md for the full discussion.

*The research workflow for managing thermodynamic parameters: instead of declaring NRTL pairs (...) inline in the thermoPhysPropDict, the case declares only the model and lets the simulator load the parameters from a file inside the case directory.*

*This is the pattern used when you have your own fitted parameters that you do not want to mix into the standards database. The case remains self-contained and reproducible: the parameters travel with the case.*

*When the simulator builds the NRTL model and discovers there are no inline pairs in thermoPhysPropDict, it falls back to file lookup:*

1. Case-local – <case>/constant/parameters/NRTL/ethanol-water.dat ? used here 2. Standards – \$CHOUP0\_HOME/data/standards/parameters/NRTL/ethanol-water.dat

*In this case the local file exists, so it wins. If you delete the local file, the run still succeeds because the standards file is the same DECHEMA set.*

*The result is numerically identical to flash02\_ethanol\_water (same NRTL parameters; just a different way of supplying them).*

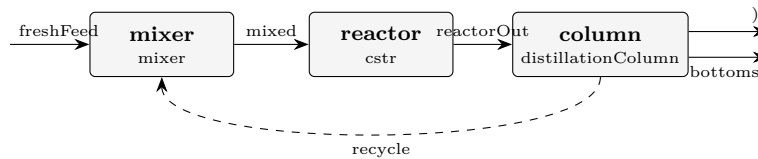
**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected         |
|--------|-------------|-----------|------------------|
| stream | feed        | F         | 0.0277777777778  |
| stream | liquid      | F         | 0.00562323865754 |
| stream | vapor       | F         | 0.0221545391202  |
| kpi    | flash01     | F_alpha   | 0.00562323865754 |
| kpi    | flash01     | F_beta    | 0.0221545391202  |
| kpi    | flash01     | F_in      | 0.0277777777778  |
| kpi    | flash01     | P         | 101325           |
| kpi    | flash01     | T         | 355              |
| kpi    | flash01     | V_over_F  | 0.797563408329   |
| kpi    | flash01     | phaseSet  | 0                |
| stream | feed        | T         | 355              |
| stream | liquid      | T         | 355              |
| stream | vapor       | T         | 355              |
| kpi    | flash01     | K_ethanol | 2.22741301451    |

... and 3 more reference values in the case's *expected*.

**process05\_isomerization\_recycle**

tutorials/steady/flowsheets/process05\_isomerization\_recycle



**What it does:** Isomerization neoPentane→nPentane: mixer + CSTR + distillation with recycle

**The story:** Reactor + distillation with recycle of the unconverted reactant — the canonical flowsheet for exercising the reports{} layer (stream table, mass balance with closure, energy balance) on a non-trivial topology with a tear stream.

freshFeed (neoPentane) [mixer] recycle (neoPentane) [reactor] neoPentane → nPentane (CSTR) [column] (distillation,  $\alpha \sim 2.5$ ) distillate bottoms (neoPentane, (nPentane, recycled) (recycle) the PRODUCT

Chemistry: neoPentane → nPentane is an isomerization — 1:1 in moles AND mass (identical MW 72.15). So at steady state the fresh feed (1 kmol/h neoPentane) equals the product (1 kmol/h nPentane in the bottoms), and the overall MASS BALANCE CLOSES TO 100 %. The recycle carries per-pass unconverted neoPentane back to the reactor, so the OVERALL conversion is ~100 % (product is 99 % nPentane) even though the per-pass conversion is only ~80 %.

Tuning notes (honest): The loop runs at 5 bar so every neoPentane/nPentane stream is a genuine LIQUID at its temperature. neoPentane's Antoine here boils at ~275 K at 1 bar ( $P_{\text{sat}}(320 \text{ K}) \sim 4.3 \text{ bar}$ ), so a 1 bar loop would flash the feed + recycle to VAPOUR — the phase is a CONSEQUENCE of (T,P,z) (Duhem), not a silent liquid default. At 5 bar the bubble point clears ~326 K, so the warm reactor (~327 K) and the recycled distillate stay liquid, as the flowsheet intends. The reaction uses a LOW activation energy on purpose — a high-Ea rate would make the per-pass conversion collapse against the recycle temperature and destabilise the loop. Low Ea decouples conversion from the recycle temperature. 'distillateRate' is FIXED here (the column's spec). It cannot perfectly track the per-pass unconverted neoPentane, so the recycle carries some nPentane (product) along — exactly why real recycle loops need a purge. Closing the recycle PURITY would be a DesignSpec manipulating distillateRate; left out here to keep the case a single pass for reviewing the reports.

This is the case to open in the GUI and to read the reports/ CSVs from: streamTable.csv, massBalance.csv (+ \_byUnit), energyBalance\_byUnit.csv.

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected          |
|--------|-------------|----------|-------------------|
| stream | bottoms     | F        | 0.000277777777778 |
| stream | freshFeed   | F        | 0.000277777777778 |
| stream | mixed       | F        | 0.000363888888889 |
| stream | reactorOut  | F        | 0.000363888888889 |
| stream | recycle     | F        | 8.61111111111e-05 |
| kpi    | column      | B        | 0.000277777777778 |
| kpi    | column      | D        | 8.61111111111e-05 |
| kpi    | column      | L_rect   | 0.000172222222222 |
| kpi    | column      | L_strip  | 0.000536111111111 |
| kpi    | column      | R        | 2                 |
| kpi    | column      | T_bottom | 364.544713604     |
| kpi    | column      | T_top    | 349.6579691       |
| kpi    | column      | V_rect   | 0.000258333333333 |
| kpi    | column      | V_strip  | 0.000258333333333 |

... and 35 more reference values in the case's expected.

**proxy01\_gas\_loop**

tutorials/steady/flowsheets/proxy01\_gas\_loop

□



**What it does:** Proxy step 1: gas WGS loop with one recycle (mixer + Gibbs reactor + splitter)

**The story:** The GUI/credo bench in one case: TWO recycles + a shaft-work energy wire (W) + a utility STREAM (cooling water). Toggle the stream classes in the canvas: energy (the W wire), recycle (the two tears), utility (the cooling-water edge).

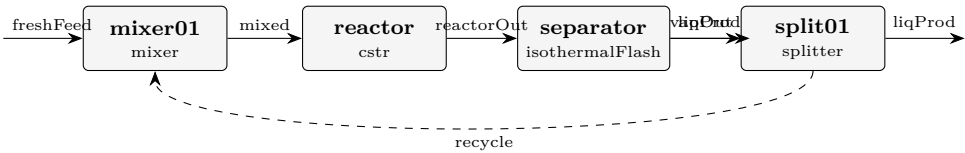
feed M1 mixer recycle1 (hot) recycle2 (cold, dry) R1 gibbsReactor (WGS) S1 splitter recycle1 (30 % hot gas back) toFlash CW (cooling water, UTILITY) HX1 warmCW (HX1: sensible cooling of the gas) F1 flash { } (no knobs: inherits T,P; condenses the water out) productWater (liquid) product dryGas (vapour) S2 splitter recycle2 (30 % dry gas back) productGas T1 turbine (power recovery) G1 electricLoad (W wire) expandedProduct

**What to expect (golden-locked):**

| kind                                                             | unit/stream     | key     | expected          |
|------------------------------------------------------------------|-----------------|---------|-------------------|
| stream                                                           | CW              | F       | 0.0166666666667   |
| stream                                                           | cooledGas       | F       | 0.000375486726088 |
| stream                                                           | dryGas          | F       | 0.000325696494167 |
| stream                                                           | expandedProduct | F       | 0.000227987545917 |
| stream                                                           | feed            | F       | 0.000277777777778 |
| stream                                                           | mixed           | F       | 0.000536409608696 |
| stream                                                           | productGas      | F       | 0.000227987545917 |
| stream                                                           | productWater    | F       | 4.97902319203e-05 |
| stream                                                           | reacted         | F       | 0.000536409608697 |
| stream                                                           | recycle1        | F       | 0.000160922882609 |
| stream                                                           | recycle2        | F       | 9.77089482502e-05 |
| stream                                                           | toFlash         | F       | 0.000375486726088 |
| stream                                                           | warmCW          | F       | 0.0166666666667   |
| kpi                                                              | F1              | F_alpha | 4.97902319203e-05 |
| ... and 86 more reference values in the case's <i>expected</i> . |                 |         |                   |

recycle\_autoinit\_tear

tutorials/steady/flowsheets/recycle\_autoinit\_tear



**What it does:** Recycle with NO authored tear guess – the solver auto-seeds the tear from the feed aggregate (announced), then converges

**The story:** same Reactor-Separator-Recycle as process03\_recycle; this case’s LESSON is the TEAR-SEED CONTRACT of the stream-state architecture:

the tear ‘recycle’ has an AUTHORED state file, 0/recycle – the seed is a first-class case file, owned by the author; completeness is FATAL before the solve: delete 0/recycle and choupoSolve REFUSES (MISSING, zero iterations run); ‘bin/choupo-init0’ refuses to INVENT a recycle seed (propagation is only honest on an acyclic graph) – it names the missing file and leaves the decision to you.

Compare the seed in 0/recycle with process03\_recycle’s: a recycle seed is a NUMERICAL aid the student owns, honest and visible ("no silent crutch", CLAUDE.md). (Historical note, marked legacy: on the retired ‘streams {}’ reader path a missing tear guess was auto-seeded from the feed aggregate, announced on the log; the 0/ contract replaced that with the authored, refuse-if-missing seed above.)

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge  
(the splitter is fed by liqProd)

Tear stream: ‘recycle’ — declared explicitly; the Flowsheet runs Wegstein on (F, z..., T) until the post-pass value matches the input guess to a tolerance.

The CSTR converts ~25% per pass (small volume); without recycle most ethanol would simply leave in liqProd. With ~80% recycle, the overall ethanol conversion approaches 100% at steady state.

What to expect (golden-locked):

| kind                                                     | unit/stream | key         | expected           |
|----------------------------------------------------------|-------------|-------------|--------------------|
| stream                                                   | freshFeed   | F           | 0.0002777777777778 |
| stream                                                   | liqOut      | F           | 0.000229958363961  |
| stream                                                   | liqProd     | F           | 9.19833455844e-05  |
| stream                                                   | mixed       | F           | 0.000415752593881  |
| stream                                                   | reactorOut  | F           | 0.000415752593881  |
| stream                                                   | recycle     | F           | 0.000137975018377  |
| stream                                                   | vapProd     | F           | 0.00018579422992   |
| kpi                                                      | mixer01     | F_out       | 0.000415752593881  |
| kpi                                                      | mixer01     | P_out       | 100000             |
| kpi                                                      | mixer01     | T_out       | 354.512898558      |
| kpi                                                      | mixer01     | n_in        | 2                  |
| kpi                                                      | mixer01     | vf          | 0                  |
| kpi                                                      | reactor     | Da_kTau     | 4.2248400393       |
| kpi                                                      | reactor     | F_in_kmol_h | 1.49670933797      |
| ... and 33 more reference values in the case’s expected. |             |             |                    |

gas-solid

**bagFilter01\_dust**

tutorials/steady/gas-solid/bagFilter01\_dust



**What it does:** Bag filter (baghouse) on a dusty gas: >99% capture + cake pressure drop

**The story:** The high-efficiency complement to cyclone01 (same dusty gas: N2 carrying 50 kg/h of silica, PSD 1..50  $\mu\text{m}$ ). Where the cyclone captured ~82 % by SIZE (fines escape), the fabric filter's dust cake captures ~99.9 % across all sizes — the price is a much larger pressure drop, dominated by the cake ( $dP = \mu V (K1 + K2 W)$ ).

Face velocity  $V = Q_{\text{gas}} / \text{filterArea}$  is the air-to-cloth ratio (~1.5 m/min here); with a cake load  $W = 0.5 \text{ kg/m}^2$  the pressure drop is ~2.7 kPa. The classic train is cyclone (cheap pre-cleaner) THEN bag filter (polish) — this case is the polish stage on the same dust.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, PSD 1..50  $\mu\text{m}$  Filter: 40  $\text{m}^2$  cloth

**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000231019341285 |
| stream | cleanGas       | F                   | 0.0416668205759   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | baghouse       | airToCloth_m_min    | 1.53857549963     |
| kpi    | baghouse       | arealDustLoad       | 0.5               |
| kpi    | baghouse       | dP_filter           | 2745.33935569     |
| kpi    | baghouse       | efficiency          | 0.999334225794    |
| kpi    | baghouse       | faceVelocity        | 0.0256429249938   |
| kpi    | baghouse       | filterArea          | 40                |
| kpi    | baghouse       | mu_gas              | 1.78433840676e-05 |
| kpi    | baghouse       | penetration         | 0.000665774205525 |
| kpi    | baghouse       | rho_gas             | 1.13794383208     |
| kpi    | baghouse       | solidsCaptured_mass | 0.0138796420244   |
| kpi    | baghouse       | solidsEscaped_mass  | 9.24686396528e-06 |

... and 4 more reference values in the case's *expected*.

**cyclone01\_dust\_removal**

tutorials/steady/gas-solid/cyclone01\_dust\_removal



**What it does:** Cyclone dust removal: silica from N2 (Lapple,  $D_c = 0.5$  m)

**The story:** A dusty gas (N2 carrying 50 kg/h of silica) passes through a cyclone. The cyclone separates BY SIZE: coarse particles spin to the wall and fall out (capturedSolids), fine particles follow the gas (cleanGas). This is the whole point of the PSD — a cyclone is a size classifier, not a simple splitter.

Lapple cut diameter  $d_{50} \sim 2.7$   $\mu\text{m}$  for this geometry/flow. The grade efficiency  $\eta(d) = 1/(1+(d_{50}/d)^2)$ :  $\sim 12\%$  at 1  $\mu\text{m}$ ,  $\sim 93\%$  at 10  $\mu\text{m}$ . Overall collection  $\sim 82\%$  of the dust mass; the escaping fines are the sub- $d_{50}$  tail (why a cyclone is a pre-cleaner, followed by a bag filter).

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica,  $\rho_p = 2650$  kg/m<sup>3</sup>, PSD 1..50  $\mu\text{m}$  Cyclone: body diameter 0.5 m, 5 effective turns

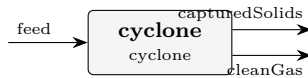
**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000189667114932 |
| stream | cleanGas       | F                   | 0.0417081728022   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | cyclone        | bodyDiameter        | 0.5               |
| kpi    | cyclone        | d50                 | 2.71094994999e-06 |
| kpi    | cyclone        | d50_micron          | 2.71094994999     |
| kpi    | cyclone        | efficiency          | 0.820454419117    |
| kpi    | cyclone        | inletVelocity       | 32.8229439921     |
| kpi    | cyclone        | mu_gas              | 1.78433840676e-05 |
| kpi    | cyclone        | numberOfTurns       | 5                 |
| kpi    | cyclone        | rho_gas             | 1.13794383208     |
| kpi    | cyclone        | rho_particle        | 2650              |
| kpi    | cyclone        | solidsCaptured_mass | 0.0113952002651   |
| kpi    | cyclone        | solidsEscaped_mass  | 0.00249368862328  |

...and 8 more reference values in the case's *expected*.

**cyclone02\_leith\_licht**

tutorials/steady/gas-solid/cyclone02\_leith\_licht



**What it does:** Cyclone dust removal – Leith-Licht back-mixing model

**The story:** The same dusty gas as cyclone01, but the grade efficiency uses the Leith-Licht back-mixing model instead of Lapple. Accounting for turbulent re-entrainment in the vortex, Leith-Licht predicts a **SMALLER** cut diameter (finer collection) for the same geometry/flow.

Leith-Licht cut diameter  $d_{50} \sim 1.11 \mu\text{m}$  here. The grade efficiency is  $\sim 48\%$  at  $1 \mu\text{m}$ ,  $\sim 94\%$  at  $10 \mu\text{m}$ . Overall collection  $\sim 88\%$  of the dust mass — higher than Lapple's  $82\%$  because more of the sub-micron tail is captured.

Gas:  $150 \text{ kmol/h N}_2$ ,  $300 \text{ K}$ ,  $1 \text{ atm}$  Dust:  $50 \text{ kg/h silica}$ ,  $\rho_p = 2650 \text{ kg/m}^3$ , PSD  $1..50 \mu\text{m}$  Cyclone: body diameter  $0.5 \text{ m}$ , 5 effective turns

**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000203443637145 |
| stream | cleanGas       | F                   | 0.04169439628     |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | cyclone        | bodyDiameter        | 0.5               |
| kpi    | cyclone        | d50                 | 1.10984825815e-06 |
| kpi    | cyclone        | d50_micron          | 1.10984825815     |
| kpi    | cyclone        | efficiency          | 0.880048347849    |
| kpi    | cyclone        | inletVelocity       | 32.8229439921     |
| kpi    | cyclone        | mu_gas              | 1.78433840676e-05 |
| kpi    | cyclone        | numberOfTurns       | 5                 |
| kpi    | cyclone        | rho_gas             | 1.13794383208     |
| kpi    | cyclone        | rho_particle        | 2650              |
| kpi    | cyclone        | solidsCaptured_mass | 0.0122228937197   |
| kpi    | cyclone        | solidsEscaped_mass  | 0.0016659951687   |

...and 8 more reference values in the case's *expected*.

**cyclone03\_iozia\_leith**

tutorials/steady/gas-solid/cyclone03\_iozia\_leith



**What it does:** Cyclone dust removal – Iozia-Leith data-fitted logistic (beta)

**The story:** The same dusty gas as cyclone01, now through the REAL Iozia-Leith (1989) FLOW-PATTERN model: the cut diameter  $d_{50}$  is NOT the Lapple residence-time surrogate but falls out of the gas flow pattern – the critical particle sits at the core radius where the tangential velocity peaks, and  $d_{50} = \sqrt{9 \mu Q / (\pi z_c \rho_p V_{t\_max}^2)}$  (their eq 4), with  $V_{t\_max}$ , the core diameter and its length regressed from the cyclone's OWN geometry (eqs 5-7). The grade curve stays logistic,  $\eta(d) = 1/(1+(d_{50}/d)^\beta)$ ,  $\beta = 3.5$ . (See the forum note: the 1989 eq-5 prefactor 6.1 is a decimal typo for 0.61 – 6.1 gives a supersonic  $V_{t\_max}$ , against the paper's own Figure 4; we use the Figure-4-consistent 0.61.)

Flow-pattern  $d_{50} \sim 1.0 \mu\text{m}$  – FINER than Lapple's  $2.71 \mu\text{m}$ : a Stairmand-shaped cyclone spins harder than the residence-time picture assumes, so it collects smaller particles. Overall collection RISES accordingly. Geometry defaults to the Stairmand high-efficiency ratios (add an operation.geometry {} block to override  $a/b/De/S/h/H/B$ ).

Gas: 150 kmol/h N<sub>2</sub>, 300 K, 1 atm Dust: 50 kg/h silica,  $\rho_p = 2650 \text{ kg/m}^3$ , PSD 1..50  $\mu\text{m}$  Cyclone: body diameter 0.5 m, 5 effective turns,  $\beta = 3.5$

**What to expect (golden-locked):**

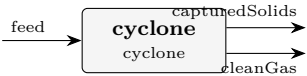
| kind   | unit/stream    | key           | expected          |
|--------|----------------|---------------|-------------------|
| stream | capturedSolids | F             | 0.000223552490081 |
| stream | cleanGas       | F             | 0.0416742874271   |
| stream | feed           | F             | 0.0418978399171   |
| kpi    | cyclone        | Q_gas         | 1.02571699975     |
| kpi    | cyclone        | bodyDiameter  | 0.5               |
| kpi    | cyclone        | d50           | 9.86636018962e-07 |
| kpi    | cyclone        | d50_micron    | 0.986636018962    |
| kpi    | cyclone        | dP_cyclone    | 4903.83536027     |
| kpi    | cyclone        | efficiency    | 0.96703441953     |
| kpi    | cyclone        | eulerNumber   | 8                 |
| kpi    | cyclone        | inletVelocity | 32.8229439921     |
| kpi    | cyclone        | loading       | 0.0118992372584   |
| kpi    | cyclone        | mu_gas        | 1.78433840676e-05 |
| kpi    | cyclone        | numberOfTurns | 5                 |

... and 8 more reference values in the case's *expected*.



cyclone04\_high\_loading

tutorials/steady/gas-solid/cyclone04\_high\_loading



**What it does:** Cyclone high dust loading – Muschelknautz mass-loading effect

**The story:** The same cyclone geometry as cyclone01, but with a HEAVY dust load — 2000 kg/h of silica (inlet loading ~ 0.476 kg dust / kg gas, ~40x the ~0.012 kg/kg of cyclone01). At high loading the Muschelknautz model captures the mass-loading effect: above a critical loading the dust forms a wall strand that sweeps out particles independently of the single-particle cut, so the collection efficiency RISES sharply.

Muschelknautz overall collection ~ 98.4 % of the dust mass — far above the ~82 % a dilute Lapple cyclone gives, because the high solids loading itself enhances separation. Grade efficiency ~ 90 % even at 1 um.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 2000 kg/h silica, ?\_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns, cLimit = 0.05

**What to expect (golden-locked):**

| kind                                                           | unit/stream    | key           | expected          |
|----------------------------------------------------------------|----------------|---------------|-------------------|
| stream                                                         | capturedSolids | F             | 0.00909835478473  |
| stream                                                         | cleanGas       | F             | 0.0418152419014   |
| stream                                                         | feed           | F             | 0.0509135966861   |
| kpi                                                            | cyclone        | bodyDiameter  | 0.5               |
| kpi                                                            | cyclone        | d50           | 1e-09             |
| kpi                                                            | cyclone        | d50_micron    | 0.001             |
| kpi                                                            | cyclone        | dP_cyclone    | 4903.83536027     |
| kpi                                                            | cyclone        | efficiency    | 0.983932479818    |
| kpi                                                            | cyclone        | eulerNumber   | 8                 |
| kpi                                                            | cyclone        | inletVelocity | 32.8229439921     |
| kpi                                                            | cyclone        | loading       | 0.475969490367    |
| kpi                                                            | cyclone        | mu_gas        | 1.78433840676e-05 |
| kpi                                                            | cyclone        | numberOfTurns | 5                 |
| kpi                                                            | cyclone        | rho_gas       | 1.13794383208     |
| ...and 8 more reference values in the case's <i>expected</i> . |                |               |                   |

**cyclone05\_dirgo\_leith\_validation**

tutorials/steady/gas-solid/cyclone05\_dirgo\_leith\_validation



**What it does:** VALIDATION vs Dirgo & Leith (1985): a Stairmand high-efficiency cyclone ( $D=0.305$  m,  $0.139$  m<sup>3</sup>/s,  $\sim 15$  m/s inlet, mineral-oil aerosol) run through the flow-pattern Iozia-Leith model. Swap the model slot to see the  $\sim 3\times$  d50 spread across correlations on ONE real cyclone – why you validate against data.

**The story:** VALIDATION against Dirgo & Leith (1985), Aerosol Sci. Technol. 4:401 – the measured fractional-efficiency baseline for a Stairmand high-efficiency cyclone. Their pilot-plant cyclone: body diameter  $D = 0.305$  m, design air flow  $0.139$  m<sup>3</sup>/s (300 cfm) → inlet velocity  $\sim 15$  m/s; test aerosol a nebulised mineral oil ( $\rho_p = 860$  kg/m<sup>3</sup>), particles  $\geq 1$   $\mu$ m.

This case reproduces THOSE conditions and runs the flow-pattern Iozia-Leith (1989) model. Re-run it with ‘model Lapple;’ / ‘LeithLicht;’ / ‘Barth;’ / ‘Muschelknautz;’ and compare each model’s d50 and grade curve against the paper’s measured fractional-efficiency curves (their Figures 5-6). THE LESSON: on this ONE real cyclone the models disagree by  $\sim 3\times$  – Iozia-Leith  $2.2$   $\mu$ m, Leith-Licht  $2.6$   $\mu$ m (the fine, high-efficiency end) Lapple / Muschelknautz  $6.2$   $\mu$ m, Barth  $6.5$   $\mu$ m (the coarse end) – which is EXACTLY why you validate against data rather than trust one correlation. Dirgo & Leith’s own finding: the theories BRACKET the measured curve and no single one is universally best (they consistently underpredicted efficiency at the highest inlet velocity,  $25$  m/s). The geometry defaults to the Stairmand high-efficiency ratios ( $a/D$   $0.5$ ,  $b/D$   $0.2$ ,  $D_e/D$   $0.5$ ,  $S/D$   $0.5$ ,  $h/D$   $1.5$ ,  $H/D$   $4$ ,  $B/D$   $0.375$ ) – exactly Dirgo & Leith’s cyclone – so the Iozia-Leith flow pattern ( $V_{t\_max}$ , core) is evaluated on the real test geometry;  $V_i = Q/(ab) \sim 15$  m/s reproduces their design point.

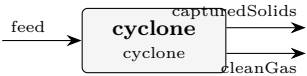
**What to expect (golden-locked):**

| kind   | unit/stream    | key           | expected          |
|--------|----------------|---------------|-------------------|
| stream | capturedSolids | F             | 6.95101380068e-05 |
| stream | cleanGas       | F             | 0.00566837296963  |
| stream | feed           | F             | 0.00573788310764  |
| kpi    | cyclone        | Q_gas         | 0.138813700633    |
| kpi    | cyclone        | bodyDiameter  | 0.305             |
| kpi    | cyclone        | d50           | 2.24296780859e-06 |
| kpi    | cyclone        | d50_micron    | 2.24296780859     |
| kpi    | cyclone        | dP_cyclone    | 648.673413872     |
| kpi    | cyclone        | efficiency    | 0.702163610002    |
| kpi    | cyclone        | eulerNumber   | 8                 |
| kpi    | cyclone        | inletVelocity | 11.9377544216     |
| kpi    | cyclone        | loading       | 0.00879253984757  |
| kpi    | cyclone        | mu_gas        | 1.78433840676e-05 |
| kpi    | cyclone        | numberOfTurns | 5                 |

...and 8 more reference values in the case’s expected.

cyclone06\_barth

tutorials/steady/gas-solid/cyclone06\_barth



**What it does:** WITNESS: the Barth (1956) equilibrium-orbit cyclone model on the Dirgo & Leith Stairmand geometry (same design point as cyclone05); self-recorded golden – exercises a registered model no case selected.

**The story:** WITNESS for the Barth (1956) cyclone model – the static-pressure / equilibrium-orbit correlation – on the SAME Stairmand high-efficiency geometry and design point as cyclone05\_dirgo\_leith\_validation ( $D = 0.305\text{ m}$ ,  $0.139\text{ m}^3/\text{s}$ ,  $V_i \sim 15\text{ m/s}$ , mineral-oil aerosol  $\rho_p = 860\text{ kg/m}^3$ ).

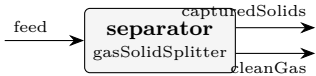
THIS IS A WITNESS, NOT A VALIDATION: its golden is self-recorded, so a PASS means Barth’s answer has not moved – never that it is right. The validation discussion (the  $\sim 3\times d_{50}$  spread across the five correlations, and Dirgo & Leith’s measured curves) lives in cyclone05’s header; this case exists because Barth was a registered model that NO case selected, which invariant I8 names as not-a-capability (2026-08-22 fresh-eyes audit). Until today its only exercise was an invitation in a comment.

**What to expect (golden-locked):**

| kind                                                   | unit/stream    | key           | expected          |
|--------------------------------------------------------|----------------|---------------|-------------------|
| stream                                                 | capturedSolids | F             | 2.14047701787e-05 |
| stream                                                 | cleanGas       | F             | 0.00571647833746  |
| stream                                                 | feed           | F             | 0.00573788310764  |
| kpi                                                    | cyclone        | Q_gas         | 0.138813700633    |
| kpi                                                    | cyclone        | bodyDiameter  | 0.305             |
| kpi                                                    | cyclone        | d50           | 6.49490874523e-06 |
| kpi                                                    | cyclone        | d50_micron    | 6.49490874523     |
| kpi                                                    | cyclone        | dP_cyclone    | 648.673413872     |
| kpi                                                    | cyclone        | efficiency    | 0.216222426411    |
| kpi                                                    | cyclone        | eulerNumber   | 8                 |
| kpi                                                    | cyclone        | inletVelocity | 11.9377544216     |
| kpi                                                    | cyclone        | loading       | 0.00879253984757  |
| kpi                                                    | cyclone        | mu_gas        | 1.78433840676e-05 |
| kpi                                                    | cyclone        | numberOfTurns | 5                 |
| ...and 8 more reference values in the case’s expected. |                |               |                   |

gasSolidSplitter01\_ideal

tutorials/steady/gas-solid/gasSolidSplitter01\_ideal



**What it does:** Ideal gas-solid splitter: specified 90% solids recovery (no efficiency physics)

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000208055925425 |
| stream | cleanGas       | F                   | 0.0416897839917   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | separator      | gasCarryover        | 0                 |
| kpi    | separator      | gasToCleanGas       | 0.0416666666667   |
| kpi    | separator      | gasToSolids         | 0                 |
| kpi    | separator      | solidsCaptured_mass | 0.0124999999995   |
| kpi    | separator      | solidsEscaped_mass  | 0.00138888888884  |
| kpi    | separator      | solidsIn_mass       | 0.0138888888884   |
| kpi    | separator      | solidsRecovery      | 0.9               |
| stream | capturedSolids | T                   | 300               |
| stream | cleanGas       | T                   | 300               |
| stream | feed           | T                   | 300               |

gasSolidSplitter02\_rosinRammler

tutorials/steady/gas-solid/gasSolidSplitter02\_rosinRammler



**What it does:** gas-solid split with a FITTED Rosin-Rammler feed PSD (the model is declared, the engine discretises)

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

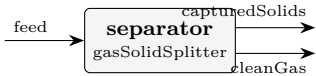
Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

What to expect (golden-locked):

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000208055925425 |
| stream | cleanGas       | F                   | 0.0416897839917   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | separator      | gasCarryover        | 0                 |
| kpi    | separator      | gasToCleanGas       | 0.0416666666667   |
| kpi    | separator      | gasToSolids         | 0                 |
| kpi    | separator      | solidsCaptured_mass | 0.0124999999995   |
| kpi    | separator      | solidsEscaped_mass  | 0.00138888888884  |
| kpi    | separator      | solidsIn_mass       | 0.0138888888884   |
| kpi    | separator      | solidsRecovery      | 0.9               |
| stream | capturedSolids | T                   | 300               |
| stream | cleanGas       | T                   | 300               |
| stream | feed           | T                   | 300               |

gasSolidSplitter03\_tabular\_psd

tutorials/steady/gas-solid/gasSolidSplitter03\_tabular\_psd



**What it does:** WITNESS: tabular (sieve-analysis) particle-size distribution on the gas-solid splitter; sibling of gasSolidSplitter02 (Rosin-Rammler).

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000208055925425 |
| stream | cleanGas       | F                   | 0.0416897839917   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | separator      | gasCarryover        | 0                 |
| kpi    | separator      | gasToCleanGas       | 0.0416666666667   |
| kpi    | separator      | gasToSolids         | 0                 |
| kpi    | separator      | solidsCaptured_mass | 0.0124999999995   |
| kpi    | separator      | solidsEscaped_mass  | 0.00138888888884  |
| kpi    | separator      | solidsIn_mass       | 0.0138888888884   |
| kpi    | separator      | solidsRecovery      | 0.9               |
| stream | capturedSolids | T                   | 300               |
| stream | cleanGas       | T                   | 300               |
| stream | feed           | T                   | 300               |

**gasSolidSplitter04\_lognormal\_psd**

tutorials/steady/gas-solid/gasSolidSplitter04\_lognormal\_psd



**What it does:** WITNESS: log-normal particle-size distribution (dgm, sigmaG) on the gas-solid splitter; sibling of gasSolidSplitter02.

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N<sub>2</sub> + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind   | unit/stream    | key                 | expected          |
|--------|----------------|---------------------|-------------------|
| stream | capturedSolids | F                   | 0.000208055925425 |
| stream | cleanGas       | F                   | 0.0416897839917   |
| stream | feed           | F                   | 0.0418978399171   |
| kpi    | separator      | gasCarryover        | 0                 |
| kpi    | separator      | gasToCleanGas       | 0.0416666666667   |
| kpi    | separator      | gasToSolids         | 0                 |
| kpi    | separator      | solidsCaptured_mass | 0.0124999999995   |
| kpi    | separator      | solidsEscaped_mass  | 0.00138888888884  |
| kpi    | separator      | solidsIn_mass       | 0.0138888888884   |
| kpi    | separator      | solidsRecovery      | 0.9               |
| stream | capturedSolids | T                   | 300               |
| stream | cleanGas       | T                   | 300               |
| stream | feed           | T                   | 300               |

**gibbs**

**The theory you need.** Equilibrium without a mechanism: minimise total Gibbs energy  $G = \sum_i n_i \mu_i$  over ALL species subject to elemental balances  $\mathbf{A} \mathbf{n} = \mathbf{b}$  (Lagrange multipliers per element – printed as  $\lambda_E$ , the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~100–250 K when they are not (dissociation is endothermic). Above ~1500 K the catalogue's polynomial  $c_p$  fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

*Equilibrium maps.* Sweeping the Gibbs solve over a  $T \times P$  grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes  $T$  and  $P$  in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach`  $\Delta T$  evaluates the *reaction* equilibrium at  $T + \Delta T$  while enthalpy and the energy balance stay at the physical  $T$  – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at  $P = 200$  bar and  $\Delta T = 0$ , find the temperature at which the equilibrium NH<sub>3</sub> mole fraction crosses 30 %; report  $T$  to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in  $T$  at fixed  $P$ . *Full derivation:* Theory Guide, the Gibbs-equilibrium chapter, *Sequilibrium maps*.

**gibbs01\_water\_gas\_shift**

tutorials/steady/gibbs/gibbs01\_water\_gas\_shift



**What it does:** Gibbs reactor: water-gas shift  $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$  at 800 K

**The story:** Water-gas shift reactor at 800 K, 1 bar.  $\text{CO} + \text{water} \rightleftharpoons \text{CO}_2 + \text{H}_2$  Equimolar feed. The Gibbs-minimisation reactor gives CO conversion ~ 67 % (CO 0.50  $\rightarrow$  0.164 mole fraction at equilibrium;  $\text{CO}_2 = \text{H}_2 = 0.336$ ), consistent with the independent  $K(800\text{ K})$  calculation in Smith, Van Ness, Abbott (Table 13.4).

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.000277777777778 |
| stream | products    | F            | 0.000277777777778 |
| kpi    | wgs         | F_in_kmol_h  | 1                 |
| kpi    | wgs         | F_out_kmol_h | 1                 |
| kpi    | wgs         | N_in_mol_s   | 0.277777777778    |
| kpi    | wgs         | N_out_mol_s  | 0.277777777778    |
| kpi    | wgs         | P            | 100000            |
| kpi    | wgs         | T            | 800               |
| kpi    | wgs         | converged    | 1                 |
| kpi    | wgs         | lambda_C     | 5549.05961227     |
| kpi    | wgs         | lambda_H     | -60095.3190791    |
| kpi    | wgs         | lambda_O     | -294758.913033    |
| kpi    | wgs         | y_CO         | 0.16409231685     |
| kpi    | wgs         | y_CO2        | 0.33590768315     |

... and 5 more reference values in the case's *expected*.



**gibbs02\_steam\_reforming**

tutorials/steady/gibbs/gibbs02\_steam\_reforming



**What it does:** Gibbs reactor: steam reforming of methane at 1000 K

**The story:** Industrial methane steam reforming feed: methane plus excess steam (steam-to-carbon ratio = 3, i.e. 1 mol CH<sub>4</sub> + 3 mol H<sub>2</sub>O per pass). Expected at 1000 K, 1 bar: near-complete CH<sub>4</sub> conversion, hydrogen yield ~ 2.7 mol H<sub>2</sub> per mol CH<sub>4</sub> (reforming + WGS combined).

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.001111111111111 |
| stream | syngas      | F            | 0.0016586091548   |
| kpi    | reformer    | F_in_kmol_h  | 4                 |
| kpi    | reformer    | F_out_kmol_h | 5.97099295728     |
| kpi    | reformer    | N_in_mol_s   | 1.11111111111     |
| kpi    | reformer    | N_out_mol_s  | 1.6586091548      |
| kpi    | reformer    | P            | 100000            |
| kpi    | reformer    | T            | 1000              |
| kpi    | reformer    | converged    | 1                 |
| kpi    | reformer    | lambda_C     | -33527.5812363    |
| kpi    | reformer    | lambda_H     | -75161.3018601    |
| kpi    | reformer    | lambda_O     | -309185.611918    |
| kpi    | reformer    | y_CH4        | 0.0024289965616   |
| kpi    | reformer    | y_CO         | 0.0979819433622   |

...and 6 more reference values in the case's *expected*.

**gibbs03\_methane\_combustion**

tutorials/steady/gibbs/gibbs03\_methane\_combustion



**What it does:** Gibbs reactor: methane combustion in stoichiometric air at 1800 K

**The story:** Stoichiometric combustion of methane in air:  $\text{CH}_4 + 2 (\text{O}_2 + 3.76 \text{ N}_2) \rightarrow \text{CO}_2 + 2 \text{ H}_2\text{O} + 7.52 \text{ N}_2$  (ideally) At 1800 K some CO and H<sub>2</sub> appear from partial dissociation.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.00292222222222  |
| stream | flueGas     | F            | 0.00292404303636  |
| kpi    | burner      | F_in_kmol_h  | 10.52             |
| kpi    | burner      | F_out_kmol_h | 10.5265549309     |
| kpi    | burner      | N_in_mol_s   | 2.92222222222     |
| kpi    | burner      | N_out_mol_s  | 2.92404303636     |
| kpi    | burner      | P            | 100000            |
| kpi    | burner      | T            | 1800              |
| kpi    | burner      | converged    | 1                 |
| kpi    | burner      | lambda_C     | -357484.903199    |
| kpi    | burner      | lambda_H     | -201140.892694    |
| kpi    | burner      | lambda_N     | -201371.845813    |
| kpi    | burner      | lambda_O     | -268835.220885    |
| kpi    | burner      | y_CH4        | 7.51542148893e-20 |

...and 9 more reference values in the case's *expected*.

gibbs04\_wgs\_temperature\_sweep

tutorials/steady/gibbs/gibbs04\_wgs\_temperature\_sweep



**What it does:** Water-gas shift equilibrium swept over T = 600..1200 K

**The story:** / flowsheetDict Water-gas shift equilibrium swept over T = 600..1200 K (outerDict sweep on the Gibbs reactor): CO conversion falls as T rises – exothermic equilibrium – reproducing the textbook K(T) curve point by point.

**What to expect (golden-locked):**

| kind                                                    | unit/stream | key          | expected           |
|---------------------------------------------------------|-------------|--------------|--------------------|
| stream                                                  | feed        | F            | 0.000277777777778  |
| stream                                                  | products    | F            | 0.0002777777777859 |
| kpi                                                     | wgs         | F_in_kmol_h  | 1                  |
| kpi                                                     | wgs         | F_out_kmol_h | 1.00000000029      |
| kpi                                                     | wgs         | N_in_mol_s   | 0.277777777778     |
| kpi                                                     | wgs         | N_out_mol_s  | 0.2777777777859    |
| kpi                                                     | wgs         | P            | 100000             |
| kpi                                                     | wgs         | Q_kW         | -2.0000513466      |
| kpi                                                     | wgs         | T            | 900                |
| kpi                                                     | wgs         | converged    | 1                  |
| kpi                                                     | wgs         | lambda_C     | -12610.3166837     |
| kpi                                                     | wgs         | lambda_H     | -69027.9700281     |
| kpi                                                     | wgs         | lambda_O     | -299577.53726      |
| kpi                                                     | wgs         | y_CO         | 0.199139146827     |
| ... and 5 more reference values in the case's expected. |             |              |                    |

**gibbs05\_adiabatic\_flame**

tutorials/steady/gibbs/gibbs05\_adiabatic\_flame



**What it does:** Adiabatic flame of lean CH<sub>4</sub> in air ( $\varphi = 0.5$ )

**The story:** Lean methane combustion at  $\varphi = 0.5$  (100 % excess air): 1 mol CH<sub>4</sub> + 4 mol O<sub>2</sub> + 15.04 mol N<sub>2</sub> Feed at ambient 298.15 K, 1 bar. Reactor finds its own adiabatic flame temperature.

**What to expect (golden-locked):**

| kind   | unit/stream | key             | expected          |
|--------|-------------|-----------------|-------------------|
| stream | feed        | F               | 0.005566666666667 |
| stream | flueGas     | F               | 0.00556667001791  |
| kpi    | burner      | F_in_kmol_h     | 20.04             |
| kpi    | burner      | F_out_kmol_h    | 20.0400120645     |
| kpi    | burner      | N_in_mol_s      | 5.56666666667     |
| kpi    | burner      | N_out_mol_s     | 5.56667001791     |
| kpi    | burner      | P               | 100000            |
| kpi    | burner      | T               | 1491.58454904     |
| kpi    | burner      | converged       | 1                 |
| kpi    | burner      | lambda_C        | -432004.832024    |
| kpi    | burner      | lambda_H        | -204860.137156    |
| kpi    | burner      | lambda_N        | -162961.775795    |
| kpi    | burner      | lambda_O        | -186456.945423    |
| kpi    | burner      | outerIterations | 3                 |

...and 9 more reference values in the case's *expected*.

**gibbs06\_h2\_flame\_radicals**

tutorials/steady/gibbs/gibbs06\_h2\_flame\_radicals



**What it does:** Stoichiometric H<sub>2</sub>/air adiabatic flame – the radical pool at equilibrium

**The story:** THE RADICAL DATABASE AT WORK: stoichiometric H<sub>2</sub>/air adiabatic flame, equilibrium by Gibbs minimisation over the FULL radical pool (H, O, OH, HO<sub>2</sub>, H<sub>2</sub>O<sub>2</sub> – GRI-Mech 3.0 / Burcat NASA-7 thermo) plus equilibrium thermal NO.

1 H<sub>2</sub> + 0.5 O<sub>2</sub> + 1.88 N<sub>2</sub> at 298.15 K, 1 atm, adiabatic.

THE ANSWER (golden-locked): T<sub>ad</sub> = 2385.8 K – the literature gives 2380-2390 K (Glassman, Combustion, 4th ed.; Law, Combustion Physics). Equilibrium pool: OH 0.55 %, H 0.18 %, O 0.06 %, NO 0.27 % – all inside the published equilibrium ranges. NO here is the Zeldovich CEILING (what kinetics approaches with residence time, never exceeds).

TWO LESSONS THIS CASE TEACHES: 1. THE RADICALS' WORK: endothermic dissociation (H<sub>2</sub>O → OH + H, O<sub>2</sub> → 2 O) soaks up enthalpy near 2400 K. Without the radical species the same reactor answers ~2640 K – the first-year textbook overshoot. 2. THE EXTRAPOLATION TRAP (why constant/components/ overrides exist here): the catalogue's poly3 Cp fits are 250-1500 K PROCESS fits; extrapolated to 2641 K, N<sub>2</sub>'s cubic turns NEGATIVE (-33 J/mol/K) and water's reads 45 instead of ~56. The case-local overrides switch the four stable species to the NASA-7 two-range form (GRI-Mech 3.0, valid 200-3500 K) – the same source as the radical entries. Run it once with the overrides removed to SEE the 250 K error a silent extrapolation buys.

**What to expect (golden-locked):**

| kind   | unit/stream | key             | expected          |
|--------|-------------|-----------------|-------------------|
| stream | burnt       | F               | 0.000808562123724 |
| stream | feed        | F               | 0.000938888888889 |
| kpi    | flame       | F_in_kmol_h     | 3.38              |
| kpi    | flame       | F_out_kmol_h    | 2.91082364541     |
| kpi    | flame       | N_in_mol_s      | 0.938888888889    |
| kpi    | flame       | N_out_mol_s     | 0.808562123724    |
| kpi    | flame       | P               | 101325            |
| kpi    | flame       | T               | 2385.83679448     |
| kpi    | flame       | converged       | 1                 |
| kpi    | flame       | lambda_H        | -240335.177161    |
| kpi    | flame       | lambda_N        | -277509.500244    |
| kpi    | flame       | lambda_O        | -344071.838893    |
| kpi    | flame       | outerIterations | 3                 |
| kpi    | flame       | y_H             | 0.00181958387904  |

...and 11 more reference values in the case's *expected*.

**gibbs06\_thermal\_nox**  
tutorials/steady/gibbs/gibbs06\_thermal\_nox



**What it does:** Methane combustion at 1800 K with thermal NOx formation

**The story:** / flowsheetDict Methane combustion at 1800 K with thermal NOx: the Gibbs reactor finds the full equilibrium including NO/NO2 – the EQUILIBRIUM ceiling on NOx, to contrast with the KINETIC path (nox01\_zeldovich\_postflame, where the rate, not equilibrium, controls).

**What to expect (golden-locked):**

| kind                                                     | unit/stream | key          | expected          |
|----------------------------------------------------------|-------------|--------------|-------------------|
| stream                                                   | feed        | F            | 0.00292222222222  |
| stream                                                   | flueGas     | F            | 0.00292416183331  |
| kpi                                                      | burner      | F_in_kmol_h  | 10.52             |
| kpi                                                      | burner      | F_out_kmol_h | 10.5269825999     |
| kpi                                                      | burner      | N_in_mol_s   | 2.92222222222     |
| kpi                                                      | burner      | N_out_mol_s  | 2.92416183331     |
| kpi                                                      | burner      | P            | 100000            |
| kpi                                                      | burner      | T            | 1800              |
| kpi                                                      | burner      | converged    | 1                 |
| kpi                                                      | burner      | lambda_C     | -355589.713611    |
| kpi                                                      | burner      | lambda_H     | -200666.346513    |
| kpi                                                      | burner      | lambda_N     | -201373.229482    |
| kpi                                                      | burner      | lambda_O     | -269787.239957    |
| kpi                                                      | burner      | y_CH4        | 9.68349314225e-20 |
| ... and 10 more reference values in the case's expected. |             |              |                   |

**gibbs07\_wgs\_cooled**  
tutorials/steady/gibbs/gibbs07\_wgs\_cooled



**What it does:** Cooled WGS Gibbs reactor (350 K): water condenses, shifting the equilibrium (multi-phase)

**The story:** multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <=> CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The ‘elementPotential’ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated ( $y_{\text{H}_2\text{O.P}} > P_{\text{sat}}(350\text{ K}) \sim 0.42\text{ bar}$ ), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas’ (the dry-ish product gas) + ‘condensate’ (liquid water).

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key                 | expected          |
|-----------------------------------------------------------------|-------------|---------------------|-------------------|
| stream                                                          | condensate  | F                   | 3.76244802832e-05 |
| stream                                                          | feed        | F                   | 0.000277777777778 |
| stream                                                          | gas         | F                   | 0.000240153299998 |
| kpi                                                             | wgs         | F_condensate_kmol_h | 0.135448129019    |
| kpi                                                             | wgs         | F_in_kmol_h         | 1                 |
| kpi                                                             | wgs         | F_out_kmol_h        | 0.864551879994    |
| kpi                                                             | wgs         | N_in_mol_s          | 0.277777777778    |
| kpi                                                             | wgs         | N_liq_mol_s         | 0.0376244802832   |
| kpi                                                             | wgs         | N_out_mol_s         | 0.277777780282    |
| kpi                                                             | wgs         | P                   | 100000            |
| kpi                                                             | wgs         | T                   | 350               |
| kpi                                                             | wgs         | converged           | 1                 |
| kpi                                                             | wgs         | lambda_C            | 50085.9480936     |
| kpi                                                             | wgs         | lambda_H            | -24736.1727481    |
| ...and 12 more reference values in the case’s <i>expected</i> . |             |                     |                   |

gibbs08\_wgs\_cooled\_reactiveflash

tutorials/steady/gibbs/gibbs08\_wgs\_cooled\_reactiveflash



**What it does:** Cooled WGS via reactiveFlash (B1) – cross-checks the elementPotential V+L result

**The story:** the gibbs07\_wgs\_cooled problem solved by ‘model reactiveFlash;’ (route B1: an equilibrium-reaction + isothermal-flash inner loop) instead of the default elementPotential (A). Cross-checks the two-phase V+L answer by an independent method.

The physics is gibbs07’s: multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <=> CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The ‘elementPotential’ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated ( $y_{\text{H}_2\text{O.P}} > P_{\text{sat}}(350\text{ K}) \sim 0.42\text{ bar}$ ), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas’ (the dry-ish product gas) + ‘condensate’ (liquid water).

**What to expect (golden-locked):**

| kind   | unit/stream | key                 | expected          |
|--------|-------------|---------------------|-------------------|
| stream | condensate  | F                   | 3.76244802827e-05 |
| stream | feed        | F                   | 0.000277777777778 |
| stream | gas         | F                   | 0.000240153299999 |
| kpi    | wgs         | F_condensate_kmol_h | 0.135448129018    |
| kpi    | wgs         | F_in_kmol_h         | 1                 |
| kpi    | wgs         | F_out_kmol_h        | 0.864551879995    |
| kpi    | wgs         | N_in_mol_s          | 0.277777777778    |
| kpi    | wgs         | N_liq_mol_s         | 0.0376244802827   |
| kpi    | wgs         | N_out_mol_s         | 0.277777780281    |
| kpi    | wgs         | P                   | 100000            |
| kpi    | wgs         | T                   | 350               |
| kpi    | wgs         | converged           | 1                 |
| kpi    | wgs         | lambda_C            | 50085.9480935     |
| kpi    | wgs         | lambda_H            | -24736.1727482    |

... and 12 more reference values in the case’s *expected*.





**gibbs10\_ammonia\_fugacity**

tutorials/steady/gibbs/gibbs10\_ammonia\_fugacity



**What it does:** Ammonia synthesis equilibrium at 200 bar with the REAL-gas fugacity correction:  $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$  at 700 K, 200 bar over an SRK equation of state. At this pressure the ideal-gas equilibrium (30.0 %  $\text{NH}_3$ ) is wrong – the Gibbs reactor now folds the SRK fugacity coefficients into the equilibrium ( $K_{\text{phi}} \sim 0.75$ , favouring the fewer-moles  $\text{NH}_3$  side), giving 32.4 %  $\text{NH}_3$ , toward the measured ~35-40 %. Switch the equationOfState to idealGas and watch  $\text{NH}_3$  drop back to 30.0 % – the fugacity correction is the whole point of a high-pressure reactor.

**The story:** the high-pressure equilibrium done RIGHT

$\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$  at 700 K and 200 bar, a stoichiometric feed. The Gibbs reactor minimises the REAL-gas Gibbs energy: it pulls the fugacity coefficients from the package EoS (SRK here) and adds  $\ln(\phi_i)$  to each species' chemical potential. At 200 bar this MATTERS – the fugacity ratio  $K_{\text{phi}} = \phi_{\text{NH}_3}^2 / (\phi_{\text{N}_2} \phi_{\text{H}_2}^3) \sim 0.75$  shifts the equilibrium toward  $\text{NH}_3$  (the fewer-moles side): 30.0 % (ideal)  $\rightarrow$  32.4 % (SRK), toward the measured ~35-40 %. Set the package EoS to idealGas and  $\text{NH}_3$  falls back to 30.0 %. (A cornerstone of the classic ammonia synthesis loop: the reactor is only quantitative at pressure once fugacity is in.)

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.000277777777778 |
| stream | products    | F            | 0.000209823048593 |
| kpi    | ammonia     | F_in_kmol_h  | 1                 |
| kpi    | ammonia     | F_out_kmol_h | 0.755362974933    |
| kpi    | ammonia     | N_in_mol_s   | 0.277777777778    |
| kpi    | ammonia     | N_out_mol_s  | 0.209823048593    |
| kpi    | ammonia     | P            | 20000000          |
| kpi    | ammonia     | Q_kW         | -3.68175302075    |
| kpi    | ammonia     | T            | 700               |
| kpi    | ammonia     | converged    | 1                 |
| kpi    | ammonia     | lambda_H     | -34945.3954982    |
| kpi    | ammonia     | lambda_N     | -59416.3950887    |
| kpi    | ammonia     | y_H2         | 0.507099864715    |
| kpi    | ammonia     | y_N2         | 0.169033288238    |

... and 3 more reference values in the case's *expected*.

**heat**

**The theory you need.** Heat exchange couples streams without mixing them:  $Q = U A \Delta T_{\text{lm}}$ , effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

*Exchanger DESIGN vs rating.* A shell-and-tube poses two inverse questions: RATE it (geometry given  $\rightarrow$  find  $U$ , area,  $\Delta P$ , duty – model geometry;) or DESIGN it (duty given  $\rightarrow$  size the geometry – model design;).  $U$  is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty`  $\rightarrow$  `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Heat-exchanger design.*

**condenser01\_film\_nusselt**

tutorials/steady/heat/condenser01\_film\_nusselt



**What it does:** Horizontal-tube steam condenser: Nusselt film HTC  $\rightarrow$  duty (geometry mode)

**The story:** A horizontal-tube steam condenser whose duty is NOT specified – it EMERGES from the physics of the condensing film, exactly as a real condenser's does.

Saturated steam at 1 bar enters a bundle of horizontal tubes cooled by cooling water. The 'phaseChanger' runs in 'model geometry;' (the seventh resolution): a Nusselt (1916) LAMINAR FILM coefficient  $h_{\text{cond}}$  is computed on the tube, the wall conducts, and the cooling-water film carries the heat away. Because the condensing-film coefficient itself depends on the driving  $dT_{\text{film}} = T_{\text{sat}} - T_{\text{wall}}$ , the unit solves a 1-D energy balance at the wall ( $q_{\text{cond}} == q_{\text{cool}}$ ) – the wall temperature is the unknown. Then the duty is  $Q = h_{\text{cond}} * A * dT_{\text{film}}$  and the outlet is the dome-flash at that duty.

steam (saturated vapour, 1 bar,  $vf = 1$ )  $\rightarrow$  condenser (model geometry: NusseltFilm + cooling-water film)  $\rightarrow$  condensate (two-phase,  $vf \sim 0.55$ ) +  $Q$  (RESULT)

PARTIAL CONDENSATION: with this tube area the duty  $Q \sim -507$  kW removes only part of the latent heat, so the outlet is NOT fully condensed — it leaves at the dome as a wet two-phase stream,  $vf \sim 0.55$  (about 45 % of the vapour condensed) at  $T_{\text{sat}}$ . Fully condensing it would need more area; here the area is fixed and the outlet quality is the result.

Method (glass-box):  $h_{\text{cond}}$  : Nusselt film, horizontal tube, coefficient  $0.729 h = 0.729 [k_l^3 \rho_l / (\rho_l - \rho_v) g h_{\text{fg}}' / (\mu_l D dT_{\text{film}})]^{1/4} h_{\text{fg}}' = h_{\text{fg}} + 0.68 cp_l dT_{\text{film}}$  (Rohsenow subcooling corr.) (Incropera & DeWitt, Fund. of Heat & Mass Transfer, Ex. 10.4) wall : carbon-steel cylindrical resistance  $r_o \ln(r_o/r_i)/k$  coolant: a cooling-water film coefficient  $h$  (Dittus-Boelter order) the three resistances + which CONTROLS are printed; the duty is cross-checked against  $U * A * dT$ .

All film/saturation properties are IAPWS-IF97 water. The case carries its own constant/components/water.dat (case-local copy). ASSUMPTION stated plainly: the coolant is treated as ISOTHERMAL at its inlet  $T$  (its  $\sim 10$  K heat-up is neglected, so  $dT = T_{\text{sat}} - T_{\text{in}}$  slightly OVERSTATES the true LMTD – the computed area errs a few % small). The utility allocation still reports the real cooling-water flow and duty.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected          |
|--------|-------------|-----------|-------------------|
| stream | condensate  | F         | 0.0277777777778   |
| stream | steam       | F         | 0.0277777777778   |
| kpi    | condenser   | F         | 0.0277777777778   |
| kpi    | condenser   | P         | 100000            |
| kpi    | condenser   | Q         | -506700.51677     |
| kpi    | condenser   | Q_UAdT_kW | 506.617311982     |
| kpi    | condenser   | Q_kW      | -506.70051677     |
| kpi    | condenser   | R_coolant | 0.00014880952381  |
| kpi    | condenser   | R_film    | 0.000349003587291 |
| kpi    | condenser   | R_wall    | 4.35883467862e-05 |
| kpi    | condenser   | Re_film   | 15.1166609728     |
| kpi    | condenser   | T_in      | 372.76            |
| kpi    | condenser   | T_out     | 372.755918611     |
| kpi    | condenser   | T_wall    | 325.855275434     |

... and 14 more reference values in the case's *expected*.

**coolingTower01\_merkel**

tutorials/steady/heat/coolingTower01\_merkel



**What it does:** Water cooling tower, Merkel enthalpy driving force – rating: the packing is given, the cold-water T is the result

**The story:** A counter-current evaporative cooling tower in RATING mode: the packing characteristic is declared ( $Me = KaV/L = 1.5$ ) and the cold-water temperature is the RESULT. Hot water at 45 C meets ambient air at 25 C ( $Y = 0.01286$  kg/kg dry, about 62 % RH on the N2 carrier),  $L/G \sim 0.99$ .

The whole construction lives on the Merkel diagram the unit publishes as its profile: air enthalpy  $h$  against water temperature  $T$ , the saturation curve  $h^*(T)$  above, the operating line  $h_1 + (L\,cpL / G)(T - T_{out})$  below, and  $Me$  the integral of  $cpL\,dT$  over the gap between them.

Method hypotheses (each announced at run time): Lewis = 1; the water loss is neglected INSIDE the integral but reported and honoured at the boundary (the coolWater outlet carries  $L_{in}$  minus the evaporation); the exit air is assumed saturated;  $cp$  at mean film temperatures.

What to look at in the output:  $T_{water\_out}$  against the wet-bulb floor ( $T_{wb} \sim 293$  K): the approach is the tower's real currency – a tower can never cool BELOW  $T_{wb}$ . merkelNumber\_chebyshev4 beside the converged integral: the CTI four-point hand method against Simpson on 200 intervals. The gap is the price of the hand method, visible, not hidden. evaporation\_kg\_s = makeup\_kg\_s: the water the cooling COSTS. A tower cools  $\sim 1$  % of  $L$  per  $\sim 7$  K of range by evaporating it.

**What to expect (golden-locked):**

| kind   | unit/stream | key                  | expected        |
|--------|-------------|----------------------|-----------------|
| stream | air         | F                    | 0.0184166666667 |
| stream | coolWater   | F                    | 0.027110090426  |
| stream | hotWater    | F                    | 0.0277777777778 |
| stream | humidAir    | F                    | 0.0190843540184 |
| kpi    | tower01     | L_over_G             | 0.98937581178   |
| kpi    | tower01     | Q_kW                 | 36.4315830585   |
| kpi    | tower01     | T_air_out            | 307.613817692   |
| kpi    | tower01     | T_water_out          | 300.778649138   |
| kpi    | tower01     | T_wb_in              | 293.021697982   |
| kpi    | tower01     | Y_in                 | 0.0128618855531 |
| kpi    | tower01     | Y_out                | 0.0366432593178 |
| kpi    | tower01     | approach_K           | 7.75695115581   |
| kpi    | tower01     | evaporation_kg_s     | 0.0120283876424 |
| kpi    | tower01     | evaporation_pct_of_L | 2.40367446642   |

...and 8 more reference values in the case's *expected*.

heatExchanger01\_water\_water

tutorials/steady/heat/heatExchanger01\_water\_water



**What it does:** Two-stream heat exchanger (eps-NTU): hot/cold water recovery

**The story:** Two-stream heat exchanger by the effectiveness-NTU method (the built-in ‘heatExchanger‘). A hot water stream is cooled by a cold water stream in a counter-current unit — the canonical heat-recovery problem.

Hardware (the credo): area + U. The outlet temperatures and the duty are RESULTS. "I need T\_hot\_out = X" would be a DesignSpec on \$area.

Hot : 100 kmol/h water at 360 K Cold : 120 kmol/h water at 300 K Hardware: A = 10 m<sup>2</sup>, U = 500 W/(m<sup>2</sup>.K), counter-current

Expected (eps-NTU, Cp\_water ~ 75 J/mol/K): C\_hot = 100/3.6 \* 75 ~ 2083 W/K (= C\_min) C\_cold = 120/3.6 \* 75 ~ 2500 W/K NTU = U.A/C\_min ~ 2.4, C\_r ~ 0.83, effectiveness ~ 0.75 Q ~ 93 kW T\_hot\_out ~ 315 K (cooled ~45 K) T\_cold\_out ~ 337 K (heated ~37 K) LMTD ~ 22 K, and U.A.LMTD ~ Q (cross-check) energy balance: hot stream loses what the cold stream gains.

**What to expect (golden-locked):**

| kind   | unit/stream | key        | expected         |
|--------|-------------|------------|------------------|
| stream | cold        | F          | 0.03333333333333 |
| stream | coldOut     | F          | 0.03333333333333 |
| stream | hot         | F          | 0.02777777777778 |
| stream | hotOut      | F          | 0.02777777777778 |
| kpi    | hx          | C_cold     | 2516.66666667    |
| kpi    | hx          | C_hot      | 2097.22222222    |
| kpi    | hx          | C_r        | 0.833333333333   |
| kpi    | hx          | LMTD       | 18.7584806629    |
| kpi    | hx          | NTU        | 2.38410596026    |
| kpi    | hx          | Q          | 93792.4033146    |
| kpi    | hx          | Q_kW       | 93.7924033146    |
| kpi    | hx          | T_cold_in  | 300              |
| kpi    | hx          | T_cold_out | 337.268504628    |
| kpi    | hx          | T_hot_in   | 360              |

...and 10 more reference values in the case's *expected*.

## heatExchanger02\_geometry\_U

tutorials/steady/heat/heatExchanger02\_geometry\_U



**What it does:** Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m<sup>2</sup>.K), Kern / Sinnott). Tube-side  $h_i$  via Gnielinski (Incropera & DeWitt 6th ed.); shell-side  $h_o$  via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

**The story:** A shell-and-tube heat exchanger whose overall coefficient U and area are COMPUTED from the bundle geometry plus per-side convective correlations (the 'model geometry;' path), instead of being specified. This is the glass-box answer to "where does U come from?".

Hot water (shell side) is cooled by cold water (tube side), clean service.

Method: tube-side  $h_i$ : Gnielinski  $Nu = (f/8)(Re-1000)Pr/[1+12.7 \sqrt{f/8}(Pr^{2/3}-1)]$  (Incropera & DeWitt, Fund. of Heat and Mass Transfer, 6th ed.) shell-side  $h_o$ : Kern method  $Nu = 0.36 Re_s^{0.55} Pr^{1/3}$  (Kern, Process Heat Transfer, 1950) on the bundle equivalent diameter  $D_e$  and crossflow mass flux  $G_s$ . U on the OUTSIDE area:  $1/U = 1/h_o + r_o \ln(r_o/r_i)/k_{wall} + r_o/(r_i h_i)$  area =  $\pi d_o L n_{tubes}$  PRESSURE DROP (Kern, both sides): tube  $f = 0.079 Re^{-0.25}$  (Blasius),  $dP = (4 f L n_p/d_i + 4 n_p)(\rho u^2/2)$  - friction + turnaround losses; shell  $f = \exp(0.576 - 0.19 \ln Re_s)$ ,  $dP = f G_s^2 D_s (N_b+1) / (2 \rho D_e)$  over the  $N_b$  baffles. (Kern, Process Heat Transfer, 1950.)

Properties are IF97 water ( $\rho$ ,  $\mu$ ,  $\lambda$ ) at each stream temperature.

Expected order of magnitude (clean water-water):  $U \sim 2000-3000$  W/(m<sup>2</sup>.K), the SHELL side controlling, both sides turbulent ( $Re \sim 2-4 \times 10^4$ ); tube-side  $dP \sim 9$  kPa, shell-side  $dP \sim 35$  kPa (shell > tube: baffled crossflow is more dissipative – the duty-vs-pumping trade-off you design around, NOT just the area). The exact numbers are pinned by –record – they are RESULTS.

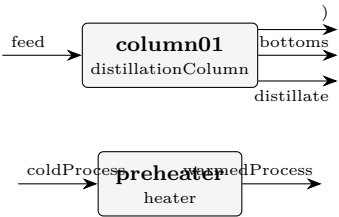
### What to expect (golden-locked):

| kind   | unit/stream | key      | expected       |
|--------|-------------|----------|----------------|
| stream | coldOut     | F        | 2.22027777778  |
| stream | coldTube    | F        | 2.22027777778  |
| stream | hotOut      | F        | 1.66527777778  |
| stream | hotShell    | F        | 1.66527777778  |
| kpi    | hx          | C_cold   | 167630.972222  |
| kpi    | hx          | C_hot    | 125728.472222  |
| kpi    | hx          | C_r      | 0.750031277368 |
| kpi    | hx          | LMTD     | 15.5339177262  |
| kpi    | hx          | NTU      | 1.26058061182  |
| kpi    | hx          | Nu_shell | 171.621364606  |
| kpi    | hx          | Nu_tube  | 287.732215939  |
| kpi    | hx          | Pr_shell | 3.16405821902  |
| kpi    | hx          | Pr_tube  | 5.87094963164  |
| kpi    | hx          | Q        | 2065604.42087  |

... and 27 more reference values in the case's *expected*.

heatlink01\_condenser\_to\_heater

tutorials/steady/heat/heatlink01\_condenser\_to\_heater



**What it does:** Forward heat-link: column condenser preheats a downstream stream

**The story:** feed column01 distillate bottoms (condenser heat) preheater warmedProcess coldProcess

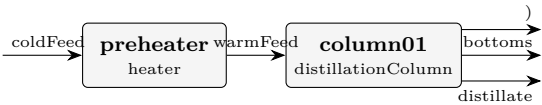
The preheater is heated by the column’s rejected condenser duty (~1.28 MW), warming a 1500 kmol/h benzene stream from 320 K by ~23 K. column01 is listed FIRST so it solves before the preheater reads its condenser KPI (forward heat-link).

**What to expect (golden-locked):**

| kind                                                    | unit/stream   | key            | expected        |
|---------------------------------------------------------|---------------|----------------|-----------------|
| stream                                                  | bottoms       | F              | 0.0138888888889 |
| stream                                                  | coldProcess   | F              | 0.416666666667  |
| stream                                                  | distillate    | F              | 0.0138888888889 |
| stream                                                  | feed          | F              | 0.0277777777778 |
| stream                                                  | warmedProcess | F              | 0.416666666667  |
| kpi                                                     | column01      | B              | 0.0138888888889 |
| kpi                                                     | column01      | D              | 0.0138888888889 |
| kpi                                                     | column01      | L_rect         | 0.0277777777778 |
| kpi                                                     | column01      | L_strip        | 0.0555555555556 |
| kpi                                                     | column01      | Q_condenser_kW | -1281.04348002  |
| kpi                                                     | column01      | Q_reboiler_kW  | 1279.30480005   |
| kpi                                                     | column01      | R              | 2               |
| kpi                                                     | column01      | T_bottom       | 382.893530951   |
| kpi                                                     | column01      | T_top          | 354.21089412    |
| ...and 26 more reference values in the case's expected. |               |                |                 |

heatlink02\_condenser\_feed\_preheat

tutorials/steady/heat/heatlink02\_condenser\_feed\_preheat



**What it does:** Feedback heat-link (energy tear): column condenser preheats its OWN feed

**The story:** FEEDBACK heat-link == an ENERGY TEAR.

coldFeed preheater column01 distillate bottoms (condenser heat)

The preheater (UPSTREAM of the column) is heated by the column’s CONDENSER duty (DOWN-STREAM) → circular. The Flowsheet detects the feedback, turns the recycle outer loop on, and converges the duty by successive substitution (Wegstein). Sized so the ~1 MW condenser preheats the large 1000 kmol/h feed by a modest ~25 K (stays liquid).

**What to expect (golden-locked):**

| kind                                                    | unit/stream | key            | expected        |
|---------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                  | bottoms     | F              | 0.263888888889  |
| stream                                                  | coldFeed    | F              | 0.277777777778  |
| stream                                                  | distillate  | F              | 0.0138888888889 |
| stream                                                  | warmFeed    | F              | 0.277777777778  |
| kpi                                                     | column01    | B              | 0.263888888889  |
| kpi                                                     | column01    | D              | 0.0138888888889 |
| kpi                                                     | column01    | L_rect         | 0.0208333333333 |
| kpi                                                     | column01    | L_strip        | 0.298611111111  |
| kpi                                                     | column01    | Q_condenser_kW | -1067.88365467  |
| kpi                                                     | column01    | Q_reboiler_kW  | 1846.42629755   |
| kpi                                                     | column01    | R              | 1.5             |
| kpi                                                     | column01    | T_bottom       | 365.99791143    |
| kpi                                                     | column01    | T_top          | 354.558892551   |
| kpi                                                     | column01    | V_rect         | 0.0347222222222 |
| ...and 24 more reference values in the case’s expected. |             |                |                 |



**hx01\_heater**

tutorials/steady/heat/hx01\_heater



**What it does:** Heater: 50/50 ethanol/water from 300 K to ~338 K (sensible)

**The story:** Stand-alone heater — credo-pure (v0.34+).

Hardware spec:  $Q = 100$  kW (the absolute thermal power delivered by the heater – think of it as the electrical rating).

Streams IN → OP (Q only) → Streams OUT

Feed is 100 kmol/h of 50/50 ethanol/water at 300 K. The simulator solves the enthalpy balance  $H_{\text{out}} = H_{\text{in}} + Q/F$  to compute  $T_{\text{out}}$  as a RESULT. With  $Q = 100$  kW the outlet temperature lands near 338 K ( $C_{p,\text{mix}} \sim 94$  J/(mol\*K) at this composition); the exact value falls out of the Newton-1D in T.

NOTE (duty reduced 160 → 100 kW). At 160 kW the outlet reached ~360 K, ABOVE the 50/50 ethanol/water bubble point at 1 atm – the old single-phase heater silently reported a physically-impossible "superheated liquid". The new saturation guard refuses that (a heater is sensible-only); a duty that boils belongs to a 'phaseChanger' / 'boiler' (see steady/heat/phasechange01\_partial\_condenser and steady/power/rankine02\_water). 100 kW keeps this the clean sensible-heat lesson it was meant to be.

If you instead need a specific  $T_{\text{out}}$  (say "I want exactly 338 K"), wrap the heater in a DesignSpec outerDict that manipulates \$Q with a target on 'heater01.T\_out'. The unit op itself never specifies its outlet state – that is the credo we adopted in v0.33.

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected         |
|---------|-------------|-------------------------|------------------|
| stream  | feed        | F                       | 0.02777777777778 |
| stream  | hotStream   | F                       | 0.02777777777778 |
| kpi     | heater01    | F                       | 0.02777777777778 |
| kpi     | heater01    | P                       | 101325           |
| kpi     | heater01    | Q                       | 100000           |
| kpi     | heater01    | Q_kW                    | 100              |
| kpi     | heater01    | Q_per_mol               | 3600             |
| kpi     | heater01    | T_in                    | 300              |
| kpi     | heater01    | T_out                   | 338.338658147    |
| kpi     | heater01    | dT                      | 38.338658147     |
| utility | heater01    | heating.steamLP.duty_kW | 100              |
| utility | heater01    | heating.steamLP.T       | 338.338658146    |
| utility | heater01    | heating.steamLP.kg_s    | 0.0457456541629  |
| utility | heater01    | heating.steamLP.eur_h   | 4.32             |

...and 3 more reference values in the case's *expected*.

## hxWorkflow1\_design\_from\_duty

tutorials/steady/heat/hxWorkflow1\_design\_from\_duty



**What it does:** PART 1 of 2 – DESIGN: the DUTY TARGET is given (cool the hot shell 330→315 K) and the exchanger GEOMETRY is SOLVED for (model design;). Glass-box Kern sizing: fix the tube choices, then find the tube count (and the shell diameter it implies, Kern/Sinnott bundle) so  $U \cdot A = Q_{\text{req}}/\text{LMTD}$ ;  $U/dP$  then rated with Gnielinski (tube) + Kern (shell). The detailed-design (sizing) step, glass-box. Part 2 (hxWorkflow2\_rate\_designed) rates the bundle this case produces.

**The story:** PART 1 of 2: DESIGN from the duty

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 1. DESIGN mode ('model design;') – detailed sizing, glass-box. Instead of GIVING the tube count and shell diameter (that is 'model geometry;'), the RATING problem: geometry  $\rightarrow U$ , here we give the DUTY TARGET (cool the hot shell stream from 330 K to 315 K) and the engine SOLVES for the geometry: fix the tube choices (OD/ID/length/pitch/passes/material), then find the number of tubes – and the shell diameter it implies (Kern/Sinnott bundle correlation, the chosen pattern + passes) so the exchanger DELIVERS THE DUTY – the design loop sizes  $N$  by the eps-NTU duty (incl. the 1-2 penalty for a 2-pass U-tube), not merely  $U \cdot A = Q/\text{LMTD}$ , so the outlet actually reaches the target. It then rates that designed bundle exactly like a given one (same Gnielinski tube + Kern shell correlations) and reports  $U$ , area, and the pressure drop.

DO I HAVE TO SPECIFY EVERYTHING? No – and this is the whole point. A HX poses two OPPOSITE questions, and this is the DESIGN one:

RATING ('model geometry;'): you GIVE the full hardware (tube count, shell diameter, ...) and ask "what outlet T do I get?". The temperature is a RESULT. DESIGN ('model design;'), THIS case): you GIVE the outlet T you NEED and ask "what hardware delivers it?". The GEOMETRY is the result.

So HOW does the outlet reach the temperature you need? You do NOT tune the geometry by hand until it works. You DECLARE the target and the engine sizes the hardware to hit it. Look at the two blocks below:

```

'design { spec { stream hotShell; T 315 K; } }' <- WHAT YOU WANT: the required outlet temperature.
THIS is where the final temperature is set. 'geometry { tubeID ...; passes 2; tubePattern ...; wallMaterial ...; }' <- the FIXED CHOICES a designer commits to (a standard tube, the number of passes, the layout, the metal). Notice what is MISSING: there is NO 'nTubes' and NO 'shellID'. Those are exactly what the engine SOLVES.

```

So the student specifies the REQUIREMENT (the outlet T) + the design CHOICES (tube size, passes, pattern, material) – NOT the answer. The engine returns the tube count, the shell diameter,  $U$ , the pressure drop, and confirms the outlet lands on the target. You said "cool this to 315 K with 1/2-inch steel tubes in a 2-pass U-tube"; it answers "then you need 105 tubes in a 0.31 m shell – here is  $U$ , the  $dP$ , and yes it leaves at 315 K".

This is the workflow that matches industrial practice: establish the DUTY, THEN design the exchanger from it – you do NOT invent the geometry up front. (Standard practice splits this into a shortcut that gets the duty + a detailed-design step that gets the geometry; Choupo's design mode folds both into one honest step – it computes the duty from your target and ANNOUNCES it, then sizes.) PART 2 (hxWorkflow2\_rate\_designed) then RATES the bundle this case produced – a detailed-rating step – to verify it and probe off-design (turn the coolant down and watch the outlet drift off the target; the geometry is now FIXED, only the performance moves).

Water-water, clean service, 2-pass U-tube (the general arrangement). Expect: ~100 tubes, a ~0.3 m shell; rated  $U \sim 4000 \text{ W}/(\text{m}^2 \cdot \text{K})$  (2 tube passes  $\rightarrow$  high tube velocity), and the delivered hot outlet 315 K (the target, reproduced by the 1-2 eps-NTU on the designed geometry). Numbers are RESULTS pinned by –record. Single-phase; the 1-shell/2-tube-pass eps-NTU carries the built-in LMTD F-correction ( $F < 1$ ), so a 2-pass needs more area than pure counter-current.

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected       |
|--------|-------------|----------|----------------|
| stream | coldOut     | F        | 2.22027777778  |
| stream | coldTube    | F        | 2.22027777778  |
| stream | hotOut      | F        | 1.66527777778  |
| stream | hotShell    | F        | 1.66527777778  |
| kpi    | hx          | C_cold   | 167630.972222  |
| kpi    | hx          | C_hot    | 125728.472222  |
| kpi    | hx          | C_r      | 0.750031277368 |
| kpi    | hx          | LMTD     | 16.7868881533  |
| kpi    | hx          | NTU      | 1.0053658663   |
| kpi    | hx          | Nu_shell | 195.331265319  |
| kpi    | hx          | Nu_tube  | 416.603846461  |
| kpi    | hx          | Pr_shell | 3.16405821902  |
| kpi    | hx          | Pr_tube  | 5.87094963164  |
| kpi    | hx          | Q        | 1888506.69951  |

...and 27 more reference values in the case's *expected*.

**hxWorkflow2\_rate\_designed**

tutorials/steady/heat/hxWorkflow2\_rate\_designed



**What it does:** Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band ( $\sim 2000\text{--}3000\text{ W/(m}^2\text{K)}$ ), Kern / Sinnott). Tube-side  $h_i$  via Gnielinski (Incropera & DeWitt 6th ed.); shell-side  $h_o$  via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

**The story:** PART 2 of 2: RATE the designed exchanger

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 2.

Part 1 (hxWorkflow1\_design\_from\_duty) gave the engine a DUTY TARGET (cool the hot shell  $330 \rightarrow 315\text{ K}$ ) and it SIZED the bundle: 105 tubes, shell ID 0.306 m (Kern/Sinnott, 2-pass U-tube). That is the detailed-design step.

Part 2 – THIS case – is the detailed-rating / checking step: take the DESIGNED geometry as a FIXED given ('model geometry;') and RATE it – confirm it delivers the target outlet, and read the honest pressure drop. This is the verification a real project always does after sizing: the design loop used a bundle correlation for the shell diameter and rounded the tube count UP, so the rated exchanger is slightly OVER-surfaced – the outlet comes out a touch below 315 K (good: margin), and that is what you check here.

OFF-DESIGN (try it): drop the cold-tube flow F to  $\sim 6000\text{ kmol/h}$  and re-run – less coolant  $\rightarrow$  the hot stream leaves WARMER than 315 K, and the shell dP falls. That is the rating question ("will my exchanger cope when the plant turns down?") the design step cannot answer. The geometry is now FIXED; only the performance moves.

Numbers are RESULTS pinned by –record. Same clean water-water service, same Gnielinski (tube) + Kern (shell) correlations, single-phase, 2-pass U-tube (1-2 eps-NTU).

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected       |
|--------|-------------|----------|----------------|
| stream | coldOut     | F        | 2.22027777778  |
| stream | coldTube    | F        | 2.22027777778  |
| stream | hotOut      | F        | 1.66527777778  |
| stream | hotShell    | F        | 1.66527777778  |
| kpi    | hx          | C_cold   | 167630.972222  |
| kpi    | hx          | C_hot    | 125728.472222  |
| kpi    | hx          | C_r      | 0.750031277368 |
| kpi    | hx          | LMTD     | 16.7876807682  |
| kpi    | hx          | NTU      | 1.00522927337  |
| kpi    | hx          | Nu_shell | 195.283521062  |
| kpi    | hx          | Nu_tube  | 416.603846461  |
| kpi    | hx          | Pr_shell | 3.16405821902  |
| kpi    | hx          | Pr_tube  | 5.87094963164  |
| kpi    | hx          | Q        | 1888394.47754  |

... and 27 more reference values in the case's *expected*.

**mheatx01\_two\_hot\_one\_cold**

tutorials/steady/heat/mheatx01\_two\_hot\_one\_cold



**What it does:** Multi-stream heat exchanger: 2 hot + 1 cold, balance closes

**The story:** Multi-stream heat exchanger (the commercial multi-stream detailed-rating analogue): TWO hot water streams are cooled by ONE cold water stream inside a single adiabatic shell — the canonical multi-stream heat-recovery problem.

THE SPEC (glass-box, "information follows the streams"): every inlet carries its OWN outlet target temperature. The unit does NOT rate hardware; it VERIFIES the first law:

$Q_k = F_k (h_{k,out} - h_{k,in})$  per stream  $Q_{hot} = \sum Q_k$  over the streams that cool ( $< 0$ , released)  
 $Q_{cold} = \sum Q_k$  over the streams that heat ( $> 0$ , absorbed) imbalance =  $\sum Q_k$  over ALL streams (external duty)

For a truly adiabatic shell the imbalance is ~0; a non-zero value is the external duty the topology would have to supply, REPORTED honestly (never a silently-invented duty). The unit also builds the hot/cold composite curves and reports the INTERNAL minimum-approach temperature (the in-unit pinch).

Streams (water, liquid): hot1 : 100 kmol/h, 380 → 330 K (gives up ~105 kW) hot2 : 60 kmol/h, 360 → 330 K (gives up ~38 kW) cold : 340 kmol/h, 300 → 320 K (takes up ~143 kW)

The cold flow is sized so the bundle is adiabatic: cold absorbs what the two hot streams release. Both hot outlets (330 K) stay above the cold outlet (320 K), so the composite curves never cross —  $dT_{min} > 0$ .

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key           | expected           |
|-----------------------------------------------------------------|-------------|---------------|--------------------|
| stream                                                          | cold        | F             | 0.0944444444444444 |
| stream                                                          | coldOut     | F             | 0.0944444444444444 |
| stream                                                          | hot1        | F             | 0.0277777777777778 |
| stream                                                          | hot1Out     | F             | 0.0277777777777778 |
| stream                                                          | hot2        | F             | 0.0166666666666667 |
| stream                                                          | hot2Out     | F             | 0.0166666666666667 |
| kpi                                                             | mhx         | Q_cold        | 142611.111111      |
| kpi                                                             | mhx         | Q_cold_kW     | 142.611111111      |
| kpi                                                             | mhx         | Q_hot         | -142611.111111     |
| kpi                                                             | mhx         | Q_hot1_kW     | -104.861111111     |
| kpi                                                             | mhx         | Q_hot2_kW     | -37.75             |
| kpi                                                             | mhx         | Q_hot_kW      | -142.611111111     |
| kpi                                                             | mhx         | balanceCloses | 1                  |
| kpi                                                             | mhx         | dT_min        | 30                 |
| ...and 10 more reference values in the case's <i>expected</i> . |             |               |                    |

**phasechange01\_partial\_condenser**

tutorials/steady/heat/phasechange01\_partial\_condenser



**What it does:** Partial condenser: benzene/toluene vapour cooled to quality 0.4

**The story:** A superheated benzene/toluene vapour enters a condenser that is asked to land its outlet at a vapour QUALITY of 0.4 (40 % of the moles stay vapour, 60 % condense). The duty Q is the RESULT – the unit solves it as a built-in self-target (no outerDict, no variables block) and ANNOUNCES that it did so. Because the mixture is multicomponent, the condensed liquid is toluene-RICH and the remaining vapour benzene-RICH – the partial- condensation lesson: a partial condenser is also a SEPARATION.

feedVapour (superheated, 1 bar, 390 K,  $vf = 1$ ) → condenser (outletQuality 0.4) → wetOut (two-phase,  $0 < vf < 1$ )

Same outlet vocabulary as rankine02\_water's IF97 boiler/condenser, on the GENERIC (Antoine + Raoult) backend.

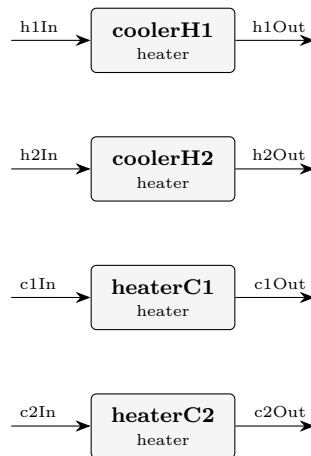
**What to expect (golden-locked):**

| kind    | unit/stream | key                          | expected        |
|---------|-------------|------------------------------|-----------------|
| kpi     | condenser   | vf_out                       | 0.399999998746  |
| kpi     | condenser   | T_out                        | 367.494289759   |
| kpi     | condenser   | Q_kW                         | -630.508868406  |
| kpi     | condenser   | Tsat                         | 352.826063711   |
| stream  | wetOut      | F                            | 0.0277777777778 |
| stream  | wetOut      | vf                           | 0.399999998746  |
| utility | condenser   | cooling.coolingWater.duty_kW | -630.508868406  |
| utility | condenser   | cooling.coolingWater.T       | 367.494289759   |
| utility | condenser   | cooling.coolingWater.kg_s    | 15.0839442202   |
| utility | condenser   | cooling.coolingWater.eur_h   | 0.907932770504  |
| stream  | feedVapour  | F                            | 0.0277777777778 |
| stream  | feedVapour  | T                            | 390             |
| stream  | wetOut      | T                            | 367.494289759   |
| kpi     | condenser   | F                            | 0.0277777777778 |

... and 6 more reference values in the case's *expected*.

**pinch01\_four\_stream\_classic**

tutorials/steady/heat/pinch01\_four\_stream\_classic



**What it does:** Pinch P1 targets: classic 4-stream problem table (dTmin 20 K)

**The story:** Four independent duty units realise the classic four-stream pinch population. The working fluid (hxFluid, case-local SYNTHETIC teaching fluid) has a CONSTANT liquid cp = 100 J/(mol K), so each stream's CP = F\*cp is exact and the targets are hand-checkable:

stream F [mol/s] CP [kW/K] T\_in [C] T\_out [C] Q [kW] H1 hot 20 2.0 150 60 -180.0 H2 hot 80 8.0 90 60 -240.0 C1 cold 25 2.5 20 125 +262.5 C2 cold 30 3.0 25 100 +225.0

Problem table at dTmin = 20 K (shift hot -10 K, cold +10 K; shifted boundaries in C, descending: 140 135 110 80 50 35 30; a positive dH is a heat SURPLUS cascading downward):

interval T\* CP\_hot CP\_cold dH [kW] cascade [kW] 140 → 135 2 0 +10.0 +10.0 135 → 110 2 2.5 -12.5 -2.5 110 → 80 2 5.5 -105.0 -107.5 <- minimum 80 → 50 10 5.5 +135.0 +27.5 50 → 35 0 5.5 -82.5 -55.0 35 → 30 0 2.5 -12.5 -67.5

Targets: Q\_H,min = -min(cascade) = 107.5 kW, Q\_C,min = final + Q\_H,min = -67.5 + 107.5 = 40 kW, pinch at T\* = 80 C → hot 90 C / cold 70 C.

Each unit is a 'heater' with a SIGNED hardware duty Q; the outlet T is the enthalpy-balance RESULT (with cp constant it lands exactly on the table's target temperatures). The four units are thermally independent here – the pinchPass only TARGETS and ANALYSES; the network is never rewritten.

P2 candidate-match table (hand-worked; pinch hot 90 C / cold 70 C, segments clipped at the pinch; Q\_max = min(Q\_hot, Q\_cold, min(CP\_h, CP\_c)\*(Th\_hi - Tc\_lo - dTmin))):

region hot cold CP rule at the pinch Q\_max [kW] limited by above H1 C1 2.0 ≤ 2.5 holds 120 approach (and H1's 120 kW) above H1 C2 2.0 ≤ 3.0 holds 90 C2's above-duty below H1 C1 2.0 ≥ 2.5 VIOLATED 60 H1's below-duty (away from pinch only) below H1 C2 2.0 ≥ 3.0 VIOLATED 60 H1's below-duty (away from pinch only) below H2 C1 8.0 ≥ 2.5 holds 125 approach == C1's below-duty below H2 C2 8.0 ≥ 3.0 holds 135 approach == C2's below-duty

Violation diagnostics (each unit is utility-served here, so every duty on the wrong side of the pinch is a coursework violation): cooler above the pinch: H1's 120 kW above 90 C; heaters below the pinch: C1's 125 kW + C2's 135 kW below 70 C. Identity check: 120 + 260 = 380 kW = Q\_heat\_current - Q\_H\_min = 487.5 - 107.5 – the excess over target IS the sum of the violations.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected |
|--------|-------------|-----------|----------|
| stream | c1In        | F         | 0.025    |
| stream | c1Out       | F         | 0.025    |
| stream | c2In        | F         | 0.03     |
| stream | c2Out       | F         | 0.03     |
| stream | h1In        | F         | 0.02     |
| stream | h1Out       | F         | 0.02     |
| stream | h2In        | F         | 0.08     |
| stream | h2Out       | F         | 0.08     |
| kpi    | coolerH1    | F         | 0.02     |
| kpi    | coolerH1    | P         | 101325   |
| kpi    | coolerH1    | Q         | -180000  |
| kpi    | coolerH1    | Q_kW      | -180     |
| kpi    | coolerH1    | Q_per_mol | -9000    |
| kpi    | coolerH1    | T_in      | 423.15   |

...and 66 more reference values in the case's *expected*.



**reboiler\_water\_copper**

tutorials/steady/heat/reboiler\_water\_copper



**What it does:** Steam-heated copper-surface pool reboiler: Rohsenow nucleate flux → duty, Zuber CHF ceiling (geometry mode)

**The story:** A steam-heated copper-surface POOL REBOILER whose duty is NOT specified – it EMERGES from the physics of NUCLEATE BOILING, exactly as a real reboiler’s does.

Saturated-liquid water at 1 bar sits in a pool boiling on a bundle of horizontal copper tubes heated INSIDE by condensing steam. The ‘phaseChanger’ runs in ‘model geometry;’ with a ‘boiling { ... }’ block (the boiling resolution): a Rohsenow (1952) NUCLEATE-boiling flux  $q''$  is computed on the tube surface, the copper wall conducts, and the heating-medium film delivers the heat. Because the nucleate flux itself depends on the wall superheat  $dT_{\text{excess}} = T_{\text{surface}} - T_{\text{sat}}$  (and on its CUBE), the unit solves a 1-D energy balance at the surface ( $q_{\text{boil}} == q_{\text{heat}}$ ) – the surface temperature is the unknown. Then the duty is  $Q = q'' * A$  and the outlet is the dome-flash at that duty.

water (saturated liquid, 1 bar,  $vf = 0$ ) → reboiler (model geometry: Rohsenow nucleate flux + steam film) → wet vapour (partially boiled) +  $Q$  (RESULT)

HONESTY-AS-STRUCTURE (the binding constraint): the nucleate flux is +/-100% uncertain –  $C_{\text{sf}}$  is surface-FINISH lab data (polish, oxidation), NOT a fluid property. There is NO default  $C_{\text{sf}}$  and NO catalogue in v1: it is STUDENT-SUPPLIED here (water-copper 0.013, polished) WITH a mandatory citation. the result carries the  $[q/2, 2q]$  scatter band ( $C_{\text{sf}}$  +/-25%, cubed). the RELIABLE figure is the Zuber (1959) CHF ceiling, printed FIRST. A design above CHF is HARD-REFUSED (burnout) – a pool boiler cannot operate above the critical heat flux.

Method (glass-box):  $q_{\text{boil}}$  : Rohsenow nucleate flux, water-copper,  $C_{\text{sf}}$  0.013,  $s$  1.0  $q'' = \mu_l h_{\text{fg}} \sqrt{g(\rho_l - \rho_v)/\sigma} [cp_l dT_{\text{excess}} / (C_{\text{sf}} h_{\text{fg}} Pr_l^s)]^3$  (Rohsenow 1952; Incropera & DeWitt, Ch.10 Table 10.1)  $q_{\text{CHF}}$  : Zuber 1959,  $q''_{\text{max}} = 0.149 h_{\text{fg}} \rho_v^{0.5} [\sigma g (\rho_l - \rho_v)]^{0.25}$  (~1.26 MW/m<sup>2</sup> for water at 1 atm) wall : copper cylindrical resistance  $r_o \ln(r_o/r_i)/k$  medium : a condensing-steam film coefficient  $h$  (high) the CHF margin SAFE/BURNOUT + the cube sensitivity are printed.

All film/saturation properties are IAPWS-IF97 water (incl.  $\sigma$  via R1-76). The case carries its own constant/components/water.dat (case-local copy).

**What to expect (golden-locked):**

| kind   | unit/stream | key        | expected         |
|--------|-------------|------------|------------------|
| stream | feed        | F          | 0.02777777777778 |
| stream | wetVapour   | F          | 0.02777777777778 |
| kpi    | reboiler    | F          | 0.02777777777778 |
| kpi    | reboiler    | P          | 100000           |
| kpi    | reboiler    | Q          | 111804.644945    |
| kpi    | reboiler    | Q_kW       | 111.804644945    |
| kpi    | reboiler    | T_in       | 372              |
| kpi    | reboiler    | T_out      | 372.755918611    |
| kpi    | reboiler    | T_surface  | 381.390508143    |
| kpi    | reboiler    | Tsat       | 372.755918611    |
| kpi    | reboiler    | area       | 1.25663706144    |
| kpi    | reboiler    | chf_margin | 0.0709724846287  |
| kpi    | reboiler    | dT_excess  | 8.6345895313     |
| kpi    | reboiler    | q_CHF      | 1253960.82217    |

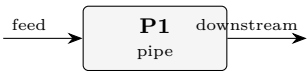
... and 10 more reference values in the case’s *expected*.

## hydraulics

**The theory you need.** Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves' intersection.

pipe01\_water\_line

tutorials/steady/hydraulics/pipe01\_water\_line



**What it does:** Pipe pressure drop: 2000 kmol/h water in a 0.1 m x 50 m commercial-steel line (Churchill friction)

**The story:** A single PIPE element. Water enters at 5 bar and flows through 50 m of 0.1 m commercial-steel pipe with two elbows, a globe valve, and a 2 m rise. The pressure DROP is a COMPUTED RESULT of the geometry and the flow — the pipe has NO pressure knob (information follows the streams: a pipe only dissipates, a pump would raise P, a valve would set a let-down P).

feed (5 bar liquid, 2000 kmol/h) → [P1 pipe] D, L, roughness, dz, fittings → ΔP (result) → downstream (P reduced by the computed ΔP, T held)

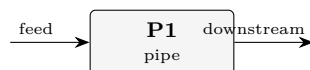
Friction factor: Churchill (default) — one explicit correlation across laminar / transition / turbulent. Re ~ 1.5e5 here, so the flow is turbulent and all three ΔP terms (friction, fittings, elevation) matter.

**What to expect (golden-locked):**

| kind                                                           | unit/stream | key            | expected        |
|----------------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                         | downstream  | F              | 0.555555555556  |
| stream                                                         | feed        | F              | 0.555555555556  |
| kpi                                                            | P1          | F              | 0.555555555556  |
| kpi                                                            | P1          | P_in           | 500000          |
| kpi                                                            | P1          | P_out          | 462986.302076   |
| kpi                                                            | P1          | dP_elevation   | 19520.1545706   |
| kpi                                                            | P1          | dP_fittings    | 9626.38419279   |
| kpi                                                            | P1          | dP_friction    | 7867.15916089   |
| kpi                                                            | P1          | deltaP         | 37013.6979243   |
| kpi                                                            | P1          | density        | 995.250904772   |
| kpi                                                            | P1          | frictionFactor | 0.0192870918591 |
| kpi                                                            | P1          | head_loss_m    | 1.79235705234   |
| kpi                                                            | P1          | regime         | 2               |
| kpi                                                            | P1          | reynolds       | 149291.31749    |
| ...and 4 more reference values in the case's <i>expected</i> . |             |                |                 |

**pipe02\_airwater\_twophase**

tutorials/steady/hydraulics/pipe02\_airwater\_twophase



**What it does:** Gas-liquid two-phase pressure drop: air (N<sub>2</sub>) + water in a horizontal 0.05 m x 10 m line, Lockhart-Martinelli. The two-phase path activates automatically because the feed flashes two-phase ( $0 < V/F < 1$ ).

**The story:** GAS-LIQUID two-phase pressure drop

feed (air N<sub>2</sub> + water, two-phase at 2 bar / 25 °C) → [P1 pipe] D, L, roughness, horizontal → ΔP (result)  
→ downstream (P reduced by the two-phase ΔP)

The two-phase path activates AUTOMATICALLY: the feed flashes two-phase at (T, P) (N<sub>2</sub> is supercritical → all vapour; water → mostly liquid), so  $0 < V/F < 1$  and the pipe routes to the gas-liquid models instead of the single-phase liquid path. Model: Lockhart-Martinelli (Chisholm multiplier + Butterworth holdup). Try 'model homogeneous;' in twoPhase to compare.

The Lockhart-Martinelli correlation was developed and VALIDATED on isothermal air-water (and similar gas-liquid) pipe data – Lockhart & Martinelli, Chem. Eng. Prog. 45 (1949) 39 – so air-water is exactly the case it is trusted on.

Basis: N<sub>2</sub> 3 kmol/h (the fast light gas) + water 200 kmol/h (the slow heavy liquid) → a liquid-dominated two-phase flow ( $j_G \sim$  a few m/s,  $j_L \sim 0.5$  m/s).

*Air (N<sub>2</sub>) + water flowing through a horizontal 0.05 m × 10 m commercial-steel line. The pipe's two-phase path activates automatically: the feed flashes two-phase at 2 bar / 25 °C (N<sub>2</sub> is supercritical → all vapour; water → mostly liquid), so  $0 < V/F < 1$  and the pipe routes to the gas-liquid models instead of the single-phase liquid path.*

*The two teaching points: 1. The multiplier  $\varphi_L^2 = 8.6$ : two-phase friction is far above either phase alone — the classic Lockhart-Martinelli amplification, visible in the printout via X and the Chisholm C. 2. Slip → holdup: the homogeneous (no-slip) model would put the liquid fraction at  $j_L/(j_L+j_G) \approx 0.12$ ; L-M gives 0.36 because the liquid moves slower than the gas and accumulates. Switch twoPhase { model homogeneous; } to see the difference for yourself.*

*The Lockhart-Martinelli correlation (Chem. Eng. Prog. 45 (1949) 39) was developed and validated on isothermal air-water (and similar gas-liquid) pipe data — so air-water is exactly the system it is trusted on. The model structure here (X,  $\varphi_L^2$ , the tt/vt/tv/vv C-selection, the Butterworth 1975 void fraction) reproduces that framework.*

*Switch twoPhase { model ...; } and compare — the spread IS a lesson (two-phase ΔP correlations routinely disagree by ~50 %):*

*Layer 2 (Friedel, BeggsBrill) needs surface tension — supplied here by transport { surfaceTension { model BrockBird; } } (from water's Tc/Pc/Tb). Beggs-Brill also classifies the flow pattern (segregated / intermittent / distributed) and corrects holdup for pipe inclination (here horizontal, so  $\psi = 1$ ).*

*- Adiabatic: the inlet vapour fraction is held along the pipe (no flashing / condensation in the line). Flashing flow is a future increment. - Liquid density:  $\rho_L = 877$  kg/m<sup>3</sup> comes from the Rackett branch, ~12 % low for water (a known anomaly, also in pipe01\_water\_line). It biases the \*absolute\* ΔP; the correlation \*structure\* is unaffected. - Liquid viscosity via operation { viscosity ...; } (water's value): the flash leaves a trace of dissolved N<sub>2</sub> in the liquid and the Vogel model has no N<sub>2</sub> entry.*

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected        |
|--------|-------------|---------------|-----------------|
| stream | downstream  | F             | 0.0563888888889 |
| stream | feed        | F             | 0.0563888888889 |
| kpi    | P1          | F             | 0.0563888888889 |
| kpi    | P1          | P_in          | 200000          |
| kpi    | P1          | P_out         | 193989.686272   |
| kpi    | P1          | Re_Gs         | 26035.4493901   |
| kpi    | P1          | Re_Ls         | 28789.8441141   |
| kpi    | P1          | X_martinelli  | 2.63535511116   |
| kpi    | P1          | dP_elevation  | 0               |
| kpi    | P1          | dP_friction   | 6010.31372773   |
| kpi    | P1          | deltaP        | 6010.31372773   |
| kpi    | P1          | holdup_liquid | 0.357791422881  |
| kpi    | P1          | jG            | 4.07152675186   |
| kpi    | P1          | jL            | 0.514023769077  |

...and 8 more reference values in the case's *expected*.

## pumpSystem01\_operating\_point

tutorials/steady/hydraulics/pumpSystem01\_operating\_point



**What it does:** Pump vs system curve: sweep the feed flow, the operating point is the intersection

**The story:** A pump and its piping system IN SERIES, swept over the feed flow (see system/outerDict). The classical operating-point construction:

PUMP characteristic – the shaft delivers a FIXED 2 kW at eta 0.65, so the pressure rise FALLS with flow:  $dP_{\text{pump}} = \eta W / Q_{\text{vol}}$ . This is the constant-shaft-power characteristic of the pump MODEL, not a manufacturer's tested  $H(Q)$  curve – the engine says what it models, never pretends to a datasheet it was not given.

SYSTEM curve – the same flow through 50 m of 0.1 m commercial-steel pipe with two elbows, a globe valve and a 2 m rise:  $dP_{\text{system}} = \text{friction}(Q^2) + \text{fittings}(Q^2) + \rho g dz$ .

Where the two curves CROSS, the pump's rise exactly pays the system's demand – the operating point. Hand estimate (water,  $v \sim 1.807\text{e-}5 \text{ m}^3/\text{mol}$ ,  $A = 7.854\text{e-}3 \text{ m}^2$ ,  $f \sim 0.018$ , sum  $K = 11.8$ ):

pump:  $dP \sim 1300 / (F \cdot 1.807\text{e-}5) \sim 7.19\text{e}4 / F \text{ Pa}$  system:  $dP \sim 5.45\text{e}4 F^2 + 1.956\text{e}4 \text{ Pa}$  ( $F$  in  $\text{kmol/s}$ )  
cross:  $F^* \sim 0.99 \text{ kmol/s}$  ( $\sim 3.6\text{e}3 \text{ kmol/h}$ ),  $dP^* \sim 0.73 \text{ bar}$

The sweep straddles that crossing (0.4..1.6  $\text{kmol/s}$ ); each point is a complete converged pass, and the curves are read off the pump's and the pipe's own published KPIs – nothing is fitted or interpolated here.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | delivered   | F              | 1               |
| stream | feed        | F              | 1               |
| stream | mid         | F              | 1               |
| kpi    | P1          | F              | 1               |
| kpi    | P1          | P_in           | 371942.446043   |
| kpi    | P1          | P_out          | 297097.627131   |
| kpi    | P1          | dP_elevation   | 19553.4351728   |
| kpi    | P1          | dP_fittings    | 31136.3992358   |
| kpi    | P1          | dP_friction    | 24154.9845033   |
| kpi    | P1          | deltaP         | 74844.818912    |
| kpi    | P1          | density        | 996.947743257   |
| kpi    | P1          | frictionFactor | 0.0183083994383 |
| kpi    | P1          | head_loss_m    | 5.65541381864   |
| kpi    | P1          | regime         | 2               |

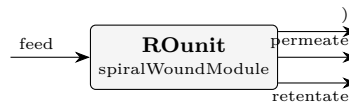
...and 23 more reference values in the case's *expected*.

## membranes

**The theory you need.** Solution-diffusion (Wijmans & Baker 1995): inside a dense membrane the pressure is uniform, so permeation is concentration-driven. Linearised:  $J_w = A_w(\Delta P - \Delta\pi)$  and  $J_s = B_s(c_m - c_p)$  – water driven by NET pressure, solute blind to pressure; ( $A_w, B_s$ ) are MATERIAL constants from `data/standards/assets/`. Concentration polarisation (film model) makes the wall concentration  $c_m = c_b e^{J_w/k}$  exceed the bulk – the flux–polarisation–osmotic feedback is THE nonlinearity of RO/NF, solved per module segment. Expect: rejection improving with pressure ( $c_p = B_s c_m / (J_w + B_s)$ ), polarisation eating driving force at high flux, and charged-membrane (Donnan) effects on multivalent ions that plain solution-diffusion cannot see – that is what the DSPM-DE sibling model is for. *Full derivation: Theory Guide, the membrane chapter.*

**membrane01\_RO\_NaCl\_seawater**

tutorials/steady/membranes/membrane01\_RO\_NaCl\_seawater



**What it does:** Spiral-wound RO of 35 g/L NaCl feed at 55 bar (canonical seawater desalination)

**The story:** the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With  $c_{\text{NaCl}} = 35/58.44 = 0.599$  kmol/m<sup>3</sup> and  $c_{\text{water}}$  in seawater  $\approx 1025$  kg/m<sup>3</sup> ? 18 kg/kmol = 56.94 kmol/m<sup>3</sup>, the mole fractions are  $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$  and  $z_{\text{water}} = 0.9896$ . Feed molar flow  $F = 575$  kmol/h ( $\approx 10$  m<sup>3</sup>/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m<sup>2</sup> active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome  $\sim 30$  bar of osmotic pressure at 35 g/L plus channel  $\Delta P$  plus a net driving force of  $\sim 22$  bar). Permeate pressure 1 bar.

Mass transfer:  $k_{\text{film}} = 5\text{e-}5$  m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives  $\sim 5 \times 10^{??}$  m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

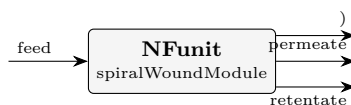
Expected (with the SW30HR parameters): water recovery  $\approx 18$  % (single pass) observed NaCl rej.  $\approx 99.5$  % (typical for SW30HR) average water flux  $\approx 20\text{--}25$  LMH feed  $\Delta P = 2$  bar (set linearly via  $dP_{\text{feed\_total}}$ )

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| kpi    | ROunit      | R_obs_NaCl       | 0.997859093278    |
| kpi    | ROunit      | water_recovery   | 0.157762204919    |
| kpi    | ROunit      | J_w_avg          | 1.32725173967e-05 |
| stream | feed        | F                | 0.159722222222    |
| stream | feed        | T                | 298.15            |
| stream | permeate    | F                | 0.0257849252597   |
| stream | permeate    | T                | 298.15            |
| stream | retentate   | F                | 0.133938394975    |
| stream | retentate   | T                | 298.15            |
| kpi    | ROunit      | dP_feed          | 200000            |
| kpi    | ROunit      | mass_closure_rel | 1.50817535928e-16 |

**membrane02\_NF\_sugar**

tutorials/steady/membranes/membrane02\_NF\_sugar



**What it does:** NF270 partial rejection of glucose at 10 bar (canonical NF: 10 g/L sugar feed)

**The story:** loose-NF separation of glucose from water, the canonical NF test that pedagogically distinguishes NF from RO:

RO tight membrane, high pressure, near-100 % rejection NF (here) loose membrane, low pressure, PARTIAL rejection (~50 %)

Feed: 10 g/L glucose (a typical pharma / food-processing feed).  $c_{\text{glucose}} = 10 / 180.16 = 0.0555 \text{ kmol/m}^3$   
 With pure-water  $c_{\text{water}} = 1000/18 = 55.56 \text{ kmol/m}^3$ ,  $z_{\text{glucose}} = 0.0555 / (0.0555 + 55.56) = 9.97\text{e-}4$ ,  
 $z_{\text{water}} = 0.999003$ . Feed  $F = 200 \text{ kmol/h}$  ( $\approx 3.6 \text{ m}^3/\text{h}$  of dilute solution).

Module: NF270-style spiral-wound element,  $35 \text{ m}^2$  area, 1 m length. Feed pressure 10 bar (an order of magnitude less than the SW30HR seawater case — the price of loose rejection is that membrane permeability  $A_w$  is  $10\times$  higher, so the same operating flux is achieved with  $10\times$  less driving pressure). Permeate at atmospheric. Same  $k_{\text{film}} = 5\text{e-}5 \text{ m/s}$ .

Expected (with the NF270 parameters): water recovery  $\approx 20\text{-}25 \%$  observed glucose rej.  $\approx 50\text{-}65 \%$  (? the SW30HR's 99 %+ !) average water flux  $\approx 30\text{-}40 \text{ LMH}$

The partial rejection is the defining NF feature: the membrane is not tight enough to fully reject glucose (MW 180, NF270 nominal MWCO 200 Da), so a fraction of the sugar leaks through with the water — and that leakage is what makes NF useful for fractionation rather than just total rejection.

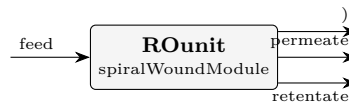
**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.05555555555556  |
| stream | permeate    | F                | 0.0149607433177   |
| stream | retentate   | F                | 0.0405951326871   |
| kpi    | NFunit      | J_w_avg          | 5.40656892958e-05 |
| kpi    | NFunit      | R_obs_glucose    | 0.662821422531    |
| kpi    | NFunit      | dP_feed          | 100000            |
| kpi    | NFunit      | water_recovery   | 0.267701129953    |
| stream | feed        | T                | 298.15            |
| stream | permeate    | T                | 298.15            |
| stream | retentate   | T                | 298.15            |
| kpi    | NFunit      | mass_closure_rel | 2.19886558047e-16 |



## membrane03\_concentration\_polarization

tutorials/steady/membranes/membrane03\_concentration\_polarization



**What it does:** Concentration polarization study: sweep  $k_{\text{film}}$  over a SW30HR seawater RO case

**The story:** Same SW30HR seawater RO case as membrane01\_RO\_NaCl\_seawater, but here we use the SweepDriver (see outerDict) to vary the mass-transfer coefficient  $k_{\text{film}}$  over a decade and plot how observed rejection and average flux change.

The physics:  $c_m / c_b = \exp ( J_w / k_{\text{film}} )$  (film model)

When  $k_{\text{film}}$  is large (good cross-flow mixing, e.g. high-velocity channel + good spacer), the boundary layer is thin and the concentration at the membrane wall is close to the bulk:  $c_m \approx c_b$ .  $\Delta\pi$  then reflects the bulk concentration, fluxes are high, and observed rejection is close to the intrinsic value.

When  $k_{\text{film}}$  is small (poor mixing, low velocity, dead zones), the solute concentrates at the wall:  $c_m \gg c_b$ .  $\Delta\pi$  rises sharply,  $J_w$  collapses, and a larger fraction of the wall solute leaks through, dropping observed rejection.

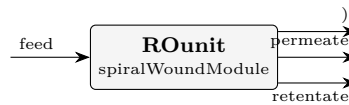
This sweep illustrates why membrane-module designers obsess over spacer geometry and feed velocity — they are not adjusting the membrane itself, they are adjusting  $k_{\text{film}}$ .

**What to expect (golden-locked):**

| kind   | unit/stream | key                         | expected          |
|--------|-------------|-----------------------------|-------------------|
| stream | feed        | F                           | 0.159722222222    |
| stream | permeate    | F                           | 0.0312284897591   |
| stream | retentate   | F                           | 0.128495408945    |
| kpi    | ROunit      | $J_w_{\text{avg}}$          | 1.60743521195e-05 |
| kpi    | ROunit      | $R_{\text{obs\_NaCl}}$      | 0.998356903981    |
| kpi    | ROunit      | $dP_{\text{feed}}$          | 200000            |
| kpi    | ROunit      | $\text{mass\_closure\_rel}$ | 3.01635071855e-16 |
| kpi    | ROunit      | $\text{water\_recovery}$    | 0.191065881266    |
| stream | feed        | T                           | 298.15            |
| stream | permeate    | T                           | 298.15            |
| stream | retentate   | T                           | 298.15            |

**membrane04\_spacer\_schock\_miquel**

tutorials/steady/membranes/membrane04\_spacer\_schock\_miquel



**What it does:** Seawater RO with  $k_{\text{film}}$  computed from spacer-channel hydraulics (Schock-Miquel Sherwood);  $k_{\text{film}}$  falls  $\sim 6.6$  to  $\sim 5.6 \times 10^{-5}$  m/s along the channel

**The story:** the same seawater RO as membrane01, but  $k_{\text{film}}$  is COMPUTED from the spacer-channel hydraulics (Schock-Miquel Sherwood correlation) instead of being a fixed number. The ‘massTransfer’ block makes  $k_{\text{film}}$  a selectable model:  $Sh = 0.065 Re^{0.875} Sc^{0.25}$  with  $Re$  from the local crossflow velocity  $u = Q_b(z)/A_{\text{channel}}$  — so  $k_{\text{film}}$  FALLS along the channel as the retentate flow drops (the profile column ‘ $k_{\text{film}}$ ’ shows this). At the inlet  $u \sim 0.26$  m/s gives  $k_{\text{film}} \sim 7 \times 10^{-5}$  m/s, a touch above membrane01’s fixed  $5 \times 10^{-5}$  (better mixing  $\rightarrow$  less polarisation  $\rightarrow$  slightly higher recovery/rejection). Declare a bare ‘ $k_{\text{film}} < \text{value} >$ ’; (no massTransfer block) to recover membrane01’s fixed-coefficient behaviour — the ONE way to pin it (the redundant ‘model constant;’ route was removed 2026-08-23).

membrane01\_RO\_NaCl\_seawater — the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With  $c_{\text{NaCl}} = 35/58.44 = 0.599$  kmol/m<sup>3</sup> and  $c_{\text{water}}$  in seawater  $\approx 1025$  kg/m<sup>3</sup>  $\cdot 18$  kg/kmol = 56.94 kmol/m<sup>3</sup>, the mole fractions are  $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$  and  $z_{\text{water}} = 0.9896$ . Feed molar flow  $F = 575$  kmol/h ( $\approx 10$  m<sup>3</sup>/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m<sup>2</sup> active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome  $\sim 30$  bar of osmotic pressure at 35 g/L plus channel  $\Delta P$  plus a net driving force of  $\sim 22$  bar). Permeate pressure 1 bar.

Mass transfer:  $k_{\text{film}} = 5 \times 10^{-5}$  m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives  $\sim 5 \times 10^{??}$  m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

Expected (with the SW30HR parameters): water recovery  $\approx 18$  % (single pass) observed NaCl rej.  $\approx 99.5$  % (typical for SW30HR) average water flux  $\approx 20$ –25 LMH feed  $\Delta P = 2$  bar (set linearly via  $dP_{\text{feed\_total}}$ )

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.159722222222    |
| stream | permeate    | F                | 0.0286373914761   |
| stream | retentate   | F                | 0.131086212764    |
| kpi    | ROunit      | J_w_avg          | 1.47407124897e-05 |
| kpi    | ROunit      | R_obs_NaCl       | 0.99810213206     |
| kpi    | ROunit      | dP_feed          | 26065.9064084     |
| kpi    | ROunit      | water_recovery   | 0.175213731874    |
| stream | feed        | T                | 298.15            |
| stream | permeate    | T                | 298.15            |
| stream | retentate   | T                | 298.15            |
| kpi    | ROunit      | mass_closure_rel | 7.54087679638e-16 |

**membrane05\_train**

tutorials/steady/membranes/membrane05\_train



**What it does:** Multi-element membrane train: 3 SW30HR elements in series (seawater RO)

**The story:** a MULTI-ELEMENT TRAIN: 3 SW30HR spiral-wound elements in series, seawater RO (the same feed + per-element geometry as membrane04, which is the single element). ‘elements 3;’ chains them: the retentate of element 1 is the feed of element 2, and so on; the permeate is collected over the whole train. Both the spacer  $k_{\text{film}}$  and the spacer pressure drop (Schock-Miquel) are computed from the local hydraulics, so they fall element by element as the flow drops.

The pedagogical point is the per-element DEGRADATION reported in the KPIs ( $\text{recovery\_elemK}$ ,  $\text{c\_ret\_elemK}$ ): element 1 sees fresh feed (high  $P$ , low  $c$ ) and recovers the most; the last element sees concentrated, lower-pressure brine ( $P$  dropped by friction + the inter-element connector loss) and recovers far less. The train recovery is the cumulative sum — higher than a single element, but with diminishing returns down the train.

Feed: 35 g/L NaCl seawater,  $z_{\text{NaCl}} = 0.01040$ ,  $F = 575 \text{ kmol/h}$  ( $\sim 10 \text{ m}^3/\text{h}$ ), 55 bar, 298.15 K — the same feed as membrane01/04. Each of the 3 elements is  $35 \text{ m}^2 / 1 \text{ m}$ , SW30HR, permeate at 1 bar, with a 0.2 bar inter-element connector loss.

What this run produces (golden master): train water recovery 35.3 % (3 elements, cumulative) observed NaCl rejection 99.68 % train-average water flux 34.85 LMH ( $9.68\text{e-}6 \text{ m/s}$ ) total feed-side  $\Delta P \sim 0.99 \text{ bar}$  (Schock-Miquel friction + 2 connector losses)

The per-element DEGRADATION (the teaching point) reads off the KPIs:  $\text{recovery\_elem1}$  17.2 %  $\text{c\_ret\_elem1}$  0.697 (fresh feed)  $\text{recovery\_elem2}$  28.4 %  $\text{c\_ret\_elem2}$  0.805 (cumulative)  $\text{recovery\_elem3}$  35.3 %  $\text{c\_ret\_elem3}$  0.891 (cumulative) The retentate concentration climbs element to element while each successive element adds less recovery: a single SW30HR element at this feed recovers  $\sim 17 \%$ , the second adds  $\sim 11 \%$ , the third only  $\sim 7 \%$  — the diminishing return of stacking elements in series at fixed feed.

NOTE (known issue): this train case currently has an open mass-balance discrepancy under investigation; it is run-only in the regression and the KPIs above are the present golden values, not a validated material balance.

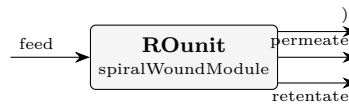
**What to expect (golden-locked):**

| kind   | unit/stream | key                      | expected                   |
|--------|-------------|--------------------------|----------------------------|
| stream | feed        | F                        | 0.159722222222             |
| stream | permeate    | F                        | 0.0590874153104            |
| stream | retentate   | F                        | 0.100637449271             |
| kpi    | ROunit      | $J_{\text{w\_avg}}$      | $1.01384186683\text{e-}05$ |
| kpi    | ROunit      | $R_{\text{obs\_NaCl}}$   | 0.99694467035              |
| kpi    | ROunit      | $\text{c\_ret\_elem1}$   | 0.68349184216              |
| kpi    | ROunit      | $\text{c\_ret\_elem2}$   | 0.793343136295             |
| kpi    | ROunit      | $\text{c\_ret\_elem3}$   | 0.881961644632             |
| kpi    | ROunit      | $\text{dP\_feed}$        | 101143.55107               |
| kpi    | ROunit      | elements                 | 3                          |
| kpi    | ROunit      | $\text{recovery\_elem1}$ | 0.175213731874             |
| kpi    | ROunit      | $\text{recovery\_elem2}$ | 0.289843097471             |
| kpi    | ROunit      | $\text{recovery\_elem3}$ | 0.361527335552             |
| kpi    | ROunit      | water_recovery           | 0.361527335552             |

... and 4 more reference values in the case's *expected*.

**membrane06\_pitzer**

tutorials/steady/membranes/membrane06\_pitzer



**What it does:** Seawater RO with the Pitzer osmotic model (vs membrane01 van't Hoff)

**The story:** the canonical seawater RO (same as membrane01) but with the PITZER osmotic-pressure model instead of van't Hoff. van't Hoff ( $\phi = 1$ ) over-estimates the osmotic pressure of a real brine: for NaCl the osmotic coefficient is  $\phi \sim 0.92$  at seawater concentration (and dips further before rising at high molality). The Pitzer ion-interaction model captures  $\phi(I)$ , so the osmotic back-pressure is  $\sim 7\text{--}8\%$  LOWER  $\rightarrow$  a slightly HIGHER water flux and recovery than membrane01 (15.55 %). The 'osmotic' block makes it a selectable model with the NaCl coefficients (calibratable); remove the block to recover van't Hoff.

membrane01\_RO\_NaCl\_seawater – the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With  $c_{\text{NaCl}} = 35/58.44 = 0.599$  kmol/m<sup>3</sup> and  $c_{\text{water}}$  in seawater  $\approx 1025$  kg/m<sup>3</sup>  $\div 18$  kg/kmol = 56.94 kmol/m<sup>3</sup>, the mole fractions are  $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$  and  $z_{\text{water}} = 0.9896$ . Feed molar flow  $F = 575$  kmol/h ( $\approx 10$  m<sup>3</sup>/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m<sup>2</sup> active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome  $\sim 30$  bar of osmotic pressure at 35 g/L plus channel  $\Delta P$  plus a net driving force of  $\sim 22$  bar). Permeate pressure 1 bar.

Mass transfer:  $k_{\text{film}} = 5\text{e-}5$  m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives  $\sim 5 \times 10^{??}$  m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

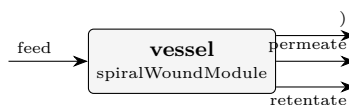
Expected (with the SW30HR parameters): water recovery  $\approx 18\%$  (single pass) observed NaCl rej.  $\approx 99.5\%$  (typical for SW30HR) average water flux  $\approx 20\text{--}25$  LMH feed  $\Delta P = 2$  bar (set linearly via  $dP_{\text{feed\_total}}$ )

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.159722222222    |
| stream | permeate    | F                | 0.0284588599363   |
| stream | retentate   | F                | 0.13126472432     |
| kpi    | ROunit      | J_w_avg          | 1.46488544181e-05 |
| kpi    | ROunit      | R_obs_NaCl       | 0.997986472373    |
| kpi    | ROunit      | dP_feed          | 200000            |
| kpi    | ROunit      | water_recovery   | 0.174121871794    |
| stream | feed        | T                | 298.15            |
| stream | permeate    | T                | 298.15            |
| stream | retentate   | T                | 298.15            |
| kpi    | ROunit      | mass_closure_rel | 6.03270143711e-16 |

**membrane07\_scaling\_si**

tutorials/steady/membranes/membrane07\_scaling\_si



**What it does:** Brackish RO, 6 x 8-inch modules: per-module scaling audit (SI at wall vs bulk)

**The story:** WHERE along the vessel does scaling start?

Brackish RO on the SAME representative groundwater analysis as tutorials/props/electrolyte/scaling\_ro\_brackish (~1500 mg/L TDS):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026  
Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003

Here the analysis is a PROCESS STREAM: the ions are flowsheet components (case-local constant/components/, MW from ions.dat) and the mg/L totals become mole fractions against water ( $x_i = m_i / 55.55$ ;  $m_i = (\text{mg/L})/\text{MW}$  at  $\rho \sim 1 \text{ kg/L}$ ):

Ca 3.7731e-5 Mg 1.9998e-5 Na 2.8425e-4 K 5.5253e-6 Cl 2.5897e-4 SO4 4.6852e-5 HCO3 5.3989e-5 water = balance

HARDWARE BY NOMINAL DIAMETER (no 'area'): one pressure vessel with 6 x 8-inch standard 40-inch elements – 'moduleDiameter 8 in; nModules 6;'. The per-module area (37.2 m<sup>2</sup>, 400 ft<sup>2</sup> class) comes from the engine's nominal table and the derivation is announced: "6 x 37.2 m<sup>2</sup> (8in/40in standard element) = 223.2 m<sup>2</sup>".

THE SCALING AUDIT ('scaling {}'): at each module exit the local BULK concentrate AND the membrane-WALL composition ( $c_m$  from the module's own film-model polarisation) are speciated (Davies) and the calcite / gypsum saturation indices recorded. The WALL crosses SI=0 modules before the bulk does – concentration polarisation, normally an invisible flux penalty, is seen here as THE scaling driver.

pH 7.0 GIVEN, not solved: the raw well water is already calcite- supersaturated (scaling\_ro\_brackish solves the raw feed to pH 7.72, SI\_calcite = +0.24 – it would scale in the PIPE, let alone the vessel). Real brackish plants dose acid for exactly that reason; here the feed is dosed and CONTROLLED at pH 7.0, and the audit asks the honest follow-up: with the dosing in place, where along the vessel does the concentrate STILL reach calcite saturation? ('pH solve;' is also accepted – it re-solves electroneutrality per module, the undosed scenario.)

Expected story (KPIs locked by the golden master; recovery 55 %): calcite WALL crosses SI=0 at module 3 of 6 ( $r_{cum} = 0.31$ ); calcite BULK is still undersaturated there (-0.27) and only crosses in module 6 (+0.03) – the wall leads the bulk by three full modules: polarisation IS the scaling driver; gypsum stays undersaturated everywhere (max wall SI = -0.73).

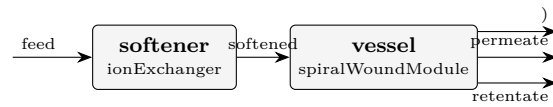
**What to expect (golden-locked):**

| kind   | unit/stream | key                  | expected          |
|--------|-------------|----------------------|-------------------|
| stream | feed        | F                    | 0.194444444431    |
| stream | permeate    | F                    | 0.107212342592    |
| stream | retentate   | F                    | 0.0872324772734   |
| kpi    | vessel      | J_w_avg              | 8.65338211369e-06 |
| kpi    | vessel      | R_obs_Ca             | 0.999474633761    |
| kpi    | vessel      | R_obs_Cl             | 0.994770268574    |
| kpi    | vessel      | R_obs_HCO3           | 0.996072731843    |
| kpi    | vessel      | R_obs_K              | 0.993471097928    |
| kpi    | vessel      | R_obs_Mg             | 0.999605925236    |
| kpi    | vessel      | R_obs_Na             | 0.994770268574    |
| kpi    | vessel      | R_obs_SO4            | 0.999737250092    |
| kpi    | vessel      | SI0_module_calcite   | 3                 |
| kpi    | vessel      | SI0_module_gypsum    | -1                |
| kpi    | vessel      | SI0_recovery_calcite | 0.30517794604     |

...and 27 more reference values in the case's *expected*.

**membrane08\_softened\_scaling**

tutorials/steady/membranes/membrane08\_softened\_scaling



**What it does:** Softener + brackish RO: ion-exchange removes the scaling driver upstream of the vessel

**The story:** SOFTEN the feed, and the scaling is GONE.

THE CONTRAST (read membrane07\_scaling\_si first – the BEFORE):

membrane07 the SAME brackish groundwater (~1500 mg/L TDS, Ca 84 / Mg 27 mg/L) goes straight into a 6 x 8-inch brackish-RO vessel. As the concentrate builds along the train the membrane-WALL composition crosses calcite SI = 0 at MODULE 3 of 6 (r\_cum = 0.31) – concentration polarisation is THE scaling driver; the vessel would foul.

membrane08 here a cation-exchange SOFTENER ('ionExchanger', SAC resin in the Na form, Gaines-Thomas) sits UPSTREAM of the SAME vessel. Ca and Mg drop ~99% onto the resin (Na rises eq-for-eq – the salt penalty), so the water entering the vessel is essentially hardness-free. With Ca gone the calcite IAP collapses and the per-module WALL SI\_calcite stays NEGATIVE through ALL 6 modules – the scaling driver is removed at source.

ZERO change to the membrane. The spiralWoundModule is byte-for-byte the membrane07 vessel (module-Diameter 8 in; nModules 6; the same scaling{} audit); it simply receives a BETTER feed. Removing the hardness, not chasing the supersaturation in the vessel, is the cure – the engineering lesson.

HONESTY (printed, non-suppressible): the softener returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass, NOT a bed in service (breakthrough / regeneration are TRANSIENT and not modelled). See the banner on every run.

Data: curated USGS PHREEQC records (speciation, minerals, ions, exchange; public domain) resolved from data/standards/ at runtime + case-local ion .dats (constant/components/), the SAC\_Na resin nameplate and the SW30HR membrane (constant/membranes/) – the same records as membrane07 / softener\_brackish; self-DESCRIBING (inline manifest), seal with bin/choupo-import for a runtime-self-contained copy.

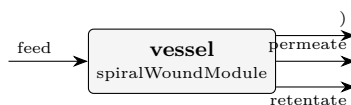
Expected story (golden master): softener removes ~99.99% hardness (Ca/Mg leakage ~ 1e-2 mg/L, Na added ~ 6.7 meq/L at the given pH 7.0 closure – the eq-for-eq salt penalty for the full hardness load); the vessel's maxSI\_calcite\_wall is NEGATIVE (-3.68: no SI = 0 crossing in any module, SI0\_module\_calcite = -1), vs +0.27 / crossing at module 3 in membrane07.

**What to expect (golden-locked):**

| kind                                                     | unit/stream | key                    | expected         |
|----------------------------------------------------------|-------------|------------------------|------------------|
| stream                                                   | feed        | F                      | 0.194444444431   |
| stream                                                   | permeate    | F                      | 0.105450929615   |
| stream                                                   | retentate   | F                      | 0.0890050816188  |
| stream                                                   | softened    | F                      | 0.19445566796    |
| kpi                                                      | softener    | CEC_utilised_pct       | 100              |
| kpi                                                      | softener    | Ca_leakage_mgL         | 0.00999051830577 |
| kpi                                                      | softener    | I                      | 0.024397591267   |
| kpi                                                      | softener    | Mg_leakage_mgL         | 0.00549118647587 |
| kpi                                                      | softener    | Na_added_meqL          | 6.71875413026    |
| kpi                                                      | softener    | Na_added_mgL           | 154.464157455    |
| kpi                                                      | softener    | bedVolume_m3           | 1.5              |
| kpi                                                      | softener    | dH_exchange_Jkgw       | 21.9905023247    |
| kpi                                                      | softener    | hardness_in_mgL_CaCO3  | 320.962820378    |
| kpi                                                      | softener    | hardness_out_mgL_CaCO3 | 0.0475632817167  |
| ... and 44 more reference values in the case's expected. |             |                        |                  |

## membrane09\_index\_vs\_rigorous

tutorials/steady/membranes/membrane09\_index\_vs\_rigorous



**What it does:** Seawater RO at high recovery: industry calcite indices (LSI / Stiff-Davis / Ryznar) vs the rigorous activity SI at the membrane WALL

**The story:** the bag-of-indices industry shortcut is NOT safe at the membrane wall in a high-recovery concentrate.

THE LESSON (activity vs concentration – the whole point of the case):

The industry calcite-scaling indices read every plant data-sheet ships – LSI Langelier saturation index (Langelier 1936) Stiff-Davis brine-corrected LSI (Stiff & Davis 1952) RSI Ryznar stability index (Ryznar 1944) are built from ION CONCENTRATIONS plus an EMPIRICAL ionic-strength correction (a published pK-shift / chart-fit K). Choupo's rigorous SI\_calcite instead forms real ION ACTIVITIES  $a_i = \gamma_i m_i$  from the Davies model:

$LSI = pH - \log K_{cc}(T) - p[Ca^{2+}] - p[HCO_3^-]$  (CONCENTRATIONS)  $SI = pH - \log K_{cc}(T) - p\{a_{Ca^{2+}}\} - p\{a_{HCO_3^-}\}$  (ACTIVITIES)

They share the SAME thermodynamic anchor  $\log K_{cc}(T)$  (the  $HCO_3$ -basis calcite  $\log K$ ,  $= pK_2 - pK_{sp}$ ) – so the ONLY difference is concentration vs activity. At LOW ionic strength  $\gamma \rightarrow 1$  and  $LSI == SI$  (validation); as the activity coefficients fall away from 1 the two DIVERGE.

WHERE IT BITES – THE WALL. Concentration polarisation piles solute against the membrane wall: the LOCAL ionic strength at the wall runs far above the bulk. There the empirical index over-predicts calcite saturation (it cannot see the gamma reduction), while the rigorous activity SI does not. Reading the index off the bulk analysis – as the industry shortcut does – therefore MISLEADS at exactly the place scaling actually starts.

THE CASE: standard surface seawater ( $S = 35$ , the major-ion subset), one pressure vessel of 7 x 8-inch SW elements, pushed to high recovery. Acid is dosed and CONTROLLED at pH 7.0 (real SWRO practice). At each module exit the BULK concentrate AND the membrane-WALL composition are speciated (Davies) and BOTH the rigorous SI\_calcite AND the empirical LSI / Stiff-Davis / Ryznar are recorded, bulk + wall.

THE STORY (locked by the golden master): - This is already a SEAWATER concentrate (bulk  $I \sim 0.68$  mol/kg at module 1, far above the  $I \rightarrow 0$  regime where  $LSI == SI$ ), so the empirical index OVER-PREDICTS calcite saturation at EVERY module, including the first: at module 1  $LSI(wall) = +0.98$  vs rigorous  $SI\_calcite(wall) = +0.20$  (gap  $\sim 0.78$ );  $LSI(bulk) = +0.56$  vs  $SI(bulk) = -0.28$  (gap  $\sim 0.84$ ). At this ionic strength there is no point where index and activity-SI agree to hundredths – to SEE that limit you must drop to a brackish / low-I feed (the index family is only validated there). - the gap widens further into the vessel: at the deepest module  $LSI(wall)$  over-predicts the rigorous  $SI\_calcite(wall)$  by  $\sim 0.78$  – the gap ( $LSI - SI$ ) printed in the audit footer is the headline number. Stiff-Davis pulls PART of the way back toward the rigorous SI (its empirical brine K), but not all the way – it is still a chart fit, not the real gammas. - Ryznar ( $RSI = 2$  pHs - pH) reads the same saturation pHs as LSI on the practical 0..14 stability scale.

Engineering take-away (Choupo's transparency differentiator): the concentration-based index family is convenient but UNSAFE for high-recovery membrane scaling – the rigorous activity SI, visible term-by-term, is the honest call.

Sibling cases: membrane07\_scaling\_si (bulk vs wall SI, brackish); tutorials/props/electrolyte/pitzer\_vs\_davies\_ro\_concent (Davies vs Pitzer activity fork). Index kernel + primary sources: src/thermo/electrolyte/ScalingIndices.{H,cpp}.

PRIMARY SOURCES: Langelier, W.F. (1936) J. AWWA 28(10) 1500-1521. Stiff, H.A. & Davis, L.E. (1952) Trans. AIME 195, 213-216. Ryznar, J.W. (1944) J. AWWA 36(4) 472-486.



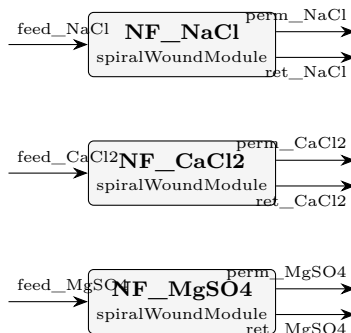
**What to expect (golden-locked):**

| kind   | unit/stream | key                  | expected          |
|--------|-------------|----------------------|-------------------|
| stream | feed        | F                    | 0.2500000000004   |
| stream | permeate    | F                    | 0.067554824405    |
| stream | retentate   | F                    | 0.182446484542    |
| kpi    | vessel      | J_w_avg              | 4.67400706828e-06 |
| kpi    | vessel      | LSI_calcite_bulk_end | 0.765353874066    |
| kpi    | vessel      | LSI_calcite_wall_end | 0.985330125213    |
| kpi    | vessel      | RSI_calcite_wall_end | 5.02933974957     |
| kpi    | vessel      | R_obs_Ca             | 0.999255215369    |
| kpi    | vessel      | R_obs_Cl             | 0.992602201838    |
| kpi    | vessel      | R_obs_HCO3           | 0.994441275819    |
| kpi    | vessel      | R_obs_K              | 0.990769979906    |
| kpi    | vessel      | R_obs_Mg             | 0.999441306528    |
| kpi    | vessel      | R_obs_Na             | 0.992602201838    |
| kpi    | vessel      | R_obs_SO4            | 0.99962746766     |

...and 29 more reference values in the case's *expected*.

**membrane10\_dspmde\_divalent**

tutorials/steady/membranes/membrane10\_dspmde\_divalent



**What it does:** DSPM-DE NF: divalent (MgSO<sub>4</sub>) > mixed (CaCl<sub>2</sub>) > monovalent (NaCl) rejection at 10 bar

**The story:** the charge-aware NF signature, glass-box.

THREE parallel NF270 elements, one per salt, all on the SAME membrane (NF270\_dspmde, with the DSPM-DE charged-pore tier) at the SAME 10 bar / 5 mM feed. The point is the REJECTION ORDERING that the Donnan-Steric Pore Model with Dielectric Exclusion produces from first principles:

R<sub>obs</sub>(MgSO<sub>4</sub>) > R<sub>obs</sub>(CaCl<sub>2</sub>) > R<sub>obs</sub>(NaCl) (2:2 divalent) (2:1 mixed) (1:1 monovalent)

WHY (Bowen-Welfoot 2002): the membrane carries a NEGATIVE fixed charge ( $X_d = -40$  mol/m<sup>3</sup>). The CO-ION (the anion, here Cl<sup>-</sup> / SO<sub>4</sub><sup>-</sup>) is excluded by Donnan; a DIVALENT co-ion (SO<sub>4</sub><sup>-</sup>) is excluded far more strongly than a monovalent one ( $z^2$  in the Donnan + Born partition). So MgSO<sub>4</sub> – whose co-ion is divalent – is rejected most; NaCl – whose co-ion is the monovalent Cl<sup>-</sup> – least. This is the classic NF fingerprint that plain solution-diffusion (one B<sub>s</sub> per salt) cannot predict mechanistically.

Data: the salts (case-local constant/components/) carry an electrolyte{ } ion decomposition and the membrane (constant/membranes/) carries the poreModel{ } block; the ions' radius/D0 transport tier resolves from data/standards/ at runtime – self-DESCRIBING (inline manifest), seal with bin/choupo-import for a runtime-self-contained copy.

‘transport DSPM-DE;’ selects the charged-pore law; drop it (or set ‘transport solutionDiffusion;’) to compare against the baseline.

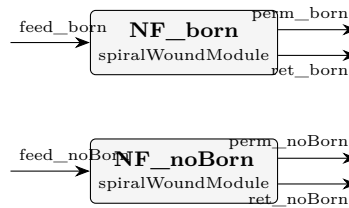
**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected          |
|--------|-------------|-------------|-------------------|
| stream | feed_CaCl2  | F           | 0.05555555555556  |
| stream | feed_MgSO4  | F           | 0.05555555555556  |
| stream | feed_NaCl   | F           | 0.05555555555556  |
| stream | perm_CaCl2  | F           | 0.0191775929422   |
| stream | perm_MgSO4  | F           | 0.020040469745    |
| stream | perm_NaCl   | F           | 0.0208297354228   |
| stream | ret_CaCl2   | F           | 0.0363780274704   |
| stream | ret_MgSO4   | F           | 0.0355151987788   |
| stream | ret_NaCl    | F           | 0.034725847724    |
| kpi    | NF_CaCl2    | J_w_avg     | 6.91063443929e-05 |
| kpi    | NF_CaCl2    | R_obs_CaCl2 | 0.704588151782    |
| kpi    | NF_CaCl2    | R_obs_MgSO4 | 1                 |
| kpi    | NF_CaCl2    | R_obs_NaCl  | 1                 |
| kpi    | NF_CaCl2    | dP_feed     | 100000            |

... and 25 more reference values in the case's *expected*.

**membrane11\_dspmde\_born\_toggle**

tutorials/steady/membranes/membrane11\_dspmde\_born\_toggle



**What it does:** DSPM-DE NF: Born dielectric-exclusion toggle (same NaCl feed, poreDielectric on vs off)

**The story:** SEE the dielectric-exclusion (Born) term.

The SAME NaCl feed (50 mM, 10 bar) through TWO NF270 elements that differ in ONE thing only:

NF\_born : membrane NF270\_dspmde – poreDielectric 35 (Born ON) NF\_noBorn : membrane NF270\_dspmde\_noBorn – poreDielectric absent (Born OFF)

Everything else – pore radius, porosity/thickness, fixed charge  $X_d$ , feed, pressure, area,  $k_{\text{film}}$  – is IDENTICAL. The ONLY physics that changes is the Born dielectric-exclusion term

$$dW_i = (z_i^2 e^2 / 8 \pi \epsilon_0 r_i) (1/\epsilon_p - 1/\epsilon_b)$$

(Bowen & Welfoot 2002, Eq. 16). With  $\epsilon_p < \epsilon_b$  an ion pays an extra solvation-energy penalty to enter the low-dielectric pore, so it is excluded MORE strongly. The student reads the Born contribution straight off the two  $R_{\text{obs}}(\text{NaCl})$  KPIs:  $R_{\text{obs}}(\text{NF\_born}) > R_{\text{obs}}(\text{NF\_noBorn})$ .

This is the pedagogical core of DSPM-"DE": the DE (dielectric exclusion) is what distinguishes the 2002 Bowen-Welfoot model from the 1997 steric-Donnan DSPM. Toggling poreDielectric is a single, visible switch.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected         |
|--------|-------------|----------------|------------------|
| stream | feed_born   | F              | 0.05555555555556 |
| stream | feed_noBorn | F              | 0.05555555555556 |
| stream | perm_born   | F              | 0.0103299082671  |
| stream | perm_noBorn | F              | 0.0112254914132  |
| stream | ret_born    | F              | 0.0452257284146  |
| stream | ret_noBorn  | F              | 0.0443301389741  |
| kpi    | NF_born     | J_w_avg        | 3.7229087272e-05 |
| kpi    | NF_born     | R_obs_NaCl     | 0.861027895843   |
| kpi    | NF_born     | dP_feed        | 100000           |
| kpi    | NF_born     | water_recovery | 0.185615581216   |
| kpi    | NF_noBorn   | J_w_avg        | 4.0473218085e-05 |
| kpi    | NF_noBorn   | R_obs_NaCl     | 0.65954044001    |
| kpi    | NF_noBorn   | dP_feed        | 100000           |
| kpi    | NF_noBorn   | water_recovery | 0.201790063872   |

...and 8 more reference values in the case's *expected*.

**optimisation**

**The theory you need.** The outer driver wraps the whole simulation in  $f(\mathbf{x})$ : sweeps map it, Nelder–Mead descends it, parameter estimation fits it to data (Levenberg–Marquardt, with identifiability diagnostics –  $\chi_{\text{red}}^2$ , correlation, conditioning). Expect every function evaluation to be a full, honest flowsheet pass.

**designSpec01\_triple\_equal\_areas**

tutorials/steady/optimisation/designSpec01\_triple\_equal\_areas



**What it does:** DesignSpec: equal-area triple-effect sugar evaporator (manipulate \$A so  $L_3 = 4500$  kg/h)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. ('sucrose.dat' does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

Same physics as evaporator02\_triple\_effect\_sugar (textbook Example 10.4 — three cocurrent effects on a 22 500 kg/h, 5 wt% sucrose feed). The twist: instead of HARD-CODING each effect's area at  $100 \text{ m}^2$ , we declare  $A$  as a CASE-LEVEL VARIABLE and use the outer DesignSpec driver to find the equal-areas value that delivers  $L_3 \sim 4500$  kg/h (25 wt% sucrose), the textbook design target.

This tutorial showcases two new pieces of v0.30 machinery:

1. \$references in the dict ('variables { ... }' block + '\$A' syntax inside any scalar). The same  $A$  is shared by the three effects without copy-paste — which makes it trivially a single degree of freedom for the outer solver.
2. The 'designSpec' outer driver: a Newton-ND wrap on Flowsheet::solve, manipulating the case's \$variables to satisfy named targets. Here it is a 1-D Newton in  $A$  on the residual  $L_3.F - 0.001249 \text{ kmol/s}$ . Convergence in 4-5 evaluations.

$F=22\,500 \text{ kg/h}$   $L_1$   $L_2$   $L_3$  (target  $\sim 4500 \text{ kg/h}$ )  $x=0.05 \rightarrow [1] \longrightarrow [2] \longrightarrow [3] \longrightarrow \hat{\hat{\hat{}}}$  |  $V_1$   $V_2$   $V_3$   
 $\rightarrow$  condenser Steam (drives 2) (drives 3) 200 kPa  $P_1 = 80$   $P_2 = 45$   $P_3 = 15$  kPa

Constraints we leave fixed: -  $P_i$  pressures (textbook cascade) -  $U_i$  overall HTC's per effect (textbook values) - Feed and steam conditions - Equal areas: enforced trivially by the single  $A$  variable

Read the run log to see the Newton converge in  $<10$  evaluations. At the optimum, the case ends with the converged final state on stdout, exactly like a hand-tuned evaporator02 run — but the area was DERIVED, not specified.

Initial guess  $A_0 = 100 \text{ m}^2$  (the educated guess from evaporator02); the solver should home in on  $\sim 110$ - $120 \text{ m}^2$  to hit the target.

Target tolerance:  $\pm 5 \text{ kg/h}$  on  $L_3$ , encoded via  $\text{tol} = 1.4\text{e-}6 \text{ kmol/s}$  ( $= 5 \text{ kg/h} ? 3600 \text{ s/h} ? 18 \text{ kg/kmol}$ ).

**What to expect (golden-locked):**

| kind                                                                  | unit/stream         | key  | expected        |
|-----------------------------------------------------------------------|---------------------|------|-----------------|
| stream                                                                | L1                  | F    | 0.242853616254  |
| stream                                                                | L2                  | F    | 0.15012886479   |
| stream                                                                | L3                  | F    | 0.0529527028804 |
| stream                                                                | V1                  | F    | 0.0876457894447 |
| stream                                                                | V2                  | F    | 0.0927247514636 |
| stream                                                                | V3                  | F    | 0.0971761619099 |
| stream                                                                | cond1               | F    | 0.13885829138   |
| stream                                                                | cond2               | F    | 0.0876457894447 |
| stream                                                                | cond3               | F    | 0.0927247514636 |
| stream                                                                | feed                | F    | 0.330499405699  |
| stream                                                                | utilitySteam_200kPa | F    | 0.13885829138   |
| kpi                                                                   | effect1             | A    | 108.00857667    |
| kpi                                                                   | effect1             | A_m2 | 108.00857667    |
| kpi                                                                   | effect1             | BPE  | 0.107441424817  |
| <i>...and 92 more reference values in the case's <b>expected</b>.</i> |                     |      |                 |

**fitNRTL01\_ethanol\_water**

tutorials/steady/optimisation/fitNRTL01\_ethanol\_water

**fit\_NRTL\_ethanol\_water**  
 fitParameters

**What it does:** Fit NRTL ethanol/water pair against 1 atm bubble-T data (Levenberg-Marquardt)

**The story:** fitNRTL01 – Levenberg-Marquardt regression of the ethanol-water NRTL pair against 1 atm bubble-T data, on the CANONICAL fit engine (fitParameters, choupProps). This case previously ran the condemned FitBinaryPair outer driver in choupSolve, which abused a dummy flowsheet and steered the feed composition through the legacy streams{ } block per data row – the design forum (#53) ordered it moved here, WITH a golden: its  $\chi^2$  once degraded 100x with nothing going red, because a fit without numerical verification does not belong in a green corpus.

Data: transcribed from constant/experiments/ethanol-water-101kPa (the raw- data home; DECHEMA / Gmehling-Onken Vol. I Part 1, pedagogical subset). Starting point: the catalogue NRTL values in constant/thermoPhysPropDict.

*The parameter-estimation workflow: regress the four NRTL binary interaction parameters of the ethanol-water pair ( $a_{ij}$ ,  $b_{ij}$ ,  $a_{ji}$ ,  $b_{ji}$ ;  $\alpha$  held fixed) against experimental bubble-temperature data at 1.01325 bar, using hand-rolled Levenberg-Marquardt inside the fitParameters engine (successor of the retired fitBinaryPair driver).*

*\* See an outer Marquardt loop (3-up/5-down trust region on  $\lambda$ ) wrapped around an inner Newton-1D solving  $\sum K x = 1$  for each  $(x_i, T)$  pair. \* Watch  $\lambda$  oscillate as the algorithm switches between Gauss-Newton ( $\lambda$  small) and gradient-descent ( $\lambda$  large) regimes. \* Watch some LM steps get rejected —  $\chi^2$  gets worse, the step is reversed,  $\lambda$  is multiplied by 5, retry. The whole "trust region" idea visible in one column of the log.*

*Note how minimal the forward case is: one stream, one bubbleT unit. The outer driver is what does the work — iterating over the dataset internally and looping LM over the four parameters.*

*Four keywords (plus optional maxIter, tolerance, fdStep, lambda0). The driver:*

*1. locates the pair in thermoPhysPropDict.activityModel.pairs by name, 2. reads the dataset (a two-column flat list of  $x_i$   $T_i$ ), 3. finds the bubbleT unit's input stream from the flowsheetDict, 4. for each LM step: - 8 forward evals to build a central-difference Jacobian over 4 params  $\times$  N data points, - solve the  $4 \times 4$  normal equations with Marquardt damping, - try the step; accept if  $\chi^2$  decreased, otherwise raise  $\lambda$  and retry.*

*After the run header you get (the iteration trace below is illustrative – it shows the SHAPE of an LM run, accept/reject and  $\lambda$  adaptation; the actual run converges to  $\chi^2 = 1.0426$ , RMS = 0.3079 K with fitted  $a_{ij} = 9.328$ ,  $b_{ij} = -3609.18$ ):*

**What to expect (golden-locked):**

| kind | unit/stream            | key             | expected    |
|------|------------------------|-----------------|-------------|
| diag | fit_NRTL_ethanol_water | chi2            | 1.04279     |
| diag | fit_NRTL_ethanol_water | chi2_reduced    | 0.14897     |
| diag | fit_NRTL_ethanol_water | cond_JtJ        | 3.55369e+11 |
| diag | fit_NRTL_ethanol_water | converged       | 1           |
| diag | fit_NRTL_ethanol_water | corr_computed   | 1           |
| diag | fit_NRTL_ethanol_water | dof             | 7           |
| diag | fit_NRTL_ethanol_water | identifiable    | 0           |
| diag | fit_NRTL_ethanol_water | invertible      | 1           |
| diag | fit_NRTL_ethanol_water | iter            | 24          |
| diag | fit_NRTL_ethanol_water | max_abs_corr    | 0.999967    |
| diag | fit_NRTL_ethanol_water | max_abs_resid   | 0.625729    |
| diag | fit_NRTL_ethanol_water | n_data          | 11          |
| diag | fit_NRTL_ethanol_water | n_params        | 4           |
| diag | fit_NRTL_ethanol_water | params_at_bound | 0           |

*... and 6 more reference values in the case's expected.*

**fitNRTL02\_thermoml\_isobars**

tutorials/steady/optimisation/fitNRTL02\_thermoml\_isobars

|                                         |                                              |                                                        |
|-----------------------------------------|----------------------------------------------|--------------------------------------------------------|
| <b>fit_NRTL_across</b><br>fitParameters | <b>score_catalogue_pair</b><br>fitParameters | <b>score_catalogue_pair_at_vacuum</b><br>fitParameters |
|-----------------------------------------|----------------------------------------------|--------------------------------------------------------|

**What it does:** Fit the ethanol/water NRTL pair across three measured isobars, then predict a fourth held out from a different laboratory

**The story:** fitNRTL02 – A GOOD FIT THAT PREDICTS WORSE. The ethanol/water NRTL pair regressed across three measured vacuum isobars, then asked about a fourth isobar, at atmospheric pressure, from a different laboratory.

THE RESULT FIRST, BECAUSE IT IS NOT THE ONE THE CASE WAS BUILT EXPECTING. The regression converges in 19 iterations to a reduced chi-square of 0.023 over 45 points – an in-sample rms of 0.144 K, which is at the scatter of the measurements. Every number on the fit's own report says it went well. Then the fitted pair meets 21 points it never saw, at 101.3 kPa, and its average absolute deviation is 0.386 K. The pair the case STARTED from – the catalogue's DECHEMA set, adjusted by nothing – manages 0.082 K on the same 21 points. The fit made the model 4.7 times worse where it was not looking.

The two control operations below make that symmetric rather than a one-sided accusation. All four numbers, each measured by this run:

vacuum data atmospheric data (Voutsas, 13-33 kPa) (Kamihama, 101.3 kPa) catalogue pair rms 0.1619 K  
aad 0.0821 K pair fitted here rms 0.1438 K (in-sample) aad 0.3857 K

Each pair is at its best where its evidence was taken. The regression bought an 11 % improvement where it looked, and lost a factor of 4.7 where it did not. A student who reads only 'converged 1', 'chi2\_reduced 0.023' and 'rms 0.144 K' would ship the new pair; the held-out column is the only thing in the run that says not to.

WHY IT HAPPENS, AND THE SECOND LESSON. NRTL's interaction is  $\tau = a + b/T$ , so identifying 'a' and 'b' separately needs data that span  $1/T$ . Three isobars from 13 to 33 kPa span about 28 K around 320 K, which moves  $1/T$  by roughly 9 % – better than the 12 K of a single isobar (fitNRTL01), and it shows: the condition number of J'J falls from  $3.6e11$  to  $5.6e9$ , two orders of magnitude. But 'max|correlation|' is still 1.000 and the run still reports 'identifiable 0'. Two orders of magnitude of conditioning did NOT buy identifiability, and the confidence intervals say why:  $b_{ij} = -539 \pm 456$ ,  $b_{ji} = 682 \pm 768$ . The parameters are a correlated pair that happens to describe the vacuum data; nothing pins either one, so nothing constrains what they do at 101 kPa. The catalogue record's own declared validity window is 298-373 K – it was fitted where it is being asked, and this fit was not.

THE EVIDENCE, AND WHERE IT CAME FROM. Both datasets were read out of the NIST/TRC ThermoML Archive by 'bin/choupo-thermoml extract-vle' and carry the citation of the article they were measured in, as a field the engine reads rather than a comment it discards:

FIT Voutsas, Pamouktsis, Argyris & Pappa, Fluid Phase Equilib. (2011), doi:10.1016/j.fluid.2011.06.009 – 45 points on three isobars, 13.15 / 19.71 / 32.86 kPa.

HELD OUT Kamihama, Matsuda, Kurihara, Tochigi & Oba, J. Chem. Eng. Data (2012), doi:10.1021/je2008704 – 21 points at 101.3 kPa.

The held-out set is not a slice withheld from the same measurement: it is a DIFFERENT LABORATORY, a different journal, a different year, at a pressure a full decade above anything the fitter was shown. That is what makes its number mean something the in-sample one cannot. The identity collision the engine CAN settle (the same DOI in both roles) is settled by construction here, because the DOIs differ; whether two studies are scientifically independent stays a human judgement, and the run announces what it knows rather than adjudicating it.

ITS NEIGHBOUR, AND WHAT IS DIFFERENT. 'tutorials/props/curation/curate02\_vle\_heldout\_ethanol\_water' already holds out three of eleven measured points on this same system, and already reports the same identifiability finding – it was the first place in the corpus where a fitted binary pair met real evidence it had not been handed, and this case does not claim that. What is new here is the SHAPE of the test: curate02 withholds points from ONE series at ONE pressure, so it asks whether the model interpolates

within a study. This case withholds an ENTIRE INDEPENDENT STUDY at a pressure a decade away, which asks whether the model transfers at all – and the two questions get different answers. Read curated02 first; it is the cleaner statement of the contract.

NO ACCEPTANCE BAND IS DECLARED, deliberately. This run REPORTS the held-out residual and claims no pass or fail on it, because a limit chosen after seeing the residual is not a limit. Every number quoted above is pinned in ‘expected’, so none of them can move unnoticed – including the ones that make the fit look bad. Nothing here was tuned to produce this outcome: the fit is the catalogue pair as its starting point, the bounds are fitNRTL01’s, and the two datasets were the only ethanol+water bubble-temperature studies in the mirror that suited the roles.

WHAT THE RUN’S TWO CAVEATS ARE ABOUT, said plainly. The end-of-run block reports ethanol’s Antoine correlation evaluated at 371.8 K and 372.8 K, outside its declared 273-369 K window. Both are the bubble-point Newton’s STARTING temperature – the mole-fraction-weighted normal boiling point of the first datum in each set – and neither is a state any published answer contains. The mechanism that would say so (‘AdvisoryFrame’) is opened around this operation’s parameter search but NOT yet inside the bubble-point Newton itself, so those two lines are reported as though they qualified the answer. That gap is named in docs/design/advisory-attribution.md rather than papered over here.

*Regress the ethanol/water NRTL pair against 45 bubble points measured on three vacuum isobars, then ask the fitted pair about 21 points at atmospheric pressure, measured by a different laboratory in a different year. Both datasets come out of the NIST/TRC ThermoML Archive and carry the citation of the article they were published in.*

*The regression converges in 19 iterations to a reduced chi-square of 0.023 over 45 points — an in-sample rms of 0.144 K, which is at the scatter of the measurements. Everything on the fit’s own report says it went well.*

*The regression bought an 11 % improvement where it looked, and lost a factor of 4.7 where it did not. A reader who saw only converged 1, chi2\_reduced 0.023 and rms 0.144 K would ship the new pair.*

*Each pair is at its best where its evidence was taken. The two evaluate operations in system/propsDict are what make that symmetric rather than an accusation: they score the \*catalogue\* pair on both datasets, so all four numbers come from this one run.*

*NRTL’s interaction is  $\tau = a + b/T$ , so separating  $a$  from  $b$  needs data that span  $1/T$ . Three isobars from 13 to 33 kPa span about 28 K around 320 K — better than the 12 K of the single-isobar twin (fitNRTL01), and it shows: the condition number of  $J'J$  falls from  $3.6e11$  to  $5.6e9$ . But  $\max$  is still 1.000 and the run still reports identifiable 0. Two orders of magnitude of conditioning did not buy identifiability, and the confidence intervals say why —  $b_{ij} = -539 \pm 456$ ,  $b_{ji} = 682 \pm 768$ . The parameters are a correlated pair that happens to describe the vacuum data; nothing pins either one, so nothing constrains what they do at 101 kPa.*

*The catalogue record’s own declared validity window is 298–373 K. It was fitted where it is being asked; this fit was not.*

#### What to expect (golden-locked):

| kind                                                             | unit/stream             | key             | expected        |
|------------------------------------------------------------------|-------------------------|-----------------|-----------------|
| diag                                                             | fit_NRTL_across_isobars | aad_heldout_K   | 0.3857          |
| diag                                                             | fit_NRTL_across_isobars | aad_heldout_pct | 0.1088          |
| diag                                                             | fit_NRTL_across_isobars | chi2            | 0.9303          |
| diag                                                             | fit_NRTL_across_isobars | chi2_in_sample  | 0.9303          |
| diag                                                             | fit_NRTL_across_isobars | chi2_reduced    | 0.0227          |
| diag                                                             | fit_NRTL_across_isobars | cond_JtJ        | 5604973889.6716 |
| diag                                                             | fit_NRTL_across_isobars | converged       | 1.0000          |
| diag                                                             | fit_NRTL_across_isobars | corr_computed   | 1.0000          |
| diag                                                             | fit_NRTL_across_isobars | dof             | 41.0000         |
| diag                                                             | fit_NRTL_across_isobars | identifiable    | 0.0000          |
| diag                                                             | fit_NRTL_across_isobars | invertible      | 1.0000          |
| diag                                                             | fit_NRTL_across_isobars | iter            | 19.0000         |
| diag                                                             | fit_NRTL_across_isobars | max_abs_corr    | 0.9995          |
| diag                                                             | fit_NRTL_across_isobars | max_abs_resid   | 0.3678          |
| ... and 32 more reference values in the case’s <i>expected</i> . |                         |                 |                 |



optim01\_column\_reflux

tutorials/steady/optimisation/optim01\_column\_reflux



**What it does:** Optimise reflux ratio + feedStage of benzene/toluene column to minimise reboiler vapour V\_strip

**The story:** Benzene/toluene distillation, used by the optimisation driver. Same column as sensitivity01\_column\_reflux; we add looser tolerances on the outer loop so the simplex can probe odd feedStages without blowing up.

**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected        |
|--------|-------------|-----------|-----------------|
| stream | bottoms     | F         | 0.0138888888889 |
| stream | distillate  | F         | 0.0138888888889 |
| stream | feed        | F         | 0.0277777777778 |
| kpi    | column01    | B         | 0.0138888888889 |
| kpi    | column01    | D         | 0.0138888888889 |
| kpi    | column01    | L_rect    | 0.0416666666667 |
| kpi    | column01    | L_strip   | 0.0694444444444 |
| kpi    | column01    | R         | 3               |
| kpi    | column01    | T_bottom  | 383.299688974   |
| kpi    | column01    | T_top     | 353.770171691   |
| kpi    | column01    | V_rect    | 0.0555555555556 |
| kpi    | column01    | V_strip   | 0.0555555555556 |
| kpi    | column01    | feedStage | 9               |
| kpi    | column01    | nStages   | 15              |

...and 15 more reference values in the case's expected.

**optim02\_process\_cost**

tutorials/steady/optimisation/optim02\_process\_cost



**What it does:** Minimise total module Guthrie cost of reactor + heater by varying reactor  $V_R$  and heater  $Q$

**The story:** Same reactor → heater → separator flowsheet as process02\_with\_design. The optimisation driver varies two design variables:

reactor  $V_R$  in  $[0.002, 0.010]$  m<sup>3</sup> heater  $Q$  in  $[0, 500]$  W (credo-pure: hardware, not outlet  $T$ )

In v0.34 the heater is credo-pure: it specifies  $Q$  (the absolute thermal power, hardware), not  $T_{out}$  (which would be a DesignSpec disguised as a unit-op parameter). The Nelder-Mead simplex therefore varies the heater's  $Q$  directly; the heater's  $T_{out}$  becomes a KPI the student can plot, not a control variable.

Initial guesses are placeholders – overwritten by Nelder-Mead.

*A small reactor → heater → separator process. The OptimizationDriver (Nelder-Mead) tunes two design variables to minimise the total Guthrie module cost of the two costed units (reactor + heater):*

*The heater is credo-pure: it specifies  $Q$  (the thermal power, hardware), not an outlet  $T$ .  $T_{out}$  is a reported KPI, not a control variable.*

*The optimum is not at the lower corner. Nelder-Mead converges (8 moves, simplexInit 0.15) to:*

*Increasing  $Q$  monotonically lowers the objective: the simplex walks the cost from  $\sim 2.0e6$  EUR at  $Q \approx 170$  W down to  $\sim 9.35e5$  EUR at  $Q = 500$  W.*

*The dominant cost is the heat exchanger ( $\sim 916$  k of the  $\sim 935$  k total). Its area follows  $A = Q / (U \cdot \text{LMTD}) = Q / 6000$  m<sup>2</sup>, so  $Q = 500$  W gives only  $A \approx 0.083$  m<sup>2</sup>. The Turton shell-and-tube correlation used by the Guthrie model is valid for  $A \in [10, 1000]$  m<sup>2</sup>; our exchanger is two orders of magnitude below  $S_{min}$ . In that extrapolated tail the correlation*

*turns \*upward\* as  $A$  shrinks (the quadratic term dominates for  $\log_{10} A < 0$ ), so a larger area is cheaper. The optimiser correctly exploits this: pushing  $Q$  to its upper bound grows  $A$  toward the correlation's valid range and slides down the cost curve. The reactor volume settles at an interior  $0.00807$  m<sup>3</sup> — large enough for adequate conversion ( $X \approx 0.64$ ), small enough that its vessel cost ( $\sim 19$  k) stays a rounding error next to the exchanger.*

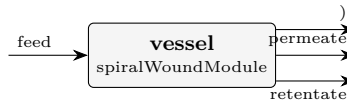
**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected           |
|--------|-------------|-----------|--------------------|
| stream | feed        | F         | 0.0002777777777778 |
| stream | hotStream   | F         | 0.0002777777777778 |
| stream | liqProd     | F         | 0.000257419353963  |
| stream | reactorOut  | F         | 0.0002777777777778 |
| stream | vapProd     | F         | 2.03584238147e-05  |
| kpi    | heater      | F         | 0.0002777777777778 |
| kpi    | heater      | P         | 101325             |
| kpi    | heater      | Q         | 500                |
| kpi    | heater      | Q_kW      | 0.5                |
| kpi    | heater      | Q_per_mol | 1800               |
| kpi    | heater      | T_in      | 350                |
| kpi    | heater      | T_out     | 361.533466984      |
| kpi    | heater      | dT        | 11.5334669843      |
| kpi    | heater      | vf_out    | 0.0112269153032    |

... and 37 more reference values in the case's *expected*.

## optim04\_membrane\_recovery\_scaling

tutorials/steady/optimisation/optim04\_membrane\_recovery\_scaling



**What it does:** Brackish RO: maximise water recovery up to the scaling limit (SQP, scaling constraints active at the optimum)

**The story:** HOW HARD can we push recovery before the membrane scales?

The flagship CONSTRAINED-OPTIMISATION case. Same brackish-RO vessel and scaling physics as tutorials/steady/membranes/membrane07\_scaling\_si (6 x 8-inch elements, the ~1500 mg/L groundwater analysis, the bulk-vs-wall saturation-index audit), but now we ASK the engine the design question the audit only diagnosed:

maximise water\_recovery varying feed pressure and the (continuous) module count subject to NO mineral scaling at the membrane wall:  $\text{maxSI\_gypsum\_wall} \leq 0$   $\text{maxSI\_calcite\_wall} \leq 0$  (rigorous Davies saturation index – smoother than the Stiff-Davis/Langelier chart fit)

Recovery and wall saturation pull AGAINST each other: every extra % recovery concentrates the retentate and drives the WALL composition (bulk x polarisation) further into supersaturation. The optimum therefore sits ON the scaling limit – at least one of the two constraints is ACTIVE, with a strictly positive Lagrange multiplier (the shadow price: how much more recovery one unit of scaling head-room would buy). That is exactly what the SQP trace prints.

GOLDEN-CHECKED regression: the SQP now converges to a CERTIFIED KKT optimum ( $\|\text{grad } L\|_{\text{inf}} \sim 3\text{e-}8$ , all feasibility 0, complementarity  $\sim 1\text{e-}8$ , calcite-wall constraint ACTIVE with  $\lambda \sim 1.14 > 0$ ). Because this is a certified stationary point – not a guessed answer – its KPIs are pinned in ‘expected’, which ALSO guards against a regression to the singular-QP failure a readiness review caught earlier (a "QP subproblem failed: KKT system singular – LICQ failure" at iteration 1 that left the design infeasible). The KPIs still inherit the polarisation + Davies model assumptions; the golden locks the answer THIS model gives at the certified optimum, not an independent ground truth. (If the engine ever fails to converge again, the optimum block prints an explicit "NO KKT certificate ... NOT a certified optimum" block – never the misleading all-zeros certificate it used to forge on the failure path.)

Two design variables, written through \$references so the SQP can drive them and the trace shows their canonical-SI values.

### What to expect (golden-locked):

| kind   | unit/stream | key                  | expected          |
|--------|-------------|----------------------|-------------------|
| stream | feed        | F                    | 0.194444444431    |
| stream | permeate    | F                    | 0.0842121209871   |
| stream | retentate   | F                    | 0.110232483957    |
| kpi    | vessel      | J_w_avg              | 5.94313791976e-06 |
| kpi    | vessel      | LSI_calcite_bulk_end | 0.408647968153    |
| kpi    | vessel      | LSI_calcite_wall_end | 0.592350216873    |
| kpi    | vessel      | RSI_calcite_wall_end | 5.81529956625     |
| kpi    | vessel      | R_obs_Ca             | 0.999395465329    |
| kpi    | vessel      | R_obs_Cl             | 0.993987079839    |
| kpi    | vessel      | R_obs_HCO3           | 0.995483580386    |
| kpi    | vessel      | R_obs_K              | 0.992495032466    |
| kpi    | vessel      | R_obs_Mg             | 0.999546531069    |
| kpi    | vessel      | R_obs_Na             | 0.993987079839    |
| kpi    | vessel      | R_obs_SO4            | 0.999697642081    |

... and 29 more reference values in the case's *expected*.

**optim05\_reactor\_npv**

tutorials/steady/optimisation/optim05\_reactor\_npv



**What it does:** Esterification CSTR sized & operated for MAX NPV by hand-rolled SQP, subject to a reactor-size cap and a capital-budget cap (active-constraint shadow prices)

**The story:** CONSTRAINED ECONOMIC optimisation: size & operate an esterification CSTR plant for MAXIMUM NPV.

Same esterification plant as the economics01 flagship –

feed → reactor (CSTR) → heater → separator (flash) → vapProd (ethyl-acetate-rich | overhead, the prime | product @ 1.55 EUR/kg) → liqProd (water/acid-rich bottoms, low-value byproduct @ 0.30 EUR/kg)

– but instead of a FIXED design we ask the engine the design question:

maximise economics.NPV varying  $V_R$  (reactor volume,  $m^3$ )  $T_{\text{feed}}$  (feed/reactor T, K) subject to  $\text{reactor.V}_R \leq V_{\text{max}}$  (a physical SIZE cap)  $\text{economics.FCI} \leq \text{FCI}_{\text{max}}$  (a CAPITAL-BUDGET cap)

THE ECONOMIC TRADE-OFF (why this is a genuine optimum, not a corner)

The heater runs at a FIXED partial duty, so the downstream flash SPLITS the reactor effluent into a light, ethyl-acetate-rich VAPOUR overhead (sold high) and a heavy, water/acid-rich LIQUID bottoms (sold cheap). Esterification conserves total mass; what conversion buys you is the RESHUFFLING of that mass toward the valuable overhead. Both knobs raise conversion – and the CSTR is ISOTHERMAL at the feed temperature (CSTR.cpp:  $T = \text{feed.T}$ ), so:

$V_R$  (reactor volume) – bigger  $V_R$  raises the residence time  $\tau = V_R/Q$ , so conversion  $X = k\tau/(1 + k\tau)$  climbs, more acetic acid + ethanol turn into ethyl acetate, and the priced overhead grows and enriches (REVENUE up). But a bigger vessel costs more steel: the StirredTank sizing → Guthrie costing drives FCI (CAPEX) UP. Revenue rises with DIMINISHING returns as  $X$  saturates toward 1, while CAPEX keeps climbing – so an NPV maximum exists.

$T_{\text{feed}}$  (feed/reactor temperature) – higher  $T$  raises the Arrhenius rate constant  $k = A \cdot \exp(-E_a/RT)$ , so  $X$  climbs too.  $T_{\text{feed}}$  is the cheap conversion ACCELERATOR; its bounds are kept inside the window where the flash splits cleanly (below the temperature that boils the bottoms dry).

We deliberately set the size cap TIGHTER than where the NPV-greedy design would otherwise stop, so at the optimum the reactor-size constraint BINDS.

HOW TO READ THE SHADOW PRICES (the whole point of the tutorial)

The SQP prints, at the certified optimum, a Lagrange multiplier ( $\lambda$ ) for each inequality. A constraint with  $\lambda > 0$  is ACTIVE (binding) and its multiplier is the SHADOW PRICE – the marginal NPV you would gain by relaxing that cap by one unit:

reactor. $V_R$  cap ACTIVE →  $\lambda = d(\text{NPV})/d(V_{\text{max}})$  [EUR of NPV per extra  $m^3$  of reactor head-room]. economics.FCI cap →  $\lambda = d(\text{NPV})/d(\text{FCI}_{\text{max}})$  [EUR of NPV per extra EUR of capital you may spend].

A constraint with  $\lambda \sim 0$  is SLACK (not binding) – relaxing it buys nothing. Here the FCI cap is deliberately generous: it stays SLACK so the student sees BOTH an active and an inactive constraint side by side. The trace marks the active set and the KKT certificate proves the point is a true constrained optimum ( $\| \text{grad } L \| \sim 0$ , complementarity  $\sim 0$ ).

The two design variables are case-level \$references so the SQP can drive them through ‘variables.<name>’ and the trace shows their canonical-SI values.

THE ECONOMICS INVERTED ON 2026-08-09, AND THE OPTIMISER DID NOT

Everything above about the SQP is unchanged and still true: the run converges, the reactor-size cap is ACTIVE with a positive multiplier, the FCI cap stays SLACK, and the KKT certificate closes. What changed is the SIGN of the money.

The ‘heater’ used to invert its fixed 200 kW on a SENSIBLE enthalpy relation, which walked the stream 11.8 K to 364.27 K without paying any latent heat. The flash then found 98.8 % vapour, and the ethyl-acetate-rich overhead – the whole revenue of this case – existed because the heater had crossed the bubble point for free. The heater now solves for the outlet STATE whose enthalpy is  $H_{in} + Q$ : 200 kW reaches 356.55 K and 7.7 % vapour, the flash splits V/F 0.077, revenue falls 321.7  $\rightarrow$  88.2 M EUR against an essentially unchanged 221 M EUR cost of manufacture, and the optimum NPV is -646 M EUR (IRR and payback are UNDEFINED and published as null – the project never breaks even).

So this case still demonstrates constrained SQP, shadow prices and a KKT certificate, and it now demonstrates them on a project that loses money. Restoring the intended economics is a DESIGN decision, not a code fix: the duty (or the pressure, or the product spec) has to be chosen so the overhead is actually made. It is deliberately NOT chosen here – picking a number to make the old answer come back would be fitting the case to a result nobody re-derived. Record: docs/design/tp-stream-energy-coherence.md.

*Take the economics01 ethyl-acetate plant and stop \*fixing\* the design — let the engine choose it. The hand-rolled SQP sizes and operates the reactor to maximise the net present value, subject to inequality caps, and prints the shadow prices (Lagrange multipliers) of the binding caps.*

*The objective economics.NPV is a KPI published by the EconomicsPass in the system/postDict chain (sizing  $\rightarrow$  costing  $\rightarrow$  economics). The optimiser runs that whole post chain on every evaluation, so the economics scalars (economics.NPV, economics.FCI, ?) are reachable as ordinary objective/constraint paths — exactly like a sweep reads them.*

*The heater runs at a fixed partial duty, so the flash \*splits\* the reactor effluent into a light ethyl-acetate-rich vapour overhead (the prime product at 1.55 ?/kg) and a heavy water/acid-rich liquid bottoms (a low-value byproduct at 0.30 ?/kg). Esterification conserves total mass; what conversion buys you is the reshuffling of that mass toward the valuable overhead.*

*\*  $V_R$  — a bigger reactor raises the residence time  $\tau = V_R/Q$ , so the isothermal-CSTR conversion  $X = k\tau/(1+k\tau)$  climbs and the priced overhead grows. But a bigger vessel costs more steel — the StirredTank sizing  $\rightarrow$  Guthrie costing drives FCI (CAPEX) up. Revenue rises with diminishing returns ( $X$  saturates toward 1) while CAPEX keeps climbing  $\rightarrow$  an NPV maximum exists. \*  $T_{feed}$  — a higher feed temperature raises the Arrhenius rate  $k = A \cdot \exp(-E^\ddagger/RT)$ , the cheap conversion \*accelerator\*. Its bounds stay inside the window where the flash splits cleanly (below the  $\sim 353$  K edge that boils the bottoms dry).*

*We deliberately set the size cap tighter than where the NPV-greedy design would otherwise stop, so the reactor-size constraint binds at the optimum.*

*The optimiser drives the money-losing initial design ( $V_R = 4 \text{ m}^3$ ,  $T_{feed} = 350 \text{ K} \rightarrow \text{NPV} \approx -164 \text{ M?}$ ) to a profitable, certified optimum:*

#### What to expect (golden-locked):

| kind   | unit/stream | key     | expected       |
|--------|-------------|---------|----------------|
| stream | feed        | F       | 0.138888888889 |
| stream | hotStream   | F       | 0.138888888889 |
| stream | liqProd     | F       | 0.126262849909 |
| stream | reactorOut  | F       | 0.138888888889 |
| stream | vapProd     | F       | 0.01262603898  |
| kpi    | economics   | COM_d   | 221248623.305  |
| kpi    | economics   | FCI     | 522154.982187  |
| kpi    | economics   | NPV     | -632834040.588 |
| kpi    | economics   | TCI     | 600478.229515  |
| kpi    | economics   | WC      | 78323.2473281  |
| kpi    | economics   | revenue | 90983516.5487  |
| kpi    | heater      | F       | 0.138888888889 |
| kpi    | heater      | P       | 101325         |
| kpi    | heater      | Q       | 200000         |

*... and 43 more reference values in the case's **expected**.*

optim06\_integer\_stages

tutorials/steady/optimisation/optim06\_integer\_stages



**What it does:** The classic N-vs-R trade solved in ONE run: nStages declared integer → exhaustive enumeration, one full SQP per stage count, ALL values on the table

**The story:** sensitivity01\_column\_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T\_top, T\_bottom, x\_D\_LK, x\_B\_HK and V\_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected        |
|--------|-------------|----------------|-----------------|
| stream | bottoms     | F              | 0.0138888888889 |
| stream | distillate  | F              | 0.0138888888889 |
| stream | feed        | F              | 0.0277777777778 |
| kpi    | column01    | B              | 0.0138888888889 |
| kpi    | column01    | D              | 0.0138888888889 |
| kpi    | column01    | L_rect         | 0.0313732166828 |
| kpi    | column01    | L_strip        | 0.0591509944605 |
| kpi    | column01    | Q_condenser_kW | -1391.66557074  |
| kpi    | column01    | Q_reboiler_kW  | 1389.90800504   |
| kpi    | column01    | R              | 2.25887160116   |
| kpi    | column01    | T_bottom       | 382.835800828   |
| kpi    | column01    | T_top          | 354.273383646   |
| kpi    | column01    | V_rect         | 0.0452621055716 |
| kpi    | column01    | V_strip        | 0.0452621055716 |

...and 18 more reference values in the case's expected.

pareto01\_purity\_energy

tutorials/steady/optimisation/pareto01\_purity\_energy



**What it does:** The purity-energy PARETO FRONT by epsilon-constraint: minimise V\_strip subject to x\_D\_LK >= eps, eps swept 0.96..0.994 – each front point one auditable SQP run

**The story:** sensitivity01\_column\_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T\_top, T\_bottom, x\_D\_LK, x\_B\_HK and V\_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

What to expect (golden-locked):

| kind                                                    | unit/stream | key            | expected        |
|---------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                  | bottoms     | F              | 0.0138888888889 |
| stream                                                  | distillate  | F              | 0.0138888888889 |
| stream                                                  | feed        | F              | 0.0277777777778 |
| kpi                                                     | column01    | B              | 0.0138888888889 |
| kpi                                                     | column01    | D              | 0.0138888888889 |
| kpi                                                     | column01    | L_rect         | 0.052352377807  |
| kpi                                                     | column01    | L_strip        | 0.0801301555848 |
| kpi                                                     | column01    | Q_condenser_kW | -2035.41756863  |
| kpi                                                     | column01    | Q_reboiler_kW  | 2033.86732819   |
| kpi                                                     | column01    | R              | 3.7693712021    |
| kpi                                                     | column01    | T_bottom       | 383.489767144   |
| kpi                                                     | column01    | T_top          | 353.562992445   |
| kpi                                                     | column01    | V_rect         | 0.0662412666959 |
| kpi                                                     | column01    | V_strip        | 0.0662412666959 |
| ...and 18 more reference values in the case's expected. |             |                |                 |

sensitivity01\_column\_reflux

tutorials/steady/optimisation/sensitivity01\_column\_reflux



**What it does:** Sweep reflux ratio  $R = 1.5..5.0$  (10 pts) across the column01 case; responses:  $x_D$  purity +  $V_{strip}$  (reboiler-duty proxy)

**The story:** the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over  $R = 1.5 \dots 5.0$  (10 points) and reads  $T_{top}$ ,  $T_{bottom}$ ,  $x_D_{LK}$ ,  $x_B_{HK}$  and  $V_{strip}$  (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per  $R$ .

**What to expect (golden-locked):**

| kind                                                    | unit/stream | key            | expected        |
|---------------------------------------------------------|-------------|----------------|-----------------|
| stream                                                  | bottoms     | F              | 0.0138888888889 |
| stream                                                  | distillate  | F              | 0.0138888888889 |
| stream                                                  | feed        | F              | 0.0277777777778 |
| kpi                                                     | column01    | B              | 0.0138888888889 |
| kpi                                                     | column01    | D              | 0.0138888888889 |
| kpi                                                     | column01    | L_rect         | 0.0424382716049 |
| kpi                                                     | column01    | L_strip        | 0.0702160493827 |
| kpi                                                     | column01    | Q_condenser_kW | -1730.95769425  |
| kpi                                                     | column01    | Q_reboiler_kW  | 1729.37389301   |
| kpi                                                     | column01    | R              | 3.05555555556   |
| kpi                                                     | column01    | T_bottom       | 383.383107764   |
| kpi                                                     | column01    | T_top          | 353.679332066   |
| kpi                                                     | column01    | V_rect         | 0.0563271604938 |
| kpi                                                     | column01    | V_strip        | 0.0563271604938 |
| ...and 15 more reference values in the case's expected. |             |                |                 |



**sensitivity02\_feasibility\_map**

tutorials/steady/optimisation/sensitivity02\_feasibility\_map



**What it does:** Feasibility MAP: the reflux sweep with declared constraints – every row marked (value, bound, residual, satisfied), nothing filtered; the feasible window between the purity spec and the boilup budget read off the CSV

**The story:** sensitivity01\_column\_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over  $R = 1.5 \dots 5.0$  (10 points) and reads  $T_{\text{top}}$ ,  $T_{\text{bottom}}$ ,  $x_{\text{D\_LK}}$ ,  $x_{\text{B\_HK}}$  and  $V_{\text{strip}}$  (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per  $R$ .

**What to expect (golden-locked):**

| kind   | unit/stream | key            | expected         |
|--------|-------------|----------------|------------------|
| stream | bottoms     | F              | 0.01388888888889 |
| stream | distillate  | F              | 0.01388888888889 |
| stream | feed        | F              | 0.02777777777778 |
| kpi    | column01    | B              | 0.01388888888889 |
| kpi    | column01    | D              | 0.01388888888889 |
| kpi    | column01    | L_rect         | 0.0424382716049  |
| kpi    | column01    | L_strip        | 0.0702160493827  |
| kpi    | column01    | Q_condenser_kW | -1730.95769425   |
| kpi    | column01    | Q_reboiler_kW  | 1729.37389301    |
| kpi    | column01    | R              | 3.055555555556   |
| kpi    | column01    | T_bottom       | 383.383107764    |
| kpi    | column01    | T_top          | 353.679332066    |
| kpi    | column01    | V_rect         | 0.0563271604938  |
| kpi    | column01    | V_strip        | 0.0563271604938  |

... and 18 more reference values in the case's *expected*.

**power**

**The theory you need.** Gas (Brayton) and steam (Rankine) cycles, alone and combined. Compressor and turbine hardware is specified by SHAFT WORK and isentropic efficiency – outlet pressure is a RESULT, not an input (the anti-DesignSpec credo). Expect: pressure ratio setting Brayton efficiency, the combined cycle's gain coming from bottoming the gas turbine's exhaust exergy, and every energy flow visible in the ledger. *Full derivation: Theory Guide, the power-cycle chapter.*

brayton01\_gas\_turbine

tutorials/steady/power/brayton01\_gas\_turbine



**What it does:** Closed Brayton (gas-turbine) cycle: compress air → combust (gas-aware heater) → expand to feed P

**The story:** a closed-loop Brayton (gas-turbine) cycle, the showcase for the gas-aware heater (v0.65).

feed (air, 1 bar, 298 K) → compressor (+10 kW, eta 0.80) : 1 → ~9.9 bar, 298 → 633 K → combustor (heater, +20 kW) : add heat at constant P; the air is gas (vf=1) so the heater integrates the ideal-gas H<sub>ig</sub> (N<sub>2</sub>/O<sub>2</sub> carry no liquid Cp) → 633 → ~1246 K → turbine (-W<sub>turb</sub>, eta 0.85) : expand back to 1.013 bar

The turbine shaft power is NOT guessed: a DesignSpec (system/outerDict) manipulates \$W<sub>turb</sub> until the turbine discharges at the feed pressure, closing the cycle. Net shaft work and thermal efficiency are computed \$variables (v0.64), evaluated after the solve from the unit KPIs.

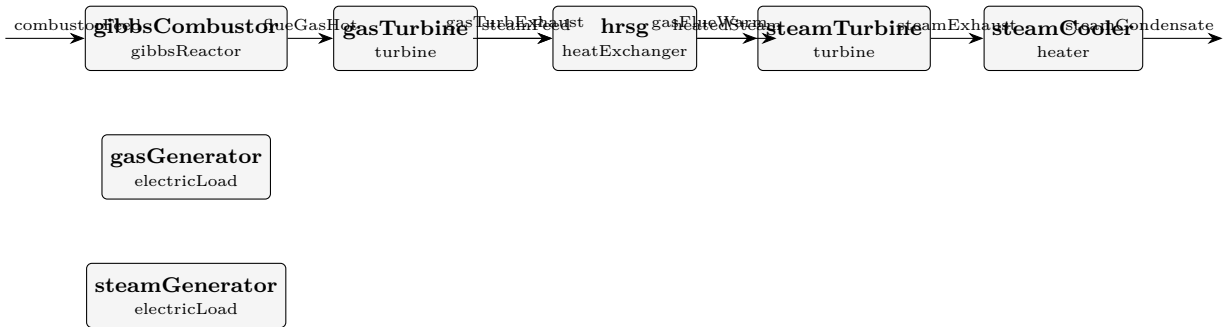
**What to expect (golden-locked):**

| kind   | unit/stream | key       | expected      |
|--------|-------------|-----------|---------------|
| stream | compressed  | F         | 0.001         |
| stream | exhaust     | F         | 0.001         |
| stream | feed        | F         | 0.001         |
| stream | hot         | F         | 0.001         |
| kpi    | combustor   | F         | 0.001         |
| kpi    | combustor   | P         | 991200.306811 |
| kpi    | combustor   | Q         | 20000         |
| kpi    | combustor   | Q_kW      | 20            |
| kpi    | combustor   | Q_per_mol | 20000         |
| kpi    | combustor   | T_in      | 633.374172563 |
| kpi    | combustor   | T_out     | 1246.40394079 |
| kpi    | combustor   | dT        | 613.029768231 |
| kpi    | compressor  | F         | 0.001         |
| kpi    | compressor  | P_in      | 100000        |

...and 38 more reference values in the case's *expected*.

combined01\_brayton\_rankine

tutorials/steady/power/combined01\_brayton\_rankine



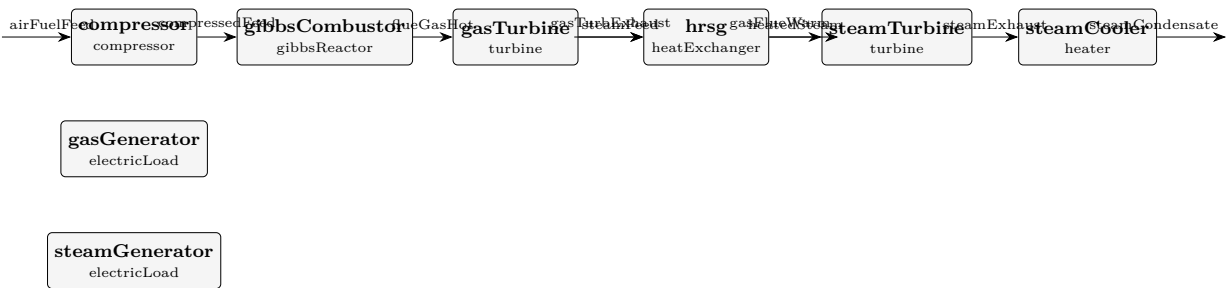
**What it does:** Combined-cycle: Brayton (CH4 + air, Gibbs combustor) + HRSG + Rankine + 2 generators

**What to expect (golden-locked):**

| kind                                                            | unit/stream     | key           | expected |
|-----------------------------------------------------------------|-----------------|---------------|----------|
| stream                                                          | combustorFeed   | F             | 0.1025   |
| stream                                                          | flueGasHot      | F             | 0.1025   |
| stream                                                          | gasFlueWarm     | F             | 0.1025   |
| stream                                                          | gasTurbExhaust  | F             | 0.1025   |
| stream                                                          | heatedSteam     | F             | 0.01     |
| stream                                                          | steamCondensate | F             | 0.01     |
| stream                                                          | steamExhaust    | F             | 0.01     |
| stream                                                          | steamFeed       | F             | 0.01     |
| kpi                                                             | gasGenerator    | W_electric    | 582000   |
| kpi                                                             | gasGenerator    | W_electric_kW | 582      |
| kpi                                                             | gasGenerator    | W_losses      | 18000    |
| kpi                                                             | gasGenerator    | W_shaft       | 600000   |
| kpi                                                             | gasGenerator    | W_shaft_kW    | 600      |
| kpi                                                             | gasGenerator    | eta_generator | 0.97     |
| ...and 98 more reference values in the case's <i>expected</i> . |                 |               |          |

combined02\_brayton\_rankine\_shaft

tutorials/steady/power/combined02\_brayton\_rankine\_shaft



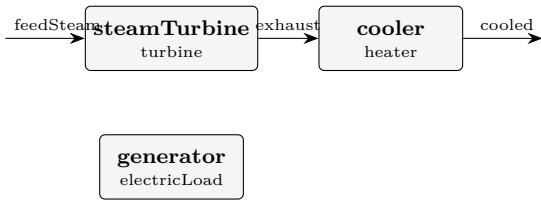
**What it does:** Combined cycle with SHAFT-SPLIT: gas turbine drives the compressor + net to the generator

**What to expect (golden-locked):**

| kind   | unit/stream     | key   | expected      |
|--------|-----------------|-------|---------------|
| stream | airFuelFeed     | F     | 0.1025        |
| stream | compressedFeed  | F     | 0.1025        |
| stream | flueGasHot      | F     | 0.1025        |
| stream | gasFlueWarm     | F     | 0.1025        |
| stream | gasTurbExhaust  | F     | 0.1025        |
| stream | heatedSteam     | F     | 0.01          |
| stream | steamCondensate | F     | 0.01          |
| stream | steamExhaust    | F     | 0.01          |
| stream | steamFeed       | F     | 0.01          |
| kpi    | compressor      | F     | 0.1025        |
| kpi    | compressor      | P_in  | 100000        |
| kpi    | compressor      | P_out | 828817.033759 |
| kpi    | compressor      | T_in  | 300           |
| kpi    | compressor      | T_out | 592.090873473 |

... and 115 more reference values in the case's expected.

**rankine01\_water**  
tutorials/steady/power/rankine01\_water



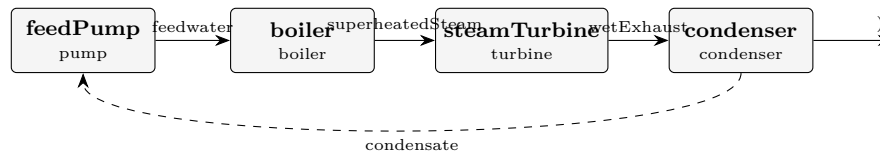
**What it does:** Simplified gas-phase Rankine: superheated steam → turbine (-10 kW) → cooler (gas-only, no phase change)

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key           | expected       |
|-----------------------------------------------------------------|-------------|---------------|----------------|
| stream                                                          | cooled      | F             | 0.001          |
| stream                                                          | exhaust     | F             | 0.001          |
| stream                                                          | feedSteam   | F             | 0.001          |
| kpi                                                             | cooler      | F             | 0.001          |
| kpi                                                             | cooler      | P             | 52827.3976295  |
| kpi                                                             | cooler      | Q             | -5000          |
| kpi                                                             | cooler      | Q_kW          | -5             |
| kpi                                                             | cooler      | Q_per_mol     | -5000          |
| kpi                                                             | cooler      | T_in          | 530.878445241  |
| kpi                                                             | cooler      | T_out         | 388.087132345  |
| kpi                                                             | cooler      | dT            | -142.791312896 |
| kpi                                                             | generator   | W_electric    | 9700           |
| kpi                                                             | generator   | W_electric_kW | 9.7            |
| kpi                                                             | generator   | W_losses      | 300            |
| ...and 30 more reference values in the case's <i>expected</i> . |             |               |                |

**rankine02\_water**

tutorials/steady/power/rankine02\_water



**What it does:** Closed IF97 Rankine cycle: pump → boiler → turbine → condenser → recycle

**The story:** a CLOSED Rankine steam cycle on IAPWS-IF97

recycle (condensate) [feedPump] [boiler] [steamTurbine] [condenser] raise P cross the extract work cross the dome to 80 bar dome up, (W\_shaft, eta), DOWN, land on (liquid) superheat exhaust ~0.1 bar saturated liquid

A real Rankine: the working fluid crosses the saturation dome TWICE – up in the boiler (subcooled liquid → superheated vapour) and down in the condenser (wet vapour → saturated liquid). Pure water on IF97 supplies the steam-table h/s/v surface, so:

the boiler is a dome-crossing phaseChanger driven to superheatedVapour (its duty Q\_boiler is the result), the turbine expands on the IF97 entropy surface (H\_real/S\_real routed), the condenser is a dome-crossing phaseChanger driven to saturatedLiquid (its duty Q\_condenser is the result), the pump returns the condensate to boiler pressure (closed-form v\*dP).

The loop is CLOSED: the condensate is the tear stream, recycled to the pump. Wegstein/Newton drives the post-pass condensate to match the guess.

First-law check (KPIs): Q\_boiler + Q\_condenser + W\_pump + W\_turbine ~ 0.

Contrast: rankine01\_water runs the SAME architecture on the ideal gas, in the gas phase only (no liquid leg, no phase change) – the pedagogical baseline. rankine02 is the steam-table truth.

**What to expect (golden-locked):**

| kind    | unit/stream      | key                          | expected       |
|---------|------------------|------------------------------|----------------|
| kpi     | boiler           | T_out                        | 773.15         |
| kpi     | boiler           | vf_out                       | 1.0            |
| kpi     | boiler           | Tsat                         | 568.159121229  |
| kpi     | boiler           | Q_kW                         | 47.8559413747  |
| kpi     | condenser        | vf_out                       | 0.0            |
| kpi     | condenser        | Tsat                         | 446.515894355  |
| kpi     | condenser        | Q_kW                         | -39.017165456  |
| kpi     | steamTurbine     | P_out                        | 858590.397228  |
| kpi     | steamTurbine     | W_gen_kW                     | 9.0            |
| stream  | superheatedSteam | vf                           | 1.0            |
| stream  | condensate       | vf                           | 0.0            |
| utility | boiler           | heating.-.unserved           | 1              |
| utility | condenser        | cooling.coolingWater.duty_kW | -39.0172479639 |
| utility | condenser        | cooling.coolingWater.T       | 446.515894355  |

... and 55 more reference values in the case's *expected*.

**reactors**

**The theory you need.** Batch reactors integrate  $dn_i/dt = \nu_i r V$  from a charged initial state – no in/out flows; time replaces residence time. Adiabatic runs couple the energy balance  $\sum_i n_i \bar{c}_{p,i} dT/dt = -\Delta H_{rxn} r V$ , so composition and temperature evolve together. Stiffness appears the moment fast and slow reactions coexist (radical chains:  $\mu s$  radical lifetimes vs ms induction) – the explicit RK4 integrator then needs  $\Delta t$  at the FASTEST scale or blows up, while Rosenbrock23 (L-stable, linearly implicit) steps at the scale of the SOLUTION. The stiffness ratio is printed; watch it. *Full derivation: Theory Guide, the batch/ODE chapter.*

**acetone02\_luyben\_reactor**

tutorials/steady/reactors/acetone02\_luyben\_reactor



**What it does:** Luyben's acetone reactor: IPA dehydrogenation at 90 % conversion, 623 K – outlet composition and duty against the published Rout

**The story:** the FIRST PROCESS slice of the acetone plant

W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a – the Turton base case of his Figure 1.

ONE UNIT, chosen because it is the piece of the flowsheet whose answer does NOT depend on the mixture thermodynamics. At 623 K and 2.6 atm everything is vapour; no activity model is consulted. What is exercised is stoichiometry, the elements-datum heat of reaction and the energy balance – and those can be held against published numbers without first solving the harder problem the separations pose (acetone01\_ipa\_water\_azeotrope measures how hard).

THE FEED IS DERIVED FROM THE PAPER'S OWN STREAM TABLE, and the derivation is a check in itself. Luyben gives  $R_{in} = 57.83$  kmol/h at 389 K, 2.6 atm and  $R_{out} = 92.6$  kmol/h with H<sub>2</sub> 0.3755, acetone 0.3755, IPA 0.0417, water 0.2073. Water is inert and hydrogen is produced one-for-one with acetone, so:

H<sub>2</sub> produced =  $92.6 \times 0.3755 = 34.77$  kmol/h = IPA converted IPA out =  $92.6 \times 0.0417 = 3.86$  kmol/h  
IPA in =  $34.77 + 3.86 = 38.63$  kmol/h water in =  $92.6 \times 0.2073 = 19.19$  kmol/h (inert) total in =  $38.63 + 19.19 = 57.82$  kmol/h vs the stated 57.83 conversion =  $34.77 / 38.63 = 90.0$  % vs the stated 90 %

Two independent closures on a table transcribed by hand. The feed in 0/ is those numbers; the conversion spec is the paper's 90 %.

WHAT IS BEING COMPARED (see 'expected'): the outlet composition, four mole fractions from Luyben's Rout; the reactor DUTY against his 0.960 MW.

THE DUTY IS THE INTERESTING ONE. Choupo derives the heat of reaction from each species' own formation enthalpy – it is never read from the reactions dict – and isopropanol's dHf here is a JOBACK ESTIMATE, because Choupo has no curated isopropanol. So the published duty tests a chain that runs from a group-contribution estimate of one molecule all the way to a megawatt of furnace heat. That is the whole point of building this rather than asserting it: a case says where in the chain the error entered.

NOT MODELLED, and named rather than implied: the reactor's SIZE. Luyben specifies 450 tubes with a 624 K molten-salt jacket, and the paper does not give the tube dimensions this record was built from, so a 'pfr' on his kinetics cannot be dimensioned from what is in hand. A 'conversionReactor' at his stated 90 % asks the thermodynamic question without pretending to the kinetic one. The kinetics (Luyben Table 1) are transcribed in the design record and remain unused here, on purpose.

**What to expect (golden-locked):**

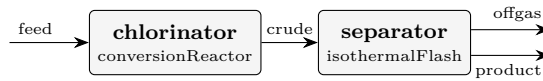
| kind   | unit/stream | key              | expected         |
|--------|-------------|------------------|------------------|
| stream | Rin         | F                | 0.01606111111111 |
| stream | Rout        | F                | 0.02571861111111 |
| kpi    | reactor     | F_in_kmol_h      | 57.82            |
| kpi    | reactor     | F_out_kmol_h     | 92.587           |
| kpi    | reactor     | Q_kW             | 840.263234652    |
| kpi    | reactor     | T                | 623              |
| kpi    | reactor     | conversion       | 0.9              |
| kpi    | reactor     | dHrxn_kJ_per_mol | 48.7173210118    |
| kpi    | reactor     | extent_kmol_h    | 34.767           |
| stream | Rout        | isopropanol      | 0.0417229200644  |
| stream | Rout        | acetone          | 0.375506280579   |
| stream | Rout        | H2               | 0.375506280579   |
| stream | Rout        | water            | 0.207264518777   |
| kpi    | reactor     | Q_reaction_kW    | 470.487527672    |

...and 3 more reference values in the case's *expected*.



**chlorophenol01\_chlorination**

tutorials/steady/reactors/chlorophenol01\_chlorination



**What it does:** Chlorophenol: classic phenol chlorination ( $\text{PhOH} + \text{Cl}_2 \rightarrow 4\text{-CP} + \text{HCl}$ ) + HCl flash

**The story:** the CLASSIC chlorophenol process.

Direct chlorination of phenol with chlorine gas – the industrial route to the chlorophenols (here the mono, 4-chlorophenol):

chlorinator :  $\text{C}_6\text{H}_5\text{OH} + \text{Cl}_2 \rightarrow \text{C}_6\text{H}_4\text{ClOH} + \text{HCl}$  (exothermic,  $\sim 130$  kJ/mol) separator : a flash at 80 C, 1 atm splits the volatile HCl (+ excess  $\text{Cl}_2$ ) off the liquid phenol / chlorophenol.

Feed: 1.1 kmol/h phenol + 1.0 kmol/h  $\text{Cl}_2$  (slight phenol excess  $\rightarrow$   $\text{Cl}_2$  limiting). phenol + 4-chlorophenol are JOBACK ESTIMATES (glass-box, unvalidated).

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected           |
|--------|-------------|---------------|--------------------|
| stream | crude       | F             | 0.0005833333333333 |
| stream | feed        | F             | 0.0005833333333333 |
| stream | offgas      | F             | 0.000275331266235  |
| stream | product     | F             | 0.000308002067098  |
| kpi    | chlorinator | F_in_kmol_h   | 2.1                |
| kpi    | chlorinator | F_out_kmol_h  | 2.1                |
| kpi    | chlorinator | T             | 353                |
| kpi    | chlorinator | conversion    | 0.9                |
| kpi    | chlorinator | extent_kmol_h | 0.9                |
| kpi    | separator   | F_alpha       | 0.000308002067098  |
| kpi    | separator   | F_beta        | 0.000275331266235  |
| kpi    | separator   | F_in          | 0.0005833333333333 |
| kpi    | separator   | K_Cl2         | 26.0852015979      |
| kpi    | separator   | K_HCl         | 122.317563764      |

... and 12 more reference values in the case's *expected*.

**conversion01\_hda**

tutorials/steady/reactors/conversion01\_hda



**What it does:** Conversion reactor: HDA toluene+H2  $\rightarrow$  benzene+CH4 at 75%/pass (gas-basis, no kinetics, no Vliq)

**The story:** A single gas-phase conversion reactor: toluene + H2  $\rightarrow$  benzene + CH4, 75 % of the limiting reactant (toluene) per pass. H2 is fed in excess (3:1), as in the real HDA.

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected       |
|--------|-------------|------------------|----------------|
| stream | feed        | F                | 0.111111111111 |
| stream | product     | F                | 0.111111111111 |
| kpi    | reactor     | F_in_kmol_h      | 400            |
| kpi    | reactor     | F_out_kmol_h     | 400            |
| kpi    | reactor     | Q_kW             | -1007.67392146 |
| kpi    | reactor     | T                | 900            |
| kpi    | reactor     | conversion       | 0.75           |
| kpi    | reactor     | dHrxn_kJ_per_mol | -48.3683482303 |
| kpi    | reactor     | extent_kmol_h    | 75             |
| kpi    | reactor     | Q_reaction_kW    | -1007.67392146 |
| kpi    | reactor     | Q_sensible_kW    | 0              |
| stream | feed        | T                | 900            |
| stream | product     | T                | 900            |

**conversion02\_parallel\_selectivity**

tutorials/steady/reactors/conversion02\_parallel\_selectivity



**What it does:** Conversion reactor, PARALLEL network  $A \rightarrow B$  /  $A \rightarrow C$ : selectivity as a SPEC (75 %)

**The story:** a stoichiometric (conversion) reactor with a PARALLEL network,  $A \rightarrow B$  /  $A \rightarrow C$ .

Multi-reaction grammar 'reactions ( a\_to\_b a\_to\_c );'. Each reaction's extent is a SPEC, not a result: 'conversions ( { reaction ...; conversion ...; } ... )' gives the fraction of the FEED A that each leg consumes. 60 % of A goes the wanted way and 20 % the side way, so:

total A conversion = 80 % selectivity  $B/(B+C) = 0.36 / 0.48 = 75$  %

This is the honest reactor when you have neither rate data (cstr/pfr) nor an equilibrium (equilibriumReactor) – just a catalyst whose split you measured. A single-reaction conversionReactor cannot express a selectivity at all.

**What to expect (golden-locked):**

| kind   | unit/stream | key                  | expected           |
|--------|-------------|----------------------|--------------------|
| stream | feed        | F                    | 0.0002777777777778 |
| stream | product     | F                    | 0.0001888888888889 |
| kpi    | reactor     | F_in_kmol_h          | 1                  |
| kpi    | reactor     | F_out_kmol_h         | 0.68               |
| kpi    | reactor     | T                    | 350                |
| kpi    | reactor     | extent_a_to_b_kmol_h | 0.12               |
| kpi    | reactor     | extent_a_to_c_kmol_h | 0.04               |
| kpi    | reactor     | nReactions           | 2                  |
| stream | feed        | T                    | 350                |
| stream | product     | T                    | 350                |

**cstr01\_first\_order**  
tutorials/steady/reactors/cstr01\_first\_order



**What it does:** CSTR with pseudo-first-order Arrhenius esterification ( $\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$ ), ~53% conversion of ethanol

**The story:** CSTR single-reaction tutorial. The reaction kinetics live in constant/reactions; here the unit references it by name.

**What to expect (golden-locked):**

| kind                                                           | unit/stream | key             | expected          |
|----------------------------------------------------------------|-------------|-----------------|-------------------|
| kpi                                                            | reactor     | X_limiting      | 0.52590711876     |
| kpi                                                            | reactor     | T               | 350               |
| kpi                                                            | reactor     | k               | 0.00357499942014  |
| stream                                                         | feed        | F               | 0.000277777777778 |
| stream                                                         | feed        | T               | 350               |
| stream                                                         | reactorOut  | F               | 0.000277777777778 |
| stream                                                         | reactorOut  | T               | 350               |
| kpi                                                            | reactor     | Da_kTau         | 1.10929132154     |
| kpi                                                            | reactor     | F_in_kmol_h     | 1                 |
| kpi                                                            | reactor     | F_out_kmol_h    | 1                 |
| kpi                                                            | reactor     | Q_kW            | -1.87948262728    |
| kpi                                                            | reactor     | V_R             | 0.005             |
| kpi                                                            | reactor     | minus_r_ethanol | 14.6085310767     |
| kpi                                                            | reactor     | tau_s           | 310.291329081     |
| ...and 1 more reference values in the case's <i>expected</i> . |             |                 |                   |

**cstr02\_reversible\_wgs**

tutorials/steady/reactors/cstr02\_reversible\_wgs



**What it does:** Reversible water-gas shift ( $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ ) in a CSTR

**The story:** water-gas-shift reaction in a CSTR with reverse rate from thermodynamic  $K_{\text{eq}}$ .

Feed: equimolar  $\text{CO} + \text{H}_2\text{O}$  at 600 K, 10 bar. The forward Arrhenius kinetics drives a high conversion in the early residence times; the reverse term (auto-derived from  $g_{\text{pure\_ig}}$  of the four species at  $T = 600$  K) caps the conversion at the Le Chatelier equilibrium.

At  $T = 600$  K:  $dG_{\text{rxn}} = g_{\text{CO}_2} + g_{\text{H}_2} - g_{\text{CO}} - g_{\text{H}_2\text{O}} \approx -16.3$  kJ/mol  $K_{\text{eq}} = \exp(16300 / (R \cdot 600)) \approx 26.5$  For equimolar  $\text{CO} + \text{H}_2\text{O}$ ,  $K_{\text{eq}} = X^2 / (1-X)^2 \rightarrow X_{\text{eq}} \approx 0.84$

$V_R$  is picked large enough ( $1\text{e-}3$  m<sup>3</sup> for a 1 kmol/h feed) that the kinetics are fast relative to residence; we see the conversion approach (but not exceed)  $X_{\text{eq}} \approx 0.84$  — the thermodynamic ceiling.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected           |
|--------|-------------|--------------|--------------------|
| stream | feed        | F            | 0.0002777777777778 |
| stream | feed        | vf           | 1                  |
| stream | products    | F            | 0.0002777777777778 |
| stream | products    | vf           | 1                  |
| kpi    | wgs         | Da_kTau      | 145.857299066      |
| kpi    | wgs         | F_in_kmol_h  | 1                  |
| kpi    | wgs         | F_out_kmol_h | 1                  |
| kpi    | wgs         | T            | 600                |
| kpi    | wgs         | V_R          | 0.001              |
| kpi    | wgs         | X_limiting   | 0.841431955275     |
| kpi    | wgs         | k            | 1.08521882096      |
| kpi    | wgs         | tau_s        | 134.403584096      |
| kpi    | wgs         | xi_mol_per_s | 0.116865549344     |
| stream | feed        | T            | 600                |

... and 3 more reference values in the case's *expected*.

**cstr03\_series\_selectivity**

tutorials/steady/reactors/cstr03\_series\_selectivity



**What it does:** CSTR with a SERIES network  $A \rightarrow B \rightarrow C$ : multi-reaction, the selectivity problem

**The story:** a CSTR with a SERIES reaction network,  $A \rightarrow B \rightarrow C$ , solved as ONE coupled system.

The multi-reaction grammar ‘reactions ( a\_to\_b b\_to\_c );’ names both steps from constant/reactions; the reactor solves the R coupled design equations  $xi\_j = r\_j(xi)$   $V\_R$  together by the multivariate Newton. A single-reaction CSTR cannot express this at all: the intermediate B is BOTH a product and a reactant, so its yield peaks in residence time and SELECTIVITY becomes the design question.

Feed: pure A at 350 K. Try sweeping  $V\_R$  (an outerDict sweep) and watch the B yield rise, peak, and fall as the over-reaction  $B \rightarrow C$  takes over.

**What to expect (golden-locked):**

| kind   | unit/stream | key                 | expected           |
|--------|-------------|---------------------|--------------------|
| stream | feed        | F                   | 0.0002777777777778 |
| stream | product     | F                   | 0.000199218293117  |
| kpi    | reactor     | F_in_kmol_h         | 1                  |
| kpi    | reactor     | F_out_kmol_h        | 0.717185855222     |
| kpi    | reactor     | T                   | 350                |
| kpi    | reactor     | V_R                 | 0.005              |
| kpi    | reactor     | k_a_to_b            | 0.00214522777035   |
| kpi    | reactor     | k_b_to_c            | 0.000536306942589  |
| kpi    | reactor     | nReactions          | 2                  |
| kpi    | reactor     | newtonResidual      | 6.93889390391e-18  |
| kpi    | reactor     | tau_s               | 225                |
| kpi    | reactor     | xi_a_to_b_mol_per_s | 0.0392797423303    |
| kpi    | reactor     | xi_b_to_c_mol_per_s | 0.0126884461066    |
| stream | feed        | T                   | 350                |

... and 2 more reference values in the case's *expected*.

**cstr04\_adiabatic**

tutorials/steady/reactors/cstr04\_adiabatic



**What it does:** ADIABATIC CSTR:  $T_{\text{out}}$  solved WITH the extents;  $Q=0$  by construction (elements datum)

**The story:** an ADIABATIC continuous stirred-tank reactor.

The outlet temperature is an UNKNOWN, solved WITH the reaction extents:

$x_{i,j} = r_j(x_i, T) V_R$  (R design equations)  $H_{\text{out}}(T, x_i) = H_{\text{in}}$  (one enthalpy balance)

There is NO  $dH_{\text{rxn}}$  source term. With the enthalpy on the ELEMENTS datum, the heat of reaction already lives inside H: as reactants become products the formation enthalpies differ and  $T$  rises on its own.  $Q_{\text{kW}} = 0$  by construction.

A non-isothermal CSTR can have up to THREE steady states (multiplicity). The Newton lands on ONE of them, selected by ' $T_{\text{guess}}$ ' – sweep it to find the ignited and extinguished branches. That honesty is the point: the reactor reports which root it found, it does not pretend there is only one.

**What to expect (golden-locked):**

| kind   | unit/stream | key                               | expected          |
|--------|-------------|-----------------------------------|-------------------|
| stream | feed        | F                                 | 0.000277777777778 |
| stream | reactorOut  | F                                 | 0.000277777777778 |
| kpi    | reactor     | $F_{\text{in\_kmol\_h}}$          | 1                 |
| kpi    | reactor     | $F_{\text{out\_kmol\_h}}$         | 1                 |
| kpi    | reactor     | $Q_{\text{kW}}$                   | 4.70379291073e-12 |
| kpi    | reactor     | T                                 | 454.152948902     |
| kpi    | reactor     | $T_{\text{guess}}$                | 350               |
| kpi    | reactor     | $T_{\text{out}}$                  | 454.152948902     |
| kpi    | reactor     | $V_R$                             | 0.005             |
| kpi    | reactor     | $dT_{\text{rise}}$                | 104.152948902     |
| kpi    | reactor     | $k_{\text{esterification\_etac}}$ | 0.88934531923     |
| kpi    | reactor     | nReactions                        | 1                 |
| kpi    | reactor     | newtonResidual                    | 3.5527136788e-15  |
| kpi    | reactor     | $\tau_{\text{s}}$                 | 310.291329081     |

... and 4 more reference values in the case's *expected*.

**cstr05\_multiplicity**

tutorials/steady/reactors/cstr05\_multiplicity



**What it does:** CSTR multiplicity: three steady states – the reactor prints them all, T\_guess picks the branch

**The story:** an ADIABATIC CSTR with THREE steady states.

Eliminate the extents (at a fixed T the design equations are monotone) and what remains is ONE equation in the temperature:

$$\phi(T) = H_{\text{out}}(T) - H_{\text{in}} - Q_{\text{ext}}(T) = 0$$

the textbook "heat generated equals heat removed". The generation curve  $G(T)$  is a sigmoid (the rate saturates once the reactant is gone); the removal line  $R(T) = F c_p (T - T_{\text{in}})$  is straight (adiabatic: the only sink is the sensible heat that leaves with the product). A straight line can cut a sigmoid THREE times: the extinguished state, an unstable middle state, and the ignited state.

The feed enters COLD (300 K), where the rate is negligible. Nothing forces this reactor to ignite – and nothing forces it to stay cold either.

Choupo scans, finds them all, prints them, and reports the one nearest 'T\_guess'. Change T\_guess and the reactor moves to another branch – the same equipment, the same feed, a different answer. That is not a numerical defect; it is the physics, and it is why real reactors need a start-up procedure.

**What to expect (golden-locked):**

| kind   | unit/stream | key                   | expected          |
|--------|-------------|-----------------------|-------------------|
| stream | feed        | F                     | 0.000277777777778 |
| stream | reactorOut  | F                     | 0.000277777777778 |
| kpi    | reactor     | F_in_kmol_h           | 1                 |
| kpi    | reactor     | F_out_kmol_h          | 1                 |
| kpi    | reactor     | Q_kW                  | 8.68283223099e-12 |
| kpi    | reactor     | T                     | 405.903598573     |
| kpi    | reactor     | T_guess               | 380               |
| kpi    | reactor     | T_out                 | 405.903598573     |
| kpi    | reactor     | V_R                   | 0.005             |
| kpi    | reactor     | dT_rise               | 105.903598573     |
| kpi    | reactor     | k_esterification_etac | 0.406108484365    |
| kpi    | reactor     | nReactions            | 1                 |
| kpi    | reactor     | newtonResidual        | 8.32667268469e-16 |
| kpi    | reactor     | steadyStates          | 3                 |

... and 4 more reference values in the case's *expected*.



ctr06\_jacketed

tutorials/steady/reactors/ctr06\_jacketed



**What it does:** JACKETED CSTR: the coolant carries the reaction heat away –  $Q = UA (T_c - T_{out})$

**The story:** the same esterification as ctr04, now with a COOLING JACKET instead of an adiabatic wall.

The energy balance gains one term, and only one:

$$H_{out}(T) = H_{in} + UA (T_{coolant} - T_{out})$$

There is still NO dH\_rxn source: on the elements datum the heat of reaction is already inside H. What the jacket does is remove some of it, so the outlet settles BELOW the adiabatic 454.2 K of ctr04.

Check it by hand from the printed KPIs: Q\_kW should equal UA (T\_c - T\_out), and it is NEGATIVE – heat leaves the reactor. Cool harder (raise UA, lower T\_coolant) and the outlet slides down the generation curve; cool too hard and the reactor extinguishes. Nothing here is switched on by a flag; it is the same one equation in T, with one more term.

What to expect (golden-locked):

| kind   | unit/stream | key                   | expected          |
|--------|-------------|-----------------------|-------------------|
| stream | feed        | F                     | 0.000277777777778 |
| stream | reactorOut  | F                     | 0.000277777777778 |
| kpi    | reactor     | F_in_kmol_h           | 1                 |
| kpi    | reactor     | F_out_kmol_h          | 1                 |
| kpi    | reactor     | Q_kW                  | -1.85832125175    |
| kpi    | reactor     | T                     | 394.33285007      |
| kpi    | reactor     | T_guess               | 400               |
| kpi    | reactor     | T_out                 | 394.33285007      |
| kpi    | reactor     | V_R                   | 0.005             |
| kpi    | reactor     | dT_rise               | 44.3328500695     |
| kpi    | reactor     | k_esterification_etac | 0.0534254733127   |
| kpi    | reactor     | nReactions            | 1                 |
| kpi    | reactor     | newtonResidual        | 1.38777878078e-16 |
| kpi    | reactor     | steadyStates          | 1                 |

... and 4 more reference values in the case's expected.

**cstr07\_lhhw\_methylAcetate**

tutorials/steady/reactors/cstr07\_lhhw\_methylAcetate



**What it does:** LHHW on activities: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

**The story:** an isothermal CSTR packed with Amberlyst 15.

The rate constants are reported per GRAM of dry catalyst, so the reactor needs to know how much catalyst a cubic metre of it holds: ‘catalystLoading’. It is a declared, visible number, not a fudge – change the packing and the reactor changes with it.

Run it and read the KPI ‘k\_esterification\_meoac’ against the conversion: the forward rate constant says one thing, the adsorption denominator says another, and the outlet is where they settle.

This reactor is KINETICALLY limited, which is the point – if it sat at equilibrium the denominator would be invisible. Sweep V\_R and watch it climb to the activity-based equilibrium and stop:

V\_R = 0.005 m3 X = 49.02 % <- this case V\_R = 0.020 m3 X = 63.74 % V\_R = 0.100 m3 X = 72.92 %  
V\_R = 1.000 m3 X = 76.18 % <- equilibrium

The plateau is not asserted anywhere in the dict. It is where  $k_1 a'_{\text{HOAc}} a'_{\text{MeOH}} = k_{-1} a'_{\text{MeOAc}} a'_{\text{H}_2\text{O}}$ , i.e. exactly the  $K_a = 25.50$  that the authors’ own Eq 17 recovers from the same four rate constants and four adsorption constants at 333.15 K. Two independent equations of the paper agreeing is a cheap and honest check that the parameters were entered right.

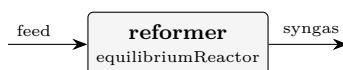
**What to expect (golden-locked):**

| kind   | unit/stream | key                               | expected           |
|--------|-------------|-----------------------------------|--------------------|
| stream | feed        | F                                 | 0.0002777777777778 |
| stream | reactorOut  | F                                 | 0.0002777777777778 |
| kpi    | reactor     | F_in_kmol_h                       | 1                  |
| kpi    | reactor     | F_out_kmol_h                      | 1                  |
| kpi    | reactor     | Q_kW                              | -2.09513270004     |
| kpi    | reactor     | T                                 | 333.15             |
| kpi    | reactor     | T_out                             | 333.15             |
| kpi    | reactor     | V_R                               | 0.005              |
| kpi    | reactor     | k_esterification_meoac            | 0.00280769603355   |
| kpi    | reactor     | nReactions                        | 1                  |
| kpi    | reactor     | newtonResidual                    | 1.2490009027e-16   |
| kpi    | reactor     | tau_s                             | 366.524129505      |
| kpi    | reactor     | xi_esterification_meoac_mol_per_s | 0.0680930737737    |
| stream | feed        | T                                 | 333.15             |

...and 1 more reference values in the case’s *expected*.

**equil01\_reforming**

tutorials/steady/reactors/equil01\_reforming



**What it does:** Equilibrium reactor (REquil): SMR + WGS to SIMULTANEOUS equilibrium at 1100 K

**The story:** the EQUILIBRIUM reactor (REquil) with MULTIPLE simultaneous reactions.

Steam-methane reforming: two independent reactions are driven to SIMULTANEOUS chemical equilibrium at 1100 K, 1 bar – no kinetics, no residence time. The two extents are a RESULT of each reaction's  $K_p(T)$  (from the standardThermochemistry data), solved together by the multivariate Newton:

smr :  $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3 \text{H}_2$  (strongly endothermic) wgs :  $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$  (mildly exothermic)

The stoichiometry lives in constant/reactions and is referenced BY NAME – the same ‘reactions ( r1 r2 );’ grammar the batch / dynamic reactors take.

Feed: 1 kmol/h  $\text{CH}_4$  + 3 kmol/h steam (steam-to-carbon = 3). At 1100 K, 1 bar the reforming is near-complete (Le Chatelier favours the mole-increasing SMR at low pressure); the shift then sets the CO/CO<sub>2</sub> split. This is the classic syngas problem – the reactor a ‘conversionReactor’ cannot do (it needs the extents GIVEN) and a ‘cstr’/‘pfr’ cannot do without rate data.

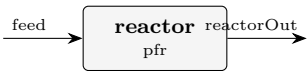
**What to expect (golden-locked):**

| kind   | unit/stream | key               | expected          |
|--------|-------------|-------------------|-------------------|
| stream | feed        | F                 | 0.001111111111111 |
| stream | syngas      | F                 | 0.00166593958576  |
| kpi    | reformer    | F_in_kmol_h       | 4                 |
| kpi    | reformer    | F_out_kmol_h      | 5.99738250873     |
| kpi    | reformer    | Kp_smr            | 312.740523794     |
| kpi    | reformer    | Kp_wgs            | 0.978425617548    |
| kpi    | reformer    | Q_kW              | 59.6424648629     |
| kpi    | reformer    | T                 | 1100              |
| kpi    | reformer    | conversion_smr    | 0.998691254363    |
| kpi    | reformer    | conversion_wgs    | 0.109768673641    |
| kpi    | reformer    | extent_smr_kmol_h | 0.998691254363    |
| kpi    | reformer    | extent_wgs_kmol_h | 0.329306020923    |
| kpi    | reformer    | nReactions        | 2                 |
| kpi    | reformer    | newtonResidual    | 3.2596148003e-13  |

... and 3 more reference values in the case's *expected*.

pfr01\_first\_order

tutorials/steady/reactors/pfr01\_first\_order



**What it does:** PFR with the same kinetics as cstr01 (RK4 integration)

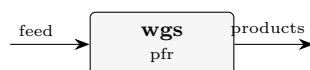
**The story:** PFR with the same kinetics and conditions as cstr01. For a first-order reaction the PFR gives  $X = 1 - \exp(-k\tau)$ . With  $k\tau = 1.10$ :  $X_{\text{PFR}} = 1 - \exp(-1.10) = 0.667$  (vs CSTR  $X = 0.524$ )

What to expect (golden-locked):

| kind   | unit/stream | key              | expected           |
|--------|-------------|------------------|--------------------|
| stream | feed        | F                | 0.0002777777777778 |
| stream | reactorOut  | F                | 0.0002777777777778 |
| stream | feed        | T                | 350                |
| stream | reactorOut  | T                | 350                |
| kpi    | reactor     | Da_kTau          | 1.10929132154      |
| kpi    | reactor     | F_out_kmol_h     | 1                  |
| kpi    | reactor     | Q_kW             | -2.39518182773     |
| kpi    | reactor     | T                | 350                |
| kpi    | reactor     | V_R              | 0.005              |
| kpi    | reactor     | X_limiting       | 0.670207404765     |
| kpi    | reactor     | dHrxn_kJ_per_mol | -25.7313020373     |
| kpi    | reactor     | tau_s            | 310.291329081      |

**pfr02\_reversible\_wgs**

tutorials/steady/reactors/pfr02\_reversible\_wgs



**What it does:** Reversible water-gas shift ( $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ ) in a PFR; approaches the same  $X_{\text{eq}}$  as the CSTR

**The story:** the reversible water-gas shift in a PLUG-FLOW reactor, the PFR sibling of cstr02\_reversible\_wgs.

Same reaction, same feed (equimolar  $\text{CO} + \text{H}_2\text{O}$  at 600 K, 10 bar), same reverse term auto-derived from `g_pure_ig`. The pedagogical point: a reversible reaction relaxes toward the THERMODYNAMIC equilibrium conversion regardless of reactor type — with enough volume the PFR approaches the very same ceiling the CSTR hit,

$K_{\text{eq}}(600 \text{ K}) \sim 26.5 \rightarrow X_{\text{eq}} \sim 0.84$  (equimolar feed),

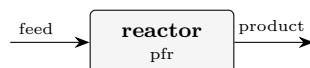
not the  $X = 1$  a forward-only PFR would chase.  $V_R$  is taken large enough that the axial profile flattens onto  $X_{\text{eq}}$  well before the exit.

**What to expect (golden-locked):**

| kind   | unit/stream | key              | expected           |
|--------|-------------|------------------|--------------------|
| stream | feed        | F                | 0.0002777777777778 |
| stream | feed        | vf               | 1                  |
| stream | products    | F                | 0.0002777777777778 |
| stream | products    | vf               | 1                  |
| stream | feed        | T                | 600                |
| stream | products    | T                | 600                |
| kpi    | wgs         | Da_kTau          | 0.00175028758879   |
| kpi    | wgs         | F_out_kmol_h     | 1                  |
| kpi    | wgs         | Q_kW             | -4.55086758232     |
| kpi    | wgs         | T                | 600                |
| kpi    | wgs         | V_R              | 0.002              |
| kpi    | wgs         | X_limiting       | 0.841431577908     |
| kpi    | wgs         | dHrxn_kJ_per_mol | -38.9410707335     |
| kpi    | wgs         | tau_s            | 268.807168191      |

**pfr03\_series\_selectivity**

tutorials/steady/reactors/pfr03\_series\_selectivity



**What it does:** PFR with the SAME series network  $A \rightarrow B \rightarrow C$ : no back-mixing  $\rightarrow$  MORE of the intermediate B

**The story:** the SAME series network  $A \rightarrow B \rightarrow C$  as cstr03, now in a PFR. Compare them.

Multi-reaction grammar 'reactions ( a\_to\_b b\_to\_c );'; the PFR marches  $dF_i/dV = \sum_j \nu_{ij} r_j(F)$  by RK4 – an IVP, no Newton. At the SAME residence time ( $\tau = 225$  s):

species CSTR PFR compA (feed) 40.8 % 14.1 % compB (want) 52.8 % 79.2 % <- the point compC (over) 6.4 % 6.8 %

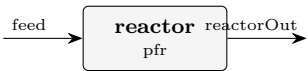
The PFR converts far more A AND keeps far more of the intermediate B: with no back-mixing, the freshly made B is not held in a vessel at low A concentration where the over-reaction  $B \rightarrow C$  dominates. For positive-order kinetics the PFR beats the CSTR – the classic result a single-reaction reactor cannot show.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected           |
|--------|-------------|--------------|--------------------|
| stream | feed        | F            | 0.0002777777777778 |
| stream | product     | F            | 0.000176612457804  |
| kpi    | reactor     | F_in_kmol_h  | 1                  |
| kpi    | reactor     | F_out_kmol_h | 0.635804848095     |
| kpi    | reactor     | T            | 350                |
| kpi    | reactor     | V_R          | 0.005              |
| kpi    | reactor     | k_a_to_b     | 0.00214522777035   |
| kpi    | reactor     | k_b_to_c     | 0.000536306942589  |
| kpi    | reactor     | nReactions   | 2                  |
| kpi    | reactor     | nSteps       | 200                |
| kpi    | reactor     | tau_s        | 225                |
| stream | feed        | T            | 350                |
| stream | product     | T            | 350                |
| kpi    | reactor     | T_out        | 350                |

pfr04\_adiabatic

tutorials/steady/reactors/pfr04\_adiabatic



**What it does:** ADIABATIC PFR: the exothermic esterification heats its own stream – T(V) is a RESULT

**The story:** an ADIABATIC plug-flow reactor. The temperature is a RESULT, marched with the species by the same RK4:

$$dF_i/dV = \sum_j \nu_{ij} r_j(F,T) \quad dT/dV = \sum_j (-\Delta H_j(T)) r_j / \sum_i F_i c_{p,i}(T)$$

The exothermic esterification heats its own stream, which accelerates the Arrhenius rate, which releases more heat – the positive feedback every reactor engineer must see. T(V) is in the axial profile: plot it.

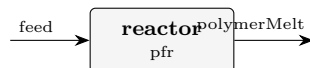
‘thermalMode’ takes ‘isothermal’ (default – T imposed, duty a result), ‘adiabatic’ (here), or ‘heatExchange’ (a jacket: U, areaPerVolume, T\_coolant). A non-isothermal reactor needs the heat of reaction, so every reacting species must carry ‘standardThermochemistry’ – a fictitious toy species cannot be used.

What to expect (golden-locked):

| kind                                                    | unit/stream | key                   | expected           |
|---------------------------------------------------------|-------------|-----------------------|--------------------|
| stream                                                  | feed        | F                     | 0.0002777777777778 |
| stream                                                  | reactorOut  | F                     | 0.0002777777777778 |
| kpi                                                     | reactor     | F_in_kmol_h           | 1                  |
| kpi                                                     | reactor     | F_out_kmol_h          | 1                  |
| kpi                                                     | reactor     | Q_kW                  | -1.42108547152e-14 |
| kpi                                                     | reactor     | T                     | 350                |
| kpi                                                     | reactor     | T_max                 | 454.513818186      |
| kpi                                                     | reactor     | T_min                 | 350                |
| kpi                                                     | reactor     | T_out                 | 454.513818186      |
| kpi                                                     | reactor     | V_R                   | 0.005              |
| kpi                                                     | reactor     | dT_rise               | 104.513818186      |
| kpi                                                     | reactor     | hotSpot_dT            | 104.513818186      |
| kpi                                                     | reactor     | k_esterification_etac | 0.00357499942014   |
| kpi                                                     | reactor     | nReactions            | 1                  |
| ... and 4 more reference values in the case's expected. |             |                       |                    |

**pfr\_polyesterification**

tutorials/steady/reactors/pfr\_polyesterification



**What it does:** PFR step-growth polyesterification — Carothers/Flory chain statistics

**The story:** Step-growth polyesterification in a PFR. A molten AB-type hydroxy-acid monomer self-condenses to polycaprolactone, releasing water. As the monomer conversion  $p$  rises along the reactor, the Carothers relation  $X_n = 1/(1-p)$  makes the average chain length climb steeply, and the polydispersity  $PDI = 1+p$  approaches 2 (the most-probable distribution).

Try a ‘sweep’ over operation.  $V_R$  and watch  $X_n$ ,  $M_n$  and  $PDI$  climb and the Flory-Schulz distribution broaden – the whole story in one glass-box run.

*A molten AB-type hydroxy-acid monomer (6-hydroxyhexanoic acid) self- condenses to polycaprolactone in a plug-flow reactor, releasing one water per ester bond:*

*As the monomer conversion  $p$  climbs along the reactor, the Carothers relation makes the average chain length rise steeply and the polydispersity approach 2 — the most-probable (Flory–Schulz) distribution.*

*and emits the Flory–Schulz distribution as a profile (chainLength vs  $n_x$ ,  $w_x$ ) so the GUI plots it:  $n(x)=(1-p)p^{x-1}$  decays monotonically,  $w(x)=x(1-p)^2 p^{x-1}$  peaks near  $x \approx X_n$ .*

*Run a sweep over operation.  $V_R$  (reactor volume  $\rightarrow$  residence time  $\rightarrow p$ ) and watch  $PDI$  climb toward 2,  $X_n$  and  $M_n$  shoot up, and the weight distribution broaden. The whole Carothers/Flory story in one glass-box run.*

*Lives inside the reaction. Absent ? the reactor behaves as an ordinary PFR.  $p$  is taken from the reactor’s own conversion of limitingReactant (single source of truth). The distribution +  $PDI = 1+p$  are valid because a PFR has a uniform residence time; on a CSTR the residence-time distribution broadens both, so the distribution is gated to PFR/batch only.*

- The chain-statistics math is exact and standard: P.J. Flory, *\*Principles of Polymer Chemistry\**, Cornell Univ. Press (1953), Ch. VIII–IX (the most-probable distribution and  $PDI = 1+p$ ); G. Odian, *\*Principles of Polymerization\**, 4th ed., Ch. 2; H.S. Fogler, *\*Elements of Chemical Reaction Engineering\**, 5th ed., ?9. - The monomer/repeat component properties are ILLUSTRATIVE (clearly labelled in each .dat) — plausible round numbers for a teaching case, not curated literature values. The Arrhenius rate is illustrative, tuned so  $Da = k\tau \approx 4$  gives  $p \approx 0.98$ . Curate these before any quantitative use.

**What to expect (golden-locked):**

| kind   | unit/stream | key          | expected          |
|--------|-------------|--------------|-------------------|
| stream | feed        | F            | 0.000277777777778 |
| stream | polymerMelt | F            | 0.000550395491925 |
| kpi    | reactor     | Da_kTau      | 3.98587252785     |
| kpi    | reactor     | F_out_kmol_h | 1.98142377093     |
| kpi    | reactor     | Mn_kg_kmol   | 6144.41174045     |
| kpi    | reactor     | Mw_kg_kmol   | 12174.6834809     |
| kpi    | reactor     | PDI          | 1.98142377093     |
| kpi    | reactor     | T            | 450               |
| kpi    | reactor     | V_R          | 0.2               |
| kpi    | reactor     | X_limiting   | 0.981423770929    |
| kpi    | reactor     | Xn           | 53.8322388335     |
| kpi    | reactor     | Xw           | 106.664477667     |
| kpi    | reactor     | p_conversion | 0.981423770929    |
| kpi    | reactor     | tau_s        | 5829.95951417     |

... and 4 more reference values in the case’s *expected*.



**rotating**

**The theory you need.** Compressors, turbines, pumps: specified by shaft work  $W_s$  and isentropic efficiency  $\eta_s$ ; the outlet state follows from  $h_{\text{out}} = h_{\text{in}} + W_s/\dot{n}$  with the isentropic reference path. Outlet pressure is a RESULT (hardware, not a spec).

**compressor01\_air**

tutorials/steady/rotating/compressor01\_air



**What it does:** Adiabatic air compressor: 10 kW shaft, eta 0.75 — P\_out and T\_out are RESULTS

**The story:** The simplest credo-pure rotating-equipment case. An adiabatic compressor takes ambient air (1 bar, 25 degC) and a fixed shaft power of 10 kW with isentropic efficiency 0.75.

Hardware (the only things you specify): W\_shaft = 10 kW motor rating delivered to the gas eta = 0.75 isentropic efficiency (machine characteristic)

RESULTS the simulator computes: P\_out the discharge pressure the machine reaches T\_out the discharge temperature (real, with losses) ratio P\_out / P\_in T\_out\_isen isentropic discharge T (diagnostic) dS\_gen entropy generated by the irreversibility

Hand check (ideal gas, Cp ~ 29.2 J/(mol\*K), gamma ~ 1.40): F = 3.6 kmol/h = 1 mol/s w\_real = 10000 W / 1 mol/s = 10000 J/mol w\_isen = 0.75 \* 10000 = 7500 J/mol T\_out\_isen = 298.15 + 7500/29.2 ~ 555 K ratio = (T\_isen/T\_in)^(gamma/(gamma-1)) = (555/298)^3.5 ~ 8.8 T\_out = 298.15 + 10000/29.2 ~ 640 K

"I need exactly 8 bar at discharge" is a DESIGN question — it goes in a DesignSpec that manipulates \$W\_shaft with target compressor.P\_out.

**What to expect (golden-locked):**

| kind   | unit/stream  | key        | expected      |
|--------|--------------|------------|---------------|
| stream | feed         | F          | 0.001         |
| stream | out          | F          | 0.001         |
| kpi    | compressor01 | F          | 0.001         |
| kpi    | compressor01 | P_in       | 100000        |
| kpi    | compressor01 | P_out      | 890211.642995 |
| kpi    | compressor01 | T_in       | 298.15        |
| kpi    | compressor01 | T_out      | 633.374206884 |
| kpi    | compressor01 | T_out_isen | 551.37943898  |
| kpi    | compressor01 | W_isen     | 7500          |
| kpi    | compressor01 | W_isen_kW  | 7.5           |
| kpi    | compressor01 | W_shaft    | 10000         |
| kpi    | compressor01 | W_shaft_kW | 10            |
| kpi    | compressor01 | dS_gen     | 4.22615345389 |
| kpi    | compressor01 | dT         | 335.224206884 |

... and 9 more reference values in the case's *expected*.

**compressor02\_designspec\_pout**

tutorials/steady/rotating/compressor02\_designspec\_pout



**What it does:** Air compressor sized to hit  $P_{out} = 8$  bar exactly, via DesignSpec on  $W_{shaft}$

**The story:** Same air compressor as compressor01, but now answering the DESIGN question: "what shaft power do I need to discharge at exactly 8 bar?"

The credo forbids specifying  $P_{out}$  on the unit op (it's a result). Instead the outer DesignSpec manipulates the hardware variable  $W_{shaft}$  until the computed  $compressor.P_{out}$  hits the 8-bar target. This is the rotating-equipment analogue of sizing an evaporator's area to hit a product concentration.

Converges to  $W_{shaft} \sim 9.36$  kW (slightly below compressor01's 10 kW, which over-shot to 8.90 bar). At the solution  $T_{out} \sim 612$  K (339 degC) and  $W_{isen} \sim 7.02$  kW ( $\eta * W_{shaft}$ ,  $\eta = 0.75$ ).

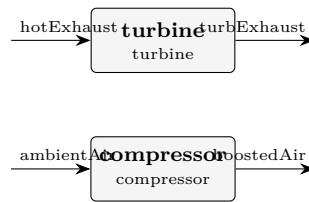
**What to expect (golden-locked):**

| kind   | unit/stream | key             | expected      |
|--------|-------------|-----------------|---------------|
| stream | feed        | F               | 0.001         |
| stream | out         | F               | 0.001         |
| kpi    | compressor  | F               | 0.001         |
| kpi    | compressor  | $P_{in}$        | 100000        |
| kpi    | compressor  | $P_{out}$       | 800004.793643 |
| kpi    | compressor  | $T_{in}$        | 298.15        |
| kpi    | compressor  | $T_{out}$       | 612.41049696  |
| kpi    | compressor  | $T_{out\_isen}$ | 535.384868081 |
| kpi    | compressor  | $W_{isen}$      | 7017.30522226 |
| kpi    | compressor  | $W_{isen\_kW}$  | 7.01730522226 |
| kpi    | compressor  | $W_{shaft}$     | 9356.40696301 |
| kpi    | compressor  | $W_{shaft\_kW}$ | 9.35640696301 |
| kpi    | compressor  | $dS_{gen}$      | 4.08116534481 |
| kpi    | compressor  | $dT$            | 314.26049696  |

... and 9 more reference values in the case's *expected*.

**compressorTurbine01\_turbo\_booster**

tutorials/steady/rotating/compressorTurbine01\_turbo\_booster

**What it does:** Turbo-booster: turbine shaft drives a compressor (energy wire)**The story:** turbo-booster: turbine drives compressor through an ENERGY WIRE (v0.80)

Two physically independent gas streams, one shared shaft:

hotExhaust → turbine —shaft—+

ambientAir → compressor &lt;—+

Hardware specs (the only knobs): turbine  $W_{\text{shaft}} = -5$  kW (NEGATIVE: work extracted)  $\eta = 0.80$   
 compressor  $W_{\text{shaft}}$  is OMITTED (the wire provides it)  $\eta = 0.75$

The wire injects  $+W_{\text{turb}}$  into the compressor before its solve(), so the compressor's  $W_{\text{shaft}}$  is whatever the turbine produces — here, 5 kW (sign flipped:  $-W_{\text{shaft}}$  on the producer's expression → the consumer reads  $+W_{\text{shaft}}$ , the usual sign convention).

Hand check (ideal gas,  $C_p \sim 29$  J/(mol\*K),  $\gamma \sim 1.40$ ): Turbine: 1 mol/s, 800 K, 5 bar → extract 5 kW (= 5000 J/mol)  $w_{\text{isen}} = w_{\text{real}} / \eta = 5000 / 0.80 = 6250$  J/mol  $T_{\text{isen}} = 800 - 6250/29 \sim 585$  K → ratio =  $(585/800)^{3.5} \sim 0.35$   $P_{\text{out}} \sim 5 * 0.35 \sim 1.74$  bar  $T_{\text{out}} (\text{real}) = 800 - 5000/29 \sim 627$  K (warmer than  $T_{\text{isen}}$ : the losses stay as heat in the gas)

Compressor: 0.5 mol/s, 298 K, 1 bar, receiving 5 kW  $w_{\text{real}} = 5000 / 0.5 = 10000$  J/mol  $w_{\text{isen}} = 0.75 * 10000 = 7500$  J/mol  $T_{\text{isen}} = 298 + 7500/29 \sim 557$  K → ratio =  $(557/298)^{3.5} \sim 8.7$   $P_{\text{out}} \sim 8.7$  bar  $T_{\text{out}} (\text{real}) = 298 + 10000/29 \sim 643$  K

The wire makes the energy flow visible at the case level — a single edge in the flowsheet that the GUI can render dashed.

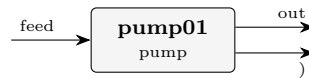
**What to expect (golden-locked):**

| kind   | unit/stream | key        | expected      |
|--------|-------------|------------|---------------|
| stream | ambientAir  | F          | 0.0005        |
| stream | boostedAir  | F          | 0.0005        |
| stream | hotExhaust  | F          | 0.001         |
| stream | turbExhaust | F          | 0.001         |
| kpi    | compressor  | F          | 0.0005        |
| kpi    | compressor  | P_in       | 100000        |
| kpi    | compressor  | P_out      | 890211.642995 |
| kpi    | compressor  | T_in       | 298.15        |
| kpi    | compressor  | T_out      | 633.374206884 |
| kpi    | compressor  | T_out_isen | 551.37943898  |
| kpi    | compressor  | W_isen     | 3750          |
| kpi    | compressor  | W_isen_kW  | 3.75          |
| kpi    | compressor  | W_shaft    | 5000          |
| kpi    | compressor  | W_shaft_kW | 5             |

... and 24 more reference values in the case's *expected*.

**pump01\_water**

tutorials/steady/rotating/pump01\_water



**What it does:** Water pump: 2 kW shaft, eta 0.65 — incompressible, P\_out / T\_out are RESULTS

**The story:** An incompressible-liquid pump. Liquid water enters at 1 bar, 298 K; the pump shaft delivers 2 kW with efficiency 0.65.

Hardware (the only OP parameters): W\_shaft = 2 kW motor power delivered to the liquid eta = 0.65 pump efficiency

Because the liquid is incompressible ( $v \sim \text{const}$ ), the discharge is closed-form — no EoS, no Newton:  $w_{\text{real}} = W_{\text{shaft}} / F_{\text{mol\_s}}$   $w_{\text{isen}} = \eta * w_{\text{real}}$  (the useful  $v*dP$  part)  $dP = w_{\text{isen}} / v_{\text{molar}}$   $P_{\text{out}} = P_{\text{in}} + dP$  (RESULT)  $dT = (1 - \eta) * w_{\text{real}} / C_{p\_liq}$  (dissipation)

Hand check (water:  $v \sim 1.8\text{e-}5 \text{ m}^3/\text{mol}$ ,  $C_{p\_liq} \sim 75 \text{ J}/(\text{mol}\cdot\text{K})$ ):  $F = 36 \text{ kmol/h} = 10 \text{ mol/s}$   $w_{\text{real}} = 2000 \text{ W} / 10 \text{ mol/s} = 200 \text{ J/mol}$   $w_{\text{isen}} = 0.65 * 200 = 130 \text{ J/mol}$   $dP = 130 / 1.8\text{e-}5 \sim 7.2\text{e}6 \text{ Pa} = 72 \text{ bar}$ ... (so  $P_{\text{out}} \sim 73 \text{ bar}$ )  $dT = 0.35 * 200 / 75 \sim 0.9 \text{ K}$  (tiny — pumps barely heat)

"Pump to exactly 50 bar" is a DesignSpec on \$W\_shaft targeting pump.P\_out (converges in one step — the relation is linear).

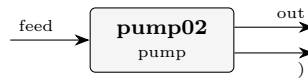
**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected       |
|--------|-------------|-------------|----------------|
| stream | feed        | F           | 0.01           |
| stream | out         | F           | 0.01           |
| kpi    | pump01      | F           | 0.01           |
| kpi    | pump01      | P_in        | 100000         |
| kpi    | pump01      | P_out       | 7294244.60432  |
| kpi    | pump01      | T_in        | 298.15         |
| kpi    | pump01      | T_out       | 299.077152318  |
| kpi    | pump01      | W_hydraulic | 1300           |
| kpi    | pump01      | W_shaft     | 2000           |
| kpi    | pump01      | W_shaft_kW  | 2              |
| kpi    | pump01      | dP          | 7194244.60432  |
| kpi    | pump01      | dT          | 0.927152317881 |
| kpi    | pump01      | eta         | 0.65           |
| kpi    | pump01      | head_m      | 735.848502288  |

...and 7 more reference values in the case's *expected*.

**pump02\_pressure\_spec**

tutorials/steady/rotating/pump02\_pressure\_spec



**What it does:** Water pump specified by DISCHARGE PRESSURE (P\_out 30 bar) — the shaft power W\_shaft is the RESULT (closed-form inverse of pump01)

**The story:** The SAME incompressible-liquid pump as pump01, specified the OTHER way: by the discharge pressure it must reach. In a real plant you rarely know a pump's shaft power up front — you know the HEADER it feeds (here 30 bar). So you state P\_out and the shaft power is the RESULT.

This is NOT a DesignSpec. For an incompressible liquid the pump relation is invertible in CLOSED FORM,

$$\Delta P = \eta \cdot W_{\text{shaft}} / Q_{\text{vol}} \Leftrightarrow W_{\text{shaft}} = Q_{\text{vol}} \cdot \Delta P / \eta ,$$

with  $Q_{\text{vol}} = F \cdot v$  set by the feed. Give P\_out, get W\_shaft — one shot, no Newton, no outer loop. (pump01 gives W\_shaft and gets P\_out; the two cases are exact inverses of one another.)

Specify EXACTLY ONE of { W\_shaft | P\_out | dP }. A pressure let-down (P\_out < P\_in) is not a pump — use a 'valve'.

Hand check (water:  $v \sim 1.8\text{e-}5 \text{ m}^3/\text{mol}$ ):  $F = 36 \text{ kmol/h} = 10 \text{ mol/s}$   $Q_{\text{vol}} = 10 * 1.8\text{e-}5 = 1.8\text{e-}4 \text{ m}^3/\text{s}$   
 $dP = 30 - 1 \text{ bar} = 2.9\text{e}6 \text{ Pa}$   $W_{\text{shaft}} = Q_{\text{vol}} * dP / \eta = 1.8\text{e-}4 * 2.9\text{e}6 / 0.65 \sim 803 \text{ W} \sim 0.80 \text{ kW}$

**What to expect (golden-locked):**

| kind   | unit/stream | key         | expected       |
|--------|-------------|-------------|----------------|
| stream | feed        | F           | 0.01           |
| stream | out         | F           | 0.01           |
| kpi    | pump02      | F           | 0.01           |
| kpi    | pump02      | P_in        | 100000         |
| kpi    | pump02      | P_out       | 3000000        |
| kpi    | pump02      | T_in        | 298.15         |
| kpi    | pump02      | T_out       | 298.523735099  |
| kpi    | pump02      | W_hydraulic | 524.03         |
| kpi    | pump02      | W_shaft     | 806.2          |
| kpi    | pump02      | W_shaft_kW  | 0.8062         |
| kpi    | pump02      | dP          | 2900000        |
| kpi    | pump02      | dT          | 0.373735099338 |
| kpi    | pump02      | eta         | 0.65           |
| kpi    | pump02      | head_m      | 296.620531272  |

... and 7 more reference values in the case's *expected*.

**turbine01\_air**

tutorials/steady/rotating/turbine01\_air



**What it does:** Adiabatic gas turbine: extract 10 kW from hot air, eta 0.85 — P<sub>out</sub> / T<sub>out</sub> are RESULTS

**The story:** The mirror of compressor01. Hot compressed air (10 bar, 900 K) drives an adiabatic turbine; the shaft extracts 10 kW with isentropic efficiency 0.85.

Hardware (the only things you specify): W<sub>shaft</sub> = -10 kW shaft power extracted (NEGATIVE = out) eta = 0.85 isentropic efficiency

Sizing note: at 1 mol/s and 900 K inlet, the enthalpy available down to ambient bounds the extractable work to ~15 kW. The 10 kW asked for here expands the gas to ~1.32 bar (ratio ~7.6); asking for much more (e.g. 30 kW) would drive the isentropic discharge below 0 K — physically impossible, and the solver says so.

RESULTS the simulator computes (this run): P<sub>out</sub> ~1.32 bar the discharge pressure the gas expands to T<sub>out</sub> ~583 K the (real) exhaust temperature T<sub>out\_isen</sub> ~524 K the ideal (isentropic) discharge T ratio ~7.59 P<sub>in</sub> / P<sub>out</sub> (expansion ratio) W<sub>generated</sub> +10 kW what the shaft delivers to the grid

Per the credo, P<sub>out</sub> is a RESULT. If the turbine must exhaust at a fixed back-pressure (a condenser, the atmosphere), wrap the case in a DesignSpec that manipulates \$W<sub>shaft</sub> until turbine.P<sub>out</sub> hits it.

**What to expect (golden-locked):**

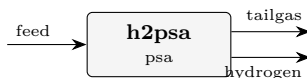
| kind   | unit/stream | key                    | expected       |
|--------|-------------|------------------------|----------------|
| stream | exhaust     | F                      | 0.001          |
| stream | hotAir      | F                      | 0.001          |
| kpi    | turbine01   | F                      | 0.001          |
| kpi    | turbine01   | P <sub>in</sub>        | 1000000        |
| kpi    | turbine01   | P <sub>out</sub>       | 131729.126973  |
| kpi    | turbine01   | T <sub>in</sub>        | 900            |
| kpi    | turbine01   | T <sub>out</sub>       | 582.935206477  |
| kpi    | turbine01   | T <sub>out_isen</sub>  | 524.463853734  |
| kpi    | turbine01   | W <sub>gen_kW</sub>    | 10             |
| kpi    | turbine01   | W <sub>generated</sub> | 10000          |
| kpi    | turbine01   | W <sub>shaft</sub>     | -10000         |
| kpi    | turbine01   | dS <sub>gen</sub>      | 3.19034018436  |
| kpi    | turbine01   | dT                     | -317.064793523 |
| kpi    | turbine01   | dT <sub>isen</sub>     | -375.536146266 |

... and 5 more reference values in the case's *expected*.

**separation**

**psa01\_h2\_psa**

tutorials/steady/separation/psa01\_h2\_psa



**What it does:** H2-PSA: 20-bar syngas (H<sub>2</sub>/CO<sub>2</sub>/CH<sub>4</sub>/CO/N<sub>2</sub>) over zeolite 5A → high-purity H<sub>2</sub> raffinate + CO<sub>2</sub>-rich tail gas (equilibrium cyclic-steady-state shortcut)

**The story:** the canonical first PSA case: hydrogen recovery from steam-methane-reformer / shifted syngas.

Feed: a typical PSA feed gas (mol fractions) H<sub>2</sub> 0.70 (the LIGHT product → raffinate) CO<sub>2</sub> 0.18 (the strongest adsorbate → tail gas) CH<sub>4</sub> 0.05 CO 0.04 N<sub>2</sub> 0.03 F = 1000 kmol/h, 20 bar, 313 K.

Adsorbent: zeolite 5A (Ca-LTA), the classic H<sub>2</sub>-PSA sieve. CO<sub>2</sub> » CH<sub>4</sub> > CO ~ N<sub>2</sub> » H<sub>2</sub> on the competitive (extended) Langmuir isotherm, so the heavy impurities load onto the bed at P<sub>high</sub> (20 bar) and are blown down with the tail at P<sub>low</sub> (1.2 bar), while H<sub>2</sub> barely adsorbs and sweeps through as the high-pressure raffinate product.

Operation: P<sub>high</sub> 20 bar / P<sub>low</sub> 1.2 bar swing; isothermal 313 K bed; purgeRatio 0.15 (a Skarstrom-style countercurrent light-product sweep that cleans the regeneration at the cost of product); eta 0.80 (the equilibrium-utilisation / LUB derating — the equilibrium ceiling is eta = 1); bedCapacity 2.0 kg adsorbent per mol of feed throughput (the lumped throughput knob; cycle time / bed volume are deliberately collapsed into it by the equilibrium shortcut).

Expected (this model, golden-checked): H<sub>2</sub> recovery ~ 74 % (single train, with the 0.15 purge sweep) H<sub>2</sub> purity ~ 98-99 % in the raffinate tail gas CO<sub>2</sub>-enriched ~ 18 % → ~ 38 %

The run header prints, out loud, what the equilibrium shortcut does NOT model (breakthrough / cycle time, thermal swing, blowdown gas loss, hydraulics, isotherm-validity window) — "all models are wrong, some useful".

**What to expect (golden-locked):**

| kind   | unit/stream | key           | expected         |
|--------|-------------|---------------|------------------|
| stream | feed        | F             | 0.277777777778   |
| stream | hydrogen    | F             | 0.145276064457   |
| stream | tailgas     | F             | 0.13250171332    |
| kpi    | h2psa       | Dq_CH4        | 0.0260273101613  |
| kpi    | h2psa       | Dq_CO         | 0.0367785976251  |
| kpi    | h2psa       | Dq_CO2        | 1.74280135323    |
| kpi    | h2psa       | Dq_H2         | 0.0186248522203  |
| kpi    | h2psa       | Dq_N2         | 0.00987823175641 |
| kpi    | h2psa       | eta_used      | 0.8              |
| kpi    | h2psa       | pressureRatio | 16.6666666667    |
| kpi    | h2psa       | productivity  | 0.0715324198173  |
| kpi    | h2psa       | purgeRatio    | 0.15             |
| kpi    | h2psa       | purity_H2     | 0.984779152498   |
| kpi    | h2psa       | recovery_H2   | 0.735762032407   |

... and 8 more reference values in the case's *expected*.



tsa01\_co2\_twin\_bed

tutorials/steady/separation/tsa01\_co2\_twin\_bed



**What it does:** Twin-bed TSA shortcut: continuous CO2 removal from helium by alternating adsorption at 298 K and regeneration at 398 K

**The story:** cycle-averaged twin-bed thermal swing.

One column adsorbs while the other regenerates. tCycle is the switch interval and therefore the adsorption duration of one column. The unit refuses a bed that cannot hold every adsorbate mole arriving in that time.

This teaching case uses a CASE-LOCAL synthetic property anchor: cpSolid = 920 J/(kg.K), explicitly assumed for this numerical gate. It is not promoted to the global zeolite13X catalogue.

**What to expect (golden-locked):**

| kind                                                            | unit/stream     | key                       | expected         |
|-----------------------------------------------------------------|-----------------|---------------------------|------------------|
| stream                                                          | feed            | F                         | 2.77777777778e-4 |
| stream                                                          | cleanHe         | F                         | 2.475e-4         |
| stream                                                          | regenerationGas | F                         | 3.02777777778e-5 |
| kpi                                                             | co2tsa          | recovery_He               | 0.9              |
| kpi                                                             | co2tsa          | purity_He                 | 1.0              |
| kpi                                                             | co2tsa          | Dq_CO2                    | 0.408156018947   |
| kpi                                                             | co2tsa          | capacity_utilisation_CO2  | 0.680567638068   |
| kpi                                                             | co2tsa          | Q_sensible_per_cycle_kJ   | 276.0            |
| kpi                                                             | co2tsa          | Q_desorption_per_cycle_kJ | 37.5             |
| kpi                                                             | co2tsa          | Q_regeneration_kW         | 1.045            |
| kpi                                                             | co2tsa          | Q_adsorption_cooling_kW   | -1.045           |
| stream                                                          | cleanHe         | T                         | 298              |
| stream                                                          | feed            | T                         | 298              |
| stream                                                          | regenerationGas | T                         | 298              |
| ... and 8 more reference values in the case's <i>expected</i> . |                 |                           |                  |

**tsa02\_refused\_capacity**

tutorials/steady/separation/tsa02\_refused\_capacity



**What it does:** TSA refusal lesson: one undersized column cannot hold the CO<sub>2</sub> loaded during the declared switch interval

**solids**

**The theory you need.** Solids carry mass and enthalpy without VLE: crystallisers, dryers, cyclones, gas–solid splitters. A dryer’s air is a REAL stream (two inlets, adiabatic mixing, outlet humidity from the psychrometrics); a cyclone’s cut size comes from the selected model (Lapple...Muschelknautz). Expect solid streams flagged separately ( $F_{\text{solid}}$ ) and energy balances that close across phase change.

**pneumaticConveyor01\_sand**

tutorials/steady/solids/pneumaticConveyor01\_sand

**What it does:** Dilute-phase pneumatic conveying of sand in nitrogen: the glass-box dP breakdown + the saltation-margin safety check. Particle velocity from a mechanistic FORCE BALANCE (drag = wall friction + gravity); solids friction from Yang (1974, AIChE J 20:605); Rizk (1973) saltation. Loading ~2 kg/kg (in the Yang voidage range); solids friction DOMINATES the dP – conveying solids IS a pressure-drop problem.  $u_g/u_{salt} = 1.72$  (safe margin above plugging).

**The story:** / flowsheetDict (single unit)

Operation block is HARDWARE only (the credo): pipe diameter, length, rise. Every velocity, the suspended density and the pressure drop are RESULTS.

*A single pneumaticConveyor unit: silica sand (~200  $\mu\text{m}$ ) carried by nitrogen up a 100 mm line, 50 m long, with a 10 m vertical rise.*

*The total conveying-line pressure drop as a sum of physical contributions (glass-box) — Rhodes, \*Introduction to Particle Technology\*, Ch. 8:*

*- superficial gas velocity  $u_g = 21.9$  m/s, - particle velocity  $u_p = 21.1$  m/s (Hinkle slip), - single-particle terminal velocity  $u_t = 1.03$  m/s, - saltation velocity  $u_{salt} = 3.6$  m/s (Rizk) —  $u_g$  is well above it, so transport is stable. The conveyor warns aloud (verbosity  $\geq 1$ ) if  $u_g$  ever falls below  $u_{salt}$  (risk of solids settling and blocking the line) — a first-class, announced design check, never a silent clip.*

*Solids loading = 0.20 kg solid / kg gas (dilute phase), suspended-solids density  $\rho_s = 0.47$  kg/m<sup>3</sup>.*

*- particle velocity: Hinkle  $u_p = u_g(1 - 0.68 d_p^{0.92} \rho_p^{0.5} \rho_g^{-0.2} D^{-0.54})$ , with an honest  $u_g - u_t$  fallback if the correlation leaves the dilute regime; - gas friction: Fanning (Blasius turbulent /  $16/\text{Re}$  laminar); - solids friction: Hinkle  $f_p = (3/8)(\rho_g/\rho_p) C_D (D/d_p)((u_g - u_p)/u_p)^2$ ,  $C_D$  from Schiller–Naumann at the slip Reynolds number; - saltation: Rizk (1973).*

*Operation block is hardware only (the credo): pipe diameter  $D$ , length  $L$ , vertical rise  $dz$ . Every velocity, the suspended density and the pressure drop are results. Particle density comes from the solid component's solid {  $\rho_p$ ; } (silica = 2650 kg/m<sup>3</sup>); the size from the inlet PSD (Sauter mean for the drag). The carrier-gas viscosity needs a transport { model Chung; } block in the thermoPhysPropDict (for the friction Reynolds numbers).*

**What to expect (golden-locked):**

| kind   | unit/stream | key                   | expected        |
|--------|-------------|-----------------------|-----------------|
| stream | Feed        | F                     | 0.0268345909167 |
| stream | conveyed    | F                     | 0.0268345909167 |
| kpi    | node        | P_out                 | 234394.335965   |
| kpi    | node        | accelerationLength    | 16.8207521299   |
| kpi    | node        | deltaP                | 65605.6640351   |
| kpi    | node        | deltaP_bends          | 0               |
| kpi    | node        | deltaP_gasAccel       | 361.754574941   |
| kpi    | node        | deltaP_gasFriction    | 1492.10736304   |
| kpi    | node        | deltaP_solidsAccel    | 1383.86777284   |
| kpi    | node        | deltaP_solidsFriction | 61751.3873821   |
| kpi    | node        | deltaP_static         | 616.546942187   |
| kpi    | node        | nBends                | 0               |
| kpi    | node        | particleWallFriction  | 0.4             |
| kpi    | node        | rho_gas               | 3.39180174393   |

*... and 12 more reference values in the case's expected.*

## pneumaticConveyor02\_bends

tutorials/steady/solids/pneumaticConveyor02\_bends

**What it does:** A SHORT (8 m) conveying line routed through 6 bends – the lesson: A FITTING ADDS PRESSURE DROP WITHOUT ADDING LENGTH. With long-radius sweeps the 6 bends cost ~10% of the total dP; swap them for blind tees (type blindTee;) and their share jumps to ~25% – the erosion-vs-pressure trade made visible. (On this short line the straight-pipe solids friction still carries most of the dP; the bends’ share GROWS as lines get shorter and bendier.) Same gas/solids as pneumaticConveyor01 (which runs 30 m / 6 m rise, bend-free) – compare the PER-METRE dP, not the totals.

**The story:** a fitting adds pressure drop without adding length

The same sand-in-N<sub>2</sub> service as pneumaticConveyor01 (loading ~2), on a SHORTER line: 100 mm, 8 m, 2 m rise, routed through 6 long-radius bends. (Case 01 runs 30 m / 6 m rise, bend-free – compare dP PER METRE, not totals.) Each bend decelerates the solids to ~0.75  $u_p$ ; the gas re-accelerates them downstream – the same momentum work as the line entry, once per bend:  $dP_{bend} = m\dot{S} (u_p - u_{p,bend}) / A$ . Run both cases and compare the dP breakdown: the bends add ~10% (long-radius) to ~25% (blind tees) of the dP while adding NO length. Swap a bend ‘type longRadius;’ for ‘type blindTee;’ and watch its re-acceleration cost jump (ratio 0.75 → 0.25) – the blind tee trades pressure for near-zero erosion. Solids friction: Yang (1974); particle velocity: force balance; Rizk saltation.

**What to expect (golden-locked):**

| kind                                                             | unit/stream | key                   | expected        |
|------------------------------------------------------------------|-------------|-----------------------|-----------------|
| stream                                                           | Feed        | F                     | 0.0268345909167 |
| stream                                                           | conveyed    | F                     | 0.0268345909167 |
| kpi                                                              | node        | P_out                 | 279210.165227   |
| kpi                                                              | node        | accelerationLength    | 15.5453372639   |
| kpi                                                              | node        | deltaP                | 20789.8347732   |
| kpi                                                              | node        | deltaP_bends          | 2070.63576095   |
| kpi                                                              | node        | deltaP_gasAccel       | 361.754574941   |
| kpi                                                              | node        | deltaP_gasFriction    | 397.89529681    |
| kpi                                                              | node        | deltaP_solidsAccel    | 1380.42384063   |
| kpi                                                              | node        | deltaP_solidsFriction | 16373.2628921   |
| kpi                                                              | node        | deltaP_static         | 205.862407813   |
| kpi                                                              | node        | nBends                | 6               |
| kpi                                                              | node        | particleWallFriction  | 0.4             |
| kpi                                                              | node        | rho_gas               | 3.39180174393   |
| ... and 12 more reference values in the case’s <i>expected</i> . |             |                       |                 |

## thermo

**The theory you need.** Consistency tests interrogate DATA, not models: the van Ness point test checks measured  $(x, y, T, P)$  sets against the Gibbs–Duhem constraint (residuals in  $\ln(\gamma_1/\gamma_2)$ ); Herington’s area test is its integral cousin. A dataset that fails is not “bad” – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

**air01\_heater\_expand**

tutorials/steady/thermo/air01\_heater\_expand



**What it does:** Predefined mixture: heat a stream fed with ‘components ( air )’, which expands to N2/O2/Ar at load time

**The story:** showcase for TRACK A predefined-mixture expansion.

The thermoPhysPropDict lists ‘components ( air )’. At load time the loader splices ‘air’ into N2 + O2 + Ar (see constant/thermoPhysPropDict). A single heater then adds 5 kW to a 1 mol/s air stream at 1 bar, 298.15 K. The air is gas ( $vf = 1$ ), so the heater integrates the ideal-gas mixture  $H_{ig}$  over the three real components → a correct MULTICOMPONENT result.

Z-SEEDING NOTE (glass-box honesty): The expansion captures the member mole fractions (N2 0.7808 / O2 0.2095 / Ar 0.0093) as a per-component seed on the ThermoPackage (‘mixtureSeedVector()’). The feed below STATES the same fractions explicitly so the case is self-evident and the result is unambiguous – a stream that declares its own composition always wins over the seed. The seed exists so a future feed that OMITTS its composition can be filled from the mixture defaults; that wiring lives in the stream-construction path (Flowsheet / Composition), not in this case.

**What to expect (golden-locked):**

| kind    | unit/stream | key                     | expected         |
|---------|-------------|-------------------------|------------------|
| stream  | feed        | F                       | 0.001            |
| stream  | hot         | F                       | 0.001            |
| kpi     | airHeater   | F                       | 0.001            |
| kpi     | airHeater   | P                       | 100000           |
| kpi     | airHeater   | Q                       | 5000             |
| kpi     | airHeater   | Q_kW                    | 5                |
| kpi     | airHeater   | Q_per_mol               | 5000             |
| kpi     | airHeater   | T_in                    | 298.15           |
| kpi     | airHeater   | T_out                   | 468.478847995    |
| kpi     | airHeater   | dT                      | 170.328847995    |
| utility | airHeater   | heating.steamHP.duty_kW | 5                |
| utility | airHeater   | heating.steamHP.T       | 468.478847995    |
| utility | airHeater   | heating.steamHP.kg_s    | 0.00291715285881 |
| utility | airHeater   | heating.steamHP.eur_h   | 0.324            |

...and 3 more reference values in the case’s *expected*.

**basis01\_two\_unit\_chain**

tutorials/steady/thermo/basis01\_two\_unit\_chain



**What it does:** Basis reconciliation, the two-unit chain: a speciating flash hands its brine to a unit running a DIFFERENT thermo world, which transports the species basis as matter instead of re-asking the chemistry

**The story:** THE TWO-UNIT CHAIN (basis-reconciliation spike S2, ratified 2026-08-03).

Topology only; state lives in 0/.

feed → [speciator: isothermalFlash] → offgas

brine (carries BOTH bases: the apparent components | that ARE the state, and the species the | network resolved in them) v [transporter: heater, LOCAL molecular thermo world]

WHAT THE CHAIN IS FOR. The ‘speciator’ runs the case’s electrolyte world and SOLVES the chemistry: its brine outlet carries a speciation block stamped ‘origin "solved:speciator"’. The ‘transporter’ runs a DIFFERENT world – a plain molecular gammaPhi declared in its own ‘thermo {}’ block, which resolves no ions at all. It changes T (a duty) and nothing else: one material in, one material out, F and z untouched.

The question the chain asks: what happens to the species basis at that boundary?

Three answers were available and only one is honest. (a) Re-derive the block with the GLOBAL package. That is a chemistry the unit’s own world does not do, attributed to the unit – silent respeciation across a model boundary. This is what the engine did before the spike, and it is what the transport contract now stops. (b) Drop the block. The outlet then reports one basis where its inlet reported two, and a student comparing them sees chemistry vanish across a heater. (c) CARRY it. The heater moved no atoms and no charge; the ions in the brine are the same ions, in the same amounts, at a higher T. The block crosses as MATTER, stamped ‘origin "transported:transporter"’ so the reader can see it was carried and not re-solved.

(c) is the contract. ‘converged/warmBrine’ is where it is visible.

AND THE PART THAT TOOK A SECOND OPINION TO GET RIGHT. A heater is a heater: the outlet is at 330.81 K while the speciation it carries was solved at 313.15 K. The first version of this case stamped ‘solvedAtT’ and considered the matter closed – the file said where the block came from, so the reader could not be misled. That was wrong, and the counsel of 2026-08-03 named why:

AN EQUILIBRIUM IS NOT A COMPOSITION. It is a RELATION between a composition and a thermodynamic state. A composition solved at 313 K does not become the equilibrium composition at 331 K by being carried there with a note saying where it came from. Stamping the provenance makes the datum TRACEABLE; it does not make it VALID.

So the two claims are now separate, and only one of them travels:

the AMOUNTS are a material inventory – the transport conserved every species exactly, and they stand; the EQUILIBRIUM is a claim about this state, and it dies here.

‘converged/warmBrine’ therefore carries ‘equilibriumValidHere false’, both temperatures, and NO pH – the pH is WITHHELD rather than inherited, because inheriting it would publish a number nothing here computed. The same withholding applies to any activity, saturation index or conditional constant that would rest on the old equilibrium.

Staleness is DECIDED, not assumed: the block goes stale because T actually moved. A pump carrying the same brine at constant T and P across the same boundary would leave the equilibrium intact and the pH reported – the boundary alone does not invalidate anything; the change of state does. A reader who wants the equilibrium AT the outlet must put a unit there that can solve one; the molecular world cannot, which is the whole point of the boundary.

THE IDENTITY SECOND WITNESS. Delete the ‘thermo {}’ block below: the transporter then runs the GLOBAL electrolyte world, which CAN re-solve the chemistry, and does – at 332.23 K, giving a different (and locally correct) block. What must be identical either way is the INTERMEDIATE stream ‘brine’:

what unit 2 is cannot reach back and change what unit 1 solved. That is the invariant the gate probes; the differing outlet is not a failure, it is the boundary being visible.

AND SINCE 2026-08-09 IT IS VISIBLE IN THE TEMPERATURE TOO, which is worth reading slowly. A ‘heater’ now solves for the outlet STATE whose enthalpy is  $H_{in} + Q$  – in ITS OWN declared world (R-E1: an unpinned stream means the equilibrium of (T, P, z) in the CONSUMING unit’s thermo world). The molecular world below finds this brine 1.3 % vapour at ~331 K, so 40 kW buys 17.66 K there; the global electrolyte world holds the same volatiles as ions, finds essentially no vapour, and the same 40 kW buys 19.08 K to 332.23 K. Same enthalpy, two temperatures – exactly the settled rule that H is the conserved truth and T is the model-dependent readout. The consequence a reader will meet: the energy-balance report prices EVERY stream in the GLOBAL package, so this unit’s RAW closure reads 92.54 % rather than 100 %. That residual is the model boundary, not a first-law error – and SINCE THE SAME DAY THE REPORT SAYS SO. It was a named gap for exactly one afternoon (the report had no notion of a per-unit thermo world, recorded in docs/design/tp-stream-energy-coherence.md), and it was closed by the model-boundary LEDGER:

raw imbalance -2.9851 kW (dH 37.0149 vs 40.0000 kW of duty) boundary step -2.9851 kW (independently computed, see below) remaining  $9.4\text{e-}9$  kW → adjusted closure 100.00 %, no alarm

reports/balances/energyBalance\_byUnit.csv carries all three, side by side and never collapsed, plus a ledger row naming BOTH worlds by their declared models. The step is not taken on trust: the report assembles this unit’s declared world ITSELF, prices both streams in both packages at the same (T, P, z), and credits the step only if its own number reproduces the imbalance – an auditor that reuses the auditee’s arithmetic checks nothing. See docs/design/model-boundary-energy-ledger.md.

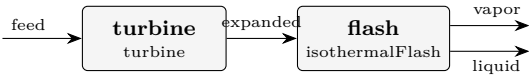
Before the 2026-08-09 heater fix the sensible inversion happened to track the global datum and the gap read 100 %, which was luck, not agreement.

#### What to expect (golden-locked):

| kind   | unit/stream | key       | expected          |
|--------|-------------|-----------|-------------------|
| stream | branchA     | F         | 0.00689112315881  |
| stream | branchB     | F         | 0.0206733694764   |
| stream | brine       | F         | 0.0275644926352   |
| stream | feed        | F         | 0.0277222222222   |
| stream | offgas      | F         | 0.000157729586982 |
| stream | warmBrine   | F         | 0.0275644926352   |
| kpi    | speciator   | F_alpha   | 0.0275644926352   |
| kpi    | speciator   | F_beta    | 0.000157729586982 |
| kpi    | speciator   | F_in      | 0.0277222222222   |
| kpi    | speciator   | K_CO2     | 4.78835784059     |
| kpi    | speciator   | K_N2      | 105036.780664     |
| kpi    | speciator   | K_NH3     | 0.777804159627    |
| kpi    | speciator   | K_ethanol | 0.180718643037    |
| kpi    | speciator   | K_water   | 0.0721940189402   |

... and 34 more reference values in the case’s *expected*.

**perUnitThermo01\_srknrtl**  
tutorials/steady/thermo/perUnitThermo01\_srknrtl



**What it does:** Per-unit thermo: turbine SRK (real gas) → flash NRTL (non-ideal liquid)

**The story:** Demonstrates PER-UNIT thermo models: the global package (Raoult) is the default, but each unit can override it with a ‘thermo { ... }’ block that replaces the model it cares about. Components stay global.

turbine → equationOfState SRK (real-gas expansion, high P) flash → activityModel NRTL (non-ideal ethanol/water liquid)

Pedagogical caveat: the stream that crosses from the turbine (SRK) into the flash (NRTL + idealGas) is re-interpreted by a different model – a small thermodynamic inconsistency at the boundary. Real practice keeps one consistent method per section; mixing is an advanced, deliberate act.

**What to expect (golden-locked):**

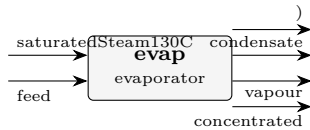
| kind                                                             | unit/stream | key       | expected         |
|------------------------------------------------------------------|-------------|-----------|------------------|
| stream                                                           | expanded    | F         | 0.0138888888889  |
| stream                                                           | feed        | F         | 0.0138888888889  |
| stream                                                           | liquid      | F         | 0.00854830793289 |
| stream                                                           | vapor       | F         | 0.005340580956   |
| kpi                                                              | flash       | F_alpha   | 0.00854830793289 |
| kpi                                                              | flash       | F_beta    | 0.005340580956   |
| kpi                                                              | flash       | F_in      | 0.0138888888889  |
| kpi                                                              | flash       | K_ethanol | 1.4838682811     |
| kpi                                                              | flash       | K_water   | 0.647356121192   |
| kpi                                                              | flash       | P         | 100000           |
| kpi                                                              | flash       | Q         | -413177.313739   |
| kpi                                                              | flash       | Q_kW      | -413.177313739   |
| kpi                                                              | flash       | T         | 353              |
| kpi                                                              | flash       | V_over_F  | 0.384521828832   |
| ... and 30 more reference values in the case's <i>expected</i> . |             |           |                  |

**thermoTest**



model1\_lumped\_evaporator

tutorials/steady/thermoTest/model1\_lumped\_evaporator



**What it does:** thermoTest model 1 – NaCl as a LUMPED molecular component: the boiling-point elevation is the IDEAL  $K_b \cdot m$  (van’t Hoff), BPE ~ 2.11 K. Compare model2 (same evaporator, Pitzer  $a_w \rightarrow$  BPE ~ 5.07 K): the REPRESENTATION axis, lumped vs dissociated, on one operation.

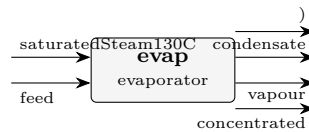
**The story:** thermoTest model1\_lumped\_evaporator / flowsheetDict NaCl treated as a LUMPED molecular component: the boiling-point elevation is the IDEAL van’t Hoff  $K_b \cdot m$  (~2.11 K). Model 1 of the thermoTest ladder – the deliberately-too-simple baseline the pitzer (model 2) and eNRTL rungs correct. Same evaporator, climbing thermo.

What to expect (golden-locked):

| kind                                                            | unit/stream        | key           | expected         |
|-----------------------------------------------------------------|--------------------|---------------|------------------|
| stream                                                          | concentrated       | F             | 0.020084561515   |
| stream                                                          | condensate         | F             | 0.0125           |
| stream                                                          | feed               | F             | 0.0277777777778  |
| stream                                                          | saturatedSteam130C | F             | 0.0125           |
| stream                                                          | vapour             | F             | 0.00769321626283 |
| kpi                                                             | evap               | A             | 20               |
| kpi                                                             | evap               | A_m2          | 20               |
| kpi                                                             | evap               | BPE           | 2.11524224539    |
| kpi                                                             | evap               | F_conc        | 0.020084561515   |
| kpi                                                             | evap               | F_conc_kmol_h | 72.3044214538    |
| kpi                                                             | evap               | F_conc_mass   | 0.417969209025   |
| kpi                                                             | evap               | F_cond        | 0.2251875        |
| kpi                                                             | evap               | F_cond_kg_h   | 810.675          |
| kpi                                                             | evap               | F_in          | 0.0277777777778  |
| ...and 24 more reference values in the case's <i>expected</i> . |                    |               |                  |

**model2\_pitzer\_evaporator**

tutorials/steady/thermoTest/model2\_pitzer\_evaporator



**What it does:** thermoTest model 2 – NaCl as a PITZER electrolyte (Na+ + Cl-, aqueous-inf-dilution reference): the boiling-point elevation comes from the real water activity  $a_w$ , BPE ~ 5.07 K vs the ideal 2.11 K of model1.

**The story:** BPE NOTE (2026-07-02, a story in three acts): (1) frozen 25 C surface → BPE 5.07 K; (2) LINEAR calorimetric T-slots (Parker fit) → 5.58 K – the 25 C derivative extrapolated straight to the boil; (3) the FULL Silvester & Pitzer 1977 T-functions (eqs 30-32, Table IV verbatim in Na-Cl.dat) +  $A_{\phi}(T)$  curated against their Table II → BPE 5.06 K.  $\beta_0(T)$  has a MAXIMUM near 150 C (their Fig 4): the linear slope overshoots where the real curve has already bent over. The lesson: a first derivative is not a function – validity windows are data.

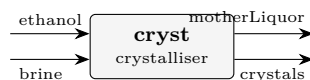
**What to expect (golden-locked):**

| kind   | unit/stream        | key           | expected         |
|--------|--------------------|---------------|------------------|
| stream | concentrated       | F             | 0.0203403273092  |
| stream | condensate         | F             | 0.0125           |
| stream | feed               | F             | 0.0277777777778  |
| stream | saturatedSteam130C | F             | 0.0125           |
| stream | vapour             | F             | 0.00743745046862 |
| kpi    | evap               | A             | 20               |
| kpi    | evap               | A_m2          | 20               |
| kpi    | evap               | BPE           | 5.0621680882     |
| kpi    | evap               | F_conc        | 0.0203403273092  |
| kpi    | evap               | F_conc_kmol_h | 73.225178313     |
| kpi    | evap               | F_conc_mass   | 0.422576829808   |
| kpi    | evap               | F_cond        | 0.2251875        |
| kpi    | evap               | F_cond_kg_h   | 810.675          |
| kpi    | evap               | F_in          | 0.0277777777778  |

... and 26 more reference values in the case's *expected*.

**model3\_enrtl\_antisolvent\_crystalliser**

tutorials/steady/thermoTest/model3\_enrtl\_antisolvent\_crystalliser



**What it does:** Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises  $\gamma_{\text{pm}}$ , and DROPS the saturation molality  $m_{\text{sat}}$  – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product  $K_{\text{sp}}$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\text{pm}}$ .

**The story:** thermoTest/model3\_enrtl\_antisolvent\_crystalliser – step 3 of the thermoTest arc (same {NaCl, CaSO<sub>4</sub>, ethanol, water} system, five property models): here the package selects eNRTL MIXED-SOLVENT, the only model in the arc that can reach the ethanol-lowered dielectric. The flowsheet is crystalliser06\_nacl\_antisolvent's, unchanged:

ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity  $\gamma_{\text{pm}}$  (eNRTL mixed-solvent + Born), so the saturation molality  $m_{\text{sat}}$  COLLAPSES (6.14 → ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product  $K_{\text{sp}} = (\gamma_{\text{pm}} * m_{\text{sat}})^2$  anchored in pure water, solved with the mixed-solvent  $\gamma_{\text{pm}}$  at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine ( $x_{\text{water}}$  0.900,  $x_{\text{NaCl}}$  0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol → salt-free  $x_{\text{ethanol}} \sim 0.40$ . Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

**What to expect (golden-locked):**

| kind   | unit/stream  | key               | expected          |
|--------|--------------|-------------------|-------------------|
| stream | brine        | F                 | 0.02777777777778  |
| stream | crystals     | F                 | 0.00202035108762  |
| stream | ethanol      | F                 | 0.01666666666667  |
| stream | motherLiquor | F                 | 0.0424240933568   |
| kpi    | cryst        | Ksp_activity      | 34.8439846653     |
| kpi    | cryst        | Q_kW              | -6.55987377671    |
| kpi    | cryst        | Q_removed         | 6559.87377671     |
| kpi    | cryst        | T_op              | 298.15            |
| kpi    | cryst        | antisolvent_xfrac | 0.399920015997    |
| kpi    | cryst        | c_sat             | 0.0517444790334   |
| kpi    | cryst        | cakeLiquidFlow    | 0.000329661969914 |
| kpi    | cryst        | cakeMoisture      | 0.1               |
| kpi    | cryst        | cakeWetness_pct   | 9.09090909091     |
| kpi    | cryst        | crystal_flow      | 0.0988038720389   |

... and 13 more reference values in the case's *expected*.

**model5\_nrtl\_flash**

tutorials/steady/thermoTest/model5\_nrtl\_flash



**What it does:** thermoTest model 5 – water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model. This is the antisolvent (ethanol) and the solvent (water) in their own role – the same substances that dissolve/precipitate NaCl in models 2-4, here separated by distillation.

**The story:** thermoTest model5\_nrtl\_flash / flowsheetDict Water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model, selected through the ethanolWater\_NRTL property package. Model 5 of the thermoTest ladder – the molecular gamma-phi anchor beside the electrolyte rungs.

**What to expect (golden-locked):**

| kind   | unit/stream | key      | expected        |
|--------|-------------|----------|-----------------|
| stream | feed        | F        | 0.0277777777778 |
| stream | liquid      | F        | 0.0170966158658 |
| stream | vapor       | F        | 0.010681161912  |
| kpi    | flash       | F_alpha  | 0.0170966158658 |
| kpi    | flash       | F_beta   | 0.010681161912  |
| kpi    | flash       | F_in     | 0.0277777777778 |
| kpi    | flash       | P        | 100000          |
| kpi    | flash       | Q        | 0               |
| kpi    | flash       | Q_kW     | 0               |
| kpi    | flash       | T        | 353             |
| kpi    | flash       | V_over_F | 0.384521828832  |
| kpi    | flash       | phaseSet | 0               |
| stream | feed        | T        | 353             |
| stream | liquid      | T        | 353             |

...and 3 more reference values in the case's *expected*.

**userops**

**The theory you need.** User-defined operations: the extension seam. A custom property op or unit compiled into the binary via the explicit factory – the tutorial shows the registration pattern (`registerBuiltins`) end to end.

userOp01\_yield\_reactor

tutorials/steady/userops/userOp01\_yield\_reactor



**What it does:** User-defined yieldReactor (case-local code/): neoPentane → nPentane, X=0.8

**The story:** A single unit op — but a USER-DEFINED one ('yieldReactor'), written in this case's code/ folder, not in the Choupo source tree. Build + run it with:

bin/buildCode tutorials/steady/userops/userOp01\_yield\_reactor

(NOT runCase: the type 'yieldReactor' does not exist in the stock binary; buildCode compiles the case's code/ against libchoupo.so.)

The op converts a SPECIFIED fraction of the reactant to the product (no kinetics) — a "yield reactor", which the built-in CSTR/PFR are not. Here: 80 % of neoPentane isomerised to nPentane. Because the two are isomers (same MW), the mass balance closes to 100 %.

What to expect (golden-locked):

| kind   | unit/stream | key              | expected          |
|--------|-------------|------------------|-------------------|
| stream | feed        | F                | 0.000277777777778 |
| stream | out         | F                | 0.000277777777778 |
| kpi    | reactor     | F_in_kmol_h      | 1                 |
| kpi    | reactor     | F_out_kmol_h     | 1                 |
| kpi    | reactor     | T                | 320               |
| kpi    | reactor     | X                | 0.8               |
| kpi    | reactor     | converted_kmol_h | 0.8               |
| stream | feed        | T                | 320               |
| stream | out         | T                | 320               |

userOp02\_component\_splitter

tutorials/steady/userops/userOp02\_component\_splitter



**What it does:** User-defined componentSplitter (case-local code/): shortcut benzene/toluene split

**The story:** A USER-DEFINED unit op (‘componentSplitter’), written in this case’s code/ folder. Build + run with:

```
runCase tutorials/steady/userops/userOp02_component_splitter
```

(runCase sees the code/ folder and compiles it first.)

The op is a SHORTCUT separator: a per-component recovery to the overhead. Here a benzene/toluene feed is split so 95 % of the benzene and only 8 % of the toluene leave overhead — an enriched light cut, the kind of stand-in you place before sizing a real column. Two outlet streams (overhead, bottoms); mass closes to 100 % (every component is conserved across the split).

**What to expect (golden-locked):**

| kind   | unit/stream | key               | expected        |
|--------|-------------|-------------------|-----------------|
| stream | bottoms     | F                 | 0.0134722222222 |
| stream | feed        | F                 | 0.0277777777778 |
| stream | overhead    | F                 | 0.0143055555556 |
| kpi    | splitter    | F_bottoms_kmol_h  | 48.5            |
| kpi    | splitter    | F_overhead_kmol_h | 51.5            |
| kpi    | splitter    | overhead_fraction | 0.515           |
| stream | bottoms     | T                 | 360             |
| stream | feed        | T                 | 360             |
| stream | overhead    | T                 | 360             |

**utilities**

**The theory you need.** Utility streams (steam levels, cooling water, refrigerants, electricity) carry duty into and out of the plant at declared temperature levels and prices; allocation is by *T*-level feasibility, announced.

utility01\_dowtherm\_preheat

tutorials/steady/utilities/utility01\_dowtherm\_preheat



**What it does:** Dowtherm A loop preheats a hydrocarbon feed (eps-NTU)

**What to expect (golden-locked):**

| kind   | unit/stream | key        | expected        |
|--------|-------------|------------|-----------------|
| stream | hotOil      | F          | 0.0416666666667 |
| stream | oilReturn   | F          | 0.0416666666667 |
| stream | processFeed | F          | 0.0111111111111 |
| stream | processOut  | F          | 0.0111111111111 |
| kpi    | preheater   | C_cold     | 2992.03981719   |
| kpi    | preheater   | C_hot      | 17325.8215856   |
| kpi    | preheater   | C_r        | 0.17269252153   |
| kpi    | preheater   | LMTD       | 59.6763812217   |
| kpi    | preheater   | NTU        | 5.01330226751   |
| kpi    | preheater   | Q          | 895145.718325   |
| kpi    | preheater   | Q_kW       | 895.145718325   |
| kpi    | preheater   | T_cold_in  | 320             |
| kpi    | preheater   | T_cold_out | 619.175737295   |
| kpi    | preheater   | T_hot_in   | 623.15          |

...and 10 more reference values in the case's expected.

utility02\_hitec\_csp\_heater

tutorials/steady/utilities/utility02\_hitec\_csp\_heater



**What it does:** HITEC molten salt loop sensibly heats pressurised (40 bar) boiler feedwater to ~244 C, just below T<sub>sat</sub>, for a downstream steam generator (eps-NTU, no phase change), ~208 kW duty

**What to expect (golden-locked):**

| kind                                                             | unit/stream | key        | expected        |
|------------------------------------------------------------------|-------------|------------|-----------------|
| stream                                                           | boilerFeed  | F          | 0.0222222222222 |
| stream                                                           | boilerOut   | F          | 0.0222222222222 |
| stream                                                           | boilerOut   | T          | 517.400926398   |
| stream                                                           | saltReturn  | F          | 0.0833333333333 |
| stream                                                           | saltSupply  | F          | 0.0833333333333 |
| kpi                                                              | boilerHX    | C_cold     | 1677.77777778   |
| kpi                                                              | boilerHX    | C_hot      | 12795.4797468   |
| kpi                                                              | boilerHX    | C_r        | 0.131122694184  |
| kpi                                                              | boilerHX    | LMTD       | 306.566828203   |
| kpi                                                              | boilerHX    | NTU        | 0.405298013245  |
| kpi                                                              | boilerHX    | Q          | 208465.443178   |
| kpi                                                              | boilerHX    | Q_kW       | 208.465443178   |
| kpi                                                              | boilerHX    | T_cold_in  | 393.15          |
| kpi                                                              | boilerHX    | T_cold_out | 517.400926398   |
| ... and 10 more reference values in the case's <i>expected</i> . |             |            |                 |



utility03\_glycol\_brine\_chiller

tutorials/steady/utilities/utility03\_glycol\_brine\_chiller



**What it does:** PG-30 brine sensibly cools an ethanol vapour stream from 330 K to ~273 K (eps-NTU, no phase change), ~20.8 kW duty

**What to expect (golden-locked):**

| kind                                                            | unit/stream   | key        | expected         |
|-----------------------------------------------------------------|---------------|------------|------------------|
| stream                                                          | brineReturn   | F          | 0.416666666667   |
| stream                                                          | brineSupply   | F          | 0.416666666667   |
| stream                                                          | ethanolOut    | F          | 0.00555555555556 |
| stream                                                          | ethanolVapour | F          | 0.00555555555556 |
| kpi                                                             | condChiller   | C_cold     | 57599.6888818    |
| kpi                                                             | condChiller   | C_hot      | 623.88888889     |
| kpi                                                             | condChiller   | C_r        | 0.0108314628256  |
| kpi                                                             | condChiller   | LMTD       | 12.519257418     |
| kpi                                                             | condChiller   | NTU        | 4.48797862867    |
| kpi                                                             | condChiller   | Q          | 35053.9207703    |
| kpi                                                             | condChiller   | Q_kW       | 35.0539207703    |
| kpi                                                             | condChiller   | T_cold_in  | 273.15           |
| kpi                                                             | condChiller   | T_cold_out | 273.758578301    |
| kpi                                                             | condChiller   | T_hot_in   | 330              |
| ...and 10 more reference values in the case's <i>expected</i> . |               |            |                  |

## Category II: The property bench (choupoProps)

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Questions asked of the thermodynamics BEFORE a simulation uses it – CREATE, SEE, TEST, FIT, PROMOTE.

### adsorption

**The theory you need.** The isotherm bench: evaluating a curated adsorption-equilibrium record (Henry-limit, saturation, pin and unit-invariance gates run aloud) and REGRESSING one from data – the multi-temperature van’t Hoff Langmuir fit with identifiability verdicts: single- $T$  data refuses  $\Delta H_{\text{ads}}$ , Henry-regime data yields only the product  $q_{\text{max}}b$ , and a non-converged fit refuses its verdict by name. *Full derivation: Theory Guide, the adsorption chapter.*

**isotherm01\_eval\_13x\_co2**

tutorials/props/adsorption/isotherm01\_eval\_13x\_co2

**eval\_13x\_co2**  
 isothermEval

**What it does:** Adsorption isotherm bench: CO2/zeolite13X q(T,p) grid + Henry/saturation/pin/unit-invariance gates

**The story:** One isothermEval operation: the curated CO2/zeolite13X langmuir record on a 3-T x 4-p grid (K, Pa – grid values are RAW canonical SI), with the unit-invariance twin (gate d) declared case-locally in constant/. Temperatures match the Jaschik 2020 dataset the FIT tutorial (isotherm02\_fit\_jaschik) regresses – eval here, fit there.

**What to expect (golden-locked):**

| kind | unit/stream  | key               | expected  |
|------|--------------|-------------------|-----------|
| diag | eval_13x_co2 | T0                | 2.931e+02 |
| diag | eval_13x_co2 | T1                | 3.131e+02 |
| diag | eval_13x_co2 | T2                | 3.331e+02 |
| diag | eval_13x_co2 | b_T0              | 1.227e+01 |
| diag | eval_13x_co2 | b_T1              | 3.772e+00 |
| diag | eval_13x_co2 | b_T2              | 1.337e+00 |
| diag | eval_13x_co2 | henry_T0          | 6.133e-04 |
| diag | eval_13x_co2 | henry_T1          | 1.886e-04 |
| diag | eval_13x_co2 | henry_T2          | 6.683e-05 |
| diag | eval_13x_co2 | qsat              | 5.000e+00 |
| diag | eval_13x_co2 | tRef              | 2.981e+02 |
| diag | eval_13x_co2 | gate_henry        | 1.000e+00 |
| diag | eval_13x_co2 | gate_henry_maxdev | 1.227e-07 |
| diag | eval_13x_co2 | gate_saturation   | 1.000e+00 |

*...and 9 more reference values in the case's **expected**.*

**isotherm02\_fit\_jaschik**

tutorials/props/adsorption/isotherm02\_fit\_jaschik

fit\_jaschik  
fitParametersrecovery\_multiT  
fitParametersgate\_singleT  
fitParametersgate\_henry  
fitParameters

**What it does:** Isotherm regression: CO2/13X Jaschik 2020 (CC BY) multi-T Langmuir fit + recovery/single-T/Henry identifiability gates

**The story:** Four fitParameters(kind isotherm) operations: the REAL Jaschik fit plus the three identifiability gates on zero-noise synthetics generated from a KNOWN surface ( $q_{\text{max}}$  4.6 mol/kg,  $b_{\text{ref}}$  3.0e-4 1/Pa @ 298.15 K,  $dH_{\text{ads}}$  -38000 J/mol) – so every gate has an exact truth to recover or to honestly refuse. Inline synthetic data is canonical SI (K, Pa, mol/kg); the vendored Jaschik dataset declares its own units (bar, mol/kg) and is converted at the data boundary.

**What to expect (golden-locked):**

| kind | unit/stream | key                    | expected     |
|------|-------------|------------------------|--------------|
| diag | fit_jaschik | converged              | 1.00000e+00  |
| diag | fit_jaschik | identifiable           | 1.00000e+00  |
| diag | fit_jaschik | n_data                 | 1.32000e+02  |
| diag | fit_jaschik | n_params               | 3.00000e+00  |
| diag | fit_jaschik | dof                    | 1.29000e+02  |
| diag | fit_jaschik | single_T               | 0.00000e+00  |
| diag | fit_jaschik | henry_regime           | 0.00000e+00  |
| diag | fit_jaschik | endothermic_suspicious | 0.00000e+00  |
| diag | fit_jaschik | fit.q_max.value        | 4.31448e+00  |
| diag | fit_jaschik | fit.q_max.stderr       | 5.17246e-02  |
| diag | fit_jaschik | fit.b_ref.value        | 4.06758e-04  |
| diag | fit_jaschik | fit.b_ref.stderr       | 2.90989e-05  |
| diag | fit_jaschik | fit.dH_ads.value       | -3.73835e+04 |
| diag | fit_jaschik | fit.dH_ads.stderr      | 1.90453e+03  |

...and 69 more reference values in the case's *expected*.

**isotherm03\_fit\_refusals**

tutorials/props/adsorption/isotherm03\_fit\_refusals

|                                          |                                    |                                             |
|------------------------------------------|------------------------------------|---------------------------------------------|
| gate_henry_nonconverged<br>fitParameters | gate_nonconverged<br>fitParameters | gate_langfull_nonconverged<br>fitParameters |
|------------------------------------------|------------------------------------|---------------------------------------------|

**What it does:** Fit refusal gates: non-converged fits refuse their verdicts by name

**The story:** Two ADVERSARIAL fitParameters(kind isotherm) operations – see the controlDict header. Synthetic data generated EXACTLY from the known surface (q\_max 4.6 mol/kg, b\_298 3.0e-4 1/Pa, dH\_ads -38000 J/mol, tRef 298.15 K) – zero noise, so the ONLY failure mode under test is the strangled optimiser, never the data.

**compare**

**The theory you need.** Method comparison runs the SAME property through different models side-by-side – UNIFAC vs NRTL, Joback vs measured, ideal vs EOS. The gap between curves IS the modelling uncertainty; picking a method is a decision the case author owns, made with eyes open. *Full derivation: Theory Guide, the Props Guide.*

## acetone01\_ipa\_water\_azeotrope

tutorials/props/compare/acetone01\_ipa\_water\_azeotrope

bp\_ipa  
propertyPoint

bp\_water  
propertyPoint

azeo\_luyben\_x  
propertyPoint

txy\_unifac  
propertyScan1D

**What it does:** IPA/water T-x-y at 1 atm from PREDICTIVE UNIFAC on a lake-estimated isopropanol, held against Luyben's published azeotrope

**The story:** THE FIRST RUNNABLE SLICE of the acetone-from-2-propanol reference case (docs/design/acetone-ipa-reference-case.md). It answers ONE question, and the question is chosen because it can be answered honestly today:

how close does a component Choupo has never curated, priced by a model that saw no data for it, land on a published azeotrope?

WHAT IS BEING TESTED, and it is a stack of three things at once – said up front, because a single number that agrees can hide two errors that cancel:

1. ISOPROPANOL IS A LAKE ESTIMATE. It is NOT in data/standards/. The record in this case's constant/components/ came from data/groupEstimative/, where every NUMBER is a Joback / Lee-Kesler estimate from the molecular structure – including Tb, Tc, Pc, omega and hence the whole Ambrose-Walton vapour pressure. Its estimated Tb is 359.98 K; the measured normal boiling point Luyben quotes is 355.4 K. The pure component is already 4.6 K out before any mixture model is asked anything. 2. UNIFAC IS PREDICTIVE. No pair was fitted, because Choupo ships no UNIQUAC or NRTL pair for IPA-water and inventing one would convert 'unsourced' into 'falsely sourced'. gamma comes from the groups. 3. WATER IS CURATED. It is the one leg of the three that rests on reviewed data, which is what makes the comparison legible at all.

THE COMPARISON POINTS: W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. He states the IPA/water azeotrope as 67.32 mol % IPA at 353.4 K and 1 atm, and normal boiling points IPA 355.4 K, water 373 K. Those numbers come out of his UNIQUAC model, fitted to VLE data neither this case nor its components have ever seen – so for the predictive route here they are an INDEPENDENT target.

A NOTE ON WHAT KIND OF ROW PINS THIS, because it is not an 'anchor'. Luyben's azeotrope is his UNIQUAC model's value, not a measurement, and the AAD computed here is derived from that number and this model TOGETHER – so nobody published it and no primary can be quoted for it. It is pinned as a self-recorded 'aad' row: what the suite holds is that the agreement has not MOVED. Anchoring properly needs a bubble temperature published as an operation diagnostic, which is a separate gap – see the note at the end.

WHAT THIS CASE IS NOT. It is not a reproduction of Luyben's flowsheet, and it makes no claim about the acetone process. The full reproduction needs three curated UNIQUAC pairs that do not exist; the design record's section 4 lists them and section 5 says which case category that would be. This is the bottom rung: the thermodynamics, alone, against three published points.

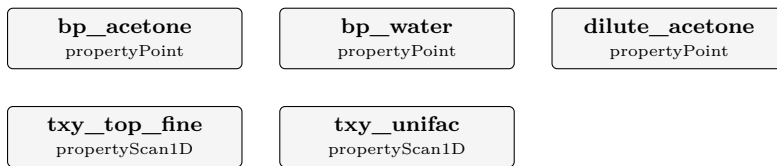
A GAP FOUND BUILDING THIS, and left visible rather than worked around. The natural shape for this case is three 'propertyPoint' ops asking for 'T\_bubble' at the three compositions, because a propertyPoint publishes into the operation diagnostics where a golden row can read it. It does not work: 'PropertyPoint' never reads its own 'properties ( ... )' list – it emits a fixed set and discards what the case asked for. Both the class header ("properties (all); // or pick a subset") and the published JSON schema (which names 'T\_bubble' among the accepted keywords) say otherwise. That is a parsed-and-dropped field, the shape this project has paid for three times, and it is why the comparison below rides the overlay instead.

What to expect (golden-locked):

| kind                                                            | unit/stream | key      | expected     |
|-----------------------------------------------------------------|-------------|----------|--------------|
| diag                                                            | bp_ipa      | Cp_ig    | 99.0857      |
| diag                                                            | bp_ipa      | H_R      | 0.0000       |
| diag                                                            | bp_ipa      | H_ig     | -257924.6674 |
| diag                                                            | bp_ipa      | H_real   | -257924.6674 |
| diag                                                            | bp_ipa      | P        | 101325.0000  |
| diag                                                            | bp_ipa      | S_R      | 0.0000       |
| diag                                                            | bp_ipa      | S_ig     | 329.0377     |
| diag                                                            | bp_ipa      | S_real   | 329.0377     |
| diag                                                            | bp_ipa      | T        | 350.0000     |
| diag                                                            | bp_ipa      | T_bubble | 359.9075     |
| diag                                                            | bp_ipa      | Z        | 1.0000       |
| diag                                                            | bp_ipa      | gamma    | 1.0916       |
| diag                                                            | bp_ipa      | v_molar  | 0.0287       |
| diag                                                            | bp_water    | Cp_ig    | 34.0552      |
| ...and 30 more reference values in the case's <i>expected</i> . |             |          |              |

**acetone04\_acetone\_water\_vle**

tutorials/props/compare/acetone04\_acetone\_water\_vle



**What it does:** Acetone/water T-x-y at 1 atm on PREDICTIVE UNIFAC – both components curated, so this isolates the mixture model

**The story:** MEASURE THE THERMODYNAMICS BEFORE BUILDING THE UNIT THAT RESTS ON IT.

Luyben’s absorber recovers acetone from the reactor’s hydrogen-rich offgas by washing it with water, and it is the next unit of the flowsheet to build (docs/design/acetone-ipa-reference-case.md). What that unit costs, and whether it can work at all, is set by ONE thing: how volatile acetone is out of a dilute aqueous solution. So that is measured here first – the same order that made acetone01 useful before acetone03 was written.

WHAT MAKES THIS A CLEANER TEST THAN acetone01. Both components are CURATED: acetone and water are in data/standards/, neither comes from the estimate lake. acetone01 had to measure a mixture model and a lake-estimated component at once and separate them afterwards; here there is nothing to separate. Whatever deviation appears is UNIFAC, plus whatever the pure vapour pressures carry – and the two pure boiling points below say how much that second part is.

WHAT LUYBEN GIVES, and it is less than for the IPA binary: acetone’s normal boiling point 329.4 K, and the qualitative statement that acetone/water has NO azeotrope but a pinch at the high-acetone end. A qualitative claim is still falsifiable – a model that invents an azeotrope here would be wrong in a way no AAD would show.

NOTE THE WATER LEG IS KNOWN TO BE 0.70 K LOW before it is measured again here: Choupo’s curated water record puts the normal boiling point at 372.4536 K against the definition’s 373.15 K, because its Antoine set is being used at the very top of its own declared window (see acetone01\_ipa\_water\_azeotrope/CLAUDE.md). It is repeated in this case rather than assumed, because a number quoted from another case is a number with two homes.

**What to expect (golden-locked):**

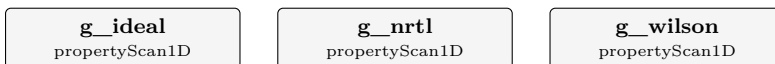
| kind | unit/stream | key      | expected     |
|------|-------------|----------|--------------|
| diag | bp_acetone  | Cp_ig    | 79.3862      |
| diag | bp_acetone  | H_R      | 0.0000       |
| diag | bp_acetone  | H_ig     | -214666.2971 |
| diag | bp_acetone  | H_real   | -214666.2971 |
| diag | bp_acetone  | P        | 101325.0000  |
| diag | bp_acetone  | S_R      | 0.0000       |
| diag | bp_acetone  | S_ig     | 302.9409     |
| diag | bp_acetone  | S_real   | 302.9409     |
| diag | bp_acetone  | T        | 330.0000     |
| diag | bp_acetone  | T_bubble | 329.4179     |
| diag | bp_acetone  | Z        | 1.0000       |
| diag | bp_acetone  | gamma    | 1.1170       |
| diag | bp_acetone  | v_molar  | 0.0271       |
| diag | bp_water    | Cp_ig    | 34.2177      |

... and 33 more reference values in the case’s *expected*.



**compare\_activity\_etoh\_water**

tutorials/props/compare/compare\_activity\_etoh\_water



**What it does:** Compare 3 activity models (ideal / NRTL / Wilson) for  $\gamma_{\text{ethanol}}$  in equimolar mixture with water vs T

**The story:** COMPARE 3 activity-coefficient models for  $\gamma_{\text{ethanol}}$  in an equimolar ethanol/water mixture, T = 280 K to 360 K, P = 1 atm.

ideal :  $\gamma = 1$  always (Raoult's law assumption). NRTL : strong non-ideality,  $\gamma_{\text{ethanol}} > 1$  (azeotrope behaviour). Wilson : similar to NRTL for this binary, slightly different curve.

Pedagogical point: a student studying ethanol/water VLE will see that "ideal" is a bad approximation here ( $\gamma_{\text{ethanol}}$  grows to ~2-3 at low T), and that NRTL/Wilson give similar non-ideal corrections. The choice between the two is then decided by which fits the project's experimental data better (workflow Mode B, see tutorials/props/README.md).

Each op overrides the global activityModel block via 'thermo {}'. The three CSVs (gamma\_ideal.csv, gamma\_nrtl.csv, gamma\_wilson.csv) are auto-overlaid by the GUI.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | g_ideal     | n_emptyCurves | 0        |
| diag | g_ideal     | n_failures    | 0        |
| diag | g_ideal     | n_points      | 41       |
| diag | g_nrtl      | n_emptyCurves | 0        |
| diag | g_nrtl      | n_failures    | 0        |
| diag | g_nrtl      | n_points      | 41       |
| diag | g_wilson    | n_emptyCurves | 0        |
| diag | g_wilson    | n_failures    | 0        |
| diag | g_wilson    | n_points      | 41       |

**compare\_eos\_co2**

tutorials/props/compare/compare\_eos\_co2

|                                  |                                |                               |
|----------------------------------|--------------------------------|-------------------------------|
| <b>z_ideal</b><br>propertyScan1D | <b>z_srk</b><br>propertyScan1D | <b>z_pr</b><br>propertyScan1D |
|----------------------------------|--------------------------------|-------------------------------|

**What it does:** Compare 3 EoS (idealGas / SRK / PR) for Z of CO2 near critical (T=310 K, P from 1 to 100 bar)

**The story:** COMPARE 3 equations of state for the same property at the same conditions. This is what choupProps is *\*for\**: helping a student or researcher CHOOSE the model that best describes the system at hand — not just look at pretty Z curves of one EoS at a time.

System: pure CO2 at T = 310 K (just above its critical temperature  $T_c = 304.13$  K), pressure swept from 1 bar to 100 bar. Near the critical pressure ( $P_c = 7.38$  MPa) the compressibility plunges as the fluid turns liquid-like; the three curves read (at P = 73.7 bar, the swept point nearest  $P_c$ ):

idealGas: Z = 1 always (textbook reference; ignores attraction). SRK : Z = 0.538 at  $P_c$ . PR : Z = 0.509 at  $P_c$  — systematically lower than SRK (more attraction), and essentially on the experimental value. NIST : the *\*experimental\** Z near the critical point is ~0.49.

The lesson: at and just above  $T_c$ , BOTH cubics track CO2 well — PR sits almost exactly on the ~0.49 datum and SRK is only ~0.05 high. This is a regime where cubic EoS are at their best, not their worst. Both Z curves keep falling past  $P_c$  to a shallow minimum (SRK ~0.30, PR ~0.275 near 98 bar) before rising again; the SRK-vs-PR gap is the diagnostic a student should read off the overlay.

Each operation overrides the global thermoPhysPropDict with a different EoS via the ‘thermo { }’ per-op block. The three CSVs are independent, so the GUI overlays them on the same Plotly canvas (same X axis “P”, same Y “Z”). The result is one figure with three curves, ready for the project report.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | z_ideal     | n_emptyCurves | 0        |
| diag | z_ideal     | n_failures    | 0        |
| diag | z_ideal     | n_points      | 50       |
| diag | z_srk       | n_emptyCurves | 0        |
| diag | z_srk       | n_failures    | 0        |
| diag | z_srk       | n_points      | 50       |
| diag | z_pr        | n_emptyCurves | 0        |
| diag | z_pr        | n_failures    | 0        |
| diag | z_pr        | n_points      | 50       |

**compare\_kinetics\_order**

tutorials/props/compare/compare\_kinetics\_order

**rate\_order1**  
 kinetics1D

**rate\_order2**  
 kinetics1D

**What it does:** Rate-law order from DATA: 1st- vs 2nd-order fits overlaid on measured  $c(t)$  — the data picks the order

**The story:** The "see, then decide" of CHEMICAL KINETICS: fit a 1st- and a 2nd-order rate law to the SAME measured  $c(t)$  and overlay both on the data. No 'k' is given, so each op REGRESSES its own k from the points (provenance = fitted). The data here are second-order – the 1st-order curve misses the tail, the 2nd lands on the points. The student picks the order from the evidence.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected    |
|------|-------------|---------------|-------------|
| diag | rate_order1 | R2            | 0.973397    |
| diag | rate_order1 | chi2          | 0.0105136   |
| diag | rate_order1 | fitted        | 1           |
| diag | rate_order1 | k             | 0.00065871  |
| diag | rate_order1 | max_abs_resid | 0.0544611   |
| diag | rate_order1 | n_data        | 8           |
| diag | rate_order1 | n_points      | 61          |
| diag | rate_order1 | order         | 1           |
| diag | rate_order1 | rms           | 0.0362518   |
| diag | rate_order2 | R2            | 0.999421    |
| diag | rate_order2 | chi2          | 0.000228631 |
| diag | rate_order2 | fitted        | 1           |
| diag | rate_order2 | k             | 0.000992909 |
| diag | rate_order2 | max_abs_resid | 0.00829088  |

*...and 8 more reference values in the case's **expected**.*

**compare\_psat\_ambrose\_walton**

tutorials/props/compare/compare\_psat\_ambrose\_walton

**psat\_aw\_vs\_antoine**  
 propertyScan1D

**What it does:** Psat: Ambrose-Walton corresponding-states PREDICTION vs fitted Antoine DATA, for benzene (non-polar, good) and ethanol (associating, degrades) — see the error, then decide

**The story:** Ambrose-Walton corresponding states (Ambrose & Walton 1989; Poling et al. 5th ed., Eq. 7-2.7) predicts vapour pressure from  $T_c$ ,  $P_c$  and omega ALONE – exactly the constants a Joback + Lee-Kesler estimate produces. So an estimated component becomes flashable with NO measured Psat data. But it is an ESTIMATE, and the glass-box rule is: SEE the error before you trust it.

This sweeps  $T = 270 \rightarrow 360$  K and tabulates four curves: Psat\_benzene – fitted Antoine (catalogue DATA) } non-polar pair Psat\_benzeneAW – Ambrose-Walton PREDICTION } Psat\_ethanol – fitted Antoine (catalogue DATA) } associating pair Psat\_ethanolAW – Ambrose-Walton PREDICTION }

Open ‘psat\_aw\_vs\_antoine.csv’ (GUI plot, or gnuplot). What you SEE: Near the normal boiling point (~350-360 K) BOTH agree to ~1 % – corresponding states is anchored there and is excellent. Towards the cold end (270 K) the curves separate, and MORE for the associating ethanol than for non-polar benzene: benzene AW vs Antoine: +3.3 % average, +8.9 % at 270 K ethanol AW vs Antoine: +5.2 % average, -16 % at 270 K So corresponding states is a solid FREE first predictor (it makes an estimated component flashable with no data), best near  $T_b$  and degrading at low reduced temperature – the degradation worse for hydrogen-bonding species. The lesson: prediction gets you started; for an associating species, or far below  $T_b$ , fit Antoine to data (choupoProps vaporPressureFit) before any design number.

**What to expect (golden-locked):**

| kind | unit/stream        | key           | expected |
|------|--------------------|---------------|----------|
| diag | psat_aw_vs_antoine | n_emptyCurves | 0        |
| diag | psat_aw_vs_antoine | n_failures    | 0        |
| diag | psat_aw_vs_antoine | n_points      | 46       |

**compare\_transport\_water**

tutorials/props/compare/compare\_transport\_water

mu\_iapws  
propertyScan1Dmu\_generic  
propertyScan1Dk\_iapws  
propertyScan1Dk\_generic  
propertyScan1D

**What it does:** Compare water transport: IAPWS (IF97 R12-08/R15-11) vs generic correlations (Andrade / SatoRiedel), 280-370 K

**The story:** COMPARE water transport properties computed TWO WAYS on the SAME water, over the saturated-liquid window  $T = 280\text{--}370\text{ K}$ :

(a) IAPWS reference – water flagged ‘pureFluids { water { method IF97; } }’ so the transport accessors route to the matching IAPWS releases: viscosity  $\rightarrow$  R12-08 (industrial) thermal conductivity  $\rightarrow$  R15-11 (background term) evaluated at the saturation state (the legacy accessors carry no P, so  $p = p_{\text{sat}}(T)$  – saturated liquid). These are THE international standard reference values (verified to  $1\text{e-}8$  by IAPWSTransport::verify(), which fires on every run).

(b) generic correlations – the same property KEY, but with a generic transport block and NO pureFluids flag, so the accessors fall to: viscosity  $\rightarrow$  Andrade  $\ln(\mu) = A + B/T$  thermal conductivity  $\rightarrow$  SatoRiedel fitted from the per-component .dat ( $T_c/T_b +$  the Andrade pair).

Because BOTH ops emit the same property key (‘viscosity\_liquid’, ‘thermal\_conductivity\_liquid’) the two curves overlay directly – the student SEES where the cheap generic correlation deviates from the IAPWS reference.

THE LESSON (the actual computed gaps): VISCOSITY – Andrade tracks IAPWS within a few percent over the window ( $\sim 7\%$  at 280 K,  $\sim 2\text{--}4\%$  mid-range,  $\sim 3\%$  at 370 K). The 2-parameter  $\ln(\mu)=A+B/T$  cannot follow the full curvature, but a few % is fine for a Reynolds number; it is NOT good enough for a precision steam duty. CONDUCTIVITY – SatoRiedel is a corresponding-states estimate and it FAILS BADLY for water: it underpredicts IAPWS by  $\sim 50\text{--}60\%$  across the window. Water is the textbook SatoRiedel failure (strong hydrogen bonding, anomalous k); this is exactly why "SEE before you commit" matters – a blind generic pick would have halved the heat-transfer coefficient. For water conductivity use IAPWS R15-11, full stop. Generic correlations are CHEAP and GENERAL (any compound with the handful of .dat parameters); IAPWS is EXACT but water-only. For a water-dominated process use the IF97 + IAPWS route; for a screening sweep over many fluids the generic correlations are the right tool. SEE the gap, then decide – do not pick blindly.

The GUI renders the CSVs via the existing compare machinery (overlay of the \*\_iapws / \*\_generic curves vs T) – no new Explorer kind.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | mu_iapws    | n_failures    | 0        |
| diag | mu_iapws    | n_points      | 46       |
| diag | mu_generic  | n_failures    | 0        |
| diag | mu_generic  | n_points      | 46       |
| diag | k_iapws     | n_failures    | 0        |
| diag | k_iapws     | n_points      | 46       |
| diag | k_generic   | n_failures    | 0        |
| diag | k_generic   | n_points      | 46       |
| diag | mu_iapws    | n_emptyCurves | 0        |
| diag | mu_generic  | n_emptyCurves | 0        |
| diag | k_iapws     | n_emptyCurves | 0        |
| diag | k_generic   | n_emptyCurves | 0        |

**compare\_visc\_liquid\_ammonia**

tutorials/props/compare/compare\_visc\_liquid\_ammonia

**mu\_andrade**  
propertyScan1D

**What it does:** Liquid-viscosity Andrade for ammonia, 235-300 K (cross-check mu@239.85K ~ 0.255 mPa.s)

**The story:** Liquid-viscosity Andrade correlation for pure ammonia (NH<sub>3</sub>), 235-300 K. Coefficients live in data/standards/components/NH3.dat (PROVISIONAL — a 2-point Andrade fit; see that file's provenance block).

Validation anchor: mu\_NH3(liquid) @ 239.85 K (normal b.p., -33.3 degC) is approx 0.255 mPa.s. This 2-point PROVISIONAL Andrade fit gives ~0.277 mPa.s at 240 K – about 9% ABOVE the anchor, i.e. it does NOT yet pass through the cross-check point. That gap is the honest state of a provisional fit (see the provenance block in NH3.dat); refitting against more low-T data would close it. Note this case scans a SINGLE method (Andrade) – it is a single-correlation curve, not a multi-method comparison. Ammonia is liquid only under pressure in this T window; viscosity\_liquid is a pure-T correlation, so the state P is immaterial to the value.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | mu_andrade  | n_emptyCurves | 0        |
| diag | mu_andrade  | n_failures    | 0        |
| diag | mu_andrade  | n_points      | 27       |

**compare\_visc\_liquid\_water**

tutorials/props/compare/compare\_visc\_liquid\_water

**mu\_andrade**  
 propertyScan1D

**mu\_vogel**  
 propertyScan1D

**What it does:** Compare 2 liquid viscosity correlations (Andrade vs Vogel) for water, 280-360 K

**The story:** COMPARE 2 liquid-viscosity correlations for pure water,  $T = 280\text{-}360\text{ K}$ .

Andrade:  $\ln(\mu) = A + B/T$  — 2-parameter, simplest fit, OK over a narrow  $T$  window. Vogel :  $\ln(\mu) = A + B/(T + C)$  — 3-parameter, captures the curvature better, recommended for wider  $T$  ranges.

Pedagogical point: for a project using water transport in a non-trivial  $T$  window (e.g.  $25\text{-}90\text{ }^{\circ}\text{C}$  for a heated reactor), the student should know WHICH correlation he is committing to. Both correlations give viscosity FALLING with temperature (the physical trend):  $\sim 1.3\text{-}1.4\text{ mPa}\cdot\text{s}$  near 280 K down to  $\sim 0.32\text{ mPa}\cdot\text{s}$  at 360 K. This case shows the two models disagree MOST at the cold end ( $\sim 6\%$  at 280 K), are within  $\sim 2\%$  at room  $T$ , and agree BEST at the hot end ( $\sim 1\%$  at 360 K) — the 2-parameter Andrade fit drifts where it is extrapolated coolest. Validate against the textbook value:  $\mu_{\text{water}} @ 298\text{ K} \approx 8.9 \times 10^{-4}\text{ Pa}\cdot\text{s}$  — Vogel reads  $8.93 \times 10^{-4}$  (essentially on it), Andrade  $9.09 \times 10^{-4}$  ( $\sim 2\%$  high).

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | mu_andrade  | n_emptyCurves | 0        |
| diag | mu_andrade  | n_failures    | 0        |
| diag | mu_andrade  | n_points      | 41       |
| diag | mu_vogel    | n_emptyCurves | 0        |
| diag | mu_vogel    | n_failures    | 0        |
| diag | mu_vogel    | n_points      | 41       |

compare\_vle\_etoh\_water

tutorials/props/compare/compare\_vle\_etoh\_water

txy\_ideal  
propertyScan1D

txy\_nrtl  
propertyScan1D

txy\_unifac  
propertyScan1D

consistency\_etoh\_water  
vleConsistency

txy\_wilson  
propertyScan1D

**What it does:** Ethanol/water T-x-y at 1 atm: ideal vs NRTL model curves OVERLAID with measured VLE data — see the data, pick the model

**The story:** The canonical "SEE, then DECIDE" props case: ethanol/water T-x-y at 1 atm, with two MODEL curves and the MEASURED data on ONE plot.

Piece 1 – two scan operations, identical but for the activity model: txy\_ideal Raoult ( $\gamma = 1$ ) → a smooth lens, NO azeotrope txy\_nrtl NRTL → the real minimum-boiling azeotrope Each emits the (x, T\_bubble, y\_eq) trio → the T-x-y / x-y / T-x / T-y toggle (x-y is the McCabe-Thiele diagram).

Piece 2 – the ‘experimental {}’ overlay: lists the two ops as ‘models’, and points ‘dataset’ at REAL measured points (constant/experiments/...). The GUI draws both model curves + the lab points together; the student SEES that the ideal model misses the azeotrope and NRTL captures it, and picks NRTL from evidence – not from a recommended-by badge.

What to expect (golden-locked):

| kind                                                     | unit/stream            | key           | expected  |
|----------------------------------------------------------|------------------------|---------------|-----------|
| diag                                                     | txy_ideal              | n_emptyCurves | 0         |
| diag                                                     | txy_ideal              | n_failures    | 0         |
| diag                                                     | txy_ideal              | n_points      | 51        |
| diag                                                     | txy_nrtl               | n_emptyCurves | 0         |
| diag                                                     | txy_nrtl               | n_failures    | 0         |
| diag                                                     | txy_nrtl               | n_points      | 51        |
| diag                                                     | txy_unifac             | n_emptyCurves | 0         |
| diag                                                     | txy_unifac             | n_failures    | 0         |
| diag                                                     | txy_unifac             | n_points      | 51        |
| diag                                                     | consistency_etoh_water | gamma1_inf    | 4.68506   |
| diag                                                     | consistency_etoh_water | gamma2_inf    | 2.27627   |
| diag                                                     | consistency_etoh_water | gd_max_resid  | 0.0964023 |
| diag                                                     | consistency_etoh_water | gd_mean_resid | 0.05208   |
| diag                                                     | consistency_etoh_water | herington_D   | 12.0482   |
| ... and 14 more reference values in the case's expected. |                        |               |           |

curation



**curate01\_water\_evidence\_partition**

tutorials/props/curation/curate01\_water\_evidence\_partition

**psat\_water\_validated**  
 vaporPressureFit

**psat\_water\_refused**  
 vaporPressureFit

**What it does:** The FIT/HELD-OUT partition contract: the same property fitted twice, differing only in whether evidence was held out – one VALIDATED, one VALIDATION REFUSED

**The story:** THE FIT / HELD-OUT VALIDATION CONTRACT, and its refusal, side by side.

TWO OPERATIONS. SAME COMPONENT. SAME PROPERTY. SAME TWO DATASETS. The ONLY thing that differs is whether any evidence was held out – which is exactly the variable the contract governs, and isolating it is why this witness does not use two different properties.

psat\_water\_validated lowT ‘role fit’ + highT ‘role validation’ psat\_water\_refused lowT ‘role fit’ + highT ‘role fit’

The first can report a held-out residual because the fitter never received the 325-370 K points. The second cannot, and says so by name instead of presenting its in-sample R2 as though it were an external check:

VALIDATION REFUSED: No independent experimental evidence remains after fitting.

THE ANCHOR IS ENFORCED BY CONSTRUCTION. The partition is parsed and frozen before any fitting code runs, and the fitter is handed only the ‘role fit’ points. It is not forbidden to move a held-out point into the fit – it is never given one. A rule can be broken by the next edit; a data-flow fact cannot. ‘role’ has NO default, because a defaulted role would make every dataset a fitting dataset and bring the in-sample problem back invisibly.

WHAT THIS CASE DOES NOT SHOW. Both datasets are generated from the SAME curated Antoine correlation (see either dataset header), so the held-out residual is arithmetic rather than evidence about the physics. This is a STRUCTURAL witness – no number in it is validated against measurement.

Contract: src/propertyOps/EvidencePartition.H

**What to expect (golden-locked):**

| kind | unit/stream          | key                | expected    |
|------|----------------------|--------------------|-------------|
| diag | psat_water_validated | A                  | 5.40221     |
| diag | psat_water_validated | B                  | 1838.68     |
| diag | psat_water_validated | C                  | -31.737     |
| diag | psat_water_validated | R2                 | 1           |
| diag | psat_water_validated | aad_heldout_pct    | 5.95532e-08 |
| diag | psat_water_validated | dHvap_kJ_per_mol   | 43.3787     |
| diag | psat_water_validated | n_heldout          | 10          |
| diag | psat_water_validated | n_points           | 10          |
| diag | psat_water_validated | rms_heldout_log10P | 2.76024e-10 |
| diag | psat_water_refused   | A                  | 5.40221     |
| diag | psat_water_refused   | B                  | 1838.68     |
| diag | psat_water_refused   | C                  | -31.737     |
| diag | psat_water_refused   | R2                 | 1           |
| diag | psat_water_refused   | dHvap_kJ_per_mol   | 42.1162     |

... and 2 more reference values in the case's *expected*.

**curate02\_vle\_heldout\_ethanol\_water**

tutorials/props/curation/curate02\_vle\_heldout\_ethanol\_water

 nrtl\_etoh\_water\_heldout  
 fitParameters

**What it does:** A binary NRTL pair regressed on eight MEASURED bubble temperatures and scored on three it never saw – the first held-out validation in the corpus against real evidence

**The story:** curate02 – THE FIT / HELD-OUT CONTRACT ON A BINARY PHASE EQUILIBRIUM.

The NRTL ethanol-water pair is regressed against eight measured bubble temperatures at 101.325 kPa, and then asked to predict THREE it never saw.

WHY THIS CASE EXISTS BESIDE curate01. curate01 proves the machinery: its two halves come from the same Antoine equation, so its held-out residual is round-off and its acceptance band is a numerical criterion, which its own ‘acceptance.origin’ says out loud. Here the evidence is MEASURED, so the held-out residual is a statement about the MODEL – the first place in the corpus where a fitted binary pair is scored on real evidence it was never handed.

THE SPLIT IS DECLARED AND ARITHMETIC. Every third point of the eleven-point series ( $x_1 = 0.20, 0.50, 0.80$ ) is held out, spread across the composition range rather than cut contiguously, because a model can interpolate well and extrapolate badly and a contiguous cut tests only one of those. The rule was fixed before anything was fitted.

THE ACCEPTANCE BAND IS DECLARED BEFORE THE FIT, WITH ITS ORIGIN, AND IS NOT ADJUSTED AFTERWARDS. Whatever verdict falls out of it is the result. A band chosen after seeing the residual would make the whole contract decorative.

Read the run’s four sections in order: FIT QUALITY is what the model does on what it saw; HELD-OUT EVIDENCE is what it did on what it did not.

TWO THINGS THIS CASE DOES NOT SHOW, BOTH VISIBLE IN ITS OWN GOLDEN.

(1) THE PARAMETERS ARE NOT IDENTIFIABLE. ‘identifiable 0’, ‘cond\_JtJ ~4e11’ and pairwise correlations of 0.9999 between each a/b pair: eight points at ONE pressure cannot separate the entropic and enthalpic halves of an NRTL parameter, because they only ever appear together over a narrow T span. The model PREDICTS well and the four numbers are individually meaningless – a validated prediction is not a determined parameter set, and a student must be able to read both facts off the same run. This is why the verdict is scoped ‘binaryVLE.T\_bubble’ in the dossier and not to the pair itself.

(2) THE HELD-OUT MARGIN IS NOT LARGE. 0.0737 % against a 0.1 % band is a pass by a factor of 1.4, not by an order of magnitude. The band was declared before the fit and is not adjusted; the margin is reported rather than described as comfortable.

The run also announces an extrapolation: ethanol’s vapour pressure is evaluated at 372 K, outside its declared Trange (273-369). Announced, not judged – it rides the caveat surface like every other extrapolation.

**What to expect (golden-locked):**

| kind | unit/stream             | key             | expected          |
|------|-------------------------|-----------------|-------------------|
| diag | nrtl_etoh_water_heldout | aad_heldout_pct | 0.0737            |
| diag | nrtl_etoh_water_heldout | chi2            | 0.8097            |
| diag | nrtl_etoh_water_heldout | chi2_in_sample  | 0.8097            |
| diag | nrtl_etoh_water_heldout | chi2_reduced    | 0.2024            |
| diag | nrtl_etoh_water_heldout | cond_JtJ        | 394331596991.7449 |
| diag | nrtl_etoh_water_heldout | converged       | 1.0000            |
| diag | nrtl_etoh_water_heldout | corr_computed   | 1.0000            |
| diag | nrtl_etoh_water_heldout | dof             | 4.0000            |
| diag | nrtl_etoh_water_heldout | identifiable    | 0.0000            |
| diag | nrtl_etoh_water_heldout | invertible      | 1.0000            |
| diag | nrtl_etoh_water_heldout | iter            | 29.0000           |
| diag | nrtl_etoh_water_heldout | max_abs_corr    | 0.9999            |
| diag | nrtl_etoh_water_heldout | max_abs_resid   | 0.5825            |
| diag | nrtl_etoh_water_heldout | n_data          | 8.0000            |

*...and 8 more reference values in the case's expected.*

**curate03\_thermoml\_fixture\_bubble**

tutorials/props/curation/curate03\_thermoml\_fixture\_bubble

|                                                 |
|-------------------------------------------------|
| <b>thermoml_fixture_bubble</b><br>fitParameters |
|-------------------------------------------------|

**What it does:** ThermoML  $\rightarrow$  extract  $\rightarrow$  partition  $\rightarrow$  fit  $\rightarrow$  held-out  $\rightarrow$  dossier, on a SYNTHETIC structural fixture: the plumbing proven end to end, and no physical claim anywhere

**The story:** curate03 – THE THERMOML PATH, END TO END, ON A STRUCTURAL FIXTURE.

ThermoML .xml  $\rightarrow$  extract-vle  $\rightarrow$  EvidencePartition  $\rightarrow$  FIT only  $\rightarrow$  T\_bubble regression  $\rightarrow$  held-out  $\rightarrow$  dossier

EVERY NUMBER HERE IS INVENTED. Both datasets were extracted by ‘bin/choupo-thermoml extract-vle’ from synthetic fixtures under tutorials/props/curation/fixtures/thermoml/, and each carries ‘provenance { source synthetic; }’ – so the run ANNOUNCES that its evidence was generated, and the dossier records it per dataset. A reader cannot arrive at the verdict without having been told what the numbers are.

WHAT THIS CASE PROVES: that a ThermoML file crosses into the evidence contract without losing its identity – archive file, DOI, system, pressure and checksum all survive into the dossier – and that the partition, the regression and the verdict work on evidence that arrived through the parser rather than being typed into a case.

WHAT IT DOES NOT PROVE: anything about ethanol, water, or NRTL. The acceptance band below is a PLUMBING tolerance and says so. The scientific witness on measured evidence is curate02, next door.

**What to expect (golden-locked):**

| kind | unit/stream             | key             | expected          |
|------|-------------------------|-----------------|-------------------|
| diag | thermoml_fixture_bubble | aad_heldout_pct | 0.0529            |
| diag | thermoml_fixture_bubble | chi2            | 0.6692            |
| diag | thermoml_fixture_bubble | chi2_in_sample  | 0.6692            |
| diag | thermoml_fixture_bubble | chi2_reduced    | 0.2231            |
| diag | thermoml_fixture_bubble | cond_JtJ        | 272780104373.5624 |
| diag | thermoml_fixture_bubble | converged       | 1.0000            |
| diag | thermoml_fixture_bubble | corr_computed   | 1.0000            |
| diag | thermoml_fixture_bubble | dof             | 3.0000            |
| diag | thermoml_fixture_bubble | identifiable    | 0.0000            |
| diag | thermoml_fixture_bubble | invertible      | 1.0000            |
| diag | thermoml_fixture_bubble | iter            | 42.0000           |
| diag | thermoml_fixture_bubble | max_abs_corr    | 1.0000            |
| diag | thermoml_fixture_bubble | max_abs_resid   | 0.6562            |
| diag | thermoml_fixture_bubble | n_data          | 7.0000            |

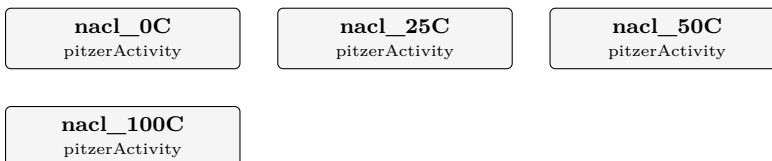
...and 8 more reference values in the case's *expected*.

**electrolyte**

**The theory you need.** Ions interact at long range (Debye–Hückel,  $\sqrt{I}$  law) and short range (ion–ion, ion–solvent). Choupo’s ladder: Davies (charge-only,  $I \leq 0.5$ ), single-salt Pitzer ( $\beta_0, \beta_1, C^\varphi$ ; to  $I \sim 6$ , and with the Silvester–Pitzer  $T$ -functions to 300°C), multi-ion Pitzer–HMW (per-ion, mixing terms  $\theta, \psi$  – seawater-class brines), and eNRTL (local-composition; single-salt, mixed-solvent, and the 2009 symmetric multi-salt with both published and Gibbs–Duhem-consistent formulations). Speciation solves mass action + charge balance by Newton in log-molality space; saturation indices  $SI = \log(IAP/K_{sp})$  flag scaling. Expect: models to DISAGREE above their validity ( $\gamma$  divergence is the lesson, e.g. Davies vs Pitzer at RO recoveries), and every activity model to announce itself. *Full derivation: Theory Guide, the electrolyte + speciation chapters.*

**archer01\_nacl\_cold\_to\_hot**

tutorials/props/electrolyte/archer01\_nacl\_cold\_to\_hot



**What it does:** NaCl Pitzer 0-100 C vs Archer 1992 – the cold edge the freezing curve stands on

**The story:** The catalogue's NaCl Pitzer surface (Appelo 25 C parameters + SP77 T-functions) against Archer, J. Phys. Chem. Ref. Data 21 (1992) 793, DOI 10.1063/1.555915, Tables 9 (phi) and 10 (gamma\_pm) – HIS fitted equation's check values ("presented only as a means to test the validity of coding", the paper's own words), at 273, 298, 323 and 373 K.

**WHY THE 273 K ROW IS THE POINT:** the SP77 temperature extension was fitted ABOVE 25 C, and the freezing-point curve (fpd01\_nacl\_freezing) leans on that surface BELOW 0 C as an announced extrapolation. This case measures what the extrapolation costs at the last anchor before the ice field – the op announces the T-window violation at 273 K, and the AAD against Archer says whether the announced risk is a real one at the cold edge.

**WHAT THIS COMPARISON IS,** said precisely: Archer's tables are values of an independent CRITICAL FIT (his global equation, 250-600 K), not raw measurements, and his database overlaps the corpora SP77 drew on. Agreement here is CROSS-EVALUATION CONSISTENCY between two independent fits of overlapping data – stronger than fit-consistency (our parameters never saw Archer's equation), weaker than agreement with an independent measurement. The Hamer & Wu case (pitzer\_gamma\_hamer\_wu) plays the same role at 25 C; this case extends it across T.

Temperatures are the table's printed values (273 K, not 273.15). If Archer meant T+0.15 K the shift in gamma\_pm is below 2e-4 – inside the quoted AADs either way, and stated here so the reading is a recorded choice, not an accident.

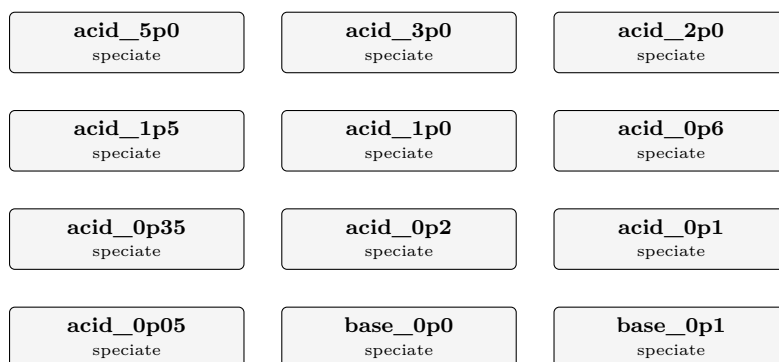
**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | nacl_0C     | aw_1m         | 0.9674   |
| diag | nacl_0C     | gamma_AAD_pct | 1.0546   |
| diag | nacl_0C     | gamma_nMeas   | 5.0000   |
| diag | nacl_0C     | gamma_pm_1m   | 0.6418   |
| diag | nacl_0C     | gamma_pm_3m   | 0.6649   |
| diag | nacl_0C     | phi_1m        | 0.9204   |
| diag | nacl_0C     | phi_AAD_pct   | 0.3706   |
| diag | nacl_0C     | phi_nMeas     | 5.0000   |
| diag | nacl_25C    | aw_1m         | 0.9668   |
| diag | nacl_25C    | gamma_AAD_pct | 0.1865   |
| diag | nacl_25C    | gamma_nMeas   | 5.0000   |
| diag | nacl_25C    | gamma_pm_1m   | 0.6571   |
| diag | nacl_25C    | gamma_pm_3m   | 0.7138   |
| diag | nacl_25C    | phi_1m        | 0.9363   |

...and 18 more reference values in the case's *expected*.

**bjerrum01\_carbonate\_pH**

tutorials/props/electrolyte/bjerrum01\_carbonate\_pH



**What it does:** Carbonate speciation across the pH range – the Bjerrum plot, titrated

**The story:** THE SPECIATION OF A CARBONATE SOLUTION ACROSS THE WHOLE pH RANGE – the diagram every textbook draws, built the way Choupo has to build it.

THE ONE THING THAT MAKES THIS DIFFERENT FROM THE TEXTBOOK VERSION: here the pH is NOT an axis you set – and that is a CHOICE this case makes, in one word, which you can read three screens down. A textbook draws the fractions against pH by substituting pH into the mass-action expressions, which is arithmetic and always works. Choupo can do that too: ‘pH 7.8;’ is a legal declaration, it fixes [H+], and it DROPS the electroneutrality row – the run then reports the net charge the composition carries (‘pH GIVEN – this is the net charge the analysis carries’) instead of forcing it to zero. That mode exists because a laboratory sheet reports a MEASURED pH, which is a datum. It is not what this case wants. Every op below declares ‘pH solve;’ instead, so [H+] joins the unknowns and the charge balance is the equation that closes the system. The price of that choice is the whole lesson: each point had to be a beaker somebody could weigh out, the 44 of them differ only in what is IN the beaker, and the x-axis is a RESULT, exactly as it is in a laboratory.

WHAT IS ACTUALLY IN EACH BEAKER. Total carbonate is held at 1 mmol/kg throughout. The 44 points differ only in how much strong acid or strong base sits beside it, and each one is a real neutral mixture somebody could weigh out:

Cl / C\_T = 5 ... 0.05 hydrochloric acid added to dissolved CO<sub>2</sub> Na / C\_T = 0 dissolved CO<sub>2</sub> alone (H<sub>2</sub>CO<sub>3</sub>) Na / C\_T = 1 sodium bicarbonate (NaHCO<sub>3</sub>) Na / C\_T = 2 sodium carbonate (Na<sub>2</sub>CO<sub>3</sub>) Na / C\_T = 2 ... 8 sodium carbonate + caustic soda

That is what ‘totalsBasis stoichiometric;’ declares below: these totals are neutral-molecule feeds, so the water-analysis charge-imbalance check – which is right for a LAB SHEET, where an imbalance means somebody mismeasured – does not apply. Here the imbalance IS the titrant, and the electroneutrality equation IS the weak-acid pH. Declaring it changes no number: it changes what the engine says about them.

WHAT TO LOOK FOR, in the order a student should look:

1. THE TWO CROSSEOVERS LAND ON THE TWO pK's, AND NOBODY PUT THEM THERE. CO<sub>2</sub>(aq) and HCO<sub>3</sub><sup>-</sup> are equal at pH 6.341; the record's log K<sub>1</sub> is 6.352. HCO<sub>3</sub><sup>-</sup> and CO<sub>3</sub><sup>-</sup> are equal at pH 10.262; log K<sub>2</sub> is 10.329. Both crossover pH's are INTERPOLATED between the two beakers that bracket them – the axis is 44 sampled points, not a continuum, and no beaker lands exactly on a crossing. Neither pK was used as an input to any point – they are the mass- action constants inside the network, and the crossover is where the titration happens to arrive. The small shift DOWN in both is the activity coefficients: gamma\_CO<sub>3</sub> is 0.90 at the first crossover and 0.81 at the second, while a textbook plot takes every gamma as 1.
2. THE THREE FRACTIONS DO NOT SUM TO ONE. They dip to 0.99966 through the middle of the range, and the missing carbon is NOT round-off: it is ‘NaHCO<sub>3</sub>aq’, the sodium-bicarbonate ION PAIR, which the curated chemistry carries and the three-curve drawing has no place for. The titrant is not a spectator. Look at the ‘m\_NaHCO<sub>3</sub>aq’ diagnostic – and note where it is ABSENT: the two beakers with no

sodium in them (the acid side, and  $\text{Na}/\text{C\_T} = 0$ ) have no such species to report, and the engine REFUSES to report a species that is not in the table rather than printing a zero. Absent and zero are different claims.

3. THE ACTIVITY COEFFICIENT OF THE DIVALENT ION MAKES A U.  $\gamma_{\text{CO}_3}$  runs 0.74 at the acid end, up to 0.98 near  $\text{Na}/\text{C\_T} = 0$  where the ionic strength is lowest, and back down to 0.67 at the alkaline end – because BOTH ends of a titration are loaded with strong electrolyte and only the middle is dilute. A textbook diagram takes every  $\gamma$  as 1 at every pH, which is the one assumption that is worst exactly where the diagram is usually read.

4. THE SATURATION INDICES ARE THERE ALL ALONG. Every point reports  $\text{SI}_{\text{nahcolite}} / \text{SI}_{\text{natron}} / \text{SI}_{\text{trona}}$  – how far the solution is from precipitating a sodium carbonate mineral. At 1 mmol/kg they stay far below zero, which is why this stays a clean acid-base diagram; make  $\text{C\_T}$  a hundred times larger and the diagram acquires a ceiling.

THE CONSTANTS, and their lineage:  $\log K_1 = 6.352$ ,  $\log K_2 = 10.329$  and  $K_w = 1\text{e-}14$  at 25 C, from the USGS PHREEQC database (public domain), which is where `data/standards/chemistry/{CO2aq,CO3}-formation.dat` and `water-dissociation.dat` come from. Each carries the van't Hoff  $dH$  and the full analytic  $\log K(T)$ , so this whole diagram can be redrawn at another temperature and both  $pK$ 's will move – change 'temperature' below and watch which crossover moves further.

NOT VALIDATED AGAINST MEASUREMENT. This case is a STRUCTURAL witness: it shows that the engine's speciation network, its activity model and its charge balance compose into the diagram the chemistry predicts. No titration data is compared against; the two crossovers landing near the two  $pK$ 's is an internal consistency check on the assembly, not agreement with an experiment.

#### What to expect (golden-locked):

| kind | unit/stream | key                             | expected     |
|------|-------------|---------------------------------|--------------|
| diag | acid_5p0    | I                               | 0.00499268   |
| diag | acid_5p0    | aw                              | 0.999802     |
| diag | acid_5p0    | chargeImposed                   | 1            |
| diag | acid_5p0    | chargeResidual                  | -3.02883e-16 |
| diag | acid_5p0    | $\gamma_{\text{CO}_2\text{aq}}$ | 1            |
| diag | acid_5p0    | $\gamma_{\text{CO}_3}$          | 0.738626     |
| diag | acid_5p0    | $\gamma_{\text{HCO}_3}$         | 0.927056     |
| diag | acid_5p0    | $m_{\text{CO}_2\text{aq}}$      | 0.000999896  |
| diag | acid_5p0    | $m_{\text{CO}_3}$               | 1.31694e-15  |
| diag | acid_5p0    | $m_{\text{HCO}_3}$              | 1.03591e-07  |
| diag | acid_5p0    | pH                              | 2.33456      |
| diag | acid_3p0    | I                               | 0.0029974    |
| diag | acid_3p0    | aw                              | 0.999874     |
| diag | acid_3p0    | chargeImposed                   | 1            |

*...and 635 more reference values in the case's expected.*

**composition01\_nacl**

tutorials/props/electrolyte/composition01\_nacl

**naclWater**  
 speciate

**What it does:** Apparent-salt input: the analysis is given as composition { NaCl 2.0 mol/kg }, expanded to Na + Cl ion totals through NaCl's dissociatesTo, with electroneutrality validated (Sum nu\*charge = 0). Byte-identical to writing totals { Na 2.0; Cl 2.0 } by hand.

*The speciate op expands each salt to ion totals through its dissociatesTo (NaCl → Na 1, Cl 1), and validates electroneutrality (Sum nu\*charge = 0) so a formulated-salts input can never silently unbalance charge. This is byte-identical to writing the ions by hand (totals { Na 2.0; Cl 2.0 }): SI\_halite -0.953, feed charge imbalance 0.00%.*

*composition{ } is the apparent-salt input mode; analyticalTotals{ } / totals{ } stay as the mode for waters measured directly in ions/masters (roadmap Phase B).*

**What to expect (golden-locked):**

| kind | unit/stream | key            | expected     |
|------|-------------|----------------|--------------|
| diag | naclWater   | I              | 2            |
| diag | naclWater   | SI_halite      | -0.953442    |
| diag | naclWater   | aw             | 0.930475     |
| diag | naclWater   | gamma_Cl       | 1.01683      |
| diag | naclWater   | gamma_Na       | 1.01683      |
| diag | naclWater   | m_Cl           | 2            |
| diag | naclWater   | m_Na           | 2            |
| diag | naclWater   | pH             | 7.13303      |
| diag | naclWater   | chargeImposed  | 1            |
| diag | naclWater   | chargeResidual | -6.39009e-17 |



**davies01\_band\_edge**

tutorials/props/electrolyte/davies01\_band\_edge

|                          |                          |                         |
|--------------------------|--------------------------|-------------------------|
| davies_m0p05<br>speciate | hmw_m0p05<br>speciate    | davies_m0p1<br>speciate |
| hmw_m0p1<br>speciate     | davies_m0p25<br>speciate | hmw_m0p25<br>speciate   |
| davies_m0p5<br>speciate  | hmw_m0p5<br>speciate     | davies_m1p0<br>speciate |
| hmw_m1p0<br>speciate     | davies_m2p0<br>speciate  | hmw_m2p0<br>speciate    |

**What it does:** The Davies band edge, MEASURED:  $\gamma_{\text{pm}}$  and  $a_{\text{w}}$  for NaCl under Davies beside PitzerHMW on identical brines,  $m = 0.05..6$  mol/kg – the cross-model curve behind every 'indicative beyond  $I \sim 0.5$ ' advisory

**The story:** THE DAVIES BAND EDGE, measured instead of remembered.

Same NaCl brine, same totals, same solved pH – speciated twice per rung: once under Davies (charge-only, the teaching rung;  $\phi = 1$  water activity) and once under PitzerHMW (per-ion virials; rigorous  $\phi$ ). What this measures is CROSS-MODEL divergence between two of Choupo's own engines – weaker than agreement with an independent measurement, and said so; the Pitzer surface itself is anchored elsewhere (pitzer01\_nacl, archer01\_nacl\_cold\_to\_hot, pitzer\_gamma\_hamer\_wu).

WHY IT EXISTS: every electrolyte run above  $I \sim 0.5$  mol/kg carries the advisory "Davies ... INDICATIVE"; the marcilla01/02 anchor contacts put numbers on what that costs at saturation (halite at 3.65 vs the measured 6.14 mol/kg). This case turns the advisory into a CURVE – per rung,  $\gamma_{\text{Na}}$ ,  $\gamma_{\text{Cl}}$  and  $a_{\text{w}}$  under both closures, side by side. No threshold is invented: the divergence grows smoothly and there is no principled cut, so the table reports and the reader judges (the same posture as the neglected-dCp advisory in SolidPhase).

*Same NaCl brine speciated twice per rung: Davies (teaching rung, charge-only,  $\phi = 1$ ) beside PitzerHMW (per-ion virials), 298.15 K. Cross-model divergence between two in-tree engines — weaker than an independent measurement and said so; the Pitzer surface itself is anchored elsewhere (pitzer01\_nacl, archer01\_nacl\_cold\_to\_hot, pitzer\_gamma\_hamer\_wu).*

*\* The " $I \sim 0.5$  trust band" every advisory quotes is real: the  $\gamma \pm$  error crosses  $\sim 2-8$  % over  $m = 0.25-0.5$  and explodes beyond it (Davies'  $+0.3 \cdot I$  term has no physics above the band — it is the fit's tail). \* At  $m = 6$ , Davies reports  $SI_{\text{halite}} = +1.10$  where PitzerHMW reports  $-0.022$  — the rigorous rung sits essentially AT saturation beside Archer's peritectic (6.096 mol/kg), and its  $a_{\text{w}} = 0.7592$  beside the literature  $\approx 0.753$ . This is the whole quantitative story of the marcilla01/02 anchor gaps (Davies saturating at 3.65 vs 6.14 mol/kg; the too-wet organic from  $a_{\text{w}} \approx 0.88$ ): the model, not the machinery.*

What to expect (golden-locked):

| kind | unit/stream  | key       | expected  |
|------|--------------|-----------|-----------|
| diag | davies_m0p05 | I         | 0.0500001 |
| diag | davies_m0p05 | SI_halite | -4.34316  |
| diag | davies_m0p05 | aw        | 0.9982    |
| diag | davies_m0p05 | gamma_Cl  | 0.821203  |
| diag | davies_m0p05 | gamma_Na  | 0.821203  |
| diag | davies_m0p05 | m_Cl      | 0.05      |
| diag | davies_m0p05 | m_Na      | 0.05      |
| diag | davies_m0p05 | pH        | 7.00292   |
| diag | hmw_m0p05    | I         | 0.0500001 |
| diag | hmw_m0p05    | SI_halite | -4.34351  |
| diag | hmw_m0p05    | aw        | 0.998302  |
| diag | hmw_m0p05    | gamma_Cl  | 0.820867  |
| diag | hmw_m0p05    | gamma_Na  | 0.820867  |
| diag | hmw_m0p05    | m_Cl      | 0.05      |

...and 146 more reference values in the case's *expected*.

**edwards01\_sour\_water\_activity**

tutorials/props/electrolyte/edwards01\_sour\_water\_activity

sourWater\_edwards  
speciatesourWater\_davies  
speciate

**What it does:** Edwards et al. (1978) activity model on a sour-water NH<sub>3</sub>/CO<sub>2</sub> brine – the molecular solutes get a real gamma from the model their parameters were regressed in, where Davies can only return 1.

**The story:** the WITNESS for the ‘edwardsPitzer’ activity model.

**WHAT THIS CASE IS FOR.** Choupo already carries two aqueous activity models, and each fails this fluid in its own way:

‘davies’ is a charge-only correlation. Every singly-charged ion gets the same gamma and every NEUTRAL solute gets gamma = 1 – exactly, by construction, at every ionic strength. A sour-water stripper’s whole difficulty is the volatility of dissolved NH<sub>3</sub> and CO<sub>2</sub>, i.e. of two NEUTRAL molecules, and Davies cannot say anything about either. ‘pitzerHMW’ is the Harvie-Moller-Weare parameterisation: a DIFFERENT truncation, with a different parameter set (beta2, C-phi, theta, psi), fitted to strong electrolytes. Its numbers are not Edwards’ numbers, and Edwards’ numbers are not valid inside its equation.

‘edwardsPitzer’ is the truncation Edwards, Maurer, Newman & Prausnitz (AIChE J. 24(6):966-976, 1978) REGRESSED HIS NH<sub>3</sub>/CO<sub>2</sub>/H<sub>2</sub>S/SO<sub>2</sub> PARAMETERS IN: Pitzer’s expansion cut after the second virial coefficient, extended to molecular solutes. Using his parameters anywhere else would be quoting his numbers under someone else’s equation.

THE COMPOSITION is the sour-water region his Table 7 anchors: ammonia in large excess over carbon dioxide, the carbonate and carbamate networks both live, at an ionic strength around 1 mol/kg where a dilute correlation has no business being trusted.

**WHAT TO READ IN THE OUTPUT.** Run the two operations side by side and compare ‘gamma\_NH3aq’ and ‘gamma\_CO2aq’:

under ‘davies’ they are 1.000000 and 1.000000. Not approximately – exactly, because the model’s gamma is a function of charge alone and the charge is zero. A student who has only ever seen Davies may not realise this is a STRUCTURAL silence rather than a small correction. under ‘edwardsPitzer’ they are 1.0224 and 1.0101: both SALTED OUT, NH<sub>3</sub> rather more than CO<sub>2</sub>. Both rise because both Table 4 self-terms are positive at 25 C – Eq 14 gives beta0(NH<sub>3</sub>,NH<sub>3</sub>) = 0.01522 and beta0(CO<sub>2</sub>,CO<sub>2</sub>) = 0.00822 kg/mol – and NH<sub>3</sub> rises further because it is the more abundant of the two by three orders of magnitude, so its own self-term dominates its Eq 8 sum.

THE SPECIATION MOVES TOO, and by far more than the gammas: the answer is I = 2.78 against Davies’ 1.78, with m\_NH3aq 0.41 against 1.01. That is not a molecular effect at all – it is the IONIC one. Davies at I ~ 2 returns gamma\_NH<sub>4</sub> = 0.956, because its bracket saturates and then turns back up; Edwards’ Eq 8 returns 0.301 at the same charge. The NH<sub>4</sub><sup>+</sup>/NH<sub>3</sub>aq equilibrium reads those gammas, so the split follows. The lesson is the one the advisory prints on the Davies run: a dilute correlation used far outside its band does not merely lose accuracy, it changes the answer’s SHAPE.

Neither number here is validated against measurement – see below.

**WHAT THIS CASE IS \*NOT\*.** It is not the Table 7 anchor. Reproducing Table 7 needs the vapour side – Henry’s constants on the paper’s own convention, the two-phase equilibrium, the total pressure – and that slice is BUILT: ‘edwards02\_table7\_vle’ (2026-08-23; record in docs/design/sour-water-stripper-scope.md section 6c). This case validates the ACTIVITY SURFACE alone, which is the piece the model owns.

**WHAT IS AND IS NOT VALIDATED HERE.** This case is a STRUCTURAL witness: it proves the model is wired, that its five parameter rules fire and are labelled, and that a neutral solute gets a gamma. It does NOT validate any number against data – there is no measurement in it. The kernel itself is separately checked against two closed forms the paper supplies (the Eq 8a low-dissociation limit and the Debye-Huckel limiting law), and the measured-vs-computed comparison is Table 7 – edwards02\_table7\_vle. A green run here means "the machine is right", never "the answer is right".

HONESTY ABOUT THE PARAMETERS. Edwards measured exactly one family: beta0 for LIKE molecule pairs (Eq 14, Table 4). Every other entry in the matrix Eq 8 sums over is ESTIMATED from those by his Eqs 21-24, and the paper calls both of its parameter tables "preliminary results subject to change as more and better data become available". The engine prints, at verbosity  $\geq 3$ , WHICH RULE produced each number – one measured family and four estimation routes – so nothing here can be mistaken for a measured constant.

DATA SCOPE: SEALED (bin/choupo-import). The case carries its own species, reactions and EdwardsPitzer parameter records under constant/, each named and sha256-verified by constant/propertyManifest, and the runtime is FORBIDDEN the installation catalogue.

Sealing this case is where the importer's dependency closure was found not to know the EdwardsPitzer parameter home. The first seal kept 9 of the 28 pair parameters – the Bronsted like-sign zeros, which need no record at all – and the case still RAN, still converged, and answered with 19 terms missing: gamma\_HCO3 and gamma\_NH4 identical (the charge-only Debye-Huckel fallback) and both neutral gammas exactly 1, which is what a Davies run prints. Nothing refused. The importer now compares the staged sealed run against this case's own 'expected' golden, because running is not agreeing.

#### What to expect (golden-locked):

| kind | unit/stream       | key         | expected    |
|------|-------------------|-------------|-------------|
| diag | sourWater_edwards | I           | 2.77652     |
| diag | sourWater_edwards | aw          | 0.954591    |
| diag | sourWater_edwards | gamma_CO2aq | 1.01011     |
| diag | sourWater_edwards | gamma_CO3   | 0.0181147   |
| diag | sourWater_edwards | gamma_HCO3  | 0.227994    |
| diag | sourWater_edwards | gamma_NH3aq | 1.02239     |
| diag | sourWater_edwards | gamma_NH4   | 0.30091     |
| diag | sourWater_edwards | m_CO2aq     | 0.000484785 |
| diag | sourWater_edwards | m_CO3       | 0.537903    |
| diag | sourWater_edwards | m_HCO3      | 0.911612    |
| diag | sourWater_edwards | m_NH3aq     | 0.41019     |
| diag | sourWater_edwards | m_NH4       | 2.48981     |
| diag | sourWater_edwards | pH          | 9           |
| diag | sourWater_davies  | I           | 1.77822     |

... and 16 more reference values in the case's *expected*.

**edwards02\_table7\_vle**

tutorials/props/electrolyte/edwards02\_table7\_vle

table7\_row1  
speciatetable7\_row2  
speciatetable7\_row3  
speciaterow1\_nocarbamate  
speciate

**What it does:** Edwards (1978) Table 7 anchor: NH<sub>3</sub>-CO<sub>2</sub>-H<sub>2</sub>O VLE at 100 degC – the paper’s own predictions reproduced from its own equations, the experiment quoted as context.

**The story:** the LITERATURE ANCHOR for the edwardsPitzer model: Table 7 of Edwards, Maurer, Newman & Prausnitz, AIChE J. 24(6):966-976 (1978) – NH<sub>3</sub>-CO<sub>2</sub>-H<sub>2</sub>O at 100 degC, three points.

edwards01 is the STRUCTURAL witness: it proves the activity surface is wired, and its own header says none of its numbers is validated against anything. This case is the validation it points at. For each Table 7 point the paper states the total molalities of ammonia and carbon dioxide in the LIQUID and predicts the speciation (molecular NH<sub>3</sub> and CO<sub>2</sub>, ionic strength) and the VAPOUR above it (y\_NH<sub>3</sub>, y\_CO<sub>2</sub>, total pressure). The whole model stack here is the paper’s own:

liquid activities Eq 8/10 (edwardsPitzer, the case surface) equilibria K(T) Eq 7 / Table 1 + Eq 20 (case-local chemistry/) Henry’s constants Eq 13 / Table 3 (parameters/EdwardsPitzer/henry-\*) pressure correction Eq 11 / Table 2 (vInfinity in the same records) vapour side Eq 5/6 (the speciate op’s vapour{} block)

WHAT IS COMPARED, AND AGAINST WHAT. Table 7 carries TWO different columns and conflating them would be dishonest:

the paper’s own PREDICTIONS (column I) – a code-to-code comparison. Those are the golden values here, with ONE declared gap: the paper computes vapour-phase fugacity coefficients from a model its transcription does not carry, so this case declares ‘fugacity ideal;’ ( $\phi = 1$ ) and the residual disagreement is expected to be of the size of that correction (a few percent at 2-3 atm), plus the numerical difference between two implementations of Eqs 7-24.

the EXPERIMENT (Otsuka et al. 1960) – quoted below as CONTEXT, never as a target. The paper’s OWN agreement with it is rough (first row: y\_NH<sub>3</sub> 0.066 measured vs 0.123 predicted – a factor of two) and the paper says so plainly. A student reading this case sees a correlation reproduced closely and its real-world agreement stated honestly, which is the opposite of a tool that prints one number.

Table 7, the paper’s column I (the golden targets here): row mNH<sub>3</sub>tot mCO<sub>2</sub>tot m\_NH<sub>3</sub> m\_CO<sub>2</sub> I y\_NH<sub>3</sub> y\_CO<sub>2</sub> P[atm]  
 1 2.90 1.45 1.245 0.014 1.50 0.123 0.494 2.51 2 3.71 1.14 2.279 0.0056 1.26 0.293 0.230 1.98  
 3 4.30 0.907 3.087 0.0030 1.01 0.407 0.116 1.97 The experiment (Otsuka 1960, context only): 1 0.066 0.506 3.15 2 0.274 0.202 2.08 3 0.355 0.095 2.00

THE CARBAMATE ENTERS THE CATALOGUE HERE (case-local): NH<sub>2</sub>COO- with Eq 20’s equilibrium. Without it, an ammonia-rich carbonated water reports molecular molalities the paper’s own predictions contradict – it holds a substantial share of the dissolved CO<sub>2</sub> at these conditions. Eq 20 is fitted 20-60 degC and used at 100 degC exactly as the paper uses it, with the extrapolation ANNOUNCED and the paper’s own sensitivity argument in the record header (halving K<sub>20</sub> moves P by under 2 %).

DECLARED APPROXIMATIONS, each announced by the run: ‘fugacity ideal;’ –  $\phi = 1$  on all vapour species (see above); the solvent Poynting factor of Eq 6 is omitted (< 0.2 % below 4 atm); Eq 20 extrapolated 60 → 100 degC (the paper’s own move); water’s Psat comes from the case catalogue’s own curated Antoine, not from a value the paper does not print.

DATA SCOPE: sealed (bin/choupo-import); every record named and verified by constant/propertyManifest.

**What to expect (golden-locked):**

| kind | unit/stream | key            | expected   |
|------|-------------|----------------|------------|
| diag | table7_row1 | I              | 1.547      |
| diag | table7_row1 | P_atm          | 2.51337    |
| diag | table7_row1 | aw             | 0.946304   |
| diag | table7_row1 | chargeImposed  | 1          |
| diag | table7_row1 | chargeResidual | 2.5541e-13 |
| diag | table7_row1 | gamma_CO2aq    | 0.919147   |
| diag | table7_row1 | gamma_CO3      | 0.022685   |
| diag | table7_row1 | gamma_HCO3     | 0.268386   |
| diag | table7_row1 | gamma_NH2COO   | 0.528413   |
| diag | table7_row1 | gamma_NH3aq    | 0.989831   |
| diag | table7_row1 | gamma_NH4      | 0.318905   |
| diag | table7_row1 | m_CO2aq        | 0.0141073  |
| diag | table7_row1 | m_CO3          | 0.0555284  |
| diag | table7_row1 | m_HCO3         | 1.09498    |

*...and 99 more reference values in the case's **expected**.*

**edwards03\_table8\_vle**

tutorials/props/electrolyte/edwards03\_table8\_vle

table8\_row1  
speciatetable8\_row2  
speciatetable8\_row3  
speciate

**What it does:** Edwards (1978) Table 8 anchor: NH<sub>3</sub>-H<sub>2</sub>S-H<sub>2</sub>O VLE at 80 degC – the paper’s own predictions reproduced from its own equations, the experiment quoted as context.

**The story:** the SECOND literature anchor for the Edwards (1978) model stack: Table 8 of the paper, ammonia + hydrogen sulfide in water at 80 degC, against the paper’s OWN prediction column I.

**WHY A SECOND ANCHOR.** edwards02 (Table 7) pins the NH<sub>3</sub>-CO<sub>2</sub> system, where the difficulty is the carbonate/carbamate network. Table 8 exercises what Table 7 cannot: the SULFIDE family (the mi-H<sub>2</sub>Saq-\* molecule-ion records curated 2026-08-24 from the paper’s Table 6, including its two TABULATED ZEROS), a different temperature (80 degC, so every T-polynomial is read off its 25 degC calibration point), and ionic strengths up to 7.8 mol/kg – beyond the 6-molal Pitzer validity the paper itself quotes. Rows 2 and 3 are code-to-code targets and NOT physical claims; the paper runs them and says so ("if ionization is extensive, the upper limit is necessarily lower").

**WHAT IS COMPARED, AND AGAINST WHAT.** Exactly the edwards02 posture:

the paper’s own PREDICTIONS (column I: all beta0 from Tables 4 and 6, the configuration these curated records carry) are the anchor targets; the EXPERIMENT (Miles & Wilson 1975) is CONTEXT, never a target. The paper’s own worst multisolute disagreement lives here: rows 2 and 3 measure 16.50 and 13.20 atm against predictions of 12.91 and 12.64 atm (22-28 %), and its Table 9 sensitivity analysis shows the liquid-composition error needed to close such gaps.

**THE CHEMISTRY IS THE PAPER’S OWN:** case-local constant/chemistry/ carries Table 1 (H<sub>2</sub>O, NH<sub>3</sub> and H<sub>2</sub>S rows) in the formation direction, each header recording the derivation. The H<sub>2</sub>S formation logK25 is 7.0033 against the independently curated PHREEQC 6.994 – two primaries 0.009 apart, a finding, not an error.

**DECLARED GAPS,** the same list as edwards02 plus none: ‘fugacity ideal;’ – the paper’s own vapour-phi model is not transcribed; the residual is expected to carry that correction (largest on rows 2-3 at 13-16 atm); the solvent Poynting factor of Eq 6 is omitted; water’s Psat comes from the case catalogue’s own curated Antoine.

**DATA SCOPE:** sealed (bin/choupo-import); every record named and verified by constant/propertyManifest.

**What to expect (golden-locked):**

| kind | unit/stream | key            | expected     |
|------|-------------|----------------|--------------|
| diag | table8_row1 | I              | 1.13632      |
| diag | table8_row1 | P_atm          | 1.10768      |
| diag | table8_row1 | aw             | 0.896601     |
| diag | table8_row1 | chargeImposed  | 1            |
| diag | table8_row1 | chargeResidual | -4.35684e-17 |
| diag | table8_row1 | gamma_H2Saq    | 1.08209      |
| diag | table8_row1 | gamma_HS       | 0.516824     |
| diag | table8_row1 | gamma_NH3aq    | 1.06377      |
| diag | table8_row1 | gamma_NH4      | 0.531883     |
| diag | table8_row1 | m_H2Saq        | 0.0038303    |
| diag | table8_row1 | m_HS           | 1.13617      |
| diag | table8_row1 | m_NH3aq        | 3.97368      |
| diag | table8_row1 | m_NH4          | 1.13632      |
| diag | table8_row1 | pH             | 8.67302      |

...and 64 more reference values in the case’s *expected*.

**enrtl\_mixed\_nacl\_ethanol\_esteso**

tutorials/props/electrolyte/enrtl\_mixed\_nacl\_ethanol\_esteso

**esteso**  
 enrtlMixedSolvent

**What it does:** VALIDATION of the generalized segment-based eNRTL (Chen & Song 2004, AIChE J 50:1928, DOI 10.1002/aic.10151) against the PRIMARY measured  $\gamma_{\pm}$  of NaCl in ethanol-water at 25 C (Esteso et al., J. Solution Chem. 18 (1989) 277, emf cell, 0-90 wt% ethanol). ZERO refit: every  $\tau/X:Z$  is the published Table 2/4/7 value;  $A_{\phi}$  per composition is Esteso's measured Table II. Headline AAD ~4.3% over 47 points (paper quotes RMS 0.026 on  $\gamma_{\pm}$ ); ~1% in pure water, ~1.2% at 70 wt%. The +8% at 80/90 wt% is DECLARED physics the complete-dissociation model cannot carry: Esteso measures ion association  $K_A = 30\text{--}46$  there (his Table V).

**The story:** can eNRTL predict a salt's activity in the antisolvent mixture?

The SAME physics the antisolvent crystalliser leans on –  $\gamma_{\pm}$  of NaCl as ethanol pushes the water out – validated against the PRIMARY dataset:

MODEL Chen & Song, "Generalized electrolyte-NRTL model for mixed-solvent electrolyte systems", AIChE J 50(8):1928-1941 (2004), DOI 10.1002/aic.10151. Ethanol = 1.811 (-C<sub>2</sub>H<sub>4</sub>-) + 0.609 (-OH) segments (Table 4);  $\text{ion} \leftrightarrow \text{OH}$   $\tau$ s = the aqueous NaCl pair (Table 2);  $\text{ion} \leftrightarrow \text{C}_2\text{H}_4$   $\tau$ s from Table 7. ZERO refit here. DATA Esteso, Gonzalez-Diaz, Hernandez-Luis & Fernandez-Merida, "Activity coefficients for NaCl in ethanol-water mixtures at 25 C", J. Solution Chem. 18(3):277-288 (1989). Table I (emf, molal scale, per-solvent reference).  $A_{\phi}$  per composition = Esteso Table II (measured dielectric + density). 0 wt% gate: Robinson & Stokes, Electrolyte Solutions, 2nd ed. (1959), NaCl appendix table;  $A_{\phi}(\text{water}, 25\text{ C}) = 0.391$ .

WHAT THE NUMBERS SAY (locked by the golden): total AAD ~4.3% over 47 points; pure water ~1.0%; 70 wt% ethanol ~1.2%. 80/90 wt% sit +8% systematically – Esteso's own chemical-model fit finds ion-pair association  $K_A = 30\text{--}46$  there (Table V), physics a complete- dissociation model cannot represent. Declared, not tuned away.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | esteso      | AAD_total_pct | 4.3      |
| diag | esteso      | AAD_wt0_pct   | 1        |
| diag | esteso      | AAD_wt20_pct  | 5.3      |
| diag | esteso      | AAD_wt40_pct  | 5.3      |
| diag | esteso      | AAD_wt60_pct  | 2.7      |
| diag | esteso      | AAD_wt70_pct  | 1.2      |
| diag | esteso      | AAD_wt80_pct  | 7.8      |
| diag | esteso      | AAD_wt90_pct  | 7.8      |
| diag | esteso      | n_points      | 47       |



**enrtl\_multisalt\_nacl\_kcl**

tutorials/props/electrolyte/enrtl\_multisalt\_nacl\_kcl

|                                          |                                                  |
|------------------------------------------|--------------------------------------------------|
| <b>harned_nacl_kcl</b><br>enrtlMultiSalt | <b>harned_nacl_kcl_refined</b><br>enrtlMultiSalt |
|------------------------------------------|--------------------------------------------------|

**What it does:** Multi-salt symmetric eNRTL (Song & Chen 2009) – NaCl+KCl Harned sweep at 4 mol/kg

**The story:** THE MULTI-SALT eNRTL ARRIVES: the multicomponent symmetric electrolyte NRTL of Song & Chen 2009 (Ind. Eng. Chem. Res. 48, 7788, DOI 10.1021/ie9004578) on the classic Harned experiment – NaCl + KCl sharing water at a CONSTANT 4 mol/kg total molality,  $\gamma_{\pm}$  of BOTH salts as KCl replaces NaCl. This is the paper's own demonstration system (its Figures 1-2).

What the run shows (all golden-locked):  $\gamma_{\pm}(\text{NaCl})$  slides from its pure-solution value to its TRACE value in 4 m KCl, and vice versa – both  $\ln \gamma$  curves are NEARLY LINEAR in the salt fraction: Harned's rule, EMERGING from the model rather than being assumed ( $R^2 > 0.99$  with zero electrolyte-electrolyte parameters). The PURE endpoints reproduce the validated single-salt eNRTL kernel exactly (the reduction pin  $\sim 1\text{e-}15$ ): the multicomponent formulation collapses correctly. The pair parameters are the EXACT Chen & Evans 1986 Table 1 values the paper uses (case-local constant/electrolyte/enrtl.dat, cited). The standards K-Cl.dat carries an INDEPENDENT refit to Hamer & Wu 1972 that lands within 0.2 % of the 1986 values – two roads, one answer. The 1986 paper itself (Table 2) validates  $\tau_{\text{EE}} = 0$  for NaCl-KCl:  $\sigma = 0.019\%$  on the Covington 1968 mixture osmotic data with the salt-salt energies set to zero – this case's default is the canonical choice. HONESTY, printed by the op: the published multi-electrolyte eNRTL activity-coefficient expressions are NOT exactly Gibbs-Duhem consistent (the composition-dependent mixing rules are differentiated as constants). The op's FD oracle quantifies that inconsistency at the mid-sweep state ( $\text{gibbsDuhem\_dev\_mid}$ ) next to the pure-salt state where the same oracle sits at  $\sim 1\text{e-}9$ . THE REFINED FORMULATION SHIPS TOO (second op below):  $\gamma$  as the EXACT derivative of the same  $G_{\text{ex}}$  – the chain rule kept through the composition-dependent mixing rules (the Bolas, Chen & Barton 2008 idea on the symmetric form). Its oracle closes ( $\sim 1\text{e-}9$ ) at every composition; both reduce EXACTLY to the same single-salt kernel at the endpoints. Mid-sweep they differ by  $|\text{dln } \gamma_{\pm}| \sim 0.041$  ( $\sim 4\%$  in  $\gamma$ ), and the trace limits differ visibly (NaCl trace 0.617 published vs 0.678 consistent). WHICH is closer to nature is a question for MIXTURE data (Covington 1968 / Robinson 1961) – this case states the facts and locks both; it does not pick a winner without measurements.

$\tau_{\text{EE}}$  (the same-anion electrolyte-electrolyte parameter  $\tau_{\text{NaCl,KCl}}$  of the paper's Figures 1-2) defaults to 0; set it to  $\pm 1$  to reproduce the paper's sensitivity fan.

THE EXPERIMENTAL VERDICT (Robinson 1961, J. Phys. Chem. 65, 662, DOI 10.1021/j100822a016 – isopiestic NaCl-KCl mixtures, Harned coefficients at  $M \sim 4.4$  mol/kg:  $\alpha_{\text{NaCl}} = +0.0243$ ,  $\alpha_{\text{KCl}} = -0.0091$ ; the op now emits  $\text{harned\_alpha\_1/2}$  on the same convention, golden-locked):  $\alpha_{\text{NaCl}}$ : published  $+0.0285$  ( $+17\%$ ) consistent  $+0.0180$  ( $-26\%$ )  $\alpha_{\text{KCl}}$ : published  $-0.0256$  ( $2.8\text{x}$ ) consistent  $-0.0187$  ( $2.1\text{x}$ ) With  $\tau_{\text{EE}} = 0$ , PUBLISHED wins on the dominant NaCl coefficient; BOTH formulations overstate the (small) effect on KCl.

CROSS-CHECK vs THE REFINED-eNRTL PAPER (Bolas, Chen & Barton 2008, AIChE J 54, 1608, DOI 10.1002/aic.11485 – their Figure 1 is THIS system, 4 m, same 1986 parameters, Robinson's points overlaid): their Fig 1a (original eNRTL) matches our 'published' to the pixel (NaCl solids at  $x=1$  for  $\tau = 0/0.25/0.50$ : ours  $-0.216/-0.186/-0.157$  vs read  $-0.215/-0.185/-0.15$ ; KCl dashed at  $x=0$ :  $-0.144/-0.183$  vs  $-0.145/-0.185$ ); their Fig 1b (refined) shows the SIGNATURE our 'consistent' shows: the  $\tau$  fan nearly COLLAPSES (exact derivatives are almost insensitive to  $\tau_{\text{EE}}$ : ours moves only  $-0.176 \rightarrow -0.184$  for  $\tau 0 \rightarrow 0.5$ , where published moves  $-0.216 \rightarrow -0.157$ ), and the KCl trace lands at  $\sim -0.16$  (ours  $-0.163$ ). Their refinement differentiates the 1986 UNSYMMETRIC  $G_{\text{ex}}$ ; ours differentiates the 2009 SYMMETRIC one – same idea, sibling surfaces, same qualitative physics, near-identical aqueous numbers. A single fitted  $\tau_{\text{EE}}$  cannot close both (best compromise  $\tau \sim 0.29$  trades the NaCl error up to  $-23\%$  to halve the KCl one) – the residual is structural, not a missing parameter. Note the consistency of the record: Chen & Evans 1986 validated  $\tau_{\text{EE}} = 0$  against mixture OSMOTIC data ( $\sigma 0.019\%$ ) – the solvent sees the mixture mean, where the individual- $\gamma$  asymmetries largely cancel; Robinson's per-salt  $\gamma$ s are where they show.

FIGURE-1 REPRODUCTION – TO THE PIXEL (checked 2026-07-02, figure read at 200 dpi): the paper's y-axis says "ln gamma" but the numbers are LOG10 ( $\log_{10}$  of the pure-NaCl-4m  $\gamma$  is  $-0.106 =$  the figure's  $-0.105$ ;  $\ln$  would be  $-0.24$ , off the plotted range). With the 1986 parameters this kernel reproduces

the full tau fan to  $\leq 0.001$  in  $\log_{10}$ :  $x_{\text{KCl}}=0$  (all curves): -0.106 (figure  $\sim -0.105$ )  $x_{\text{KCl}}=1$ :  $\tau=0 \rightarrow -0.216$  (figure  $\sim -0.215$ )  $\tau=+1 \rightarrow$  flat, dip to -0.101, back to -0.106 (the figure's circle-marked curve shows the same shallow dip)  $\tau=-1 \rightarrow -0.364$  (figure  $\sim -0.365$ )

**What to expect (golden-locked):**

| kind                                                            | unit/stream     | key                 | expected  |
|-----------------------------------------------------------------|-----------------|---------------------|-----------|
| diag                                                            | harned_nacl_kcl | consistent          | 0.00e+00  |
| diag                                                            | harned_nacl_kcl | formulation_gap_mid | 4.13e-02  |
| diag                                                            | harned_nacl_kcl | gamma1_pure         | 8.01e-01  |
| diag                                                            | harned_nacl_kcl | gamma1_trace        | 6.17e-01  |
| diag                                                            | harned_nacl_kcl | gamma2_pure         | 5.76e-01  |
| diag                                                            | harned_nacl_kcl | gamma2_trace        | 7.29e-01  |
| diag                                                            | harned_nacl_kcl | gibbsDuhem_dev_mid  | 4.72e-01  |
| diag                                                            | harned_nacl_kcl | harned_R2_min       | 9.99e-01  |
| diag                                                            | harned_nacl_kcl | harned_alpha_1      | 2.85e-02  |
| diag                                                            | harned_nacl_kcl | harned_alpha_2      | -2.56e-02 |
| diag                                                            | harned_nacl_kcl | harned_slope_1      | -2.62e-01 |
| diag                                                            | harned_nacl_kcl | harned_slope_2      | -2.36e-01 |
| diag                                                            | harned_nacl_kcl | oracle_pure_dev     | 4.03e-09  |
| diag                                                            | harned_nacl_kcl | reduction_pin_dev   | 1.11e-15  |
| ...and 18 more reference values in the case's <i>expected</i> . |                 |                     |           |

## enthalpy\_naoh\_water

tutorials/props/electrolyte/enthalpy\_naoh\_water

**gamma\_naoh**  
pitzerActivity

**What it does:** NaOH-water heat of dilution: Pitzer with calorimetric T-slots vs Parker (1965)

**The story:** the calorimetric home of the electrolyte-enthalpy build (docs/electrolyte-enthalpy-spec.md).

constant/experiments/phiL\_25C\_parker1965.dat carries the MEASURED  $\Phi_L$  (Parker NSRDS-NBS 2, 1965) used by the curation refit (bin/curate/fit\_lphi\_pitzer.py) that fitted the Na-OH  $\delta\beta/\delta T$  T-slots in the standard catalogue (the FIT's own rel-AAD 4.97%, gate <5%; the in-engine  $L_\phi$  scan pinned by this case lands at rel-AAD 4.77% – two measurements, not one). The op below sweeps the ACTIVITY surface whose T-derivative is  $L_\phi$ ; the in-engine  $L_\phi$  scan + AAD overlay land with slices 4-5 (the 3-curve chart).

**What to expect (golden-locked):**

| kind | unit/stream | key                     | expected   |
|------|-------------|-------------------------|------------|
| diag | gamma_naoh  | Lphi_AAD_abs_Jmol       | 22.9776    |
| diag | gamma_naoh  | Lphi_AAD_rel_pct        | 4.7712     |
| diag | gamma_naoh  | Lphi_extrap_maxdev_Jmol | 15774.0028 |
| diag | gamma_naoh  | Lphi_nMeas              | 29.0000    |
| diag | gamma_naoh  | aw_1m                   | 0.9665     |
| diag | gamma_naoh  | gamma_pm_1m             | 0.6679     |
| diag | gamma_naoh  | gamma_pm_3m             | 0.7838     |
| diag | gamma_naoh  | phi_1m                  | 0.9471     |

**farelo\_licl\_range**

tutorials/props/electrolyte/farelo\_licl\_range

licl\_1.0  
speciatelicl\_6.0  
speciatelicl\_13.0  
speciatelicl\_19.7  
speciatefarelo\_LiCl\_sat  
speciate

**What it does:** LiCl Pitzer REFITTED to Hamer-Wu + Farelo saturation:  $\gamma_{\pm}$  to ~11% over 0.5-19 mol/kg (was 774 at 19.7m), and the LiCl.H<sub>2</sub>O hydrate Ksp reproduces Farelo's 19.7 mol/kg via the HMW osmotic water activity ( $a_w=0.11$ ).

*Prof. F?tima Farelo measured LiCl solubility at 19.7  $\rightarrow$  23.5 mol/kg (solid: LiCl.H<sub>2</sub>O) — \*J. Chem. Eng. Data\* 50 (2005) 1470, DOI 10.1021/je050111j. That is one of the most concentrated aqueous electrolytes there is.*

*The catalogue's earlier Li-Cl Pitzer parameters (Appelo, Parkhurst & Post, \*Geochim. Cosmochim. Acta\* 125 (2014) 49), fitted below ~6 mol/kg, blow up at high concentration —  $\gamma_{\pm} = 774$  at 19.7 mol/kg vs ~45 measured. The two-parameter B-term is linear in  $m$ , so  $\gamma$  grows exponentially.*

*Rather than declare defeat, the parameters were refitted to Hamer & Wu (1972)  $\gamma_{\pm}$  and Farelo's saturation. A negative  $C\varphi$  (-0.0026) tames the high- $m$  blow-up, and the three-parameter form then reaches saturation:*

*carries a water-activity leg. At 19.7 mol/kg the model gives  $a_w = 0.11$  (not 1!) — which only the rigorous Pitzer-HMW osmotic coefficient delivers, not the  $\varphi = 1$  dilute shortcut. The lithiumChlorideH<sub>2</sub>O mineral Ksp ( $\log K_{25} = 4.984$ ) is calibrated to Farelo's saturation \*through\* that  $a_w$ . The two pieces meet: the refitted  $\gamma$  and the osmotic  $a_w$  together reproduce the measured solubility.*

*- electrolyte/pairs.dat: the Li-Cl Pitzer pair, refitted to Hamer-Wu + Farelo, valid to ~19 mol/kg (supersedes the Appelo low- $m$  fit for concentrated LiCl). - electrolyte/minerals.dat: the LiCl.H<sub>2</sub>O hydrate Ksp, calibrated to Farelo. - constant/species/Li.dat: Li? aqueous formation datum ( $h_f^{\text{Aq}} = -278.49$  kJ/mol, CODATA).*

*- The full measured dataset (this and the other five ternaries) is tabulated, cited, in constant/experimental/farelo\_solubility.csv. - The equilibrate round-trip (precipitate LiCl.H<sub>2</sub>O from a supersaturated feed) diverges at  $I \approx 20$  — a known active-set /  $\gamma$  fixed-point numerical limit at extreme ionic strength, \*independent of the activity model\*. The  $SI = 0$  calibration at the saturation composition is exact; reaching it by precipitation is the convergence problem, not the thermodynamics. - The AlCl<sub>3</sub> systems from the same paper remain deferred: Al<sup>3+</sup>? hydrolyses (the paper notes  $\text{pH} < 2.75$ ) and needs an Al speciation model Choupo does not yet carry.*

**What to expect (golden-locked):**

| kind | unit/stream     | key                   | expected    |
|------|-----------------|-----------------------|-------------|
| diag | farelo_LiCl_sat | I                     | 19.7        |
| diag | farelo_LiCl_sat | SI_lithiumChlorideH2O | 1.18638e-05 |
| diag | farelo_LiCl_sat | aw                    | 0.113466    |
| diag | farelo_LiCl_sat | chargeImposed         | 1           |
| diag | farelo_LiCl_sat | chargeResidual        | 5.04184e-14 |
| diag | farelo_LiCl_sat | gamma_Cl              | 46.7905     |
| diag | farelo_LiCl_sat | gamma_Li              | 46.7905     |
| diag | farelo_LiCl_sat | m_Cl                  | 19.7        |
| diag | farelo_LiCl_sat | m_Li                  | 19.7        |
| diag | farelo_LiCl_sat | pH                    | 7.99525     |

**farelo\_nacl\_nh4cl**

tutorials/props/electrolyte/farelo\_nacl\_nh4cl

|                                 |                                 |                                 |
|---------------------------------|---------------------------------|---------------------------------|
| farelo_NaCl_pure<br>speciate    | farelo_NaCl_nh4_093<br>speciate | farelo_NaCl_nh4_187<br>speciate |
| farelo_NaCl_nh4_280<br>speciate | farelo_NH4Cl_pure<br>speciate   |                                 |

**What it does:** Pitzer-HMW vs Farelo (JCED 2005): NaCl+NH<sub>4</sub>Cl common-ion salting-out – model SI ~ 0 at each measured saturation point (NaCl 6.15→5.5→5.0; NH<sub>4</sub>Cl 7.35 mol/kg, 298 K).

*A glass-box validation of Choupo's multi-ion Pitzer electrolyte model against the solubility measurements of Prof. Fátima Farelo (Chemical Engineering Dept., IST — Técnico Lisboa).*

> M. Farelo, C. Fernandes & A. Avelino, \*Solubilities for Six Ternary Systems: > NaCl + NH<sub>4</sub>Cl + H<sub>2</sub>O, KCl + NH<sub>4</sub>Cl + H<sub>2</sub>O, NaCl + LiCl + H<sub>2</sub>O, KCl + LiCl + H<sub>2</sub>O, > NaCl + AlCl<sub>3</sub> + H<sub>2</sub>O, and KCl + AlCl<sub>3</sub> + H<sub>2</sub>O at T = (298 to 333) K\*, > J. Chem. Eng. Data 50 (2005) 1470–1477. DOI [10.1021/je050111j](https://doi.org/10.1021/je050111j).

Saturation molalities measured to ±0.0007 mol/kg (visual polythermal / isothermal methods). The pure-salt anchors at 298.15 K: NaCl 6.15, NH<sub>4</sub>Cl 7.35 mol/kg; the salting-out series (Table 1): NaCl solubility falls 6.15 → 5.5 → 5.0 → 4.95 as NH<sub>4</sub>Cl rises 0 → 0.93 → 1.87 → 2.80 mol/kg.

The model is calibrated to only the two pure-salt solubilities — halite's K<sub>sp</sub> from PHREEQC, sal-ammoniac's K<sub>sp</sub> pinned to Farelo's NH<sub>4</sub>Cl point. Everything else is a prediction. The NH<sub>4</sub>-Cl Pitzer interaction parameters come from the primary single-electrolyte fit (Pitzer & Mayorga 1973); the catalogue already carried Na-Cl and K-Cl.

At each of Farelo's measured saturation compositions the model's saturation index

should be ≈ 0 if the model reproduces the measurement. It is, to ±0.03 log units:

**What to expect (golden-locked):**

| kind | unit/stream         | key            | expected  |
|------|---------------------|----------------|-----------|
| diag | farelo_NaCl_pure    | I              | 6.15      |
| diag | farelo_NaCl_pure    | SI_halite      | 0.0168246 |
| diag | farelo_NaCl_pure    | aw             | 0.751868  |
| diag | farelo_NaCl_pure    | gamma_Cl       | 1.0105    |
| diag | farelo_NaCl_pure    | gamma_Na       | 1.0105    |
| diag | farelo_NaCl_pure    | m_Cl           | 6.15      |
| diag | farelo_NaCl_pure    | m_Na           | 6.15      |
| diag | farelo_NaCl_pure    | pH             | 7.11704   |
| diag | farelo_NaCl_nh4_093 | I              | 6.42998   |
| diag | farelo_NaCl_nh4_093 | SI_halite      | -0.022239 |
| diag | farelo_NaCl_nh4_093 | SI_salammoniac | -0.70594  |
| diag | farelo_NaCl_nh4_093 | aw             | 0.746427  |
| diag | farelo_NaCl_nh4_093 | gamma_NH4      | 0.591469  |
| diag | farelo_NaCl_nh4_093 | gamma_Na       | 1.04074   |

... and 39 more reference values in the case's *expected*.

**fpd01\_nacl\_freezing**

tutorials/props/electrolyte/fpd01\_nacl\_freezing

**fpd\_nacl**  
freezingPoint

**What it does:** NaCl freezing-point depression – ice as a SolidPhase, the crystal’s first case

**The story:** THE ICE WITNESS. The freezing-point curve of NaCl(aq), solved as chemical-potential equality between an ACTUAL SolidPhase (the crystallising mode, its first case-reachable consumer) and the Pitzer surface’s water activity:

$$f_{\text{solid}}(T) = a_w(m, T) * P_{\text{sat}}(T)$$

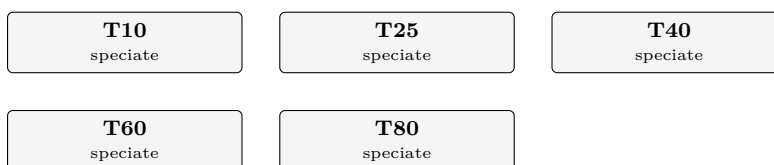
Until this case the crystal was exercised only by check\_ice\_freezing’s probe; here its refusals and announced approximations (the neglected liquid/solid heat-capacity term, the sub-freezing Psat extrapolation) all fire on a real run. The dilute-limit KPI closes the K\_f loop: the derived cryoscopic constant against the slope this same phase produces. ANCHOR HONESTY: the validation dataset carries ONE measured point with an INTERIM citation (see its header) – a thin anchor, said plainly, and the enrichment path (a primary FPD table read) is on the curation ledger.

**What to expect (golden-locked):**

| kind | unit/stream | key                   | expected  |
|------|-------------|-----------------------|-----------|
| diag | fpd_nacl    | Tf_lastM_K            | 266.325   |
| diag | fpd_nacl    | depression_lastM_K    | 6.83528   |
| diag | fpd_nacl    | slope_dilute_K_kg_mol | 3.46546   |
| diag | fpd_nacl    | slope_over_nuKf       | 0.931447  |
| diag | fpd_nacl    | theta_AAD_pct         | 0.0460433 |
| diag | fpd_nacl    | theta_nMeas           | 1         |

**ksp\_temperature**

tutorials/props/electrolyte/ksp\_temperature



**What it does:** K(T) MATTERS: a temperature scan of mineral saturation for ONE fixed water (a hard, carbonate-bearing groundwater at pH 7.6), speciated at 10, 25, 40, 60 and 80 C. The point is the SIGN of dK/dT. CALCITE dissolution is EXOTHERMIC, so its equilibrium constant FALLS as temperature rises and its saturation index SI\_calcite RISES – RETROGRADE SOLUBILITY: calcite is LESS soluble hot. This is why hot heat-exchanger surfaces and warm RO concentrate scale with calcium carbonate even when the cold feed looks safe. Contrast HALITE (NaCl): no analytic K(T) in the database, so it falls back to constant-dH van't Hoff with a small POSITIVE dH (endothermic) – normal solubility, SI\_halite drifts gently DOWN with T. At 25 C nothing in the K(T) machinery moves (every form returns logK25 exactly, byte-stable); off 25 C the run announces which K(T) form each entry uses (analytic / van't Hoff / flat) and warns if any analytic is used beyond its fitted validity range. K(T): analytic = PHREEQC -analytical\_expression anchored on logK25 ( $\log K(T) = \log K25 + [\text{ana}(T) - \text{ana}(298.15)]$ ); USGS PHREEQC, public domain.

**The story:** why hot surfaces scale with calcite (K(T) MATTERS).

ONE fixed water, speciated at five temperatures. A hard, carbonate-bearing groundwater (gypsum- and calcite-relevant) at a GIVEN pH 7.6 – holding pH and the totals fixed isolates the lesson: the only thing that changes from point to point is the equilibrium constant K(T).

ion mol/kg w ion mol/kg w Ca 0.010 Cl 0.050 Na 0.030 SO4 0.012 HCO3 0.005 pH 7.6 (given)

THE LESSON – the SIGN of dK/dT:

calcite dissolution is EXOTHERMIC ( $dH = -28.1$  kJ/mol), so log K FALLS as T rises. SI\_calcite =  $\log(IAP) - \log K(T)$  therefore RISES with temperature – RETROGRADE SOLUBILITY. Calcite is LESS soluble hot, which is exactly why heat-exchanger tubes and warm RO concentrate scale with CaCO<sub>3</sub> although the cold feed looks safe. K(T) here is the PHREEQC ANALYTIC expression (-analytical\_expression A1..A5), anchored on logK25 so  $\log K(25\text{ C}) = \log K25$  EXACTLY.

gypsum analytic too, but only mildly retrograde (its solubility peaks near 40 C) – a gentler curve for contrast.

halite NaCl carries NO analytic K(T) in phreeqc.dat (the db's -analytic line is commented out), so it falls back to constant-dH van't Hoff.  $dH = +1.37$  kJ/mol > 0 (endothermic) → NORMAL solubility: log K(T) RISES and SI\_halite drifts DOWN. The honest contrast with calcite: the scaling verdict flips with temperature for the retrograde salt, not for the normal one.

At 25 C the K(T) machinery returns logK25 for EVERY entry (byte-stable); off 25 C the run announces the form in use per entry (analytic / van't Hoff / flat) and warns loudly if an analytic is used beyond its fitted validity range. (Calcite's analytic is valid 0 - 300 C, so the whole 10 - 80 C scan is in range.)

Speciation/mineral data: case-local constant/electrolyte/ – a self-contained SNAPSHOT of the standards catalogue (USGS PHREEQC, public domain; see the .dat headers). The case files ECLIPSE data/standards/electrolyte/.

**What to expect (golden-locked):**

| kind | unit/stream | key        | expected  |
|------|-------------|------------|-----------|
| diag | T10         | I          | 0.0795197 |
| diag | T10         | SI_calcite | 0.56588   |
| diag | T10         | SI_gypsum  | -0.335309 |
| diag | T10         | SI_halite  | -4.56674  |
| diag | T10         | aw         | 0.998115  |
| diag | T10         | pH         | 7.6       |
| diag | T25         | I          | 0.0772864 |
| diag | T25         | SI_calcite | 0.789002  |
| diag | T25         | SI_gypsum  | -0.365046 |
| diag | T25         | SI_halite  | -4.60657  |
| diag | T25         | aw         | 0.998126  |
| diag | T25         | pH         | 7.6       |
| diag | T40         | I          | 0.074682  |
| diag | T40         | SI_calcite | 0.981885  |

*...and 91 more reference values in the case's **expected**.*



**overlay01\_nacl\_ksp**  
tutorials/props/electrolyte/overlay01\_nacl\_ksp

naclWater  
speciate

**What it does:** PARTIAL deep-merge overlay: a case recalibrates ONLY the halite equilibrium logK25 via a case-local overlayOf NaCl; every other datum (dH, dissolutionReaction, calorimetric, crystal) still comes from the standard components/NaCl.dat record. The [overlay] line names exactly which leaf came from the case tier.

*A case recalibrates ONLY the halite equilibrium logK25 (1.60 here) to its own value, via a case-local constant/components/NaCl.dat declaring overlayOf NaCl;.*

*The reader deep-merges this delta over the standard components/NaCl.dat: only solidPhases.halite.equilibrium.logK25 is replaced; dH, the dissolutionReaction, the calorimetric anchor and the crystal block all still resolve from the standard record. The run prints the provenance:*

*This is the roadmap Phase A acceptance test: a case ships ONLY its deltas, never a second competing complete record, and per-value provenance (tier + file) is preserved.*

**What to expect (golden-locked):**

| kind | unit/stream | key            | expected     |
|------|-------------|----------------|--------------|
| diag | naclWater   | I              | 2            |
| diag | naclWater   | SI_halite      | -0.983442    |
| diag | naclWater   | aw             | 0.930475     |
| diag | naclWater   | gamma_Cl       | 1.01683      |
| diag | naclWater   | gamma_Na       | 1.01683      |
| diag | naclWater   | m_Cl           | 2            |
| diag | naclWater   | m_Na           | 2            |
| diag | naclWater   | pH             | 7.13303      |
| diag | naclWater   | chargeImposed  | 1            |
| diag | naclWater   | chargeResidual | -6.39009e-17 |

**pb82\_calcite\_open\_co2**

tutorials/props/electrolyte/pb82\_calcite\_open\_co2

**calciteEq\_25C**  
 speciate

**What it does:** Calcite in CO<sub>2</sub>-saturated water (Plummer & Busenberg 1982): open pCO<sub>2</sub> 0.956 atm + equilibrate calcite at 25 C – the engine’s pH at calcite equilibrium beside the paper’s junction-corrected 6.004 +/- 0.005 and its model value 6.011

**The story:** THE PLUMMER & BUSENBERG (1982) EXPERIMENT, as the engine models it: calcite at equilibrium with CO<sub>2</sub>-saturated water – an OPEN system at pCO<sub>2</sub> = 0.956 atm, 25 C (their electrode runs’ mean CO<sub>2</sub> pressure), the gas pin, the full carbonate speciation, the CaHCO<sub>3</sub><sup>+</sup>/CaCO<sub>3</sub>(aq) ion pairs and the mineral ceiling all engaged at once.

PRIMARY: L. N. Plummer & E. Busenberg, *Geochim. Cosmochim. Acta* 46 (1982) 1011-1040, DOI 10.1016/0016-7037(82)90056-4. Owner-provided PDF.

THE ANCHOR, and what it independently tests: their MEASURED pH at calcite equilibrium under exactly these conditions, corrected for liquid-junction potential, is 6.004 +/- 0.005 (4 M KCl filling, p. 1022; uncorrected Orion runs read near 5.984); their own aqueous model puts the thermodynamic pH at 6.011. The engine’s number below comes out of Henry + K<sub>1</sub>/K<sub>2</sub> + the ion pairs + Davies + electroneutrality + the SI = 0 ceiling END TO END – none of which was fitted to this pH.

LINEAGE, stated honestly: the calcite log K(T), K<sub>H</sub>, K<sub>1</sub>, K<sub>2</sub> and ion-pair constants in our curated records descend from the USGS PHREEQC database, whose carbonate block IS this paper’s Table 3 and Eqs – so agreement in the CONSTANTS is lineage consistency, not validation. What the measured pH tests independently is the ASSEMBLY: that the engine composes those constants (gas pin, chained carbonate master, ion pairs, activity model, charge balance, mineral ceiling) into the same physical answer the electrode saw.

WHAT IT ACTUALLY LANDS AT, and where that is written. The engine gives pH 6.02946 – 0.025 ABOVE the measurement, 0.018 above their own model’s 6.011 – and the case attributes that offset to an activity-model difference (Davies here against their Truesdell-Jones, at I = 0.027). SINCE 2026-08-12 THAT COMPARISON IS DATA, NOT PROSE: ‘expected’ carries an ‘anchor’ row on the measured 6.004 with a band set to the explained offset, beside the self-recorded row on 6.02946 – so a drift AWAY from the electrode fails as an anchor disagreement, and an unexplained drift TOWARD it fails as a moved golden. Both ends, on purpose. This header used to end with "a wrong composition would not land within 0.01 pH", which reads as though the engine does; it does not, and the number is above.

APPROACH FROM ABOVE: the engine’s equilibrate is a precipitation CEILING (SI → 0 from supersaturation; it does not dissolve a solid it was never given). The seed is therefore a SUPERSATURATED Ca(HCO<sub>3</sub>)<sub>2</sub> solution under the pin; the ceiling relaxes it down to the same calcite saturation P&B’s dissolution runs approached from below – one fixed point at (T, pCO<sub>2</sub>), whichever side you arrive from. The HCO<sub>3</sub> seed is an initial guess only: the CO<sub>2</sub> pin REPLACES the carbonate mole balance, so dissolved carbonate is a solved outcome (announced by the run).

**What to expect (golden-locked):**

| kind | unit/stream   | key            | expected    |
|------|---------------|----------------|-------------|
| diag | calciteEq_25C | I              | 0.0265778   |
| diag | calciteEq_25C | SI_aragonite   | -0.114      |
| diag | calciteEq_25C | SI_calcite     | 3.22147e-11 |
| diag | calciteEq_25C | SI_portlandite | -13.0919    |
| diag | calciteEq_25C | aw             | 0.998916    |
| diag | calciteEq_25C | n_calcite      | 0.00543277  |
| diag | calciteEq_25C | pH             | 6.02946     |
| diag | calciteEq_25C | pH             | 6.004       |
| diag | calciteEq_25C | chargeImposed  | 1           |
| diag | calciteEq_25C | chargeResidual | 2.22179e-18 |

**pitzer01\_nacl**

tutorials/props/electrolyte/pitzer01\_nacl

**gamma\_nacl**  
 pitzerActivity

**What it does:** Pitzer mean ionic activity coefficient  $\gamma_{\text{pm}}(m)$  for NaCl in water at 25 C — validated vs Robinson & Stokes (AAD 0.17%). The glass-box electrolyte foundation: see the famous  $\gamma_{\text{pm}}$  dip ( $\sim 0.66$  near 1 m) and the rise above 1 at high concentration.

**The story:** the first electrolyte case.

$\gamma_{\text{pm}}$ , the osmotic coefficient  $\phi$  and the water activity  $a_{\text{w}}$  for NaCl, by the Pitzer ion-interaction model. The parameters are CURATED, case-local data, read from constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62 — a 3-parameter set,  $\beta_0/\beta_1/C\phi$ ). Open pitzer\_nacl.csv:  $\gamma_{\text{pm}}$  dips to  $\sim 0.657$  near 1 mol/kg, then rises above 1 by  $\sim 6$  mol/kg — the textbook NaCl signature, from a closed form.

NOTE on the two  $L_{\text{phi}}$  columns: pitzer\_nacl.csv reports  $L_{\text{phi\_fit}}$  and  $L_{\text{phi\_anchored}}$ , and their SPLIT is the lesson. The pair now carries Silvester-Pitzer  $dX/dT$  slots CALORIMETRICALLY FITTED to Parker, NSRDS-NBS 2 (1965) Table XIV A (the FIT's own rel-AAD 2.5 %, m 0.1-6; bin/curate/fit\_lphi\_pitzer.py): the fit column tracks the measured heat-of-dilution curve — the sign change near 1 mol/kg, the deep negative branch to -480 cal/mol — while the anchored column (slots zeroed, Gibbs-Helmholtz-consistent but uncalibrated) misses it entirely. The validation block below pins the IN-ENGINE AAD vs the same curated Parker series, golden-locked (rel-AAD 10.24 % on this op's scan grid — the fit tool's 2.5 % above is its own residual, a separate measurement). The  $\gamma_{\text{pm}}$  /  $\phi$  /  $a_{\text{w}}$  values come from the untouched 25 C surface and are identical in both.

**What to expect (golden-locked):**

| kind | unit/stream | key               | expected |
|------|-------------|-------------------|----------|
| diag | gamma_nacl  | aw_1m             | 0.9668   |
| diag | gamma_nacl  | gamma_pm_1m       | 0.6572   |
| diag | gamma_nacl  | gamma_pm_3m       | 0.7140   |
| diag | gamma_nacl  | phi_1m            | 0.9363   |
| diag | gamma_nacl  | Lphi_AAD_abs_Jmol | 97.6414  |
| diag | gamma_nacl  | Lphi_AAD_rel_pct  | 10.2355  |
| diag | gamma_nacl  | Lphi_nMeas        | 29.0000  |

**pitzer02\_cacl2**

tutorials/props/electrolyte/pitzer02\_cacl2

**gamma\_cacl2**  
pitzerActivity

**What it does:** Pitzer  $\gamma_{\text{pm}}(m)$  for  $\text{CaCl}_2$  (a 2:1 salt) in water at 25 C — the SAME general formula as  $\text{NaCl}$ , only the charges/stoichiometry differ. Validated vs Robinson & Stokes (AAD 0.49%, 0.1-2 m).

**The story:** the same Pitzer engine on a 2:1 electrolyte ( $\text{Ca}^{2+}$ , 2  $\text{Cl}^-$ ). Parameters curated case-local in constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62).  $\gamma_{\text{pm}}$  dips below 0.45 then climbs steeply — the divalent-cation signature. Valid to ~2 mol/kg;  $\text{CaCl}_2$  carries a non-zero  $\beta_2$  (-1.13, with  $\alpha_2 = 12$ ) on top of  $\beta_0/\beta_1/C\phi$  — the extra association term a 2:1 salt needs ( $\text{NaCl}$ , by contrast, has  $\beta_2 = 0$ ).

**What to expect (golden-locked):**

| kind | unit/stream | key         | expected |
|------|-------------|-------------|----------|
| diag | gamma_cacl2 | aw_1m       | 0.9449   |
| diag | gamma_cacl2 | gamma_pm_1m | 0.4985   |
| diag | gamma_cacl2 | gamma_pm_3m | 1.4771   |
| diag | gamma_cacl2 | phi_1m      | 1.0483   |

**pitzer02\_nacl\_package**

tutorials/props/electrolyte/pitzer02\_nacl\_package

**gamma\_nacl**  
electrolyteActivity

**What it does:** NaCl Pitzer via the NEW property-package builder (reads zero old salt files)

**The story:** No ‘cation’/‘anion’ keys: the salt is resolved by the SELECTED propertyPackage (constant/thermoPhysPropDict → aqueousNaCl\_pitzer). The op consumes the assembled package via thermo.electrolyte() and reports gamma\_pm / phi / a\_w.

**What to expect (golden-locked):**

| kind | unit/stream | key         | expected |
|------|-------------|-------------|----------|
| diag | gamma_nacl  | aw_1m       | 0.966826 |
| diag | gamma_nacl  | gamma_pm_1m | 0.657172 |
| diag | gamma_nacl  | gamma_pm_3m | 0.714026 |
| diag | gamma_nacl  | phi_1m      | 0.93634  |

**pitzer\_calcite\_brine**

tutorials/props/electrolyte/pitzer\_calcite\_brine

seawater\_pitzer  
speciateseawater\_davies  
speciate

**What it does:** Pitzer-HMW carbonate system on surface seawater – pins SI\_calcite / SI\_gypsum against the published oceanographic saturation state, captures the neutral CO<sub>2</sub> salting-out ( $\gamma_{\text{CO2aq}} > 1$  via  $\lambda$ ), and the Davies-vs-Pitzer SI divergence.

**The story:** the slice-S4 GATE for the Pitzer CARBONATE system.

S4 extends the multi-ion Pitzer-HMW model (binaries + theta/psi ternaries, S3) to the CARBONATE subsystem so calcite / gypsum scaling works in brine: the carbonate ANIONS (CO<sub>3</sub><sup>2-</sup>, HCO<sub>3</sub><sup>-</sup>) join the cation/anion ternary sums (their theta/psi imported from pitzer.dat into mixing.dat); the neutral CO<sub>2</sub>aq gets its salting-out gamma via the lambda neutral-ion term ( $\lambda_{\text{CO2,ion}}$ ), so  $\gamma_{\text{CO2aq}} > 1$  in brine (Davies has none).

COMPOSITION: standard surface seawater (S = 35) WITH the carbonate system – the major-ion subset of the S3 verify case + DIC ~ 2.3 mmol/kg as HCO<sub>3</sub> (the measured-alkalinity carrier), at the canonical surface-ocean pH 8.2. Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO<sub>4</sub>-2 0.02926 HCO<sub>3</sub>- 0.00230 (I ~ 0.68 mol/kg)

THE PUBLISHED PIN (cited; E\_theta before/after recorded below):

SI\_calcite (surface seawater, 25 C, pH 8.2) ~ +0.6 to +0.8. Surface ocean water is calcite-SUPERSATURATED: the calcite saturation ratio  $\Omega_{\text{calcite}} = \text{IAP}/K_{\text{sp}} \sim 4\text{-}6$ , i.e.  $\text{SI} = \log_{10}(\Omega) \sim +0.6$  to +0.8. Sources: Mucci, A., Am. J. Sci. 283 (1983) 780 (the seawater calcite/ aragonite solubility); Millero, F.J., Chem. Rev. 107 (2007) 308 (the seawater carbonate system); a well-documented oceanographic number (the ocean-acidification baseline). Choupo (Pitzer-HMW) computes SI\_calcite = +0.655 – INSIDE the published band. SI\_gypsum (seawater) ~ -0.9 to -0.6 (strongly UNDERsaturated – seawater is nowhere near gypsum saturation); Choupo computes -0.964 – a touch below the -0.9 floor (this carbonate-bearing brine sits at slightly higher I than the carbonate-free seawater\_verify case, which gives -0.687 in-band). Still firmly undersaturated, the physically-correct verdict.

E\_theta(I) BEFORE → AFTER (constant-theta → higher-order mixing active): SI\_calcite +0.724 → +0.655 (IN band +0.6..+0.8) – E\_theta lowers the 2+/2- ion gammas, nudging the carbonate IAP SI\_gypsum -0.875 → -0.964  $\gamma_{\text{CO2aq}}$  1.105 → 1.106 (salting-out kept)  $\gamma_{\text{Ca}}$  0.227 → 0.194 (E\_theta on Ca pairs) Davies (untouched – E\_theta is Pitzer-only): SI\_calcite +0.868, byte-ident.

The golden master locks Choupo's exact output; the band above validates it is PHYSICALLY right (a wrong pin is worse than no pin – the band is cited).

DAVIES vs PITZER (the differentiator, also pinned): the SAME composition under Davies gives SI\_calcite +0.868 and  $\gamma_{\text{CO2aq}} = 1.000$  (no neutral salting-out) – Davies is out of its trustworthy band (~0.5) at this I, overshoots the calcite SI by ~0.21 vs Pitzer-HMW-with-E\_theta and misses the CO<sub>2</sub> salting-out entirely. Pitzer is the rigorous tool for brine. Both entries run here so the divergence is VISIBLE.

DATA SCOPE: self-DESCRIBING (inline manifest); the ions, pairs, mixing WITH carbonate theta/p-si/lambda, speciation and minerals resolve from the curated data/standards/ tree at runtime – seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind | unit/stream     | key          | expected    |
|------|-----------------|--------------|-------------|
| diag | seawater_pitzer | I            | 0.682714    |
| diag | seawater_pitzer | SI_anhydrite | -1.24804    |
| diag | seawater_pitzer | SI_aragonite | 0.540696    |
| diag | seawater_pitzer | SI_calcite   | 0.654696    |
| diag | seawater_pitzer | SI_gypsum    | -0.964342   |
| diag | seawater_pitzer | SI_halite    | -2.48855    |
| diag | seawater_pitzer | SI_sylvite   | -3.51959    |
| diag | seawater_pitzer | aw           | 0.981412    |
| diag | seawater_pitzer | gamma_CO2aq  | 1.10585     |
| diag | seawater_pitzer | gamma_CO3    | 0.105443    |
| diag | seawater_pitzer | gamma_Ca     | 0.194358    |
| diag | seawater_pitzer | gamma_HCO3   | 0.659478    |
| diag | seawater_pitzer | m_CO2aq      | 1.4318e-05  |
| diag | seawater_pitzer | m_CO3        | 7.71637e-05 |

*...and 94 more reference values in the case's **expected**.*

**pitzer\_gamma\_hamer\_wu**

tutorials/props/electrolyte/pitzer\_gamma\_hamer\_wu

gamma\_nacl  
pitzerActivitygamma\_kcl  
pitzerActivitygamma\_kbr  
pitzerActivitygamma\_licl\_consistency  
pitzerActivitygamma\_hcl  
pitzerActivitygamma\_naoh  
pitzerActivity**What it does:** Pitzer gamma\_pm/phi vs Hamer & Wu 1972 – 6 salts, evaluated NBS data

**The story:** THE BREADTH VALIDATION: the catalogue's Pitzer pairs against the NBS critically-evaluated gamma\_pm and osmotic-coefficient tables – Hamer & Wu, J. Phys. Chem. Ref. Data 1 (1972) 1047 – for SIX 1-1 salts at 25 C. Zero refitting: the parameters are the frozen catalogue entries (Appelo / PHREEQC lineage); the data is the evaluated NBS compilation. Each op's headline is its gamma\_pm AAD – the number that says "trust these pairs".

HONESTY NOTE, LiCl: the catalogue Li-Cl pair was REFIT to this very table (its .dat says so), so its AAD here is a FIT-CONSISTENCY check, not independent validation – and at ~3 % it is the WORST of the six: the two-anchor compromise (this table + Farello saturation at 19 mol/kg) stretches 3 parameters past what the most non-ideal 1-1 salt allows. The five pairs that never saw this data land at 0.1-0.7 %.

Windows stay inside each pair's declared validity (KCl saturates at 4.803 mol/kg; KBr data used to 5).

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | gamma_nacl  | aw_1m         | 0.9668   |
| diag | gamma_nacl  | gamma_AAD_pct | 0.1117   |
| diag | gamma_nacl  | gamma_nMeas   | 8.0000   |
| diag | gamma_nacl  | gamma_pm_1m   | 0.6572   |
| diag | gamma_nacl  | gamma_pm_3m   | 0.7140   |
| diag | gamma_nacl  | phi_1m        | 0.9363   |
| diag | gamma_nacl  | phi_AAD_pct   | 0.0870   |
| diag | gamma_nacl  | phi_nMeas     | 8.0000   |
| diag | gamma_kcl   | aw_1m         | 0.9681   |
| diag | gamma_kcl   | gamma_AAD_pct | 0.1340   |
| diag | gamma_kcl   | gamma_nMeas   | 7.0000   |
| diag | gamma_kcl   | gamma_pm_1m   | 0.6043   |
| diag | gamma_kcl   | gamma_pm_3m   | 0.5698   |
| diag | gamma_kcl   | phi_1m        | 0.8987   |

...and 49 more reference values in the case's *expected*.



**pitzer\_hmw\_hot\_agreement**

tutorials/props/electrolyte/pitzer\_hmw\_hot\_agreement

hmw\_nacl\_100C

speciate

kernel\_nacl\_100C

pitzerActivity

**What it does:** HMW vs single-salt Pitzer at 100 C, same NaCl 4 m state: the two surfaces must AGREE now that both apply the SP77 T-functions

**The story:** one salt, one temperature, TWO Pitzer engines: they must agree (forum 100.2/#101, the HMW-virials(T) fix).

NaCl at 4 mol/kg, 100 C. Op 1 drives the MULTI-ION HMW speciation engine (activityModel pitzerHMW); op 2 drives the SINGLE-SALT kernel (pitzerActivity). Both now read the SAME Na-Cl.dat record: the 25 C virials PLUS the full Silvester-Pitzer 1977 T-functions (sp\_q1..q14) and the same A\_phi(T). For a lone 1:1 salt the HMW sum collapses to the same B/C terms as the kernel, so

$$\text{gamma\_pm}(\text{HMW}) = \sqrt{\text{gamma\_Na} * \text{gamma\_Cl}} == \text{gamma\_pm}(\text{kernel})$$

to numerical noise. BEFORE the fix the HMW held its virials at 25 C and this case disagreed in the second decimal – the adversarial regression this golden forbids forever.

(The kernel's own hot-brine validation against SP77 Table V lives in pitzer\_nacl\_sp77\_hot; this case is the ENGINE-AGREEMENT pin.)

**What to expect (golden-locked):**

| kind | unit/stream      | key            | expected |
|------|------------------|----------------|----------|
| diag | hmw_nacl_100C    | I              | 4.0000   |
| diag | hmw_nacl_100C    | SI_halite      | -0.7734  |
| diag | hmw_nacl_100C    | aw             | 0.8533   |
| diag | hmw_nacl_100C    | gamma_Cl       | 0.7309   |
| diag | hmw_nacl_100C    | gamma_Na       | 0.7309   |
| diag | hmw_nacl_100C    | m_Cl           | 4.0000   |
| diag | hmw_nacl_100C    | m_Na           | 4.0000   |
| diag | hmw_nacl_100C    | pH             | 7.0000   |
| diag | kernel_nacl_100C | aw_1m          | 0.9670   |
| diag | kernel_nacl_100C | gamma_pm_1m    | 0.6221   |
| diag | kernel_nacl_100C | gamma_pm_3m    | 0.6760   |
| diag | kernel_nacl_100C | phi_1m         | 0.9327   |
| diag | hmw_nacl_100C    | chargeImposed  | 0.0000   |
| diag | hmw_nacl_100C    | chargeResidual | -0.0000  |

**pitzer\_nacl\_sp77\_hot**

tutorials/props/electrolyte/pitzer\_nacl\_sp77\_hot

nacl\_100C

pitzeActivity

nacl\_150C

pitzeActivity

nacl\_200C

pitzeActivity

**What it does:** NaCl Pitzer to 200 C – Silvester & Pitzer 1977 T-functions, validated vs their Table V

**The story:** HOT BRINE: the NaCl Pitzer surface from 25 to 200 C on the FULL Silvester & Pitzer 1977 temperature functions (J. Phys. Chem. 81, 1822 – eqs 30-32, Table IV coefficients VERBATIM in the standards Na-Cl.dat), applied as a deviation from the validated 25 C surface, with A\_phi(T) curated against the same paper’s Table II (quintic, max dev 0.17 %, 0-300 C; the old closed form drifted +67 % at 300 C).

Three ops, one per temperature (100/150/200 C), each validating gamma\_pm at the SATURATION molality against the paper’s own Table V – an m-extrapolation (m\_satd 6.7-8.0 > the 6 mol/kg weighting window), declared in the dataset header, exactly as the paper frames it.

What to look at: gamma\_pm(1 m) falls 0.657 → ~0.48 from 25 to 200 C (electrostatics intensify as eps\_w collapses – the paper’s central observation), and the validation AADs stay single-digit % even at the saturated-extrapolated edge.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | nacl_100C   | aw_1m         | 0.9670   |
| diag | nacl_100C   | gamma_AAD_pct | 0.2851   |
| diag | nacl_100C   | gamma_nMeas   | 1.0000   |
| diag | nacl_100C   | gamma_pm_1m   | 0.6221   |
| diag | nacl_100C   | gamma_pm_3m   | 0.6760   |
| diag | nacl_100C   | phi_1m        | 0.9327   |
| diag | nacl_150C   | aw_1m         | 0.9679   |
| diag | nacl_150C   | gamma_AAD_pct | 0.0425   |
| diag | nacl_150C   | gamma_nMeas   | 1.0000   |
| diag | nacl_150C   | gamma_pm_1m   | 0.5593   |
| diag | nacl_150C   | gamma_pm_3m   | 0.5736   |
| diag | nacl_150C   | phi_1m        | 0.9054   |
| diag | nacl_200C   | aw_1m         | 0.9693   |
| diag | nacl_200C   | gamma_AAD_pct | 0.4317   |

... and 4 more reference values in the case’s *expected*.

**pitzer\_seawater\_verify**

tutorials/props/electrolyte/pitzer\_seawater\_verify

reduction\_nacl  
speciateseawater  
speciate

**What it does:** Pitzer-HMW verification: (binaries + ternary theta/psi) on standard synthetic seawater ( $I \sim 0.7$ ). Pins per-ion + mean activity coefficients against PUBLISHED HMW-1984 / seawater-Pitzer values, and the single-salt reduction (ternaries  $\rightarrow 0$ ).

**The story:** the TIER-2 verification weapon for the Pitzer-HMW ternary-mixing slice (S3).

The forum gate: "NO published-seawater golden masters  $\Rightarrow$  NO-GO." This case speciates a STANDARD SYNTHETIC SEAWATER with the multi-ion Pitzer model (binaries + ternary theta/psi) and pins the per-ion + mean activity coefficients against PUBLISHED values, to literature tolerance.

SEAWATER COMPOSITION (the canonical major-ion seawater of the HMW-1984 carbonate-free major-ion subset; molality, mol/kg-water,  $S = 35$ ): Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO4-2 0.02926 giving ionic strength  $I \sim 0.70$  mol/kg – the classic seawater benchmark. (The carbonate system – HCO<sub>3</sub>/CO<sub>3</sub>,  $\sim 2.3$  mmol/kg,  $< 0.4\%$  of  $I$  – is OMITTED: it is slice S4, and its absence shifts  $I$  by  $< 0.3\%$  and the major-ion gammas by  $< 0.2\%$ , well inside the pins' tolerance.) pH is GIVEN at 8.1 (typical surface seawater) – with no carbonate in the analyticalTotals it only fixes the trace H<sup>+</sup>/OH<sup>-</sup>, which do not enter the major-ion gammas.

WHY PITZER, NOT DAVIES: at  $I \sim 0.7$  Davies is already out of its trustworthy band ( $\sim 0.5$ ). The Pitzer specific-interaction model – WITH the ternary theta/psi mixing terms AND the E\_theta(I) higher-order electrostatic mixing term (Pitzer 1975, now ACTIVE for unlike-charge same-sign pairs) – is the standard tool for seawater and brines to  $I \sim 6$ . E\_theta is a refinement: it shifts the 2:1 / 1:2 ion gammas a few % (gamma\_Ca, gamma\_SO4) and tightens the gamma\_pm(KCl) pin to 0.638 (inside 0.62-0.64); the 1:1 NaCl gamma\_pm stays  $\sim 0.66$ . (The only remaining v2 deferral is the full beta(T)/theta(T) temperature surface; 25 C base here.)

PUBLISHED PINS (each cited; deviation Choupo vs published in VALIDATION.md):

gamma\_pm(NaCl) in seawater  $\sim 0.66$ -0.68 Pitzer's seawater model / Millero & Pierrot; classic textbook value gamma\_pm(NaCl, seawater  $I \sim 0.7$ )  $\sim 0.67$ . gamma\_pm(KCl)  $\sim 0.62$ -0.64 ; gamma\_pm(MgCl<sub>2</sub>), gamma\_pm(CaCl<sub>2</sub>) etc. All within the Harvie-Moller-Weare (1984) seawater Pitzer model, the reference implementation this catalogue's theta/psi come from.

The golden master locks the Choupo-computed per-ion gammas; this header records the published comparison + the deviation. A WRONG pin is worse than no pin – the numbers here are pinned to the model's own reproducible output and CROSS-CHECKED against the published seawater range.

FINDING, 2026-08-10: "ALL PINS STILL IN BAND" WAS CHECKED ON TWO OF SIX.

The paragraph below was written when E\_theta (the I-dependent higher-order electrostatic mixing term) was activated. It re-checked the two 1-1 salts – whose gamma\_pm is a square root of two numbers sitting next to each other in the output – and concluded "all pins STILL in their published band". The four DIVALENT salts were never recomputed, and E\_theta acts precisely on unlike-charge same-sign pairs, i.e. on exactly those four.

The engine now PUBLISHES gamma\_pm (diagMeanIonic), so the comparison is no longer a hand calculation in a header. Against the bands in data/standards/parameters/electrolyte/VALIDATION.md:

NaCl 0.6646 band 0.66-0.69 IN KCl 0.6381 band 0.62-0.65 IN MgCl<sub>2</sub> 0.4608 band 0.47-0.50 BELOW CaCl<sub>2</sub> 0.4459 band 0.46-0.50 BELOW Na<sub>2</sub>SO<sub>4</sub> 0.3489 band 0.36-0.40 BELOW MgSO<sub>4</sub> 0.1460 band 0.15-0.20 BELOW

Four of six are outside, by 2-4 %. VALIDATION.md still says all six are inside, still lists the PRE-E\_theta numbers as Choupo's, and still describes E\_theta as "deferred to v2 ( $< 0.5\%$  on the major-ion gamma at this I)" – while this run announces "E\_theta now ACTIVE" and the divalent shift is  $\sim 5\%$ . That estimate was wrong by an order of magnitude on the ions it was made about.

NOT RESOLVED, and deliberately NOT TUNED. Two readings are open and this case cannot choose between them: the quoted bands are loose secondary citations ("~ 0.47-0.50", sourced to HMW 1984 with no table or page), or E\_theta is being applied where those bands' authors did not. Resolving it needs the primary tables (Harvie-Moller-Weare 1984; Millero & Pierrot), which the project does not hold. No parameter, no golden and no band was moved to make this look better; the six gamma\_pm values are pinned as goldens so the next change to them is visible, and PIN B carries no anchor row (the 'expected' file says why).

The paragraph as originally written, kept because it is the record of what was checked:

E\_theta(I) BEFORE → AFTER (constant-theta → higher-order mixing active), all pins STILL in their published band: gamma\_pm(NaCl) 0.6714 → 0.6646 (band 0.66-0.68; 1:1, E\_theta=0 for Na-Cl, shift via coupling) – IN BAND gamma\_pm(KCl) 0.6447 → 0.6381 (band 0.62-0.64) – E\_theta moved it INTO band gamma\_Ca 0.2141 → 0.1856 (2+ ion; E\_theta acts via Ca-Na/Ca-K/Ca-Mg) gamma\_SO4 0.1227 → 0.1040 (2- ion; E\_theta acts via SO4-Cl/SO4-OH) gamma\_Mg 0.2363 → 0.2049 gamma\_Na 0.6495 → 0.6390 gamma\_Cl 0.6941 → 0.6911 Single-salt reduction (PIN A, NaCl alone): gamma\_pm UNCHANGED at 0.6572 – E\_theta vanishes identically for one cation + one anion (the oracle stays 1.5658e-14 over all 54 binaries).

DATA SCOPE: the SPECIATION network is a complete case-local replace-mode snapshot (constant/-electrolyte/speciation.dat) – that claim covers the speciation data only. Ions, Pitzer pairs/mixing and minerals resolve from the curated data/standards/ tree at runtime; seal with bin/choupo-import for a runtime-self-contained copy.

#### What to expect (golden-locked):

| kind                                                             | unit/stream    | key            | expected  |
|------------------------------------------------------------------|----------------|----------------|-----------|
| diag                                                             | reduction_nacl | I              | 1         |
| diag                                                             | reduction_nacl | SI_halite      | -1.93464  |
| diag                                                             | reduction_nacl | aw             | 0.966826  |
| diag                                                             | reduction_nacl | gamma_Cl       | 0.657172  |
| diag                                                             | reduction_nacl | gamma_Na       | 0.657172  |
| diag                                                             | reduction_nacl | gamma_pm_Na_Cl | 0.657172  |
| diag                                                             | reduction_nacl | m_Cl           | 1         |
| diag                                                             | reduction_nacl | m_Na           | 1         |
| diag                                                             | reduction_nacl | pH             | 7         |
| diag                                                             | seawater       | I              | 0.720461  |
| diag                                                             | seawater       | SI_anhydrite   | -0.970682 |
| diag                                                             | seawater       | SI_gypsum      | -0.687047 |
| diag                                                             | seawater       | SI_halite      | -2.48552  |
| diag                                                             | seawater       | SI_sylvite     | -3.51313  |
| ... and 50 more reference values in the case's <i>expected</i> . |                |                |           |

**pitzer\_vs\_davies\_ro\_concentrate**

tutorials/props/electrolyte/pitzer\_vs\_davies\_ro\_concentrate

**scan\_davies**  
 scalingScan

**scan\_pitzer**  
 scalingScan

**What it does:** Davies vs Pitzer-HMW on a seawater RO concentrate: gypsum/calcite SI vs recovery from the same feed under two activity models (full narrative in propsDict).

**The story:** THE FLAGSHIP of the Pitzer x speciation feature (slice S6): the payoff that makes the whole multi-ion Pitzer-HMW build worth it. ONE feed, TWO activity models, the engineering answer FORKS.

THE QUESTION (every RO / brine-concentration designer asks it): "At what water recovery does my concentrate start to scale gypsum?" The answer DEPENDS ON THE ACTIVITY MODEL – and that is the whole point.

THE FEED: standard surface seawater ( $S = 35$ ), the major-ion subset:  $\text{Na}^+ 0.4861$   $\text{Mg}^{+2} 0.0547$   $\text{Ca}^{+2} 0.01065$   $\text{K}^+ 0.01058$  (cations)  $\text{Cl}^- 0.5658$   $\text{SO}_4^{+2} 0.02926$   $\text{HCO}_3^- 0.0024$  (anions, DIC carrier)  $I \sim 0.7$  mol/kg at the feed. Recovery  $r$  scales every total as feed/(1 -  $r$ ); scanning  $r = 0 \rightarrow 0.80$  drives  $I$  from  $\sim 0.7$  up to  $\sim 3.5$  mol/kg – DEEP into the brine regime where Davies is long out of its trust window.

THE TWO RUNS (identical feed, identical recovery grid, ONLY the gamma model differs – the controlled experiment): scan\_davies activityModel davies; (the textbook extended-DH model) scan\_pitzer activityModel pitzerHMW; (Pitzer-HMW, the brine-grade model)

THE PAYOFF ARTEFACTS (read straight off the two CSVs + the KPIs): 1. SI\_gypsum vs recovery /  $I$ , BOTH models – the HEADLINE. They track together at low  $I$  and FORK visibly past  $I \sim 1$ : Davies keeps climbing, Pitzer's gypsum SI is held DOWN by the strong Ca-SO<sub>4</sub> specific interaction ( $\beta_2 = -59.3$ , the 2:2 ion-pairing virial Davies cannot see). The gypsum-onset recovery is LATER under Pitzer. 2. SI\_calcite vs recovery /  $I$ , both models – same fork, carbonate flavour. pH is SOLVED from electroneutrality at every point (closed system; DIC concentrates with the water and the pH drifts). 3. gamma\_Ca<sup>2+</sup> and gamma\_SO<sub>4</sub><sup>2-</sup> vs  $I$ , both models – the MECHANISM plot. Davies declines monotonically (its single  $I$ -controlled curve); Pitzer turns UP at high  $I$  (the specific-interaction upturn). This is WHERE the SI divergence comes from – the gammas diverge, so the IAP diverges, so the SI forks. (scalingScan emits gamma\_<ion> columns whenever diagSpecies is given and the model is pitzer/davies.)

THE KPI THAT TELLS THE STORY: recovery\_at\_gypsum\_onset (the SI\_gypsum = 0 crossing) – ONE number per model. Davies predicts gypsum scaling at one recovery; Pitzer at another. The engineering decision – how hard can  $I$  push recovery before gypsum scales? – flips on the activity model. Endpoint diags carry SI\_gypsum and SI\_calcite at  $r_{\text{max}}$  for both models, and the two onset recoveries.

WHY PITZER IS THE ANSWER (cite the credentials, do not assert): Davies is ANNOUNCED-untrustworthy past  $I \sim 0.5$  mol/kg (the run says so). Pitzer-HMW is built for exactly this regime – valid to  $I \sim 6$  mol/kg for the seawater system, with the  $E_{\text{theta}}(I)$  higher-order electrostatic mixing term ACTIVE (Pitzer 1975) – and is CREDENTIALLED in this repo by two pinned cases: pitzer\_seawater\_verify – single-salt oracle  $1.57\text{e-}14$  + HMW seawater mean activity coefficients inside the published bands; pitzer\_calcite\_brine – SI\_calcite / SI\_gypsum on surface seawater inside the published oceanographic saturation band. So here we TRUST the Pitzer fork and read Davies as the cautionary curve.

$E_{\text{theta}}(I)$  BEFORE  $\rightarrow$  AFTER (the scan\_pitzer fork only; scan\_davies is BYTE- IDENTICAL –  $E_{\text{theta}}$  is Pitzer-only). At  $r_{\text{max}}$  ( $I \sim 3.3$  mol/kg, deep brine where higher-order mixing bites hard-est): SI\_gypsum\_rmax -0.144  $\rightarrow$  -0.317 SI\_calcite\_rmax 2.026  $\rightarrow$  1.938 SI\_gypsum\_r0 -0.877  $\rightarrow$  -0.967 SI\_calcite\_r0 1.240  $\rightarrow$  1.183  $I_{\text{rmax}}$  3.271  $\rightarrow$  3.322 gypsum stays undersaturated end-to-end The engineering verdict (gypsum the limiting scale) is UNCHANGED in direction;  $E_{\text{theta}}$  refines the SI magnitude in the high- $I$  tail.

DATA SCOPE: self-DESCRIBING (inline manifest); the ions, pairs, mixing with the carbonate theta/p-si/lambda, speciation and minerals resolve from the curated data/standards/ tree at runtime – seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind                                                                   | unit/stream | key               | expected  |
|------------------------------------------------------------------------|-------------|-------------------|-----------|
| diag                                                                   | scan_davies | I_r0              | 0.657308  |
| diag                                                                   | scan_davies | I_rmax            | 3.03589   |
| diag                                                                   | scan_davies | SI_anhydrite_r0   | -0.889841 |
| diag                                                                   | scan_davies | SI_anhydrite_rmax | 0.269649  |
| diag                                                                   | scan_davies | SI_aragonite_r0   | 1.26146   |
| diag                                                                   | scan_davies | SI_aragonite_rmax | 2.0927    |
| diag                                                                   | scan_davies | SI_calcite_r0     | 1.37546   |
| diag                                                                   | scan_davies | SI_calcite_rmax   | 2.2067    |
| diag                                                                   | scan_davies | SI_gypsum_r0      | -0.607638 |
| diag                                                                   | scan_davies | SI_gypsum_rmax    | 0.481394  |
| diag                                                                   | scan_davies | SI_halite_r0      | -2.3956   |
| diag                                                                   | scan_davies | SI_halite_rmax    | -0.45497  |
| diag                                                                   | scan_davies | SI_sylvite_r0     | -3.3946   |
| diag                                                                   | scan_davies | SI_sylvite_rmax   | -1.44942  |
| <i>...and 192 more reference values in the case's <b>expected</b>.</i> |             |                   |           |

**precipitation\_ro\_brackish**

tutorials/props/electrolyte/precipitation\_ro\_brackish

**pin\_gypsum\_amount**  
speciate

**feed**  
speciate

**feed\_eq**  
speciate

**scan\_eq**  
scalingScan

**What it does:** Equilibrium PRECIPITATION (the SI = 0 ceiling) of a brackish RO concentrate: aqueous speciation (Davies) + augmented-Newton chemical equilibrium with an announced active set. Sibling of scaling\_ro\_brackish (which reports SI only) – here named minerals are ALLOWED to precipitate to saturation and the engine returns the thermodynamic MAXIMUM amount (SI → 0, infinite time, no nucleation barrier – a CEILING, NOT a deposit forecast; real scale is kinetically limited). Story: (1) pin\_gypsum\_amount – a CaSO<sub>4</sub> water with Ca and SO<sub>4</sub> DOUBLED past gypsum saturation; the machine hands back EXACTLY the over-added excess as solid (n\_gypsum). (2) feed / feed\_eq – the brackish water raw and equilibrated side by side: calcite reaches its ceiling, releases H<sup>+</sup> (pH drops, solved closure), and consuming shared Ca pushes gypsum further from saturation. (3) scan\_eq – calcite + gypsum allowed across recovery to r = 0.85 with a feedFlow, giving a kg/day calcite curve; gypsum stays n = 0 throughout while its SI<sub>eq</sub> drifts BELOW the raw free-water SI once calcite consumes the shared Ca – the coupling, visible. Cross-check tool: PHREEQC EQUILIBRIUM\_PHASES.

**The story:** the EQUILIBRIUM-PRECIPITATION ceiling.

Sibling of scaling\_ro\_brackish (which is SI-ONLY – "how supersaturated?"). Here the 'equilibrate { minerals ( ... ); }' block names the ALLOWED solids, and the engine drives each to its SI = 0 saturation, returning the amount precipitated. THE FRAMING (printed loudly at run time, non-suppressible):

EQUILIBRIUM CEILING – not a deposit prediction. The amounts are the thermodynamic MAXIMUM (SI → 0, infinite time, no nucleation barrier). Real scale is kinetically limited (induction time, antiscalants act on KINETICS, which this calculation cannot see). Actual deposit ≤ ceiling, often far less. The safe direction is rigorous: ceiling ~ 0 => no driving force.

ALGORITHM (visible in the trace): augmented Newton in (ln m<sub>j</sub> [, ln a<sub>H</sub>]) plus one raw-linear n<sub>p</sub> per ACTIVE mineral; each master mole balance gains a sink -sum nu<sub>pj</sub> n<sub>p</sub>; one SI = 0 row per active phase; an OUTER active-set loop ADMITS the most-supersaturated allowed mineral and EVICTS any with n<sub>p</sub> < 0, re-solving between, until complementarity holds (SI ≤ 0 with n = 0, OR SI = 0 with n > 0). A carbonate mineral's H<sup>+</sup> leg lands the released H<sup>+</sup> in the electroneutrality row (pH solve) – so precipitating calcite ACIDIFIES.

CROSS-CHECK TOOL: PHREEQC EQUILIBRIUM\_PHASES (same ceiling framing). Data: curated USGS PHREEQC records (public domain) resolved from data/standards/ at runtime; self-DESCRIBING (inline manifest) – seal with bin/choupo-import for a runtime-self-contained copy.

GOLDEN PINS (locked by the golden master): pin\_gypsum\_amount Ca = SO<sub>4</sub> = 0.030542 mol/kg (DOUBLE the gypsum-saturation value 0.015271) + equilibrate gypsum → SI\_gypsum = 0.000 and n\_gypsum = 0.015271 (the machine returns EXACTLY the over-added excess). feed\_eq calcite + gypsum allowed; calcite reaches SI = 0, releases H<sup>+</sup> (pH 7.72 → ~7.50, solved), gypsum stays dissolved. scan\_eq r 0 → 0.85 with feedFlow: calcite kg/day ceiling; gypsum n = 0 throughout while SI<sub>eq</sub>\_gypsum drifts BELOW the raw SI\_gypsum (calcite eats the shared Ca).

**What to expect (golden-locked):**

| kind | unit/stream       | key          | expected    |
|------|-------------------|--------------|-------------|
| diag | pin_gypsum_amount | I            | 0.0455084   |
| diag | pin_gypsum_amount | SI_anhydrite | -0.299583   |
| diag | pin_gypsum_amount | SI_gypsum    | 1.33618e-10 |
| diag | pin_gypsum_amount | aw           | 0.99952     |
| diag | pin_gypsum_amount | n_gypsum     | 0.0152707   |
| diag | pin_gypsum_amount | pH           | 7           |
| diag | feed              | I            | 0.0266002   |
| diag | feed              | SI_anhydrite | -1.68532    |
| diag | feed              | SI_aragonite | 0.111093    |
| diag | feed              | SI_calcite   | 0.225093    |
| diag | feed              | SI_gypsum    | -1.38593    |
| diag | feed              | SI_halite    | -5.35323    |
| diag | feed              | SI_sylvite   | -6.39779    |
| diag | feed              | aw           | 0.999302    |

*...and 193 more reference values in the case's **expected**.*



**rainwater\_air**

tutorials/props/electrolyte/rainwater\_air

**air25**  
speciate**air10**  
speciate

**What it does:** WATER OPEN TO AIR: the generic multi-gas Henry pin. Pure water equilibrated with the atmosphere ( $p\text{CO}_2$  4.2e-4,  $p\text{O}_2$  0.2095,  $p\text{N}_2$  0.7808 atm) at 25 C and at 10 C. Each listed gas PINS its dissolved species by Henry's law and that master's mole balance is REPLACED by the pin – three pins at once:  $\text{CO}_2$  chains through the carbonate family ( $\text{CO}_2\text{aq}$  is a computed species, the  $\text{HCO}_3$  row is swapped), while  $\text{O}_2$  and  $\text{N}_2$  are inert neutral masters pinned directly. TWO CLASSIC NUMBERS FALL OUT: (1) unpolluted RAINWATER pH ~ 5.6 – carbonic acid from 420 ppm  $\text{CO}_2$  alone, no pollutant needed; (2) DISSOLVED-OXYGEN saturation ~ 8.5 mg/L at 25 C rising to ~ 11 mg/L at 10 C – gas solubility is RETROGRADE with temperature ( $K_H$  analytic from USGS PHREEQC PHASES, public domain), which is why cold streams hold more oxygen than warm ones. The 'totals' entries for  $\text{HCO}_3$  /  $\text{O}_2$  /  $\text{N}_2$  are initial guesses only (announced): in an OPEN system each pinned total is a SOLVED OUTCOME of gas-liquid equilibrium.

**The story:** water open to AIR: the generic multi-gas Henry pin.

Pure water equilibrated with the ATMOSPHERE – three gases pinned at once:

atmosphere {  $p\text{CO}_2$  4.2e-4 atm;  $\text{O}_2$  0.2095 atm;  $\text{N}_2$  0.7808 atm; }

Each gas pins its dissolved species by Henry's law,  $a = K_H(T) * p$ , and the pinned master's mole balance is REPLACED by the pin (the OPEN system of SpeciationSolver, generalised beyond  $\text{CO}_2$ ). Two pin shapes appear side by side, which is the point of the case:

$\text{CO}_2$  dissolves into a COMPUTED species ( $\text{CO}_2\text{aq}$ , formed from the  $\text{HCO}_3$  master): the pin swaps the  $\text{HCO}_3$  row and the whole carbonate family ( $\text{CO}_2\text{aq}$  /  $\text{HCO}_3^-$  /  $\text{CO}_3^{2-}$ ) equilibrates with the gas – DIC is a solved OUTCOME;  $\text{O}_2, \text{N}_2$  dissolve into INERT NEUTRAL MASTERS (species/aqueous  $\text{O}_2$ ,  $\text{N}_2$  – identity records, charge 0): the pin replaces each master's own row directly – the dissolved concentration is set by the gas phase alone.

THE TWO CLASSIC NUMBERS (textbook checks, no fitting anywhere):

pH ~ 5.6 unpolluted rainwater: carbonic acid from 420 ppm  $\text{CO}_2$  only (pH solve; –  $\text{H}^+$  from electroneutrality against  $\text{HCO}_3^-$ ); DO 8.5 → 11 mg/L dissolved-oxygen saturation, 25 C → 10 C:  $K_H(T)$  analytic (USGS PHREEQC PHASES) makes gas solubility RETROGRADE – cold water holds more oxygen.  $\text{N}_2$  saturates near 14 mg/L (25 C) for the same reason.

The 'totals' entries below are INITIAL GUESSES only (the run announces it): in an OPEN system every pinned total is a solved outcome. The feed charge-imbalance warning is EXPECTED here – an  $\text{HCO}_3$  guess with no cation is not a measured analysis; the solved  $\text{H}^+$  is its physical counter-ion.

Diagnostics locked by the golden master: pH, I,  $m_{\text{CO}_2\text{aq}}$ , and the dissolved  $\text{O}_2$  /  $\text{N}_2$  molalities at both temperatures.

What to expect (golden-locked):

| kind                                                            | unit/stream | key         | expected    |
|-----------------------------------------------------------------|-------------|-------------|-------------|
| diag                                                            | air25       | I           | 2.52807e-06 |
| diag                                                            | air25       | aw          | 0.999986    |
| diag                                                            | air25       | gamma_CO2aq | 1           |
| diag                                                            | air25       | gamma_OH    | 0.998138    |
| diag                                                            | air25       | m_CO2aq     | 1.42971e-05 |
| diag                                                            | air25       | m_OH        | 3.97038e-09 |
| diag                                                            | air25       | pH          | 5.59803     |
| diag                                                            | air10       | I           | 2.79349e-06 |
| diag                                                            | air10       | aw          | 0.999981    |
| diag                                                            | air10       | gamma_CO2aq | 1           |
| diag                                                            | air10       | gamma_OH    | 0.998089    |
| diag                                                            | air10       | m_CO2aq     | 2.2591e-05  |
| diag                                                            | air10       | m_OH        | 1.04436e-09 |
| diag                                                            | air10       | pH          | 5.55469     |
| ...and 12 more reference values in the case's <i>expected</i> . |             |             |             |

**scaling\_ro\_brackish**

tutorials/props/electrolyte/scaling\_ro\_brackish

pin\_kw  
speciatepin\_gypsum  
speciatepin\_neutral  
speciatefeed  
speciatescan  
scalingScanscan\_open  
scalingScan

**What it does:** RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices  $SI = \log_{10}(IAP/K)$ , scanned over water recovery to  $r = 0.85$  (concentrate =  $\text{feed}/(1-r)$ ) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere ( $a(\text{CO}_2\text{aq})$  pinned by Henry at  $p\text{CO}_2 = 4.2\text{e-}4$  atm, DIC a solved outcome – this  $\text{CO}_2$ -rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces  $K_w$  ( $m_{\text{OH}} \sim 1\text{e-}7$ ), a back-computed gypsum-saturated  $\text{CaSO}_4$  solution gives  $SI_{\text{gypsum}} \sim 0.00$ , and pure water with pH solved lands at exactly 7.000.

**The story:** will my RO concentrate scale?

A REPRESENTATIVE brackish groundwater (generic textbook-style composition, ~1500 mg/L TDS; no licensed table reproduced):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026 Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003 lab pH 7.8 (nominal – see below)

(totals are TOTAL element amounts: the  $\text{HCO}_3$  total is total dissolved inorganic carbon – at pH 7.8 the reported  $\text{HCO}_3$  is ~97% of it. Cl is 510 mg/L so the analysis is genuinely consistent with its nominal pH 7.8: an analysis balances charge only once the carbonate split at the lab pH is counted.)

pH IS SOLVED, NOT GIVEN ('pH solve;'): H+ joins the Newton unknowns and ELECTRONEUTRALITY  $\sum z_i m_i = 0$  closes the system. Real analyses are never perfectly balanced, so the engine ANNOUNCES the feed charge imbalance first (PHREEQC convention,  $200|S+ - S-|/(S+ + S-)$ ; here 0.35%) – the solved pH absorbs exactly that error, and above ~5% the solver warns loudly that it is not trustworthy. This feed solves to pH ~7.72, near the lab's nominal 7.8: the analysis is consistent.

UNITS: every 'totals' entry carries its basis (bare numbers are refused). The feed + scans take the analysis EXACTLY as the lab sheet reports it – mg/L – converted through the ion MW (ions.dat) at  $\rho \sim 1$  kg/L (dilute aqueous, announced once per run). The analytic pins stay in explicit mol/kg (back-computed equilibria, not analyses).

THE CLOSED-vs-OPEN LESSON (two scans, same feed): scan CLOSED system – every total (DIC included) concentrates as  $\text{feed}/(1-r)$ ; the charge-balance ratio is scale-invariant, so the pH only drifts gently ( $7.72 \rightarrow 7.58$ , the Davies gammas at work) while calcite SI climbs  $0.23 \rightarrow 1.38$ . scan\_open OPEN to the atmosphere ('atmosphere {  $p\text{CO}_2$  4.2e-4 atm; }') –  $a(\text{CO}_2\text{aq})$  is PINNED by Henry, the  $\text{HCO}_3$  mole balance is REPLACED by the pin, and DIC becomes a solved OUTCOME. This groundwater is ~8x  $\text{CO}_2$ -supersaturated vs the atmosphere, so opening it DEGASES  $\text{CO}_2$ : the pH jumps to 8.54 and climbs to 9.08, and calcite SI lands ABOVE the closed curve (2.59 vs 1.38 at  $r = 0.85$ ) – why concentrate pH control (acid dosing) matters. Compare the two CSVs (both carry a pH column right after I); the contrast IS the lesson.

VALIDATION PINS (locked by the golden master): pin\_kw pure water, pH 7 GIVEN  $\rightarrow m_{\text{OH}} = K_w a_w/(a_H \gamma_{\text{OH}}) \sim 1.0004\text{e-}7$  (unchanged) pin\_gypsum totals  $\text{Ca} = \text{SO}_4 = 0.015271$  mol/kg, pH 7 GIVEN, BACK-COMPUTED from the catalogue's  $\log K_{25}(\text{gypsum}) = -4.55 + \text{CaSO}_4\text{aq} \log K_{25} = 2.14$  + Davies at the resulting  $I = 0.0455$  (26% ion pairing) so that  $SI_{\text{gypsum}} = 0.00 \pm 0.001$  (unchanged) pin\_neutral pure water, pH SOLVED  $\rightarrow$  electroneutrality  $m_H = m_{\text{OH}}$  must give pH 7.000 (locks the new charge-balance closure)

Speciation/mineral/gas data: case-local constant/electrolyte/ (self-contained SNAPSHOT of the standards catalogue, restricted to this master set; the case files ECLIPSE data/standards/electrolyte/ – the network is taken whole. Provenance: USGS PHREEQC, public domain – see the .dat headers; the  $\text{CO}_2(\text{g})$  Henry constant is the db's PHASES  $\text{CO}_2(\text{g})$  entry).

**What to expect (golden-locked):**

| kind | unit/stream | key          | expected     |
|------|-------------|--------------|--------------|
| diag | pin_kw      | I            | 1.00037e-07  |
| diag | pin_kw      | aw           | 1            |
| diag | pin_kw      | m_OH         | 1.00037e-07  |
| diag | pin_kw      | pH           | 7            |
| diag | pin_gypsum  | I            | 0.0455076    |
| diag | pin_gypsum  | SI_anhydrite | -0.299593    |
| diag | pin_gypsum  | SI_gypsum    | -1.04349e-05 |
| diag | pin_gypsum  | aw           | 0.99952      |
| diag | pin_gypsum  | m_CaSO4aq    | 0.0038941    |
| diag | pin_gypsum  | pH           | 7            |
| diag | pin_neutral | I            | 1.00037e-07  |
| diag | pin_neutral | aw           | 1            |
| diag | pin_neutral | pH           | 7            |
| diag | feed        | I            | 0.0266002    |

*...and 244 more reference values in the case's **expected**.*

**softener\_brackish**

tutorials/props/electrolyte/softener\_brackish

raw  
scalingScansoftened  
exchangesoftened\_scaling  
scalingScan

**What it does:** Ion-exchange softening of a brackish RO feed: removes the calcite-scaling driver (raw vs softened vs softened-scaling; full narrative in propsDict).

**The story:** ion-exchange SOFTENING removes the scaling driver.

A three-act story on the SAME brackish RO feed as scaling\_ro\_brackish / precipitation\_ro\_brackish:

(1) raw scalingScan of the raw water: calcite SI > 0 already at the feed and climbing with recovery – the SCALING PROBLEM (a CaCO<sub>3</sub>-supersaturated concentrate). (2) softened the ‘exchange’ op equilibrates the SAME water with a strong-acid cation resin (SAC\_Na, Gaines-Thomas). Ca and Mg drop ~99% onto the resin; Na rises eq-for-eq (the salt penalty, printed in meq/L); hardness\_removal\_pct ~ 99%. (3) softened\_scaling feed the SOFTENED composition (Ca/Mg gone, Na up) into a scalingScan: SI\_calcite is now NEGATIVE across the whole recovery range – the scaling DRIVER is gone. Removing the hardness, not chasing the supersaturation, is the cure.

MODEL (visible in the verbosity>=3 trace): the SAME log-space Newton as speciate, PLUS one master (the free resin site X-), one balance (CEC capacity = sum of equivalents on the sites), and the exchange half-reactions  $\text{Me}(z+) + z \text{X}^- = \text{MeX}$  (Gaines-Thomas equivalent-fraction activity for the bound species, Davies for the aqueous ions). Divalent Ca<sup>2+</sup> (logK 0.8) is equivalent-fraction-favoured and outcompetes Na<sup>+</sup>; watch the per-iteration selectivity ratio  $\beta(\text{CaX}_2)/\beta(\text{NaX})^2$  climb.

CROSS-CHECK TOOL: PHREEQC EXCHANGE (same Gaines-Thomas half-reactions).

THE FRESH-RESIN CAVEAT (printed loudly, non-suppressible): the ‘exchange’ op returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass. A real fixed bed leaks rising hardness as the exchange front migrates to breakthrough, then must be regenerated; cycle length and breakthrough are TRANSIENT and not modelled here.

Brackish water (I ~ 0.03-0.06): Davies-valid, no Pitzer needed; the honesty banner + the I>0.5 Davies advisory cover the boundary.

Data: curated USGS PHREEQC records (public domain) resolved from data/standards/ at runtime, + the SAC\_Na resin nameplate; self-DESCRIBING (inline manifest) – seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind | unit/stream | key               | expected  |
|------|-------------|-------------------|-----------|
| diag | raw         | I_r0              | 0.0266002 |
| diag | raw         | I_rmax            | 0.165631  |
| diag | raw         | SI_anhydrite_r0   | -1.68532  |
| diag | raw         | SI_anhydrite_rmax | -0.646016 |
| diag | raw         | SI_aragonite_r0   | 0.111093  |
| diag | raw         | SI_aragonite_rmax | 1.26704   |
| diag | raw         | SI_calcite_r0     | 0.225093  |
| diag | raw         | SI_calcite_rmax   | 1.38104   |
| diag | raw         | SI_gypsum_r0      | -1.38593  |
| diag | raw         | SI_gypsum_rmax    | -0.349983 |
| diag | raw         | SI_halite_r0      | -5.35323  |
| diag | raw         | SI_halite_rmax    | -3.82475  |
| diag | raw         | SI_sylvite_r0     | -6.39779  |
| diag | raw         | SI_sylvite_rmax   | -4.87569  |

... and 193 more reference values in the case's *expected*.

**tartaricAcid\_acidulation**

tutorials/props/electrolyte/tartaricAcid\_acidulation

acidulation\_pitzer  
speciateacidulation\_davies  
speciategypsum\_ceiling  
speciatepKa2\_davies  
speciatepKa2\_pitzer  
speciate

**What it does:** Tartaric-acid acidulation liquor: rigorous Pitzer brine chemistry (Ca-SO<sub>4</sub>-K-Cl) + the gypsum precipitation ceiling, with tartrate honestly on the Debye-Hueckel limit (no published Pitzer params). Davies vs Pitzer shown side by side.

**What to expect (golden-locked):**

| kind | unit/stream        | key          | expected  |
|------|--------------------|--------------|-----------|
| diag | acidulation_pitzer | I            | 1.6414    |
| diag | acidulation_pitzer | SI_anhydrite | 1.07386   |
| diag | acidulation_pitzer | SI_arcanite  | -2.80366  |
| diag | acidulation_pitzer | SI_gypsum    | 1.36221   |
| diag | acidulation_pitzer | SI_sylvite   | -3.52403  |
| diag | acidulation_pitzer | aw           | 0.986675  |
| diag | acidulation_pitzer | gamma_Ca     | 0.0716778 |
| diag | acidulation_pitzer | gamma_H2Tart | 1         |
| diag | acidulation_pitzer | gamma_HTart  | 0.409404  |
| diag | acidulation_pitzer | gamma_SO4    | 0.0662673 |
| diag | acidulation_pitzer | gamma_Tart   | 0.036756  |
| diag | acidulation_pitzer | m_Ca         | 0.397714  |
| diag | acidulation_pitzer | m_H2Tart     | 0.0656066 |
| diag | acidulation_pitzer | m_HTart      | 0.0314016 |

... and 74 more reference values in the case's *expected*.

**estimate**

**The theory you need.** When a compound is missing, group contribution builds it: Joback sums first-order group increments into  $T_b, T_c, P_c, V_c, \Delta H_f^\circ, \Delta H_{\text{vap}}, c_p^{\text{ig}}(T)$ ; Lee–Kesler closes  $\omega$  from  $(T_b, T_c, P_c)$ ; Ambrose–Walton gives  $P^{\text{sat}}(T)$  from corresponding states; Rackett the liquid volume. The estimate is a PROPOSAL: a **.dat** you review against any reference data and promote by hand – the engine never estimates at runtime. Molecular structure is **IDENTITY**: components carry **jobackGroups{}** in their own record, and **bin/estimate <name>** is the one-command curation act. Expect few-percent errors on criticals (and the guide's worked example shows them), weaker  $T_b$  for hydrogen-bonded species, and polymers via van Krevelen / Yang on the repeat unit. *Full derivation: Theory Guide, the estimation chapter of the Props Guide.*

**estimate\_acetone**

tutorials/props/estimate/estimate\_acetone

**estimate\_acetone**  
 estimateComponent

**What it does:** CREATE a component from its groups: Joback estimate of acetone (2 CH<sub>3</sub> + 1 ketone), validated against NIST — see the estimate built up from groups

**The story:** The FIRST stage of the props lifecycle (CREATE → curate → wire → simulate): build a pure-component property set from its molecular GROUPS by Joback group contribution, and SEE each group's contribution add up.

Acetone CH<sub>3</sub>-CO-CH<sub>3</sub> = 2 x CH<sub>3</sub> + 1 x ketone (>C=O).

The reference block validates the estimate against NIST: Joback lands within ~2% on T<sub>c</sub>/P<sub>c</sub>/V<sub>c</sub> and ~0.3% on dH<sub>f</sub> – and you SEE the (small) error, the whole point of "estimation, then trust". omega comes from (T<sub>b</sub>,T<sub>c</sub>,P<sub>c</sub>) by Lee-Kesler, not a group sum.

**What to expect (golden-locked):**

| kind | unit/stream      | key            | expected |
|------|------------------|----------------|----------|
| diag | estimate_acetone | Cp298          | 74.97    |
| diag | estimate_acetone | Hvap_kJmol     | 27.53    |
| diag | estimate_acetone | MW             | 58.08    |
| diag | estimate_acetone | Pc_bar         | 48.02    |
| diag | estimate_acetone | Psat_298_bar   | 0.39     |
| diag | estimate_acetone | Psat_Tb_bar    | 0.98     |
| diag | estimate_acetone | Tb_K           | 322.11   |
| diag | estimate_acetone | Tc_K           | 500.56   |
| diag | estimate_acetone | Vc_cm3mol      | 209.50   |
| diag | estimate_acetone | Vliq298_cm3mol | 82.11    |
| diag | estimate_acetone | dGf_kJmol      | -154.54  |
| diag | estimate_acetone | dHf_kJmol      | -217.83  |
| diag | estimate_acetone | omega          | 0.30     |

**estimate\_ethanol\_benzene**

tutorials/props/estimate/estimate\_ethanol\_benzene

**estimate\_ethanol**  
 estimateComponent

**estimate\_benzene**  
 estimateComponent

**What it does:** Joback estimates: ethanol (H-bonding → Tb weak, the honest limit) and benzene (aromatic groups → good) — see where group contribution wins and loses

**The story:** Two Joback estimates, side by side, to SEE the method's spread:

ethanol (1 CH<sub>3</sub> + 1 CH<sub>2</sub> + 1 OH) – a strong H-bonder; Joback's Tb comes out ~14 K LOW (337 vs 351 K). The honest limit, in plain sight. benzene (6 arCH) – aromatic ring; Joback Tc/Tb within ~1.5%. Group contribution at its best.

The lesson is the CONTRAST: an estimate is a starting point whose error you must SEE, not a number to trust blindly.

**What to expect (golden-locked):**

| kind | unit/stream      | key            | expected |
|------|------------------|----------------|----------|
| diag | estimate_ethanol | Cp298          | 64.62    |
| diag | estimate_ethanol | Hvap_kJmol     | 36.73    |
| diag | estimate_ethanol | MW             | 46.07    |
| diag | estimate_ethanol | Pc_bar         | 57.57    |
| diag | estimate_ethanol | Psat_298_bar   | 0.16     |
| diag | estimate_ethanol | Psat_Tb_bar    | 0.97     |
| diag | estimate_ethanol | Tb_K           | 337.54   |
| diag | estimate_ethanol | Tc_K           | 499.41   |
| diag | estimate_ethanol | Vc_cm3mol      | 166.50   |
| diag | estimate_ethanol | Vliq298_cm3mol | 57.70    |
| diag | estimate_ethanol | dGf_kJmol      | -170.86  |
| diag | estimate_ethanol | dHf_kJmol      | -236.84  |
| diag | estimate_ethanol | omega          | 0.57     |
| diag | estimate_benzene | Cp298          | 81.82    |

*...and 12 more reference values in the case's expected.*



**estimate\_petroleum\_cut**

tutorials/props/estimate/estimate\_petroleum\_cut

|                                                    |
|----------------------------------------------------|
| <b>estimate_petroleum_cut</b><br>estimateComponent |
|----------------------------------------------------|

**What it does:** ESTIMATE a petroleum PSEUDO-COMPONENT by name from (Tb, SG) – Riazi-Daubert; see the SCREAMING ESTIMATE provenance and that the lump FLASHES

**The story:** CREATE a PETROLEUM PSEUDO-COMPONENT from the two bulk measurements every crude assay reports: the normal boiling point Tb and the specific gravity SG. The Riazi-Daubert (1987) correlations turn (Tb, SG) into the corresponding-states constants (MW, Tc, Pc, Vc); omega follows by Lee-Kesler, and the ideal-gas Cp by Kesler-Lee (1976).

This is the SCALAR-input sibling of the Joback group path: a petroleum cut is a LUMP of hundreds of species with NO single molecular structure, so it has NO group decomposition – it is anchored on bulk (Tb, SG), not groups. The estimator is a registered ConstantEstimator ('model RiaziDaubert;'), chosen exactly like Joback – the SAME explicit factory.

SEE in the console: the (Tb, SG) anchors and the Watson K factor; the constants built from them, validated against n-decane (a real species at this Tb/SG: Tc ~ exact, omega within ~2 %, Pc/MW within the correlation's stated ~5-7 %); the proposal .dat whose provenance SCREAMS status ESTIMATE / lumped true / "NOT a real species".

Promote with the printed 'mv', then it FLASHES (see the sibling tutorial steady/flash/flash\_pseudoComponent\_petCut, which ships this very lump and closes an energy balance with it).

**What to expect (golden-locked):**

| kind | unit/stream            | key            | expected |
|------|------------------------|----------------|----------|
| diag | estimate_petroleum_cut | Cp298          | 248.06   |
| diag | estimate_petroleum_cut | MW             | 151.51   |
| diag | estimate_petroleum_cut | Pc_bar         | 19.78    |
| diag | estimate_petroleum_cut | Psat_Tb_bar    | 0.97     |
| diag | estimate_petroleum_cut | SG             | 0.73     |
| diag | estimate_petroleum_cut | Tb_K           | 447.30   |
| diag | estimate_petroleum_cut | Tc_K           | 617.80   |
| diag | estimate_petroleum_cut | Vc_cm3mol      | 627.46   |
| diag | estimate_petroleum_cut | Vliq298_cm3mol | 203.43   |
| diag | estimate_petroleum_cut | omega          | 0.48     |
| diag | estimate_petroleum_cut | Psat_298_bar   | 0.00     |

**userEstimate01\_linear\_tb**

tutorials/props/estimate/userEstimate01\_linear\_tb

tb\_acetone  
linearTbEstimate

**What it does:** User-code property estimation: a student's OWN linear group-contribution Tb estimator (linearTbEstimate), compiled from code/ and linked into choupProps.

**The story:** A student's OWN property-estimation op (linearTbEstimate), defined in code/ and linked into choupProps by bin/buildCode. Tb = base + sum(count\*incr). Here: 198.2 + 2\*23.58 + 1\*76.75 = 322.11 K (the Joback CH3,CH3,ketone sum). —

**What to expect (golden-locked):**

| kind | unit/stream | key  | expected |
|------|-------------|------|----------|
| diag | tb_acetone  | Tb_K | 322.11   |

**vanKrevelen01\_polystyrene\_density**

tutorials/props/estimate/vanKrevelen01\_polystyrene\_density

**vanKrevelen01\_polystyrene\_density**  
 estimateComponent

**What it does:** ESTIMATE the density of polystyrene by Van Krevelen group contribution on the repeat unit – glass-box additive sum, Vw from Bondi 1964 (licence-clean via UNIFAC R)

**The story:** ESTIMATE a POLYMER's repeat-unit DENSITY by group contribution – the polymer sibling of the Joback small-molecule path. Polystyrene's repeat unit  $-\text{[CH}_2\text{-CH(C}_6\text{H}_5\text{)]-}$  is decomposed into groups; each adds its molar mass and its Bondi van der Waals volume Vw; the sums give

$M0 = \sum n_i MW_i$  repeat-unit molar mass  $Vw = \sum n_i Vw_i$  van der Waals volume (Bondi 1964)  $V = k * Vw$  molar volume (k = packing factor)  $\rho = M0 / V$  density

GLASS-BOX: the console prints every group's contribution – you can redo the whole estimate by hand. M0 is an exact mass-balance fact (104.15 g/mol); rho carries the packing-factor uncertainty (the teaching point below).

LICENCE (the honest part): the Vw group values are Bondi, J. Phys. Chem. 68 (1964) 441 – the PRIMARY per value – derived from the UNIFAC R\_k volume parameters Choupo already ships ( $Vw = R_k \times 15.17$ ), NOT copied from the Van Krevelen book table (copyright). See data/standards/parameters/vanKrevelen.dat. Slice 1 ships DENSITY only; Tg / solubility-parameter / Tm are DEFERRED (their open parameters are a separate, correctly-attributed thing).

THE TEACHING POINT – the packing factor k: rho depends on  $k = V/Vw$ , the fraction of space the chains do NOT pack into. Default k = 1.60 (amorphous/glassy); crystalline packs denser (~1.43). k is YOURS: change 'polymer { packing k; }' and watch rho move. Atactic PS is amorphous (rho ~ 1.05) – the estimate lands there; a crystalline polymer would need the smaller k. That k is visible (never hidden) is the lesson.

M0 = 104.15 also feeds the step-growth reactor KPIs ( $Mn = M0/(1-p)$ ) – the same repeat-unit mass, two places.

**What to expect (golden-locked):**

| kind | unit/stream                       | key            | expected |
|------|-----------------------------------|----------------|----------|
| diag | vanKrevelen01_polystyrene_density | M0_g_per_mol   | 104      |
| diag | vanKrevelen01_polystyrene_density | V_cm3_per_mol  | 101      |
| diag | vanKrevelen01_polystyrene_density | Vw_cm3_per_mol | 62.9     |
| diag | vanKrevelen01_polystyrene_density | density_g_cm3  | 1.04     |
| diag | vanKrevelen01_polystyrene_density | packing_k      | 1.6      |

**yang2020\_Tg\_polystyrene**

tutorials/props/estimate/yang2020\_Tg\_polystyrene

|                                                     |
|-----------------------------------------------------|
| <b>yang2020_Tg_polystyrene</b><br>estimateComponent |
|-----------------------------------------------------|

**What it does:** ESTIMATE the glass-transition Tg of polystyrene by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH<sub>2</sub>-CH(phenyl)- gives Tg(inf)=444 K, an HONEST +19% over-estimate of the measured ~373 K (a teaching point about group-contribution limits)

**The story:** Yang-2020 Tg estimate + an HONEST limitation

ESTIMATE the GLASS-TRANSITION temperature Tg of polystyrene by the YANG 2020 modified group-contribution scheme. The repeat unit -CH<sub>2</sub>-CH(phenyl)- decomposes into two Yang main-chain groups:

CH<sub>2</sub> : MW 14 Yg 4.026 CHphenyl: MW 90 Yg 42.153

$M0 = 14 + 90 = 104 \text{ g/mol}$   $Yg(inf) = 4.026 + 42.153 = 46.179 \text{ (1e3 g.K/mol)}$   $Tg(inf) = 46.179 \times 10^3 / 104 = 444 \text{ K}$

»> THE HONEST TEACHING POINT «< Measured polystyrene Tg ~ 373 K (atactic, high Mw). Yang 2020 gives 444 K – a +19% OVER-ESTIMATE. This is NOT a bug: it is the model telling the truth about itself. The CHphenyl contribution (Yg 42.153) was regressed across the whole training set, and the bulky monosubstituted phenyl pendant is one where an additive main-chain scheme over-predicts the limiting Tg. Box: "all models are wrong, some are useful". The student SEES the deviation (the console prints the per-cent error against the measured anchor) and learns where a group-contribution method earns trust (PVC: 351 vs 354 K, -0.8%) and where it does not (polystyrene: 444 vs 373 K, +19%).

Contrast the flagship 'yang2020\_Tg\_pvc', where Tg(inf)=351 K reproduces the paper's Fig 6 value EXACTLY – that case goldens Tg as validated physics. Here Tg is SHOWN (the machinery is identical) but the case exists to expose the over-estimate, so only M0 (an exact mass-balance fact) is goldened, not Tg.

This is Yang 2020 – NOT bare Van Krevelen. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer.

SOURCE + LICENCE: Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0; values cited per group as "Yang 2020", never "Van Krevelen".

**What to expect (golden-locked):**

| kind | unit/stream             | key            | expected |
|------|-------------------------|----------------|----------|
| diag | yang2020_Tg_polystyrene | M0_g_per_mol   | 104      |
| diag | yang2020_Tg_polystyrene | Tg_K           | 444      |
| diag | yang2020_Tg_polystyrene | YgSum_1e3gKmol | 46.18    |

**yang2020\_Tg\_pvc**

tutorials/props/estimate/yang2020\_Tg\_pvc

|                                             |
|---------------------------------------------|
| <b>yang2020_Tg_pvc</b><br>estimateComponent |
|---------------------------------------------|

**What it does:** ESTIMATE the glass-transition Tg of poly(vinyl chloride) (PVC) by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH<sub>2</sub>-CHCl- gives Tg(inf)=351 K, EXACT vs the paper (Fig 6)

**The story:** FLAGSHIP Yang-2020 glass-transition estimate

ESTIMATE the GLASS-TRANSITION temperature Tg of poly(vinyl chloride) (PVC) by the YANG 2020 modified group-contribution scheme – the polymer sibling of the Joback small-molecule path, for Tg. The repeat unit -CH<sub>2</sub>-CHCl- is decomposed into two Yang main-chain groups; each adds its limiting Tg contribution Yg(inf):

$M0 = \sum n_i MW_i$  repeat-unit molar mass [g/mol]  $Yg(inf) = \sum n_i Yg_i(inf) (+ Nb * Ygb)$  limiting Tg function [1e3 g.K/mol]  $Tg(inf) = Yg(inf) * 1e3 / M0$  Tg at infinite Mw [K]

THE ADDITIVE SUM (shown on the console): CH<sub>2</sub> : MW 14 Yg 4.026 CHCl: MW 48.5 Yg 17.911

$M0 = 14 + 48.5 = 62.5$  g/mol  $Yg(inf) = 4.026 + 17.911 = 21.937$  (1e3 g.K/mol)  $Tg(inf) = 21.937 * 1e3 / 62.5 = 351$  K <- EXACT vs Yang 2020, Fig 6

This is Yang 2020 – NOT bare Van Krevelen. The scheme predicts the INFINITE- Mw limit Tg(inf); the molecular-weight dependence  $Tg(Mn) = Tg(inf) - K/Mn$  is a later slice. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer, so the bracket is just the sum of the two main-chain group contributions.

SOURCE + LICENCE (the honest part): Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0, so the parameter values ARE redistributable. Cited per value as "Yang 2020", never "Van Krevelen".

THE GOLDEN: M0 (an exact mass-balance fact) AND Tg(inf)=351 K are checked – Tg here is VALIDATED PHYSICS (it reproduces the paper's Fig 6 value exactly), not just a mass balance. The measured anchor (PVC Tg ~ 354 K) confirms it.

**What to expect (golden-locked):**

| kind | unit/stream     | key            | expected |
|------|-----------------|----------------|----------|
| diag | yang2020_Tg_pvc | M0_g_per_mol   | 62.5     |
| diag | yang2020_Tg_pvc | Tg_K           | 351      |
| diag | yang2020_Tg_pvc | YgSum_1e3gKmol | 21.937   |

**fit**

**cp01\_polynomial\_recovery**

tutorials/props/fit/cp01\_polynomial\_recovery

**cp\_dce\_liquid\_fit**  
 heatCapacityFit

**What it does:** Quadratic Cp fit on data generated from a quadratic – exact recovery, with the Kirchhoff Cp<sub>ig</sub> - Cp<sub>liq</sub> cross-check that ties Cp to dHvap

**The story:** A LINEAR FIT, AND THE PHYSICS THAT CHECKS IT.

‘heatCapacityFit’ regresses  $C_p(T) = A_0 + A_1 T + A_2 T^2$  by linear least squares – normal equations, no iteration. Fed data generated FROM a quadratic, it must return that quadratic, so the expected file pins the coefficients rather than a tolerance band.

The interesting number is not the fit. It is KIRCHHOFF:

$$d(dHvap)/dT = C_{p\_ig} - C_{p\_liq}$$

The op takes the FITTED liquid Cp and the component’s CATALOGUE ideal-gas Cp and reports that difference at the mid temperature. It should be NEGATIVE – the enthalpy of vaporisation falls as T rises, to zero at the critical point – and its size is the slope of dHvap(T). That is the link tying Cp to dHvap to the Antoine fit next door (psat01\_antoine\_recovery, which reports dHvap from B): three ops, one consistency network, and a student can check one against the other by hand.

Why the case exists: ‘heatCapacityFit’ had NO case and no gate. Writing the first one for its sibling vaporPressureFit segfaulted choupProps twice and found a thermoPhysPropDict being silently ignored.

**What to expect (golden-locked):**

| kind | unit/stream       | key                | expected   |
|------|-------------------|--------------------|------------|
| diag | cp_dce_liquid_fit | A0                 | 194        |
| diag | cp_dce_liquid_fit | A1                 | -0.618063  |
| diag | cp_dce_liquid_fit | A2                 | 0.00170318 |
| diag | cp_dce_liquid_fit | Cp_at_Tmid         | 166.076    |
| diag | cp_dce_liquid_fit | Cp_ig_at_Tmid      | 84.2326    |
| diag | cp_dce_liquid_fit | R2                 | 1          |
| diag | cp_dce_liquid_fit | dHvap_dT_kirchhoff | -81.8435   |
| diag | cp_dce_liquid_fit | degree             | 2          |
| diag | cp_dce_liquid_fit | n_points           | 21         |

**psat01\_antoine\_recovery**

tutorials/props/fit/psat01\_antoine\_recovery

**psat\_water\_fit**  
 vaporPressureFit

**What it does:** Antoine fit on data generated from Choupo's own water correlation: the coefficients must come back exactly, so any deviation is the op's arithmetic

**The story:** FITTING ANTOINE, AND KNOWING THE ANSWER BEFOREHAND.

'vaporPressureFit' regresses  $\log_{10}(\text{Psat}[\text{bar}]) = A - B/(T + C)$  on measured  $\text{Psat}(T)$ . The fit is linear in (A, B) at fixed C, so the op golden-sections C and solves (A, B) by least squares at each one – derivative-free, and worth reading in the source.

This case feeds it data generated FROM THE SAME EQUATION, by evaluating Choupo's own curated water correlation at twenty temperatures. That is deliberate. Against laboratory data a fit can only be judged to the scatter of the measurements; against data the equation itself produced, the fit must RETURN THE COEFFICIENTS IT WAS GIVEN – so any deviation is the op's arithmetic and nothing else, and the expected file can pin it to the digit.

Why the case exists: 'vaporPressureFit' had NO case and no gate. It could have been broken for a year and the suite would have stayed green; the first to find out would be a student fitting their own data. The psychrometric chart was in exactly that position last week, and writing its first test found it reporting nothing.

What to read in the output: A, B, C – the recovered coefficients, against 5.40221 / 1838.675 / -31.737; R2 – 1 to floating point, because the data has no scatter; dHvap – the Clausius-Clapeyron reading of B WITH the  $(T/(T+C))^2$  correction, at the highest T. Compare it to water's tabulated enthalpy of vaporisation: this is the physical cross-check on a fit that is otherwise just a straight line through transformed numbers.

**What to expect (golden-locked):**

| kind | unit/stream    | key              | expected |
|------|----------------|------------------|----------|
| diag | psat_water_fit | A                | 5.40221  |
| diag | psat_water_fit | B                | 1838.68  |
| diag | psat_water_fit | C                | -31.737  |
| diag | psat_water_fit | R2               | 1        |
| diag | psat_water_fit | dHvap_kJ_per_mol | 42.1162  |
| diag | psat_water_fit | n_points         | 20       |

**gibbs**

**The theory you need.** Equilibrium without a mechanism: minimise total Gibbs energy  $G = \sum_i n_i \mu_i$  over ALL species subject to elemental balances  $\mathbf{A} \mathbf{n} = \mathbf{b}$  (Lagrange multipliers per element – printed as  $\lambda_E$ , the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~100–250 K when they are not (dissociation is endothermic). Above ~1500 K the catalogue's polynomial  $c_p$  fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

*Equilibrium maps.* Sweeping the Gibbs solve over a  $T \times P$  grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes  $T$  and  $P$  in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach`  $\Delta T$  evaluates the *reaction* equilibrium at  $T + \Delta T$  while enthalpy and the energy balance stay at the physical  $T$  – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at  $P = 200$  bar and  $\Delta T = 0$ , find the temperature at which the equilibrium  $\text{NH}_3$  mole fraction crosses 30%; report  $T$  to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in  $T$  at fixed  $P$ . *Full derivation:* Theory Guide, the Gibbs-equilibrium chapter, *Sequilibrium maps*.

**nh3\_equilibrium\_map**

tutorials/props/gibbs/nh3\_equilibrium\_map

**haber\_map**  
 gibbsMap

**What it does:** Ammonia equilibrium map (T x P): why Haber runs at 400-500 C and 150-300 bar

**The story:** THE HABER MAP – the forum-ratified Gibbs equilibrium map (docs/design/gibbs-map-forum-2026-07-02.md) on its canonical pilot:  $\text{N}_2 + 3 \text{H}_2$  at stoichiometric feed, equilibrium  $\text{NH}_3$  mole fraction contoured over 300-700 C x 1-300 bar. The map shows the whole Haber story at a glance: equilibrium loves COLD and PRESSURE (exothermic,  $\Delta_n < 0$ ); the industrial window (400-500 C, 150-300 bar) sits where the catalyst can actually work – the tension IS the lesson.

VALIDATION ANCHORS (golden-locked; verified 2026-07-03): the classic measurements of Larson & Dodge, J. Am. Chem. Soc. 45 (1923) 2918 – 400 C / 10 atm: engine 4.09 % vs L&D 3.85 % (+6 %) 450 C / 10 atm: engine 2.22 % vs L&D 2.11 % (+5 %) The ~5 % offset is consistent with propagating the JANAF formation-data uncertainty of  $\text{NH}_3$  into  $K_p$  ( $\exp(-dG/RT)$  amplifies ~0.2 kJ/mol into ~5 %) – honest agreement for an equilibrium constant spanning decades. IDEAL-GAS VALIDITY: the kernel is ideal-gas, so pins stay at  $\leq 50$  atm, 350-500 C (the forum’s chair: at 300 bar a temperatureApproach would silently absorb the missing fugacity corrections – announced, and the high-P corner of this map is drawn as EQUILIBRIUM SHAPE, not quantitative prediction).

temperatureApproach: 0 here (true equilibrium). Re-run with e.g. ‘temperatureApproach 25;’ to see the contours slide – the empirical closeness-to-equilibrium parameter industry quotes in C (announced loud, the anchor KPI names then carry  $\_dT25$  so the golden breaks visibly).

**What to expect (golden-locked):**

| kind | unit/stream | key                  | expected  |
|------|-------------|----------------------|-----------|
| diag | haber_map   | metric_T400C_P10atm  | 0.0409234 |
| diag | haber_map   | metric_T400C_P50atm  | 0.157785  |
| diag | haber_map   | metric_T450C_P10atm  | 0.0221983 |
| diag | haber_map   | metric_T500C_P100atm | 0.104969  |
| diag | haber_map   | n_cells              | 625       |
| diag | haber_map   | n_unconverged        | 0         |

**heat**

**The theory you need.** Heat exchange couples streams without mixing them:  $Q = U A \Delta T_{lm}$ , effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

*Exchanger DESIGN vs rating.* A shell-and-tube poses two inverse questions: RATE it (geometry given  $\rightarrow$  find  $U$ , area,  $\Delta P$ , duty – **model geometry**;) or DESIGN it (duty given  $\rightarrow$  size the geometry – **model design**;)  $U$  is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty`  $\rightarrow$  `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Sheat-exchanger design.*



**htbench01\_correlations**

tutorials/props/heat/htbench01\_correlations

**bench**  
 heatTransferBench

**What it does:** Heat-transfer correlation bench – each convective correlation (Dittus-Boelter, Gnielinski, Kern) verified against its published Nusselt anchor. Dittus-Boelter and Gnielinski are pinned to Incropera & DeWitt 6th ed. Example 8.4; Kern’s closed  $j_H$  form is self-consistency-checked (its original pin is a chart). AAD tolerance 5% honours textbook rounding + the correlations’ own scatter.

**The story:** the heat-transfer correlation self-check bench.

Mirrors the membrane mass-transfer story (Sherwood) on the heat side (Nusselt): each registered convective correlation RUNS its `verify()` against a published Nu anchor and the deviation is printed + locked.

DittusBoelter  $Nu = 0.023 Re^{0.8} Pr^{0.4}$  (Incropera Ex 8.4,  $Nu=205$ ) Gnielinski  $Nu = (f/8)(Re-1000)Pr/[..]$  (Incropera Ex 8.4 alt,  $Nu=224$ ) Kern (shell)  $Nu = 0.36 Re_s^{0.55} Pr^{1/3}$  (Kern PHT; closed-form check)

The per-correlation deviation diags are the golden KPIs.

**What to expect (golden-locked):**

| kind | unit/stream | key                        | expected    |
|------|-------------|----------------------------|-------------|
| diag | bench       | Nu_choupo_DittusBoelter    | 200.475     |
| diag | bench       | Nu_choupo_Gnielinski       | 223.734     |
| diag | bench       | Nu_choupo_Kern             | 97.5647     |
| diag | bench       | Nu_published_DittusBoelter | 205         |
| diag | bench       | Nu_published_Gnielinski    | 224         |
| diag | bench       | Nu_published_Kern          | 97.5647     |
| diag | bench       | allPass                    | 1           |
| diag | bench       | dev_DittusBoelter          | 0.0220718   |
| diag | bench       | dev_Gnielinski             | 0.00118956  |
| diag | bench       | dev_Kern                   | 0           |
| diag | bench       | dev_NusseltFilm            | 0.0245797   |
| diag | bench       | dev_Rohsenow               | 0.000270621 |
| diag | bench       | dev_ZuberCHF               | 0.000684704 |
| diag | bench       | h_choupo_NusseltFilm       | 6680.26     |

...and 5 more reference values in the case’s *expected*.

**psychro01\_n2\_water**

tutorials/props/heat/psychro01\_n2\_water

**airWater**  
 psychrometricChart

**What it does:** Psychrometric chart for N2/water at 1 atm, 0-60 C: saturation + four RH lines + three adiabatic-saturation lines, from the vapour-pressure model a flash would use

**The story:** THE PSYCHROMETRIC CHART, computed rather than looked up.

Nitrogen + water at 1 atm, 0 → 60 C: the saturation curve, four relative-humidity lines and three adiabatic-saturation (wet-bulb) lines, all from the SAME vapour-pressure model a flash would use. Nothing here is a chart digitised from a handbook – the curve is  $Y_{\text{sat}}(T)$  evaluated point by point, so the student can change the model and watch the chart move.

Why this case exists at all: ‘psychrometricChart’ is reachable from the GUI’s Property Explorer, and until now NOTHING exercised it. It could break and the whole suite would stay green; the first to find out would be a student opening the Explorer. An op a user can reach and a test cannot is a promise with no guard behind it.

What to read in the output (long-format CSV: T\_C, Y, curve): ‘saturation’ –  $Y_{\text{sat}} = 0.622 * P_{\text{sat}} / (P - P_{\text{sat}})$ , the chart’s ceiling; ‘RH=...’ – the same expression at a fraction of  $P_{\text{sat}}$ , so the lines fan out and steepen with T exactly as the printed chart does; ‘wetBulb=...’ – the adiabatic-saturation lines, which are NOT parallel to the RH lines: they carry the latent heat of the water evaporated, so their slope is set by  $H_{\text{vap}}/c_p$  of the moist air, not by geometry.

**What to expect (golden-locked):**

| kind | unit/stream | key                | expected   |
|------|-------------|--------------------|------------|
| diag | airWater    | Y_adiabatic_last   | 0.0150791  |
| diag | airWater    | Y_sat_first        | 0.00390022 |
| diag | airWater    | Y_sat_last         | 0.158606   |
| diag | airWater    | nRhCurves          | 4          |
| diag | airWater    | nSaturationPoints  | 61         |
| diag | airWater    | Lewis_last         | 0.816578   |
| diag | airWater    | nAdiabaticCurves   | 3          |
| diag | airWater    | nTrueWetBulbCurves | 3          |

**psychro02\_air\_water**

tutorials/props/heat/psychro02\_air\_water

**airWater**  
 psychrometricChart

**What it does:** Psychrometric chart with dry AIR as the carrier (the textbook object): saturation, RH, adiabatic-saturation and true wet-bulb families at 101.325 kPa, Lewis  $\sim 0.851$  – the case-local air component deliberately shadows the catalogue mixture alias

**The story:** THE TEXTBOOK CHART: dry AIR as the carrier, not a stand-in. Twin of psychro01\_n2\_water with the REAL carrier – the lumped pseudo-component air (MW 28.96, ChemSep v8.3), which this case declares CASE-LOCALLY in constant/components/air.dat. That declaration is load-bearing, not a convenience: the catalogue also holds mixtures/air.dat (an expansion table,  $\text{air} \rightarrow \text{N}_2 + \text{O}_2 + \text{Ar}$ ), and the settled precedence – the case always overrides – is what makes the token load as the COMPONENT here; the run announces the shadowed expansion. Until 2026-08-31 that tie went to the mixture unconditionally and this exact case refused with "carrier 'air' is not a component".

What the twin buys over psychro01:  $Y_{\text{sat}}(60\text{ C}) = 0.1534$  against the  $\sim 0.152$  a printed air chart reads (N2 gives 0.1586 – the 3.4 % MW error the registry note predicted), and Lewis = 0.851, the handbook air/water value the  $\text{Le}^{(2/3)}$  wet-bulb family exists to teach.

**What to expect (golden-locked):**

| kind | unit/stream | key                | expected   |
|------|-------------|--------------------|------------|
| diag | airWater    | Lewis_last         | 0.850543   |
| diag | airWater    | Y_adiabatic_last   | 0.0146521  |
| diag | airWater    | Y_sat_first        | 0.00377268 |
| diag | airWater    | Y_sat_last         | 0.15342    |
| diag | airWater    | nAdiabaticCurves   | 3          |
| diag | airWater    | nRhCurves          | 4          |
| diag | airWater    | nSaturationPoints  | 61         |
| diag | airWater    | nTrueWetBulbCurves | 3          |

**hydraulics**

**The theory you need.** Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves' intersection.

**moody01\_friction\_correlations**

tutorials/props/hydraulics/moody01\_friction\_correlations

moody  
frictionBench

**What it does:** The Moody chart's four correlations, each verified against its own published anchor and then all asked the same question – so the disagreement is visible instead of hidden behind a default

**The story:** moody01 – FOUR CORRELATIONS, ONE QUESTION, FOUR ANSWERS.

The Darcy friction factor is the most-used correlated quantity in chemical engineering, and every simulator computes it. What none of them shows is that the answer **DEPENDS ON WHICH CORRELATION YOU PICKED**.

Choupo has computed Blasius, Colebrook-White, Haaland and Churchill since the pipe unit was written – correctly, with their citations in comments, as free functions nobody could name or reach. This case is what happens when they become objects: each states the window its own authors gave it, each carries its primary citation, each reproduces a published anchor on every run, and all four can be asked the same question at once.

**WHAT THE RUN DOES, IN TWO PARTS.**

(1) **VERIFY.** Each correlation reproduces its own anchor, and the anchors are deliberately of **DIFFERENT KINDS**, because what can be checked differs:

Blasius against its own closed form. This checks the **ARITHMETIC** and nothing else, and the bench says so rather than letting a 0.000 % deviation look like a validation. Colebrook against the fully-rough von Karman limit, where the implicit equation has a closed form. This checks that the **FIXED-POINT ITERATION** reaches the answer it must. Haaland against **COLEBROOK** at the same point – because Haaland's own published claim is "I reproduce that equation within about 2 %". Anchoring it to anything else would be checking a claim nobody made. Churchill against the **EXACT** laminar law  $f = 64/Re$ , which its single expression contains as its own low- $Re$  limit. This is the strongest of the four: a mis-typed exponent fails here while the turbulent branch still looks plausible.

(2) **COMPARE.** All four are evaluated at  $Re = 1e5$ ,  $eps/D = 1e-3$  – an ordinary turbulent water line in a commercial steel pipe – and the spread is published as a number.

**WHAT TO READ IN THE OUTPUT.** Blasius is the one to watch: it carries **NO** roughness term at all, so at  $eps/D = 1e-3$  it answers as though the wall were smooth, and it is simply wrong. It says so on its own line. That is the whole lesson of a validity window – not that the correlation is bad, but that it was asked a question it was never fitted for, and only the window can tell you.

**WHAT THIS CASE DOES NOT DO.** It does not say which correlation is right. That depends on the pipe – its roughness, whether the flow is developed, whether the duct is even circular – and a bench that ranked them would hide the choice the engineer has to make, which is exactly what a default does today in every tool that has one.

**What to expect (golden-locked):**

| kind | unit/stream | key                 | expected    |
|------|-------------|---------------------|-------------|
| diag | moody       | allPass             | 1           |
| diag | moody       | dev_Blasius         | 0           |
| diag | moody       | dev_Churchill       | 1.73472e-16 |
| diag | moody       | dev_Colebrook       | 1.61329e-05 |
| diag | moody       | dev_Haaland         | 0.0134393   |
| diag | moody       | f_Blasius           | 0.01777     |
| diag | moody       | f_Churchill         | 0.0223432   |
| diag | moody       | f_Colebrook         | 0.0221745   |
| diag | moody       | f_Haaland           | 0.0219662   |
| diag | moody       | f_max               | 0.0223432   |
| diag | moody       | f_min               | 0.01777     |
| diag | moody       | n_outside_window    | 1           |
| diag | moody       | spread_inWindow_pct | 1.71637     |
| diag | moody       | spread_pct          | 25.7358     |

**kinetics**

**rate01\_arrhenius\_recovery**

tutorials/props/kinetics/rate01\_arrhenius\_recovery

|                                     |
|-------------------------------------|
| <b>rate_arrhenius</b><br>kinetics1D |
|-------------------------------------|

**What it does:** Second-order decay at three temperatures, generated from  $k_0 = 4e6$ ,  $E_a = 60$  kJ/mol: the Arrhenius line must give both back

**The story:** THE ACTIVATION ENERGY COMES OUT OF A SLOPE.

‘kinetics1D’ is a PROPERTY evaluation, not a reactor: it uses the integrated rate law in closed form, so a student can overlay competing orders on the same lab data and SEE which one is straight. Given isotherms at more than one temperature it goes further – fits  $k$  on each one, then the line

$$\ln k = \ln k_0 - (E_a/R) (1/T)$$

and  $E_a$  falls out of the SLOPE. That is the whole of Arrhenius, and it is worth watching happen on data whose answer is known.

So the data here is generated from  $k_0 = 4.0e6$  L/(mol s),  $E_a = 60$  kJ/mol, second order, at 40/60/80 C. The fit must return those two numbers; the expected file pins them, and  $R^2 = 1$  because the points carry no scatter.

What to read: the per-isotherm  $k$  values ( $3.93e-4$ ,  $1.57e-3$ ,  $5.34e-3$ ), which span a factor of 13.6 over 40 K – the reason a rate constant is never quoted without its temperature.

Why the case exists: ‘kinetics1D’ had NO case and no gate. Its sibling vaporPressureFit was in the same position, and writing the first case for THAT one segfaulted choupops twice.

**What to expect (golden-locked):**

| kind | unit/stream    | key           | expected |
|------|----------------|---------------|----------|
| diag | rate_arrhenius | Ea            | 60000    |
| diag | rate_arrhenius | Ea_kJ_per_mol | 60       |
| diag | rate_arrhenius | R2            | 1        |
| diag | rate_arrhenius | arrhenius     | 1        |
| diag | rate_arrhenius | fitted        | 1        |
| diag | rate_arrhenius | k0            | 4e+06    |
| diag | rate_arrhenius | n_temps       | 3        |
| diag | rate_arrhenius | order         | 2        |

**molecular**

**The theory you need.** Molecular activity models on display: NRTL/Wilson/UNIQUAC correlate binary  $\gamma$ ’s from pair parameters; UNIFAC PREDICTS them from group counts. Expect UNIFAC within a few percent of fitted models for families it was trained on – and to be honestly worse outside them. *Full derivation: Theory Guide, the activity-model chapter.*

cosmoSAC01\_water\_ethanol

tutorials/props/molecular/cosmoSAC01\_water\_ethanol

**pure\_water**  
propertyScan1D

**bin\_wet**  
propertyScan1D

**ternary**  
propertyScan1D

**water\_hexane**  
propertyScan1D

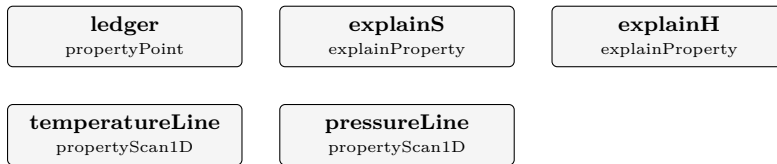
**What it does:** COSMO-SAC 2002 (NIST benchmark, VT-2005 profiles): pure→lngamma=0, binary/ternary multicomponent gammas, no NaN.

**What to expect (golden-locked):**

| kind | unit/stream  | key           | expected |
|------|--------------|---------------|----------|
| diag | pure_water   | n_failures    | 0        |
| diag | pure_water   | n_points      | 2        |
| diag | bin_wet      | n_failures    | 0        |
| diag | bin_wet      | n_points      | 2        |
| diag | ternary      | n_failures    | 0        |
| diag | ternary      | n_points      | 2        |
| diag | water_hexane | n_failures    | 0        |
| diag | water_hexane | n_points      | 2        |
| diag | pure_water   | n_emptyCurves | 0        |
| diag | bin_wet      | n_emptyCurves | 0        |
| diag | ternary      | n_emptyCurves | 0        |
| diag | water_hexane | n_emptyCurves | 0        |

**entropy01\_air\_ledger**

tutorials/props/molecular/entropy01\_air\_ledger



**What it does:** The entropy ledger of 79/21 N<sub>2</sub>/O<sub>2</sub> at 400 K, 2 bar on SRK: every line of s(T,P,y) published separately, so a reader can rebuild the total by hand

**The story:** THE ENTROPY LEDGER, published line by line – the witness behind the "What is entropy?" EduTool page.

The engine's mixture entropy is assembled in ThermoPackage::S<sub>ig</sub> (src/thermo/ThermoPackage.cpp) as  $S_{ig}(T,P,y) = \sum_i y_i * s_{ig\_i}(T)$  (pure-state line) -  $R * \sum_i y_i * \ln y_i$  (mixing line) -  $R * \ln(P / 1 \text{ bar})$  (pressure line)

with each  $s_{ig\_i}(T) = s_{298\_i} + \text{integral } Cp\_i/T \text{ d}T$  off the component's own .dat (Component::s<sub>formation</sub>), and  $S_{real} = S_{ig} + S_{residual}(\text{SRK})$  (the model line, Sandler eq. 6.4-31). The 'ledger' op below publishes the per-component  $s_{ig\_i}$  AND the assembled  $S_{ig} / S_R / S_{real}$  at one state, so a reader – or the EduTool page in the browser – can rebuild the total from the lines and check it against the engine's own number. The two scans draw the temperature line (integral  $Cp/T$  at fixed P) and the pressure line ( $-R \ln P$  at fixed T, with the residual gap growing).

Nothing here is a validation against measurement: every number is the engine's own answer, decomposed. The value of the case is that the decomposition CANNOT drift from the simulator, because it is published by the same call that computes it.

**What to expect (golden-locked):**

| kind | unit/stream | key                | expected    |
|------|-------------|--------------------|-------------|
| diag | ledger      | Cp <sub>ig</sub>   | 29.4637     |
| diag | ledger      | H <sub>R</sub>     | -5.4858     |
| diag | ledger      | H <sub>ig</sub>    | 2986.1689   |
| diag | ledger      | H <sub>real</sub>  | 2980.6831   |
| diag | ledger      | P                  | 200000.0000 |
| diag | ledger      | S <sub>R</sub>     | -0.0192     |
| diag | ledger      | S <sub>ig</sub>    | 201.5778    |
| diag | ledger      | S <sub>real</sub>  | 201.5586    |
| diag | ledger      | T                  | 400.0000    |
| diag | ledger      | Z                  | 1.0007      |
| diag | ledger      | gamma              | 1.3931      |
| diag | ledger      | s <sub>ig_N2</sub> | 200.1887    |
| diag | ledger      | s <sub>ig_O2</sub> | 213.8979    |
| diag | ledger      | v <sub>molar</sub> | 0.0166      |

... and 36 more reference values in the case's *expected*.



**envelope01\_natural\_gas**  
tutorials/props/molecular/envelope01\_natural\_gas

envelope

phaseEnvelope

**What it does:** P-T phase envelope of a lean natural gas: the dew branch over its cricondenthem, and a bubble branch that never starts

**What to expect (golden-locked):**

| kind | unit/stream | key                 | expected |
|------|-------------|---------------------|----------|
| diag | envelope    | bubble_points       | 0        |
| diag | envelope    | bubble_stopped_bar  | 1        |
| diag | envelope    | dew_cricondenthem_K | 274.705  |
| diag | envelope    | dew_points          | 40       |

envelope02\_root\_jump

tutorials/props/molecular/envelope02\_root\_jump

envelope  
phaseEnvelope

**What it does:** P-T envelope where the saturation solve jumps root: both branches stop AT the jump and report it as one, and no cricondentherm is claimed

**What to expect (golden-locked):**

| kind | unit/stream | key                | expected |
|------|-------------|--------------------|----------|
| diag | envelope    | bubble_points      | 17       |
| diag | envelope    | bubble_stopped_bar | 18       |
| diag | envelope    | dew_points         | 25       |
| diag | envelope    | dew_stopped_bar    | 26       |

**eval\_nrtl\_catalogue**

tutorials/props/molecular/eval\_nrtl\_catalogue

|                                                |
|------------------------------------------------|
| <b>catalogue_as_assembled</b><br>fitParameters |
|------------------------------------------------|

**What it does:** EVALUATE mode: score the CATALOGUE NRTL pair against your own data – the scale before any fit

**The story:** ‘mode evaluate’: the SCALE, not the fit.

ONE operation, one question: how good is the ethanol-water NRTL pair against this 1-atm bubble-T dataset? Nothing is adjusted, nothing is written. The thermoPhysPropDict declares NO inline pairs, so what gets scored is a REAL provenance-bearing pair resolved through the search path (from this case’s own vendored constant/parameters/ copy, identical parameters and full primary citation) – not numbers typed inline into the dict.

Read the verdict (rms/aad in kelvin) and THEN decide whether a refit is worth it: compare with fitNRTL01’s fitted rms (0.31 K) on this same data. Failures are honest: a non-converged point becomes status=nonconverged in the parity CSV and never enters the statistics; no chi2\_reduced is printed – without sigmas and degrees of freedom from THIS act, the name would lie.

The pinned-what-if grammar and the leakage sentinel live in the sibling case eval\_nrtl\_pinned (a pinned path must EXIST in the dict, so what-ifs run on an INLINE scaffold and say so).

**What to expect (golden-locked):**

| kind | unit/stream            | key                | expected |
|------|------------------------|--------------------|----------|
| diag | catalogue_as_assembled | aad                | 2.0400   |
| diag | catalogue_as_assembled | chi2               | 60.1945  |
| diag | catalogue_as_assembled | max_abs_resid      | 3.4379   |
| diag | catalogue_as_assembled | max_resid_x1       | 0.5000   |
| diag | catalogue_as_assembled | mode_evaluate      | 1.0000   |
| diag | catalogue_as_assembled | n_data             | 11.0000  |
| diag | catalogue_as_assembled | n_feasible         | 11.0000  |
| diag | catalogue_as_assembled | partial            | 0.0000   |
| diag | catalogue_as_assembled | penalised_points   | 0.0000   |
| diag | catalogue_as_assembled | rms                | 2.3393   |
| diag | catalogue_as_assembled | sse                | 60.1945  |
| diag | catalogue_as_assembled | validity_declared  | 1.0000   |
| diag | catalogue_as_assembled | n_outside_validity | 0.0000   |

**eval\_nrtl\_pinned**

tutorials/props/molecular/eval\_nrtl\_pinned

scaffold\_as\_is

fitParameters

rough\_paper\_values

fitParameters

scaffold\_again

fitParameters

**What it does:** EVALUATE a pinned what-if: a pinned run scores the package but mutates NOTHING (the leakage sentinel)

**The story:** the pinned WHAT-IF, and the LEAKAGE SENTINEL.

TRANSITIONAL SCAFFOLD: pinning by dict path requires an inline pair; the permanent surface (ParameterOverrides on the RESOLVED parameter registry, design forum #67/#69) replaces this case when it lands.

Three evaluations of the same dataset: 1. `scaffold_as_is` – the inline pair, untouched (the baseline). 2. `rough_paper_values` – the SAME dict with four coefficients PINNED to deliberately rough values. The pin operates on a DEEP COPY (Dictionary::deepCopy, forum #63-7); nothing on disk or in memory survives the op. 3. `scaffold_again` – repeats 1 AFTER the pinned run. Its verdict must be IDENTICAL to 1's: the golden pins both, so the working-tree isolation is guarded by the corpus forever. Before deepCopy existed, the old alias bug would make THIS operation report the rough values' score.

**What to expect (golden-locked):**

| kind | unit/stream        | key              | expected |
|------|--------------------|------------------|----------|
| diag | scaffold_as_is     | aad              | 2.0400   |
| diag | scaffold_as_is     | chi2             | 60.1945  |
| diag | scaffold_as_is     | max_abs_resid    | 3.4379   |
| diag | scaffold_as_is     | max_resid_x1     | 0.5000   |
| diag | scaffold_as_is     | mode_evaluate    | 1.0000   |
| diag | scaffold_as_is     | n_data           | 11.0000  |
| diag | scaffold_as_is     | n_feasible       | 11.0000  |
| diag | scaffold_as_is     | partial          | 0.0000   |
| diag | scaffold_as_is     | penalised_points | 0.0000   |
| diag | scaffold_as_is     | rms              | 2.3393   |
| diag | scaffold_as_is     | sse              | 60.1945  |
| diag | rough_paper_values | aad              | 1.5942   |
| diag | rough_paper_values | chi2             | 61.7621  |
| diag | rough_paper_values | max_abs_resid    | 5.5866   |

...and 21 more reference values in the case's *expected*.

**exergy01\_air\_dead\_state**

tutorials/props/molecular/exergy01\_air\_dead\_state

**b\_state**  
 exergy

**b\_dead**  
 exergy

**What it does:** Physical exergy of 79/21 N<sub>2</sub>/O<sub>2</sub> at 400 K, 2 bar against a DECLARED 298.15 K / 1 bar dead state –  $b = (h-h_0) - T_0(s-s_0)$  from the engine's own H\_real/S\_real, with the structural zero at the dead state itself as the second row

**The story:** PHYSICAL EXERGY, the number the entropy ledger buys – the witness behind the exergy work that follows the "What is entropy?" page.

$b = (h - h_0) - T_0(s - s_0)$ , every term the engine's own H\_real/S\_real at the SAME composition, so the enthalpy datum, the s\_298 anchors, the mixing term and the reference pressure ALL cancel: the exergy is datum-independent. The dead state is DECLARED (an exergy is a statement about a state AND an environment; the engine refuses to assume one).

Two rows on purpose: the same state the entropy01 ledger decomposes, and the dead state itself – whose exergy is 0 identically, by construction, which the golden pins as the structural zero.

Nothing here is validated against measurement: every number is the engine's own answer, decomposed. The published dh and T<sub>0</sub>ds legs re-add to b on the reader's desk.

**What to expect (golden-locked):**

| kind | unit/stream | key        | expected |
|------|-------------|------------|----------|
| diag | b_state     | P          | 200000   |
| diag | b_state     | P0         | 100000   |
| diag | b_state     | T          | 400      |
| diag | b_state     | T0         | 298.15   |
| diag | b_state     | T0ds       | 850.601  |
| diag | b_state     | b_physical | 2136.71  |
| diag | b_state     | dh         | 2987.32  |
| diag | b_state     | h          | 2980.68  |
| diag | b_state     | h0         | -6.63211 |
| diag | b_state     | s          | 201.559  |
| diag | b_state     | s0         | 198.706  |
| diag | b_dead      | P          | 100000   |
| diag | b_dead      | P0         | 100000   |
| diag | b_dead      | T          | 298.15   |

*...and 8 more reference values in the case's **expected**.*

flash01\_operating\_line

tutorials/props/molecular/flash01\_operating\_line

txy  
propertyScan1D

**What it does:** The equilibrium curve a flash operating line is drawn on

**The story:** THE CURVE, AND ONLY THE CURVE.

A binary flash is fixed by two of {T, P, V/F} – Duhem – so at a frozen pressure the equilibrium curve  $y^*(x)$  is frozen too. Everything the flash tool draws on top of it (the operating line, its intersection, the lever rule) is GRAPHICAL construction on this one table, which is why turning a knob there redraws at once instead of re-solving.

So this case publishes exactly one thing: the curve. It does NOT flash anything. The flash the student watches is the intersection, performed on these rows – and the corpus already carries the ENGINE’s own answer for the same pair (tutorials/steady/flash/flash01\_benzene\_toluene,  $V/F = 0.30398$  at 370 K), which is the number to check the construction against.

51 points is not a rounding of taste: the construction interpolates linearly between rows (no spline – an overshoot near an azeotrope would invent equilibrium the engine never computed), so the sampling IS the resolution of every answer read off it.

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | txy         | n_emptyCurves | 0        |
| diag | txy         | n_failures    | 0        |
| diag | txy         | n_points      | 51       |

**mixrules01\_natural\_gas**

tutorials/props/molecular/mixrules01\_natural\_gas

**mix\_30bar**  
 mixingRules

**mix\_1bar**  
 mixingRules

**What it does:** The van der Waals one-fluid mixing rule, decomposed: SRK  $a_{\text{mix}}/b_{\text{mix}}$  of a 90/5/5 CH<sub>4</sub>/N<sub>2</sub>/CO<sub>2</sub> gas at 250 K, pair by pair – the one curated kij (N<sub>2</sub>-CH<sub>4</sub>, DECHEMA) beside the pairs that run the kij = 0 default, and the same decomposition at 1 bar showing  $a_{\text{mix}}$  is a (T, y) fact while only A, B, Z carry the pressure

**The story:** THE MIXING RULE, DECOMPOSED – the multicomponent lesson. A cubic prices a MIXTURE through two numbers, and the one-fluid rule says how n pure components and  $n(n-1)/2$  binary corrections build them:

$$a_{\text{mix}}(T) = \sum_i \sum_j y_i y_j (1 - kij) \sqrt{a_i a_j} \quad b_{\text{mix}} = \sum_i y_i b_i$$

Every term is on the table: the six pair terms of this ternary re-add to the  $a_{\text{mix}}$  the flash would run (the op REFUSES if they do not), the curated N<sub>2</sub>-CH<sub>4</sub> kij sits beside the two CO<sub>2</sub> pairs running the announced kij = 0 default, and the pair table is where a student sees that a ternary needs THREE binary decisions – there is no ternary parameter.

Two states on purpose, same composition and temperature:  $a_{\text{mix}}$  and  $b_{\text{mix}}$  are IDENTICAL at 30 bar and 1 bar (the rule sees only T and y), while A, B and Z carry the pressure – at 1 bar the same gas is nearly ideal ( $Z \sim 0.997$ ) and at 30 bar it is not ( $Z \sim 0.93$ ), priced by the SAME two numbers through the cubic.

Nothing here is validated against measurement: every number is the engine's own arithmetic, decomposed. The one cited constant is the N<sub>2</sub>-CH<sub>4</sub> kij record (Knapp et al., DECHEMA VI, 1982).

**What to expect (golden-locked):**

| kind | unit/stream | key      | expected     |
|------|-------------|----------|--------------|
| diag | mix_30bar   | A        | 0.140523     |
| diag | mix_30bar   | B        | 0.0428434    |
| diag | mix_30bar   | P_Pa     | 3e+06        |
| diag | mix_30bar   | T_K      | 250          |
| diag | mix_30bar   | Z_vapour | 0.901046     |
| diag | mix_30bar   | a_CH4    | 0.200506     |
| diag | mix_30bar   | a_CO2    | 0.429536     |
| diag | mix_30bar   | a_N2     | 0.0837883    |
| diag | mix_30bar   | a_mix    | 0.202383     |
| diag | mix_30bar   | b_CH4    | 2.98484e-05  |
| diag | mix_30bar   | b_CO2    | 2.96742e-05  |
| diag | mix_30bar   | b_N2     | 2.6754e-05   |
| diag | mix_30bar   | b_mix    | 2.9685e-05   |
| diag | mix_30bar   | dadT_mix | -0.000515994 |

... and 46 more reference values in the case's *expected*.

**nrtl01\_ethanol\_water\_package**  
tutorials/props/molecular/nrtl01\_ethanol\_water\_package

**g\_nrtl**

activityCoefficients

**What it does:** ethanol-water NRTL via the SAME property-package builder (molecular U4)

**What to expect (golden-locked):**

| kind | unit/stream | key           | expected |
|------|-------------|---------------|----------|
| diag | g_nrtl      | gamma_ethanol | 1.26245  |
| diag | g_nrtl      | gamma_water   | 1.4349   |



**pcsaft01\_pure\_nhexane**

tutorials/props/molecular/pcsaft01\_pure\_nhexane

**liquid\_298K\_1bar**  
 propertyPoint

**What it does:** PC-SAFT pure n-hexane: Z + liquid molar volume at 298 K, 1 bar

**The story:** the NON-ASSOCIATING PC-SAFT core, validated on a pure n-alkane. n-hexane's segment parameters (m, sigma, epsilon/k) are the Gross & Sadowski 2001 Table 1 fit; the case evaluates the compressibility factor Z and the liquid molar volume at 298.15 K / 1 bar and locks them as a golden. The engine PROVES the model self-consistent on every run (PCSAFT::verify: a\_res  $\rightarrow$  0 in the ideal limit, and the residual chemical potentials reproduce the bulk residual Gibbs energy to  $\sim 1e-9$ ). Physical cross-check (see README): the PC-SAFT liquid density here is  $\sim 650$  kg/m<sup>3</sup>, within  $\sim 0.8$  % of the  $\sim 655$  kg/m<sup>3</sup> experimental value.

**What to expect (golden-locked):**

| kind | unit/stream      | key         | expected     |
|------|------------------|-------------|--------------|
| diag | liquid_298K_1bar | Cp_ig       | 1.43106e+02  |
| diag | liquid_298K_1bar | H_R         | -3.91460e+02 |
| diag | liquid_298K_1bar | H_ig        | -1.67200e+05 |
| diag | liquid_298K_1bar | H_real      | -1.67591e+05 |
| diag | liquid_298K_1bar | P           | 1.00000e+05  |
| diag | liquid_298K_1bar | S_R         | -3.54309e-01 |
| diag | liquid_298K_1bar | S_ig        | 3.88510e+02  |
| diag | liquid_298K_1bar | S_real      | 3.88156e+02  |
| diag | liquid_298K_1bar | T           | 2.98150e+02  |
| diag | liquid_298K_1bar | Z           | 9.42530e-01  |
| diag | liquid_298K_1bar | Z_liquid    | 5.35107e-03  |
| diag | liquid_298K_1bar | gamma       | 1.06168e+00  |
| diag | liquid_298K_1bar | v_molar     | 2.33649e-02  |
| diag | liquid_298K_1bar | v_molar_liq | 1.32651e-04  |

*...and 1 more reference values in the case's expected.*

**pcsaft02\_states\_crosscheck**

tutorials/props/molecular/pcsaft02\_states\_crosscheck

CH4\_vapour\_pcsaft  
propertyPointCH4\_vapour\_srk  
propertyPointpropane\_liquid  
propertyPointnButane\_liquid  
propertyPoint**What it does:** PC-SAFT across states: CH4 Z vs SRK cross-check + propane/n-butane liquid density vs handbook**The story:** PC-SAFT validated ACROSS STATES and AGAINST AN INDEPENDENT MODEL, before the model is trusted in a flowsheet.

Three probes, all goldened: (1) CH4 vapour Z at 300 K / 50 bar – PC-SAFT vs SRK at the SAME state (the per-op ‘thermo {}’ override): two independently-validated models must agree closely for a simple near-spherical vapour, and the run prints both so the student SEES the agreement (not told to assume it). (2) propane compressed liquid at 298.15 K / 15 bar – rho\_liq vs the saturated-liquid handbook value (0.493 g/mL at 25 degC). (3) n-butane compressed liquid at 298.15 K / 5 bar – rho\_liq vs 0.573 g/mL at 25 degC. Deviations sit at the 1-2 % liquid-density AAD Gross & Sadowski 2001 (Table 2) report for the n-alkane series – these parameters PREDICT the density; it was never refitted here.

**What to expect (golden-locked):**

| kind | unit/stream       | key     | expected     |
|------|-------------------|---------|--------------|
| diag | CH4_vapour_pcsaft | Cp_ig   | 3.56607e+01  |
| diag | CH4_vapour_pcsaft | H_R     | -8.28601e+02 |
| diag | CH4_vapour_pcsaft | H_ig    | -7.44541e+04 |
| diag | CH4_vapour_pcsaft | H_real  | -7.52827e+04 |
| diag | CH4_vapour_pcsaft | P       | 5.00000e+06  |
| diag | CH4_vapour_pcsaft | S_R     | -1.24811e+00 |
| diag | CH4_vapour_pcsaft | S_ig    | 1.53964e+02  |
| diag | CH4_vapour_pcsaft | S_real  | 1.52716e+02  |
| diag | CH4_vapour_pcsaft | T       | 3.00000e+02  |
| diag | CH4_vapour_pcsaft | Z       | 9.12141e-01  |
| diag | CH4_vapour_pcsaft | gamma   | 1.30404e+00  |
| diag | CH4_vapour_pcsaft | v_molar | 4.55038e-04  |
| diag | CH4_vapour_srk    | Cp_ig   | 3.56607e+01  |
| diag | CH4_vapour_srk    | H_R     | -8.31158e+02 |

... and 42 more reference values in the case's *expected*.

**pcsaft03\_association\_pure**

tutorials/props/molecular/pcsaft03\_association\_pure

**water\_liquid\_298K**  
 propertyPoint

**ethanol\_liquid\_298K**  
 propertyPoint

**What it does:** PC-SAFT association (Wertheim TPT1): 2B water + 2B ethanol, pure liquids

**The story:** PC-SAFT WITH ASSOCIATION (Wertheim TPT1; Gross & Sadowski, Ind. Eng. Chem. Res. 41 (2002) 5510) – the hydrogen-bonded liquids the non-associating core cannot represent, as PURE liquids at 298.15 K, 1 bar.

The association parameters live on the component records (2B water, 2B ethanol – both G&S 2002 Table 1, fitted WITH the association term; the paper lists water with TWO sites, and the scheme is PART of the fit: this same set mis-curated as 4C passed the pure-density check by coincidence and collapsed the ethanol/water mixture flash); the case only declares ‘model PCSAFT;’, and the engine announces the roster it found on the records:

[pcsaft] associating: water (2B), ethanol (2B); cross: Wolbach-Sandler

Liquid densities PREDICTED here (golden), vs the CRC Handbook 25 degC values – each deviation stated against what the SET is fitted for:

water  $v_{\text{molar\_liq}} \rightarrow \rho_{\text{liq}} = 921.9 \text{ kg/m}^3$  vs 997.05 (-7.5 % – the PUBLISHED trade-off of the 2-site water fit, which is  $P_{\text{sat}}$ -accurate instead: -0.2 % at 358 K, +4 % at 298 K) ethanol  $v_{\text{molar\_liq}} \rightarrow \rho_{\text{liq}} = 779.8 \text{ kg/m}^3$  vs 789.3 (-1.2 %)

What the student sees at verbosity 3 (the widened validation, ratified 2026-08-02): the construction-time `verify()` now ALSO echoes, per associating component as a pure fluid, the converged site fractions  $X^D / X^A$ , the  $a_{\text{assoc}}$  share, and the CLOSED-FORM-vs-ITERATIVE oracle deviation (for  $n_D = n_A$  the mass-action closure collapses to a quadratic; the damped fixed point must land on its root – observed machine zero).  $X^A(\text{water}) \sim 0.039$ : 96 % of water’s sites are hydrogen-bonded at ambient conditions – the physics the term adds.

The two ops publish the standard vapour-root family ( $Z, v_{\text{molar}}$ ) plus the labelled liquid root ( $v_{\text{molar\_liq}}$ ); at 1 bar both fluids carry a metastable vapour root beside the stable liquid.

**What to expect (golden-locked):**

| kind                                                             | unit/stream         | key     | expected     |
|------------------------------------------------------------------|---------------------|---------|--------------|
| diag                                                             | water_liquid_298K   | Cp_ig   | 3.36596e+01  |
| diag                                                             | water_liquid_298K   | H_R     | -2.18780e+03 |
| diag                                                             | water_liquid_298K   | H_ig    | -2.41826e+05 |
| diag                                                             | water_liquid_298K   | H_real  | -2.44014e+05 |
| diag                                                             | water_liquid_298K   | P       | 1.00000e+05  |
| diag                                                             | water_liquid_298K   | S_R     | -5.61640e+00 |
| diag                                                             | water_liquid_298K   | S_ig    | 1.88834e+02  |
| diag                                                             | water_liquid_298K   | S_real  | 1.83218e+02  |
| diag                                                             | water_liquid_298K   | T       | 2.98150e+02  |
| diag                                                             | water_liquid_298K   | Z       | 9.00945e-01  |
| diag                                                             | water_liquid_298K   | gamma   | 1.32805e+00  |
| diag                                                             | water_liquid_298K   | v_molar | 2.23340e-02  |
| diag                                                             | ethanol_liquid_298K | Cp_ig   | 6.54261e+01  |
| diag                                                             | ethanol_liquid_298K | H_R     | -4.01997e+03 |
| ... and 16 more reference values in the case’s <i>expected</i> . |                     |         |              |

**solubility01\_hildebrand\_ladder**

tutorials/props/molecular/solubility01\_hildebrand\_ladder

**What it does:** Hildebrand: one number for like-dissolves-like, and where it stops*Seven liquids on a ladder of cohesive energy density, from hexane to water.* *$\delta$  is derived, never stored:  $\sqrt{(\Delta H_{\text{vap}}(T) - RT)/V_m}$ , with the latent heat from the Watson correlation the enthalpy legs already use. Every input is already in the component record, so a stored  $\delta$  would be a second home for a fact the record fixes — free to drift from it.**The published column is an `*anchor*`, not an input. It is what ChemSep computed independently, and its only job is to give the derivation something to be wrong against. Worst deviation here is 0.85 %.**That matters because the three ways this arithmetic fails are all silent:**\* the  $RT$  term dropped —  $\delta$  a few per cent high, and a solubility parameter has no intuitive magnitude to make that look wrong; \*  $V_m$  in the wrong unit; \*  $\text{Pa}^{0.5}$  reported where the literature uses  $\text{MPa}^{0.5}$  — a factor of 31.6, and the number still reads as a solubility parameter.**Same device as the  $Z_c$  column in a property data bank: a redundant determination turns a plausible wrong number into a visible disagreement.***What to expect (golden-locked):**

| kind | unit/stream           | key               | expected    |
|------|-----------------------|-------------------|-------------|
| diag | solubilityParameter_0 | delta_acetone     | 19673.2     |
| diag | solubilityParameter_0 | delta_benzene     | 18648.2     |
| diag | solubilityParameter_0 | delta_cyclohexane | 16733.6     |
| diag | solubilityParameter_0 | delta_ethanol     | 26301.9     |
| diag | solubilityParameter_0 | delta_nHexane     | 14859       |
| diag | solubilityParameter_0 | delta_toluene     | 18183.8     |
| diag | solubilityParameter_0 | delta_water       | 48266.4     |
| diag | solubilityParameter_0 | dev_acetone       | 0.00287938  |
| diag | solubilityParameter_0 | dev_benzene       | 0.00277184  |
| diag | solubilityParameter_0 | dev_cyclohexane   | 0.000380181 |
| diag | solubilityParameter_0 | dev_ethanol       | 0.00619352  |
| diag | solubilityParameter_0 | dev_nHexane       | 0.000736807 |
| diag | solubilityParameter_0 | dev_toluene       | 0.003629    |
| diag | solubilityParameter_0 | dev_water         | 0.00849075  |

*...and 6 more reference values in the case's `expected`.***reactor**

**The theory you need.** Three idealised steady reactors. Conversion reactor: stoichiometry only,  $n_{\text{out}} = n_{\text{in}} + \nu \xi$ , duty from the ONE enthalpy base  $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$  (formation datum – no separate “heat of reaction” input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate  $r(c_{\text{out}}, T)$ , so  $F(c_{\text{in}} - c_{\text{out}}) = rV$ . PFR: the same rate integrated along the volume,  $dF_i/dV = \nu_i r$ . Arrhenius kinetics  $k = AT^b e^{-E_a/RT}$ , orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential  $T$  sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

**thiele01\_sphere\_first\_order**

tutorials/props/reactor/thiele01\_sphere\_first\_order

**thiele\_CO\_alumina**  
 thielePellet

**What it does:** Single catalyst pellet: intraparticle concentration field, effectiveness factor, and the eta(phi) family, each verified against the first-order isothermal closed form

**The story:** One thielePellet operation.

THE RATE CONSTANT IS DECLARED HERE, and not on the catalyst record, because kinetics are equipment- and reaction-specific: the asset carries the SOLID, never the chemistry run on it. Two reactors loaded with the same catalyst would otherwise carry two copies of one rate constant and drift apart.

$k = 40 \text{ 1/s}$  is a TEACHING VALUE chosen so the generalised modulus lands near 1 – the transition region, where the field has a visible gradient and eta is neither  $\sim 1$  nor  $\sim 1/\phi$ . It is not measured and is not claimed to be.

*Solves the intraparticle boundary-value problem for one porous catalyst pellet, publishes the concentration field, the effectiveness factor and the Thiele modulus, and checks every one of them against the closed form.*

*With  $\xi$  the normalised coordinate from the pellet centre (0) to its external surface (1) and  $u = c/c_s$ :*

*Both moduli are published because both conventions are in circulation and they differ by a factor of three for a sphere. The engine's own announcement in `src/unitOperations/reactor/CatalystPellet.H` uses the generalised one.*

*The equation is discretised with second-order central differences and handed to the existing solver::newtonND (finite-difference Jacobian, LU with partial pivoting) — no new dependency, no new solver, every step readable in `src/unitOperations/reactor/pellet/PelletDiffusion.cpp`.*

*$\xi$  is one column, not two aliases: it is  $r/R$  for a sphere or a cylinder and  $x/L$  for a slab.*

*First-order isothermal is the rare case that has a closed form in all three geometries, so the numerical answer is not merely produced — it is checked:*

**What to expect (golden-locked):**

| kind | unit/stream       | key            | expected    |
|------|-------------------|----------------|-------------|
| diag | thiele_CO_alumina | phi            | 1.0161e+00  |
| diag | thiele_CO_alumina | phi_char       | 3.0483e+00  |
| diag | thiele_CO_alumina | Lambda         | 5.0000e-04  |
| diag | thiele_CO_alumina | dimension      | 1.5000e-03  |
| diag | thiele_CO_alumina | epsilon_p      | 4.5000e-01  |
| diag | thiele_CO_alumina | tau            | 3.0000e+00  |
| diag | thiele_CO_alumina | rateConstant   | 4.0000e+01  |
| diag | thiele_CO_alumina | T              | 5.7315e+02  |
| diag | thiele_CO_alumina | P              | 1.01325e+05 |
| diag | thiele_CO_alumina | D_molecular    | 6.45709e-05 |
| diag | thiele_CO_alumina | D_eff          | 9.68564e-06 |
| diag | thiele_CO_alumina | eta            | 6.65742e-01 |
| diag | thiele_CO_alumina | eta_closedForm | 6.65742e-01 |
| diag | thiele_CO_alumina | eta_flux       | 6.65731e-01 |

*... and 17 more reference values in the case's expected.*

**thiele02\_prater\_temperature**

tutorials/props/reactor/thiele02\_prater\_temperature

**thiele\_CO\_alumina\_prater**  
 thielePellet

**What it does:** Single catalyst pellet: intraparticle concentration field, effectiveness factor, and the eta(phi) family, each verified against the first-order isothermal closed form

**The story:** One thielePellet operation.

THE RATE CONSTANT IS DECLARED HERE, and not on the catalyst record, because kinetics are equipment- and reaction-specific: the asset carries the SOLID, never the chemistry run on it. Two reactors loaded with the same catalyst would otherwise carry two copies of one rate constant and drift apart.

$k = 40 \text{ 1/s}$  is a TEACHING VALUE chosen so the generalised modulus lands near 1 – the transition region, where the field has a visible gradient and eta is neither  $\sim 1$  nor  $\sim 1/\phi$ . It is not measured and is not claimed to be.

*The SAME pellet as thiele01\_sphere\_first\_order, with one block added, so the difference between the two runs is the temperature.*

*Everything else — catalyst, species,  $T$ ,  $P$ , rate constant, diffusion, grid, verification tolerances — is thiele01 unchanged. Run both and diff the output: nothing else moves.*

*The reaction is exothermic, so the pellet is hottest where the reactant is most depleted, which is the centre:*

*The field CSV gains two columns,  $T_{\text{over}} T_s$  and  $T_K$ , one value per node — so the temperature is a FIELD, not a headline number, and the EduTool paints the same discs by it.*

*The species and energy balances inside a pellet carry the same reaction term. Eliminate it between them and, for constant  $D_{\text{eff}}$  and  $k_{\text{eff}}$  and any kinetics at all, what is left is a first integral — Prater's relation:*

*So the temperature field costs no second boundary-value problem. It is algebra over the concentration field the operation already solved.*

**What to expect (golden-locked):**

| kind | unit/stream              | key                 | expected    |
|------|--------------------------|---------------------|-------------|
| diag | thiele_CO_alumina_prater | D_eff               | 9.68564e-06 |
| diag | thiele_CO_alumina_prater | D_molecular         | 6.45709e-05 |
| diag | thiele_CO_alumina_prater | Lambda              | 0.0005      |
| diag | thiele_CO_alumina_prater | P                   | 101325      |
| diag | thiele_CO_alumina_prater | T                   | 573.15      |
| diag | thiele_CO_alumina_prater | T_centre_K          | 588.722     |
| diag | thiele_CO_alumina_prater | c_centre            | 0.289874    |
| diag | thiele_CO_alumina_prater | c_centre_closedForm | 0.289873    |
| diag | thiele_CO_alumina_prater | deltaT_centre_K     | 15.5718     |
| diag | thiele_CO_alumina_prater | dimension           | 0.0015      |
| diag | thiele_CO_alumina_prater | epsilon_p           | 0.45        |
| diag | thiele_CO_alumina_prater | eta                 | 0.665742    |
| diag | thiele_CO_alumina_prater | eta_closedForm      | 0.665742    |
| diag | thiele_CO_alumina_prater | eta_flux            | 0.665731    |

...and 23 more reference values in the case's *expected*.

**scan**

**The theory you need.** Property scans sweep a model across  $T$ ,  $P$  or composition and plot what the thermodynamics DOES:  $P^{\text{sat}}(T)$  curves,  $\gamma_{\pm}(m)$ , VLE envelopes, ternary diagrams, phase boundaries. Nothing is fitted – these ops answer “how does the selected model behave?”, the SEE step of the props loop (CREATE → SEE → TEST → FIT → PROMOTE). Expect model disagreements to be VISIBLE (overlay two methods and look), and every curve exportable as CSV next to the case. *Full derivation: Theory Guide, the Props Guide.*

**binary01\_lle\_water\_nbutanol**

tutorials/props/scan/binary01\_lle\_water\_nbutanol

lle  
propertyScanBinary

**What it does:** Binary LLE  $g_{\text{mix}}$  + common-tangent (water/1-butanol) at 298.15 K via propertyScanBinary, cited Winkelman 2009 UNIQUAC

**The story:** Binary liquid-liquid teaching map – water / 1-butanol at 25 C.

The molar Gibbs energy of mixing  $g_{\text{mix}}(x_1)$  swept across composition PLUS the two coexisting liquid compositions from ONE liquid-liquid IsothermalFlash (PhaseSet::LL). The two binodal points lie on the  $g_{\text{mix}}$  curve and share its common tangent – the student reads the split straight off the diagram.

$\gamma$  comes from the CITED Winkelman et al. (2009) UNIQUAC LLE set (Table 2; structural  $r, q$  from Table 1), so the  $g_{\text{mix}}$  curve goes NON-CONVEX and the flash emits the real water/1-butanol binodal: ~1.9 mol% butanol in the water-rich phase and ~49 mol% butanol in the butanol-rich phase (matching their Figs 1-2 at 298 K). See README.md.

The flash feed is set to  $z(\text{water}) = 0.80$  (20 mol% butanol), INSIDE the asymmetric gap; an equimolar feed would lie outside it (both binodal branches sit below 50 mol% butanol). Run-only smoke (the CSV is a teaching curve, not pinned KPIs).

*Water + 1-butanol (n-butanol) is \*the\* classic partially-miscible binary. At 25 °C it splits into a water-rich liquid (~2 mol % butanol) and a butanol-rich liquid (~50 mol % butanol). A  $g_{\text{mix}}(x_{\text{water}})$  scan should therefore show a clearly non-convex region: two binodal points sharing a common tangent, which a student reads the split straight off.*

*and binaryLle.csv is a  $g_{\text{mix}}(x_1)$  curve that goes non-convex, plus the two binodal points the liquid-liquid flash locates on it:*

*The agreement is essentially exact: the cited UNIQUAC fit reproduces the published mutual solubilities.*

- J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, \* "Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"\*, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013.

- Interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation  $A_{ij}(T) = a + b \cdot T + c \cdot T^2$  (Eq. 10),  $\tau_{ij} = \exp(-A_{ij}/T)$  (Eq. 7): -  $A_{1,3}$  (butanol→water):  $a = 155.31$ ,  $b = 1.0822$ ,  $10^3 \cdot c = -43.711$  -  $A_{3,1}$  (water→butanol):  $a = -579.36$ ,  $b = 2.7517$ ,  $10^3 \cdot c = -6.7700$  - Structural  $r, q$  — their Table 1 (p.72): 1-butanol  $r = 3.9243$ ,  $q = 3.668$ ; water  $r = 0.9200$ ,  $q = 1.400$ .

*The numbers are inline here, fully provenanced above; the record is STAGED CASE-LOCALLY, not curated: it lives at tutorials/steady/flash/marcilla02\_lls\_ternary\_invariant/constant/parameters/UNIQUAC/nButanol-water.dat with reviewStatus interim, and promotion into data/standards/ is the maintainer's batch review. This README used to claim the numbers already lived under data/standards/ – asserting a curation status that had been deliberately withheld, which is a worse error than saying nothing.*

**What to expect (golden-locked):**

| kind | unit/stream | key   | expected |
|------|-------------|-------|----------|
| diag | lle         | T     | 298.15   |
| diag | lle         | split | 1        |

**hxy01\_ethanol\_water\_1atm**

tutorials/props/scan/hxy01\_ethanol\_water\_1atm

**hxy**  
 enthalpyConcentration

**What it does:** Enthalpy-concentration locus (h-x-y) of ethanol/water at 1 atm on the elements datum – the data a Ponchon-Savarit construction is drawn on

**The story:** The enthalpy-concentration locus of ethanol/water at 1 atm.

McCabe-Thiele carries equilibrium and material balance under CONSTANT MOLAR OVERFLOW – one mole condensing releases exactly what one mole vaporising absorbs, whatever the composition. This case publishes the quantity that assumption is about: the saturated-liquid enthalpy  $h(x)$  on its bubble line, the saturated-vapour enthalpy  $H(y)$  on its dew line, and the equilibrium tie between them, all on Choupo's ONE enthalpy datum (the elements at 298.15 K).

$\lambda(x) = H(y^*) - h(x)$  is the molar heat of vaporisation ALONG the locus. CMO says it is a constant. Read the two  $\lambda$  diagnostics and decide for yourself.

Run-only + KPI golden; see README.md for the provenance of every number and for what is NOT validated.

*The saturated-liquid enthalpy on its bubble line, the saturated-vapour enthalpy on its dew line, and the equilibrium tie between them — the data an enthalpy-concentration (Ponchon–Savarit) construction is drawn on, and the first thing the engine publishes that carries enthalpy against composition.*

*enthalpyConcentration.csv — a regular table, one header, homogeneous rows:*

*Every row is one equilibrium. It carries \*both\* ends of a tie line — the liquid ( $x1$ ,  $h\_liquid$ ) and the vapour ( $y1$ ,  $H\_vapour$ ) that are in equilibrium with each other — at the one temperature  $T\_K$  they share. So nothing downstream pairs rows by index and nothing interpolates one end of a tie. role says only which composition was the uniform grid node:*

*The saturated-liquid curve is ( $x1$ ,  $h\_liquid$ ) over all rows; the saturated-vapour curve is ( $y1$ ,  $H\_vapour$ ) over all rows. status is ok or names a refusal; a refused node keeps its grid coordinate and leaves the solved fields empty, so a gap is visible rather than silently bridged.*

*enthalpyConcentration.meta — a key,value sidecar, written on every run. It states the two components, the pressure, which enthalpy datum the two columns are on, which kernel computed each, the tie contract, whether the heat of mixing is included, and the reason for every refused node. The construction's lever arms are enthalpy \*differences\*, so a constant offset cancels in the geometry — which is exactly why the file has to say what its axis means instead of letting the next reader assume.*

*Both enthalpy columns are absolute on Choupo's one enthalpy base, the elements at 298.15 K, from the same two kernels every energy balance and duty in the engine is priced on:*

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected  |
|------|-------------|--------------------------|-----------|
| diag | hxy         | H_vapour_pure1_J_per_mol | -231233.2 |
| diag | hxy         | H_vapour_pure2_J_per_mol | -239304.0 |
| diag | hxy         | P_Pa                     | 101325.0  |
| diag | hxy         | T_bubble_pure1_K         | 351.5     |
| diag | hxy         | T_bubble_pure2_K         | 372.5     |
| diag | hxy         | h_liquid_pure1_J_per_mol | -271912.8 |
| diag | hxy         | h_liquid_pure2_J_per_mol | -280791.7 |
| diag | hxy         | lambda_max_J_per_mol     | 45052.6   |
| diag | hxy         | lambda_min_J_per_mol     | 40679.6   |
| diag | hxy         | n_nodes                  | 82.0      |
| diag | hxy         | n_refused                | 0.0       |



**phase01\_water\_full**  
tutorials/props/scan/phase01\_water\_full

pd  
purePhaseDiagram

**What it does:** Complete P-T phase diagram of water (purePhaseDiagram, solid+liquid+vapour)

**The story:** Complete pure-compound P-T phase diagram of water: liquid-vapour saturation (to the critical point) + solid-vapour sublimation + solid-liquid fusion (the negative-slope ice line) + triple and critical points. Clapeyron physics in the purePhaseDiagram engine op; cited solid data in the solid{} block (IAPWS/CRC, see solidPhaseData.ts). Run-only smoke.

**What to expect (golden-locked):**

| kind | unit/stream | key       | expected |
|------|-------------|-----------|----------|
| diag | pd          | Pc_bar    | 220.64   |
| diag | pd          | Tc        | 647.14   |
| diag | pd          | has_solid | 1        |

**scan2d01\_co2\_compressibility**

tutorials/props/scan/scan2d01\_co2\_compressibility

**Z\_co2\_TP**  
 propertyScan2D

**What it does:** Z(T,P) for CO2 on Peng-Robinson over a 15 x 20 grid that crosses the critical point: the compressibility surface, one row per node

**The story:** THE COMPRESSIBILITY SURFACE, NOT A CURVE.

‘propertyScan2D’ sweeps TWO state variables over a rectangle and reports the property at every node – long-form CSV, one row per (T, P), ready to pivot into a heat map. Here it is Z(T, P) for CO2 on Peng-Robinson, 280-420 K by 1-200 bar: a 15 x 20 grid that walks straight over the critical point (304.13 K, 73.77 bar).

What to read: along an isotherm below Tc the surface DROPS steeply and then turns up – the liquid-like branch; above Tc the same isotherm is smooth: the discontinuity is a coexistence line, and the grid crosses where it ends;  $Z \rightarrow 1$  in the whole low-P edge, which is the ideal-gas limit falling out of a cubic rather than being assumed.

The op reports ‘failures’ – nodes where the property could not be evaluated – as a diagnostic, so a grid that quietly returned fewer points than it promised is visible in the payload rather than only in the CSV’s line count. The expected file pins it at zero: on this rectangle every node must solve.

Why the case exists: ‘propertyScan2D’ had NO case and no gate, though ‘propertyScan1D’, ‘propertyScan-Binary’ and ‘propertyScanTernary’ all have one. Its sibling vaporPressureFit was in the same position, and writing the first case for THAT one segfaulted choupProps twice.

**What to expect (golden-locked):**

| kind | unit/stream | key      | expected |
|------|-------------|----------|----------|
| diag | Z_co2_TP    | failures | 0        |
| diag | Z_co2_TP    | n_points | 300      |
| diag | Z_co2_TP    | n_x      | 15       |
| diag | Z_co2_TP    | n_y      | 20       |

ternary01\_audit\_vlle

tutorials/props/scan/ternary01\_audit\_vlle

lle

propertyScanTernary

**What it does:** Ternary VLLE phase map (compA/compB/compC) at 350 K, 1 bar via propertyScanTernary

**The story:** ILLUSTRATIVE / SYNTHETIC DATA. compA / compB / compC are NOT real species: they are artificial components (case-local constant/components/ + inline NRTL pairs in thermoPhysPropDict) deliberately tuned to guarantee a ternary VLLE region, so this case exercises the propertyScanTernary classifier on a known-3-phase map. Architecture (see thermoPhysPropDict): A and B mutually immiscible (LL gap on the A-B axis), C a volatile bridge soluble in both liquids. At 350 K, 1 bar the equimolar feed gives vapour rich in C, L\_alpha rich in A, L\_beta rich in B.

Expected map (14 intervals/edge, 78 nodes): VLLE 36, LL 29, VL 0, 1-phase 13, 22 tie-lines, 0 failures. The point to SEE: a healthy spread of 3-phase (VLLE) and LL nodes with zero solver failures — the classifier resolves the engineered gap node-by-node.

What to expect (golden-locked):

| kind | unit/stream | key        | expected |
|------|-------------|------------|----------|
| diag | lle         | failures   | 0        |
| diag | lle         | n_ll       | 29       |
| diag | lle         | n_nodes    | 78       |
| diag | lle         | n_onephase | 13       |
| diag | lle         | n_tielines | 22       |
| diag | lle         | n_vl       | 0        |
| diag | lle         | n_vlle     | 36       |

**ternary02\_bubbleT\_benzene\_toluene\_water**  
tutorials/props/scan/ternary02\_bubbleT\_benzene\_toluene\_water

bt  
propertyScanTernary

**What it does:** Ternary T\_bubble surface (benzene/toluene/water) at 1 bar via propertyScanTernary mode bubbleT

**The story:** Ternary boiling-temperature surface (T\_bubble over the composition triangle) via propertyScanTernary mode bubbleT — reuses BubblePoint per simplex node. Works with ideal binaries (only the 3 Psat are needed). Run-only smoke.

**What to expect (golden-locked):**

| kind | unit/stream | key      | expected |
|------|-------------|----------|----------|
| diag | bt          | failures | 0        |
| diag | bt          | n_nodes  | 150      |

**ternary03\_lle\_water\_ethanol\_benzene**

tutorials/props/scan/ternary03\_lle\_water\_ethanol\_benzene

|                     |
|---------------------|
| lle                 |
| propertyScanTernary |

**What it does:** Ternary LLE solubility (water/ethanol/benzene) at 298 K via UNIFAC + propertyScanTernary mode lle

**The story:** Ternary LIQUID-LIQUID solubility map (water / ethanol / benzene) — the classic extraction triangle. Activity from UNIFAC (group contribution): NO fitted binary pairs needed — gamma is predicted from molecular groups, so a real system shows a real miscibility gap with no curated pair data. mode lle uses the liquid-liquid flash (PhaseSet::LL): clean binodal + tie-lines, no spurious vapour. Run-only smoke (a phase map, not KPIs).

**What to expect (golden-locked):**

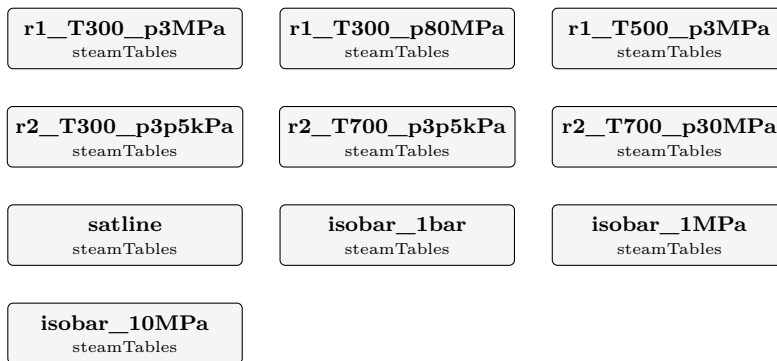
| kind | unit/stream | key        | expected |
|------|-------------|------------|----------|
| diag | lle         | failures   | 0        |
| diag | lle         | n_ll       | 69       |
| diag | lle         | n_nodes    | 105      |
| diag | lle         | n_onephase | 36       |
| diag | lle         | n_tielines | 18       |
| diag | lle         | n_vl       | 0        |
| diag | lle         | n_vlle     | 0        |

**steam**

**The theory you need.** Water/steam properties from IAPWS-IF97: regions (liquid, vapour, supercritical),  $T_{\text{sat}}(P)$ , and  $h(T, P)/s(T, P)$  from the dimensionless Gibbs functions. Power cycles are enthalpy bookkeeping around them: pump work  $\approx v \Delta P$ , turbine work from isentropic expansion  $\times$  efficiency, cycle efficiency  $\eta = W_{\text{net}}/Q_{\text{in}}$  bounded by Carnot. *Full derivation: Theory Guide, the steam/power chapter.*

**steam01\_if97\_verification**

tutorials/props/steam/steam01\_if97\_verification



**What it does:** IAPWS-IF97 steam tables (R7-97(2012)) – the verification case. The six computer-program verification states of the release (Table 5 region 1, Table 15 region 2) evaluated as points, the saturation line 0.01-350 degC, and three isobars whose Tsat crossings pin the release’s Table 36 backward values. The published numbers ARE the golden master (reltol 1e-8).

**The story:** the IAPWS-IF97 weapon case.

The release (IAPWS R7-97(2012)) publishes computer-program verification tables; this case evaluates EVERY one of them this slice covers and locks the published numbers as golden KPIs at reltol 1e-8 – a single wrong digit in a transcribed coefficient fails the regression:

Table 5 – region 1 at (300 K, 3 MPa), (300 K, 80 MPa), (500 K, 3 MPa) Table 15 – region 2 at (300 K, 3.5 kPa), (700 K, 3.5 kPa), (700 K, 30 MPa) Table 35 – psat at 300 / 500 K (the psat\_MPa diag of the point ops) Table 36 – Tsat at 0.1 / 1 / 10 MPa (the Tsat\_K diag of the isobars)

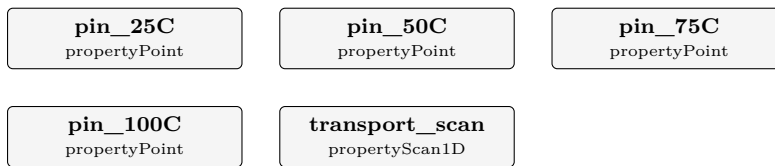
Plus the working artefacts: the saturated-steam table 0.01-350 degC (saturation.csv) and three isobars (h, s, v, cp vs T) crossing the saturation line – the crossing is ANNOUNCED in the log.

**What to expect (golden-locked):**

| kind                                                     | unit/stream    | key      | expected       |
|----------------------------------------------------------|----------------|----------|----------------|
| diag                                                     | r1_T300_p3MPa  | region   | 1              |
| diag                                                     | r1_T300_p3MPa  | v        | 0.100215168e-2 |
| diag                                                     | r1_T300_p3MPa  | h        | 0.115331273e3  |
| diag                                                     | r1_T300_p3MPa  | u        | 0.112324818e3  |
| diag                                                     | r1_T300_p3MPa  | s        | 0.392294792    |
| diag                                                     | r1_T300_p3MPa  | cp       | 0.417301218e1  |
| diag                                                     | r1_T300_p3MPa  | w        | 0.150773921e4  |
| diag                                                     | r1_T300_p3MPa  | psat_MPa | 0.353658941e-2 |
| diag                                                     | r1_T300_p80MPa | region   | 1              |
| diag                                                     | r1_T300_p80MPa | v        | 0.971180894e-3 |
| diag                                                     | r1_T300_p80MPa | h        | 0.184142828e3  |
| diag                                                     | r1_T300_p80MPa | u        | 0.106448356e3  |
| diag                                                     | r1_T300_p80MPa | s        | 0.368563852    |
| diag                                                     | r1_T300_p80MPa | cp       | 0.401008987e1  |
| ... and 47 more reference values in the case’s expected. |                |          |                |

**steam02\_water\_transport**

tutorials/props/steam/steam02\_water\_transport



**What it does:** IAPWS water transport properties – viscosity (R12-08), thermal conductivity (R15-11) and surface tension (R1-76). Water flagged as a pure fluid (method IF97) routes the transport accessors to the matching IAPWS releases: the density comes from the IF97 thermodynamic kernel at the saturation state, fed to the (rho,T) transport kernels; surface tension is the R1-76 saturation-line form. A coarse 4-point scan (25/50/75/100 degC) pinned against the published IAPWS values, plus a fine sweep for the plot.

**The story:** the IAPWS water transport route.

Water flagged ‘pureFluids { water { method IF97; } }’ routes the transport accessors to the matching IAPWS releases:

viscosity → R12-08 (industrial, critical enhancement off) thermal\_conductivity → R15-11 (background, critical enhancement off) surface\_tension → R1-76 (saturation line)

The (rho,T) viscosity / conductivity kernels get their density from the IF97 thermodynamic kernel at the SATURATION state (the legacy transport accessors carry no pressure, so  $p = p_{\text{sat}}(T)$  is the physical convention – saturated liquid / saturated vapour). Surface tension is a saturation-line property and takes only T.

‘pins’ evaluates the three properties at 25/50/75/100 degC; the golden master locks them at reltol 1e-5 (the formulation is verified to 1e-8 by IAPWSTransport::verify(), which fires on every run). ‘scan’ writes a fine sweep for the plot.

**What to expect (golden-locked):**

| kind | unit/stream    | key                  | expected     |
|------|----------------|----------------------|--------------|
| diag | pin_25C        | surface_tension      | 0.0720       |
| diag | pin_25C        | thermal_conductivity | 0.6065       |
| diag | pin_25C        | viscosity            | 0.0009       |
| diag | pin_50C        | surface_tension      | 0.0679       |
| diag | pin_50C        | thermal_conductivity | 0.6406       |
| diag | pin_75C        | surface_tension      | 0.0636       |
| diag | pin_75C        | thermal_conductivity | 0.6635       |
| diag | pin_100C       | surface_tension      | 0.0589       |
| diag | pin_100C       | thermal_conductivity | 0.6772       |
| diag | transport_scan | n_points             | 46           |
| diag | transport_scan | n_failures           | 0            |
| diag | pin_25C        | Cp_ig                | 33.6596      |
| diag | pin_25C        | H_R                  | 0.0000       |
| diag | pin_25C        | H_ig                 | -241826.0000 |

... and 49 more reference values in the case’s *expected*.

**thermo**

**The theory you need.** Consistency tests interrogate DATA, not models: the van Ness point test checks measured  $(x, y, T, P)$  sets against the Gibbs–Duhem constraint (residuals in  $\ln(\gamma_1/\gamma_2)$ ); Herington’s area test is its integral cousin. A dataset that fails is not “bad” – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

**temperature01\_gas\_thermometer**

tutorials/props/thermo/temperature01\_gas\_thermometer

**Z\_of\_P**  
 propertyScan1D

**nearly\_ideal**  
 propertyPoint

**laboratory\_pressure**  
 propertyPoint

**What it does:** The gas thermometer: why the ideal gas is a limit, not a substance

**The story:** A constant-volume gas thermometer reads a temperature off  $PV = nRT$ . A real gas does not obey that, so the reading depends on the PRESSURE it was taken at – and only in the limit  $P \rightarrow 0$  does it become the thermodynamic temperature. This case measures that departure, at the temperature the accompanying notes are about.

500.012 K is not a special temperature. It is the number the notes use to ask what the last digits of a temperature actually assert, and the scan is run there so the two are talking about the same state.

*> \* "Thermodynamics is a funny subject. The first time you go through it, you > don't understand it at all. The second time you go through it, you think you > understand it, except for one or two small points. The third time you go > through it, you know you don't understand it, but by that time you are so > used to it, it doesn't bother you any more." \* > > — attributed to Arnold Sommerfeld. PROVENANCE UNVERIFIED: Sommerfeld > published no such sentence; it reaches us by report, and this repository does > not quote a source it has not read. It is here as an epigraph, marked, and > must be either sourced or dropped before it appears in any published guide.*

*What does the 012 assert? Not "roughly five hundred kelvin" — three digits past the decimal point is a specific claim about the world. Whose claim, and resting on what?*

*The kelvin is no longer defined by a substance. It is defined by fixing the Boltzmann constant:*

*Not measured. Fixed. So a temperature is a conversion factor between energy and degrees, and nothing else. The triple point of water — which \*defined\* the kelvin for sixty-five years — became, overnight, a quantity that is measured, with an uncertainty.*

*That is the first dogma, and it is worth sitting with: the definition tells you what a kelvin \*means\* and gives you no way whatsoever to \*measure\* one.*

*Nobody puts the Boltzmann constant inside a thermometer. To read a temperature you need a \*realisation\*: a chain of reproducible fixed points and an instrument that interpolates between them. In practice that is ITS-90, with a platinum resistance thermometer doing the interpolating. At 500 K you are between the triple point of water and the freezing point of zinc.*

**What to expect (golden-locked):**

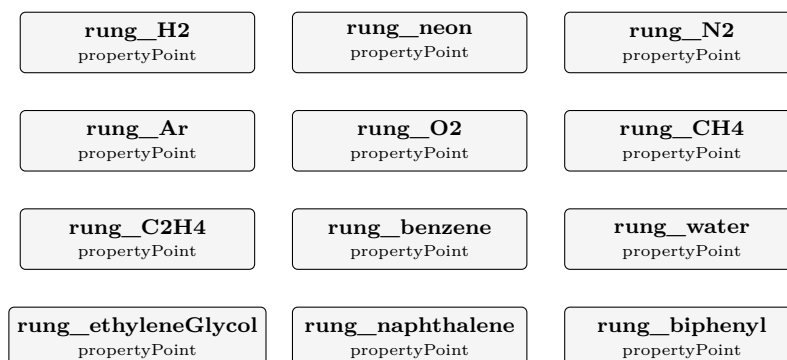
| kind | unit/stream  | key           | expected  |
|------|--------------|---------------|-----------|
| diag | Z_of_P       | n_emptyCurves | 0.0000    |
| diag | Z_of_P       | n_failures    | 0.0000    |
| diag | Z_of_P       | n_points      | 21.0000   |
| diag | nearly_ideal | Cp_ig         | 29.6051   |
| diag | nearly_ideal | H_R           | -0.0040   |
| diag | nearly_ideal | H_ig          | 5915.3500 |
| diag | nearly_ideal | H_real        | 5915.3460 |
| diag | nearly_ideal | P             | 1000.0000 |
| diag | nearly_ideal | S_R           | -0.0000   |
| diag | nearly_ideal | S_ig          | 245.0414  |
| diag | nearly_ideal | S_real        | 245.0413  |
| diag | nearly_ideal | T             | 500.0120  |
| diag | nearly_ideal | Z             | 1.0000    |
| diag | nearly_ideal | gamma         | 1.3905    |

*... and 13 more reference values in the case's **expected**.*



**temperature02\_engineers\_ladder**

tutorials/props/thermo/temperature02\_engineers\_ladder



**What it does:** The range a chemical engineer actually works in

**The story:** THE LADDER A CHEMICAL ENGINEER ACTUALLY CLIMBS, and the engine CHECKS it rather than quoting it.

Each rung asks one question: at the temperature this substance's record calls its normal boiling point, what does its VAPOUR-PRESSURE CORRELATION say the pressure is? The answer should be one atmosphere – that is what a normal boiling point means – and how close it lands is a measure of the record, published rather than assumed.

THE OTHER WAY ROUND WOULD HAVE BEEN NICER AND DOES NOT WORK. Asking for 'T\_bubble' at 1 atm solves for the temperature, which is the more direct question; 'BubblePoint::compute' carries its own initial guess and does not converge at 20 K, a gap this repository already has on record (see docs/design/held-out-pressure.md). Verifying the pressure at a declared temperature asks the same physics of the same correlation and is robust, so that is what runs here – said plainly rather than papered over.

WHY THE LADDER STOPS AT GLYCEROL, and it is not because the list got boring. Above roughly 600 K an ordinary organic does not have a normal boiling point to check: it CRACKS first, so there is no one-atmosphere equilibrium to ask about. The scale itself changes character there – ITS-90 stops interpolating on vapour pressure and gas thermometry altogether and realises its upper range on the FREEZING POINTS OF METALS, then on Planck radiation above the silver point. Those fixed points are declared values of the scale, not something this engine computes, and the page says so where it lists them.

DOWTHERM A IS ABSENT AND THAT IS A FINDING, not an oversight. The hot-oil fluid an engineer actually runs a 530 K loop on is curated here with 'role nonvolatile' and carries NO vapour-pressure correlation, so this case cannot ask it the question it asks the others. Named rather than quietly skipped: an omission a reader cannot see is the kind of gap this project refuses to leave.

*Eight rungs, from liquid hydrogen to boiling water, checked by the engine rather than quoted at you.*

*At the temperature each substance's record calls its normal boiling point, what does its vapour-pressure correlation say the pressure is?*

*It should be one atmosphere — that is what a normal boiling point \*means\*. How close it lands measures the record, and the answer is published rather than assumed.*

*The substance every engineer trusts most is the worst rung on the ladder.*

*Water's Antoine fit declares Trange (273 373). Its normal boiling point is 373.15 K. The fit stops 0.15 K before the temperature everybody uses it at, and evaluated there it returns 1.026 atm instead of 1.000 — 2.6 % out, on the one substance whose boiling point every reader could recite.*

*Seven cryogenics the reader has no intuition for land inside 1.4 %. The familiar one does not. Familiarity is not accuracy, and a declared validity range is worth more than a feeling.*

What to expect (golden-locked):

| kind                                                             | unit/stream | key     | expected    |
|------------------------------------------------------------------|-------------|---------|-------------|
| diag                                                             | rung_H2     | Cp_ig   | 27.3234     |
| diag                                                             | rung_H2     | H_R     | 0.0000      |
| diag                                                             | rung_H2     | H_ig    | -7841.8140  |
| diag                                                             | rung_H2     | H_real  | -7841.8140  |
| diag                                                             | rung_H2     | P       | 101325.0000 |
| diag                                                             | rung_H2     | Psat_H2 | 101850.7659 |
| diag                                                             | rung_H2     | S_R     | 0.0000      |
| diag                                                             | rung_H2     | S_ig    | 55.7336     |
| diag                                                             | rung_H2     | S_real  | 55.7336     |
| diag                                                             | rung_H2     | T       | 20.3900     |
| diag                                                             | rung_H2     | Z       | 1.0000      |
| diag                                                             | rung_H2     | gamma   | 1.4374      |
| diag                                                             | rung_H2     | v_molar | 0.0017      |
| diag                                                             | rung_neon   | Cp_ig   | 20.7861     |
| ...and 155 more reference values in the case's <i>expected</i> . |             |         |             |

thermoTest

**atoms01\_formula\_parser**

tutorials/props/thermoTest/atoms01\_formula\_parser

|                                      |
|--------------------------------------|
| <b>atoms</b><br>elementalComposition |
|--------------------------------------|

**What it does:** The shared formula parser, opened up: nested groups, both hydrate spellings, an ion charge that is not an element

**The story:** THE FORMULA PARSER, WITH ITS LID OFF.

The element balance is only as good as the thing that reads a formula, so that reader is ONE piece of code (src/thermo/ElementComposition) and ‘elementalComposition’ is its glass-box surface: give it a formula, see the atoms it counted.

The formulas here are chosen to be the awkward ones, because those are where a hand-rolled parser goes wrong:

C<sub>2</sub>H<sub>6</sub>O plain, the control; Ca(OH)<sub>2</sub> a nested group with a multiplier – 2 O and 2 H, not 1; CaSO<sub>4</sub>·2H<sub>2</sub>O a HYDRATE, the colon form: 6 O in total, 4 from the sulfate and 2 from the two waters; MgSO<sub>4</sub>·7H<sub>2</sub>O the same salt, colon form: Mg 1, S 1, O 11, H 14; MgSO<sub>4</sub>·7H<sub>2</sub>O THE SAME SALT AS A CHEMIST WRITES IT, and the one formula here that is REFUSED. The ASCII ‘.’ is ambiguous, and Choupo resolved it in favour of the decimal subscript – see sepiolite below – so this spelling would silently count the seven waters into the oxygen: O 5.7 and H 2 where the salt has O 11 and H 14. A wrong element balance from a spelling nobody would flag by eye. It refuses and names both unambiguous forms; Mg<sub>2</sub>Si<sub>3</sub>O<sub>7</sub>·5OH·3H<sub>2</sub>O why the dot cannot simply BE the hydrate separator: this is curated sepiolite, and its 7.5 is a real half-integer; SO<sub>4</sub><sup>-2</sup> an ION: the charge is not an element and must not become one; (NH<sub>4</sub>)<sub>2</sub>SO<sub>4</sub> a nested group of a nested group’s worth of hydrogen – 8 H.

A parser that silently mis-reads any of these produces an element balance that closes on the wrong atoms, which is worse than one that refuses.

**What to expect (golden-locked):**

| kind | unit/stream | key                               | expected |
|------|-------------|-----------------------------------|----------|
| diag | atoms       | C <sub>2</sub> H <sub>6</sub> O_C | 2        |
| diag | atoms       | C <sub>2</sub> H <sub>6</sub> O_H | 6        |
| diag | atoms       | C <sub>2</sub> H <sub>6</sub> O_O | 1        |
| diag | atoms       | CO <sub>2</sub> _C                | 1        |
| diag | atoms       | CO <sub>2</sub> _O                | 2        |
| diag | atoms       | ethanol_C                         | 2        |
| diag | atoms       | ethanol_H                         | 6        |
| diag | atoms       | ethanol_O                         | 1        |
| diag | atoms       | formulas_available                | 10       |
| diag | atoms       | formulas_total                    | 11       |
| diag | atoms       | water_H                           | 2        |
| diag | atoms       | water_O                           | 1        |

## h\_surface\_identity

tutorials/props/thermoTest/h\_surface\_identity

**h\_identity**  
hConsistency

**What it does:** The state identities of the canonical enthalpy surface:  $dh/dT == Cp_{phase}$ , the 298 K vaporisation anchor, and Kirchhoff on  $Hvap\_state$  – pinned to round-off

**The story:** thermodynamics as a REGRESSION TEST.

A surface that anchors each phase leg on the 298.15 K formation datum and integrates the TARGET phase's own  $Cp$  obeys, by construction:

(1)  $dh_{liq}/dT = Cp_{liquid}(T)$  (2)  $dh_{gas}/dT = Cp_{idealGas}(T)$  (3)  $h_g(298.15) - h_{liq}(298.15) = Hvap(298.15)$  (the anchor) (4)  $d[h_g - h_{liq}]/dT = Cp_g - Cp_{liq}$  (Kirchhoff)

This op checks all four by central finite difference for benzene, toluene, water and ethanol at 350 K and REFUSES above  $1e-6$  relative error – so no future "convenient" leg (a Watson-slope liquid, a mixed-datum gas) can re-enter the balances unnoticed. The golden pins every per-species error.

**What to expect (golden-locked):**

| kind | unit/stream | key                        | expected  |
|------|-------------|----------------------------|-----------|
| diag | h_identity  | benzene_anchor298_rel      | 0.000e+00 |
| diag | h_identity  | benzene_dhgas_vs_cpig_rel  | 7.289e-11 |
| diag | h_identity  | benzene_dhliq_vs_cplic_rel | 4.729e-13 |
| diag | h_identity  | benzene_kirchhoff_rel      | 2.058e-10 |
| diag | h_identity  | ethanol_anchor298_rel      | 1.694e-16 |
| diag | h_identity  | ethanol_dhgas_vs_cpig_rel  | 3.826e-11 |
| diag | h_identity  | ethanol_dhliq_vs_cplic_rel | 1.907e-11 |
| diag | h_identity  | ethanol_kirchhoff_rel      | 1.287e-10 |
| diag | h_identity  | toluene_anchor298_rel      | 0.000e+00 |
| diag | h_identity  | toluene_dhgas_vs_cpig_rel  | 6.085e-11 |
| diag | h_identity  | toluene_dhliq_vs_cplic_rel | 1.854e-13 |
| diag | h_identity  | toluene_kirchhoff_rel      | 2.214e-10 |
| diag | h_identity  | water_anchor298_rel        | 6.529e-16 |
| diag | h_identity  | water_dhgas_vs_cpig_rel    | 3.627e-11 |

... and 3 more reference values in the case's *expected*.

**model4\_speciation\_scaling**

tutorials/props/thermoTest/model4\_speciation\_scaling

brine  
speciateconcentrate  
scalingScan

**What it does:** RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices  $SI = \log_{10}(IAP/K)$ , scanned over water recovery to  $r = 0.85$  (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere ( $a(\text{CO}_2\text{aq})$  pinned by Henry at  $p\text{CO}_2 = 4.2\text{e-}4$  atm, DIC a solved outcome – this  $\text{CO}_2$ -rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces  $K_w$  ( $m_{\text{OH}} \sim 1\text{e-}7$ ), a back-computed gypsum-saturated  $\text{CaSO}_4$  solution gives  $SI_{\text{gypsum}} \sim 0.00$ , and pure water with pH solved lands at exactly 7.000.

**The story:** thermoTest / model4\_speciation\_scaling

The SAME system {NaCl,  $\text{CaSO}_4$ , water} in its THIRD role: a MULTI-ION broth. Here we no longer think of "NaCl" or " $\text{CaSO}_4$ " – the aqueous speciation engine reasons in the shared ions  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Ca}^{2+}$ ,  $\text{SO}_4^{2-}$  (plus the neutral ion pair  $\text{CaSO}_4\text{aq}$ ), and reports SATURATION INDICES  $SI = \log_{10}(IAP/K_{\text{sp}})$  for every solid the ions can form: gypsum ( $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ) and halite (NaCl).

$\text{Na}^+$  comes from NaCl,  $\text{Cl}^-$  from NaCl,  $\text{Ca}^{2+}/\text{SO}_4^{2-}$  from  $\text{CaSO}_4$  – but once dissolved there are no salt boundaries, only ions. This is why a "salt" cannot carry its own activity model: the ion pool is shared (the species/ tier of the tree).

MODEL: Pitzer-HMW is MANDATORY here. The ionic strength is  $I \sim 3.3$  mol/kg (NaCl-dominated); the Davies law is only trustworthy to  $I \sim 0.5$  m, so the 'activityModel' is declared explicitly (no silent default) – with Davies the saturation indices at this  $I$  are quantitatively meaningless.

The brine is NaCl-dominated (3 mol/kg, near seawater) with  $\text{CaSO}_4$  (0.10 mol/kg) right at gypsum saturation ( $SI \sim 0$ ). Charge balances exactly ( $\text{Na} + 2 \text{Ca} = \text{Cl} + 2 \text{SO}_4$ ), so pH is pinned at 7. The scan concentrates the brine (evaporation;  $r$  = fraction of water removed): gypsum scales first, at  $r \sim 0.035$  (it starts saturated), halite only near  $r \sim 0.50$ . Note  $m_{\text{CaSO}_4\text{aq}} \sim 0.005$ : the neutral ion pair is a real fraction of the calcium – "dissociation" is partial.

**What to expect (golden-locked):**

| kind | unit/stream | key                               | expected   |
|------|-------------|-----------------------------------|------------|
| diag | brine       | I                                 | 3.26782    |
| diag | brine       | $SI_{\text{anhydrite}}$           | -0.22134   |
| diag | brine       | $SI_{\text{gypsum}}$              | -0.0221674 |
| diag | brine       | $SI_{\text{halite}}$              | -0.910511  |
| diag | brine       | $SI_{\text{mirabilite}}$          | -2.13799   |
| diag | brine       | $SI_{\text{thenardite}}$          | -2.98985   |
| diag | brine       | aw                                | 0.890402   |
| diag | brine       | $\gamma_{\text{CaSO}_4\text{aq}}$ | 1          |
| diag | brine       | $m_{\text{CaSO}_4\text{aq}}$      | 0.00466294 |
| diag | brine       | pH                                | 7          |
| diag | concentrate | $I_{\text{r0}}$                   | 3.26782    |
| diag | concentrate | $I_{\text{rmax}}$                 | 15.6285    |
| diag | concentrate | $SI_{\text{anhydrite\_r0}}$       | -0.22134   |
| diag | concentrate | $SI_{\text{anhydrite\_rmax}}$     | 1.39807    |

... and 26 more reference values in the case's expected.

## Category III: Batch and time-dependent (choupoBatch)

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$d\mathbf{Y}/dt = f(\mathbf{Y})$  integrated in real time, with recipes for discrete events; trajectories, not steady states.

### adsorber

**The theory you need.** Adsorption in a vessel or a bed: the isotherm  $q^*(T, p)$  (Langmuir  $q = q_{\max}bp/(1+bp)$ , van't Hoff  $b(T)$ ) is the curated PAIR datum; the LDF rate  $dq/dt = k(q^* - q)$  carries the EQUIPMENT datum  $k$  (this particle, this bed – it lives in the case, never the catalogue). Work the cases as a staircase. **batch09:** a closed vessel relaxes to the inventory equilibrium – one algebraic root, pinned independently of the run. **batch10:** two species compete for the same sites (extended Langmuir) – the selectivity is an equilibrium statement, and the model's Gibbs–Duhem inconsistency for unequal  $q_{\max}$  is declared, not hidden. **batch11:** in the Henry limit the whole vessel is LINEAR and the integrator must match  $q_{\infty}(1 - e^{-k_{\text{eff}}t})$  to  $10^{-8}$  – the analytic gate. **batch12:** declaring the headspace pressure AND the inventory over-specifies a closed ideal-gas vessel – the contradiction FATALs;  $P$  is a result. **batch13:** the fixed bed – the stoichiometric time  $t_{\text{st}} = (L/u)(\varepsilon + \rho_b q^*/c_{\text{in}})$  is announced BEFORE integration, breakthrough spreads around it, and the constant- $u$  closure's fabricated carrier is printed, pinned and discounted: the DECLARED model residual is itself the lesson. **batch14/batch15** are the numerical hygiene: pure transport vs the analytic advection–dispersion solution, and the mesh study that earns  $N = 100$ . *Full derivation: Theory Guide, the adsorption chapter.*

**batch09\_adsorber\_co2**

tutorials/batch/adsorber/batch09\_adsorber\_co2

**adsorber**  
 batchAdsorber

**What it does:** Closed isothermal CO<sub>2</sub> uptake on zeolite 13X (LDF, solid film): the vessel relaxes to the inventory equilibrium  $p_{\text{inf}} = 7229.13$  Pa,  $q_{\text{inf}} = 1.9708$  mol/kg

**The story:** closed, isothermal gas-solid uptake vessel.

2 mol of pure CO<sub>2</sub> in a 0.01 m<sup>3</sup> ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film ( $dq/dt = k(q^* - q)$ ,  $k = 0.05$  1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO<sub>2</sub>/zeolite13X:  $q_{\text{max}} = 5.0$  mol/kg,  $b_{298} = 9.0$  1/bar,  $dH_{\text{ads}} = -45$  kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with  $n_{\text{gas}} = p V / (R T)$  and  $q = q_{\text{max}} b (p/1e5) / (1 + b p/1e5)$ ,

$$p V / (R T) + m_{\text{ads}} * q_{\text{max}} * b * (p/1e5) / (1 + b(p/1e5)) = n_0$$

whose root at 298.15 K is (a2\_vectors.py, 10 digits)

$p_{\text{inf}} = 7229.128209$  Pa  $q_{\text{inf}} = 1.970838025$  mol/kg  $n_{\text{gas}} = 0.02916197467$  mol uptake = 1.970838025 mol (98.54 %)  $\tau$  (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY  $q$ ;  $n_{\text{gas}}$  is algebraic ( $n_{\text{tot}} - m_{\text{ads}} q$ ), so the per-species closure column closure\_worst\_rel in trajectory.csv is exact by construction ( $\leq 1e-13$ ). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected        |
|------|-------------|---------------------------|-----------------|
| kpi  | adsorber    | P_final                   | 7229.128209     |
| kpi  | adsorber    | Q_ads_isosteric_kJ        | -88.68771114    |
| kpi  | adsorber    | T_final                   | 298.15          |
| kpi  | adsorber    | closure_worst_rel         | 0               |
| kpi  | adsorber    | n_CO2_final               | 2.916197467e-05 |
| kpi  | adsorber    | q_CO2_final               | 1.970838025     |
| kpi  | adsorber    | t_end                     | 600             |
| kpi  | adsorber    | totalMoles_final          | 2.916197467e-05 |
| kpi  | adsorber    | uptake_CO2_mol            | 1.970838025     |
| kpi  | campaign    | element_C_closure_rel     | 0               |
| kpi  | campaign    | element_O_closure_rel     | 0               |
| kpi  | campaign    | element_worst_closure_rel | 0               |
| kpi  | campaign    | energy_balance_available  | 0               |
| kpi  | campaign    | energy_records_invalid    | 0               |

... and 12 more reference values in the case's *expected*.

**batch10\_adsorber\_co2\_n2**

tutorials/batch/adsorber/batch10\_adsorber\_co2\_n2

**adsorber**  
 batchAdsorber

**What it does:** Closed isothermal competitive CO<sub>2</sub>/N<sub>2</sub> uptake on zeolite 13X: extended-Langmuir site competition drives the equilibrium selectivity  $S = 71.43$

**The story:** COMPETITIVE binary uptake, closed vessel.

0.5 mol CO<sub>2</sub> + 1.5 mol N<sub>2</sub> in a 0.01 m<sup>3</sup> ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Both species carry an isotherm record (extended Langmuir: CO<sub>2</sub>  $q_{\max} 5.0 / b_{298} 9.0$  1/bar; N<sub>2</sub>  $3.5 / 0.18$  1/bar) and both carry a declared LDF constant – so both integrate, and the strong adsorbate suppresses the weak one through the SHARED site denominator  $1 + \sum_j b_j p_j$ .

The final state solves the 2-species inventory system  $n_{\text{gas}_i} + m_{\text{ads}} * q_i(p_{\text{CO}_2}, p_{\text{N}_2}) = n0_i, p_i = n_{\text{gas}_i} R T / V$  whose root (a2\_vectors.py, Newton residual 0):

CO<sub>2</sub>:  $p_{\text{inf}} = 0.01611568794$  bar  $q = 0.4934990047$  mol/kg N<sub>2</sub>:  $p_{\text{inf}} = 1.802652166$  bar  $q = 0.7728183086$  mol/kg  $P_{\text{total}} = 181876.7854$  Pa selectivity  $S = (q_{\text{CO}_2}/q_{\text{N}_2})/(p_{\text{CO}_2}/p_{\text{N}_2}) = 71.42857143 = (q_{\max} * b)_{\text{CO}_2}/(q_{\max} * b)_{\text{N}_2}$  (extended-Langmuir identity)

The golden pins the COMPETITIVE path: get the shared denominator wrong and every one of these numbers moves. closure\_worst\_rel stays exact by construction (state = q only; n\_gas algebraic).

**What to expect (golden-locked):**

| kind | unit/stream | key                   | expected          |
|------|-------------|-----------------------|-------------------|
| kpi  | adsorber    | P_final               | 181876.7854       |
| kpi  | adsorber    | Q_ads_isosteric_kJ    | -36.1181845604    |
| kpi  | adsorber    | T_final               | 298.15            |
| kpi  | adsorber    | closure_worst_rel     | 0                 |
| kpi  | adsorber    | n_CO2_final           | 6.500995276e-06   |
| kpi  | adsorber    | n_N2_final            | 7.271816914e-04   |
| kpi  | adsorber    | q_CO2_final           | 0.4934990047      |
| kpi  | adsorber    | q_N2_final            | 0.7728183086      |
| kpi  | adsorber    | t_end                 | 600               |
| kpi  | adsorber    | totalMoles_final      | 0.000733682686703 |
| kpi  | adsorber    | uptake_CO2_mol        | 0.4934990047      |
| kpi  | adsorber    | uptake_N2_mol         | 0.7728183086      |
| kpi  | campaign    | element_C_closure_rel | 0                 |
| kpi  | campaign    | element_N_closure_rel | 0                 |

...and 16 more reference values in the case's *expected*.



**batch11\_adsorber\_henry**

tutorials/batch/adsorber/batch11\_adsorber\_henry

**adsorber**  
 batchAdsorber

**What it does:** Closed isothermal Henry-limit uptake (linear isotherm): the coupled gas+solid system is LINEAR and the run must match  $q(t) = q_{\text{inf}}(1 - \exp(-k_{\text{eff}} t))$  to 1e-8

**The story:** the EXACT-solution gate (G1a of the A2 spec).

1 mol of CO<sub>2</sub> in a 0.01 m<sup>3</sup> ideal-gas headspace over 1 kg of a SYNTHETIC Henry-limit sorbent (linear isotherm  $q^* = K_H p$ ,  $K_H = 0.4$  mol/(kg bar) at 298.15 K – a teaching record, vendored case-local). With a linear isotherm and one species the coupled vessel is LINEAR and closes in analytic form:

$\alpha = K_H[\text{mol}/(\text{kg Pa})] * R * T / V = 0.9915828118$  1/kg  $k_{\text{eff}} = k * (1 + \alpha * m_{\text{ads}}) = 0.09957914059$  1/s  
 $q_{\text{inf}} = \alpha * n_0 / (1 + \alpha * m) = 0.4978868094$  mol/kg  $q(t) = q_{\text{inf}} * (1 - \exp(-k_{\text{eff}} * t))$  ( $q_0 = 0$ )

$q(5) = 0.1952670621$   $q(10) = 0.3139520079$   $q(30) = 0.4727835287$   $q(60) = 0.4966211107$  mol/kg

The pedagogical point the closed vessel adds over the fixed- $q^*$  packet formula:  $k_{\text{eff}} > k$  – gas depletion ACCELERATES the approach to equilibrium (the packet's  $q^* = \text{const}$  law is the  $V_{\text{gas}} \rightarrow \text{infinity}$  limit of this same solution). The trajectory rows at  $t = 5/10/30/60$  s must match the table above to 1e-8 mol/kg (RK4  $dt = 0.01$  s); the golden pins the  $t = 60$  s state.

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key                       | expected        |
|-----------------------------------------------------------------|-------------|---------------------------|-----------------|
| kpi                                                             | adsorber    | P_final                   | 124785.4636     |
| kpi                                                             | adsorber    | Q_ads_isosteric_kJ        | -9.93242221329  |
| kpi                                                             | adsorber    | T_final                   | 298.15          |
| kpi                                                             | adsorber    | closure_worst_rel         | 0               |
| kpi                                                             | adsorber    | n_CO2_final               | 5.033788893e-04 |
| kpi                                                             | adsorber    | q_CO2_final               | 0.4966211107    |
| kpi                                                             | adsorber    | t_end                     | 60              |
| kpi                                                             | adsorber    | totalMoles_final          | 5.033788893e-04 |
| kpi                                                             | adsorber    | uptake_CO2_mol            | 0.4966211107    |
| kpi                                                             | campaign    | element_C_closure_rel     | 0               |
| kpi                                                             | campaign    | element_O_closure_rel     | 0               |
| kpi                                                             | campaign    | element_worst_closure_rel | 0               |
| kpi                                                             | campaign    | energy_balance_available  | 0               |
| kpi                                                             | campaign    | energy_records_invalid    | 0               |
| ...and 12 more reference values in the case's <i>expected</i> . |             |                           |                 |

**batch12\_adsorber\_refused\_P**

tutorials/batch/adsorber/batch12\_adsorber\_refused\_P

**adsorber**  
 batchAdsorber

**What it does:** REFUSAL lesson: declared initial.P (10 bar) contradicts  $nR^*T/V$  (4.958 bar) – the honest cross-check FATALs; P of a closed ideal-gas headspace is a result, not an input

**The story:** the REFUSAL lesson (deliberate, must FAIL).

Identical to batch09\_adsorber\_co2 EXCEPT initial.P declares 10 bar while  $nR^*T/V = 4.9579$  bar. P of a closed ideal-gas headspace is a RESULT (n, T, V fix it – Duhem); a declared P is honoured only as an honest cross-check, and a relative deviation  $> 1e-3$  REFUSES, printing both numbers. Expected FATAL (exit != 0):

batchAdsorber 'adsorber': declared P inconsistent with  $nR^*T/V$  – initial.P =  $1e+06$  Pa (10 bar) but  $nR^*T/V = 495791$  Pa (4.95791 bar), relative deviation  $1.01698 > 1e-3$ . ...

Marked .expect-nonconvergence: counted EXPECTED-FAIL by runTests, never green, never recordable. Base case (the passing twin):

2 mol of pure CO<sub>2</sub> in a 0.01 m<sup>3</sup> ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film ( $dq/dt = k(q^* - q)$ ,  $k = 0.05$  1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO<sub>2</sub>/zeolite13X:  $q_{\max} = 5.0$  mol/kg,  $b_{298} = 9.0$  1/bar,  $dH_{\text{ads}} = -45$  kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with  $n_{\text{gas}} = p V / (R T)$  and  $q = q_{\max} b (p/1e5) / (1 + b (p/1e5))$ ,

$$p V / (R T) + m_{\text{ads}} * q_{\max} * b * (p/1e5) / (1 + b * (p/1e5)) = n_0$$

whose root at 298.15 K is (a2\_vectors.py, 10 digits)

$p_{\text{inf}} = 7229.128209$  Pa  $q_{\text{inf}} = 1.970838025$  mol/kg  $n_{\text{gas}} = 0.02916197467$  mol uptake = 1.970838025 mol (98.54 %)  $\tau$  (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY  $q$ ;  $n_{\text{gas}}$  is algebraic ( $n_{\text{tot}} - m_{\text{ads}} * q$ ), so the per-species closure column closure\_worst\_rel in trajectory.csv is exact by construction ( $\leq 1e-13$ ). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

**batch13\_breakthrough\_co2**

tutorials/batch/adsorber/batch13\_breakthrough\_co2

**bed**  
 fixedBedAdsorber

**What it does:** CO2 15%/He breakthrough on zeolite 13X – 1-D isothermal fixed bed (A3):  $t_{st} = 3040.5$  s announced pre-run,  $t_b(5\%) = 2890.7$  s, ledger closure at machine level

**The story:** CO2/He breakthrough on a zeolite-13X bed.

THE A3 anchor case (spec 2026-07-12): a 0.5 m isothermal fixed bed of zeolite 13X, fed 15% CO2 in helium at  $u = 0.05$  m/s superficial, 1 bar, 298 K. Helium carries NO isotherm record on 13X → announced inert CARRIER, transported but never adsorbed, closed by difference from the constant  $c_{tot} = P/(RT)$ . At 15% CO2 this is NOT a dilute approximation: the constant-( $u, P, T$ ) closure fabricates He at the net uptake rate and the campaign material balance SHOWS that residual (KPIs carrier\_fabricated\_mol, physical\_mass\_closure\_rel) – the named error bites, by design (forum #119); the genuinely clean transport anchor is batch14. The engine announces, BEFORE integrating, where the front must arrive:

$$q*(c_{in}) = 5*9*0.15/(1 + 9*0.15) = 2.872340426 \text{ mol/kg } R_f = \epsilon_p + \rho_b q* / c_{in} = 304.0512509$$

$$t_{st} = (L/u) R_f = 3040.512509 \text{ s } u_{sh} = u/R_f = 1.644459605e-4 \text{ m/s}$$

Independent gates this run must reproduce (anchors\_a3.py, FV N = 100):

$$t_b(5\%) = 2890.669098 \text{ s } t_{50} = 3036.448293 \text{ s } t_b(95\%) = 3204.166058 \text{ s } \int (1 - c_{out}/c_{in}) dt = t_{st}$$

to  $2.2e-10$  (telescopic FV conservation) ledger mass closure  $\leq 1e-10$  (IN/OUT integrated as rows of the SAME ODE)

Numerics: conservative FV, nCells = 100 DECLARED (do the 25/50/100/200 mesh study – batch15 publishes it: first-order upwind, observed order ~1.03-1.05), Danckwerts inlet as the imposed face flux  $u*c_{in}$ , outlet  $dc/dz = 0$ .  $u, P, T$  are DECLARED constants of this phase, and the price is said OUT LOUD: the real velocity drops ~15% across the uptake zone, so the constant- $u$  closure FABRICATES carrier (He) at exactly the net uptake rate (9.19 mol here) – announced in the run header, pinned as the KPI carrier\_fabricated\_mol, discounted in the campaign inventory; Ergun and the energy balance land in A4 (the energy gap is NAMED).

Log: run header prints the hypotheses, the derived  $\rho_p$ ,  $Pe = 250$ , the Gershgorin step bound term by term and the stiffness verdict; profiles land in  $\langle t \rangle / \text{bed.profile}$  every 100 s; trajectory.csv carries  $c_{out}$ ,  $y_{out}$ ,  $q_{bar}$  and the per-species ledger closure.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | bed         | E_adsorption_seg0_kJ      | -413.617021183    |
| kpi  | bed         | E_adsorption_total_kJ     | -413.617021183    |
| kpi  | bed         | P_final                   | 100000            |
| kpi  | bed         | Pe                        | 250               |
| kpi  | bed         | T_final                   | 298               |
| kpi  | bed         | c_out_over_cin_final_CO2  | 0.999999989455    |
| kpi  | bed         | carrier_fabricated_mol    | 9.19148935961     |
| kpi  | bed         | integral_anchor_CO2       | 3040.51245606     |
| kpi  | bed         | mass_closure_CO2          | 8.94231742202e-15 |
| kpi  | bed         | n_CO2_final               | 1.21079552014e-05 |
| kpi  | bed         | n_He_final                | 6.86117461711e-05 |
| kpi  | bed         | physical_mass_closure_rel | 0.108446875314    |
| kpi  | bed         | qbar_CO2_final            | 2.87234042488     |
| kpi  | bed         | retention_factor_CO2      | 304.051250948     |

... and 29 more reference values in the case's *expected*.

**batch14\_transport\_only**

tutorials/batch/adsorber/batch14\_transport\_only

**bed**  
 fixedBedAdsorber

**What it does:** Pure transport ( $k = 0$ ) on the batch13 bed,  $N = 200$ :  $t_{50}$  vs the Ogata-Banks analytic ADE solution (gate G1) under fixed RK4 with the Gershgorin bound printed term by term

**The story:** the  $k = 0$  transport gate (A3 spec, gate G1).

Same bed, same feed, same mesh family as batch13 – but the LDF constant is DECLARED ZERO: the solid is frozen and CO<sub>2</sub> crosses the bed as a pure tracer. What is left is exactly the advection-dispersion equation with a Danckwerts inlet, and THAT has an analytic yardstick (Ogata-Banks, semi-infinite ADE; the finite-domain difference is  $O(1/Pe) = 0.4\%$  at  $Pe = 250$ ):

$u_i = u/eps = 0.125$  m/s  $D_i = Dax/eps = 2.5e-4$  m<sup>2</sup>/s  $c/c_0$  (L, 3.5 s) = 0.07340274358  $c/c_0$  (L, 4.0 s) = 0.5178057707  $c/c_0$  (L, 4.5 s) = 0.913645218  $t_{50}$  analytic = 3.984074227 s

FV yardsticks pinned by this case (anchors\_a3.py,  $N = 200$ ):

$t_{50}$  numeric = 3.975611088 s (0.21% below analytic – first-order upwind; the deviation HALVES with each mesh doubling: 1.40% at  $N=25$ , 0.74% at 50, 0.40% at 100, 0.21% at 200, 0.11% at 400)  $INT(1 - c_{out}/c_{in})dt = eps L / u = 4.0$  s (telescopic conservation: exact identity once saturated) ledger closure  $\sim 1e-14$  ( $k = 0$ )

A declared zero is not a silent default: the unit announces "k = 0 DECLARED: transport-only diagnostic, the solid is frozen (gate G1)". Note the He carrier fabrication of batch13 vanishes here (no uptake):  $carrier\_fabricated\_mol = 0$ .

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | bed         | E_adsorption_seg0_kJ      | 0                 |
| kpi  | bed         | E_adsorption_total_kJ     | 0                 |
| kpi  | bed         | P_final                   | 100000            |
| kpi  | bed         | Pe                        | 250               |
| kpi  | bed         | T_final                   | 298               |
| kpi  | bed         | c_out_over_cin_final_CO2  | 0.999999999893    |
| kpi  | bed         | carrier_fabricated_mol    | 0                 |
| kpi  | bed         | integral_anchor_CO2       | 3.99999999999     |
| kpi  | bed         | mass_closure_CO2          | 4.09039795713e-14 |
| kpi  | bed         | n_CO2_final               | 1.21079552058e-05 |
| kpi  | bed         | n_He_final                | 6.86117461666e-05 |
| kpi  | bed         | physical_mass_closure_rel | 0                 |
| kpi  | bed         | qbar_CO2_final            | 0                 |
| kpi  | bed         | t_50_CO2                  | 3.97561121931     |

... and 23 more reference values in the case's *expected*.

**batch15\_mesh\_study**

tutorials/batch/adsorber/batch15\_mesh\_study

**bed**  
 fixedBedAdsorber

**What it does:** Mesh-study member  $N = 25$  of the batch13 breakthrough:  $t_b(5\%)$  converges monotonically  $2732 \rightarrow 2839 \rightarrow 2891 \rightarrow 2916$  s with observed order 1.03-1.05; full table in reports/meshStudy.md

**The story:** the declared-mesh lesson (A3 spec, gate G6).

The axial mesh is a DECLARED modelling choice (nCells; absent = FATAL), and this case is the receipt: the batch13 breakthrough run on the 25/50/100/200 family, everything else identical. First-order upwind advection dominates the error, so  $t_b(5\%)$  must converge MONOTONICALLY with observed order  $p \sim 1$ :

$N = 25$   $t_{b5} = 2732.027658$  s (this case)  $N = 50$   $t_{b5} = 2838.542518$  s  $N = 100$   $t_{b5} = 2890.669098$  s (batch13, the production mesh)  $N = 200$   $t_{b5} = 2915.764796$  s  $p(25/50/100) = 1.031$   $p(50/100/200) = 1.055$  Richardson ( $p = 1$ ):  $t_{b5}(\text{inf}) \sim 2940.86$  s  $\rightarrow N = 100$  sits 1.7% low

The full table, the four overlaid breakthrough curves and the verdict are PUBLISHED in reports/mesh-Study.md (generated from the four runs; the report header cites mesh, tolerances and integrator). The ledger mass closure stays  $\leq \sim 1e-11$  on EVERY mesh – conservation is telescopic in the FV form, it does not improve with refinement because it was never discretisation-limited (gate G3).

This case pins the  $N = 25$  member; its numbers are DELIBERATELY the worst of the family ( $t_{b5}$  5.8% below batch13) – coarse-mesh optimism made visible, the point of the lesson.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | bed         | E_adsorption_seg0_kJ      | -413.616606535    |
| kpi  | bed         | E_adsorption_total_kJ     | -413.616606535    |
| kpi  | bed         | P_final                   | 100000            |
| kpi  | bed         | Pe                        | 250               |
| kpi  | bed         | T_final                   | 298               |
| kpi  | bed         | c_out_over_cin_final_C02  | 0.999973635802    |
| kpi  | bed         | carrier_fabricated_mol    | 9.19148014522     |
| kpi  | bed         | integral_anchor_C02       | 3040.50945345     |
| kpi  | bed         | mass_closure_C02          | 7.40535661511e-15 |
| kpi  | bed         | n_C02_final               | 1.21079316034e-05 |
| kpi  | bed         | n_He_final                | 6.86117697691e-05 |
| kpi  | bed         | physical_mass_closure_rel | 0.108446766597    |
| kpi  | bed         | qbar_C02_final            | 2.87233754538     |
| kpi  | bed         | retention_factor_C02      | 304.051250948     |

... and 29 more reference values in the case's *expected*.

**batch16\_ergun\_profile**

tutorials/batch/adsorber/batch16\_ergun\_profile

|                                |
|--------------------------------|
| <b>bed</b><br>fixedBedAdsorber |
|--------------------------------|

**What it does:** A4 F1: pure-He Ergun profile against the closed-form P-squared solution

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | bed         | P_first_cell_Pa           | 99999.4670134     |
| kpi  | bed         | P_mid_cell_Pa             | 99946.1540012     |
| kpi  | bed         | P_last_cell_Pa            | 99893.8796441     |
| kpi  | bed         | P_out_boundary_Pa         | 99893.3459257     |
| kpi  | bed         | carrier_fabricated_mol    | 0                 |
| kpi  | bed         | physical_mass_closure_rel | 0                 |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | bed         | E_adsorption_seg0_kJ      | 0                 |
| kpi  | bed         | E_adsorption_total_kJ     | 0                 |
| kpi  | bed         | P_final                   | 99946.6820773     |
| kpi  | bed         | T_final                   | 298.15            |
| kpi  | bed         | deltaP_first_to_out_Pa    | 106.12034902      |
| kpi  | bed         | gasInventory_mol          | 0.0806360746763   |
| kpi  | bed         | n_He_final                | 8.06360746763e-05 |

... and 22 more reference values in the case's *expected*.

**batch17\_dilute\_ergun\_limit**

tutorials/batch/adsorber/batch17\_dilute\_ergun\_limit

**legacy**  
fixedBedAdsorber

**ergun**  
fixedBedAdsorber

**What it does:** A4 F2: 400 ppm uptake, Ergun must approach the A3 constant-velocity limit

**What to expect (golden-locked):**

| kind                                                             | unit/stream | key                                   | expected          |
|------------------------------------------------------------------|-------------|---------------------------------------|-------------------|
| kpi                                                              | campaign    | element_C_closure_rel                 | 5.32078747864e-13 |
| kpi                                                              | campaign    | element_He_closure_rel                | 0.000139137956493 |
| kpi                                                              | campaign    | element_O_closure_rel                 | 5.32078747864e-13 |
| kpi                                                              | campaign    | element_worst_closure_rel             | 0.000139137956493 |
| kpi                                                              | campaign    | element_worst_closure_vs_declared_rel | 5.32078747864e-13 |
| kpi                                                              | campaign    | energy_balance_available              | 0                 |
| kpi                                                              | campaign    | energy_records_invalid                | 0                 |
| kpi                                                              | campaign    | energy_records_valid                  | 2                 |
| kpi                                                              | campaign    | mass_closure_rel                      | 0.000138528772162 |
| kpi                                                              | campaign    | mass_kg_external_out                  | 0.161636709221    |
| kpi                                                              | campaign    | mass_kg_final                         | 0.000558961907879 |
| kpi                                                              | campaign    | mass_kg_initial                       | 0.162173202362    |
| kpi                                                              | campaign    | mass_residual_kg                      | 2.24687671715e-05 |
| kpi                                                              | campaign    | moles_kmol_external_out               | 0.0403339312462   |
| ... and 53 more reference values in the case's <i>expected</i> . |             |                                       |                   |

**batch18\_ergun\_conservation**

tutorials/batch/adsorber/batch18\_ergun\_conservation

|                                |
|--------------------------------|
| <b>bed</b><br>fixedBedAdsorber |
|--------------------------------|

**What it does:** A4 F3: 15 percent CO2 with explicit carrier continuity and Ergun velocity

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | bed         | E_adsorption_seg0_kJ     | -413.699637907    |
| kpi  | bed         | E_adsorption_total_kJ    | -413.699637907    |
| kpi  | bed         | P_final                  | 100047.159082     |
| kpi  | bed         | P_first_cell_Pa          | 100092.418138     |
| kpi  | bed         | P_last_cell_Pa           | 100001.886961     |
| kpi  | bed         | P_mid_cell_Pa            | 100047.162776     |
| kpi  | bed         | P_out_boundary_Pa        | 100000            |
| kpi  | bed         | Pe                       | 250               |
| kpi  | bed         | T_final                  | 298.15            |
| kpi  | bed         | c_out_over_cin_final_CO2 | 1.00000763107     |
| kpi  | bed         | carrier_fabricated_mol   | -6.8212102633e-13 |
| kpi  | bed         | deltaP_first_to_out_Pa   | 92.4181379529     |
| kpi  | bed         | integral_anchor_CO2      | 3028.46636949     |
| kpi  | bed         | mass_closure_CO2         | 1.84877618489e-14 |

... and 39 more reference values in the case's *expected*.



**batch19\_feed\_switch\_purge**

tutorials/batch/adsorber/batch19\_feed\_switch\_purge

**bed**  
 fixedBedAdsorber

**What it does:** A5 feed switch: CO<sub>2</sub>/He load to 3600 s (batch13 anchors), then pure-He purge to 9000 s – amended feed commitment ledgered, frozen breakthrough KPIs, desorption duty as the same state difference

**The story:** THE A5 WITNESS: load, then switch the feed.

batch13's bed (CO<sub>2</sub> 15% in He on zeolite 13X, the A3 anchor), run through a TWO-STEP cycle in one campaign:

0 .. 3600 s LOAD – identical to batch13's feed; the front arrives on the batch13 anchors ( $t_{.5} = 2890.7$ ,  $t_{.50} = 3036.4$ ,  $t_{.95} = 3204.2$  s) 3600 s SWITCH – recipe setParameter bed.feed.CO<sub>2</sub> = 0: the feed becomes pure He (concentration- swing regeneration) 3600 .. 9000 s PURGE –  $q^*(c \rightarrow 0) = 0$ , the LDF drives the bed to desorb; the favourable (Langmuir) isotherm makes desorption the UNFAVOURABLE direction – watch the long tail on  $c_{out}$

What the switch does, and where each consequence is VISIBLE:

The feed commitment is DECLARED matter (the campaign datum owns  $A*u*c_{in}*(endTime - t)$  of future feed). Re-declaring the feed at  $t = 3600$  s AMENDS that commitment: the un-fed CO<sub>2</sub> leaves the datum and the extra He enters it, priced and recorded as 'feedAmendment' ledger records against the external boundary – the campaign mass/element balances read the records and still close (KPIs mass\_kg\_amend\_in/out, element\_worst\_closure\_vs\_declared\_rel  $\sim 1e-14$ ).

The breakthrough sampler's reference basis ( $c_{out}/c_{in}$ ) dies with the old feed: crossings and integral FREEZE at their pre-switch values (announced), so  $t_{.5}/t_{.50}/t_{.95}$  here must MATCH batch13's anchors – the frozen claim describes the first feed step. The desorption wave lives on in the trajectory's  $c_{out\_CO2}/qbar\_CO2$  columns.

The adsorption duty stays an EXACT state difference on the adsorbed inventory:  $E_{adsorption} = dH_{ads} * (n_{ads}(end) - n_{ads}(0))$ . batch13 (load only) ledgers -413.6 kJ; here the purge takes part of the loading back OUT, so  $|E|$  is SMALLER – desorption is endothermic and the ledger shows the heat returning, with no new formula anywhere.

$u$ ,  $P$ ,  $T$  stay DECLARED CONSTANTS: asking the bed for a thermal swing (feed.T), a flow transient ( $u$ ) or a blowdown ( $P$ ) is REFUSED by name (fired by bin/curate/check\_feed\_switch.py, batch12-style honesty).

The carrier-fabrication caveat of the constant- $(u, P, T)$  closure applies unchanged (forum #119): the declared residual explains the He element closure, shown and never discounted.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | bed         | E_adsorption_seg0_kJ      | -23.659607554     |
| kpi  | bed         | E_adsorption_total_kJ     | -23.659607554     |
| kpi  | bed         | P_final                   | 100000            |
| kpi  | bed         | Pe                        | 250               |
| kpi  | bed         | T_final                   | 298               |
| kpi  | bed         | carrier_fabricated_mol    | 0.525769056755    |
| kpi  | bed         | integral_anchor_CO2       | 3040.50572088     |
| kpi  | bed         | mass_closure_CO2          | 1.23888355951e-14 |
| kpi  | bed         | n_CO2_final               | 3.21638319252e-07 |
| kpi  | bed         | n_He_final                | 8.03980630532e-05 |
| kpi  | bed         | physical_mass_closure_rel | 0.00289489594687  |
| kpi  | bed         | qbar_CO2_final            | 0.164302830236    |
| kpi  | bed         | t_50_CO2                  | 3036.44839075     |
| kpi  | bed         | t_breakthrough_5pct_CO2   | 2890.66919752     |

... and 31 more reference values in the case's expected.

**batch20\_thermal\_breakthrough**

tutorials/batch/adsorber/batch20\_thermal\_breakthrough

**bed**  
 fixedBedAdsorber

**What it does:** A5-T1 adiabatic CO<sub>2</sub>/He breakthrough (ergun, one T per cell): thermal wave + earlier front vs batch18, anchors `u_th` and `DT_ad` announced pre-run, campaign energy on vessel enthalpy alone

**The story:** THE A5-T1 WITNESS: the adiabatic bed.

batch18's ergun-mode bed (15% CO<sub>2</sub>/He,  $N = 25$ ) with the energy balance DECLARED: 'energyBalance adiabatic;' – one temperature per cell, the isosteric heat staying IN the bed as a travelling thermal wave instead of being removed by an unnamed coolant. Design record: docs/design/fixed-bed-thermal-a5.md.

What the engine announces BEFORE integrating (equilibrium theory):

pure thermal-wave velocity  $u_{th} = u_{c\_tot} c_{pg} / (\epsilon c_{tot} c_{pg} + \rho_b c_{p\_s})$  and the time  $L/u_{th}$  the heat needs to LEAVE the bed; the adiabatic LIMIT  $DT_{ad} = q*(-dH_{ads})/(c_{p\_s} + \epsilon c_{tot} c_{pg}/\rho_b)$  – the all-heat-retained BOUND. Here  $u_{th} < u_{sh}$ : the heat LAGS the front, so the observed outlet-T rise sits well BELOW the bound ( $\sim +38$  K vs the 140 K limit) while the WARM bed breaks through far earlier than batch18's isothermal front ( $t_{50} \sim 806$  s vs 3042 s –  $q^*(T)$  falls via the van't Hoff  $b(T)$ , the SAME datum the energy source term uses; one  $dH_{ads}$ , three readers).

The golden pins  $T_{out}/T_{max}$ , both anchors, the early crossings, and the campaign ENERGY closure – claimed, adiabatically, on vessel enthalpy against the boundary streams alone (per-step raffinate packages carry the outlet cell's own T).

$c_{p\_s}$  enters as DECLARED case data (the kLDF pattern – scope + source say where the number came from; curating a primary-cited cp onto the adsorbent asset record is a separate curation act). The adsorbed-phase heat capacity is neglected, declared. Feed enters at the declared T; the raffinate package leaves at the OUTLET CELL's own T so the material ledger carries the thermal wave's enthalpy.

Adiabatic means NO energy record: nothing is exchanged with the environment; the campaign energy balance closes on vessel enthalpy (per-cell pricing: gas + adsorbed at  $T_j$ , solid sensible  $c_{p\_s}(T_j - T_0)$ ) against the boundary streams alone.

**What to expect (golden-locked):**

| kind | unit/stream | key                                   | expected          |
|------|-------------|---------------------------------------|-------------------|
| kpi  | bed         | <code>P_final</code>                  | 100050.377191     |
| kpi  | bed         | <code>P_first_cell_Pa</code>          | 100096.839501     |
| kpi  | bed         | <code>P_last_cell_Pa</code>           | 100002.080676     |
| kpi  | bed         | <code>P_mid_cell_Pa</code>            | 100050.98468      |
| kpi  | bed         | <code>P_out_boundary_Pa</code>        | 100000            |
| kpi  | bed         | <code>Pe</code>                       | 250               |
| kpi  | bed         | <code>T_final</code>                  | 315.614986735     |
| kpi  | bed         | <code>T_max_final_K</code>            | 334.272787919     |
| kpi  | bed         | <code>T_out_final_K</code>            | 334.272787919     |
| kpi  | bed         | <code>c_out_over_cin_final_CO2</code> | 0.822859498342    |
| kpi  | bed         | <code>carrier_fabricated_mol</code>   | 2.54374299402e-11 |
| kpi  | bed         | <code>dT_adiabatic_CO2_K</code>       | 140.405462904     |
| kpi  | bed         | <code>deltaP_first_to_out_Pa</code>   | 96.8395014859     |
| kpi  | bed         | <code>integral_anchor_CO2</code>      | 2850.55000879     |

... and 37 more reference values in the case's *expected*.

**batch21\_tsa\_hot\_purge**

tutorials/batch/adsorber/batch21\_tsa\_hot\_purge

|                                |
|--------------------------------|
| <b>bed</b><br>fixedBedAdsorber |
|--------------------------------|

**What it does:** T1.5 TSA half-cycle: adiabatic load to 3600 s, then 400 K clean He purge – feed.T as the hot-purge control, the re-declared commitment ledgered OUT at 298 K / IN at 400 K, energy balance claimed across the swing

**The story:** THE T1.5 WITNESS: a real TSA half-cycle.

batch20's adiabatic bed run as LOAD → HOT CLEAN PURGE in one campaign:

0 .. 3600 s LOAD – 15% CO<sub>2</sub>/He at 298.15 K (batch20's feed; the warm front is through by ~806 s) 3600 s SWITCH – recipe, two setParameter events: feed.CO<sub>2</sub> = 0 (clean He purge) feed.T = 400 K (the HOT purge) 3600 .. 12000 s PURGE – the hot clean carrier sweeps the bed: q\*(T) collapses via the van't Hoff b(T) and desorption runs FAR faster than batch19's cold concentration swing

This is the step the old refusal named: "thermal swing needs the non-isothermal bed energy balance". A5-T1 built that balance; feed.T is now the hot-purge CONTROL (in thermal mode ONLY – the isothermal bed still refuses it, by name).

The LEDGER story is the point. A feed at a new temperature re-declares the WHOLE remaining commitment – at the pinned feed pressure the total concentration drops to P/(R\*400) (composition preserved) and even unchanged matter changes enthalpy – so the switch retires the old commitment (one OUT feedAmendment package, all species, at 298.15 K) and declares the new one (one IN package at 400 K): the enthalpy repricing rides in the packages' own T, and the campaign energy balance stays CLAIMED across the swing, boundary streams alone.

Breakthrough crossings/integral freeze at the switch (they describe the load step); the desorption story lives in c\_out/qbar/T\_out. The frozen pre-run anchors (u\_th, DT\_ad bound) keep their initialise values – a later switch never rewrites an announced claim.

**What to expect (golden-locked):**

| kind | unit/stream | key                    | expected           |
|------|-------------|------------------------|--------------------|
| kpi  | bed         | P_final                | 100030.435428      |
| kpi  | bed         | P_first_cell_Pa        | 100063.521237      |
| kpi  | bed         | P_last_cell_Pa         | 100001.142356      |
| kpi  | bed         | P_mid_cell_Pa          | 100028.717422      |
| kpi  | bed         | P_out_boundary_Pa      | 100000             |
| kpi  | bed         | Pe                     | 250                |
| kpi  | bed         | T_final                | 350.354335031      |
| kpi  | bed         | T_max_final_K          | 399.999997842      |
| kpi  | bed         | T_out_final_K          | 306.850590216      |
| kpi  | bed         | carrier_fabricated_mol | -2.52953213931e-12 |
| kpi  | bed         | dT_adiabatic_CO2_K     | 140.405462904      |
| kpi  | bed         | deltaP_first_to_out_Pa | 63.521236595       |
| kpi  | bed         | integral_anchor_CO2    | 1312.9982971       |
| kpi  | bed         | mass_closure_CO2       | 3.75114008529e-15  |

... and 39 more reference values in the case's *expected*.

**batch22\_wall\_cooled**

tutorials/batch/adsorber/batch22\_wall\_cooled

**bed**  
 fixedBedAdsorber

**What it does:** T2 wall-cooled CO<sub>2</sub>/He breakthrough ( $h=5$  W/m<sup>2</sup>K,  $T_{\text{wall}}=298.15$  K): the bed BETWEEN its limits –  $t_{50}$  bracketed by the adiabatic 806 s and isothermal 3036 s,  $Q_{\text{wall}}$  as a state row of the same ODE, wallHeat record read from the state

**The story:** THE T2 WITNESS: the bed between its two limits.

batch20's adiabatic bed with 'energyBalance wallCooled'; – the energy row gains  $-h a_w (T_j - T_{\text{wall}})$  and the bed sits BETWEEN the limits it must contain:

$h \rightarrow 0$  batch20 (adiabatic:  $t_{50} = 806$  s,  $T_{\text{max}} \sim 336$  K)  $h \rightarrow \text{infinity}$  batch18 (isothermal:  $t_{50} = 3036$  s,  $T = T_{\text{wall}}$ )

so the golden pins the BRACKETING:

$t_{50}(\text{adiabatic}) < t_{50}(\text{wallCooled}) < t_{50}(\text{isothermal})$   $T_{\text{wall}} < T_{\text{max}}(\text{wallCooled}) < T_{\text{max}}(\text{adiabatic})$

at  $h = 5$  W/(m<sup>2</sup> K),  $T_{\text{wall}} = 298.15$  K, dBed 0.113 m DECLARED (never derived from the cross-section by a silent circle assumption); the engine announces  $a_w = 4/\text{dBed}$  and the wall NTU before integrating.

The LEDGER is the design point (docs/design/fixed-bed-thermal-a5.md section 6): the removed heat integrates as ONE MORE STATE ROW of the same ODE – the  $M_{\text{in}}/M_{\text{out}}$  ledger-row pattern – so the 'wallHeat' energy record READS a state ( $E = -Q_{\text{wall}}$ ,  $T_{\text{service}} = T_{\text{wall}}$ ) and the campaign closure  $dH_{\text{vessels}} = Q_{\text{ledger}} + H_{\text{external}}$  holds with every term a state function. Guards: 'adiabatic' + a wallHeatTransfer block refuses (contradiction), 'wallCooled' without one refuses,  $h \leq 0$  refuses (adiabatic is the way to SAY zero) – gate check\_thermal\_bed.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | bed         | E_wallHeat_seg0_kJ       | -184.059495315    |
| kpi  | bed         | E_wallHeat_total_kJ      | -184.059495315    |
| kpi  | bed         | P_final                  | 100048.41634      |
| kpi  | bed         | P_first_cell_Pa          | 100094.212734     |
| kpi  | bed         | P_last_cell_Pa           | 100001.948175     |
| kpi  | bed         | P_mid_cell_Pa            | 100048.658554     |
| kpi  | bed         | P_out_boundary_Pa        | 100000            |
| kpi  | bed         | Pe                       | 250               |
| kpi  | bed         | Q_wall_removed_kJ        | 184.059495315     |
| kpi  | bed         | T_final                  | 306.055276997     |
| kpi  | bed         | T_max_final_K            | 312.299755078     |
| kpi  | bed         | T_out_final_K            | 312.299755078     |
| kpi  | bed         | c_out_over_cin_final_CO2 | 0.890450653484    |
| kpi  | bed         | carrier_fabricated_mol   | 2.41868747253e-11 |

... and 44 more reference values in the case's *expected*.

**batch23\_tsa\_cycles**

tutorials/batch/adsorber/batch23\_tsa\_cycles

**bed**  
 fixedBedAdsorber

**What it does:** Toward A6: three hand-declared TSA cycles (cold load / 400 K purge) in one campaign – eleven ledgered feed amendments, balances closed, final KPIs at the cycle boundary; the CSS verdict machinery stays a named proposal

**The story:** toward A6: three FULL TSA cycles, hand-declared.

batch21's half-cycle repeated three times in ONE campaign, with the recipe grammar that already exists – nothing new in the engine:

cycle k (k = 0,1,2), each 7200 s: t = 7200k feed.T = 298.15 K (cold again) [k>0] feed.CO2 = 0.15 (feed returns) [k>0] t = 7200k + 3600 feed.CO2 = 0 (clean purge) feed.T = 400 K (hot purge)

ending EXACTLY at the third cycle boundary (21600 s), so the FINAL KPIs are end-of-cycle values. Every switch re-declares the feed commitment through the T1.5 ledger machinery (the T switches retire and redeclare the WHOLE commitment; the composition switches amend by delta) – FIFTEEN feedAmendment records (three per switching instant: the composition delta + the T retire/declare pair), and the campaign mass/element/energy balances still close on records, never jumps.

The A6 QUESTION this case makes visible (and does not yet verdict): does the bed state at successive cycle ends approach a fixed point – the cyclic steady state? MEASURED (this case's own run): qbar\_CO2 at the three cycle ends is 0.3151 → 0.2415 → 0.1748 mol/kg – a slow geometric approach (ratio ~0.9; each purge leaves residual heat, the next load starts warmer and holds less), NOT yet converged in three cycles. That unfinished convergence IS the pedagogical point: eyes on a trajectory can see it, and only the A6 verdict machinery could CLAIM it. The golden pins the third-cycle end. A first-class CSS verdict (declared cycles, per- cycle state snapshots, an invariance tolerance the case declares) is an ARCHITECTURE feature – proposed in docs/design/fixed-bed-thermal-a5.md section 7, awaiting Vitor's decision; this case is deliberately its cheap, grammar-free witness.

**What to expect (golden-locked):**

| kind | unit/stream | key                    | expected          |
|------|-------------|------------------------|-------------------|
| kpi  | bed         | P_final                | 100030.094422     |
| kpi  | bed         | P_first_cell_Pa        | 100060.3912       |
| kpi  | bed         | P_last_cell_Pa         | 100001.192563     |
| kpi  | bed         | P_mid_cell_Pa          | 100030.206567     |
| kpi  | bed         | P_out_boundary_Pa      | 100000            |
| kpi  | bed         | Pe                     | 250               |
| kpi  | bed         | T_final                | 329.119724972     |
| kpi  | bed         | T_max_final_K          | 399.878003769     |
| kpi  | bed         | T_out_final_K          | 315.723301673     |
| kpi  | bed         | carrier_fabricated_mol | 9.60653778748e-12 |
| kpi  | bed         | dT_adiabatic_CO2_K     | 140.405462904     |
| kpi  | bed         | deltaP_first_to_out_Pa | 60.3911997653     |
| kpi  | bed         | integral_anchor_CO2    | 1312.99822496     |
| kpi  | bed         | mass_closure_CO2       | 6.54546762708e-14 |

... and 43 more reference values in the case's *expected*.

**batch24\_blowdown\_inert**

tutorials/batch/adsorber/batch24\_blowdown\_inert

**bed**  
 fixedBedAdsorber

**What it does:** P-swing witness A: inert isothermal blowdown,  $n/n_0 = P/P_0$

**The story:** P-SWING WITNESS A: the inventory ratio.

The MATERIAL half of a blowdown, isolated from every thermal effect so it cannot pass by coincidence (design note fixed-bed-thermal-a5.md, section 8/9 amended battery, second-opinion review 2026-08-03):

ideal gas, INERT bed (kLDF = 0: the solid is frozen, pure transport), pure He; ISOTHERMAL ergun mode – T is a DECLARED constant of this teaching slice, so the energy question is out of frame BY DECLARATION (the thermal half is witness B, batch25); at  $t = 200$  s the recipe closes the feed valve (P2, ‘setParameter u = 0’ – the whole remaining feed commitment retires as a ledgered feedAmendment record) and drops the outlet boundary (P1, ‘setParameter P\_out’  $1.0 \rightarrow 0.5$  bar); the bed drains through the SAME Ergun face machinery every run already uses; the vented gas rides the telescopic M\_out ledger rows.

THE ANCHORS (ideal-gas exactness, hand-computable):

$n_{\text{end}} / n_0 = P_{\text{end}} / P_0 = 0.5$   $n(P) = \text{eps A L P} / (R T) = 0.0807$  mol at 1 bar, 0.0403 at 0.5 bar  
 vented = inventory drop (the mass-closure KPI is the proof)

The gate (check\_pswing) recomputes both analytically and pins the ‘gasInventory\_mol’ KPI + the closure against them.

**What to expect (golden-locked):**

| kind | unit/stream | key                    | expected           |
|------|-------------|------------------------|--------------------|
| kpi  | bed         | E_adsorption_seg0_kJ   | 0                  |
| kpi  | bed         | E_adsorption_total_kJ  | 0                  |
| kpi  | bed         | P_final                | 49999.9998988      |
| kpi  | bed         | P_first_cell_Pa        | 49999.9998996      |
| kpi  | bed         | P_last_cell_Pa         | 49999.9998972      |
| kpi  | bed         | P_mid_cell_Pa          | 49999.999899       |
| kpi  | bed         | P_out_boundary_Pa      | 50000              |
| kpi  | bed         | Pe                     | 0                  |
| kpi  | bed         | T_final                | 298.15             |
| kpi  | bed         | carrier_fabricated_mol | -8.881784197e-16   |
| kpi  | bed         | deltaP_first_to_out_Pa | -0.000100418335933 |
| kpi  | bed         | gasInventory_mol       | 0.0403395454642    |
| kpi  | bed         | mass_closure_CO2       | 0                  |
| kpi  | bed         | n_CO2_final            | 0                  |

... and 30 more reference values in the case's *expected*.

**batch24\_tsa\_until\_css**

tutorials/batch/adsorber/batch24\_tsa\_until\_css

**bed**  
 fixedBedAdsorber

**What it does:** Toward A6: three hand-declared TSA cycles (cold load / 400 K purge) in one campaign – eleven ledgered feed amendments, balances closed, final KPIs at the cycle boundary; the CSS verdict machinery stays a named proposal

**The story:** A6 item 4: the cyclic steady state as the STOPPING CONDITION, not a count.

batch23 beside this one declares ‘repeat 3;’ and reports honestly that three cycles did not settle – it makes the question visible and does not answer it. This case asks what an engineer actually wants to ask of a TSA bed: not "run three cycles" but "cycle until it repeats, and tell me how many that took".

cycle { period 7200; repeat untilCSS; cssTolerance 3e-2; maxCycles 6; ... }

The declared cap is SIX and the run uses TWO, announcing why it stopped.

TWO KEYS ARE MANDATORY under ‘untilCSS’ and the engine refuses without either: the tolerance, because whether a bed has converged is a modelling judgement the engine must never invent; and the cap, because a bed that never settles has to stop somewhere and the run must be able to say it stopped for want of cycles rather than because it arrived. Reaching the cap is NOT an error – the trajectory is a physically valid campaign and every other KPI in it is real – but it means the case’s own question went unanswered, so it is published (‘css\_stopped\_at\_cap’) and announced.

WHAT BUILDING IT FOUND, and it outlives the feature: the CSS norm and the loading are NOT the same quantity. See the measured numbers beside ‘cssTolerance’ below. A tolerance chosen without looking can certify a bed whose loading is still visibly marching, because the norm over the whole packed state is dominated by its largest components. The verdict is only as meaningful as the number the case declares – which is the argument for the engine refusing to supply one.

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key                    | expected          |
|-----------------------------------------------------------------|-------------|------------------------|-------------------|
| kpi                                                             | bed         | P_final                | 100029.553817     |
| kpi                                                             | bed         | P_first_cell_Pa        | 100059.686721     |
| kpi                                                             | bed         | P_last_cell_Pa         | 100001.177433     |
| kpi                                                             | bed         | P_mid_cell_Pa          | 100029.506183     |
| kpi                                                             | bed         | P_out_boundary_Pa      | 100000            |
| kpi                                                             | bed         | Pe                     | 250               |
| kpi                                                             | bed         | T_final                | 324.614533765     |
| kpi                                                             | bed         | T_max_final_K          | 399.878004394     |
| kpi                                                             | bed         | T_min_final_K          | 297.926151955     |
| kpi                                                             | bed         | T_out_final_K          | 310.618231369     |
| kpi                                                             | bed         | carrier_fabricated_mol | 1.35003119794e-11 |
| kpi                                                             | bed         | dT_adiabatic_CO2_K     | 140.405462904     |
| kpi                                                             | bed         | deltaP_first_to_out_Pa | 59.6867210897     |
| kpi                                                             | bed         | gasInventory_mol       | 0.0747102437107   |
| ...and 45 more reference values in the case’s <i>expected</i> . |             |                        |                   |

**batch25\_blowdown\_thermal**

tutorials/batch/adsorber/batch25\_blowdown\_thermal

**bed**  
 fixedBedAdsorber

**What it does:** P-swing witness B: adiabatic blowdown COOLS (expansion work)

**The story:** P-SWING WITNESS B: the cooling.

The THERMAL half of a blowdown, isolated from adsorption so the only heat effect in frame is the expansion work itself (design note fixed-bed-thermal-a5.md, section 8/9 amended battery):

same inert pure-He ergun bed as witness A (batch24), but ‘energyBalance adiabatic;’ – the section-9 general energy row (cv accumulation + the explicit eps R T dc\_tot/dt source) is live; at t = 200 s the recipe closes the feed valve (P2) and halves the outlet boundary (P1) – the bed drains, dc\_tot/dt < 0, and the expansion source pulls T DOWN; the solid heat capacity is a declared SYNTHETIC LOW value (9.2 J/(kg K), scope says so): the real ~920 of a zeolite pellet would thermally bury the gas-phase cooling under the solid’s mass (dT ~ -0.03 K); at the teaching value the drop is visible (~ -3 K class). Same honest-teaching-value discipline as the hxFluid of the pinch witness – the physics on display is the GAS’s, and the header says which knob made it visible.

THE ANCHOR: T\_min\_final\_K sits BELOW the declared 298.15 K by a visible margin – a depressurising cell COOLS, which is exactly what the pre-widening cp-form could not show (the sabotage of the gate restores that form and this witness fails). The material anchors stay witness A’s business; this case pins the sign and visibility of the thermal effect, and the campaign energy claim stays closed.

**What to expect (golden-locked):**

| kind | unit/stream | key                    | expected          |
|------|-------------|------------------------|-------------------|
| kpi  | bed         | P_final                | 49999.9997347     |
| kpi  | bed         | P_first_cell_Pa        | 49999.9996936     |
| kpi  | bed         | P_last_cell_Pa         | 49999.999898      |
| kpi  | bed         | P_mid_cell_Pa          | 49999.9997123     |
| kpi  | bed         | P_out_boundary_Pa      | 50000             |
| kpi  | bed         | Pe                     | 0                 |
| kpi  | bed         | T_final                | 294.923818642     |
| kpi  | bed         | T_max_final_K          | 295.066878523     |
| kpi  | bed         | T_min_final_K          | 294.887442943     |
| kpi  | bed         | T_out_final_K          | 295.066878523     |
| kpi  | bed         | carrier_fabricated_mol | 1.15463194561e-14 |
| kpi  | bed         | deltaP_first_to_out_Pa | -0.00030635814619 |
| kpi  | bed         | gasInventory_mol       | 0.0407808219207   |
| kpi  | bed         | mass_closure_CO2       | 0                 |

...and 32 more reference values in the case’s *expected*.

**combustion**

**The theory you need.** Radical chain chemistry: initiation creates the first radicals (slow –  $\text{H}_2 + \text{M}$ ,  $E_a \sim 105 \text{ kcal}$ ), branching multiplies them ( $\text{H} + \text{O}_2 \rightarrow \text{O} + \text{OH}$ : one in, two out), termination removes them. Ignition is the race between branching and termination; the induction time is set by the SLOW build-up, then heat release runs away. Rate constants  $k = AT^b e^{-E_a/RT}$  come VERBATIM from cited mechanisms (GRI-Mech 3.0, Ó Conaire 2004, Glarborg 2018) in their native units – the engine converts and announces. Reverse rates from equilibrium: the kinetic end-state must land on the Gibbs equilibrium (the two-engine handshake the NOx tutorial pins at 98.9%). Expect stiffness ratios of  $10^6$ – $10^8$  (use Rosenbrock23), ignition delays in  $\mu\text{s}$  validated against shock-tube data, and thermal NO forming over SECONDS – the timescale gap behind quench-based NOx control. *Full derivation: Theory Guide, the combustion/kinetics chapter.*



**ignition01\_h2o2**

tutorials/batch/combustion/ignition01\_h2o2

**bomb**  
 batchReactor

**What it does:** Adiabatic constant-V H<sub>2</sub>/O<sub>2</sub> ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); Rosenbrock23 stiff solver; reports ignition delay

**The story:** closed rigid vessel, stoichiometric H<sub>2</sub>/O<sub>2</sub> in N<sub>2</sub> at 1100 K.

H<sub>2</sub> : O<sub>2</sub> : N<sub>2</sub> = 2 : 1 : 4 (stoichiometric H<sub>2</sub> + 1/2 O<sub>2</sub> → H<sub>2</sub>O, 57% N<sub>2</sub>)

mode adiabatic + energy gasConstantV: the first law  $dU = 0$  in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction  $\Delta H$  is supplied. Temperature, and the radicals H/O/OH/HO<sub>2</sub>, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01\_h2o2\_rk4 sibling) and the same step blows up.

*A closed rigid vessel holds a stoichiometric H<sub>2</sub>/O<sub>2</sub> charge diluted in N<sub>2</sub> at 1100 K and ~5 bar. A tiny H-radical seed starts chain branching; after a short induction the radical pool explodes and the mixture ignites — a sharp temperature jump from 1100 K to ~2440 K.*

*and plot trajectory.csv (T and the radicals H/O/OH vs time). You will see:*

*The ignition delay  $\tau_{\text{ign}}$  (the time of maximum  $dT/dt$ ) is  $\approx 0.9$  ms here.*

*The radical lifetimes are  $\sim 10^{-12}$ – $10^{-11}$  s while the induction runs over  $\sim 10^{-3}$  s — a timescale separation of  $\sim 10^2$ . That is stiffness. An explicit method (RK4) is stability-limited to a step of order the \*fastest\* scale, so it cannot take the  $\sim 10^{-11}$  s steps this case uses. The solver here is Rosenbrock23, an L-stable linearly-implicit method that steps by \*accuracy\*, not stability; at verbosity 3 it prints its stiffness verdict and step counts (the no-silent- crutch credo).*

*See it fail on purpose — the sibling case runs the same problem with explicit RK4 at the same step:*

*It blows up to NaN within a few steps. (That case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail.) Switch the integrator back to Rosenbrock23 and the same step converges. \*See, then decide\*, applied to numerics.*

**What to expect (golden-locked):**

| kind | unit/stream | key                   | expected          |
|------|-------------|-----------------------|-------------------|
| kpi  | bomb        | P_final               | 500000            |
| kpi  | bomb        | T_final               | 2641.48171412     |
| kpi  | bomb        | n_H2O2_final          | 2.59406092875e-11 |
| kpi  | bomb        | n_H2_final            | 2.44308154218e-06 |
| kpi  | bomb        | n_HO2_final           | 1.97492765295e-10 |
| kpi  | bomb        | n_H_final             | 1.4513832323e-06  |
| kpi  | bomb        | n_N2_final            | 2.857145e-05      |
| kpi  | bomb        | n_O2_final            | 1.02931747994e-06 |
| kpi  | bomb        | n_OH_final            | 1.08725431027e-06 |
| kpi  | bomb        | n_O_final             | 5.68688863565e-07 |
| kpi  | bomb        | n_water_final         | 1.05706749995e-05 |
| kpi  | bomb        | t_end                 | 0.0002            |
| kpi  | bomb        | totalMoles_final      | 4.57220738612e-05 |
| kpi  | campaign    | element_H_closure_rel | 6.76051276934e-15 |

*... and 17 more reference values in the case's expected.*

**ignition01\_h2o2\_eulersi**

tutorials/batch/combustion/ignition01\_h2o2\_eulersi

**bomb**  
batchReactor

**What it does:** WITNESS + EXPECTED FAILURE: semi-implicit Euler on the H<sub>2</sub>/O<sub>2</sub> ignition – A-stable but first-order, goes unphysical at this step and the engine refuses by name (see .expect-nonconvergence).

**The story:** WITNESS for the EulerSI integrator (semi-implicit Euler) – an EXPECTED FAILURE: see .expect-nonconvergence for why first order at this step goes unphysical and how the engine now refuses instead of exiting 0. Registered model no case selected until 2026-08-22 (invariant I8).

*A closed rigid vessel holds a stoichiometric H<sub>2</sub>/O<sub>2</sub> charge diluted in N<sub>2</sub> at 1100 K and ~5 bar. A tiny H-radical seed starts chain branching; after a short induction the radical pool explodes and the mixture ignites — a sharp temperature jump from 1100 K to ~2440 K.*

*and plot trajectory.csv (T and the radicals H/O/OH vs time). You will see:*

*The ignition delay  $\tau_{\text{ign}}$  (the time of maximum  $dT/dt$ ) is  $\approx 0.9$  ms here.*

*The radical lifetimes are  $\sim 10^{-12}$ – $10^{-13}$  s while the induction runs over  $\sim 10^{-3}$  s — a timescale separation of  $\sim 10^9$ . That is stiffness. An explicit method (RK4) is stability-limited to a step of order the \*fastest\* scale, so it cannot take the  $\sim 10^{-3}$  s steps this case uses. The solver here is Rosenbrock23, an L-stable linearly-implicit method that steps by \*accuracy\*, not stability; at verbosity 3 it prints its stiffness verdict and step counts (the no-silent- crutch credo).*

*See it fail on purpose — the sibling case runs the same problem with explicit RK4 at the same step:*

*It blows up to NaN within a few steps. (That case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail.) Switch the integrator back to Rosenbrock23 and the same step converges. \*See, then decide\*, applied to numerics.*

**ignition01\_h2o2\_rk4**

tutorials/batch/combustion/ignition01\_h2o2\_rk4

**bomb**  
 batchReactor

**What it does:** Adiabatic constant-V H<sub>2</sub>/O<sub>2</sub> ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); EXPLICIT RK4 – fails on the stiff chemistry (the lesson)

**The story:** ignition01\_h2o2 – closed rigid vessel, stoichiometric H<sub>2</sub>/O<sub>2</sub> in N<sub>2</sub> at 1100 K.

H<sub>2</sub> : O<sub>2</sub> : N<sub>2</sub> = 2 : 1 : 4 (stoichiometric H<sub>2</sub> + 1/2 O<sub>2</sub> → H<sub>2</sub>O, 57% N<sub>2</sub>)

mode adiabatic + energy gasConstantV: the first law  $dU = 0$  in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction  $\Delta H$  is supplied. Temperature, and the radicals H/O/OH/HO<sub>2</sub>, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01\_h2o2\_rk4 sibling) and the same step blows up.

*This is the failure half of the stiffness lesson. It runs the exact same adiabatic constant-volume H<sub>2</sub>/O<sub>2</sub> ignition as the sibling case [ignition01\_h2o2](../ignition01\_h2o2/README.md), but with the explicit RK4 integrator instead of the stiff Rosenbrock23 — at the same fixed step ( $\Delta t = 1e-5$  s).*

*trajectory.csv is a column of nan from the very first output step ( $t = 1e-4$  s) onward — there is no clean induction-then-jump, just immediate divergence:*

*That immediate blow-up is the result. The radical lifetimes ( $\sim 10^{-10}$ – $10^{-12}$  s) are six orders of magnitude shorter than the induction timescale ( $\sim 10^{-3}$  s) — the chemistry is stiff. An explicit method like RK4 is stability-limited to a step of order the \*fastest\* scale, so the  $\sim 10^{-5}$  s step this case uses is far too large: the integration is murdered by stability long before accuracy matters.*

*Compare side-by-side with the stiff sibling: switch to (or open) ignition01\_h2o2, whose L-stable adaptive sub-stepping integrates the \*same\* chemistry to a clean burned state ( $\sim 2440$  K). \*See, then decide\*, applied to numerics.*

*This case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail, and it would otherwise trip the NaN guard.*

*Same as the sibling: the thermodynamics is real curated Choupo data, but the reaction rates are synthetic (hand-chosen Arrhenius constants reproducing the chain-branching structure, not a published mechanism). Do not quote any number from this case as quantitative.*

**ignition02\_h2air\_slack**

tutorials/batch/combustion/ignition02\_h2air\_slack

**bomb**  
 batchReactor

**What it does:** Stoich H<sub>2</sub>/air ignition, 1100 K / 2 atm (Slack shock tube; O Conaire 2004 kinetics)

**The story:** / flowsheetDict Stoichiometric H<sub>2</sub>/air autoignition at 1100 K / 2 atm against the Slack shock-tube point (O Conaire 2004 kinetics): the radical pool (H, O, OH, HO<sub>2</sub>) builds exponentially through the induction time, then runaway. A stiff-integrator showcase – compare the ignition delay with the measured shock-tube value.

*Stoichiometric H<sub>2</sub>/air, 1100 K, 2 atm, constant volume — the reflected-shock condition of Slack (1977) as compiled in ? Conaire et al. 2004 (Fig 8). Kinetics: the ? Conaire H<sub>2</sub>/O<sub>2</sub> mechanism, Table I verbatim (19 reactions; two Troe falloffs; two summed pairs; A in cm<sup>3</sup>-mol-s, Ea in kcal — the engine converts and announces every one).*

*Result:  $\tau_{\text{ign}}$  ( $\max dT/dt$ ) = 43  $\mu\text{s}$  vs the measured ~40–60  $\mu\text{s}$  window.*

*Two lessons: 1. No seed. This mechanism self-initiates (R5:  $\text{H}_2 + \text{M} = 2\text{H}$ ,  $E_a = 105 \text{ kcal/mol}$  — slow, but that slowness IS the induction chemistry). With the ignition01-style trace-H seed the delay collapses to 12  $\mu\text{s}$ : initial conditions are physics, not knobs. 2. The engine now prints  $\tau_{\text{ign}}$  in the epilogue (time of  $\max dT/dt$  on the write grid) whenever a vessel ignites — no CSV spelunking.*

*Run ignition01\_h2o2 (GRI kinetics, 5 bar) for the sibling case and the stiffness lesson (\_rk4).*

**What to expect (golden-locked):**

| kind | unit/stream | key                   | expected          |
|------|-------------|-----------------------|-------------------|
| kpi  | bomb        | P_final               | 202650            |
| kpi  | bomb        | T_final               | 3041.74576271     |
| kpi  | bomb        | n_H2O2_final          | 3.94467753319e-11 |
| kpi  | bomb        | n_H2_final            | 9.82828113933e-07 |
| kpi  | bomb        | n_HO2_final           | 4.27531541611e-10 |
| kpi  | bomb        | n_H_final             | 3.53516153966e-07 |
| kpi  | bomb        | n_N2_final            | 1.391515e-05      |
| kpi  | bomb        | n_O2_final            | 3.90750822458e-07 |
| kpi  | bomb        | n_OH_final            | 4.44937008544e-07 |
| kpi  | bomb        | n_O_final             | 1.5493529764e-07  |
| kpi  | bomb        | n_water_final         | 6.00759209227e-06 |
| kpi  | bomb        | t_end                 | 0.0003            |
| kpi  | bomb        | totalMoles_final      | 2.22501764671e-05 |
| kpi  | campaign    | element_H_closure_rel | 3.43861059252e-15 |

*... and 17 more reference values in the case's expected.*

**nox01\_zeldovich\_postflame**

tutorials/batch/combustion/nox01\_zeldovich\_postflame

**postflame**  
 batchReactor

**What it does:** Thermal NO<sub>x</sub> in post-flame H<sub>2</sub>/air gas, 2125 K / 1 atm (Glarborg 2018 N chemistry)

**The story:** / flowsheetDict Thermal NO<sub>x</sub> in post-flame H<sub>2</sub>/air gas at 2125 K / 1 atm: the Zeldovich N-chemistry (Glarborg 2018 rate set) integrated in a batch reactor – NO grows on the SECONDS scale while the O/H radical pool stays equilibrated, the classic separation of scales that makes thermal NO<sub>x</sub> a post-flame problem.

*Hot burnt gas (lean H<sub>2</sub>/air, 2125 K, 1 atm — Homer & Sutton Flame 1 as compiled in Glarborg 2018) held isothermal, starting from the equilibrated H/O radical pool with all nitrogen still as N<sub>2</sub>. The Glarborg 2018 nitrogen chemistry (extended Zeldovich + NNH + N<sub>2</sub>O routes, Tables 2/5 verbatim) then forms NO.*

*1. The cross-engine pin. The kinetic plateau lands on the Gibbs equilibrium computed by a \*different engine\* (steady Gibbs minimisation vs stiff batch integration). Kinetics choose the path and the clock; the destination belongs to the thermodynamics. Reverse rates come from the same equilibrium constants — so this agreement is a genuine consistency check of the mechanism transcription + thermo, not a tautology. 2. NO<sub>x</sub> forms in engineering time. Ignition takes microseconds (ignition02: 43 μs); thermal NO takes seconds at 2125 K. That timescale gap is the whole basis of NO<sub>x</sub> control by quench: cool the gas before the Zeldovich clock runs.*

*The burnt-gas initial pool was computed with this same thermo by a 14-species gibbsReactor at 2125 K/1 atm (25 % H<sub>2</sub> / 17.5 % O<sub>2</sub> / 57.5 % N<sub>2</sub> feed); the equilibrium NO share was returned to N<sub>2</sub>/O<sub>2</sub> elementally.*

**What to expect (golden-locked):**

| kind | unit/stream | key          | expected          |
|------|-------------|--------------|-------------------|
| kpi  | postflame   | P_final      | 101325            |
| kpi  | postflame   | T_final      | 2125              |
| kpi  | postflame   | n_H2O2_final | 1.05023679159e-12 |
| kpi  | postflame   | n_H2_final   | 4.85355503797e-09 |
| kpi  | postflame   | n_HO2_final  | 1.61492364784e-11 |
| kpi  | postflame   | n_H_final    | 6.02372866884e-10 |
| kpi  | postflame   | n_N2O_final  | 1.37258448136e-12 |
| kpi  | postflame   | n_N2_final   | 3.72505182272e-06 |
| kpi  | postflame   | n_NH_final   | 2.50521342942e-15 |
| kpi  | postflame   | n_NNH_final  | 1.01561106781e-15 |
| kpi  | postflame   | n_NO_final   | 2.94524618017e-08 |
| kpi  | postflame   | n_N_final    | 2.26160617192e-14 |
| kpi  | postflame   | n_O2_final   | 3.06851934926e-07 |
| kpi  | postflame   | n_OH_final   | 2.0389175016e-08  |

...and 24 more reference values in the case's *expected*.

**crystallisation**

**The theory you need.** Crystallisation removes solute when the solution crosses saturation:  $m > m_{\text{sat}}(T)$ , with  $K_{\text{sp}}(T)$  from  $\Delta G_{\text{diss}}$  and its  $T$ -dependence from  $\Delta H_{\text{diss}}$  (van 't Hoff). The heat of crystallisation is ION-DERIVED –  $\Delta H_{\text{cryst}} = -(\Delta H_{\text{soln}}^{\infty} + \bar{L}_2(m_{\text{sat}}))$  – never a stored constant (a solid's formation enthalpy is  $\sum \nu_i h_{f,i}^{\text{aq}} - \Delta H_{\text{soln}}$ , derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant,  $\gamma_{\pm}$  rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

**batch13\_nacl\_pitzer**

tutorials/batch/crystallisation/batch13\_nacl\_pitzer

**cryst**  
 batchCrystalliser

**What it does:** Batch NaCl crystalliser: saturation from the Pitzer ion activity, not a solubility curve

**The story:** One closed vessel. The unit declares no saturation of its own: it asks the thermodynamics the case declares, and the case declares Pitzer.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_Cl_closure_rel    | 9.08953975799e-16 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_Na_closure_rel    | 9.08953975799e-16 |
| kpi  | campaign    | element_O_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 9.08953975799e-16 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 1                 |
| kpi  | campaign    | mass_closure_rel          | 3.49878229369e-16 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |
| kpi  | campaign    | mass_kg_final             | 1299.73034249     |
| kpi  | campaign    | mass_kg_initial           | 1299.73034249     |
| kpi  | campaign    | mass_residual_kg          | 4.54747350886e-13 |
| kpi  | campaign    | moles_kmol_external_out   | 0                 |

*...and 17 more reference values in the case's **expected**.*

**batch14\_nacl\_pitzer\_antisolvent**

tutorials/batch/crystallisation/batch14\_nacl\_pitzer\_antisolvent

**cryst**  
 batchCrystalliser

**What it does:** Batch NaCl crystalliser with ethanol present: a pairwise-aqueous model announces the mixed-solvent term it does not carry

**The story:** One closed vessel, identical to batch13 except for the ethanol in the charge. The unit declares no saturation of its own: it asks the thermodynamics the case declares, and the case declares Pitzer – which has nothing to say about a second solvent, and says so.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_Cl_closure_rel    | 6.05235197043e-16 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_Na_closure_rel    | 6.05235197043e-16 |
| kpi  | campaign    | element_O_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 6.05235197043e-16 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 1                 |
| kpi  | campaign    | mass_closure_rel          | 1.3612844745e-16  |
| kpi  | campaign    | mass_kg_external_out      | 0                 |
| kpi  | campaign    | mass_kg_final             | 1670.28772973     |
| kpi  | campaign    | mass_kg_initial           | 1670.28772973     |
| kpi  | campaign    | mass_residual_kg          | 2.27373675443e-13 |

...and 19 more reference values in the case's *expected*.

**drying**

**dryer01\_sucrose\_tray**

tutorials/batch/drying/dryer01\_sucrose\_tray

**dryer**  
 batchDryer

**What it does:** Tray drying of wet sucrose into declared 60 C air: the classical constant-rate / falling-rate drying curve, with the wet bulb and the GAB equilibrium moisture as results

**The story:** the DRYING CURVE, both periods, in one batch.

A tray holding 2.0 kg of dry sucrose wet to  $X_0 = 0.30$  kg water / kg dry solid, exposed over 0.5 m<sup>2</sup> to air of DECLARED, CONSTANT condition: 60 C (333.15 K) and  $Y = 0.010$  kg water / kg dry N<sub>2</sub>. Nothing about that air is integrated – it is an ENVIRONMENT, the FixedBedAdsorber's constant-carrier posture, and the water that leaves the solid leaves the campaign across that boundary (reported as the unit's DECLARED material residual, never as a silent leak).

THE HAND ESTIMATE, computed from this case's own curated records (water Antoine + Watson H<sub>vap</sub> + the two ideal-gas C<sub>p</sub> polynomials, the case-local GAB isotherm on sucrose) – the run must reproduce it:

air water activity  $p_v = P Y / (Y + M_v / M_c) = 1551$  Pa  $a_w = p_v / P_{\text{sat}}(333.15 \text{ K}) = 0.0774$  GAB equilibrium  $K a_w = 0.0743$   $X_{\text{eq}} = X_m C K a / ((1 - K a)(1 - K a + C K a)) = 0.01580$  kg/kg wet bulb (Lewis = 1)  $T_{\text{wb}} = 300.53$  K (27.4 C, a 32.6 K depression) driving force  $Y_{\text{sat}}(T_{\text{wb}}) - Y = 0.014011$  kg/kg constant-rate flux  $R_c = k_Y dY = 0.05 * 0.014011 = 7.0057\text{e-}4$  kg/(m<sup>2</sup> s)  $R_c A = 3.5028\text{e-}4$  kg/s latent load  $R_c A \lambda(T_{\text{wb}}) = 0.8645$  kW (the AIR's, not the campaign's – KPI only, never ledgered) break time  $t_c = m_s (X_0 - X_c) / (R_c A) = 1027.7$  s falling-rate constant  $\tau = m_s (X_c - X_{\text{eq}}) / (R_c A) = 595.0$  s end of run  $X(3000 \text{ s}) = X_{\text{eq}} + (X_c - X_{\text{eq}}) \exp(-(3000 - t_c) / \tau) = 0.019584$  kg/kg

What to look at in the output: trajectory.csv columns dryer.X\_kg\_kg and dryer.R\_kg\_m2\_s: the drying curve and the rate curve, engine columns, not a post-processing. R is FLAT at 7.0057e-4 until X falls through  $X_c = 0.12$ , then decays linearly in  $(X - X_{\text{eq}})$  – the textbook break, visible as a corner. the KPI  $t_{\text{critical}}$  is OBSERVED (the interpolated crossing between two accepted states), not the analytic 1027.7 s above; they agree because  $dX/dt$  is exactly constant on that leg. THE CAMPAIGN MASS BALANCE DOES NOT CLOSE, AND IT SHOULD NOT. Look at it, because it is the first thing an examiner will point at: it reports 2.600000 kg in against 2.039168 kg out, a 21.6 % shortfall. That shortfall is the DRIED WATER. It left the solid and went into the air – and the air here is a DECLARED ENVIRONMENT (a temperature and a humidity), not a stream this campaign owns, so there is no vessel on the books for the water to arrive in. The vessel's inventory genuinely shrinks.

What makes that a stated model boundary rather than a leak is that the unit PREDICTS the gap independently, from its own evaporation history, and the campaign checks the two against each other: the observed shortfall matches the declared prediction to 7e-17 (element\_worst\_closure\_vs\_declared\_rel). The raw numbers are never discounted – 21.6 % is what the balance says and what the balance reports. If this dryer ever lost matter for a REAL reason, that cross-check is what would move, and it is pinned in 'expected'.

So the honest sentence to give at a viva is: "the balance is open at the air boundary by construction, the opening is the 0.5608 kg of water evaporated, and the model reproduces that number from the other side." Closing it for real means making the drying air a ledgered stream, which this unit deliberately does not do.

the campaign energy balance is UNAVAILABLE, and says why in one sentence: the drying heat comes from air this campaign never owned. The number is not hidden – it is the latentDuty\_kW KPI.

Method hypotheses, each announced at run time: Lewis = 1 in the adiabatic-saturation equation; the humid heat evaluated at the air's own temperature; the surface held at  $T_{\text{wb}}$  while  $X > X_c$ ; the LINEAR falling-rate law (a modelling CHOICE, not physics); the solid's own temperature NOT integrated.



**What to expect (golden-locked):**

| kind | unit/stream | key                                   | expected          |
|------|-------------|---------------------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel                 | 0                 |
| kpi  | campaign    | element_H_closure_rel                 | 0.319043490708    |
| kpi  | campaign    | element_O_closure_rel                 | 0.319043490708    |
| kpi  | campaign    | element_worst_closure_rel             | 0.319043490708    |
| kpi  | campaign    | element_worst_closure_vs_declared_rel | 7.11118003994e-17 |
| kpi  | campaign    | energy_balance_available              | 0                 |
| kpi  | campaign    | energy_records_invalid                | 0                 |
| kpi  | campaign    | energy_records_valid                  | 0                 |
| kpi  | campaign    | mass_closure_rel                      | 0.215704585481    |
| kpi  | campaign    | mass_kg_external_out                  | 0                 |
| kpi  | campaign    | mass_kg_final                         | 2.03916808853     |
| kpi  | campaign    | mass_kg_initial                       | 2.60000001375     |
| kpi  | campaign    | mass_residual_kg                      | 0.560831925217    |
| kpi  | campaign    | moles_kmol_external_out               | 0                 |

...and 21 more reference values in the case's *expected*.

**membrane**

**diafilter01\_nf\_desalting**

tutorials/batch/membrane/diafilter01\_nf\_desalting

**retentate**  
 batchDiafilter

**What it does:** Constant-volume NF diafiltration: NaCl washed out of an MgSO<sub>4</sub> retentate, with the OBSERVED rejection published beside the constant-R washout law it violates

**The story:** constant-volume diafiltration on a loose NF membrane, and the exponential washout law failing in a way you can watch.

**THE OPERATION.** 10 litres of a brine carrying TWO salts sit behind 5 m<sup>2</sup> of NF270 at 10 bar. Solvent is made up as fast as it permeates, so the volume is held and the run is measured in DIAVOLUMES – how many vessel volumes of wash water have passed through. NaCl is poorly rejected by this membrane and leaves; MgSO<sub>4</sub> is well rejected and stays. That is the whole point of a nanofiltration desalting step, and it is the same arithmetic whether the retained species is a divalent salt, a sugar or a protein.

**THE LESSON.** Derive the washout by hand. Over a constant volume  $V$ , with the permeate carrying  $c_p = (1 - R) c$ ,

$$V \, dc/dt = -Q_p (1 - R) c, \quad N = Q_p t / V \Rightarrow c/c_0 = \exp(-(1 - R) N)$$

which is a straight line on a log axis against  $N$ . The derivation needs  $R$  to be CONSTANT. It is not:

as NaCl leaves, the osmotic pressure of the retentate falls, so the driving force rises and so does the water flux; a higher flux sweeps more solute to the wall, so the polarisation factor rises and the WALL concentration is no longer the bulk one; the observed rejection is measured against the bulk, so it moves even when the membrane's own permeability has not changed.

The run therefore publishes  $R_{obs}$  for each solute at every instant, beside the diavolumes, and – because this is constant-volume mode – two KPIs per solute: ‘washoutActual\_<name>’ (what the vessel did) and ‘washoutIdeal\_<name>’ (what  $\exp(-(1-R)N)$  predicts using THIS run's own initial  $R$ ). The idealisation is computed from the run, not declared, so the gap between them cannot have been arranged.

Read them together. Where the two agree, the hand derivation was safe. Where they part, the  $R_{obs}$  column says which assumption failed and when.

**WHAT IS PINNED AND WHAT IS NOT.** The golden pins the answer this model gives; NOTHING here is validated against a measured diafiltration. The NF270 permeabilities are the case-fitted pedagogical set the record declares (‘provenance fittedToCase’, fitted against membrane02\_NF\_sugar), so the rejections are representative of a loose NF membrane and are not a claim about a particular element. Fouling is NOT modelled – the flux in a real rig would decline with time and this one does not.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_Cl_closure_rel    | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 1.23912409123e-16 |
| kpi  | campaign    | element_Mg_closure_rel    | 0                 |
| kpi  | campaign    | element_Na_closure_rel    | 0                 |
| kpi  | campaign    | element_O_closure_rel     | 1.23884758239e-16 |
| kpi  | campaign    | element_S_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 1.23912409123e-16 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 0                 |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_amend_in          | 54.5637624826     |
| kpi  | campaign    | mass_kg_amend_out         | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 54.5637624826     |

*... and 34 more reference values in the case's expected.*

**diafilter02\_fouling\_decline**

tutorials/batch/membrane/diafilter02\_fouling\_decline

**retentate**  
 batchDiafilter

**What it does:** The same diafiltration with a DECLARED Hermia cake-law decline: what fouling costs a washout schedule

**The story:** the same diafiltration, with the membrane losing permeance as it works.

WHY THIS CASE EXISTS. diafilter01's flux RISES during the run (233 → 256 LMH): the salt leaves, the osmotic pressure falls, the driving force grows. That is the physics the model has. It is not what a rig does. On real equipment the flux falls, because the membrane fouls, and every schedule a student will ever write is sized on that decline.

THE MODEL. Resistance in series on the permeance the unit hands the transport law:

$$1 / A_{\text{eff}} = 1 / A_w + r_f(v), v = V_{\text{permeated}} / A$$

with  $r_f$  growing by one of HERMIA'S BLOCKING LAWS (Hermia, Trans. IChemE 60:183-187, 1982). Here 'cake':  $r_f = k v$ , the deposited-layer form. Applied where the transport CONTEXT is assembled, so solution-diffusion and DSPM-DE are served alike and neither is modified.

THE CONSTANT IN THIS CASE IS HYPOTHETICAL AND IS NOT A FITTED VALUE FOR THIS SYSTEM. \*\* A brine of NaCl and MgSO<sub>4</sub> is fully dissolved; there is no suspended matter to deposit, so 'cake' is not a mechanism this feed can honestly claim. What this case demonstrates is the CONSEQUENCE of a declared decline on a diafiltration schedule – how many more minutes the same washout costs – on a system whose other numbers are already pinned by diafilter01, so the difference between the two runs is the fouling and nothing else. It is a STRUCTURAL witness, in the same sense as edwards01\_sour\_water\_activity: no number in it is validated against a measured fouling experiment, and the tree carries no fouling data at all.

A fouling constant belongs to a RIG AND A MATERIAL, not to a membrane, so it is declared here in the case and no catalogue record is touched.

WHAT TO READ. 'A\_eff\_over\_A\_w' in the trajectory is the permeance the membrane still has, as a fraction of clean. Put diafilter01 and diafilter02 side by side: same charge, same area, same pressure, and the diavolume count at  $t = 160$  s is the whole lesson.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_Cl_closure_rel    | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 4.17214965497e-16 |
| kpi  | campaign    | element_Mg_closure_rel    | 0                 |
| kpi  | campaign    | element_Na_closure_rel    | 0                 |
| kpi  | campaign    | element_O_closure_rel     | 4.17110477583e-16 |
| kpi  | campaign    | element_S_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 4.17214965497e-16 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 0                 |
| kpi  | campaign    | mass_closure_rel          | 3.70318765373e-16 |
| kpi  | campaign    | mass_kg_amend_in          | 47.5260962641     |
| kpi  | campaign    | mass_kg_amend_out         | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 47.5260962641     |

... and 35 more reference values in the case's *expected*.

**reactor**

**The theory you need.** Three idealised steady reactors. Conversion reactor: stoichiometry only,  $n_{\text{out}} = n_{\text{in}} + \nu \xi$ , duty from the ONE enthalpy base  $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$  (formation datum – no separate “heat of reaction” input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate  $r(c_{\text{out}}, T)$ , so  $F(c_{\text{in}} - c_{\text{out}}) = rV$ . PFR: the same rate integrated along the volume,  $dF_i/dV = \nu_i r$ . Arrhenius kinetics  $k = AT^b e^{-E_a/RT}$ , orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential  $T$  sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

**batch01\_first\_order**

tutorials/batch/reactor/batch01\_first\_order

**reactor**  
 batchReactor

**What it does:** Isothermal batch Fischer esterification ( $\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$ ) at 350 K, pseudo-first-order in ethanol; analytical exp-decay comparison

**The story:** isothermal pseudo-first-order batch reactor.

Single vessel, pseudo-first-order esterification ( $\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$ , rate =  $k \cdot [\text{ethanol}]$ ). Initial charge is an equimolar ethanol + acetic acid mix; kinetics from the reactions library (Arrhenius  $A=1\text{e}8$  1/s,  $E_a=70$  kJ/mol). At  $T = 350$  K:

$$k(T) = 1\text{e}8 \cdot \exp(-70000 / (8.314 \cdot 350)) \approx 3.57\text{e-}3 \text{ s}^{-1} \quad \tau = 1/k \approx 280 \text{ s}$$

Analytical solution for first-order in a constant-volume batch:

$$n_A(t) = n_A(0) \cdot \exp(-k t)$$

The simulator integrates  $dn_A/dt = -k \cdot (n_A / V) \cdot V = -k n_A$  with RK4 at  $dt = 1$  s; final  $t = 600$  s gives  $X_A \approx 1 - \exp(-k \cdot 600) \approx 1 - \exp(-2.14) \approx 0.882$

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 3.75075346157e-16 |
| kpi  | campaign    | element_H_closure_rel     | 3.00060276926e-16 |
| kpi  | campaign    | element_O_closure_rel     | 2.50050230771e-16 |
| kpi  | campaign    | element_worst_closure_rel | 3.75075346157e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | -210.150638715    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 0                 |
| kpi  | campaign    | energy_dH_vessels_kJ      | -210.150638715    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 1                 |
| kpi  | campaign    | energy_residual_kJ        | 0                 |
| kpi  | campaign    | mass_closure_rel          | 4.5240474843e-16  |
| kpi  | campaign    | mass_kg_external_out      | 0                 |

... and 25 more reference values in the case's *expected*.

**batch01b\_first\_order\_eulersi**

tutorials/batch/reactor/batch01b\_first\_order\_eulersi

|                                |
|--------------------------------|
| <b>reactor</b><br>batchReactor |
|--------------------------------|

**What it does:** GREEN WITNESS: semi-implicit Euler (EulerSI) on batch01 mild isothermal esterification – first order suffices off-stiffness; twin of the ignition01\_h2o2\_eulersi expected failure.

**The story:** GREEN WITNESS for the EulerSI integrator: batch01's mild isothermal esterification, integrated by semi-implicit (linearly-implicit) Euler instead of the RK4 default. On a non-stiff problem the first-order method simply works, and the golden pins that its answer agrees with the physics batch01 already pins. Its EXPECTED-FAILURE twin is ignition01\_h2o2\_eulersi (stiff nonlinear ignition, where first order goes unphysical and the engine refuses by name). Registered model no case selected until 2026-08-22 (invariant I8).

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 3.75075346157e-16 |
| kpi  | campaign    | element_H_closure_rel     | 4.50090415389e-16 |
| kpi  | campaign    | element_O_closure_rel     | 2.50050230771e-16 |
| kpi  | campaign    | element_worst_closure_rel | 4.50090415389e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | -210.043853155    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 0                 |
| kpi  | campaign    | energy_dH_vessels_kJ      | -210.043853155    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 1                 |
| kpi  | campaign    | energy_residual_kJ        | 0                 |
| kpi  | campaign    | mass_closure_rel          | 5.65505935537e-16 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |

...and 25 more reference values in the case's *expected*.

**batch02\_adiabatic**

tutorials/batch/reactor/batch02\_adiabatic

**reactor**  
 batchReactor

**What it does:** Adiabatic batch esterification ( $\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$ ), exothermic, starting cold (320 K); thermal runaway to ~428 K then plateau as ethanol is consumed

**The story:** exothermic pseudo-first-order esterification ( $\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$ ) in an adiabatic batch reactor. Designed to show thermal runaway:

$T_0 = 320$  K (kinetics very slow:  $k_0 \sim 3.7\text{e-}4 \text{ s}^{-1}$ ,  $\tau_0 \sim 2700$  s).  $dH_{\text{rxn}} = -26.06$  kJ/mol of ethanol, computed from the species' standardThermochemistry (elements datum); lifts  $T \sim 108$  K at full conversion (320  $\rightarrow$  428 K), enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

$T$  | \_\_\_\_\_ plateau (~428 K) | \_/ | \_/ | \_/ | \_/ | /  $T_0$  | \_/ + \_\_\_\_\_  $\rightarrow$   $t$  induction runaway

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant- $C_p$  here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at 428 K.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | reactor     | P_final                   | 101300            |
| kpi  | reactor     | T_final                   | 428.2212252       |
| kpi  | reactor     | n_aceticAcid_final        | 9.87885051063e-31 |
| kpi  | reactor     | n_ethanol_final           | 9.87885051063e-31 |
| kpi  | reactor     | n_ethylAcetate_final      | 0.00925           |
| kpi  | reactor     | n_water_final             | 0.00925           |
| kpi  | reactor     | t_end                     | 600               |
| kpi  | reactor     | totalMoles_final          | 0.0185            |
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_O_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 0                 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |

... and 12 more reference values in the case's *expected*.

**batch03\_consecutive**

tutorials/batch/reactor/batch03\_consecutive

**reactor**  
 batchReactor

**What it does:** Isothermal batch reactor: consecutive  $A \rightarrow B \rightarrow C$  with  $k_1 > k_2$ ; intermediate B peaks then decays

**The story:** two first-order reactions in series:

with  $k_1 > k_2$ . This is the textbook series-reaction setup: B is produced from A faster than it is consumed to C, so  $n_B$  grows at first, reaches a maximum, then decays as A runs out.

Analytical solution (constant V, isothermal, first-order):

$$n_A(t) = n_{A0} * \exp(-k_1 t) \quad n_B(t) = n_{A0} * k_1 / (k_2 - k_1) * [\exp(-k_1 t) - \exp(-k_2 t)] \quad n_C(t) = n_{A0} - n_A(t) - n_B(t)$$

$$t_{\max} = \ln(k_1 / k_2) / (k_1 - k_2)$$

With  $k_1 = 0.01 \text{ 1/s}$  and  $k_2 = 0.002 \text{ 1/s}$ :

$$t_{\max} = \ln(5) / 0.008 \sim 201 \text{ s} \quad n_{B_{\max}} / n_{A0} = (k_1/k_2)^{k_2/(k_2-k_1)} = 5^{(-0.25)} \sim 0.669$$

The case demonstrates: 1. multi-reaction support in BatchReactor (v0.21); 2. time-dependent selectivity — harvesting B at  $t \sim 200 \text{ s}$  maximises yield, waiting longer wastes B to C.

No species supplies activation energy: both  $A_{\text{pre}}$  are quoted with  $E_a = 0$  so that the rate constants are exactly the values above regardless of T (the case is isothermal anyway).

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | campaign    | energy_balance_available | 0                 |
| kpi  | campaign    | energy_records_invalid   | 1                 |
| kpi  | campaign    | energy_records_valid     | 0                 |
| kpi  | campaign    | mass_closure_rel         | 5.55111512313e-16 |
| kpi  | campaign    | mass_kg_external_out     | 0                 |
| kpi  | campaign    | mass_kg_final            | 0.5               |
| kpi  | campaign    | mass_kg_initial          | 0.5               |
| kpi  | campaign    | mass_residual_kg         | 2.77555756156e-16 |
| kpi  | campaign    | transfers_H_invalid      | 0                 |
| kpi  | campaign    | transfers_H_valid        | 0                 |
| kpi  | campaign    | transfers_logged         | 0                 |
| kpi  | reactor     | P_final                  | 101300            |
| kpi  | reactor     | T_final                  | 350               |
| kpi  | reactor     | n_compA_final            | 3.05902320887e-09 |

...and 7 more reference values in the case's *expected*.



**batch04\_reversible\_wgs**

tutorials/batch/reactor/batch04\_reversible\_wgs

**reactor**  
 batchReactor

**What it does:** Batch reversible water-gas shift ( $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ ) at 600 K; conversion relaxes to  $X_{\text{eq}}$

**The story:** the reversible water-gas shift in a closed BATCH reactor (choupoBatch), the time-domain sibling of cstr02 / pfr02.

Equimolar  $\text{CO} + \text{H}_2\text{O}$  charged at 600 K. The forward Arrhenius rate drives conversion up; the reverse term ( $k_{\text{rev}} = k_{\text{fwd}} / K_{\text{eq}}$ ,  $K_{\text{eq}}$  from the four species'  $g_{\text{pure\_ig}}$ , re-evaluated each step) brakes it. The batch trajectory relaxes onto the SAME thermodynamic ceiling the CSTR and PFR reached —  $X \rightarrow X_{\text{eq}} \sim 0.84$  — then sits there (net rate  $\rightarrow 0$ ), instead of running to  $X = 1$ .

The forward  $A$  is the illustrative value tuned for stable RK4 (see constant/reactions); the equilibrium  $X_{\text{eq}}$  is  $A$ -independent.

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key                       | expected          |
|-----------------------------------------------------------------|-------------|---------------------------|-------------------|
| kpi                                                             | reactor     | P_final                   | 1000000           |
| kpi                                                             | reactor     | T_final                   | 600               |
| kpi                                                             | reactor     | n_CO2_final               | 0.000420715762123 |
| kpi                                                             | reactor     | n_CO_final                | 7.92842378768e-05 |
| kpi                                                             | reactor     | n_H2_final                | 0.000420715762123 |
| kpi                                                             | reactor     | n_water_final             | 7.92842378768e-05 |
| kpi                                                             | reactor     | t_end                     | 1200              |
| kpi                                                             | reactor     | totalMoles_final          | 0.001             |
| kpi                                                             | campaign    | element_C_closure_rel     | 1.95156391047e-15 |
| kpi                                                             | campaign    | element_H_closure_rel     | 1.95156391047e-15 |
| kpi                                                             | campaign    | element_O_closure_rel     | 1.95156391047e-15 |
| kpi                                                             | campaign    | element_worst_closure_rel | 1.95156391047e-15 |
| kpi                                                             | campaign    | energy_H_external_kJ      | 0                 |
| kpi                                                             | campaign    | energy_Q_ledger_kJ        | -0.299204643882   |
| ...and 25 more reference values in the case's <i>expected</i> . |             |                           |                   |

batch04\_transient\_dirs

tutorials/batch/reactor/batch04\_transient\_dirs

reactor

batchReactor

**What it does:** Adiabatic batch esterification with OpenFOAM-style REAL-TIME instant dirs: watch conversion climb 0 → 1 and T runaway 320 → ~428 K by scrubbing 0/ 30/ 60/ ... s

**The story:** the batch02\_adiabatic runaway re-run with REAL-TIME instant directories: the integrator dumps 0/ 30/ 60/ ... s snapshots (OpenFOAM-style) so the trajectory – conversion 0 → 1, T 320 → ~428 K – can be scrubbed frame by frame in the GUI.

The physics below is inherited from batch02\_adiabatic, unchanged: exothermic pseudo-first-order esterification (AcOH + EtOH -> EtOAc + H2O) in an adiabatic batch reactor. Designed to show thermal runaway:

T\_0 = 320 K (kinetics very slow:  $k_0 \sim 3.7\text{e-}4 \text{ s}^{-1}$ ,  $\tau_0 \sim 2700 \text{ s}$ ).  $dH_{\text{rxn}} = -26.06 \text{ kJ/mol}$  of ethanol, computed from the species' standardThermochemistry (elements datum); lifts T ~108 K at full conversion, enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

T | \_\_\_\_\_ plateau (~428 K) | \_/ | \_/ | \_/ | \_/ | / T\_0 | \_/ | + \_\_\_\_\_ → t induction runaway

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at ~430 K.

What to expect (golden-locked):

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | reactor     | P_final                   | 101300            |
| kpi  | reactor     | T_final                   | 428.2212252       |
| kpi  | reactor     | n_aceticAcid_final        | 9.87885051063e-31 |
| kpi  | reactor     | n_ethanol_final           | 9.87885051063e-31 |
| kpi  | reactor     | n_ethylAcetate_final      | 0.00925           |
| kpi  | reactor     | n_water_final             | 0.00925           |
| kpi  | reactor     | t_end                     | 600               |
| kpi  | reactor     | totalMoles_final          | 0.0185            |
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_O_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 0                 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |

...and 12 more reference values in the case's *expected*.

**batch05\_crystalliser**

tutorials/batch/reactor/batch05\_crystalliser

|                                   |
|-----------------------------------|
| <b>cryst</b><br>batchCrystalliser |
|-----------------------------------|

**What it does:** Batch crystalliser: supersaturated sugar relaxes, PBE by method of moments

**The story:** Single closed vessel. Charge: ~500 kg water + ~1250 kg dissolved sucrose ( $c_0 \sim 2.5$  kg/kg, above  $c_{\text{sat}}(20\text{ C}) \sim 2.03$  kg/kg, so  $S_0 \sim 1.23$ ). Unseeded: the moments start at zero and primary nucleation ( $j=0$ ) bootstraps the population from the initial supersaturation. Kinetics from constant/crystallisation.

**What to expect (golden-locked):**

| kind                                                            | unit/stream | key                       | expected          |
|-----------------------------------------------------------------|-------------|---------------------------|-------------------|
| kpi                                                             | campaign    | element_C_closure_rel     | 6.81588841762e-15 |
| kpi                                                             | campaign    | element_H_closure_rel     | 4.18663266355e-15 |
| kpi                                                             | campaign    | element_O_closure_rel     | 4.18663266355e-15 |
| kpi                                                             | campaign    | element_worst_closure_rel | 6.81588841762e-15 |
| kpi                                                             | campaign    | energy_balance_available  | 1                 |
| kpi                                                             | campaign    | energy_records_invalid    | 0                 |
| kpi                                                             | campaign    | energy_records_valid      | 1                 |
| kpi                                                             | campaign    | mass_closure_rel          | 5.07045231785e-15 |
| kpi                                                             | campaign    | mass_kg_external_out      | 0                 |
| kpi                                                             | campaign    | mass_kg_final             | 1748.87224776     |
| kpi                                                             | campaign    | mass_kg_initial           | 1748.87224776     |
| kpi                                                             | campaign    | mass_residual_kg          | 8.86757334229e-12 |
| kpi                                                             | campaign    | transfers_H_invalid       | 0                 |
| kpi                                                             | campaign    | transfers_H_valid         | 0                 |
| ...and 21 more reference values in the case's <i>expected</i> . |             |                           |                   |

batch06\_adaptive\_runaway

tutorials/batch/reactor/batch06\_adaptive\_runaway

reactor

batchReactor

**What it does:** Adiabatic esterification runaway (AcOH+EtOH); ADAPTIVE Rosenbrock23 time-stepping — the step grows after the fast transient (glass-box step history)

**The story:** the batch02\_adiabatic runaway integrated with ADAPTIVE Rosenbrock23 time-stepping: the step shrinks through the fast runaway and GROWS again on the plateau, and the step history is printed glass-box so the student sees the controller at work.

The physics below is inherited from batch02\_adiabatic, unchanged: exothermic pseudo-first-order esterification ( $\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$ ) in an adiabatic batch reactor. Designed to show thermal runaway:

$T_0 = 320\text{ K}$  (kinetics very slow:  $k_0 \sim 3.7\text{e-}4\text{ s}^{-1}$ ,  $\tau_0 \sim 2700\text{ s}$ ).  $dH_{\text{rxn}} = -26.06\text{ kJ/mol}$  of ethanol, computed from the species' standardThermochemistry (elements datum); lifts  $T \sim 108\text{ K}$  at full conversion, enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

$T \mid \rule{1.5cm}{0.4pt} \text{plateau } (\sim 428\text{ K}) \mid \_ / \mid \_ / \mid \_ / \mid \_ / \mid / T_0 \mid \_ / + \rule{2.5cm}{0.4pt} \rightarrow t \text{ induction runaway}$

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at  $\sim 428\text{ K}$ .

What to expect (golden-locked):

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | reactor     | P_final                   | 101300            |
| kpi  | reactor     | T_final                   | 428.221319512     |
| kpi  | reactor     | n_aceticAcid_final        | 8.88140514762e-19 |
| kpi  | reactor     | n_ethanol_final           | 1.10133235387e-34 |
| kpi  | reactor     | n_ethylAcetate_final      | 0.00925           |
| kpi  | reactor     | n_water_final             | 0.00925           |
| kpi  | reactor     | t_end                     | 600               |
| kpi  | reactor     | totalMoles_final          | 0.0185            |
| kpi  | campaign    | element_C_closure_rel     | 1.87537673079e-16 |
| kpi  | campaign    | element_H_closure_rel     | 1.50030138463e-16 |
| kpi  | campaign    | element_O_closure_rel     | 2.50050230771e-16 |
| kpi  | campaign    | element_worst_closure_rel | 2.50050230771e-16 |
| kpi  | campaign    | energy_balance_available  | 0                 |
| kpi  | campaign    | energy_records_invalid    | 0                 |

...and 12 more reference values in the case's *expected*.

**batch07\_lhhw\_methylAcetate**

tutorials/batch/reactor/batch07\_lhhw\_methylAcetate

|                                |
|--------------------------------|
| <b>reactor</b><br>batchReactor |
|--------------------------------|

**What it does:** LHHW in a BATCH reactor: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

**The story:** the SAME rate law as cstr07, in a BATCH vessel.

$\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$  over Amberlyst 15

Popken, Gotze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601, Eq 16: the adsorption-based LHHW on UNIQUAC activities. Nothing about the rate law is restated here – it lives in constant/reactions, exactly as the steady CSTR reads it, because the SAME ‘RateLaw’ evaluates it in both. A batch reactor and a continuous one cannot disagree about what an adsorption denominator means.

What changes is only what a batch vessel does with a rate: it integrates it. The charge is equimolar methanol + acetic acid; the trajectory climbs to the ACTIVITY equilibrium,  $X = 76.6\%$ , and stops. That plateau is asserted nowhere: it is where  $k_1 a'_{\text{HOAc}} a'_{\text{MeOH}} = k_{-1} a'_{\text{MeOAc}} a'_{\text{H}_2\text{O}}$ , i.e. the  $K_a = 25.50$  the authors’ independent Eq 17 recovers from the same eight constants – and the same number the steady CSTR climbs to as its volume grows. Two reactors, two integration schemes, one equilibrium.

Water is the strongest adsorber ( $K/M = 0.29$  against  $0.052$  for the acid), so as the reaction MAKES water it crowds the acid off the resin and the denominator grows. That self-inhibition is why reactive distillation, which pulls the water out as it forms, is worth its capital.

Do NOT try to see the inhibition by deleting the ‘adsorption’ block and keeping  $k_1$ . It tells you nothing: the constants were REGRESSED for this rate law, on these activities, and a rate constant without its rate law is a number without a meaning. (Delete the denominator and the reaction gets FASTER, because the adsorbed activities  $a'$  are small and dividing by their square is a multiplication. That is an arithmetic accident, not physics.) To SEE the inhibition, spike the charge with water at the same acid and alcohol fractions:

methanol 0.4, aceticAcid 0.4, water 0.0 → initial rate 58.3 mmol/(L.s) methanol 0.4, aceticAcid 0.4, water 0.2 → initial rate 29.0 mmol/(L.s)

Half the rate, with every reactant fraction unchanged. The water is not a reactant here; it is sitting on the sites.

‘catalystLoading’ (kg of dry resin per  $\text{m}^3$ ) is what turns the paper’s per-gram rate constants into a volumetric rate. With concentrations in  $\text{kmol}/\text{m}^3$  the conversion is exact and unit-free:  $\text{mol}/(\text{g.s}) \times \text{kg}/\text{m}^3 = \text{kmol}/(\text{m}^3.\text{s})$ .

The vessel runs isothermal at 333.15 K, the top of the authors’ data range. Do not extrapolate: their activation energies are APPARENT values, fitted over 303-333 K.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | reactor     | P_final                   | 101300            |
| kpi  | reactor     | T_final                   | 333.15            |
| kpi  | reactor     | n_aceticAcid_final        | 0.00216454708498  |
| kpi  | reactor     | n_methanol_final          | 0.00216454708498  |
| kpi  | reactor     | n_methylAcetate_final     | 0.00708545291502  |
| kpi  | reactor     | n_water_final             | 0.00708545291502  |
| kpi  | reactor     | t_end                     | 3600              |
| kpi  | reactor     | totalMoles_final          | 0.0185            |
| kpi  | campaign    | element_C_closure_rel     | 2.00040184617e-15 |
| kpi  | campaign    | element_H_closure_rel     | 1.87537673079e-15 |
| kpi  | campaign    | element_O_closure_rel     | 2.12542696156e-15 |
| kpi  | campaign    | element_worst_closure_rel | 2.12542696156e-15 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | -218.009898425    |

...and 25 more reference values in the case's *expected*.

**batch08\_isothermal\_duty**

tutorials/batch/reactor/batch08\_isothermal\_duty

**reactor**  
 batchReactor

**What it does:** Isothermal jacket duty from a DECLARED  $dH_{rxn}$ :  $E = dH \cdot dX_i$  via extents (lumped species, no formation data)

**The story:** one isothermal batch reactor, one declared  $dH_{rxn}$ , one hand-checkable jacket duty. See system/controlDict for what this case tests and constant/reactions for the analytic anchor ( $E = -50 \text{ kJ/mol} \times 0.9975 \text{ mol} = -49.88 \text{ kJ}$  over 600 s).

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | campaign    | energy_balance_available | 0                 |
| kpi  | campaign    | energy_records_invalid   | 0                 |
| kpi  | campaign    | energy_records_valid     | 1                 |
| kpi  | campaign    | mass_closure_rel         | 1.11022302463e-15 |
| kpi  | campaign    | mass_kg_external_out     | 0                 |
| kpi  | campaign    | mass_kg_final            | 0.05              |
| kpi  | campaign    | mass_kg_initial          | 0.05              |
| kpi  | campaign    | mass_residual_kg         | 5.55111512313e-17 |
| kpi  | campaign    | transfers_H_invalid      | 0                 |
| kpi  | campaign    | transfers_H_valid        | 0                 |
| kpi  | campaign    | transfers_logged         | 0                 |
| kpi  | reactor     | E_reaction_seg0_kJ       | -49.8760623911    |
| kpi  | reactor     | E_reaction_total_kJ      | -49.8760623911    |
| kpi  | reactor     | P_final                  | 101300            |

...and 12 more reference values in the case's *expected*.

**recipes**

**The theory you need.** A recipe is time-triggered discrete control over batch units: charge, heat/cool, hold, transfer, sample. Between events the DAE integrator runs; at each event the state jumps (a transfer moves holdup between vessels). Expect exact event times on the trajectory and holdup bookkeeping that conserves mass across transfers.

recipe01\_react\_then\_distill

tutorials/batch/recipes/recipe01\_react\_then\_distill



**What it does:** Two-vessel batch recipe: esterify ( $\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$ ) in a reactor, transfer the charge to a still, then distil

**The story:** the canonical 2-stage batch recipe.

Stage 1 ( $t = 0 \rightarrow 400$  s): A batch reactor at 350 K esterifies ethanol + acetic acid  $\rightarrow$  ethyl acetate + water (the same first-order Arrhenius reaction as batch01).  $k \approx 3.58\text{e-}3 \text{ s}^{-1}$  at 350 K,  $\tau \approx 280$  s. Over 400 s the ethanol conversion reaches  $\sim 76$  %.

Stage 2 ( $t = 400 \rightarrow 1200$  s): The recipe TRANSFERS the reactor contents to an initially empty batch still. The still then runs a Rayleigh distillation at  $F_{\text{vap}} = 2.0\text{e-}5 \text{ kmol/s}$  for 800 s, vaporising 0.016 kmol ( $\sim 85$  % of the charge) – the condensed distillate is routed, step by step, into a passive batchAccumulator (‘dischargeTo receiver;’), so the campaign ends with the ester-rich distillate IN a vessel while the ethanol-depleted residue stays in the still’s own pot (the pot IS the bottoms tank; no third transfer needed). Mass is conserved across the three vessels at every accepted step. Starting from the converted mix (ethyl acetate + water + unreacted ethanol/acid), the still progressively strips the more volatile ethyl acetate + ethanol; pot composition follows the Raoult y-x curve. (The thermoPhysPropDict here uses ideal Raoult for pedagogical simplicity.)

This case demonstrates the recipe layer (v0.23): a list of time-triggered events acting on named units. Each event in the ‘recipe’ block below has a ‘time’, an ‘action’, and action-specific keys. Actions implemented in v0.23:

transfer move all contents of ‘from’ into ‘to’ setParameter change a named operating parameter on a unit

Events fire at the next time-loop step boundary after  $t_{\text{trigger}}$ ; the trajectory CSV thus shows a single-step discontinuity in the transferred quantities.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 1.87537673079e-16 |
| kpi  | campaign    | element_H_closure_rel     | 3.00060276926e-16 |
| kpi  | campaign    | element_O_closure_rel     | 5.00100461543e-16 |
| kpi  | campaign    | element_worst_closure_rel | 5.00100461543e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | -145.006283394    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 7.81829067932e-11 |
| kpi  | campaign    | energy_dH_vessels_kJ      | -145.006282333    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 4                 |
| kpi  | campaign    | energy_residual_kJ        | 1.06158961444e-06 |
| kpi  | campaign    | mass_closure_rel          | 2.26202374215e-16 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |

... and 51 more reference values in the case’s *expected*.



**recipe02\_setpoint\_ramp**

tutorials/batch/recipes/recipe02\_setpoint\_ramp

**reactor**  
 batchReactor

**What it does:** Single-reactor recipe with T\_setpoint steps: 320 K → 350 K @ 200 s → 380 K @ 500 s

**The story:** isothermal batch reactor with a staged temperature programme controlled by the recipe layer.

The reactor starts at 320 K, where the first-order Arrhenius reaction is very slow ( $k_{320} \approx 3.7\text{e-}4 \text{ s}^{-1}$ ,  $\tau \approx 2700 \text{ s}$ ). Two recipe events ramp the setpoint upward:

$t = 200 \text{ s} \rightarrow 320 \text{ K} \rightarrow 350 \text{ K}$  ( $k$  jumps  $\sim 10\text{x}$  to  $3.58\text{e-}3 \text{ s}^{-1}$ )  $t = 500 \text{ s} \rightarrow 350 \text{ K} \rightarrow 380 \text{ K}$  ( $k$  jumps another  $\sim 7\text{x}$  to  $0.025 \text{ s}^{-1}$ )

In the trajectory CSV the temperature column shows a clean step-change at each event, and the conversion curve shows the corresponding three regimes:

$0 \leq t < 200 \text{ s}$  barely-moving plateau (slow)  $200 \leq t < 500 \text{ s}$  visible drop in  $n_{\text{ethanol}}$  (moderate)  $500 \leq t$  rapid quench to full conversion (fast)

Demonstrates the recipe layer's 'setParameter' action without inter-unit transfers, so this is the cleanest possible test of the parameter-change pathway.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 3.75075346157e-16 |
| kpi  | campaign    | element_H_closure_rel     | 3.00060276926e-16 |
| kpi  | campaign    | element_O_closure_rel     | 2.50050230771e-16 |
| kpi  | campaign    | element_worst_closure_rel | 3.75075346157e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | -104.365240634    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 0                 |
| kpi  | campaign    | energy_dH_vessels_kJ      | -104.365240634    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 5                 |
| kpi  | campaign    | energy_residual_kJ        | 0                 |
| kpi  | campaign    | mass_closure_rel          | 4.5240474843e-16  |
| kpi  | campaign    | mass_kg_external_out      | 0                 |

...and 33 more reference values in the case's *expected*.

**recipe03\_conditional\_partial**

tutorials/batch/recipes/recipe03\_conditional\_partial

**reactor**  
 batchReactor

**still**  
 batchStill

**What it does:** Conditional + partial recipe: react until  $x_{\text{ethanol}} < 0.30$ , then transfer 50% to a still

**The story:** recipe extensions (v0.50).

Two recipe features beyond the time-triggered transfer/setParameter of v0.23:

1. CONDITION-TRIGGERED event ('when { ... }' instead of 'time'): the event fires the first step its condition becomes true, evaluated on the live vessel state. Quantities: T, total,  $n_{\text{<comp>}}$ ,  $x_{\text{<comp>}}$ ; operators: gt, ge, lt, le.
2. PARTIAL transfer ('fraction'): a fraction (0..1) of the source moves; the source keeps the rest. Default 1.0 == empty the source.

Here the reactor runs the esterification (ethanol + acetic acid  $\rightarrow$  ethyl acetate + water; the batch01 first-order Arrhenius test,  $k \sim 3.58\text{e-}3 \text{ s}^{-1}$  at 350 K, so  $n_{\text{ethanol}} = n_{\text{ethanol}}(0) \cdot \exp(-k t)$ ). The charge is equimolar ( $x_{\text{ethanol}}$  starts at 0.5); when the ethanol mole fraction drops below 0.30, the recipe transfers HALF the reactor contents to a still; the reactor keeps the other half and reacts on, while the still distils its charge.

**What to expect (golden-locked):**

| kind | unit/stream | key                  | expected          |
|------|-------------|----------------------|-------------------|
| kpi  | reactor     | P_final              | 101300            |
| kpi  | reactor     | T_final              | 350               |
| kpi  | reactor     | n_aceticAcid_final   | 6.33853236066e-05 |
| kpi  | reactor     | n_ethanol_final      | 6.33853236066e-05 |
| kpi  | reactor     | n_ethylAcetate_final | 0.00456161467639  |
| kpi  | reactor     | n_water_final        | 0.00456161467639  |
| kpi  | reactor     | t_end                | 1200              |
| kpi  | reactor     | totalMoles_final     | 0.00925           |
| kpi  | still       | P_final              | 101300            |
| kpi  | still       | T_final              | 390.833339675     |
| kpi  | still       | n_aceticAcid_final   | 0                 |
| kpi  | still       | n_ethanol_final      | 0                 |
| kpi  | still       | n_ethylAcetate_final | 0                 |
| kpi  | still       | n_water_final        | 0                 |

... and 43 more reference values in the case's *expected*.

**recipe04\_charge\_enthalpy**

tutorials/batch/recipes/recipe04\_charge\_enthalpy

**hot**  
 batchAccumulator

**cold**  
 batchAccumulator

**What it does:** H-equality recipe charge:  $T_{\text{mix}}$  from the first law, not the molar average; balance CLOSES across a transfer

**The story:** mixing is a FIRST-LAW act.

Two passive tanks. ‘hot’ holds 0.5 mol of ethanol at 340 K; ‘cold’ holds 1.0 mol of water at 300 K. At  $t = 100$  s the recipe empties the hot tank into the cold one. The charge solves the mix temperature from enthalpy equality on the elements datum,

$$H(n_{\text{EtOH}} + n_{\text{W}}, T_{\text{mix}}) = H(n_{\text{EtOH}}, 340 \text{ K}) + H(n_{\text{W}}, 300 \text{ K}),$$

so heat capacity DIFFERENCES matter: ethanol carries  $\sim 112 \text{ J}/(\text{mol K})$  against water’s  $\sim 75$ , and the enthalpy-true  $T_{\text{mix}}$  ( $\sim 317 \text{ K}$ ) sits  $\sim 4 \text{ K}$  above the equal- $C_p$  molar average ( $313.3 \text{ K}$ ) a naive mixing rule gives.

Because both species price on the elements datum, no jacket acts, and nothing leaves, the campaign ENERGY BALANCE is AVAILABLE and closes:  $dH_{\text{vessels}} = 0$  to the mix-solve tolerance — the transfer moved enthalpy between tanks without creating or destroying any.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_O_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 0                 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0                 |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 2.13028428746e-12 |
| kpi  | campaign    | energy_dH_vessels_kJ      | 5.81991344006e-10 |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 0                 |
| kpi  | campaign    | energy_residual_kJ        | 5.81991344006e-10 |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |

... and 21 more reference values in the case’s *expected*.

**still**

**The theory you need.** A batch still is Rayleigh distillation:  $d(n x_i)/dt = -\dot{V} y_i$  with  $y_i$  in equilibrium with the pot – the pot composition DRIFTS as the light component leaves (contrast the steady column, where stages hold profiles constant). Expect the distillate to start rich and fade, and the recipe layer (charge, heat, switch receivers) to drive the run. *Full derivation: Theory Guide, the batch-distillation section.*

**still01\_benzene\_toluene**

tutorials/batch/still/still01\_benzene\_toluene

**still**  
 batchStill

**What it does:** Rayleigh batch distillation: benzene/toluene 50/50, 1 atm, ideal Raoult

**The story:** Rayleigh batch distillation, ideal Raoult.

Pot starts with 1 mol of equimolar benzene/toluene at 1 atm. A constant vapour offtake  $F_{\text{vap}} = 1.5\text{e-}6$  kmol/s strips the pot over 600 s — by linear mass balance the pot should hold

i.e.  $L(600\text{ s}) = 1\text{ mmol} - 0.9\text{ mmol} = 0.1\text{ mmol}$  (90 % distilled).

Benzene is the more volatile component ( $\alpha \sim 2.4$  at 1 atm), so the still pot progressively concentrates in toluene as benzene flashes off preferentially. Classical Rayleigh integration with constant  $\alpha$  gives the closed form

$$\ln(L / L_0) = (1 / (\alpha - 1)) [ \ln(x / x_0) + \alpha * \ln((1 - x_0) / (1 - x)) ]$$

where  $x = x_{\text{benzene}}$  in the still pot. Plotting  $(L/L_0)$  vs  $x_{\text{pot}}$  from trajectory.csv should track this analytical curve to RK4 precision.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 2.66880534766e-16 |
| kpi  | campaign    | element_H_closure_rel     | 1.23908819713e-16 |
| kpi  | campaign    | element_worst_closure_rel | 2.66880534766e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 38.7543769459     |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0.942780166065    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 4.58362894565e-16 |
| kpi  | campaign    | energy_dH_vessels_kJ      | -37.8115967798    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 2                 |
| kpi  | campaign    | energy_residual_kJ        | 1.7763568394e-14  |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 0.0760287317788   |
| kpi  | campaign    | mass_kg_final             | 0.00909876822121  |

...and 29 more reference values in the case's *expected*.

**still02\_ethanol\_water\_azeo**

tutorials/batch/still/still02\_ethanol\_water\_azeo

**still**  
 batchStill

**What it does:** Rayleigh batch distillation: ethanol/water 40/60 with NRTL, residue concentrates in water

**The story:** Rayleigh batch distillation of a non-ideal binary with an azeotrope.

System: ethanol / water with NRTL, pressure 1 atm. Minimum-boiling positive azeotrope at  $x_{\text{eth}} \sim 0.894$  (mole),  $T \sim 351.3$  K.

Pot starts at  $x_{\text{eth}} = 0.40$  (a "wine"-like feed — well BELOW the azeotrope). Boil-off rate  $F_{\text{vap}} = 1.5\text{e-}6$  kmol/s for 600 s consumes 90 % of the pot, leaving a 10 % residue.

Below the azeotrope,  $\gamma_{\text{ethanol}} > 1$  and  $\gamma_{\text{water}} \sim 1$ , so the vapour is markedly richer in ethanol than the liquid ( $y_{\text{eth}} \sim 0.61$  at  $x_{\text{eth}} = 0.40$ ). Ethanol therefore vaporises preferentially and the still pot progressively concentrates in WATER — i.e. the residue at end-of-run is mostly water.

The pedagogical contrast with still01 (ideal Raoult, benzene/toluene) is the SHAPE of the enrichment curve:

Raoult: smooth, alpha-driven ( $\alpha \sim 2.4$ , roughly constant). NRTL : curvature changes as  $x$  crosses concentration regimes in which  $\gamma_{\text{ethanol}}$  shrinks toward 1; the still composition curve  $x_{\text{pot}}(t)$  is noticeably non-linear vs the linear pot-volume curve  $L(t) = L_0 - F_{\text{vap}}t$ .

Crucially, this case demonstrates that a single Rayleigh still cannot cross the azeotrope. Starting below 0.894 in ethanol, the pot can only move DOWNWARD in ethanol fraction; no amount of boiling will lift the pot composition past 0.894. (To break the azeotrope one needs entrainer, pressure swing, or extractive distillation — topics for later tutorials.)

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | still       | P_final                   | 101300            |
| kpi  | still       | T_final                   | 372.446750369     |
| kpi  | still       | n_ethanol_final           | 3.38836968904e-11 |
| kpi  | still       | n_water_final             | 9.99999650927e-05 |
| kpi  | still       | t_end                     | 600               |
| kpi  | still       | totalMoles_final          | 9.9999989764e-05  |
| kpi  | campaign    | element_C_closure_rel     | 1.35525271561e-15 |
| kpi  | campaign    | element_H_closure_rel     | 9.63735264432e-16 |
| kpi  | campaign    | element_O_closure_rel     | 6.50521303491e-16 |
| kpi  | campaign    | element_worst_closure_rel | 1.35525271561e-15 |
| kpi  | campaign    | energy_H_external_kJ      | -249.391762836    |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0.490652916203    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 1.13740334008e-16 |

... and 30 more reference values in the case's *expected*.

**still03\_with\_receiver**

tutorials/batch/still/still03\_with\_receiver

**still**  
 batchStill

**receiver**  
 batchAccumulator

**What it does:** Rayleigh still distilling INTO a receiver tank (dischargeTo accumulator)

**The story:** Rayleigh batch distillation INTO a receiver (accumulator vessel + continuous dischargeTo).

Same pot as still01 (1 mol equimolar benzene/toluene,  $F_{\text{vap}} = 1.5\text{e-}6$  kmol/s, ideal Raoult), but the still's vapour is no longer lost: the 'dischargeTo receiver;' line routes the condensed distillate, step by step, into a passive 'batchAccumulator'. So the trajectory shows the POT depleting AND the RECEIVER filling — the receiver collects the total distillate (amount + composition). Mass is conserved between the two vessels: pot + receiver = 1 mol at all times. Benzene is the more volatile component, so the receiver is benzene-rich early and the cumulative receiver composition drifts toward 0.5 as distillation nears completion (it ends holding everything the pot lost).

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 2.66880534766e-16 |
| kpi  | campaign    | element_H_closure_rel     | 1.23908819713e-16 |
| kpi  | campaign    | element_worst_closure_rel | 2.66880534766e-16 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0.942780166065    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 3.00056993907e-11 |
| kpi  | campaign    | energy_dH_vessels_kJ      | 0.942780163768    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 2                 |
| kpi  | campaign    | energy_residual_kJ        | 2.2974155911e-09  |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |
| kpi  | campaign    | mass_kg_final             | 0.0851275         |

...and 35 more reference values in the case's *expected*.

**still04\_rectifier\_benzene\_toluene**

tutorials/batch/still/still04\_rectifier\_benzene\_toluene

**still**  
 batchStill

**receiver**  
 batchAccumulator

**What it does:** batch rectification: 3 stages + total condenser,  $R=3$ , with receiver tank (dischargeTo accumulator)

**The story:** the CLASSICAL batch rectification (model rectifier, design forum #85): the Rayleigh pot now feeds a column of 3 ideal stages topped by a total condenser at constant reflux  $R = 3$ .

$F_{\text{vap}}$  is the internal BOILUP  $V = 1.5\text{e-}6$  kmol/s; the distillate PRODUCT is  $D = V/(R+1) = 3.75\text{e-}7$  kmol/s and the reflux  $L = R.D$  returns to the column. The pot loses PRODUCT only:  $dn_i/dt = -D.x_{D,i}$ , with  $x_D(t)$  solved at every instant from the quasi-steady cascade (stage balances + bubble-T equilibrium + total-condenser closure – a declared residual, printed on failure).

What to watch (trajectory.csv): still.xD\_benzene starts  $\sim 0.949$  (vs  $\sim 0.71$  for the raw pot vapour – the column working) and DRIFTS DOWN as the pot strips: the constant- $R$  policy trades purity for time. The receiver's ACCUMULATED benzene fraction sits between the instantaneous  $x_D$  and the feed – two different compositions, deliberately both visible. The pot loses only  $D.dt = 0.225$  mmol over 600 s (vs 0.9 mmol in still01: same boilup, but 3/4 of it refluxes).

Physics anchors (verified at authoring):  $x_D(0)$  grows with nStages and  $R$  ( $N=1, R=1$ : 0.806;  $N=2, R=3$ : 0.909;  $N=3, R=3$ : 0.949) and the  $R \rightarrow \text{infinity}$  limit ( $R=200$ : 0.9765) lands inside the Fenske band  $\alpha^{(N+1)}$  for  $\alpha = 2.35\text{--}2.6$ . With nStages 0 and refluxRatio 0 the model reproduces still01's Rayleigh trajectory to the last digit.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 0                 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0.534303338765    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 7.08707897844e-10 |
| kpi  | campaign    | energy_dH_vessels_kJ      | 0.534303321211    |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 2                 |
| kpi  | campaign    | energy_residual_kJ        | 1.75539529579e-08 |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |
| kpi  | campaign    | mass_kg_final             | 0.0851275         |

...and 40 more reference values in the case's *expected*.

**still05\_rectifier\_constant\_xD**

tutorials/batch/still/still05\_rectifier\_constant\_xD

**still**  
 batchStill

**receiver**  
 batchAccumulator

**What it does:** batch rectification, constant-composition policy:  $R(t)$  holds  $x_D$  until  $\text{refluxMax}$ , then  $D=0$

**The story:** the SECOND classical reflux policy (design forum #85-3/4): hold the distillate composition constant by RAISING the reflux as the pot strips, and STOP when the declared ceiling can no longer reach the target.

Same pot and column as still04 (1 mol equimolar benzene/toluene,  $V = 1.5\text{e-}6$  kmol/s, 3 ideal stages + total condenser), but instead of a fixed  $R$  the policy solves  $R(t)$  each step – a BRACKETED solve over  $[0, \text{refluxMax}]$  on a feasibility map, never a bare Newton – so that  $x_D[\text{benzene}] = 0.93$  exactly. As benzene depletes,  $R$  climbs ( $\sim 2$  at the start); when even  $\text{refluxMax} = 8$  cannot reach the target the run announces ‘onRefluxLimit stopDistillation’:  $D = 0$ , the pot stops losing mass, and the campaign idles VISIBLY to  $\text{endTime}$  with  $\text{still.refluxLimitReached} = 1$  in the trajectory – never a magic termination.

What to watch (trajectory.csv):  $\text{still.xD\_benzene}$  pinned at 0.9300 while  $R$  climbs – the mirror image of still04, which held  $R$  and let purity drift;  $\text{still.R}$  rising  $2.0 \rightarrow 8$  (the ceiling), then the limit event; the receiver’s accumulated composition  $\sim 0.93$  (the whole cut is on-spec – THE selling point of this policy); pot inventory constant after  $t_{\text{refluxLimit}}$ .

Declared refusals (each with its own diagnostic, try them):  $x_D\text{-target } 0.5 \rightarrow$  BELOW the  $R=0$  floor (even zero reflux over-purifies);  $x_D\text{-target } 0.995$  with  $\text{refluxMax } 5 \rightarrow$  UNREACHABLE from the charge; ethanol/water NRTL with  $x_D\text{-target } 0.95 \rightarrow$  the 0.894 azeotrope caps the cascade at ANY reflux (the barrier is real VLE, not a message).

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | campaign    | element_C_closure_rel     | 0                 |
| kpi  | campaign    | element_H_closure_rel     | 0                 |
| kpi  | campaign    | element_worst_closure_rel | 0                 |
| kpi  | campaign    | energy_H_external_kJ      | 0                 |
| kpi  | campaign    | energy_Q_ledger_kJ        | 0.690222586439    |
| kpi  | campaign    | energy_balance_available  | 1                 |
| kpi  | campaign    | energy_closure_rel        | 8.89386285064e-10 |
| kpi  | campaign    | energy_dH_vessels_kJ      | 0.69022261392     |
| kpi  | campaign    | energy_records_invalid    | 0                 |
| kpi  | campaign    | energy_records_valid      | 2                 |
| kpi  | campaign    | energy_residual_kJ        | 2.74809703882e-08 |
| kpi  | campaign    | mass_closure_rel          | 0                 |
| kpi  | campaign    | mass_kg_external_out      | 0                 |
| kpi  | campaign    | mass_kg_final             | 0.0851275         |

...and 42 more reference values in the case’s *expected*.



**still06\_ledger\_mixed\_validity**

tutorials/batch/still/still06\_ledger\_mixed\_validity



**What it does:** Ledger H-validity is MONOTONIC: mixed valid/invalid packages never resurrect a poisoned record

**The story:** MONOTONIC enthalpy validity on the material ledger (the adversarial case of design forum #99/#101).

The campaign ledger integrates each transfer's enthalpy PACKAGE BY PACKAGE at each package's own temperature. A package containing a species with NO formation datum (here 'compA', a lumped teaching species that deliberately carries no standardThermochemistry block) cannot be priced — the record must mark H invalid and NAME the species. The contract under test is the temporal one:

once ONE package is unpriceable, the record's integrated H has lost a parcel FOREVER. Validity is MONOTONIC: a later priceable package must never resurrect it.

Two stills exercise the two orders on ONE accumulated record each:

potA (invalid → valid): starts with 10 % compA, so its distillate record potA→recA begins UNPRICEABLE. At  $t = 200$  s the recipe dumps the whole pot to 'dump' and recharges clean benzene/toluene from 'fresh'; every later distillate package is priceable. The record must STAY invalid ( $H_{\text{missing}} = \text{compA}$ ), and the discrete potA→dump record is invalid too (it carried the compA).

potB (valid → invalid): starts clean, so its distillate record potB→recB begins priceable and accumulates real enthalpy. At  $t = 200$  s the recipe spikes it with pure compA from 'spike'; every later package is unpriceable. The record flips invalid and the partial integral is WITHHELD — an emitted  $H_{\text{kJ}}$  that silently lost its tail would be a lie.

Five ledger records result: potA→recA (continuous, poisoned early), potB→recB (continuous, poisoned late), potA→dump and spike→potB (discrete, carrying compA: unpriceable), fresh→potA (discrete, clean: the ONE priceable record). The golden pins the campaign census transfers  $H_{\text{valid}} = 1 / \text{transfers\_H\_invalid} = 4$ ; the results JSON carries  $H_{\text{missing}} = [\text{compA}]$  and NO  $H_{\text{kJ}}$  key on every poisoned record.

**What to expect (golden-locked):**

| kind | unit/stream | key                  | expected          |
|------|-------------|----------------------|-------------------|
| kpi  | campaign    | mass_closure_rel     | 2.1198000621e-16  |
| kpi  | campaign    | mass_kg_external_out | 0                 |
| kpi  | campaign    | mass_kg_final        | 0.26186975        |
| kpi  | campaign    | mass_kg_initial      | 0.26186975        |
| kpi  | campaign    | mass_residual_kg     | 5.55111512313e-17 |
| kpi  | campaign    | transfers_H_invalid  | 4                 |
| kpi  | campaign    | transfers_H_valid    | 1                 |
| kpi  | campaign    | transfers_logged     | 5                 |
| kpi  | dump        | P_final              | 101300            |
| kpi  | dump        | T_final              | 372.484229602     |
| kpi  | dump        | n_benzene_final      | 0.000249727144577 |
| kpi  | dump        | n_compA_final        | 9.64864338217e-05 |
| kpi  | dump        | n_toluene_final      | 0.000353786421588 |
| kpi  | dump        | t_end                | 600               |

...and 68 more reference values in the case's *expected*.

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## Category IV: Dynamics and control (choupoCtrl)

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Continuous processes in time with feedback: PID loops, setpoint schedules, disturbances and responses.

**ctrl01\_cstr\_temp\_control**

tutorials/ctrl/ctrl01\_cstr\_temp\_control

**reactor**  
 dynamicCSTR

**What it does:** Dynamic CSTR (compA  $\rightarrow$  compB) with PID jacket-temperature control; T climbs from 320 K into the  $\pm 2\%$  setpoint band within  $\sim 20$  s (no overshoot) and tracks toward 350 K across the 150 s window

**The story:** the canonical first choupCtrl test.

**Process:** Continuous CSTR with first-order reaction compA  $\rightarrow$  compB – an ISOMERISATION: both stand-ins carry MW 30, so the pseudo-reaction conserves mass exactly and the run-end material ledger closes to machine precision. (The element balance stays honestly UNAVAILABLE: compA/compB are numerical stand-ins with no real formula, and the ledger says so by name instead of fabricating atoms.) Feed is pure compA at 320 K,  $5 \times 10^{??}$  kmol/s; reactor volume 1 L; mild exotherm ( $\Delta H_{\text{rxn}} = -50$  kJ/mol). Without control the reactor settles a few K above the inlet temperature (small reaction exotherm balanced by convective cooling), well below the desired setpoint of 350 K.

**Controller:** A PID acts on the jacket temperature ( $T_{\text{jacket}}$ ) to drive the reactor temperature (T) to a setpoint of 350 K. Tuning is pedagogical, not optimal:

$K_p = 4.0$  (proportional band  $\approx 7\text{--}8$  K)  $K_i = 0.04$  (integral time  $\tau_I = K_p/K_i \approx 100$  s)  $K_d = 0.0$  (no derivative; system is benign enough)

Jacket bounded to [280, 420] K — realistic limits of a utility loop. Anti-windup: integral does not advance when the unclamped output is outside the bounds.

**Expected trajectory (150 s window):** Open-loop: T relaxes a few K above the 320 K inlet (small exotherm balanced by convective cooling). Closed-loop: T rises fast (the jacket MV slams to its 420 K clamp, then backs off), entering the  $\pm 2\%$  setpoint band by  $\sim 20$  s, well-damped with no overshoot. By 150 s T is  $\sim 348.5$  K and still creeping up: the last  $\sim 1.5$  K rides the slow chemical tail (the exotherm's conversion,  $\tau \sim 120$  s) toward 350 K beyond the window. The jacket (MV) settles around 348.5 K to keep heating in against the cool 320 K feed. (Widen endTime to watch the final approach to 350 K; the 150 s window is sized to SHOW the control transient, not the slow tail.)

To run the OPEN-LOOP comparison, comment out the entire ‘controllers’ block below — the case will then run the bare reactor with whatever  $T_{\text{jacket}}$  the ‘operation’ block initialised it to (here 320 K, i.e. no jacket heating).

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | TC1         | MV_final                 | 348.54491648      |
| kpi  | TC1         | PV_final                 | 348.468973192     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 348.481893564     |
| kpi  | reactor     | n_compA_final            | 0.0116161140368   |
| kpi  | reactor     | n_compB_final            | 0.00038388596316  |
| kpi  | reactor     | t_end                    | 150               |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 1.77327288655e-15 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 0.225             |
| kpi  | balance     | mass_kg_initial          | 0.36              |
| kpi  | balance     | mass_kg_out_cum          | 0.225             |
| kpi  | balance     | mass_residual_kg         | 6.38378239159e-16 |

... and 4 more reference values in the case's *expected*.

ctrl02\_disturbance\_rejection

tutorials/ctrl/ctrl02\_disturbance\_rejection

reactor

dynamicCSTR

**What it does:** PID disturbance rejection: T\_in steps down then up, PID maintains T\_setpoint

**The story:** the textbook control demo.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T\_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T\_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T\_in drops 15 K; the integral action absorbs it so tightly that T\_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T\_in rises 15 K; again T\_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T\_reactor) tracks closely.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedDist    | MV_final                 | 335               |
| kpi  | FeedDist    | PV_final                 | 0                 |
| kpi  | FeedDist    | SP_final                 | 335               |
| kpi  | TC1         | MV_final                 | 347.976535585     |
| kpi  | TC1         | PV_final                 | 350.00207645      |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 350.002059115     |
| kpi  | reactor     | n_compA_final            | 0.0110806465498   |
| kpi  | reactor     | n_compB_final            | 0.000919353450202 |
| kpi  | reactor     | t_end                    | 2000              |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 2.79221090693e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 3                 |

...and 7 more reference values in the case's *expected*.

**ctrl03\_adaptive\_disturbance**

tutorials/ctrl/ctrl03\_adaptive\_disturbance

**reactor**  
 dynamicCSTR

**What it does:** PID disturbance rejection with ADAPTIVE plant integration — controller samples on the fixed 1 s grid, Rosenbrock23 sub-steps between samples (MV held)

**The story:** the ctrl02\_disturbance\_rejection demo re-integrated with an ADAPTIVE plant integrator: the PID still samples on the fixed 1 s control grid (MV held between samples) while Rosenbrock23 sub-steps the plant adaptively in between. The flowsheet below is ctrl02's textbook disturbance-rejection setup, unchanged:

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance).
2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T<sub>in</sub>:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T<sub>reactor</sub> drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T<sub>in</sub> drops 15 K; the integral action absorbs it so tightly that T<sub>reactor</sub> holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T<sub>in</sub> rises 15 K; again T<sub>reactor</sub> stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T<sub>reactor</sub>) tracks closely.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedDist    | MV_final                 | 335               |
| kpi  | FeedDist    | PV_final                 | 0                 |
| kpi  | FeedDist    | SP_final                 | 335               |
| kpi  | TC1         | MV_final                 | 347.976535626     |
| kpi  | TC1         | PV_final                 | 350.002076468     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 350.002059134     |
| kpi  | reactor     | n_compA_final            | 0.0110806465416   |
| kpi  | reactor     | n_compB_final            | 0.000919353458409 |
| kpi  | reactor     | t_end                    | 2000              |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 2.66083451568e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 3                 |

... and 7 more reference values in the case's *expected*.

ctrl03\_startup\_transient

tutorials/ctrl/ctrl03\_startup\_transient

reactor

dynamicCSTR

**What it does:** Open-loop CSTR START-UP transient with OpenFOAM-style real-time instant dirs: charged cold (320 K), no control, watch T(t) and the holdup approach the natural steady state by scrubbing 0/ 50/ 100/ ... s

**The story:** Process: A continuous CSTR (first-order compA → compB, mild exotherm) is CHARGED cold (320 K, pure compA) and then left to run with NO controller. The jacket sits at a fixed 320 K. From the cold start the reactor warms a little from the reaction exotherm, the feed/ outlet convection competes, and (T, composition) relax to the natural open-loop steady state over ~ several residence times.

What this tutorial shows: The OpenFOAM-style transient solution directories. With ‘solutionControl { write true; }’ in controlDict, the case root fills with REAL-TIME directories ‘0/ 50/ 100/ ... 1000/’. Each ‘<t>/internalState’ is the reactor HOLDUP at that instant; each ‘<t>/streamFaces’ is the instantaneous outlet face. Scrub the times and you watch T(t) and the inventory march to steady state — the same way you scrub a CFD transient in ParaView.

To compare CLOSED-LOOP control, see ctrl01\_cstr\_temp\_control (a PID drives T to a 350 K setpoint) and ctrl05\_disturbance\_transient (a PID rejecting a feed-temperature disturbance), both also writing time dirs.

What to expect (golden-locked):

| kind                                                    | unit/stream | key                      | expected           |
|---------------------------------------------------------|-------------|--------------------------|--------------------|
| kpi                                                     | reactor     | T_final                  | 320.759374011      |
| kpi                                                     | reactor     | n_compA_final            | 0.0117988900072    |
| kpi                                                     | reactor     | n_compB_final            | 0.000201109992805  |
| kpi                                                     | reactor     | t_end                    | 1000               |
| kpi                                                     | balance     | energy_balance_available | 0                  |
| kpi                                                     | balance     | mass_closure_rel         | 1.77265609598e-14  |
| kpi                                                     | balance     | mass_kg_final            | 0.36               |
| kpi                                                     | balance     | mass_kg_in_cum           | 1.5                |
| kpi                                                     | balance     | mass_kg_initial          | 0.36               |
| kpi                                                     | balance     | mass_kg_out_cum          | 1.5                |
| kpi                                                     | balance     | mass_residual_kg         | -2.65898414398e-14 |
| kpi                                                     | balance     | material_available       | 1                  |
| kpi                                                     | reactor     | F_in_final               | 5e-05              |
| kpi                                                     | reactor     | z_in_compA_final         | 1                  |
| ... and 1 more reference values in the case's expected. |             |                          |                    |

**ctrl05\_disturbance\_transient**

tutorials/ctrl/ctrl05\_disturbance\_transient

**reactor**  
dynamicCSTR

**What it does:** PID disturbance rejection with OpenFOAM-style real-time instant dirs: T\_in steps down (700 s) then up (1300 s); scrub 0/ 40/ 80/ ... s to watch the PID track the 350 K setpoint and the jacket MV move

**The story:** the textbook control demo, captured in OpenFOAM-style real-time instant directories (see controlDict's solutionControl block). Identical physics to ctrl02; the difference is that each written time drops a '<t>/internalState' + '<t>/streamFaces' you can scrub across the two disturbance steps to watch the PID reject them.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T\_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T\_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T\_in drops 15 K; the integral action absorbs it so tightly that T\_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T\_in rises 15 K; again T\_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T\_reactor) tracks closely.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedDist    | MV_final                 | 335               |
| kpi  | FeedDist    | PV_final                 | 0                 |
| kpi  | FeedDist    | SP_final                 | 335               |
| kpi  | TC1         | MV_final                 | 347.976535585     |
| kpi  | TC1         | PV_final                 | 350.00207645      |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 350.002059115     |
| kpi  | reactor     | n_compA_final            | 0.0110806465498   |
| kpi  | reactor     | n_compB_final            | 0.000919353450202 |
| kpi  | reactor     | t_end                    | 2000              |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 2.79221090693e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 3                 |

... and 7 more reference values in the case's *expected*.



**ctrl06\_sine\_disturbance**

tutorials/ctrl/ctrl06\_sine\_disturbance

**reactor**  
dynamicCSTR

**What it does:** Forced SINUSOIDAL inlet-T disturbance:  $T_{in} = 320 + 15 \sin(2 \pi t / 600)$ . The PID chases a moving target  $\rightarrow$  the reactor temperature settles into a forced sinusoidal response (attenuated + phase-lagged). Sweep the period to build a one-point Bode by trial-and-error (the frequency-response seed).

**The story:** the FREQUENCY-RESPONSE seed.

Same DynamicCSTR as ctrl01/ctrl02 (compA  $\rightarrow$  compB, mild exotherm, jacket heat transfer) and the same PID temperature controller. The NEW ingredient is the disturbance: instead of the step train of ctrl02's 'Schedule' controller, a 'Signal' controller injects a SINUSOID into the inlet temperature  $T_{in}$ :

$$T_{in}(t) = 320 + 15 * \sin(2 * \pi * t / 600) \text{ [K]}$$

mean = 320 K amplitude = 15 K period = 600 s ( $f = 1/600 = 1.667\text{e-}3$  Hz,  $\omega = 0.01047$  rad/s)

The PID controller has no foreknowledge of the forcing — it only sees  $T_{reactor}$  drift sinusoidally away from the 350 K setpoint and works the jacket to compensate. After the first  $\sim$ period the reactor temperature settles into a FORCED SINUSOIDAL RESPONSE: same period as the input, but ATTENUATED (the closed loop rejects most of the swing) and PHASE-LAGGED.

THE LESSON (Seborg/Edgar/Mellichamp/Doyle, Ch. 13-14 frequency response): read the output/input amplitude ratio [dB] and the phase lag [deg] off the steady-cycling trajectory, and you have ONE POINT of a Bode plot. Sweep the 'period' (edit one number below), re-run, and you trace the closed loop's frequency response by trial-and-error — the differentiator.

Backward-compatible by construction: ctrl02's 'type Schedule; schedule(...)' staircase still works unchanged; this case swaps it for the more general 'type Signal; signal { type sinusoidal; ... }'.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedDist    | MV_final                 | 319.842923238     |
| kpi  | FeedDist    | PV_final                 | 0                 |
| kpi  | FeedDist    | SP_final                 | 319.842923238     |
| kpi  | TC1         | MV_final                 | 350.026347162     |
| kpi  | TC1         | PV_final                 | 350.170890591     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 350.173436842     |
| kpi  | reactor     | n_compA_final            | 0.0110844237702   |
| kpi  | reactor     | n_compB_final            | 0.000915576229809 |
| kpi  | reactor     | t_end                    | 3000              |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 5.72875080707e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 4.5               |

*... and 7 more reference values in the case's expected.*

**ctrl07\_lhhw\_inhibition**

tutorials/ctrl/ctrl07\_lhhw\_inhibition

**reactor**  
 dynamicCSTR

**What it does:** Product inhibition meets a PID: an LHHW reaction whose exotherm fades as its own product covers the catalyst

**The story:** ctrl01's temperature loop, with a reaction that poisons itself.

Same vessel, same PID, same setpoint. The one change is the rate law: the product compB now ADSORBS on the catalyst, so the denominator grows as the reaction proceeds,

$$r = k(T) \cdot c_A / (1 + K_B c_B)$$

and the reaction throttles itself. A power law cannot express that at any order; only the adsorption term can. This is the same 'RateLaw' that the steady 'cstr' and the 'batchReactor' evaluate – one definition, three reactors.

What the controller sees: an exotherm that FADES as conversion proceeds, not a constant one. The heat the loop is rejecting shrinks under it, and the jacket temperature must follow. A tuning that holds against a first-order exotherm is not the same tuning that holds against a decaying one.

The adsorption constant  $K_0 = 2.0 \text{ m}^3/\text{kmol}$  is DIDACTIC – chosen so the inhibition is visible over the control horizon, not measured on any catalyst. For a real, cited LHHW see 'cstr07\_lhhw\_methylAcetate' and 'batch07\_lhhw\_methylAcetate' (Popken 2000, Eq 16, on UNIQUAC activities).

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | TC1         | MV_final                 | 349.740705746     |
| kpi  | TC1         | PV_final                 | 348.328939265     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 348.342145073     |
| kpi  | reactor     | n_compA_final            | 0.0117152677233   |
| kpi  | reactor     | n_compB_final            | 0.000284732276726 |
| kpi  | reactor     | t_end                    | 150               |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 7.70988211545e-16 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 0.225             |
| kpi  | balance     | mass_kg_initial          | 0.36              |
| kpi  | balance     | mass_kg_out_cum          | 0.225             |
| kpi  | balance     | mass_residual_kg         | 2.77555756156e-16 |

...and 4 more reference values in the case's *expected*.

**ctrl08\_prbs\_ident**

tutorials/ctrl/ctrl08\_prbs\_ident

**reactor**  
 dynamicCSTR

**What it does:** PRBS identification experiment:  $T_{in} = 320 \pm 10$  K by a maximal-length LFSR ( $n=4$ , period 15 bits  $\times$  120 s) – deterministic, seed-declared, period-VERIFIED; the broadband excitation the sine of ctrl06 cannot give.

**The story:** PRBS identification experiment (forum #100.4/#101).

The classic system-identification input: a pseudo-random binary sequence on the inlet temperature,  $T_{in} = 320 \pm 10$  K, driven by a MAXIMAL-LENGTH Fibonacci LFSR —  $n = 4$  registers, canonical taps (4 3), DECLARED seed 1, period  $2^4 - 1 = 15$  bits of 120 s each (1800 s per full period).

Why a PRBS and not ctrl06's sine: one sinusoid excites ONE frequency; the PRBS is broadband (its autocorrelation approximates an impulse), so a single record carries the information for a full low-order model fit. And unlike real "random" noise it is DETERMINISTIC: seed + taps declared in this dict reproduce the experiment bit for bit, run after run.

THE HAND-DERIVED SEQUENCE (the exact-sequence contract of #101): with the frozen convention (output =  $s_n$ ; feedback = XOR of taps; shift  $s_1 \leftarrow$  feedback; seed LSB  $\rightarrow s_1$ ),  $n=4$  / taps (4 3) / seed 1 gives

0 0 0 1 0 0 1 1 0 1 0 1 1 1 1 (bits  $k = 0..14$ , then repeats)

– walk it yourself: state 0001  $\rightarrow$  0010  $\rightarrow$  0100  $\rightarrow$  1001  $\rightarrow$  0011  $\rightarrow$  0110  $\rightarrow$  1101  $\rightarrow$  1010  $\rightarrow$  0101  $\rightarrow$  1011  $\rightarrow$  0111  $\rightarrow$  1111  $\rightarrow$  1110  $\rightarrow$  1100  $\rightarrow$  1000  $\rightarrow$  0001 (back at the seed after EXACTLY 15 steps: maximal). The run's announce line prints these same first bits; the initialiser REFUSES any tap set whose cycle is not exactly  $2^n - 1$ .

The PID (same tuning as ctrl06) holds the reactor at 350 K against the binary disturbance; the trajectory's  $T_{in}$  column IS the PRBS, and the reactor response is the identification record.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedPRBS    | MV_final                 | 330               |
| kpi  | FeedPRBS    | PV_final                 | 0                 |
| kpi  | FeedPRBS    | SP_final                 | 330               |
| kpi  | TC1         | MV_final                 | 348.579621229     |
| kpi  | TC1         | PV_final                 | 350.006520504     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | reactor     | T_final                  | 350.006465191     |
| kpi  | reactor     | n_compA_final            | 0.0110804887195   |
| kpi  | reactor     | n_compB_final            | 0.000919511280459 |
| kpi  | reactor     | t_end                    | 3600              |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 6.73638100034e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 5.4               |

... and 7 more reference values in the case's *expected*.

**ctrl09\_stream\_disturbance**

tutorials/ctrl/ctrl09\_stream\_disturbance

**reactor**  
 dynamicCSTR

**What it does:** Inlet-field disturbance: componentMolarFlow.compA steps  $3.5\text{e-}5 \rightarrow 5.25\text{e-}5$  kmol/s; F and z\_in change by the DERIVED values ( $F = 6.75\text{e-}5$ ,  $z_A = 7/9$ ) – the disturbance never hides from the other components

**The story:** the inlet-field actuator, end to end.

The Signal controller targets the reactor's INLET FACE by field name ('inletField componentMolarFlow.compA') — the only legal disturbance surface (an outlet actuator is not even expressible). The AUTHORITATIVE inlet state is the per-component molar-flow vector; F\_in and z\_in are DERIVED from it on every write, so a composition disturbance can never hide inside a silent renormalisation (forum #101):

$t < 600$  s:  $n_A = 3.5\text{e-}5$ ,  $n_B = 1.5\text{e-}5$  kmol/s  $\rightarrow F_{in} = 5.0\text{e-}5$ ,  $z_A = 0.70$ ,  $z_B = 0.30$   $t \geq 600$  s:  $n_A$  stepped to  $5.25\text{e-}5$ ;  $n_B$  UNTOUCHED at  $1.5\text{e-}5 \rightarrow F_{in} = 6.75\text{e-}5$ ,  $z_A = 5.25/6.75 = 7/9 = 0.77778$ ,  $z_B = 1.5/6.75 = 2/9 = 0.22222$  (hand-derived)

The golden pins F\_in and z\_in \* finals to these fractions, and the Signal value is the ABSOLUTE molar flow (announced at start-up). The reactor's own response (more A in, higher conversion load, T transient against the PID) is the pedagogy; the CONTRACT is the derived pair.

The 'moleFraction.<c>' sibling field instead holds F\_in constant and renormalises the OTHER components proportionally — two explicit semantics, each announced, never a guess.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected         |
|------|-------------|--------------------------|------------------|
| kpi  | FeedComp    | MV_final                 | 5.25e-05         |
| kpi  | FeedComp    | PV_final                 | 0                |
| kpi  | FeedComp    | SP_final                 | 5.25e-05         |
| kpi  | TC1         | MV_final                 | 350.204904403    |
| kpi  | TC1         | PV_final                 | 350.000124372    |
| kpi  | TC1         | SP_final                 | 350              |
| kpi  | reactor     | F_in_final               | 6.75e-05         |
| kpi  | reactor     | T_final                  | 350.000123673    |
| kpi  | reactor     | n_compA_final            | 0.00879292389459 |
| kpi  | reactor     | n_compB_final            | 0.00320707610541 |
| kpi  | reactor     | t_end                    | 1800             |
| kpi  | reactor     | z_in_compA_final         | 0.777777777778   |
| kpi  | reactor     | z_in_compB_final         | 0.222222222222   |
| kpi  | balance     | energy_balance_available | 0                |

...and 7 more reference values in the case's *expected*.

**ctrl10\_brine\_concentration**  
tutorials/ctrl/ctrl10\_brine\_concentration

tank

dynamicCSTR

**What it does:** Dynamic brine tank on a Pitzer package: the electrolyte thermodynamics through a transient

**The story:** ONE stirred vessel, NO reaction. The dynamic CSTR tolerates an empty reaction list, and that is exactly the model wanted here: the holdup is a mixing lag, and the thermodynamics – Pitzer activity coefficients on the molality basis – is what travels through it.

The jacket loop is the same PID shape as ctrl01: it acts on T\_jacket to hold the vessel temperature. A brine tank held on temperature is what sits upstream of every evaporator and crystalliser in the corpus.

**What to expect (golden-locked):**

| kind                                                    | unit/stream | key                       | expected          |
|---------------------------------------------------------|-------------|---------------------------|-------------------|
| kpi                                                     | TC1         | MV_final                  | 360               |
| kpi                                                     | TC1         | PV_final                  | 310.319973652     |
| kpi                                                     | TC1         | SP_final                  | 313.15            |
| kpi                                                     | balance     | element_Cl_closure_rel    | 7.17019034075e-16 |
| kpi                                                     | balance     | element_H_closure_rel     | 3.40011338976e-16 |
| kpi                                                     | balance     | element_Na_closure_rel    | 7.17019034075e-16 |
| kpi                                                     | balance     | element_O_closure_rel     | 3.40011338976e-16 |
| kpi                                                     | balance     | element_worst_closure_rel | 7.17019034075e-16 |
| kpi                                                     | balance     | energy_balance_available  | 1                 |
| kpi                                                     | balance     | energy_closure_rel        | 7.85289304221e-11 |
| kpi                                                     | balance     | energy_functional_H       | 1                 |
| kpi                                                     | balance     | energy_kJ_final           | -853632.741074    |
| kpi                                                     | balance     | energy_kJ_heat_cum        | 2658.78680029     |
| kpi                                                     | balance     | energy_kJ_in_cum          | -87796.8114225    |
| ...and 17 more reference values in the case's expected. |             |                           |                   |

**ctrl11\_esterification\_jacket**

tutorials/ctrl/ctrl11\_esterification\_jacket

|                               |
|-------------------------------|
| <b>reactor</b><br>dynamicCSTR |
|-------------------------------|

**What it does:** Jacketed esterification CSTR under PID temperature control: the first ctrl case whose stored enthalpy exists, so the first-law ledger claims closure instead of refusing

**The story:** One jacketed vessel and one PID loop – the same shape as ctrl01, so the two can be read side by side. What differs is the chemistry: real species with a formation datum instead of numerical stand-ins, which is the whole reason this case can carry an energy claim and ctrl01 cannot.

The reaction declares NO  $dH_{rxn}$ . On the elements datum the heat of reaction is  $\text{Sum}(\nu_i h_i(T))$  – a consequence of the same enthalpy surface the vessel stores, not a second number to keep in step with it.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | TC1         | MV_final                  | 360.338985371     |
| kpi  | TC1         | PV_final                  | 364.93860506      |
| kpi  | TC1         | SP_final                  | 365               |
| kpi  | balance     | element_C_closure_rel     | 1.32609972386e-14 |
| kpi  | balance     | element_H_closure_rel     | 1.31993181817e-14 |
| kpi  | balance     | element_O_closure_rel     | 1.32609972386e-14 |
| kpi  | balance     | element_worst_closure_rel | 1.32609972386e-14 |
| kpi  | balance     | energy_balance_available  | 1                 |
| kpi  | balance     | energy_closure_rel        | 1.42432856618e-07 |
| kpi  | balance     | energy_functional_H       | 1                 |
| kpi  | balance     | energy_kJ_final           | -28146.0243548    |
| kpi  | balance     | energy_kJ_heat_cum        | -1199.4769622     |
| kpi  | balance     | energy_kJ_in_cum          | -65562.9346941    |
| kpi  | balance     | energy_kJ_initial         | -27888.8423982    |

...and 20 more reference values in the case's *expected*.

**ctrl12\_williams\_otto**

tutorials/ctrl/ctrl12\_williams\_otto

|                                   |
|-----------------------------------|
| <b>plant</b><br>williamsOttoPlant |
|-----------------------------------|

**What it does:** Williams-Otto plant (arXiv:2004.07614 eqs. 3.6-3.11 verbatim) settling onto the published steady state  $x^* = (3.27, 7.47, 1.12, 9.81, 1.69, 0.22)$  klb

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term  $(1-\eta)\mu$ , the removed G, the  $0.1 F_{tE}$  column retention are all inside them – see the unit’s own header). The five controls sit in ‘operation’, declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper’s value in klb/h.

$u^*$  of eq. (3.20):  $F_{fA} = 10$ ,  $F_{fB} = 20$ ,  $T = 580$  degR,  $\mu = 129.5$ ,  $\eta = 0.2$ . T rides in  $0/\text{internalState}$  (322.2222 K = 580 degR exactly).

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected         |
|------|-------------|--------------------------|------------------|
| kpi  | balance     | energy_balance_available | 0                |
| kpi  | balance     | mass_closure_rel         | 4.0063377475e-10 |
| kpi  | balance     | mass_kg_final            | 10701.9131449    |
| kpi  | balance     | mass_kg_in_cum           | 5443104          |
| kpi  | balance     | mass_kg_initial          | 4989.512         |
| kpi  | balance     | mass_kg_out_cum          | 5437391.60103    |
| kpi  | balance     | mass_residual_kg         | 0.00218069130187 |
| kpi  | balance     | material_available       | 1                |
| kpi  | plant       | F_pP_klbh_final          | 3.90161524936    |
| kpi  | plant       | F_wG_klbh_final          | 1.22338475064    |
| kpi  | plant       | J_combined_klb_final     | 1071.35532885    |
| kpi  | plant       | T_final                  | 322.2222         |
| kpi  | plant       | t_end                    | 1440000          |
| kpi  | plant       | waste_klb_final          | 481.451273663    |

...and 13 more reference values in the case’s *expected*.

**ctrl13\_williams\_otto\_step**

tutorials/ctrl/ctrl13\_williams\_otto\_step

|                                   |
|-----------------------------------|
| <b>plant</b><br>williamsOttoPlant |
|-----------------------------------|

**What it does:** Williams-Otto  $F_{fB}$  step 20→21 klb/h at  $t^*=100$  h (paper Fig. 2):  $F_{pP}$  climbs  $\sim 3.90 \rightarrow \sim 4.45$  klb/h

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term  $(1-\eta)\mu$ , the removed G, the 0.1  $F_{tE}$  column retention are all inside them – see the unit's own header). The five controls sit in 'operation', declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper's value in klb/h.

u\* of eq. (3.20) plus the paper's FIRST simulation experiment (section 3.2, Fig. 2): a Schedule steps  $F_{fB}$  from 20 to 21 klb/h at  $t^* = 100$  h. The MV crosses the boundary in SI (kmol/s), so the step values below are 20/3600 and 21/3600 – the paper's numbers over the one declared conversion.

**What to expect (golden-locked):**

| kind | unit/stream | key                       | expected          |
|------|-------------|---------------------------|-------------------|
| kpi  | FeedStep    | MV_final                  | 0.0058333333      |
| kpi  | FeedStep    | PV_final                  | 0                 |
| kpi  | FeedStep    | SP_final                  | 0.0058333333      |
| kpi  | balance     | energy_balance_available  | 0                 |
| kpi  | balance     | mass_closure_rel          | 1.45035580918e-09 |
| kpi  | balance     | mass_kg_final             | 19794.220051      |
| kpi  | balance     | mass_kg_in_cum            | 2766911.20181     |
| kpi  | balance     | mass_kg_initial           | 4989.512          |
| kpi  | balance     | mass_kg_out_cum           | 2752106.49778     |
| kpi  | balance     | mass_residual_kg          | 0.00401300573503  |
| kpi  | balance     | material_available        | 1                 |
| kpi  | plant       | $F_{pP\_klbh\_final}$     | 4.44568758172     |
| kpi  | plant       | $F_{wG\_klbh\_final}$     | 1.90791543668     |
| kpi  | plant       | $J\_combined\_klb\_final$ | 529.364471374     |

...and 10 more reference values in the case's *expected*.



ctrl14\_williams\_otto\_pi

tutorials/ctrl/ctrl14\_williams\_otto\_pi

plant

williamsOttoPlant

**What it does:** Williams-Otto PI channel (F<sub>FB</sub>, F<sub>pP</sub>), paper eq. 4.6 with the section-4.1 tunings: F<sub>pP</sub> tracks 3.90 → 4.5 klb/h with the Fig. 4 overshoot

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term (1-eta)mu, the removed G, the 0.1 F<sub>tE</sub> column retention are all inside them – see the unit’s own header). The five controls sit in ‘operation’, declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper’s value in klb/h.

The paper’s control channel (F<sub>FB</sub>, F<sub>pP</sub>), eq. (4.6), with the Skogestad-IMC gains of section 4.1: K3prop = 0.069 (dimensionless – klb/h per klb/h survives the SI conversion unchanged) and K3int = 1.282 per HOUR, declared per second (1.282/3600 = 3.561111e-4). Bias = F\*<sub>FB</sub> (bumpless), setpoint F<sub>pP</sub> = 4.5 klb/h = 1.25e-3 kmol/s, output bounded to the paper’s 0 <= F<sub>FB</sub> <= 56 klb/h.

What to expect (golden-locked):

| kind                                                     | unit/stream | key                      | expected          |
|----------------------------------------------------------|-------------|--------------------------|-------------------|
| kpi                                                      | PC1         | MV_final                 | 0.00586183454679  |
| kpi                                                      | PC1         | PV_final                 | 0.00124999896025  |
| kpi                                                      | PC1         | SP_final                 | 0.00125           |
| kpi                                                      | balance     | energy_balance_available | 0                 |
| kpi                                                      | balance     | mass_closure_rel         | 2.05158151889e-09 |
| kpi                                                      | balance     | mass_kg_final            | 21425.7971042     |
| kpi                                                      | balance     | mass_kg_in_cum           | 1418738.01455     |
| kpi                                                      | balance     | mass_kg_initial          | 10701.9131449     |
| kpi                                                      | balance     | mass_kg_out_cum          | 1408014.1335      |
| kpi                                                      | balance     | mass_residual_kg         | 0.0029106566908   |
| kpi                                                      | balance     | material_available       | 1                 |
| kpi                                                      | plant       | F_pP_klbh_final          | 4.49999626428     |
| kpi                                                      | plant       | F_wG_klbh_final          | 2.00328828826     |
| kpi                                                      | plant       | J_combined_klb_final     | 255.611908728     |
| ... and 10 more reference values in the case’s expected. |             |                          |                   |

**ctrl16\_williams\_otto\_optimal**

tutorials/ctrl/ctrl16\_williams\_otto\_optimal

|                                              |
|----------------------------------------------|
| <p><b>plant</b></p> <p>williamsOttoPlant</p> |
|----------------------------------------------|

**What it does:** Williams-Otto section-5.3 combined optimisation: Nelder-Mead over two K=5 Schedule profiles, J measured against the published collocation optimum 551.8

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term  $(1-\eta)\mu$ , the removed G, the 0.1 F<sub>tE</sub> column retention are all inside them – see the unit’s own header). The five controls sit in ‘operation’, declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper’s value in klb/h.

The section-5.3 combined optimisation: system/outerDict shapes the two profiles below to maximise J = yield - waste ( $\alpha = \beta = 1$ , eq. 5.7), read straight off the plant’s J\_combined\_klb\_final KPI.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FfBprof     | MV_final                 | 0.00463988515428  |
| kpi  | FfBprof     | PV_final                 | 0                 |
| kpi  | FfBprof     | SP_final                 | 0.00463988515428  |
| kpi  | Tprof       | MV_final                 | 300.373635533     |
| kpi  | Tprof       | PV_final                 | 0                 |
| kpi  | Tprof       | SP_final                 | 300.373635533     |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 1.14401865779e-07 |
| kpi  | balance     | mass_kg_final            | 723583.798797     |
| kpi  | balance     | mass_kg_in_cum           | 2128628.70559     |
| kpi  | balance     | mass_kg_initial          | 10701.9131449     |
| kpi  | balance     | mass_kg_out_cum          | 1415747.06346     |
| kpi  | balance     | mass_residual_kg         | 0.243519095471    |
| kpi  | balance     | material_available       | 1                 |

...and 40 more reference values in the case’s *expected*.

**ctrl17\_signal\_shapes**

tutorials/ctrl/ctrl17\_signal\_shapes

**reactor**  
dynamicCSTR

**What it does:** WITNESS: ramp + pulse + staircase signals on three feed handles, with the temperature PID rejecting all three at once; the three registered shapes no case selected.

**The story:** WITNESS: the ramp, pulse and staircase SIGNALS, each on its own feed handle of the ctrl09 CSTR, with the ctrl09 temperature PID still in the loop rejecting all three at once – compA ramps +28.6 % over 1000 s, compB takes a 300 s rectangular pulse, the feed temperature walks a three-step staircase. These three registered signal shapes were selected by no case until 2026-08-23 (invariant I8). Self-recorded.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | FeedPulseB  | MV_final                 | 1.5e-05           |
| kpi  | FeedPulseB  | PV_final                 | 0                 |
| kpi  | FeedPulseB  | SP_final                 | 1.5e-05           |
| kpi  | FeedRampA   | MV_final                 | 4.5e-05           |
| kpi  | FeedRampA   | PV_final                 | 0                 |
| kpi  | FeedRampA   | SP_final                 | 4.5e-05           |
| kpi  | FeedStairT  | MV_final                 | 348               |
| kpi  | FeedStairT  | PV_final                 | 0                 |
| kpi  | FeedStairT  | SP_final                 | 348               |
| kpi  | TC1         | MV_final                 | 347.378942902     |
| kpi  | TC1         | PV_final                 | 349.993146257     |
| kpi  | TC1         | SP_final                 | 350               |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 2.98582911041e-14 |

*...and 13 more reference values in the case's expected.*

**ctrl17\_step\_reaction\_curve**

tutorials/ctrl/ctrl17\_step\_reaction\_curve

**reactor**  
dynamicCSTR

**What it does:** Open-loop process reaction curve: the jacket steps +5, -5, +20 and -20 K from one settled base, and the four responses are the identification record for the FOPDT fit in system/propsDict

**The story:** THE IDENTIFICATION, run over the trajectory the choupCtrl half of this case wrote. Run the experiment first:

```
runCase tutorials/ctrl/ctrl17_step_reaction_curve choupProps tutorials/ctrl/ctrl17_step_reaction_curve
```

TWO OPERATIONS OVER ONE EXPERIMENT, and the pair is the lesson. Both read the SAME four steps of the SAME jacket temperature; they differ only in which response they call the controlled variable.

[0] the COMPOSITION loop (jacket → product holdup). Two lags in series – the jacket heats the liquid, and the warmer liquid reacts faster – so the response is S-shaped and an FOPDT surrogate finds a real apparent dead time. All three tuning rules apply.

[1] the TEMPERATURE loop (jacket → reactor temperature). One dominant lag and no transport delay, so the identified theta comes out smaller than the trajectory's own sampling interval. The engine says so, and REFUSES every Ziegler-Nichols and Cohen-Coon setting by name: both rules put theta in a denominator, and a controller gain computed from an unresolved number is a number with nothing behind it. IMC survives, because its algebra never divides by theta – which is exactly why it is the rule of choice for a lag-dominant loop.

Nothing was tuned to make that happen. It is one plant, one experiment, and two honest answers, and the difference between them is the whole reason the reaction-curve method needs a dead time to work at all.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | JacketStep  | MV_final                 | 320               |
| kpi  | JacketStep  | SP_final                 | 320               |
| kpi  | reactor     | T_final                  | 320.759149686     |
| kpi  | reactor     | n_compA_final            | 0.0117959393424   |
| kpi  | reactor     | n_compB_final            | 0.000204060657576 |
| kpi  | reactor     | F_in_final               | 5e-05             |
| kpi  | reactor     | t_end                    | 15000             |
| kpi  | JacketStep  | PV_final                 | 0                 |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 9.91527847481e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 22.5              |
| kpi  | balance     | mass_kg_initial          | 0.36              |
| kpi  | balance     | mass_kg_out_cum          | 22.5              |

... and 4 more reference values in the case's *expected*.

**ctrl18\_rtd\_pulse\_cstr**

tutorials/ctrl/ctrl18\_rtd\_pulse\_cstr

**reactor**  
 dynamicCSTR

**What it does:** RTD by pulse tracer on a pure mixing tank: tracer substitution at constant flow,  $\tau = N/F = 240$  s; engine-measured moments against Levenspiel closed-form anchors.

**The story:** RESIDENCE-TIME DISTRIBUTION by pulse tracer – the engine’s first RTD tool (2026-08-23). A pure mixing tank (the ctrl09 vessel, reaction and controller stripped) holds  $N = 0.012$  kmol and is fed  $F = 5e-5$  kmol/s, so  $\tau = N/F = 240$  s exactly. For 30 s starting at  $t = 100$  s the feed SWAPS  $1e-5$  kmol/s of carrier for tracer – compB pulses up by exactly what compA pulses down – so the total flow never moves and the vessel is rigorously linear and time-invariant in the tracer.

THE ANCHORS ARE CLOSED FORMS (Levenspiel, Chemical Reaction Engineering, ch. 13-14 – moment additivity of a linear vessel, exact for ANY inlet shape):  $tbar\_out = tbar\_in + \tau = (100 + 15) + 240 = 355$  s  $\sigma^2\_out = \sigma^2\_in + \tau^2 = 30^2/12 + 240^2 = 57675$  s<sup>2</sup> area = amplitude x width =  $1e-5 * 30 = 3.0e-4$  kmol The engine measures the outlet response moments (rtd KPIs, trapezoid on the accepted-step grid, reports/rtd/E.csv); the arithmetic above is the case’s, stated here. Deviations from the anchors are integration truncation (run = 2600 s  $\sim 10.4$  tau past the pulse;  $e^{-10.4} \sim 3e-5$ ) plus the finite step – both stated, neither hidden.

**What to expect (golden-locked):**

| kind | unit/stream | key                      | expected          |
|------|-------------|--------------------------|-------------------|
| kpi  | CarrierDip  | MV_final                 | 5e-05             |
| kpi  | CarrierDip  | PV_final                 | 0                 |
| kpi  | CarrierDip  | SP_final                 | 5e-05             |
| kpi  | TracerPulse | MV_final                 | 0                 |
| kpi  | TracerPulse | PV_final                 | 0                 |
| kpi  | TracerPulse | SP_final                 | 0                 |
| kpi  | balance     | energy_balance_available | 0                 |
| kpi  | balance     | mass_closure_rel         | 4.73695157173e-14 |
| kpi  | balance     | mass_kg_final            | 0.36              |
| kpi  | balance     | mass_kg_in_cum           | 3.9               |
| kpi  | balance     | mass_kg_initial          | 0.36              |
| kpi  | balance     | mass_kg_out_cum          | 3.9               |
| kpi  | balance     | mass_residual_kg         | 1.84741111298e-13 |
| kpi  | balance     | material_available       | 1                 |

*...and 15 more reference values in the case’s expected.*

**ctrl19\_freq\_response\_cstr**

tutorials/ctrl/ctrl19\_freq\_response\_cstr

**reactor**  
 dynamicCSTR

**What it does:** Frequency response of the LTI mixing tank at  $\omega\tau = 1$ : engine-fitted amplitude and phase against the exact first-order closed forms ( $AR = 1/\sqrt{2}$ , phase = -45 deg).

**The story:** FREQUENCY RESPONSE of the same LTI mixing tank ( $\tau = N/F = 240$  s): the tracer feed oscillates sinusoidally at  $\omega\tau = 1$  – again as a SUBSTITUTION against the carrier, so total flow never moves and the vessel is rigorously first order in the tracer. The engine fits  $a + b \sin + c \cos$  over 4 cycles after discarding 5 (transient  $e^{-t/\tau}$  through 5 periods =  $e^{-31}$  – gone), and reports amplitude, phase and the unexplained-variance residual.

THE ANCHORS ARE THE FIRST-ORDER CLOSED FORMS (any process-control text; at  $\omega\tau = 1$  exactly):  $AR = 1/\sqrt{1 + (\omega\tau)^2} = 1/\sqrt{2} \rightarrow \text{out amplitude} = 2.0\text{e-}6 * 0.7071068 = 1.4142136\text{e-}6$  kmol/s phase =  $-\text{atan}(\omega\tau) = -45 \text{ deg} = -0.7853982 \text{ rad}$  (the drive is  $\sin(w(t-t_{\text{Start}}))$  with  $t_{\text{Start}} 0$ , so the fitted phase IS the lag). The measured amplitude lands within  $1\text{e-}6$  relative of the closed form and the fit residual is  $\sim 7\text{e-}15$  (the vessel really is LTI); the phase carries a bias of EXACTLY  $\omega\tau dt/2 = 0.00208 \text{ rad} = 0.12 \text{ deg}$  – the half-sample lag of reading accepted states on a 1 s grid, the measurement system’s own phase contribution, worth teaching in itself. The anchor band covers it and this header names it.

**What to expect (golden-locked):**

| kind | unit/stream   | key                      | expected           |
|------|---------------|--------------------------|--------------------|
| kpi  | CarrierMirror | MV_final                 | 4.41011922784e-05  |
| kpi  | CarrierMirror | PV_final                 | 0                  |
| kpi  | CarrierMirror | SP_final                 | 4.41011922784e-05  |
| kpi  | TracerSine    | MV_final                 | 5.89880772158e-06  |
| kpi  | TracerSine    | PV_final                 | 0                  |
| kpi  | TracerSine    | SP_final                 | 5.89880772158e-06  |
| kpi  | balance       | energy_balance_available | 0                  |
| kpi  | balance       | mass_closure_rel         | 2.91033739366e-15  |
| kpi  | balance       | mass_kg_final            | 0.36               |
| kpi  | balance       | mass_kg_in_cum           | 20.8095            |
| kpi  | balance       | mass_kg_initial          | 0.36               |
| kpi  | balance       | mass_kg_out_cum          | 20.8095            |
| kpi  | balance       | mass_residual_kg         | -6.05626659933e-14 |
| kpi  | balance       | material_available       | 1                  |

... and 14 more reference values in the case’s *expected*.

**ctrl19\_tanks\_in\_series**

tutorials/ctrl/ctrl19\_tanks\_in\_series



**What it does:** Tanks-in-series RTD: three equal mixing tanks chained by the ctrl router; impulse in tank 1 → Erlang-3 response at tank 3 ( $t_{bar} = 3 \tau$ ,  $\sigma^2/t_{bar}^2 = 1/3$ ), Levenspiel closed forms as anchors.

**The story:** the SECOND HALF of the RTD lesson, unlocked by the ctrl router (ruled 2026-08-24): three EQUAL mixing tanks in series, each  $\tau = n/F = 0.004/5e-5 = 80$  s, chained by the same ‘in’/‘outputs’ topology grammar every steady flowsheet already uses. The router applies each connection ONE-STEP-EXPLICITLY: at every accepted driver step the downstream tank integrates on the upstream outlet at the step’s START – a transport delay of one driver step (1 s here, against  $\tau = 80$  s), announced by the [routing] lines at startup.

THE EXPERIMENT. The tracer impulse is an INITIAL-CONDITION substitution in tank 1 only (compB 0.0002 of the 0.004 kmol holdup – substitution, not addition, so every tank holds exactly  $\tau = 80$  s and the closed forms are exact). An initial holdup  $m$  in tank 1 is mathematically a perfect impulse into tank 1 at  $t = 0$ , so tank 3’s outlet is the Erlang-3 density:

$E(t) = t^2 e^{-t/\tau} / (2 \tau^3) t_{bar} = 3 \tau = 240$  s  $\sigma^2 = 3 \tau^2 = 19200$  s<sup>2</sup> ( $\sigma^2/t_{bar}^2 = 1/3$  exactly) peak at  $2 \tau = 160$  s (visible in trajectory.csv)

Levenspiel, Chemical Reaction Engineering, ch. 14 (the tanks-in-series model): the variance ratio  $1/N$  is THE diagnostic the lesson teaches – one tank gives 1, three give  $1/3$ , a PFR gives 0. ctrl18 is the  $N = 1$  control; this case is  $N = 3$  on identical tanks.

WHAT THE ANCHORS TOLERATE, stated: the engine measures moments by trapezoid on the accepted 1 s grid and truncates at 1600 s (= 20  $\tau$  of one tank); both effects are far inside the anchor bands, and the router adds a one-step transport delay per connection (2 s total on  $t_{bar}$ ’s 240 s – inside the 1 % band, and the honest price of an explicit scheme, stated rather than hidden).

**What to expect (golden-locked):**

| kind | unit/stream | key                        | expected           |
|------|-------------|----------------------------|--------------------|
| kpi  | balance     | energy_balance_available   | 0                  |
| kpi  | balance     | mass_closure_rel           | 2.15876699233e-16  |
| kpi  | balance     | mass_kg_final              | 0.36               |
| kpi  | balance     | mass_kg_in_cum             | 7.2                |
| kpi  | balance     | mass_kg_initial            | 0.36               |
| kpi  | balance     | mass_kg_out_cum            | 7.2                |
| kpi  | balance     | mass_residual_kg           | -1.55431223448e-15 |
| kpi  | balance     | material_available         | 1                  |
| kpi  | rtd         | area_kmol                  | 0.000198752512114  |
| kpi  | rtd         | sigma2_s2                  | 19198.7859557      |
| kpi  | rtd         | tbar_s                     | 241.502454857      |
| kpi  | rtd         | truncation_final_over_peak | 1.54894525802e-06  |
| kpi  | tank1       | F_in_final                 | 5e-05              |
| kpi  | tank1       | T_final                    | 330                |

... and 23 more reference values in the case’s *expected*.

**ctrl20\_bode\_cstr**

tutorials/ctrl/ctrl20\_bode\_cstr

**reactor**  
dynamicCSTR

**What it does:** Bode table of the LTI mixing tank: outer frequency sweep (one datum, lockstep mirror) over the engine frequency-response fit; anchors are the exact first-order closed forms at each point.

**The story:** THE BODE TABLE of the LTI mixing tank ( $\tau = 240$  s): an outer sweep drives the sinusoidal tracer substitution at four frequencies,  $\omega\tau = 0.5, 2.0, 3.5, 5.0$  (linear grid in  $f$ ), and the engine's frequencyResponse fit measures amplitude and phase at each – the sweep CSV IS the Bode table. The measurement block references the drive, so each pass's fit window adapts to that pass's own period, and the sweep varies ONE datum (the drive frequency, mirrored to the carrier by linkedTargets – the same value, lockstep, never a second independent number).

ANCHORS: the exact first-order closed forms at each grid point,  $AR = 1/\sqrt{1+(wt)^2}$ ,  $\text{phase} = -\text{atan}(wt)$ :  $wt=0.5$ :  $|out| = 2e-6 \cdot 0.894427 = 1.788854e-6$ ,  $\phi = -0.463648$  rad  $wt=2.0$ :  $|out| = 2e-6 \cdot 0.447214 = 8.944272e-7$ ,  $\phi = -1.107149$  rad  $wt=3.5$ :  $|out| = 2e-6 \cdot 0.274721 = 5.494423e-7$ ,  $\phi = -1.292497$  rad  $wt=5.0$ :  $|out| = 2e-6 \cdot 0.196116 = 3.922323e-7$ ,  $\phi = -1.373401$  rad Phase bands cover the  $\omega\tau dt/2$  half-sample bias, which grows with  $\omega$  (0.10 to 1.0 mrad here) – stated, not hidden.

**What to expect (golden-locked):**

| kind | unit/stream       | key                                      | expected          |
|------|-------------------|------------------------------------------|-------------------|
| csv  | sweep_results.csv | 0:frequencyResponse.out_amplitude_kmol_s | 1.788854e-6       |
| csv  | sweep_results.csv | 1:frequencyResponse.out_amplitude_kmol_s | 8.944272e-7       |
| csv  | sweep_results.csv | 2:frequencyResponse.out_amplitude_kmol_s | 5.494423e-7       |
| csv  | sweep_results.csv | 3:frequencyResponse.out_amplitude_kmol_s | 3.922323e-7       |
| csv  | sweep_results.csv | 0:frequencyResponse.out_phase_rad        | -0.463648         |
| csv  | sweep_results.csv | 1:frequencyResponse.out_phase_rad        | -1.107149         |
| csv  | sweep_results.csv | 2:frequencyResponse.out_phase_rad        | -1.292497         |
| csv  | sweep_results.csv | 3:frequencyResponse.out_phase_rad        | -1.373401         |
| kpi  | CarrierMirror     | MV_final                                 | 4.79049069948e-05 |
| kpi  | CarrierMirror     | PV_final                                 | 0                 |
| kpi  | CarrierMirror     | SP_final                                 | 4.79049069948e-05 |
| kpi  | TracerSine        | MV_final                                 | 2.09509300524e-06 |
| kpi  | TracerSine        | PV_final                                 | 0                 |
| kpi  | TracerSine        | SP_final                                 | 2.09509300524e-06 |

... and 20 more reference values in the case's *expected*.

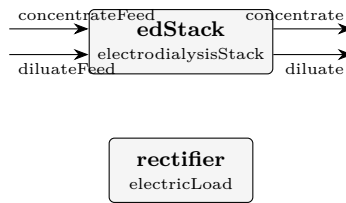


## Category V: Electrochemical systems

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**ed01\_nacl\_desalination**

tutorials/electrochem/ed01\_nacl\_desalination



**What it does:** Electrodialysis stack: NaCl diluate desalination, 100 cell pairs

**The story:** ED01 NaCl desalination by electrodialysis

An electrodialysis stack of 100 cell pairs desalinates a dilute NaCl diluate stream (0.1 mol/kg) against a more-concentrated concentrate (0.5 mol/kg). A specified stack current (BELOW the limiting current) drives Na<sup>+</sup> through the CMX (cation-exchange) membranes and Cl<sup>-</sup> through the AMX (anion-exchange) membranes, OUT of the diluate and INTO the concentrate.

diluateFeed → + ————— + → diluate (desalinated) | ED stack | concentrateFeed → | (CMX/AMX, N=100) | → concentrate (enriched) + ————— + | (power, energy wire) v rectifier (electricLoad)

The stack PRINTS (verbosity 3) the Faraday transfer, the Nernst membrane potential from REAL Davies activities, the Cowan-Brown limiting current  $i_{lim}$  (and  $i/i_{lim}$ ), the ohmic drop and the stack voltage/power.

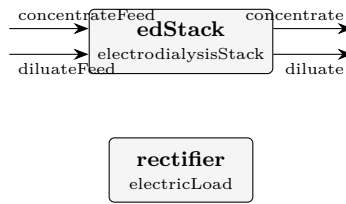
**What to expect (golden-locked):**

| kind   | unit/stream     | key                | expected         |
|--------|-----------------|--------------------|------------------|
| stream | concentrate     | F                  | 0.00501492454831 |
| stream | concentrateFeed | F                  | 0.005            |
| stream | diluate         | F                  | 0.00498507545169 |
| stream | diluateFeed     | F                  | 0.005            |
| kpi    | edStack         | E_mem_pair         | 0.0794573845707  |
| kpi    | edStack         | I                  | 8                |
| kpi    | edStack         | N_cellpairs        | 100              |
| kpi    | edStack         | R_pair             | 0.00507753048751 |
| kpi    | edStack         | U                  | 13.5077628471    |
| kpi    | edStack         | W_electric         | 108.062102777    |
| kpi    | edStack         | W_electric_kW      | 0.108062102777   |
| kpi    | edStack         | current_efficiency | 0.9              |
| kpi    | edStack         | demin_ratio        | 0.831450327631   |
| kpi    | edStack         | i_density          | 40               |

... and 17 more reference values in the case's *expected*.

**ed02\_over\_limiting\_current**

tutorials/electrochem/ed02\_over\_limiting\_current



**What it does:** Electrodialysis stack driven ABOVE its limiting current (loud warning)

**The story:** ED02 over-limiting current electrodialysis

Same stack as ED01, but run on a MORE DILUTE diluate (0.02 mol/kg) at a LOW cross-flow velocity and a HIGH current. The diluate-side limiting current density  $i_{lim}$  (Cowan-Brown) is small; the applied current density EXCEEDS it, so the stack runs in the OVER-LIMITING regime.

Choupo does NOT silently clamp the current (the "no silent crutch" credo): it computes the unclamped Faraday transfer and prints a LOUD warning that  $i > i_{lim}$  – the diluate boundary layer is ion-depleted, water splitting / pH swings set in, and the real current efficiency would collapse. The student SEES the regime and the  $i/i_{lim}$  ratio, then chooses to reduce the current or raise the flow / feed concentration.

**What to expect (golden-locked):**

| kind   | unit/stream     | key                | expected         |
|--------|-----------------|--------------------|------------------|
| stream | concentrate     | F                  | 0.00500205212539 |
| stream | concentrateFeed | F                  | 0.005            |
| stream | diluate         | F                  | 0.0199979478746  |
| stream | diluateFeed     | F                  | 0.02             |
| kpi    | edStack         | E_mem_pair         | 0.156590406589   |
| kpi    | edStack         | I                  | 5.5              |
| kpi    | edStack         | N_cellpairs        | 20               |
| kpi    | edStack         | R_pair             | 0.0130038886532  |
| kpi    | edStack         | U                  | 6.06223588363    |
| kpi    | edStack         | W_electric         | 33.34229736      |
| kpi    | edStack         | W_electric_kW      | 0.03334229736    |
| kpi    | edStack         | current_efficiency | 0.9              |
| kpi    | edStack         | demin_ratio        | 0.142508707778   |
| kpi    | edStack         | i_density          | 27.5             |

...and 17 more reference values in the case's *expected*.

## Category VI: Multi-sector plant showcases

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The fractal flowsheet at full size – read them like worked examples.

### ChemicalPlantTutorial

**The theory you need.** The full multi-sector plant showcase: feed treatment, reaction, separation, utilities – the fractal flowsheet at its largest, run-only (no golden), for reading and dissecting rather than reproducing numbers.

**CONCENTRATION**

tutorials/plant/ChemicalPlantTutorial/CONCENTRATION

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE)

Double-effect evaporator (vapour of effect 1 heats effect 2) feeding a crystalliser. Same shape as the plant flowsheetDict, one level down: children + boundary + default feeds + connections.

**DRYING**

tutorials/plant/ChemicalPlantTutorial/DRYING

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE)

Spray dryer → additional solid drying → cyclone dust separator. The solid powder goes SD → BD → product; the spray-dryer exhaust air (carrying fines) goes SD → CY, which returns clean air + recovered dust.

## FERMENTATION

tutorials/plant/ChemicalPlantTutorial/FERMENTATION

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE with INTERNAL RECYCLE)

Continuous fermentation side-loop of the sugar plant: a slip-stream of fresh juice (Must) is hydrolysed + fermented into ethanol + CO<sub>2</sub>, the Flash splits the volatiles (EthanolVapour product) from the liquid bottoms, and a Splitter recycles 80 % of the bottoms back to the Mixer so unconverted sucrose gets another shot. Pedagogical purpose:

1. Reaction (Arrhenius first-order in sucrose). 2. Vapour-liquid flash for separation. 3. Tear-stream RECYCLE that the Wegstein/Newton outer loop must iterate to a fixed point.

Must [Mixer] MixedFeed [Fermentor] FermOut [Flash] EthanolVapour Liquid Recycle [Splitter]

Tear stream: Recycle (Splitter output → Mixer input).

## JuiceSplitter

tutorials/plant/ChemicalPlantTutorial/JuiceSplitter

**The story:** / system/flowsheetDict (LEAF – plant-level juice split)

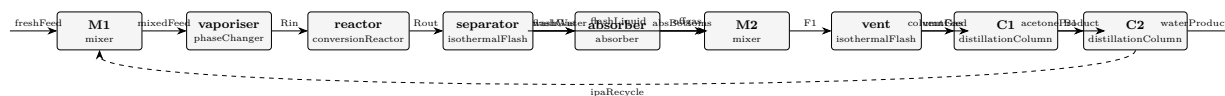
Diverts the fresh sugar-cane juice between the two production lines of the mill: 2/3 goes to the CONCENTRATION line (evaporators → crystals → dry sugar) and 1/3 to the FERMENTATION line (yeast → ethanol). This is what a real Brazilian sugar-ethanol plant does after the clarifier: the same juice header feeds both products, with the split ratio set by market price + crop quality.

Pedagogically the splitter is the single inter-sector cable that ties the two downstream branches together – without it, FERMENTATION would be a parallel island.



## acetonePlant

tutorials/plant/acetonePlant



**What it does:** Acetone from 2-propanol: Luyben's Figure 1 closed, with the IPA recycle

**The story:** ACETONE FROM 2-PROPANOL, THE CLOSED FLOWSHEET

W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a – his Figure 1 (the Turton base case), end to end:

fresh feed + IPA recycle → vaporiser → reactor → separator → absorber (offgas washed) → C1 → acetone product → C2 → water product → IPA RECYCLE (tear)

This is the case the six preceding ones were built to make possible. Each of them isolates one unit on Luyben's OWN published inlet, so that whatever it gets wrong is that unit and its thermodynamics. Here nothing is published except the fresh feed: every internal stream is the plant's own answer, and the errors compose.

WHAT THIS CASE CLAIMS, AND WHAT IT DOES NOT. It claims to be Luyben's TOPOLOGY and Luyben's EQUIPMENT SPECIFICATIONS solved on the thermodynamics Choupo actually has. It does NOT claim to reproduce his stream table – the siblings measured, in advance, exactly why it cannot, and the gap between his table and this one is the deliverable rather than a defect to be closed:

acetone01: UNIFAC puts the IPA/water azeotrope at  $x$  0.590 against his 0.673, and the lake-estimated isopropanol boils 4.5 K high. acetone04: UNIFAC INVENTS an acetone/water azeotrope at  $x$  0.978 where his UNIQUAC has none – 0.044 K deep, and decisive. acetone06: so C1 caps at 97.3 mol % acetone against his 99.9 % spec. acetone07: so C2's recycle caps at  $x_{\text{IPA}}$  0.580 against his 0.650, and 11.6 % of the isopropanol entering it leaves in the water product.

Both column ceilings were WRITTEN DOWN BEFORE the columns existed and are pinned by `check_azeotrope_ceiling`. This case is where they stop being two separate findings and become one plant.

DEPARTURES FROM HIS FIGURE 1, each argued rather than absorbed

(a) NO HYDROGEN IN THE COLUMNS. His C1 has a partial condenser and a vent (0.0465 kmol/h, carrying 0.0130 kmol/h H<sub>2</sub> and 0.0335 kmol/h acetone); Choupo's column has a total condenser and no vapour distillate. A permanent gas fed to a total condenser is asked to leave as a liquid, and the engine rightly refuses (hydrogen has no liquid heat capacity at 330 K). So the separator's LIQUID goes to C1 and the hydrogen leaves with the offgas, i.e. the columns are modelled downstream of his vent. Cost, stated: the 0.0335 kmol/h of acetone his vent loses is not lost here, which flatters this plant's acetone recovery by 0.10 %.

(b) THE SEPARATOR DOES THE COOLING. An isothermal flash at 318 K computes the duty to bring its feed there, so this one unit stands for his HX1 plus his separator and its  $Q$  is directly comparable to his 0.8993 MW. A separate 'heater' was tried in acetone03 and refused, correctly: Choupo's heater takes a DUTY and reports the temperature, because a heater with a specified outlet temperature is a design specification.

(c) THE SEPARATOR PRESSURE IS ARGUED, NOT FITTED. Luyben states 1.5 atm for the absorber gas, which is downstream; there is no compressor between reactor and separator and pressure only falls, so the separator sits at the reactor's 2.6 atm. acetone03 records the sweep that would have tempted a fitted value and why it was refused.

(d) THE ABSORBER'S SECOND OUTLET REJOINS C1's FEED, as in his Figure 1 (his F1 of 72.85 kmol/h is the flash liquid AFTER the absorber bottoms of 20.05 kmol/h rejoin it).

(e) NO VENT/PURGE ON THE RECYCLE LOOP. Luyben has none either – the loop closes through C2's overhead and nothing else – so this is his structure, not a simplification. It does mean any component

that both enters and cannot leave would accumulate without bound; here water leaves in three places and isopropanol in two, so the loop is closed and bounded.

*These are the dictionaries of the closed acetone flowsheet described in [../acetone-plant-closure-state.md](../acetone-plant-closure-state.md). It does not converge yet, and the reason is one named engine gap (F3 in that record: no unit can remove a component completely).*

*tutorials/ is the verification and regression corpus. Everything in it is run by bin/runTests and is expected to answer. Putting a case there that cannot answer would have meant claiming one of two markers, and neither is true of this case:*

*\* .expect-nonconvergence — every one of the five cases carrying it is a DELIBERATE TEACHING REFUSAL (flash15\_refused\_salt\_basis\_rank, tsa02\_refused\_capacity, ?): the refusal \*is\* the lesson, and the case is finished. This one is not refusing anything on purpose; it is unfinished. \* .known-broken — the corpus carries ZERO of these, and that zero is a claim the project makes about its own health. Spending it to park work in progress would be paying a real claim for my convenience.*

*So the case sits beside the record that explains it, makes no corpus claim, and moves to tutorials/plant/acetonePlant/ the day it converges — at which point it gets a golden, a seal and an entry in the case manifest like any other case.*

*0/ is NOT kept: it is generated by bin/choupo-init0 from the authored inlets plus the tear seed, and a stale copy of a generated state is a second home for numbers the tool derives. The tear seed itself — the one part of 0/ that is authored rather than propagated — is described in the closure record.*

*The absorber's second thermodynamic world is declared inline in system/flowsheetDict as a per-unit thermo {} block, not as a separate file, because that is where the model boundary belongs and where the run announces it.*

#### What to expect (golden-locked):

| kind   | unit/stream    | key | expected         |
|--------|----------------|-----|------------------|
| stream | B1             | F   | 0.0118946232842  |
| stream | F1             | F   | 0.0195617446099  |
| stream | Rin            | F   | 0.0160666666667  |
| stream | Rout           | F   | 0.0255036799795  |
| stream | absBottoms     | F   | 0.00608647733675 |
| stream | acetoneProduct | F   | 0.00762772388347 |
| stream | columnFeed     | F   | 0.0195223471677  |
| stream | flashGas       | F   | 0.0120284127064  |
| stream | flashLiquid    | F   | 0.0134752672731  |
| stream | freshFeed      | F   | 0.0144333333333  |
| stream | ipaRecycle     | F   | 0.00163333333333 |
| stream | mixedFeed      | F   | 0.0160666666667  |
| stream | offgas         | F   | 0.0114974909252  |
| stream | ventGas        | F   | 3.9397442245e-05 |

*... and 155 more reference values in the case's expected.*

## esterification2sector

tutorials/plant/esterification2sector

**What it does:** Esterification plant – two sectors (REACTION + SEPARATION) sharing components; per-sector constant/ data (S1 kinetics, S2 NRTL pair); ethanol-water pair inherited from the standard library

*A two-sector plant that exists to prove one thing: data lives with the sector that owns it, and a whole-plant run finds it there.*

*Components are shared across the plant. The kinetics belong to REACTION and are never copied to the root. The ethylAcetate-water NRTL pair belongs to SEPARATION — a *\*particular\** datum, fitted for this separation — while ethanol-water walks up to the standard catalogue. Three levels of ownership, one run.*

*That perNode is the whole case. The sector's pair beat the standard library, and the log says so rather than leaving you to wonder which one it used.*

*Feed 100 kmol/h equimolar acetic acid + ethanol at 360 K. The CSTR ( $V_R = 0.5 \text{ m}^3$ ) reaches 34.2 % conversion — a *PLACEHOLDER* number: the kinetics in sectors/REACTION/constant/reactions are still the pseudo-first-order stand-in, not the 2nd-order fit this case's own property evidence supports (see CLAUDE.md, "Pending"). The flash at 355 K, 1.013 bar splits the product into a vapour rich in ethyl acetate (bp 350 K) and ethanol, and a liquid rich in water and the unconverted acid. Mass closes exactly:  $34.18 + 65.82 = 100.00 \text{ kmol/h}$ .*

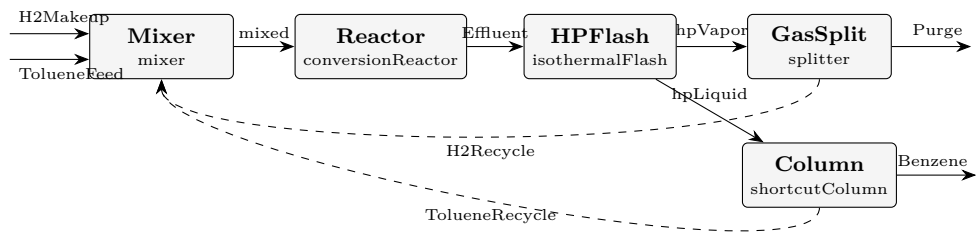
*This case shipped for weeks marked .known-broken, and it earned the mark twice over. Its flash carried  $P$  1.01325; with no unit, which is 1.01 pascal — a hard vacuum — so the  $K$ -values exploded and everything went to the vapour. And it was never wired: the sectors existed as thermo regions with the connections commented out, awaiting a "curation phase" that never ended.*

*Fixing it surfaced two real engine bugs, both of the same shape — data owned by a *\*sector\** was invisible from a whole-plant run:*

**What to expect (golden-locked):**

| kind    | unit/stream      | key                          | expected         |
|---------|------------------|------------------------------|------------------|
| stream  | Feed             | F                            | 0.02777777777778 |
| stream  | liquid           | F                            | 0.0114069310496  |
| stream  | reactorOut       | F                            | 0.02777777777778 |
| stream  | vapor            | F                            | 0.0163708467282  |
| utility | SEPARATION.flash | cooling.coolingWater.duty_kW | -229.173881338   |
| utility | SEPARATION.flash | cooling.coolingWater.T       | 355              |
| utility | SEPARATION.flash | cooling.coolingWater.kg_s    | 5.48262874014    |
| utility | SEPARATION.flash | cooling.coolingWater.eur_h   | 0.330010389127   |
| stream  | Feed             | T                            | 360              |
| stream  | liquid           | T                            | 355              |
| stream  | reactorOut       | T                            | 360              |
| stream  | vapor            | T                            | 355              |

**hda**  
tutorials/plant/hda



**What it does:** HDA plant – toluene + H2 → benzene + CH4. A single FLAT flowsheet with two plant-level recycles (an H2 gas loop and an unreacted-toluene liquid loop); cross-sector tears are unsupported, so it is not fractal. Furnace lumped into the isothermal reactor; light-gas K-values rest on Antoine extrapolation (see thermoPhysPropDict).

**What to expect (golden-locked):**

| kind                                                            | unit/stream    | key           | expected         |
|-----------------------------------------------------------------|----------------|---------------|------------------|
| stream                                                          | Benzene        | F             | 0.0308570333668  |
| stream                                                          | Effluent       | F             | 0.589422462034   |
| stream                                                          | H2Makeup       | F             | 0.0305555555556  |
| stream                                                          | H2Recycle      | F             | 0.522049697299   |
| stream                                                          | Purge          | F             | 0.0274762998579  |
| stream                                                          | TolueneFeed    | F             | 0.0277777777778  |
| stream                                                          | TolueneRecycle | F             | 0.00903943150968 |
| stream                                                          | hpLiquid       | F             | 0.0398964648764  |
| stream                                                          | hpVapor        | F             | 0.549525997157   |
| stream                                                          | mixed          | F             | 0.589422462034   |
| kpi                                                             | Column         | B             | 0.00903943150968 |
| kpi                                                             | Column         | D             | 0.0308570333668  |
| kpi                                                             | Column         | N_min         | 8.3953506183     |
| kpi                                                             | Column         | N_theoretical | 20.1717591626    |
| ...and 48 more reference values in the case's <i>expected</i> . |                |               |                  |

## lithiumBrinePlant

tutorials/plant/lithiumBrinePlant

**What it does:** lithiumBrinePlant – THE reference case: a FRACTAL lithium-carbonate plant from salar brine, WIRED into one solver. Five SECTORS, each a thermo WORLD of its own – BRINE (electrolyte/Pitzer), EXTRAC-TION (gamma-gamma NRTL LLE), CARBONATION (Li<sub>2</sub>CO<sub>3</sub> retrograde solubility), HOTAIR (Gibbs combustion), FINISHING (drying + cyclone) – connected sector to sector, each sector declaring its own propertyPackage (its world) on the global component set, with optional per-unit thermo overrides where a single unit needs a different world. The proof that the architecture is fractal: sectors are composite boxes, streams flow between them, model boundaries audited.

**The story:** Five REAL sectors. Streams are named physical graph edges. Sector names are subdomains; unit operations live inside each sector.

### What to expect (golden-locked):

| kind   | unit/stream   | key | expected          |
|--------|---------------|-----|-------------------|
| stream | carbLiquor    | F   | 0.02777777777778  |
| stream | cleanAir      | F   | 0.0320991487231   |
| stream | emulsion      | F   | 0.0380203541089   |
| stream | finest        | F   | 0                 |
| stream | fuelAir       | F   | 0.01085555555558  |
| stream | halite        | F   | 0.000868534779985 |
| stream | hotAir        | F   | 0.01085555555556  |
| stream | humidExhaust  | F   | 0.0320991487231   |
| stream | liquor        | F   | 0.0269092429978   |
| stream | loadedOrganic | F   | 0.0124069292995   |
| stream | organic       | F   | 0.01111111111111  |
| stream | product       | F   | 0.00653418461064  |
| stream | raffinate     | F   | 0.0256134248094   |
| stream | salarBrine    | F   | 0.02777777777778  |

*...and 16 more reference values in the case's **expected**.*

## polycaprolactonePlant

**The theory you need.** Ring-opening polymerisation plant: monomer to polycaprolactone with step-growth KPIs (Carothers, Flory) on the PFR, polymer properties from group contribution on the repeat unit.

## properties

tutorials/plant/polycaprolactonePlant/properties

**pclRepeat\_Tg\_yang2020**  
estimateComponent

**What it does:** Close the loop: ESTIMATE the PCL repeat-unit glass transition Tg by the Yang 2020 group-contribution scheme — the plant gives Mn/PDI, this gives the physical property from the SAME repeat unit  $-\text{O}-(\text{CH}_2)_5\text{CO}-$

**The story:** pclProperties – CLOSE THE LOOP: process  $\rightarrow$  molar mass  $\rightarrow$  physical property.

The plant (../) gave you the chain statistics from the KINETICS: Mn, Mw, PDI on the polycaprolactone it makes. Here you ESTIMATE the polymer's PHYSICAL PROPERTIES from the SAME repeat unit  $-\text{O}-(\text{CH}_2)_5\text{CO}-$  by group contribution — the bridge from "how big are the chains" to "how does the material behave".

TWO estimators, two properties, on the repeat unit pclRepeat ( $M_0 = 114.14$ ):

(A) Tg, the GLASS-TRANSITION temperature — Yang 2020 (run below). Repeat unit  $-\text{O}-(\text{CH}_2)_5\text{CO}-$  decomposes into Yang main-chain groups: O x1 : MW 16 Yg -14.718 CH<sub>2</sub> x5 : MW 14 Yg 4.026 (each) CO x1 : MW 28 Yg 4.370

$M_0 = 16 + 5 \times 14 + 28 = 114$  g/mol (matches the reactor  $M_0$ !)  $Yg(\text{inf}) = -14.718 + 5 \times 4.026 + 4.370 = 9.782$  (1e3 g.K/mol)  $Tg(\text{inf}) = 9.782 \times 10^3 / 114 \sim 86$  K (the additive estimate) (Measured PCL Tg  $\sim 213$  K / -60 C; an additive main-chain scheme is weak for a flexible aliphatic polyester — the model tells the truth about itself, which IS the lesson. See the console's per-cent error vs the anchor.)

(B) DENSITY — Van Krevelen (Bondi Vw). NOT runnable on PCL with the shipped data: the Slice-1 Van Krevelen set ships only hydrocarbon + chloro groups (CH<sub>3</sub> CH<sub>2</sub> CH C ACH AC CHCl CH<sub>2</sub>Cl); it has NO ester -O- / -CO- group, so the repeat unit  $-\text{O}-(\text{CH}_2)_5\text{CO}-$  cannot be decomposed. Running it would HARD- ERROR with "unknown polymer group 'O'" — the estimator refuses to invent a value (no silent zero). The honest path: density needs the ester groups curated into data/standards/parameters/vanKrevelen.dat first. The worked Van Krevelen DENSITY example lives in tutorials/props/estimate/vanKrevelen01\_polystyrene\_density (a pure- hydrocarbon repeat unit the Slice-1 set CAN do).

This is the same  $M_0 = 114.14$  the reactor's polymer{ } block used for  $M_n = M_0/(1-p)$  — one repeat-unit mass, two places (process AND property).

*A choupouProps sub-case of [../](../) (the polycaprolactone plant). The plant gives you the chain statistics from the kinetics (Mn / Mw / PDI); here you estimate the polymer's physical property — the glass transition Tg — from the same repeat unit  $-\text{O}-(\text{CH}_2)_5\text{CO}-$  ( $M_0 = 114.14$ , the exact number the reactor used for  $M_n = M_0/(1-p)$ ).*

*\* Tg via Yang 2020 (CC-BY) — runnable.  $Tg(\text{inf}) = 85.8$  K; the console prints the -59.7 % deviation from the measured  $\sim 213$  K (an honest limit of an additive scheme on a flexible aliphatic polyester).  $M_0 = 114$  is golden (an exact mass-balance fact); Tg is shown but not golden as validated physics. \* Density via Van Krevelen — \*not\* runnable on PCL with the shipped data (the Slice-1 group set has no ester -O-/-CO- group, so it would hard-error rather than invent a value). See the full explanation and the runnable hydrocarbon example linked from the parent [../README.md](../README.md).*

*The full teaching narrative lives in the parent README. See [system/propsDict](system/propsDict) for the exact group decomposition and the estimator invocation.*

**What to expect (golden-locked):**

| kind | unit/stream           | key            | expected |
|------|-----------------------|----------------|----------|
| diag | pclRepeat_Tg_yang2020 | M0_g_per_mol   | 114      |
| diag | pclRepeat_Tg_yang2020 | Tg_K           | 85.81    |
| diag | pclRepeat_Tg_yang2020 | YgSum_1e3gKmol | 9.782    |

## sugarPlantEconomicsSweep

tutorials/plant/sugarPlantEconomicsSweep

**What it does:** Sugar plant ECONOMICS SWEEP – IRR / payback / NPV vs RawJuice sucrose content (Phase-2 SweepDriver post-chain differentiator)

**The story:** / system/flowsheetDict (PLANT level – COMPOSITE)

The flagship sugar plant (the ChemicalPlantTutorial flowsheet, reused verbatim – sectors and units included), here driven by system/outerDict: a SweepDriver walks the RawJuice sucrose fraction and runs the FULL post chain (sizing → costing → economics, system/postDict) on every point, so IRR / NPV / payback vs feed quality come out as sweep responses.

A composite node: it has ‘children’ (sub-flowsheets, here the sectors) and ‘connections’ (the pipework between them). The same shape a sector uses to list its units – one level up. This is the fractal.

Naming convention (see docs/ai/case-layout.md): SECTORS in CAPS : CONCENTRATION, DRYING, FERMENTATION Units in PascalCase : Cryst, Evap1, Evap2 Streams in PascalCase : RawJuice, Magma, Powder, EvapVapour Components in lowercase : water, sucrose, N2

*This is the flagship sugar plant (tutorials/plant/ChemicalPlantTutorial) driven by a SweepDriver instead of a single pass. It exists to show the one thing a single CAPEX number cannot: how the project economics move when a process variable changes.*

*The driver runs the FULL post-processing chain (sizing → costing → economics, in system/postDict) on every converged point, so economics.IRR, economics.paybackYears, economics.NPV, ? resolve as ordinary KPI look-ups (the Phase-2 SweepDriver post-chain wiring). The result lands in sweep\_results.csv.*

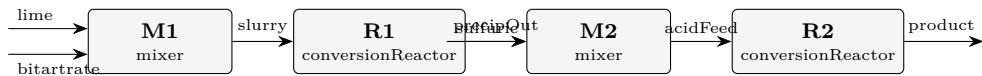
*Sweeping the RawJuice sucrose content 0.09 → 0.16 mol fraction walks the plant from sub-economic, through break-even, to clearly profitable:*

*Richer juice → more crystalline sugar per unit feed → more revenue against an almost-unchanged plant cost → the IRR climbs and the payback appears and shortens. The NPV crosses zero between 0.104 and 0.118.*

*Both use the same constant/economics prices (EU Sugar Market Observatory white sugar, ECB EUR/USD, \*Chemical Engineering\* CEPCI, US statutory tax) and the same system/postDict cost chain — only the run mode differs.*

tartaricAcid

tutorials/plant/tartaricAcid



**What it does:** Tartaric-acid plant (wine route): lime precipitation of calcium tartrate, then sulfuric acidulation to tartaric acid + gypsum. Molecular reactors; the solution chemistry sub-study is tutorials/props/electrolyte/-tartaricAcid\_acidulation. HONEST LIMITS: H2SO4 declared nonvolatile by case overlay (P<sub>sat</sub> ~1e-5 Pa at process T); precipitates ride as nonvolatile solutes with NO rigorous solid-liquid separation; reactor duties honestly UNAVAILABLE (species without ideal-gas C<sub>p</sub> – a named gap, never a fake zero).

What to expect (golden-locked):

| kind                                                            | unit/stream | key   | expected       |
|-----------------------------------------------------------------|-------------|-------|----------------|
| stream                                                          | acidFeed    | F     | 0.943266666667 |
| stream                                                          | bitartrate  | F     | 0.583333333333 |
| stream                                                          | lime        | F     | 0.166666666667 |
| stream                                                          | precipOut   | F     | 0.7766         |
| stream                                                          | product     | F     | 0.943266666667 |
| stream                                                          | slurry      | F     | 0.75           |
| stream                                                          | sulfuric    | F     | 0.166666666667 |
| kpi                                                             | M1          | F_out | 0.75           |
| kpi                                                             | M1          | P_out | 100000         |
| kpi                                                             | M1          | T_out | 333            |
| kpi                                                             | M1          | n_in  | 2              |
| kpi                                                             | M1          | vf    | 0              |
| kpi                                                             | M2          | F_out | 0.943266666667 |
| kpi                                                             | M2          | P_out | 100000         |
| ...and 20 more reference values in the case's <i>expected</i> . |             |       |                |

twoSectorDemo

**The theory you need.** The zippable two-sector demonstration plant – the case used to rehearse the online landing. Same physics as the esterification demo.



## SECTOR\_R

tutorials/plant/twoSectorDemo/SECTOR\_R



**What it does:** SECTOR\_R – the reactor sector. Runs its OWN compiled unit op (stoichReactor) on the plant-level thermo (cascaded from ../constant).

**The story:** one custom reactor (compiled from code/). Build + run with: bin/buildCode tutorials/plant/twoSectorDemo/SECTOR\_R runCase tutorials/plant/twoSectorDemo/SECTOR\_R —

## SECTOR\_S

tutorials/plant/twoSectorDemo/SECTOR\_S



**What it does:** SECTOR\_S – the separation sector. Runs its OWN compiled unit op (sharpSplitColumn) on the plant-level thermo (cascaded from ../constant).

**The story:** one custom distillation (compiled from code/). Build + run: bin/buildCode tutorials/plant/twoSectorDemo/SECTOR\_S runCase tutorials/plant/twoSectorDemo/SECTOR\_S —